Constructive Access

A Unified Reference for Inference
under Restricted Physical Access

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13 October 2026

Reference edition 10.0 • Research toolkit 0.10.0

Purpose and use

A scientific apparatus preserves some distinctions and erases others. This reference develops a unified way to determine which target properties remain identifiable, construct interventions that reveal them, and connect their probability contrasts to calibrated inference and physical expenditure.

The central object is the law of the complete accessible record under a declared experiment. General results describe targets on observational equivalence classes. Finite classical and quantum instruments admit exact algebraic certificates or finite witnesses. Further layers treat continuous time, calibration uncertainty, finite-error decisions, model adequacy and resource limits. Worked examples address hidden currents, chemical channel flux, quantum error mechanisms, Ramsey sensing and inaccessible power consumption.

The seventh edition adds a direct finite-lag dissipation inequality, joint analog acquisition, a fully covered contrast-selection library, and explicit physical attainability and temporal-aliasing results. Chapter 39 provides the entry route to these developments. The older bounds remain useful for their broader contrast and access contracts; the new result strengthens the specified three-lag case. All claims remain conditional on the declared physical reversal, stationarity and detector assumptions.

The eighth edition uses observed local diffusion geometry to close a physical minimum over compatible ordinary hidden diffusions of arbitrary finite dimension for a specified family of scalar process laws. The scalar records can themselves be reversible. A finite-resolution variational certificate, explicit bias control and frozen same-record comparisons connect the population theorem to statistical inference. Chapter 45 introduces this layer; its physical and measurement assumptions are essential.

The ninth edition adds an explicit detector layer: population recovery through known exposure and noise, a finite-data obstruction for the sharp physical minimum, an exact fixed-grid ambiguity, and matching sampling and declared measurement-energy orders for a controlled circle comparison. Chapter 52 connects these results and distinguishes ideal identification from physical resource requirements.

The tenth edition makes detector calibration part of the inference contract. It combines jointly covered nuisance parameters, finite-error diffusion certificates, an auditable virtual experiment, and exact limits on learning global calibration constants from finite passive records. Chapter 56 separates structural assumptions from measured evidence. A sharp hidden-drive theorem also shows how zero mean circulation and a reversible scalar record can coexist with an attained positive dissipation minimum over the stated compatible diffusion class.

The organizing question is what every compatible explanation must contain. The answer may be a distinguishable mechanism, a minimum classical state count, a preparation resource, or sustained dissipation. These requirements are conditional on experimental access. The reference develops both constructive certificates and countermodels that show when a stronger conclusion is impossible.

The physical development begins with an all-time finite-pulse response constraint derived from first-passage theory. It continues through robust preparation and detector contracts, sharp reset and

crossing-intensity bounds, and finite-record classical dimension witnesses. Graph access completion, shared calibration nuisance, acquisition-energy design and composition make these results usable in larger experimental workflows.

The present extension supplies a finite-data hierarchy of hidden-state requirements from one preparation, one repeated channel and one detector. A separate Erlang-clock certificate converts an established spectral obstruction into a finite-observation test without hidden-rate bounds or exact pole fitting. Explicit classical realizations identify the boundaries of both interpretations. In particular, arbitrary finite fresh-reset endpoint records can be approximated by finite irreducible detailed-balance models when hidden states, rates and preparation resources are unrestricted.

Stationary pair observations provide a different route. Reversible Markov correlations are nonnegative mixtures of decaying exponentials. Certified departures from that class impose sustained entropy-production requirements, independent of hidden-state count. A nonstationary extension quantifies the tradeoff between preparation divergence and integrated housekeeping entropy. Exact polynomial and relative-entropy certificates connect this structure to finite uncertain data. The software includes a curvature example that yields information missed by both pairwise time-reversal and monotonic-revival tests.

The virtual-laboratory extension subjects these certificates to simulated physical dynamics and apparatus imperfections. Independent trajectory and open-system solvers generate records; inference receives only its declared observations. Exact results identify which clock laws and exposure blocks preserve the comparison class. Conditional-mean confidence bounds correct correlated sampling, while fixed polynomial contrasts avoid an unnecessary simultaneous-box penalty for predeclared targets. Matched comparisons retain failed regimes, calibration costs and stronger baseline methods.

The calibrated-inference extension asks which additional measurements sharpen these conclusions. It supplies calibrated comparison classes, nonlinear information certificates, an activity-dependent entropy bound, limitations on total dissipation inference, sequential acquisition, and sparse dependent sampling. A scoped overdamped-diffusion corollary transfers the record-information bounds beyond finite jump processes. Direct bounded analog scores preserve information otherwise lost through randomized binary encoding. A public optical-tweezer dataset supplies an exploratory test of the observation assumptions and a concrete calibration gap, rather than an unsupported thermodynamic conclusion.

Worked counts are synthetic or prospective except in the explicitly identified reanalysis of previously published optical-tweezer measurements. Conservative sample bounds are reported even when large; unmatched examples are not presented as performance advantages. The earliest simple survival assay loses to an established nonlinear baseline. Core spectral, realization, combinatorial and thermodynamic ingredients are attributed to prior work. Candidate originality concerns the stated robust constructions and operational integration, and historical priority remains unresolved.

All mathematical guarantees refer to stated model and access classes. The worked studies include analytical proofs, simulated benchmarks, and explicitly labeled exploratory reanalysis of published measurements. None supplies a newly measured apparatus or an empirical entropy-production certificate. The reference has undergone internal adversarial review and reproducibility checks; it has not undergone external peer review or empirical validation on a newly measured apparatus. The release ledger distinguishes those forms of evidence.

Start here. Chapter 1 explains the question, the access contract and a complete two-state example. The subsequent chapters develop the theory. Chapter 64 gives the full target-specific workflow. The toolkit chapter supplies installation and commands. All sources, examples, verification logs and prior research provenance accompany the release.

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Notation and conventions

Symbol	Meaning
θ, h	A fixed candidate physical mechanism; h indexes a finite family.
$g(\theta)$	The scientific target: a label, parameter, current or other specified property.
π, Π_C	A permitted experimental policy; policies within a declared resource budget.
P_θ^π	Law of the complete accessible record, including observed action outcomes.
ϖ	A stationary distribution in physical network calculations. Local chapter definitions take precedence when an inherited conventional symbol is retained.
$K^{a,y}$	Classical kernel for action a, recorded outcome y, and the next hidden state.
$\mathcal{E}^{a,y}$	Completely positive quantum instrument component, including its outcome probability and state change.
$\alpha M_w \beta$	A finite linear representation of a probability difference for a chronological word w.
V, d	Accessible effect span and paired representation dimension.
C_t, $g(C_t)$	Retained model set and the corresponding set of possible target values.
$\mathrm{TV}(P, Q)$	Total variation, with convention $\frac{1}{2}\int \lvert p - q \rvert$.
η, ε	Declared calibration radius or measurement error, defined for each use.
α, β, δ	Error budgets in statistical/resource theorems; distinguished locally from linear representation vectors.
Q, G, R	Continuous-time Markov generators; their roles are specified in each model.
J_{ij}	Stationary probability current, with a stated edge orientation.
W, Σ	A declared work/expenditure variable and entropy production; never interchangeable without physical assumptions.

Classical probability distributions are row vectors and Markov generators have zero row sums. Quantum states are density operators; the finite linear representation uses an explicitly chosen real Hermitian basis. A word is chronological: its leftmost action occurs first. Exact software inputs are integers or rational strings. Floating-point approximations are labeled as numerical checks, not exact certificates.

Chapter 1

From a scientific question to an access contract

A scientist sees records, intervenes on a system, and tries to infer a property that is not directly visible. The property might be the direction of a microscopic current, an error mechanism in a quantum device, a biochemical turnover rate, or the power consumed by an inaccessible process. The same difficulty appears in each case: different underlying mechanisms can generate the same accessible evidence.

This reference develops a common way to reason about that difficulty. Its subject is *constructive access*: the relation between a specified scientific target, the physical experiments permitted by an apparatus, the distinctions those experiments preserve, and the precision and resources needed to resolve them. "Constructive" means that a positive result should provide a usable experiment or decision procedure, and a negative result should identify a reproducible obstruction. The name describes the synthesis developed here. It does not assert that a separate academic field has already been established.

The organizing question is simple:

> Given these candidate mechanisms and this apparatus, what can we infer about the property we care about, and what would change that answer?

An answer requires more than a mechanism model. It requires the preparation, controls, recorded outcomes, allowed order of operations, calibration uncertainty, decision objective, and physical resource ledger. Together these form an *access contract*. Every guarantee in the reference is conditional on an explicit contract.

1.1 The outcome scientists should expect

The framework supports four useful outcomes. An *obstruction* proves that a specified distinction is absent from every experiment in the stated access class. A *witness* supplies an experiment whose possible records have different probabilities under the proposed mechanisms. A *precision certificate* establishes how calibration and sampling affect a target estimate or decision. A *resource certificate* bounds a specifically defined expenditure under a justified physical model. These outcomes fit together, but none substitutes for the others.

For example, a nonzero probability contrast proves that information exists in principle. It does not mean a single record determines the answer. A protocol that uses the fewest control actions may require many repetitions. A low dissipation estimate for a visible subsystem does not bound an

unobserved fuel-consuming subsystem. A confidence set that retains one target does not establish that the proposed model family includes the real apparatus.

These distinctions make the framework useful across fields: the logical guarantees transfer, while the physical assumptions remain visible.

1.2 A first complete example

The initial example below introduces the access contract with two finite candidate mechanisms. Later, Chapter 7 moves beyond that finite dictionary: its response-sign theorem covers every finite classical network satisfying a local physical intervention contract. Chapter 9 derives a quantum response boundary, and Chapter 10 makes it a finite-data test. Chapters 60–62 address the additional sensors, precision, energy and interfaces needed for broader use. Chapters 12 and 15 quantify how imperfect resetting and hidden crossing rates can explain a response, including a complete counts-to-confidence calculation. These are different layers of one inference argument. No layer infers the physical validity of another layer's assumptions merely from a matrix input.

Consider a system with two hidden states, 0 and 1. Each experiment starts with probabilities $(1/2, 1/2)$. A read operation reports the current state without changing it. Under either candidate mechanism, repeated reads alone produce the same records: all zeros with probability $1/2$, or all ones with probability $1/2$. The mechanisms cannot be distinguished through this access.

Now permit a drive operation. Under mechanism A it replaces the hidden distribution by $(3/4, 1/4)$; under mechanism B it replaces it by $(1/4, 3/4)$. The drive records only a completion flag. A drive followed by a read gives

$$P_A(Y = 0) = \frac{3}{4}, \qquad P_B(Y = 0) = \frac{1}{4}.$$

The observation has become informative because the permitted experiment changed. No additional processing of the earlier passive records could have produced this distinction.

The complete contract appears in `examples/active.json`; removing the drive gives `examples/passive.json`. The two read kernels are

$$K^{\mathrm{read},0} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad K^{\mathrm{read},1} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

The drive kernels have identical rows, $(3/4, 1/4)$ for A and $(1/4, 3/4)$ for B. Kernel entries jointly describe the outcome and the resulting state. This records measurement disturbance whenever it is present; it never discards an inconvenient outcome.

With a read charge of one and drive charge of two, the least action charge of a distinguishing word is three. An independent algebraic checker verifies this claim. Fresh preparation costs an additional modeled unit per record. The complete record distributions have affinity

$$\sum_y \sqrt{P_A(y)P_B(y)} = \frac{\sqrt{3}}{2}.$$

For equal priors and independent repetitions, the optimal average error is at most half the affinity raised to the record count. Twelve records suffice for the declared 0.1 average-error target. This is a sufficient count under the specified models; the example does not claim it is the smallest possible count. Its full modeled charge includes twelve preparations as well as twelve drive–read sequences. The unit here is a counted operation, not a joule.

The example already contains the whole research problem: define the distinction, prove why the original access fails, add an allowed intervention, find a witness, compute its actual record law, and account for uncertainty and expenditure. Later chapters replace this small model by continuous-time networks, quantum instruments, and calibrated families of mechanisms.

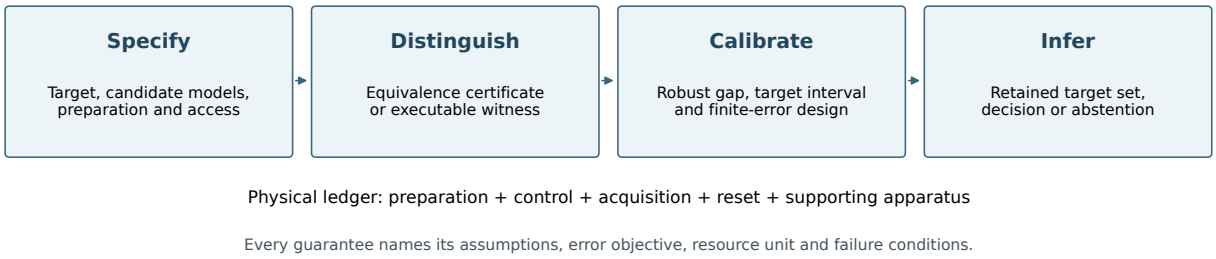


Figure 1.1: The common workflow. Physical assumptions and resource accounting accompany the entire chain, rather than being inferred from a successful algebraic test.

1.3 A target beyond a supplied list of mechanisms

The two-state example compares known alternatives. The physical results later in the reference also bound entire model classes. One example asks how many hidden classical states are needed to explain two preparation experiments. Measure their contrast C_0 at a common reference time, then measure the same contrast C_T at a uniformly randomized later time. Under common homogeneous Markov dynamics and a fixed binary detector, every classical model with at most three states satisfies $|C_0 - C_T| \leq 1$, regardless of its unknown rates, preparation distributions or detector noise. The proof and failure controls appear in Proposition 12.4.

A violation therefore supplies a lower bound on classical realization size. It does not select a unique replacement model: four classical states can explain some violating integrated records. This example illustrates the broader purpose of constructive access. A useful observation can constrain what any explanation must contain, even when reconstructing the mechanism is impossible.

The extension in Chapter 16 asks the same question about larger classical explanations and physical resources. A finite-data hierarchy forces at least five classical states in one worked design. A constructive approximation theorem explains why finite reset records cannot uniformly exclude classical models of unrestricted size and preparation cost. Stationary correlation experiments can instead impose dissipation requirements without selecting a hidden architecture. Together these results identify which added access or resource premise supports the desired conclusion.

1.4 Writing a contract before fitting data

The following questions specify the experiment rather than merely its interpretation.

1. **Target.** What quantity or decision matters? A current's sign, its magnitude, a heat rate, and the full microscopic mechanism are different targets.

2. **Candidate family.** Which systems remain possible, including nuisance parameters, unknown preparations and detector effects? A nuisance parameter fixed for an entire experiment must stay fixed when calculating its joint record law.

3. **Preparation.** What can be reset, sampled or conditioned on? A stationary initial distribution is a substantive premise. Independent repetitions require a justified reset or sampling mechanism.

4. **Operations.** Which controls and measurements can actually be performed, in what order, with what feedback and stopping rules? Unavailable controls do not belong in the witness search.

5. **Record.** What is retained, including time stamps, failed detections, aborts, and control outcomes? Postselection needs its own success probability and resource accounting.

6. **Calibration.** Which complete instrument or record laws are bounded, with what joint confidence? Matching one output marginal does not calibrate hidden-state updates.

7. **Error objective.** Is the goal a worst-case interval, an equal-prior decision error, a posterior risk, or an anytime error guarantee? These objectives need different proofs.

8. **Resources.** Which contributions to time, gross work, net dissipation, calls or storage are included? State the budget and its units, together with excluded apparatus contributions.

A useful contract also names a negative control and a condition that would cause the inference to be withheld. The software provides such outcomes directly.

1.5 The mathematical structure

The framework has one common object: the map from a hidden mechanism and a permitted protocol to a probability law of accessible records,

$$(\theta, \pi) \longmapsto P_\theta^\pi .$$

The parameter θ need not be a scalar. It may represent an entire classical or quantum model. A protocol π includes the permitted sequence of preparations, actions, observations and decisions. A target $g(\theta)$ states what the scientist wants to learn.

The questions then have a natural order. First, do two mechanisms generate identical accessible laws? Second, is the target constant across those indistinguishable mechanisms? Third, can a finite experiment reveal a relevant difference? Fourth, does the difference survive calibration and sampling? Finally, can the experiment be executed within the stated physical constraints?

The first two questions make sense for very general systems. Finite constructive algorithms require additional structure. The exact algebraic engine in this reference handles finite-dimensional linear representations, including finite classical instruments and finite quantum instruments. Continuous-time approximations, uncertainty sets, and fixed resource budgets add further assumptions. A general definition is not a universal finite algorithm.

1.6 How the reference is organized

The foundations chapter defines targets, accessible laws and equivalence. Finite access turns these definitions into certificates and witness searches. Precision and design link exact differences to robust intervals, statistical error and bounded policies. Model adequacy and physical expenditure explain how conclusions can be withheld and how resource guarantees can be composed. Continuous thermodynamics connects the abstract language to actual transition channels, current measurements and boundaries. The application chapter works through chemistry, quantum diagnostics, sensing and hidden power.

The workflow chapter adds a finite-family procedure that identifies a shared target even when several mechanisms remain indistinguishable. The toolkit chapter documents installation, contracts and independently checked outputs. The closing research map records where the results came from, their relation to established work, and the measurements needed for adoption.

A reader applying the framework can begin with this chapter, run the small example, then use the relevant physical chapter. A reader evaluating the theory should follow the proofs and the assumption tables before interpreting the numerical results. The packaged source and reproducibility driver make both paths checkable.

1.7 Scope and scientific status

The reference integrates proved finite-model results, established mathematical tools, new software interfaces, exact worked examples and internal adversarial checks. It contains no newly measured biological, materials or hardware dataset. It does not establish a physical law governing arbitrary matter, unrestricted total energy, universal quantum characterization, or autonomous improvement of arbitrary computation.

Its foundational contribution is a disciplined interface between questions that are often treated separately: access, identifiability, intervention, calibration, decisions and physical resources. The interface is useful precisely because a limitation or failure in one layer remains visible in the next. Acceptance as a scientific subject depends on independent use, criticism and successful experiments; those cannot be supplied by a manuscript alone.

Chapter 2

Scientific questions as access problems

2.1 Start with a quantity, a system, and an experiment

A scientist rarely needs every microscopic parameter of a system. A useful question is usually narrower: which direction does a current flow, how much fuel does a process consume, does a particular error mechanism occur, or which of several interventions changes a reaction pathway? The framework begins with that quantity and asks whether the available experiments can determine it. This order matters. A model can contain invisible parameters while the quantity of interest remains identifiable.

Here is a small example that will recur in more detailed physical form. Suppose two parallel channels contribute positive rates a and c to the same observed transition. A detector identifies only the total rate $k = a + c$. Replacing (a, c) by $(a + h, c - h)$, while keeping both positive, preserves every observation whose law depends only on k. The total rate is identifiable; the individual rates are not. If a calibrated intervention multiplies the first channel by a known $r \neq 1$, a second observed total $k_r = ra + c$ gives

$$a = \frac{k - k_r}{1 - r}, \qquad c = \frac{k_r - rk}{1 - r}. \tag{2.1}$$

The intervention changes what can be identified. When r is close to one, the same formulas amplify measurement and calibration error. If r is unknown, the new observation need not remove the ambiguity. Thus an experimental design requires both a model of intervention and a statement of its accuracy.

The general object is an *access specification*. It contains:

1. a candidate family Θ, including nuisance parameters and the assumptions that make it a physically admissible family;

2. a target $g : \Theta \to \mathcal{G}$, with the units and required precision or decision loss stated;

3. available preparations, controls, measurements, retained records, and the rules governing their histories;

4. a calibration model and a resource account, including preparation, reset, sensing, control, and acquisition where these are used.

This specification is a scientific input. Algebra can test its consequences; it cannot establish from notation alone that an apparatus realizes it. In particular, knowledge of the transition probabilities does not determine the energy used by an unobserved subsystem.

Preparation belongs to access. A system that starts in an unknown state can be observationally equivalent to a different system with another preparation. A known reset, an independently calibrated stationary preparation, or a recorded preparation outcome can change that conclusion. If preparation depends on an unknown physical parameter, that dependence belongs in the candidate model. It is not information available to the experimenter for free.

2.2 Transcript laws and targets

Let $\mathcal{A}$ denote an access specification and $\Pi_{\mathcal{A}}$ its permitted protocols. A protocol π selects actions causally from retained observations and any recorded or modeled randomization; it specifies when to stop. Its complete accessible transcript has probability law P_θ^π under candidate θ. A transcript may contain action choices, timestamps, detector outputs, preparation outcomes, and stopping information. Omitting one of these changes the law being compared. The same protocol is used under every candidate. A policy that secretly knows the true candidate is not an experiment for distinguishing candidates.

Definition 2.1 (Operational equivalence)**.** For a fixed access specification, write

$$\theta \sim_{\mathcal{A}} \theta' \quad \Longleftrightarrow \quad P_\theta^\pi = P_{\theta'}^\pi \quad \text{for every } \pi \in \Pi_{\mathcal{A}}. \tag{2.2}$$

The complete law profile is $\Phi_{\mathcal{A}}(\theta) = (P_\theta^\pi)_{\pi\in\Pi_{\mathcal{A}}}$. A target is *structurally identifiable under* $\mathcal{A}$ when equality of these profiles implies equality of the target.

This is identifiability from ideal experiment laws. It does not mean that a finite noisy record determines a real parameter exactly. The distinction is standard in statistical and dynamical-system identification [13, 5]. It remains essential when the word "observable" is used with different conventions in different fields.

Theorem 2.2 (Target factorization)**.** *A target g is structurally identifiable under $\mathcal{A}$ if and only if it is constant on each equivalence class of $\sim_{\mathcal{A}}$. Equivalently, there is a unique map $\bar{g}$ on the attained profiles $\Phi_{\mathcal{A}}(\Theta)$ such that*

$$g = \bar{g} \circ \Phi_{\mathcal{A}}. \tag{2.3}$$

Proof. If a factorization exists, equal profiles have equal images under $\bar{g}$ and hence equal targets. Conversely, define $\bar{g}(\Phi_{\mathcal{A}}(\theta)) = g(\theta)$. Constancy on each class makes this definition independent of the representative. Every element of the attained profile space has such a representative, proving uniqueness. □

The theorem is a set-theoretic quotient statement, not a claim that the factor map is measurable, continuous, numerically stable, or efficiently computable. Those properties need additional structure. Its practical use is to separate an experimental obstruction from a difficult calculation. An algorithm cannot reconstruct a target that takes two values on the same attained profile without using further assumptions or further access.

For example, a hidden-state model may have several coordinate descriptions related by a relabeling of latent states. Individual coordinates need not be identifiable, while predictions and invariant target quantities are. The quotient treats such representational ambiguity and physically different but observationally equivalent mechanisms by the same mathematical rule. Their scientific interpretations still differ.

2.3 Processing, comparison, and the value of additional access

For probability laws on a common measurable space define

$$\mathrm{TV}(P,Q)=\sup_B |P(B)-Q(B)|. \tag{2.4}$$

For a finite record alphabet this equals $\frac{1}{2}\sum_y |P(y)-Q(y)|$. A randomized processing rule is a Markov kernel K: it maps a complete record to a reported statistic or decision without knowing the true candidate.

Proposition 2.3 (Data processing and access refinement)**.** *For any Markov kernel K,*

$$\mathrm{TV}(KP,KQ)\le \mathrm{TV}(P,Q). \tag{2.5}$$

Suppose every protocol π available under $\mathcal{A}$ can be simulated by some protocol ρ under $\mathcal{B}$, followed by a candidate-independent kernel K, so that $P^{\pi}_{\theta}=KP^{\rho}_{\theta}$ for all θ. Then $\theta\sim_{\mathcal{B}}\theta'$ implies $\theta\sim_{\mathcal{A}}\theta'$. Every target identifiable under $\mathcal{A}$ is therefore identifiable under $\mathcal{B}$.

Proof. For any output event B, the function $z\mapsto K(z,B)$ takes values in $[0,1]$. Integrating this function against $P-Q$ has absolute value at most $\mathrm{TV}(P,Q)$, by the layer-cake representation as an integral of indicator functions. Taking the supremum over B proves contraction. If the two $\mathcal{B}$ laws coincide, applying their common kernel gives equal $\mathcal{A}$ laws for every π. Target identifiability then follows from Theorem 2.2. □

This is the familiar comparison-of-experiments principle [1], used here to compare experimental capabilities. A more accurate detector, a new control, a retained controller record, or an allowed reset can refine the equivalence classes. Conversely, discarding the action history or averaging away a temporal correlation can merge classes. The comparison needs a real simulation relation: two devices with different resolution or resource requirements are not ordered merely because one has more components.

For two fixed candidates and a declared resource budget C, let

$$D_{\mathcal{A}}(C;\theta_0,\theta_1)=\sup_{\pi\in\Pi_{\mathcal{A}}(C)}\mathrm{TV}(P^{\pi}_{\theta_0},P^{\pi}_{\theta_1}). \tag{2.6}$$

Here $\Pi_{\mathcal{A}}(C)$ must specify whether C bounds elapsed time, gross supplied energy, expected energy, operation count, or another quantity. These constraints define different feasible sets. If a simulation preserves the stated budget, Proposition 2.3 also orders the corresponding discrimination frontiers.

For known equally likely alternatives, the infimum average error over protocols and tests is

$$\varepsilon^*(C)=\frac{1-D_{\mathcal{A}}(C;\theta_0,\theta_1)}{2}. \tag{2.7}$$

Indeed, for a fixed protocol the likelihood comparison selects the larger of the two probabilities or densities at each record, giving error $(1-\mathrm{TV})/2$. Optimizing over protocols proves the expression. Suprema and infima need not be attained. In particular, $D=1$ can describe a limiting sequence of improving experiments without any single finite experiment having zero error.

If two candidates with different targets are equivalent, every randomized decision has the same distribution under both. A binary decision between the two has average success at most $1/2$ under equal priors. For a real target, every estimate $\widehat{g}$ satisfies

$$\max_{i=0,1}\mathbb{E}_{\theta_i}|\widehat{g}-g(\theta_i)|\ge\frac{|g(\theta_0)-g(\theta_1)|}{2}. \tag{2.8}$$

The expectations use the same law for $\widehat{g}$; the triangle inequality bounds the sum of the two risks below by the target separation. Unlimited reprocessing of the same accessible evidence cannot remove this obstruction. Additional physical access or a justified restriction of the candidate family can.

2.4 A common geometry for uncertainty and abstention

Exact equality is only one source of ambiguity. Calibration bounds and finite records also leave sets of compatible candidates. Let z be reported evidence and let a specified relation $\mathcal{R} \subseteq \Theta \times \mathcal{Z}$ describe what that evidence permits. Define

$$\Theta(z) = \{\theta : (\theta, z) \in \mathcal{R}\}, \qquad G(z) = \{g(\theta) : \theta \in \Theta(z)\}. \tag{2.9}$$

For example, the relation may be exact equality of an observable profile, a deterministic measurement enclosure, or a confidence construction with a separately proved coverage guarantee. These interpretations should be recorded, because they support different conclusions.

Proposition 2.4 (Compatible-target radius)**.** *Suppose g is real valued, $\Theta(z)$ is nonempty, and estimates may take any value in $\mathbb{R}$. If $G(z)$ is bounded, write $a = \inf G(z)$ and $b = \sup G(z)$. Then*

$$\inf_{t \in \mathbb{R}} \sup_{\theta \in \Theta(z)} |t - g(\theta)| = \frac{b-a}{2}, \tag{2.10}$$

attained at $t = (a+b)/2$. If $G(z)$ is unbounded, every finite estimate has infinite worst-case absolute error.

Proof. For bounded $G(z)$, the supremum loss is $\max\{|t-a|, |t-b|\}$, whether or not the endpoints are attained. Its minimum is the half-width, attained at the midpoint. If $G(z)$ is unbounded, there are target values arbitrarily far from each fixed t. □

An empty compatible set is an inconsistency between evidence and the declared model/calibration contract, rather than a uniquely identified mechanism. A nonempty set on which g is constant supports an exact target within that contract even when several mechanisms survive. A wide target set supports an interval or abstention. If $\Theta(Z)$ covers the true candidate with probability at least $1 - \delta$, the target set $G(Z)$ inherits that same coverage; the geometric proposition does not create the coverage guarantee. For categorical targets or another loss function, the appropriate decision problem replaces the real-valued midpoint.

This elementary geometry gives the common meaning of several later outputs: an exact obstruction, a calibrated target interval, and a refusal to make an unsupported decision. The linear intervention theorem specializes it to a computable global worst-case radius. Sequential model checking supplies coverage under adaptive actions and stopping. Neither step assumes that surviving candidates are the whole of nature.

2.5 When pairwise certificates give a finite scientific program

Theorem 2.5 (Finite target-separating suite)**.** *Let $\Theta = \{\theta_1, \ldots, \theta_K\}$ be a finite family. Suppose the permitted elementary experiments have finite transcript alphabets and any selected finite list can be executed on fresh preparations, independently conditional on the same fixed candidate. Assume structural target identifiability relative to these experiments. Then at most $K(K-1)/2$ experiments*

suffice to form a suite whose joint law separates every pair with different target values. If the individual models have finite linear representations, witnesses can be synthesized with the finite bound of Theorem 3.1.

Proof. For each unordered pair with distinct targets, identifiability supplies an experiment with unequal laws. On a finite alphabet, some transcript event has unequal probability. Choose one such experiment for each pair, dropping duplicates. Fresh preparation gives a product joint law for the suite under each fixed candidate. Its marginal for the chosen pair's experiment differs, so the joint laws differ for that pair. There are at most $K(K-1)/2$ pairs. The finite-representation assertion follows by applying Theorem 3.1 to each pair. □

The fresh-preparation assumption is consequential: concatenating probes on one uncontrolled evolving specimen need not preserve their earlier discriminating laws. The fixed-candidate qualification prevents a nuisance parameter from being redrawn between trials merely to simplify a likelihood. The theorem gives a finite separation program, not a finite-sample accuracy claim or an efficient selection algorithm. Positive gaps can be small. Section 64 turns this existence result into a calibration-aware experiment-cover workflow with explicit sample bounds.

2.6 What is universal here

The law-profile, comparison, and compatible-target constructions apply to any well-defined family of experiments and targets. They are universal in that mathematical sense. Finite-dimensional algorithms, Markovian dynamics, quantum instruments, calibrated interventions, and energy envelopes impose additional assumptions to make particular questions constructive.

The broad scientific contribution is an explicit chain from a question to an admissible experiment and a checked conclusion. Its components connect established statistical decision theory, system identification, control, and physical modeling. This volume does not infer new conservation laws from the shared formalism, and it does not equate an access limitation with a general limit on computation. It provides a way to locate the unresolved distinction, state the extra access that would remove it, and test whether that access is practical.

The original boundary-and-arrow question is one instance: fix the allowed local record, compare candidate dynamics with different time orientations, and ask whether orientation is constant on the resulting equivalence classes. A geometric boundary must be modeled as a physical restriction on preparation, dynamics, coupling, or measurement. It cannot be silently replaced by an unspecified loss of information. This operational entry point preserves the physical question while allowing the same tools to address reaction channels, sensor design, and quantum error hypotheses.

Chapter 3

Finite instruments and independently checkable access

3.1 Representing an experiment without discarding its disturbance

This chapter makes the abstract law profile constructive for finite classical and quantum models. The algebraic representation is shared; the conditions that make a representation physically valid differ. Throughout, states are row vectors and effects are columns. A chronological product $M_{s_1}M_{s_2}$ means that operation s_1 occurs first.

For candidate $h \in \{0, 1\}$, a classical action a has a finite recorded outcome set $\mathcal{Y}_a$ and matrices $K_h^{a,y}$ on n_h hidden states. The entry $K_h^{a,y}(x, x')$ is the joint probability of recording y and ending at x' conditional on starting at x. Physical validity requires

$$K_h^{a,y} \geq 0 \quad \text{entrywise}, \qquad \sum_{y \in \mathcal{Y}_a} K_h^{a,y}\mathbf{1} = \mathbf{1}. \tag{3.1}$$

The initial distribution x_h is nonnegative and satisfies $x_h\mathbf{1} = 1$. An abort or failed preparation is a possible recorded outcome when it can occur. Deleting its probability and renormalizing would define a different experiment, usually with an unaccounted selection and preparation cost.

For a word of action/outcome pairs $w = ((a_1, y_1), \ldots, (a_m, y_m))$, the joint event probability is

$$p_h(w) = x_h K_h^{a_1,y_1} \cdots K_h^{a_m,y_m}\mathbf{1}. \tag{3.2}$$

This expression retains both measurement probabilities and the change of hidden state caused by measurement. An outcome-only noise model generally cannot replace it when the detector disturbs subsequent dynamics.

For a quantum system of Hilbert dimension q_h, an action instead consists of completely positive trace-nonincreasing maps $\mathcal{E}_h^{a,y}$ whose sum is trace preserving. From density operator ρ_h, the probability is

$$p_h(w) = \operatorname{tr}\left[\mathcal{E}_h^{a_m,y_m} \circ \cdots \circ \mathcal{E}_h^{a_1,y_1}(\rho_h)\right]. \tag{3.3}$$

Expressing Hermitian operators in a real basis gives a row-state linear representation with dimension $n_h = q_h^2$ and a terminal column τ_h representing trace. Transfer matrices can have negative entries in that basis. Entrywise stochastic positivity is not a quantum validity test; complete positivity and trace preservation must be established from the instrument, for example through its Kraus operators.

Both cases therefore have the form

$$p_h(w) = x_h M_h^w \tau_h, \qquad M_h^w = M_h^{a_1,y_1} \cdots M_h^{a_m,y_m}, \tag{3.4}$$

on the common finite alphabet $\Sigma = \{(a, y) : y \in \mathcal{Y}_a\}$. A generic signed linear representation is an algebraic input; it does not automatically define probabilities. The exact software uses rational entries and recomputes identities with rational arithmetic. The finite-dimensional theorems also hold over the reals, but floating-point near-equality is not an exact equivalence certificate.

3.2 Finite closure decides equality of every finite record

Pair two representations by setting

$$\alpha = (x_0, -x_1), \qquad \beta = \begin{pmatrix} \tau_0 \\ \tau_1 \end{pmatrix}, \qquad M_s = \begin{pmatrix} M_0^s & 0 \\ 0 & M_1^s \end{pmatrix}, \qquad d = n_0 + n_1. \tag{3.5}$$

Their contrast is $f(w) = \alpha M_w \beta = p_0(w) - p_1(w)$. Let

$$V = \operatorname{span}\{M_w \beta : w \in \Sigma^*\}. \tag{3.6}$$

The generating columns are actual event effects; arbitrary linear combinations in V need not be effects of implementable measurements. The span is a computational device for proving statements about the generating experiments.

Theorem 3.1 (Finite effect-space criterion)**.** *For a finite alphabet and finite-dimensional representation, all contrasts vanish if and only if $\alpha V = 0$. If $\beta \neq 0$ and some contrast is nonzero, there is a distinguishing word of length at most $d - 1$. For valid instrument models this is equivalent to equality of the complete laws of every fixed finite action sequence. Such equality also implies equality under every common causal classical policy with almost surely finite stopping under both models.*

Proof. Let V_k span effects of length at most k. Then

$$V_0 = \operatorname{span}\{\beta\}, \qquad V_{k+1} = V_k + \sum_{s \in \Sigma} M_s V_k. \tag{3.7}$$

If $V_{k+1} = V_k$, the space is invariant under all letters and contains every word effect. Otherwise its dimension strictly increases. Starting from dimension one, at most $d - 1$ increases are possible. Hence $V_{d-1} = V$, proving both the criterion and the witness bound. The case $\beta = 0$ has zero contrast for every word.

For valid instruments, words with a prescribed action sequence are the atoms of its complete outcome law, so equality of all their probabilities is precisely equality of that law. Under a causal randomized policy, a complete action/outcome history has its instrument word probability multiplied by the common probabilities with which the policy chooses its actions and stopping along that history. These factors are identical under both candidates. Equality therefore holds at each finite terminal history; summing over the countable terminal histories proves equality for almost surely finite stopping. Conversely, fixed action sequences are policies. □

Finite-span equivalence and short counterexample words are established results for weighted and probabilistic automata [2, 3, 4]. Observable operator models and predictive state representations supply closely related descriptions of controlled hidden dynamics [5, 6, 7]. Quantum equivalence

also has established finite-dimensional formulations [8]. The framework uses these results as its exact algebraic foundation.

The allowed memory and control architecture must be represented. Arbitrary unmodeled ancillas, coherent feedback, or additional continuous control values are not included by calling the protocol "adaptive." A specified finite quantum memory can be included in the state dimension. Enlarging that architecture changes the access problem.

3.2.1 A certificate another implementation can check

An equivalence certificate supplies columns $b_1, \dots, b_r$, together with coefficients establishing

$$\beta \in \operatorname{span}\{b_i\}, \qquad M_s b_i \in \operatorname{span}\{b_j\} \quad \text{for all } s, i, \qquad \alpha b_i = 0. \tag{3.8}$$

Induction on word length places every $M_w\beta$ in the supplied span, so every contrast vanishes. Linear independence of the columns is useful for compact generation but unnecessary for sound checking. A non-equivalence certificate supplies an actual word and its nonzero exact contrast. Neither certificate requires the reviewer to trust the search execution.

For a minimal entry example, a one-state instrument emits a bit with probability p_h under action a. Its scalar kernels are $M_h^{a,1} = p_h$ and $M_h^{a,0} = 1 - p_h$. With $p_0 = 1/2$, $p_1 = 3/5$, the word $(a, 1)$ has contrast $-1/10$, and the bound $d - 1 = 1$ is attained. Both outcomes remain possible under both hypotheses, so this exact witness is not a perfect discriminator. Repeated sampling belongs to the statistical layer.

3.3 Constraints on experiment histories

Apparatus rules can prohibit particular action orders or require a preparation and final readout. For an action-only deterministic finite automaton, let q_0 be its initial state, F its accepting states, and $\delta(q, a)$ its possibly undefined transition. The matrix A_a has entry one at $(q, \delta(q, a))$ when that transition is defined and zero otherwise. Let v_F indicate accepting states. With S automaton states, replace

$$x_h \mapsto e_{q_0}^{\mathsf{T}} \otimes x_h, \qquad M_h^{a,y} \mapsto A_a \otimes M_h^{a,y}, \qquad \tau_h \mapsto v_F \otimes \tau_h. \tag{3.9}$$

The resulting series is the physical word probability on accepted complete action sequences and zero elsewhere. It is a masked algebraic series, not itself a normalized physical instrument on all strings. Applying Theorem 3.1 in paired dimension Sd proves equality on accepted experiments or produces an accepted witness of length at most $Sd - 1$. The lift is a standard finite-language construction, consistent with constrained switching realization theory [9, 10].

Nonaccepting intermediate grammar states are allowed. A protocol can pass through one while continuing toward a legal completed experiment. Stopping there is inadmissible. An empty accepted language gives a vacuous equality certificate. For adaptive policies the supported terminal action sequences must satisfy the grammar almost surely under both candidates.

Outcome-dependent restrictions require further care. A language that retains only selected successful outcomes can describe postselection rather than executable access. A physical specification must say what happens on the excluded outcomes and how their probability and cost are retained. The implemented grammar here is action-only; it does not silently solve that more general policy-feasibility problem.

3.4 The cheapest nonzero witness

Give every action a nonnegative rational charge $c(a)$, independent of its outcome and prior history, and define the lexicographic weight

$$\omega(w) = (c(w), |w|), \qquad c(w) = \sum_{j=1}^{|w|} c(a_j). \tag{3.10}$$

The length component selects the shortest among equal-charge witnesses. An action charge may be a calibrated upper bound on gross energy, a specified duration, or a count. The theorem optimizes the supplied additive quantity. A mean physical cost depending on the hidden state or realized path generally does not have this form.

Use a priority queue ordered by ω, with a deterministic final tie breaker. Initially it contains $(\beta, \varnothing)$. Remove the least candidate (v, w). If v is in the span of previously retained columns, discard it. Otherwise retain the column and its actual word. Return w if $\alpha v \neq 0$. If the contrast is zero, insert every $(M_s v, sw)$, where sw prepends the letter s to the chronological word. This backward construction of effects does not reverse the chronology of the resulting physical protocol.

Theorem 3.2 (Cost-priority closure)**.** *The search accepts at most d columns and removes at most $1 + |\Sigma|d$ candidates. It returns either an equivalence certificate or a minimum-charge nonzero word, shortest among all minimum-charge nonzero words. Zero-charge actions are allowed. The same result applies after the grammar lift with its enlarged dimension.*

Proof. Each retained column increases rank and only retained zero-contrast columns generate candidates. This gives both finite bounds. In particular, the search cannot loop through infinitely many zero-charge words.

A discarded candidate is a linear combination of earlier retained columns. Since prepending increases length by one and adds a nonnegative charge, queue removal weights never decrease. Every column in that combination has weight no larger than the discarded candidate. Prepending the same letter to the combination preserves the identity and the weight ordering.

For completeness of this pruning rule, fix a weight threshold T and consider the state after all generated candidates of weight below T have been processed, assuming none has nonzero contrast. Induct on word length. The empty effect has been retained or expressed in retained columns whenever its weight is below T. For $w = sv$ below T, the inductive representation of $M_v\beta$ uses columns of weight at most $\omega(v)$. Each of their s prefixes was generated and has weight at most $\omega(w) < T$. On processing, it was either retained or replaced by earlier columns of no greater weight. Thus every below-threshold word effect lies in the retained span, with no greater representative weight.

Before the returned witness, all retained effects have zero contrast. A cheaper, or equal-charge shorter, nonzero word would have an effect in their span, a contradiction. If the queue empties, the retained span contains β and every prefixed retained effect, and is annihilated by α. Equation (3.8) then certifies equivalence. □

The returned word also gives a minimum-charge fixed action sequence whose complete outcome laws differ: every outcome word for the same sequence has the same action charge. It is an event to count while executing that action sequence and retaining all outcomes. The witness does not confer a right to postselect its outcomes without accounting for the failures.

This specialization is built on established finite-rank search. Its historical priority is not established by this volume. It is useful because it aligns a qualitative witness with a declared

physical protocol charge. It does not minimize repetitions, finite-error risk, total energy, or the expected cost of adaptive stopping. A cheaper tiny gap can require much greater total acquisition than a more expensive strong measurement.

3.5 Certifying optimality without trusting the optimizer

A reviewer should be able to verify more than the fact that a supplied word distinguishes the candidates. The following certificate also proves the declared minimum charge. Let w_* be the proposed witness and $T = \omega(w_*)$. Supply a finite list of actual-word effects

$$b_i = M_{u_i}\beta, \qquad t_i = \omega(u_i) < T, \qquad \alpha b_i = 0, \tag{3.11}$$

and exact expansion coefficients with the following conditions:

$$\beta = \sum_i r_i b_i, \qquad r_i \neq 0 \implies t_i \leq (0,0), \tag{3.12}$$

$$M_s b_i = \sum_j q_{sij} b_j, \qquad q_{sij} \neq 0 \implies t_j \leq t_i + (c(s),1). \tag{3.13}$$

Require the initial expansion when $(0,0) < T$. Require the prefix expansion for every (s,i) with $t_i + (c(s),1) < T$. Recompute each effect, weight, and contrast from the declared model and word. In particular, verify $\alpha M_{w_*}\beta \neq 0$.

Theorem 3.3 (Filtered optimality certificate)**.** *A proposed nonzero word is minimum-charge and shortest among minimum-charge nonzero words if and only if a finite certificate satisfying these conditions exists. The certificate is sound with redundant columns, signed matrices, and zero-charge actions.*

Proof. For soundness, consider w with $\omega(w) < T$. Induct on its length to express $M_w\beta$ using only b_i with $t_i \leq \omega(w)$. The empty word uses Eq. (3.12). For $w = sv$, nonnegative charges and the length tie-break give $\omega(v) < \omega(sv) < T$. Every effect used for v satisfies

$$t_i + (c(s),1) \leq \omega(sv) < T. \tag{3.14}$$

The corresponding required identity (3.13) propagates the representation and its weight bound. Since $\alpha b_i = 0$, all below-threshold contrasts vanish. The supplied nonzero witness therefore has the claimed optimum. When $T = (0,0)$ there are no smaller words and no initial expansion is needed.

For completeness, run the cost-priority construction through all generated candidates below T. If w_* is optimal, every retained effect there has zero contrast. Each processed discarded effect has an expansion in earlier columns of no greater weight. The initial effect and every required prefix were processed, giving the displayed identities. At most d independent columns are needed, so this construction terminates even when some actions have zero charge. □

For full equivalence, omit the threshold, require an expansion of β and every prefix, and require zero contrast on all supplied columns. This is both sound and complete by finite closure. The checker needs only matrix multiplication, rational arithmetic, and the explicit expansion identities; it need not calculate a basis or call an optimizer.

The weight restriction in the expansion is essential. Replacing a cheap effect by an expensive representative without tracking its weight can remove a required cheap prefix from the certificate and hide a cheaper witness. Exact closure alone proves equivalence when imposed globally; partial

closure proves an optimum only when its filtration is respected. The software tests deliberately forged word costs, missing prefixes, model substitutions, and genuine but nonminimal witnesses.

A certificate must be bound to the model the reviewer intends to check. A digest of an embedded model establishes internal consistency, but an adversary could replace both model and digest. The checker therefore accepts an independently selected expected model or digest. A digest is neither a signature nor a calibration measurement. Physical validity and empirical model adequacy remain separate requirements.

3.6 What finite dimension does and does not make easy

The dimension bound concerns the supplied representation. For two generic n-qubit instruments, each real operator space has dimension 4^n and the paired dimension is $2 \cdot 4^n$. Finite linear algebra therefore does not imply a polynomial algorithm in qubit count, constant-shot process characterization, or elimination of fault-tolerance overhead. Known structure and a narrow target can reduce the represented problem, but those are additional assumptions.

The number of rank operations is finite and polynomial in the explicit matrix dimension for standard implementations. Exact arithmetic also incurs bit complexity, and certificate size includes its rational coefficients. A short word need not have a large probability contrast. Consequently a bound on witness length is not a bound on the records required for reliable inference. The statistical chapter keeps these quantities distinct.

3.6.1 Parameterized families and an algebraic route

Suppose every model has fixed representation dimension n, matrix entries are rational functions of parameters θ on a semialgebraic domain, and all denominators are nonzero there. Suppose the domain and functions are effectively given over the rationals or algebraic numbers. For a parameter pair (θ, θ'), Theorem 3.1 makes equivalence a finite collection of word-probability equalities for lengths at most $2n-1$. Clearing nonvanishing denominators gives polynomial conditions. For a rational target g, structural identifiability is the validity of

$$\forall \theta, \theta' \in \Theta : \quad \left(\bigwedge_{|w| \le 2n-1} p_\theta(w) = p_{\theta'}(w) \right) \Longrightarrow g(\theta) = g(\theta'). \tag{3.15}$$

Real quantifier elimination decides this statement in principle [12, 13]. The finite conjunction can be enormous and the decision can be computationally expensive. The observation is a route for appropriately structured symbolic problems, not an implemented solver for arbitrary continuous families. It also explains where algebraic geometry enters naturally: the fibers of the observable map are polynomial or rational constraint sets in such a model.

3.7 An exact boundary for unbounded positive-gap search

Deciding whether any word has a nonzero contrast is easier than deciding whether some arbitrarily long experiment exceeds a fixed positive discrimination threshold.

Theorem 3.4 (Strict-cutpoint boundary in instrument form)**.** *Fix a rational $\delta \in (0, 1)$. There is no algorithm that, for every pair of finite rational classical instruments with the following grammar, decides whether some admissible finite protocol has transcript total variation greater than δ: any*

finite sequence of unobserved stochastic actions, followed by exactly one terminal binary readout. The horizon is unbounded and no repeated independent preparation is available within a protocol.

Proof. Use the established undecidability of strict cutpoint emptiness for rational probabilistic finite automata, in particular the fixed cutpoint 1/7 result described and proved in Ref. [11]. Such an automaton has initial distribution μ, stochastic letter matrices T_a, and binary acceptance column v. Its acceptance probability for word w is $p(w) = \mu T_w v$. The undecidable question is whether $p(w) > 1/7$ for some finite word.

Turn each letter into a single-outcome unobserved action. After these actions allow one terminal readout with success effect $D = aI + b\,\mathrm{diag}(v)$ and failure effect $I - D$. Choose rational $b > 0$ small enough that $a = \delta - b/7 \geq 0$ and $a + b \leq 1$. These diagonal readout kernels are physically valid classical instruments. Compare against a one-state model that always reports zero. The only informative observation is Bernoulli success with probability $a + bp(w)$ versus certain failure; its total variation is $a + bp(w)$. It exceeds δ exactly when $p(w) > 1/7$.

All outcomes preceding the readout are deterministic, so feedback supplies no information for choosing the word. If the chosen word is recorded, randomization gives the average of its total variations, since the word distribution is common to both models. If that average exceeds the threshold, some deterministic word exceeds it. Thus an algorithm for the stated instrument problem would decide strict cutpoint emptiness. □

This is a translation of an established computability result, historically traced to Nasu and Honda, rather than a new impossibility theorem for physics. The reduction does not show that its stochastic matrices are embeddable continuous-time propagators, thermodynamically reversible, or available in any particular laboratory. Allowing unlimited independent resets and repeated measurements changes the problem: a nonzero single-run gap can then be amplified statistically.

A finite horizon makes the specified finite-alphabet design problem finite. A finite additive budget does so only when every repeatable action has a strictly positive minimum charge, or a separate horizon is imposed. Zero declared entropy production does not imply zero laboratory energy, and a budget on one accounting channel need not bound the number of possible operations. Bounded design, calibrated statistics, and physical accounting are therefore required layers of the constructive framework.

Chapter 4

Precision, calibration, and experimental design

Exact identifiability asks whether some accessible record law changes when the scientific target changes. Precision asks how much evidence is needed to resolve that change. These questions have different answers. A nonzero rational contrast can be arbitrarily small, an invertible response matrix can be arbitrarily poorly conditioned, and a short distinguishing experiment can require many repetitions. This chapter supplies upper and lower guarantees that connect the exact access layer to decisions a scientist can use.

Throughout, θ denotes a fixed physical model, $g(\theta)$ the target, and P_θ^π the law of the complete accessible record under a common permitted protocol π. A resource contract specifies preparation, actions, stopping, and the meaning of every budget. The general statistical ingredients below are established: binary testing, optimal recovery, concentration, and belief-state dynamic programming. Their role here is to make the interfaces explicit.

4.1 What the best possible precision means

For probability laws P, Q on a common record space, use the convention

$$\mathrm{TV}(P,Q) = \sup_A |P(A) - Q(A)| = \tfrac{1}{2}\int |p - q|\, d\mu,$$

where μ dominates both laws. If the two models are known and equally likely, the smallest average probability of choosing the wrong model is

$$r^*(P,Q) = \frac{1 - \mathrm{TV}(P,Q)}{2}. \tag{4.1}$$

Indeed, choose the first model on $\{p \geq q\}$; the resulting error is $\frac{1}{2}\int \min(p,q)\, d\mu$, and pointwise minimization shows that no other decision rule has smaller error. This is an average error for specified priors. It is not automatically a bound on each of the two conditional errors.

Let Π_C be the declared class of protocols admissible at resource budget C. With the access setting fixed, abbreviate the frontier of Equation (2.6) by

$$D_C(\theta_0, \theta_1) = \sup_{\pi \in \Pi_C} \mathrm{TV}(P_{\theta_0}^\pi, P_{\theta_1}^\pi). \tag{4.2}$$

For known equal-prior alternatives, the infimum achievable average error is $(1 - D_C)/2$. A supremum need not be attained by a protocol. If the budget classes are nested, D_C is nondecreasing. No shape, universal energy scale, or efficient algorithm for this function follows from its definition.

Proposition 4.1 (A resource-dependent lower bound for real targets)**.** *Let g be real valued, and let an estimator T use the record of any $\pi \in \Pi_C$. The policy, its randomization rule, and the estimator must be the same under both alternatives. Then*

$$\max_{j=0,1} \mathbb{E}^{\pi}_{\theta_j} |T - g(\theta_j)| \geq \frac{|g(\theta_0) - g(\theta_1)|}{2} [1 - \mathrm{TV}(P^{\pi}_{\theta_0}, P^{\pi}_{\theta_1})]. \tag{4.3}$$

Consequently, for any parameter family Θ,

$$\inf_{\pi \in \Pi_C, T} \sup_{\theta \in \Theta} \mathbb{E}^{\pi}_{\theta} |T - g(\theta)| \geq \sup_{\theta_0, \theta_1 \in \Theta} \frac{|g(\theta_0) - g(\theta_1)|}{2} [1 - D_C(\theta_0, \theta_1)]. \tag{4.4}$$

Expectations may be infinite. The bound is not asserted to be attainable.

Proof. Write $g_j = g(\theta_j)$ and $p_j = dP^{\pi}_{\theta_j}/d\mu$. The triangle inequality gives, pointwise,

$$p_0 |T - g_0| + p_1 |T - g_1| \geq \min(p_0, p_1)(|T - g_0| + |T - g_1|) \geq |g_0 - g_1| \min(p_0, p_1).$$

Integration and $\int \min(p_0, p_1) = 1 - \mathrm{TV}(P_0, P_1)$ prove (4.3). Independent estimator randomization is a common probability kernel and the same argument applies after adjoining its random seed. For every fixed pair, replacing the policy's total variation by its supremum over Π_C weakens the bound uniformly in π. Taking the infimum on the left and then the supremum over pairs proves (4.4); no minimax interchange is used. □

This is the standard two-point method associated with Le Cam [14]. It identifies an important failure case: an invisible target difference has $D_C = 0$ at every allowed budget, so more computation on the same records cannot remove its ambiguity. If invisible pairs have arbitrarily large finite target separation, the worst absolute-error risk is infinite. The conclusion follows by taking those finite pairs, without undefined products involving infinity. A lower bound caused by restricted access is different from a claim that all information about the system has a universal cap.

4.2 Linear intervention responses: from rank to stable recovery

Suppose $\theta \in \Theta \subset \mathbb{R}^p$ and a known matrix $F \in \mathbb{R}^{m \times p}$ represents intervention responses. Two physical assumptions are needed before its algebra describes the apparatus:

1. If $F\theta = F\theta'$, every permitted complete record law agrees.
2. If every permitted complete record law agrees, then $F\theta = F\theta'$.

The first is law factorization; the second is access sufficiency. A hidden generator's dependence on $F\theta$ may establish the first while leaving the second unproved for a coarse detector. Unknown preparation or a parameter-dependent detector can invalidate either identification argument.

For example, controlled continuous-time generators with known entries $q_0 + F\theta$ give factorization when preparations and readout are known and parameter independent. Access sufficiency follows in an idealized setting where every state is preparable and resolved, and transition laws are accessible at arbitrarily short times: the derivative at zero identifies each generator entry. Finite bandwidth and finite data require additional bounds; that idealization does not provide them.

Proposition 4.2 (A target-specific row-space criterion)**.** *Assume both conditions above and a nonempty open domain Θ. The linear target $g(\theta) = b^{\mathsf{T}}\theta$ is identifiable if and only if $b \in \mathrm{row}(F)$, equivalently $F^{\mathsf{T}}w = b$ for some w.*

Proof. Such a w yields $g(\theta) = w^{\mathsf{T}} F\theta$, so access sufficiency proves the claim. Conversely, if b is outside the row space, there is $h \in \ker(F)$ with $b^{\mathsf{T}} h \neq 0$. At an interior point θ_*, both $\theta_* + th$ and $\theta_* - th$ are allowed for sufficiently small $t > 0$. Factorization makes their laws identical, while their targets differ. □

On a relatively open affine parameter domain $\theta_* + V$, the correct condition is that b annihilate $V \cap \ker F$. Known constraints can therefore resolve ambiguities that remain on the full open domain. Full parameter reconstruction requires more than reconstruction of a particular scientific target.

Now suppose the measured or statistically estimated response satisfies

$$\widehat{y} = F\theta + e, \qquad |e_i| \le \epsilon_i, \tag{4.5}$$

where every ϵ_i is known and nonnegative. Any $F^{\mathsf{T}} w = b$ gives

$$|w^{\mathsf{T}} \widehat{y} - b^{\mathsf{T}} \theta| \le \sum_i \epsilon_i |w_i|. \tag{4.6}$$

No independence of coordinate errors is needed. If the entire box is certified with probability at least $1 - \delta$, this deterministic conclusion has at least that coverage. How the box is obtained remains a separate experimental question.

Theorem 4.3 (Global deterministic minimax recovery)**.** *Assume the parameter domain is the entire positive orthant $\mathbb{R}^p_{>0}$, F is known, and every error vector in the full Cartesian box $[-\epsilon, \epsilon]$ is admissible. If $b \in \operatorname{row}(F)$, then*

$$\begin{aligned} &\inf_T \sup_{\theta > 0,\, |e| \le \epsilon} |T(F\theta + e) - b^{\mathsf{T}} \theta| \\ &\qquad = \min_{F^{\mathsf{T}} w = b} \sum_i \epsilon_i |w_i| = \max_{|Fh| \le \epsilon} b^{\mathsf{T}} h. \end{aligned} \tag{4.7}$$

The extrema on the right exist, including when some error radii vanish. If $b \notin \operatorname{row}(F)$, the recovery risk is infinite, the middle problem is infeasible, and the dual supremum is unbounded.

Proof. Equation (4.6) proves the upper bound. For any dual-feasible h, choose a componentwise center $\theta_* > |h|$. The two positive parameters $\theta_* \pm h$, with errors $\mp Fh$, produce the same observation $\widehat{y} = F\theta_*$. Their target separation is $2|b^{\mathsf{T}} h|$. Every estimate at that common observation errs by at least $|b^{\mathsf{T}} h|$ on one of them. The feasible set for h is symmetric, giving the displayed dual lower bound.

Write $w = u - v$, $u, v \ge 0$. The middle value equals the linear program

$$\min_{u,v \ge 0} \epsilon^{\mathsf{T}} (u + v) \quad \text{subject to } F^{\mathsf{T}} u - F^{\mathsf{T}} v = b.$$

Its dual is exactly $\max b^{\mathsf{T}} h$ subject to $Fh \le \epsilon$ and $-Fh \le \epsilon$. The primal is feasible with a finite nonnegative infimum, and the dual is feasible at zero. Finite-dimensional linear programming duality supplies attainment and equality [15]. If b is outside the row space, scale an $h \in \ker F$ with $b^{\mathsf{T}} h > 0$ and repeat the common-observation construction to obtain arbitrarily large error lower bounds. □

An independently checkable certificate consists of w, h satisfying $F^{\mathsf{T}} w = b$, $|Fh| \le \epsilon$, and $\sum_i \epsilon_i |w_i| = b^{\mathsf{T}} h$. The checker need not trust the optimizer. This is a structured, globally worst-case specialization of the compatible-set geometry in Proposition 2.4. A particular observed $\widehat{y}$ may give a smaller feasible target interval. Bounded parameter domains or correlated error constraints can also lower the minimax value; their replacement by the full positive orthant or full error box changes the problem being solved.

4.2.1 A calibrated intervention example

Two unresolved positive channel rates have baseline response $y_0 = \theta_1 + \theta_2$. A calibrated intervention produces $y_1 = r\theta_1 + \theta_2$, with known $r \geq 0$. Thus

$$F_r = \begin{pmatrix} 1 & 1 \\ r & 1 \end{pmatrix}, \qquad \theta_1 = \frac{y_1 - y_0}{r - 1} \quad (r \neq 1).$$

At equal response-error radius ϵ, the exact global minimax error for θ_1 is $2\epsilon/|r-1|$. For the primal choose $w = (-1, 1)/(r-1)$; a matching dual is $h = \epsilon(2, -r-1)/|r-1|$. Its two response coordinates have magnitudes ϵ, and its first component equals the claimed error. At $\epsilon = 0.01$, responses $r = 1/2, 0.9, 0.99, 0.999$ give errors $0.04, 0.2, 2, 20$. At $r = 1$ the channel is unidentifiable. Exact full rank near $r = 1$ does not establish useful precision.

Adding a third response with $r = 2$ to the baseline and $r = 1/2$ gives, at the same three error radii 0.01, optimal weights $(0, -2/3, 2/3)$ and error $1/75$. The dual $h = (1/75, -1/60)$ proves optimality. This improvement presumes that the extra measurement does not worsen the existing error bounds; a fixed physical budget may require a different comparison.

Unknown selectivity is a decisive control: $(\theta_1, \theta_2, r) = (3, 1, 1/2)$ and $(2, 2, 1/4)$ both give $(y_0, y_1) = (4, 5/2)$. Inserting a nominal r into a known-matrix certificate does not validate it when r is unknown.

4.3 From one record to a finite-error experiment

4.3.1 Known laws and affinity

For known finite record laws define their Bhattacharyya affinity [16]

$$\rho(P_0, P_1) = \sum_z \sqrt{P_0(z)P_1(z)}.$$

Cauchy–Schwarz gives $0 \leq \rho \leq 1$, with equality at one exactly when the two laws agree. For M independent records under the same fixed model, the likelihood-ratio decision rule has equal-prior average error at most

$$r_M^* \leq \tfrac{1}{2}\rho(P_0, P_1)^M. \tag{4.8}$$

To prove this, use $\min(p, q) \leq \sqrt{pq}$ in Equation (4.1); affinity multiplies over product laws. An outward rational upper bound $\overline{\rho} < 1$ therefore certifies a sufficient record count. An upper enclosure at least one certifies no such count at that precision. It need not mean that the actual laws are equal.

The product-law premise requires independent preparations or a justified independent-record construction. If a nuisance parameter is fixed but unknown, it stays fixed in every factor. Averaging its law independently at every record models a different experiment. Full common support also prevents perfect finite-record classification, even when the affinity bound becomes very small.

4.3.2 A robust bounded-score witness

Let $f(z) \in [0, 1]$ be chosen before acquisition. It may be an event indicator. Suppose all nominal models in target class + satisfy $\mathbb{E}f \geq a_0$, and all in class − satisfy $\mathbb{E}f \leq b_0$. Suppose actual one-record laws are within total variation η_+ or η_- of their respective nominal laws. Since bounded-score expectations change by at most total variation, define

$$a = \max(0, a_0 - \eta_+), \quad b = \min(1, b_0 + \eta_-), \quad \gamma = a - b. \tag{4.9}$$

Proposition 4.4 (A uniform finite-record guarantee)**.** *If $\gamma > 0$, classify as $+$ when the mean of M scores exceeds $(a+b)/2$, and as $-$ otherwise. For independent records under each fixed actual model, each conditional error probability is at most*

$$\exp(-M\gamma^2/2).$$

In particular, $M \geq 2\log(1/\alpha)/\gamma^2$ gives each error at most α. The same bound holds for a dependent sequence of scores in $[0,1]$ if every full-history conditional mean is at least a under class $+$ or at most b under class $-$.

Proof. Under $+$, the rejection threshold is at least $\gamma/2$ below the mean. Hoeffding's inequality gives the result, and the argument for $-$ is symmetric [17]. For the conditional version, Hoeffding's bounded-variable moment-generating-function lemma gives, under $+$,

$$\mathbb{E}[\exp(-t(X_i - a)) \mid \mathcal{F}_{i-1}] \leq \exp(t^2/8), \qquad t \geq 0.$$

Iterate conditional expectations, apply Markov's inequality to the sum, and minimize

$$\exp(-tM\gamma/2 + Mt^2/8)$$

at $t = 2\gamma$. □

A nonpositive robust gap means that this certificate fails. It does not prove that every possible protocol fails. A one-record separating score can also fail for composite hypotheses even when repeated records identify them: a fixed Bernoulli parameter in $\{0.2, 0.8\}$ cannot be separated by a single bounded score from parameter 0.5, because the latter law is the average of the first two. Repeated independent observations of the same fixed parameter separate the classes through the sample mean.

The inverse-square bound is sufficient, not a universal sample-complexity law. For Bernoulli probabilities 0 and ε, deciding for the second model after the first success has zero error under the first and error $(1-\varepsilon)^M$ under the second. A fixed error then needs order ε^{-1} records. Interior alternatives such as $1/2 \pm \varepsilon$ behave differently.

4.4 Calibration must cover the experiment that is executed

For a classical hidden-state instrument, calibration concerns the joint distribution of the next hidden state and observed outcome. Suppose the initial state laws differ by at most η_0 in total variation and, at step k, every admissible state, retained history and selected action has joint-kernel error at most η_k. Require $0 \leq \eta_k \leq 1$, including $k = 0$; any looser total-variation bound can first be truncated at one. Under a common causal policy,

$$\mathrm{TV}(P^{\pi}_{\mathrm{actual}}, P^{\pi}_{\mathrm{nominal}}) \leq 1 - (1-\eta_0)\prod_{k=1}^{H}(1-\eta_k) \leq \eta_0 + \sum_{k=1}^{H}\eta_k. \tag{4.10}$$

Couple the initial states maximally and use the same policy randomization. As long as the histories and hidden states agree, the next coupled state/outcome agrees with conditional probability at least $1-\eta_k$. Multiplication gives the first bound; a union bound gives the second. Stopping can be padded with an identical absorbing action. Bounds on detector marginals alone are insufficient: two instruments can give the same immediate output while placing the hidden state in different conditions for the next experiment.

For quantum instruments, represent each entire outcome-retaining operation as a completely positive trace-preserving map to a classical–quantum output. An initial trace-distance bound η_0 and half-diamond-norm bounds η_k give the additive analogue of (4.10). A telescoping hybrid argument inserts one actual operation at a time; trace-distance contractivity, including ancillas and later feedback channels, bounds its contribution by η_k [18]. The classical multiplicative coupling bound is not asserted for arbitrary quantum channels. A norm on an unconstrained signed matrix representation is not by itself a physical channel calibration certificate.

There are two distinct routes for propagating repeated-record errors. A uniform one-record radius can be incorporated into the score gap before amplification. Alternatively, an independently justified whole-dataset total-variation bound Δ changes any nominal decision error by at most Δ. It is incorrect to add a one-record radius only once to a many-record decision error. Under independent product-law models a radius η yields the dataset bound $1-(1-\eta)^M \le M\eta$, which may be too loose to be useful.

4.5 Choosing a protocol for the stated decision

The shortest or lowest-charge exact witness minimizes a qualitative objective: it obtains any nonzero distinction. It need not minimize the resources needed for a prescribed error. A deeper single record may improve its contrast enough to reduce total repetitions, or its extra control and reset costs may outweigh that improvement. An experiment report should compare the full decision at matched resource budgets.

For a finite candidate family $j = 1, \dots, K$, let x_j be the unnormalized hidden-state row vector whose mass $m_j = x_j\mathbf{1}$ includes the candidate's prior and the probability of the observed history. Each candidate has a target label g_j. If an action a yields outcome y, the update is $x'_j = x_j M_{j,a,y}$. A fixed nuisance value is represented within the index j; it is never resampled between updates. Let

$$L(x) = \sum_j m_j - \max_v \sum_{j:g_j=v} m_j$$

be the Bayes error mass incurred by stopping and choosing a target label. For two single-model target classes this is $\min(m_0, m_1)$.

Proposition 4.5 (Finite-horizon Bayes design)**.** *Fix finite classical instruments with nonnegative kernels satisfying $\sum_y M_{j,a,y}\mathbf{1} = \mathbf{1}$, a common finite action/outcome library, nonnegative integer charges $c(a)$, remaining budget B, and a finite remaining action horizon h. A finite grammar state q determines legal actions, a transition $q' = \Phi(q,a)$, and whether stopping is permitted. Assign every candidate a strictly positive prior, so zero-mass branches are impossible under every candidate and may safely be ignored when enforcing feasibility. The minimum Bayes error mass satisfies*

$$V_{h,B,q}(x) = \min \left\{ \begin{array}{l} L(x) \quad \text{if stopping is permitted,} \\ \min\limits_{\substack{a \text{ legal},\ c(a)\le B \\ h>0}} \sum\limits_y V_{h-1,B-c(a),\Phi(q,a)}((x_j M_{j,a,y})_{j=1}^K) \end{array} \right\}. \tag{4.11}$$

An unavailable alternative has value $+\infty$; a zero-mass branch has value zero. The initial vectors are the supplied prior-weighted initial states.

Proof. At horizon zero only permitted stopping is possible. At a positive horizon, a causal policy first stops or selects an available action. Conditional histories of distinct outcomes are disjoint, so their unnormalized error masses add. By induction, each continuation has risk at least the corresponding displayed value. Selecting minimizing continuations attains that value. Randomizing the first action forms a convex combination and cannot lower the minimum. □

This is standard belief-state dynamic programming [19]. Rational probabilities and charges allow exact finite arithmetic. Finite horizon is essential even with zero-charge actions. Search can still grow exponentially; an implementation reaching its declared state limit has not certified an optimum. This result optimizes the given priors, model library, target loss, grammar and budget. Robust minimax design, unknown nuisance priors, new physical controls and different energy accounting are different optimization problems. An unmodeled candidate is not repaired by solving this recurrence more exactly.

4.5.1 Fisher information answers a different question

For a regular scalar Bernoulli model, local Fisher information is $I(\theta) = [p'(\theta)]^2/[p(\theta)(1-p(\theta))]$. Consider two equally costly probes, $p_A(\theta) = \theta^2$ and $p_B(\theta) = (1+\theta/2)/2$, at the target alternatives $\theta = \pm 1/2$. Probe A has Fisher information $16/3$ at either point, while probe B has $4/15$. Yet A is exactly blind to the sign, whereas B distinguishes it. Probe A resolves a local change of magnitude, so there is no contradiction. The example explains why maximizing local information, minimizing Bayes classification risk, and minimizing energy for target precision must be stated as separate objectives.

The next chapter adds two requirements for use outside a closed model exercise: the option to withhold a target when the candidate predictions fail, and a physical resource ledger whose uncertainty composes with the inference guarantee.

Chapter 5

Model adequacy, abstention, and physical resource guarantees

A decision rule that must choose between two mechanisms will still choose one when both mechanisms are wrong. A scientific workflow therefore needs an adequacy layer and an explicit abstention output. It also needs a resource ledger: knowing how many operations are executed does not determine how much energy the apparatus supplies. This chapter gives contracts under which these two layers combine with the precision results of Chapter 4.

The probability bounds are mathematical guarantees about the stated predictive and physical premises. They do not validate a detector, establish local detailed balance, measure a fuel channel, or prove that the candidate family contains the true system. Those premises must be independently justified for an application.

5.1 Conditional predictions are the adequacy contract

At round t, let $\mathcal{F}_{t-1}$ contain the full retained past, calibration information used for this prediction, the next action, and any controller randomization used to choose it. This is pre-observation information; one may formally enlarge the filtration immediately before each outcome. A binary statistic Y_t is then revealed. The statistic and event being recorded must also be chosen before seeing that outcome.

A candidate j supplies predictable envelopes $[\ell_{j,t}, u_{j,t}] \subset [0,1]$. Under a fixed true model, conditional validity means

$$\ell_{j,t} \le p_t := \mathbb{E}[Y_t \mid \mathcal{F}_{t-1}] \le u_{j,t} \quad \text{for every } t. \tag{5.1}$$

For a composite candidate, the contract must hold under each fixed member for which the candidate is asserted valid. Maximizing and minimizing correct conditional forecasts over nuisance values can produce valid but conservative envelopes. A fixed nuisance does not become independently redrawn at every round.

This contract allows dependent records, feedback and optional stopping. Its strength is also its principal burden: a stationary marginal probability or an unconditional error bound is not generally a conditional forecast. For a simple failure control, let one fair coin U be drawn and set $Y_t = U$ forever. Every marginal is Bernoulli(1/2). After the first observation, however, the next conditional probability is zero or one. Treating these outcomes as fresh fair coins can generate overwhelming, invalid rejection evidence against the true marginal description.

5.2 A sequential adequacy certificate

Before Y_t is revealed, choose a finite predictable stake $\lambda_{j,t}$. For a nonnegative stake use $b_{j,t} = u_{j,t}$, and for a negative stake use $b_{j,t} = \ell_{j,t}$. Define

$$f_{j,t}(Y_t) = 1 + \lambda_{j,t}(Y_t - b_{j,t}), \qquad M_{j,n} = \prod_{t=1}^{n} f_{j,t}(Y_t). \tag{5.2}$$

Require the two factors at $Y_t = 0$ and $Y_t = 1$ to be nonnegative. For interior endpoints this is $\lambda \le 1/u$ for positive stakes and $\lambda \ge -1/(1-\ell)$ for negative stakes; checking both factors directly handles zero and one endpoints without dividing by zero.

Proposition 5.1 (Anytime exclusion)**.** *Under (5.1), $M_{j,n}$ is a nonnegative test supermartingale starting at one. A fixed convex mixture $E_{j,n}$ of such strategies has the same property. For any $0 < \alpha_j < 1$,*

$$\Pr\{\textit{some } n:\ E_{j,n} \ge 1/\alpha_j\} \le \alpha_j. \tag{5.3}$$

Proof. The signed endpoint choice gives

$$\mathbb{E}[f_{j,t}(Y_t) \mid \mathcal{F}_{t-1}] = 1 + \lambda_{j,t}(p_t - b_{j,t}) \le 1.$$

Multiplying by past nonnegative wealth gives the supermartingale property; fixed convex combinations preserve it. For completeness, stop at the first threshold crossing or a deterministic horizon n. Optional stopping for this bounded stopping time bounds expected stopped wealth by one. On a crossing, wealth is at least $1/\alpha_j$, so its probability is at most α_j. Letting the horizon increase proves (5.3). □

This is Ville's maximal inequality and bounded betting [20, 21]. Component stakes may depend on the past. Fixed mixture weights are chosen before the tested record; replacing them after seeing the current outcome requires a separate wealth-accounting argument. The test is valid with any permitted predictable intervention policy satisfying the envelopes. Validity alone says nothing about power or acquisition efficiency.

Exclude a candidate permanently once its threshold is crossed. Denote the surviving set by S_n. Permanent exclusion records an ever-crossing event; instantaneous wealth can fall again and does not undo that event.

Theorem 5.2 (Target commitment with abstention)**.** *Suppose one fixed candidate j_* is valid throughout the run, contains the true physical model, and has the true target label g_*. Each candidate is target-homogeneous, with declared label g_j. Use level $\alpha_j = \alpha$ for every candidate. Return:*

1. `abstain_all_rejected` *if S_n is empty;*
2. `abstain_ambiguous` *if survivors assert different targets;*
3. `commit`(v) *if S_n is nonempty and every survivor asserts v.*

The probability of ever returning an empty survivor set or ever committing an incorrect target is at most α.

Proof. If j_* has not been excluded, the survivor set is nonempty and any unanimous target equals g_*. Thus either unsafe event entails exclusion of j_*. Proposition 5.1 bounds that ever-exclusion probability. □

There is no factor equal to the number of candidates in this statement: it protects one fixed valid and correctly targeted candidate and the resulting decision. To protect every valid candidate simultaneously, use levels with $\sum_j \alpha_j \le \alpha$ and apply a union bound. These are different questions. A different candidate fitting each successive prediction is not a single candidate valid throughout the run.

If a candidate has unresolved internal target values, retain its target set and return the union over survivors. Commitment to one value is justified only when that union is a singleton. Observable agreement alone does not validate an asserted hidden target: the theorem needs the true model to belong to a correctly specified candidate. If truth is outside the declared family, both wrong commitment and total rejection can occur without this guarantee. Nonrejection is not evidence that all omitted mechanisms have been ruled out. There is no general termination theorem; opposite-target models with identical accessible laws may remain ambiguous indefinitely.

5.3 Calibration uncertainty without invalid conditioning

A joint calibration statement may guarantee that all conditional prediction envelopes used during acquisition are valid with probability at least $1 - \delta$. One cannot usually prove an anytime guarantee by conditioning on this eventual good event: that conditioning can change the conditional outcome laws on which the betting calculation depends.

Proposition 5.3 (Calibration composition)**.** *Let G be any event on which all envelopes of the designated true candidate are conditionally valid. If $\Pr(G^c) \le \delta$, the unsafe-output probability in Theorem 5.2 is at most $\alpha + \delta$, capped by one. Moreover the probability of unsafe output intersected with G is at most α.*

Proof. The actual conditional mean p_t and the supplied endpoints are predictable, even if p_t is unknown to the scientist. Stop the designated candidate's wealth before the first round at which its envelope fails, replacing that and all later factors by one. The stopped wealth is a nonnegative test supermartingale. On G it equals the original wealth throughout the run. Unsafe output on G therefore entails a crossing by this valid stopped wealth, which has probability at most α. Add $\Pr(G^c)$. □

A separate 95% interval at every round does not provide one 5% failure budget for the experiment. Calibration must cover every conditional prediction used, including drift, filtering, actions and detector behavior. An interface that requires `prepare()` before `observe()` helps enforce chronology but cannot prove that its caller did not inspect future data. Scientific record retention remains part of the contract. Adding candidates invented after seeing the tested data also requires new valid tests or a justified data-reuse scheme.

5.4 What a physical energy budget must cover

An additive action charge is a bookkeeping variable until calibrated to a physical quantity. Time, supplied work, dissipated heat, net free-energy change, and operation count are different quantities. Their units and scope must be specified. In particular, a reversible added transition channel at a stationary distribution does not make the apparatus or its controller free of energy cost.

For a complete experiment, the ledger must cover preparation, acquisition, interventions, sensing and readout, controller operation, reset, intervening gaps, and calibration. Shared support can be reported separately or amortized over a declared number of experiments. Any omitted contribution

prevents a claim about total energy. A detector-law error bound does not bound energy in unobserved physical channels.

Two meanings of a budget are particularly different. A pathwise bound is never exceeded within its assumptions. An expected expenditure permits arbitrarily large realized expenditure. A finite-time continuous-time Markov process can make arbitrarily many jumps, so multiplying an expected event count by an event energy does not produce a pathwise cap. The next theorem supplies a probabilistic cap under a channel-level physical contract.

5.5 A counting-channel acquisition cap

Let N_t count supplied-energy events, starting at zero. Assume it is a simple, unit-jump, nonexplosive counting process with predictable intensity λ_t and compensator $A_t = \int_0^t \lambda_s\, ds$. Let τ be a bounded stopping time for a filtration in which this intensity statement is valid. These are assumptions on the actual process; observing N_t or knowing its exact intensity online is not required for the probability argument. A calibrated predictable intensity upper bound may replace λ_t and yields the corresponding supermartingale bound.

Theorem 5.4 (A cap valid at adaptive stopping)**.** *For $\mu > 0$ and an integer $n > \mu$,*

$$\Pr\{N_\tau \geq n,\ A_\tau \leq \mu\} \leq \exp\{-n\log(n/\mu) + n - \mu\}. \tag{5.4}$$

In particular this bounds $\Pr(N_\tau \geq n)$ whenever $A_\tau \leq \mu$ almost surely. If $A_\tau = 0$ almost surely, then $N_\tau = 0$ almost surely.

Proof. For $s > 0$, the counting-process stochastic exponential

$$Z_t = \exp\{sN_t - (e^s - 1)A_t\}$$

is a nonnegative local martingale and hence a supermartingale. This follows, for instance, by applying the jump product rule to obtain $dZ_t = Z_{t-}(e^s - 1)(dN_t - \lambda_t\, dt)$ and then localizing [22]. At the bounded stopping time, $\mathbb{E}Z_\tau \leq 1$. On the event in (5.4), $Z_\tau \geq \exp\{sn - (e^s - 1)\mu\}$. Markov's inequality bounds its probability by $\exp\{-sn + (e^s - 1)\mu\}$. Minimize at $s = \log(n/\mu) > 0$. When $\mu = 0$, use the same event bound for each $s > 0$ and let s increase to infinity for every positive integer n. □

The argument does not assume independent event increments or a Poisson count. It does require the counting-process contract. A signed transfer matrix used to compute output probabilities does not automatically supply such a physical counting instrument.

Choose n so that the right side of (5.4) is at most β. Suppose the gross work per counted event is at most $w_{\max} \geq 0$, and a separately justified bound W_{app} covers all other supplied energy. On the event where this ledger is valid,

$$W_{\text{gross}} \leq w_{\max} N_\tau + W_{\text{app}}. \tag{5.5}$$

Consequently $W_{\text{cap}} = nw_{\max} + W_{\text{app}}$ is a conservative $1 - \beta$ energy cap when the compensator and ledger premises hold surely. It is deliberately conservative by one count and is neither an exact count quantile nor a hard pathwise cap. Signed net work must not replace gross supplied work in (5.5) without an appropriate new model.

For rational μ, β and a proposed n, the cap can be checked without trusting floating-point logarithms. For rational $x \geq 1$, write $z = (x - 1)/(x + 1)$ and use

$$L_K(x) = 2\sum_{k=0}^{K-1} \frac{z^{2k+1}}{2k+1}, \qquad L_K(x) \leq \log x \leq L_K(x) + \frac{2z^{2K+1}}{(2K+1)(1-z^2)} =: U_K(x).$$

The positive series for $\log((1+z)/(1-z))$ proves the lower bound; replacing each remaining denominator by $2K+1$ gives the geometric upper tail. Range reduction by powers of two improves convergence. A rational certificate is

$$nL_K(n/\mu) - n + \mu \geq U_K(1/\beta).$$

An independent checker recomputes these inequalities. Their truth validates the probability calculation, while the laboratory remains responsible for the physical intensity, event energy and apparatus bounds.

5.6 A joint inference and energy guarantee

Theorem 5.5 (Composition without independence)**.** *Assume one fixed correctly targeted candidate is present. Let G be a joint good event of probability at least $1-\delta$ on which:*

1. *its full-history conditional prediction envelopes are valid throughout;*
2. *$A_\tau \leq \mu$ for the actual counting process;*
3. *the gross-energy ledger (5.5) holds.*

Run the target-commitment rule at level α, and choose n in Theorem 5.4 at level β. Then

$$\Pr\{\textit{ever incorrect commitment, ever all-candidate rejection, or } W_{\mathrm{gross}} > W_{\mathrm{cap}}\} \leq \alpha + \beta + \delta, \quad (5.6)$$

where inference outputs are considered through the bounded experimental stopping time. No independence of calibration, inference and energy is required.

Proof. Unsafe inference intersected with G has probability at most α by the stopped-before-invalid-forecast argument in Proposition 5.3. Energy overrun intersected with G implies $N_\tau \geq n$ and $A_\tau \leq \mu$, unless $w_{\max} = 0$, in which case it is impossible on G. Theorem 5.4 bounds that event by β. Add the two events and G^c by the union bound. □

The compensator or energy-ledger failure need not be predictable: their terminal joint events are used directly. In contrast, conditional forecasts are stopped before their first invalid factor to preserve the supermartingale. This distinction avoids an unjustified application of optional stopping or conditioning. A single joint calibration failure budget is counted once.

5.7 An end-to-end handoff from a witness to a physical guarantee

The following result gives a common handoff between the algebraic, statistical and physical layers. It uses a certified score gap; the exact witness that motivated that score need not itself optimize the final energy cost.

Corollary 5.6 (Robust score, record count, and energy cap)**.** *Fix target classes $+$ and $-$, a predeclared protocol and bounded record score, and choose constants $0 \leq b < a \leq 1$ with $\gamma = a - b > 0$. Let M be a fixed integer with $M\gamma^2/2 \geq \log(1/\alpha)$. Assume an event G of probability at least $1-\delta$ simultaneously ensures:*

1. *under the true fixed target class, every actual score's conditional mean is at least a for $+$ or at most b for $-$;*

2. the complete bounded acquisition has $A_\tau \le \mu$ and satisfies the gross-energy ledger (5.5).

The midpoint score decision after M records, with any count cap n certified at level β, obeys

$$\Pr\{\textit{wrong target or } W_{\mathrm{gross}} > nw_{\max} + W_{\mathrm{app}}\} \le \alpha + \beta + \delta \tag{5.7}$$

under every fixed model satisfying those contracts.

Proof. On G, the conditional score means obey the required one-sided bound. To avoid conditioning on G, stop at the first predictable violation of that bound and replace that and all subsequent scores by a under $+$, or by b under $-$. The modified sequence has the required conditional means everywhere and agrees with the executed sequence on G. The conditional version of Proposition 4.4 therefore bounds wrong decision intersected with G by α. Bound energy overrun intersected with G by Theorem 5.4, and add $\Pr(G^c)$. □

Independent identically prepared records from a fixed actual model are a sufficient way to obtain the score contract. The conditional formulation also states precisely what would justify using dependent records. If nominal expectation bounds and total-variation calibration produce the gap, use Equation (4.9); do not amplify an uncalibrated nominal contrast and add one-record error only at the end. If fresh reset fails or an intervention changes the nuisance between records, the model must be rebuilt.

For a simple synthetic calculation, let $a = 3/5$, $b = 2/5$, and $\alpha = 1/20$. Then $\gamma = 1/5$ and $M = 150$ certifies each conditional decision error because $M\gamma^2/2 = 3 > \log 20$. Suppose the separately specified complete physical contract has compensator cap $\mu = 150$. The rational counting checker certifies $n = 189$ at $\beta = 1/100$. With joint calibration failure $\delta = 1/100$, the probability of a wrong decision or exceeding $189w_{\max} + W_{\mathrm{app}}$ is at most $7/100$. These numbers are an arithmetic demonstration, not measured performance. If native fuel use continues during preparation or gaps, that exposure must be included in μ or bounded separately in the ledger. The declaration $\mu = 150$ is insufficient until all relevant physical time has been accounted for.

5.8 What a scientist should retain

A usable result consists of the target and candidate definition; the actual observation, preparation and intervention contract; an exact obstruction or witness where available; quantitative calibration and sampling bounds; the executed policy and complete records; and the physical resource ledger with its coverage statement. The inference should return ambiguity or failed model adequacy when those are the supported conclusions. A finite certificate is portable because another scientist can check its mathematical content. Its application becomes credible when another apparatus independently verifies the physical premises and reproduces the predicted distinction.

Chapter 6

Continuous dynamics, physical interfaces, and energy

The finite-instrument description is an interface to physical models. A waiting operation, a controlled transition channel, and a detector can be converted into instruments, but that conversion must preserve the distinction being tested and carry its approximation error. This chapter develops the conversion for finite continuous-time Markov chains. It then follows the original physical question from a driven interface to an observable arrow, showing exactly where preparation, observation, and control enter.

6.1 A source, a record, and a preparation are different objects

Let Q be an irreducible finite row generator: $Q_{ij} \geq 0$ for $i \neq j$, $Q\mathbf{1} = 0$, and $\dot{p} = pQ$. Its unique positive stationary row distribution is denoted by ϖ. Assume bidirectional support and states even under physical time reversal. The stationary reverse is

$$Q^{\dagger} = \operatorname{diag}(\varpi)^{-1} Q^{\mathsf{T}} \operatorname{diag}(\varpi). \tag{6.1}$$

This notation denotes a stationary reversal of a stochastic generator, not its Euclidean adjoint. Detailed balance means $Q = Q^{\dagger}$.

For a deterministic detector with diagonal outcome projectors D_y, an observed word has probability

$$\varpi D_{y_0} e^{t_1 Q} D_{y_1} \cdots e^{t_m Q} D_{y_m} \mathbf{1}. \tag{6.2}$$

Physical reversal reverses both the labels and the gaps. Reversing labels alone on an unequal grid generally compares different experiments. An operational stationary arrow is a difference between the record law and its physical reverse. It can coexist with constant one-time Shannon entropy. Unitary quantum evolution likewise need not increase von Neumann entropy; the existence of time is not the proposition under study.

The original interface question has an elementary source criterion. Let G be the connected undirected support graph. Partition its edges into bulk and interface edges, and suppose a positive reference weight ρ balances every bulk edge. Write

$$\mathcal{A}_{ij} = \log \frac{\rho_i Q_{ij}}{\rho_j Q_{ji}}. \tag{6.3}$$

This antisymmetric edge function vanishes in the bulk. Its sum around a cycle is the corresponding affinity. The usual cycle criterion [23, 24] states that Q is reversible exactly when every such sum vanishes.

Proposition 6.1 (Interface source capacity)**.** *If the bulk graph, including all vertices, has c connected components and there are m interface edges, the space of independent interface-supported cycle affinities has dimension $m - c + 1$. An interface-supported edge function has zero cycle affinities exactly when it is df, with $(df)_{ij} = f_j - f_i$ and f constant on each bulk component.*

Proof. On a connected graph, zero cycle sums imply that the integral of $\mathcal{A}$ along a path from a fixed root is path independent. It defines f and gives $\mathcal{A} = df$. Vanishing bulk values force f to be constant on each bulk component. The m interface edge values form $\mathbb{R}^m$; gradients of those component constants have rank $c - 1$, because an overall constant has zero gradient and the contracted graph is connected. The quotient has dimension $m - c + 1$. If $\mathcal{A} = df$, then $\varpi_i \propto \rho_i e^{f_i}$ satisfies detailed balance; the converse follows by taking $f = \log(\varpi/\rho)$. □

This is a concrete use of graph cohomology: the relevant quotient is the interface contribution to $H^1(G;\mathbb{R})$. It establishes where a stationary source can occur. It does not determine whether a particular detector sees it. If all bulk edges are balanced and the interface is the only possible drive, an observed stationary arrow requires an interface-supported cycle affinity. The converse can fail because observation discards distinctions. A driven seam on a ring is an interface in this sense; it is not a geometric outer boundary or a higher-dimensional manifold assumption.

Preparation gives a separate failure mode. Consider a reversible Q and state-dependent killing rates $k_i \geq 0$. Conditioning a path on survival to T multiplies its density by $w_T = \exp[-\int_0^T k_{X_s}\, ds]$, a reversal-invariant functional. With stationary preparation, the surviving-path law remains reversible. With a full-support initial law ρ, put $f_i = \rho_i/\varpi_i$. Its surviving law $P^{\mathrm{surv}}_{\rho,T}$ instead satisfies

$$\log \frac{dP^{\mathrm{surv}}_{\rho,T}}{d(\mathcal{R}_* P^{\mathrm{surv}}_{\rho,T})} = \log \frac{f(X_0)}{f(X_T)}, \qquad D_{\mathrm{KL}}(P^{\mathrm{surv}}_{\rho,T} \| \mathcal{R}_* P^{\mathrm{surv}}_{\rho,T}) \leq \log \frac{f_{\max}}{f_{\min}}, \tag{6.4}$$

where $\mathcal{R}$ reverses a path. Both expressions follow by comparing densities $f(X_0)w_T$ and $f(X_T)w_T$ relative to the reversible stationary path law; their normalization constants coincide. Any reversal information is then confined to the endpoints and its rate per unit time tends to zero. A finite-window arrow caused by selected preparation need not be sustained entropy production. Trials discarded during survival selection still count toward preparation and physical expenditure.

6.2 Certified finite-time instruments

For a rational generator Q, rational $t \geq 0$, and rational $\lambda \geq \max_i(-Q_{ii})$, define $P = I + Q/\lambda$. If $\lambda > 0$, P is stochastic and uniformization gives

$$e^{tQ} = e^{-x} \sum_{k=0}^{\infty} \frac{x^k}{k!} P^k, \qquad x = \lambda t. \tag{6.5}$$

This established construction represents the waiting operation as a Poisson number of stochastic updates [25]. To avoid replacing an irrational exponential by an untracked floating-point number, let

$$S_N = \sum_{k=0}^{N} \frac{x^k}{k!}, \qquad \widehat{E}_N = S_N^{-1} \sum_{k=0}^{N} \frac{x^k}{k!} P^k. \tag{6.6}$$

$\widehat{E}_N$ is an exactly rational stochastic matrix. If $x < N+2$, the omitted scalar series has the geometric bound

$$B_N = \frac{x^{N+1}}{(N+1)!}\frac{1}{1-x/(N+2)}, \qquad \epsilon_N = \frac{B_N}{S_N + B_N}. \tag{6.7}$$

The exact transition is a mixture of $\widehat{E}_N$ and a stochastic tail matrix. Its tail weight is at most ϵ_N, so every row differs from the corresponding row of $\widehat{E}_N$ by total variation at most ϵ_N. If $\lambda t = 0$, use the identity matrix and zero error.

For a fixed sequence of waiting operations, a coupling or telescoping argument adds their row errors; intervening stochastic observations and controls contract total variation. Thus, with preparation error ϵ_0,

$$\mathrm{TV}(P_{\mathrm{actual}}, P_{\mathrm{rational}}) \le \min\{1, \epsilon_0 + \textstyle\sum_j \epsilon_j\}. \tag{6.8}$$

Calibration errors in controls and detection require additional terms. For adaptive experiments the bound must hold conditionally for each history, including the updated hidden state; agreement of a single unconditional output marginal is insufficient.

If a rational event witness has gap g between two candidate laws and the two complete-record error radii are e_0, e_1, the physical gap is at least $\max(0, g - e_0 - e_1)$. A positive lower bound certifies continuous-time non-equivalence. Equality of rational approximations does *not* certify equality of the underlying continuous-time models. Exact symmetries or generator-level algebra are needed for that conclusion. Uniformization here is a classical CTMC construction; an arbitrary Lindblad generator cannot be treated as a nonnegative stochastic matrix in a Hermitian operator basis.

6.3 What a singleton detector hides

Theorem 6.2 (Stationary singleton reversal obstruction)**.** *For an irreducible finite Q started in ϖ, the complete path law of $Y_t = \mathbf{1}_{\{X_t = i\}}$ is identical under Q and $Q^\dagger$.*

Proof. The one-state case has a constant detector and is immediate. Otherwise the escape rate from i is positive by irreducibility. At every return to i, the strong Markov property restarts the process. The sojourn at i is exponential. Its duration is independent of the departure destination, and therefore of the following excursion duration. Successive sojourns and excursions are independent draws from their respective laws. A stationary alternating renewal process with these independent durations is unchanged by reversal, including its stationary age distribution. The stationary reverse of X is generated by $Q^\dagger$; applying the same detector therefore gives the same Y law. Equivalently, partition $Q = \left(\begin{smallmatrix} -\lambda & a \\ b & B \end{smallmatrix}\right)$ around i. The sojourn rate λ and excursion-return density $ae^{tB}b/\lambda$ agree under stationary reversal. These quantities and stationary ages determine the binary law. □

This standard renewal obstruction concerns orientation [26]. Waiting-time distributions may still identify a drive magnitude or rule out some equilibrium models. Nor does a single scalar detector necessarily satisfy this theorem: a scalar count can distinguish several microscopic states. A binary aggregate containing multiple states in each block need not regenerate at an observed return.

For an explicit failure of a broader "all binary records are reversible" claim, take

$$Q = \begin{pmatrix} -2 & 1 & 0 & 1 \\ 1 & -3 & 2 & 0 \\ 0 & 1 & -4 & 3 \\ 4 & 0 & 1 & -5 \end{pmatrix}, \quad \varpi = \tfrac{1}{88}(41, 18, 13, 16), \quad D = \mathrm{diag}(1, 0, 1, 0). \tag{6.9}$$

For $H(t,s) = \varpi D e^{tQ} D e^{sQ} D\mathbf{1}$, direct multiplication gives

$$H(t,s) - H(s,t) = -\frac{23}{44} ts(s-t) + O((t+s)^4). \tag{6.10}$$

Thus the two-versus-two detector sees an arrow although every singleton detector in the same stationary chain obeys the theorem.

6.4 A switched reversible channel exposes a current

Let L be a second generator reversible with respect to the *same* ϖ. For a nonnegative control amplitude η,

$$(Q+\eta L)^\dagger = Q^\dagger + \eta L, \qquad \varpi_i(Q_{ij} + \eta L_{ij}) - \varpi_j(Q_{ji} + \eta L_{ji}) = J_{ij}, \tag{6.11}$$

where $J_{ij} = \varpi_i Q_{ij} - \varpi_j Q_{ji}$. All settings preserve the stationary probabilities and net currents. Every frozen setting therefore retains singleton reversal blindness. Different independently prepared records can select frozen amplitudes adaptively without removing the obstruction: equal conditional record laws imply equal transcript laws.

Within a record, apply $Q + \eta L$ for time t followed by Q for time s. Read the singleton only at the endpoints. Let p_{AB} be the joint return probability, and let p_{BA} reverse the order of these same two stages.

Proposition 6.3 (Protocol-order current witness)**.** *With the common stationary preparation,*

$$\begin{aligned} p_Q^{AB} - p_{Q^\dagger}^{AB} &= p_Q^{AB} - p_Q^{BA} \\ &= \varpi_i [e^{t(Q+\eta L)} e^{sQ} - e^{sQ} e^{t(Q+\eta L)}]_{ii}. \end{aligned} \tag{6.12}$$

If L acts only on $i \leftrightarrow j$, then

$$\partial_t \partial_s (p_{AB} - p_{BA})|_{0,0} = -\eta L_{ji} J_{ij}. \tag{6.13}$$

Proof. The ϖ-adjoint reverses product order and preserves diagonal entries. This proves the first identity. The mixed coefficient of the commutator is $\eta[L,Q]$. Detailed balance for L then gives

$$\varpi_i [L,Q]_{ii} = \varpi_i (L_{ij} Q_{ji} - Q_{ij} L_{ji}) = -L_{ji} J_{ij}.$$

□

Comparing protocol orders can be performed on one physical system; no apparatus that reverses its hidden drive is required. This is a specific way in which histories enlarge access. The observable diagonal commutator is essential: noncommutativity elsewhere need not reach the detector. Common stationarity is also essential. Two individually reversible generators with different equilibria may show an order contrast from preparation and relaxation alone. Adding a balanced *parallel* channel is generally different from multiplying the rates of a driven channel by a common factor.

The smallest driven cycle gives a reproducible example:

$$Q_F = \begin{pmatrix} -2 & 1+F & 1-F \\ 1-F & -2 & 1+F \\ 1+F & 1-F & -2 \end{pmatrix}, \quad L = \begin{pmatrix} -1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad |F| < 1. \tag{6.14}$$

Here $\varpi = (1,1,1)/3$, $Q_F^\dagger = Q_{-F}$ and $J_{01} = 2F/3$. For the detector at state zero, the exact contrast is

$$\Delta = -\frac{2\eta}{3\sqrt{3}} e^{-3(t+s)-\eta t} \frac{\sinh(\kappa t)}{\kappa} \sin(\sqrt{3}Fs), \qquad \kappa^2 = \eta^2 - 3F^2. \tag{6.15}$$

The quotient has its entire continuation when $\kappa = 0$ or is imaginary. To derive it, write $P = \mathbf{1}\mathbf{1}^{\mathsf{T}}/3$, $R_0 = I - P$ and $C = (Q_F + 3R_0)/F$, defined by continuation at $F = 0$. On the zero-sum subspace, $C^2 = -3R_0$, $(L + R_0)^2 = R_0$, and $C(L + R_0) + (L + R_0)C = 0$. Expanding the two matrix exponentials into their even and odd parts and using $[L, C]_{00} = -2$ gives Eq. (6.15). At fixed positive lags, jointly as $(F, \eta) \to (0, 0)$, $\Delta = -(2/3)tse^{-3(t+s)}F\eta + o(F\eta)$. Full-support endpoint probabilities in this local regime then yield divergence $\Theta(F^2\eta^2)$ and simple-hypothesis fixed-error record cost $\Theta((F\eta)^{-2})$.

6.5 An autonomous equilibrium gate

An external switch is not the only source of a history. Let a finite controller have reversible generator R, positive stationary distribution ν, and nonnegative amplitude profile u_c. On joint states (c, x) set

$$G_Q = R \otimes I + I \otimes Q + U \otimes L, \qquad U = \mathrm{diag}(u), \qquad \Pi = \nu \otimes \varpi. \tag{6.16}$$

Then $\Pi G_Q = 0$ and $G_Q^\dagger = G_{Q^\dagger}$. The controller evolves independently of x, but its history modulates the system kinetics. Retaining or discarding the controller record gives different access contracts.

Suppose L acts on $i \leftrightarrow j$ with $p = L_{ij} > 0$, $q = L_{ji} > 0$, $\varpi_i p = \varpi_j q$. Define

$$\mathcal{D}_R(u) = -\sum_c \nu_c u_c (Ru)_c = \frac{1}{2} \sum_{c,d} \nu_c R_{cd} (u_d - u_c)^2. \tag{6.17}$$

This nonnegative kinetic Dirichlet form has inverse-time units when u is dimensionless. It is not an energy in joules. It vanishes exactly when u is constant on each communicating component with positive stationary weight.

Theorem 6.4 (Hidden-gate current coefficient)**.** *For $Y = \mathbf{1}_{\{X=i\}}$ in the stationary joint process, let $H_Q(t, s) = \Pr(Y_0 = Y_t = Y_{t+s} = 1)$. Then*

$$H_Q(t, s) - H_Q(s, t) = \frac{pqJ_{ij}}{6} \mathcal{D}_R(u)\, ts(s^2 - t^2) + O((t + s)^5). \tag{6.18}$$

Consequently $J_{ij} \neq 0$ and $\mathcal{D}_R(u) > 0$ suffice for the hidden-gate binary record to distinguish Q from its stationary reverse.

Proof. Let $E_n(Q)$ be the controller matrix with entries $(G_Q^n)_{(c,i),(d,i)}$, and put $M = R + Q_{ii}I - pU$. Block multiplication gives $E_0 = I$, $E_1 = M$, and

$$E_2(Q) = E_2(Q^\dagger), \qquad E_3(Q) - E_3(Q^\dagger) = K := \frac{qJ_{ij}}{\varpi_i}[R, U]. \tag{6.19}$$

For completeness, $E_2 = M^2 + \mathrm{diag}_c \sum_{k \neq i} (Q_{ik} + u_c L_{ik})(Q_{ki} + u_c L_{ki})$; each scalar product is unchanged by reversal. In the cubic expression, terms with no controller transition are diagonal entries of $(Q + u_c L)^3$ and agree under reversal. Terms with two or three controller factors depend only on the system diagonal. The remaining contrast comes from a controller transition between two system factors; its cd entry is

$$R_{cd}[(Q + u_c L)(Q + u_d L) - (Q + u_d L)(Q + u_c L)]_{ii} = \frac{qJ_{ij}}{\varpi_i} R_{cd}(u_d - u_c).$$

Writing $E_Q(t) = \sum_n E_n(Q)t^n/n!$ gives $H_Q(t,s) = \varpi_i \nu E_Q(t) E_Q(s)\mathbf{1}$. Stationary reversal makes $H_{Q^\dagger}(t,s) = H_Q(s,t)$. Terms with one zero lag degree reduce to reversible stationary binary pairs; all other terms below total degree four vanish, as does the $(2,2)$ term. Finally, $R\mathbf{1} = 0$, $\nu R = 0$ and Eq. (6.17) imply

$$\varpi_i \nu M K \mathbf{1} = pq J_{ij} \mathcal{D}_R(u), \qquad \varpi_i \nu K M \mathbf{1} = -pq J_{ij} \mathcal{D}_R(u).$$

The $(1,3)$ and $(3,1)$ Taylor factors are $1/6$, proving the result. □

For the triangle in Eq. (6.14), hidden-gate orientation is identifiable if and only if $F \neq 0$ and $\mathcal{D}_R(u) > 0$. Sufficiency follows from the theorem; if the form vanishes, conditioning on a stationary controller component reduces the system to a frozen setting, which is reversal blind. In a general network, the theorem is only a sufficient condition. A zero current on the controlled edge can coexist with higher-order signals from other cycles. With several controlled edges, different current contributions can also cancel.

For $\mathcal{D}_R(u) > 0$, the coefficient supplies a finite-lag estimator. With four readings separated by τ, define $Z_\tau = Y_0 Y_{3\tau}(Y_\tau - Y_{2\tau}) \in [-1,1]$. For any $\Lambda \geq \|G_Q\|_\infty$, the induced matrix norm, Taylor's theorem and stochastic contraction give

$$\left| \frac{\mathbb{E} Z_\tau}{pq \mathcal{D}_R(u) \tau^4} - J_{ij} \right| \leq \frac{\varpi_i (3\Lambda)^5 \tau}{60 pq \mathcal{D}_R(u)} =: B_\tau. \tag{6.20}$$

Indeed each fifth derivative of a triple probability, viewed along $(t,s) = (\lambda\tau, 2\lambda\tau)$, is at most $\varpi_i(3\Lambda\tau)^5$; the two Taylor remainders contribute a factor $2/5!$. For M independent preparations, Hoeffding gives a valid radius

$$B_\tau + \frac{\sqrt{2\log(2/\alpha)/M}}{pq \mathcal{D}_R(u) \tau^4} \tag{6.21}$$

at confidence $1-\alpha$. The coefficients, escape bound, and balance assumptions need calibration. Small lag reduces bias but sharply amplifies sampling noise. This conservative guarantee is not a claim of practical precision or minimax optimality; the archived benchmark's bias constant is 88473.6τ. Correlated windows from one trajectory need mixing or conditional-martingale bounds rather than the independent-record formula. Third-order time asymmetry itself has established spectroscopy precedents [27]; the specific content here is the edge-current coefficient and its access conditions.

6.6 Assigning physical expenditure

An action charge becomes an energy only through an implementation. Preserve distinct physical reaction channels r even if they connect the same states. Under channel-resolved local detailed balance, stationary entropy production is

$$\frac{\dot{S}_{\mathrm{tot}}}{k_B} = \sum_{i<j,r} (\varpi_i q^r_{ij} - \varpi_j q^r_{ji}) \log \frac{\varpi_i q^r_{ij}}{\varpi_j q^r_{ji}}. \tag{6.22}$$

Merging parallel channels can erase chemical cycles and lower this expression. Thermodynamic meaning requires specified reservoirs and local detailed balance; a statistically reversible fitted transition is not automatically a passive thermal device [28].

For the triangle, one realization uses degenerate molecular conformations and fueled clockwise reactions with forward/backward rates $1+F$ and $1-F$. The chemical affinity per net forward

event is $A_F = \log[(1+F)/(1-F)]$, and the added channel is a separate unfueled balanced pathway. A reversible autonomous gate can have energies $E_c = -k_B T_b \log \nu_c + \text{constant}$ in a bath at T_b. In this channel assignment,

$$\frac{\dot{S}_{\text{drive}}}{k_B} = 2F \log \frac{1+F}{1-F}, \qquad \dot{S}_{\text{gate}} = \dot{S}_{\text{added}} = 0. \tag{6.23}$$

The gate preserves the native mean drive entropy-production rate. This does not supply free preparation, detection, clocking, control, or memory reset. Nor do chemical rate ratios alone determine literal heat rather than chemical free-energy expenditure.

For M independently prepared stationary records of duration T, the mean drive free-energy loss is $E_{\text{drive}} = M k_B T_b T \, 2F A_F$. In this particular model, the KL divergence between opposite-drive laws of the full channel-labelled trajectories is exactly $E_{\text{drive}}/(k_B T_b)$. The log likelihood ratio is $A_F(N_{\text{cw}} - N_{\text{ccw}})$: initial laws and escape rates agree, while passive-channel jumps contribute zero. Its mean signed jump count is $2FMT$. Data processing and Pinsker consequently imply

$$\text{TV}(P_{\text{observed}}, P_{\text{reversed drive}}) \le \min\left\{1, \sqrt{\frac{E_{\text{drive}}}{2k_B T_b}}\right\}. \tag{6.24}$$

This is a model-specific inference–expenditure relation, not a universal replacement for Landauer's erasure bound or a guarantee that observed irreversibility equals total dissipation.

Mean expenditure and a physical safety cap also differ. If fueled events form a nonexplosive simple counting process N_t, $N_0 = 0$, with actual predictable intensity λ_t and compensator $A_t = \int_0^t \lambda_s \, ds$, then

$$Z_t(\theta) = \exp\{\theta N_t - (e^\theta - 1)A_t\}, \qquad \theta > 0, \tag{6.25}$$

is a nonnegative local martingale and hence a supermartingale. For a bounded stopping time τ,

$$\Pr(N_\tau \ge n, \ A_\tau \le \mu) \le \exp[-n \log(n/\mu) + n - \mu], \qquad n > \mu > 0. \tag{6.26}$$

On the event in question, Z_τ is at least $\exp[\theta n - (e^\theta - 1)\mu]$; optional stopping, Markov's inequality and $\theta = \log(n/\mu)$ prove the bound. If each supplied event costs at most $w_{\max}$, a gross-expenditure envelope $W \le w_{\max} N_\tau + W_{\text{apparatus}}$ converts it to an energy statement. The intensity bound and apparatus term must describe the actual experiment. A net dissipation formula cannot silently replace gross fuel supplied.

6.7 From a mathematical channel to an apparatus

The direct physical test of the gate construction has four requirements. First, measure the native and added channels separately, including their rate ratios, rather than fit only their sum. Second, verify that all gate settings share the required stationary distribution to within a declared calibration envelope. Third, measure the gate rates and profile, and test frozen-gate, constant-profile, and zero-current controls. Fourth, vary lag and controlled edge to test the predicted current coefficient and its remainder. Full expenditure must include rejected preparations, gaps between records, sensing, control, and data storage under the chosen resource boundary.

Fast hidden intermediates introduce a measurement-resolution issue. An effective added edge can be obtained by eliminating short-lived states, but agreement on a fixed finite sampling grid does not automatically imply agreement of complete, arbitrarily high-bandwidth paths. The intervention and detector bandwidth must be calibrated together. These requirements are the bridge from a usable theorem to a physical claim; the present examples provide no new measured apparatus data.

Chapter 7

A finite-time physical access theorem

An intervention library is scientifically useful only when its operations have a physical realization. The following result gives one such realization: native probability current becomes an endpoint population response under a selective kinetic intervention. The result concerns finite-state classical Markov dynamics. It does not require reconstruction of the surrounding network, infinitesimal perturbations, or an infinitesimal observation time.

Throughout this section, distributions are row vectors. A finite row generator Q has nonnegative off-diagonal entries and $Q\mathbf{1} = 0$. Let $\pi Q = 0$, with π a probability distribution, and let $i \neq j$. Write $P'(t) = e^{tQ'}$ for the semigroup after a fixed intervention. The intervention remains in place until readout at the predetermined time t.

Theorem 7.1 (Finite-amplitude endpoint response)**.** *Suppose Q' is another valid finite generator and differs from Q only in the directed rates $i \to j$, $j \to i$, and their compensating diagonal entries. Define*

$$\gamma = \pi_i(Q'_{ij} - Q_{ij}) - \pi_j(Q'_{ji} - Q_{ji}).$$

For every finite $t > 0$,

$$[\pi P'(t)]_j - \pi_j = \gamma H'_{ij}(t), \tag{7.1}$$

$$\begin{aligned} H'_{ij}(t) &= \int_0^t \left(P'_{jj}(s) - P'_{ij}(s)\right) ds \\ &= \int_0^t P'_{jj}(s)\, \overset{Q'}{\Pr_i}(\tau_j > t - s)\, ds > 0, \end{aligned} \tag{7.2}$$

where τ_j is the first hitting time of j, starting at i. Consequently the endpoint displacement has the sign of γ for every positive time, and vanishes at one such time if and only if $\gamma = 0$. Irreducibility of either generator is unnecessary.

Proof. The assumptions give $\pi Q' = \gamma(e_j - e_i)$, where e_k denotes the row point mass at k. Integrating $\frac{d}{ds}\pi P'(s) = \pi Q' P'(s)$ proves (7.1) and the first expression for H. Let $F_j(t) = \int_0^t P'_{jj}(s)\, ds$. Before τ_j, a path starting at i spends no time in j. The strong Markov property gives

$$\int_0^t P'_{ij}(s)\, ds = \mathbb{E}_i\left[\mathbf{1}_{\{\tau_j \le t\}} F_j(t - \tau_j)\right].$$

Interchanging nonnegative integrals yields (7.2). Set $\lambda_k = -Q'_{kk}$. A path remaining in its initial state gives $P'_{jj}(s) \ge e^{-\lambda_j s}$ and $\Pr_i(\tau_j > u) \ge e^{-\lambda_i u}$. Both factors are strictly positive at finite times, including when $\tau_j = \infty$ has positive probability. Their integral is positive. □

The integrand $P'_{jj}(s) - P'_{ij}(s)$ in the first line need not be positive. For example, a directed three-cycle has negative instantaneous contrast at some times. The sign theorem follows from the integrated first-hitting representation, not from pointwise transition ordering. It forbids crossing the *native baseline*; it does not assert that the displacement magnitude is monotone or cannot overshoot its eventual stationary value.

Corollary 7.2 (Local quantitative bounds)**.** *If the perturbed exit rates of i, j have upper bounds Λ_i, Λ_j, then*

$$0 < L(t) := \int_0^t e^{-\Lambda_j s - \Lambda_i (t-s)}\, ds \le H'_{ij}(t) \le t.$$

For a common upper bound Λ, this gives $te^{-\Lambda t} \le H'_{ij}(t) \le t$. For unequal bounds, $L(t) = (e^{-\Lambda_i t} - e^{-\Lambda_j t})/(\Lambda_j - \Lambda_i)$. No rate bound on other states appears.

7.1 Selective channels and native current

A physical channel e between i and j has forward and reverse rates a_e, b_e. Several physical channels may share the same state pair. The native channel current is $J_e = \pi_i a_e - \pi_j b_e$. Suppose a calibrated intervention multiplies *both* rates of this channel by $r \in (0, 1)$, leaving all other channels unchanged. Then $\gamma = -(1 - r)J_e$, so that

$$D_e(t) := \mathbb{E}_\pi[\mathbf{1}_{\{X_0 = j\}} - \mathbf{1}_{\{X_t = j\}}] = (1 - r)J_e H'_{ij}(t). \tag{7.3}$$

Thus one resolved endpoint population, compared before and after a fixed kinetic step, identifies the orientation of the native channel current in the ideal exact-law limit. The stationary distribution *after* intervention generally changes and must not be substituted for π when defining that native current.

This access genuinely separates some passively indistinguishable mechanisms. On two observed states, the two-channel rate assignments $(a_1, b_1, a_2, b_2) = (3, 1, 1, 3)$, $(2, 2, 2, 2)$, and $(1, 3, 3, 1)$ all give the same aggregate generator with rates 4 in each direction. Their first-channel native currents are $1, 0, -1$. A factor-1/2 selective intervention produces negative, zero, and positive endpoint displacements according to (7.3). A common additive state-level control applied to their identical aggregate generators cannot make that distinction.

In an Arrhenius model a barrier increase can scale the two rates by the same factor while preserving their ratio. Whether a particular chemical, transport, or electronic device provides this selective control is an empirical calibration question. A binding reporter, an arbitrary inhibitor, or a change of a shared reservoir does not automatically implement the required operation.

7.2 Which coarse detectors preserve the theorem?

Corollary 7.3 (A sufficient entrance-gateway condition)**.** *Let B contain j but not i. If every directed path from i into B under Q' first visits j, then*

$$H'_{i,j;B}(t) := \int_0^t \big(P'_j(X_s \in B) - P'_i(X_s \in B)\big)\, ds = \int_0^t P'_j(X_s \in B) \Pr_i(\tau_j > t - s)\, ds > 0.$$

Theorem 7.1 therefore also holds for the B-occupancy displacement, with this kernel. The same local lower bound applies since $P'_j(X_s \in B) \ge P'_{jj}(s)$.

The proof is the same first-hitting decomposition: occupation in B before τ_j is zero. The gateway property is checkable by removing j and testing whether i can still reach any state in B.

A more general sufficient condition is a regenerative entrance law. Let ν_A be supported outside B, let ν_B be supported inside B, and suppose $\pi Q' = \gamma(\nu_B - \nu_A)$. If, starting from ν_A, the entrance state at τ_B has law ν_B independent of the entrance time, then the response of B-occupancy is γ times

$$\int_0^t \underset{\nu_B}{\overset{Q'}{\Pr}}(X_s \in B)\, \underset{\nu_A}{\overset{Q'}{\Pr}}(\tau_B > t - s)\, ds > 0.$$

Independence here means the joint entrance law factors, including conditioning on any set of finite entrance times. It does not follow from generic coarse graining. The strong Markov property after entrance and this stated factorization prove the claim.

Without such an interface condition the sign can reverse classically. Consider the four bidirectional channels

$$(i, j, a, b) = (0, 1, 2, 1),\ (0, 2, 10, 1),\ (1, 3, 10, 1),\ (2, 3, 1, 3).$$

The native law is $(7, 4, 80, 30)/121$. Suppressing channel $0 \leftrightarrow 1$ by $1/2$ gives stationary law $(13, 6, 140, 50)/209$. For the coarse detector $B = \{1, 2\}$, the initial displacement derivative is $-5/121$ but the stationary displacement is $10/2299 > 0$. Continuity therefore forces a finite-time crossing. The resolved singleton $\{1\}$ retains the sign theorem. A measured crossing alone is consequently not a device-independent witness of quantum dynamics.

7.3 Calibration and finite independent records

For classical noninvasive paired readout, define $Z = \mathbf{1}_{\{X_0=j\}} - \mathbf{1}_{\{X_t=j\}} \in [-1, 1]$. From N independently prepared records, Hoeffding's inequality gives

$$\Pr(|\overline{Z} - \mathbb{E}Z| \geq s) \leq 2\exp(-Ns^2/2).$$

This classical paired observation is an additional access assumption; an initial quantum measurement can disturb the prepared state.

Suppose the inferred response lies in $[d_-, d_+]$, a calibrated factor lies in $[r_-, r_+] \subset (0, 1)$, and both endpoint exit rates are at most Λ. If preparation is within total variation η of native stationarity, its contribution to paired-response bias is at most 2η. Let b bound this and all other propagated response biases. Set

$$I_D = [d_- - b, d_+ + b] \cap [-1, 1], \qquad I_G = [(1 - r_+)te^{-\Lambda t},\ (1 - r_-)t].$$

An enclosing native-current interval is

$$I_J = \left[\min_{d\in\partial I_D,\ g\in\partial I_G} \frac{d}{g}, \max_{d\in\partial I_D,\ g\in\partial I_G} \frac{d}{g}\right]. \tag{7.4}$$

The denominator is strictly positive. The implementation replaces the exponential by a rational Taylor-series enclosure. If I_J contains zero, the orientation is unresolved. A joint calibration failure allowance is added to the sampling allowance once; calibration and sampling need not be independent for that union bound.

The fixed-pulse kernel interval generally leaves a nonzero magnitude uncertainty even with infinitely many records. One can calibrate the kernel separately or shorten the pulse while increasing sample size. The next bound makes the latter option explicit.

Proposition 7.4 (A constructive magnitude-precision bound)**.** *Assume exact native-stationary preparation, an exactly calibrated $a = 1 - r > 0$, endpoint exit bound $\Lambda > 0$, and $|J_e| \le J_{\max} > 0$. If $\pi_j \le p_{\max} \le 1$, then*

$$\left| \frac{\mathbb{E}Z}{at} - J_e \right| \le J_{\max}\Lambda t, \qquad \mathrm{Var}(Z) \le V(t) := \min\{1, (2p_{\max}\Lambda + aJ_{\max})t\}.$$

For target absolute error $\varepsilon > 0$, choose $t = \varepsilon/(2J_{\max}\Lambda)$ and $s = at\varepsilon/2$. The estimator $\overline{Z}/(at)$ has error at most ε with probability at least $1 - \alpha$ whenever

$$N \ge \frac{2\log(2/\alpha)}{s^2}\left(V(t) + \frac{2s}{3}\right).$$

For fixed nonzero model bounds this is a sufficient $N = O(\varepsilon^{-3})$ record count. It is not a minimax lower bound.

Proof. The response identity and $e^{-\Lambda t} \le H(t)/t \le 1$ give the bias bound, using $1 - e^{-x} \le x$. Since $Z^2 = \mathbf{1}_{\{X_0=j\}} + \mathbf{1}_{\{X_t=j\}} - 2\mathbf{1}_{\{X_0=j, X_t=j\}}$,

$$\mathbb{E}Z^2 = 2\pi_j(1 - P'_{jj}(t)) - D_e(t) \le 2p_{\max}\Lambda t + aJ_{\max}t.$$

Also $Z^2 \le 1$ and $|Z - \mathbb{E}Z| \le 2$. Bernstein's inequality bounds the sampling error by s at the stated count. Dividing by at contributes $\varepsilon/2$, and the deterministic bias contributes the remaining $\varepsilon/2$. □

The precision bound is conditional. A fixed unmodeled response bias is amplified by $1/(at)$ as pulses shorten. The appropriate finite error method must retain this bias or abstain. No rate of convergence is claimed for arbitrary imperfect hardware. If no tighter current cap is available, the same local calibration supplies $J_{\max} = \Lambda/r$: the perturbed exit rates bound ra_e and rb_e, and $|\pi_i a_e - \pi_j b_e| \le \max(a_e, b_e)$. This bound can be conservative but requires no reconstruction of the surrounding network.

7.4 Physical resources and experimental acceptance

Neither a kinetic suppression factor nor an equilibrium-preserving rate ratio is a complete energy account. An applied experiment must include native fuel consumption, stationary preparation, resets, selective control, detector operation, recording, and calibration. If independent calibration supplies a per-record upper energy bound $E_0 + P_{\max}t$, Proposition 7.4 implies the corresponding *sufficient* bound $N(E_0 + P_{\max}t)$. For small ε, its preparation term scales as $O(\varepsilon^{-3})$ and its active-pulse term as $O(\varepsilon^{-2})$. These conditional engineering bounds are not universal thermodynamic precision laws.

The physical acceptance conditions are native stationary preparation; a stable, time-homogeneous generator during each record; selective two-endpoint control with independently bounded cross-effects; a resolved endpoint or certified gateway detector; predetermined pulse duration; and readout while the perturbed generator is still applied. State-dependent stopping, feedback, releasing the perturbation before readout, uncalibrated detector latency, and arbitrary hidden-state aggregation require their own analysis. Violating any of these conditions can explain an observed failure of the sign law.

The implementation also supplies a stronger-access route based on local transition counts. Stopping at a fixed accumulated source-state exposure makes a specified transition count Poisson with parameter equal to rate times exposure. It does not make counts conditionally Poisson at fixed elapsed time. Source exposure can take arbitrarily long wall time; a deadline-limited acquisition must report incomplete exposure rather than silently condition its statistical guarantee on completion.

Chapter 8

Current measurements with imperfect physical access

The selective-channel current assay can tolerate quantified preparation, control, and readout imperfections. Its useful extension is a matched native control: preparing both arms from the same ensemble cancels a preparation penalty that otherwise grows when the pulse is shortened. Common generator drift also cancels to first order. Readout delay admits a conservative safe window expressed through local endpoint hazards.

These results concern a declared finite classical Markov model and calibrated error envelopes. They do not certify those envelopes from the response data, and do not turn a sign crossing into an unconditional quantum witness. The mathematical ingredients are standard semigroup perturbation, coupling, and oscillation contraction; the contribution here is their explicit assembly into a checkable physical current assay.

8.1 The reference experiment

Use row generators and row probability distributions. Let Q be a finite generator with stationary distribution π, and let a separately selectable bidirectional physical channel between $i \neq j$ have rates $a, b \geq 0$. Its generator L has $L_{ij} = a$, $L_{ji} = b$, $L_{ii} = -a$, $L_{jj} = -b$, with all other entries zero. Set

$$s = 1 - r \in (0, 1], \qquad Q_r = Q - sL, \qquad P(t) = e^{tQ}, \quad P_r(t) = e^{tQ_r}, \quad f = \mathbf{1}_j.$$

Both Q, Q_r must be valid generators. The native selected-channel current is $J = \pi_i a - \pi_j b$. Multiple physical channels may connect the same states; selectivity refers to the specified physical channel, not an unobserved choice of algebraic decomposition.

For a stationary preparation, the native-minus-controlled contrast obeys

$$D_\pi(t) = \pi[P(t) - P_r(t)]f = sJH(t), \qquad te^{-\Lambda_p t} \leq H(t) \leq t, \tag{8.1}$$

where Λ_p bounds the controlled exit rates of both endpoints. The positive kernel is $H(t) = \int_0^t (P_{r,jj}(u) - P_{r,ij}(u))\, du$. Its positivity follows from the first-hitting argument in the finite-pulse theorem; its integrand need not be positive. Thus the reference experiment does not require the surrounding network to be reconstructed. Current and rates have units of inverse time; H and all pulse or delay durations have units of time. A response contrast and all probability-error bounds are dimensionless.

8.2 Generator error as a coupling certificate

For a matrix E with zero row sums define

$$\|E\|_{\mathrm{rowTV}} = \max_x \frac{1}{2} \sum_y |E_{xy}|.$$

The diagonal is included. For $E = A - B$, where A, B are generators, this is

$$\max_x \max \left\{ \sum_{y \neq x} (A_{xy} - B_{xy})_+, \sum_{y \neq x} (B_{xy} - A_{xy})_+ \right\}.$$

Theorem 8.1 (Dimension-independent generator error)**.** *Let $A(u), B(u)$ be bounded measurable finite generators on a common state space, and start their processes from the same distribution. If $\|A(u) - B(u)\|_{\mathrm{rowTV}} \leq \kappa(u)$, then*

$$\mathrm{TV}(\mathcal{L}(X_t), \mathcal{L}(Y_t)) \leq 1 - \exp\left[- \int_0^t \kappa(u)\, du \right].$$

Proof. While both processes are at x, couple their equal transitions to y at rate $\min(A_{xy}, B_{xy})$. Let U, V be the total remaining jump rates of the two processes. Their remaining destinations are disjoint. Synchronize remaining jumps to distinct destinations at rate $\min(U, V)$; run the excess $|U - V|$ unilaterally. The total separation hazard is $\max(U, V)$, bounded by $\kappa(u)$. The integrated hazard bounds the probability of ever separating before t. Terminal disagreement and total variation are no larger. Complete the coupling arbitrarily after separation. □

This rate can be half the looser sum of absolute off-diagonal discrepancies. Rows $(-3, 2, 1)$ and $(-3, 1, 2)$ have row-TV discrepancy 1, while that off-diagonal sum is 2. A one-way jump against a frozen process attains $1 - e^{-\kappa t}$, so no universally smaller bound follows from κ alone. The certificate can cover a full piecewise intervention and readout history, provided its integral covers that entire history.

8.3 Matched preparation

For a real column g, write $\mathrm{osc}(g) = \max g - \min g$. A stochastic matrix contracts oscillation, and a zero-mass signed row v satisfies $|vg| \leq \mathrm{TV}(v)\, \mathrm{osc}(g)$, where $\mathrm{TV}(v) = \frac{1}{2} \sum_x |v_x|$.

Theorem 8.2 (Preparation cancellation)**.** *Suppose both arms use the same preparation ρ, with $\mathrm{TV}(\rho, \pi) \leq \eta$. Then*

$$|D_\rho(t) - D_\pi(t)| \leq \min\left\{ \eta s(a+b)t, 2\eta(1 - e^{-s \max(a,b) t}), 2\eta \right\}. \tag{8.2}$$

The same statement holds after any common stochastic terminal operation R, with f replaced by Rf.

Proof. Duhamel's identity gives

$$P(t) - P_r(t) = s \int_0^t P_r(u) L P(t-u)\, du.$$

The only nonzero entries of Lg are $a(g_j - g_i)$ at i and $-b(g_j - g_i)$ at j. Consequently $\mathrm{osc}(Lg) = (a+b)|g_j - g_i| \leq (a+b)\, \mathrm{osc}(g)$. Oscillation contraction bounds that of $[P(t) - P_r(t)]f$ by $s(a+b)t$. Apply the signed-row inequality to $\rho - \pi$. Alternatively, the coupling rate for Q, Q_r is $s \max(a, b)$, so each entry of this column has absolute value at most $1 - e^{-s \max(a,b) t}$. Its oscillation is at most twice that quantity, and is always at most 2. Finally Rf remains in $[0, 1]$. □

The coefficient $\eta s(a+b)$ is attained at first order when $\rho - \pi = (\eta, -\eta)$ on the two endpoints. Dividing by st to estimate current gives a bounded preparation error $\eta(a+b)$ as t tends to zero. Using the older response bound 2η would instead produce $2\eta/(st)$. The improvement requires actual matching of the preparation distributions, not merely the same stated preparation method.

If the controlled and native arms use ρ_c, ρ_n, one may apply the matched bound to ρ_n and add $\zeta = \mathrm{TV}(\rho_c, \rho_n)$ to the response error. That unmatched term can again be amplified by $1/t$. Random assignment to a common ensemble supports matching; it does not establish stationarity or independent records under uncontrolled drift. A steady observed mean cannot by itself calibrate the TV distance to a hidden stationary distribution.

8.4 A safe window for readout delay

After the pulse, suppose both arms evolve under the native generator for delay d, so $R(d) = e^{dQ}$. Let λ_j bound the native exit rate from j, and let β_j bound $\max_{x \neq j} Q_{xj}$. Define

$$\omega(d) = \mathrm{osc}((R(d) - I)f) \leq \bar{\omega}(d) := (1 - e^{-\lambda_j d}) + (1 - e^{-\beta_j d}). \tag{8.3}$$

Indeed, the j entry is $P_{jj}(d) - 1 \leq 0$, with magnitude at most the probability of a departure. Every other entry is $P_{xj}(d) \geq 0$, bounded by the probability of entrance to j. Before that entrance its hazard is at most β_j. Their range gives (8.3).

Theorem 8.3 (Delayed sign protection)**.** *The stationary delayed contrast is $D_\pi(t,d) = sJK(t,d)$, where*

$$K(t,d) = \int_0^t (e_j - e_i) P_r(u) R(d) f \, du,$$

and

$$\max\{-t, t(e^{-\Lambda_p t} - \bar{\omega}(d))\} \leq K(t,d) \leq t.$$

In particular, $\bar{\omega}(d) < e^{-\Lambda_p t}$ guarantees the native current sign despite the delay.

Proof. Multiplying the stationary response identity by $R(d)$ gives the expression for K. Each integrand is a difference of probabilities, so $|K| \leq t$. Furthermore

$$|K - H| \leq \int_0^t |(e_j - e_i) P_r(u)(R(d) - I)f| \, du \leq t\omega(d).$$

Combine this with (8.1). □

This is a sufficient safe window. Failure of its inequality means that the certificate abstains; it does not establish a reversal. Unrestricted readout delay can reverse a classical response, as the earlier counterexample shows.

Now let a clock select a duration $T \in [t_-, t_+]$, $t_- > 0$, independently of the hidden state and preparation, with the same law in both arms. Its distribution need not be known. For a valid uniform delay bound $\bar{\omega}$, the averaged kernel satisfies

$$L := \max\left\{-t_+, \min_{u \in \{t_-, t_+\}} u(e^{-\Lambda_p u} - \bar{\omega})\right\} \leq \mathbb{E}K(T,d) \leq t_+. \tag{8.4}$$

To see why endpoints suffice, the derivative of $u(e^{-\Lambda_p u} - \bar{\omega})$ is $e^{-\Lambda_p u}(1 - \Lambda_p u) - \bar{\omega}$. Before $1/\Lambda_p$ its positive part decreases strictly, and afterwards it is nonpositive. Thus its only possible interior extremum is a maximum; for $\Lambda_p = 0$ the function is linear. With no delay, every positive duration has a positive kernel. State-dependent stopping is a different experiment and is not covered.

8.5 An exact current interval

A common fixed binary detector has mean $o + gp_j$. The offset o cancels in contrasts. Assume $0 < g_- \le g \le g_+ \le 1$; a known negative gain can instead be handled by relabeling detector outputs. Arbitrary coarse partitions are not affine singleton detectors. Setting-dependent detector changes require their own contrast error bound ϵ_{det}.

Let η bound common preparation error, ζ its mismatch between arms, and K_n, K_c the calibrated integrated row-TV generator errors of the full native and controlled histories. With rate caps $a \le a_+$, $b \le b_+$, put

$$b_p = \min\{\eta s(a_+ + b_+)t_+, 2\eta(1 - e^{-s\max(a_+,b_+)t_+}), 2\eta\},$$
$$b = g_+\{b_p + \zeta + (1 - e^{-K_n}) + (1 - e^{-K_c})\} + \epsilon_{\mathrm{det}}.$$

If a sampling procedure supplies an interval $[D_-, D_+]$ for the actual native-minus-controlled recorded mean, widen it by b and intersect with $[-g_+, g_+]$. If the lower kernel bound L in (8.4) is positive, the ideal mean is cJ for some

$$c \in [g_- sL, g_+ st_+].$$

The four endpoint ratios enclose J. Intersect also with the physical cap $[-b_+, a_+]$. An interval excluding zero certifies current orientation; an empty interval indicates inconsistent data and calibration inputs. If $L \le 0$, this certificate does not establish a sign. Joint calibration failure probability adds once to the sampling failure probability.

The implementation uses exact rational arithmetic and positive Taylor remainder bounds for exponentials. It cannot verify an apparatus merely by receiving its asserted calibration numbers. Systematic errors do not disappear with additional records, and preparation, clocks, or generator errors cannot be treated as common unless that premise is justified.

8.6 Current measurement without stationary preparation

When stationarity cannot be established, the prepared ensemble's instantaneous current can still be a useful physical target. For an arbitrary common preparation ρ, put $J_\rho = \rho_i a - \rho_j b$.

Theorem 8.4 (Stationarity-free short-pulse assay)**.** *Let k'_i, k'_j each bound both departure from and maximal entrance to the corresponding endpoint under Q_r. Let λ_j, β_j bound native departure and entrance as above, and suppose $|J_\rho| \le J_+$. Then*

$$\left|\frac{D_\rho(t)}{st} - J_\rho\right| \le \frac{t}{2}\{ak'_i + bk'_j + J_+(\lambda_j + \beta_j)\}. \tag{8.5}$$

No stationary distribution is required.

Proof. Duhamel gives the exact identity

$$D_\rho(t) = s\int_0^t J_{\rho P_r(u)}[P_{jj}(t-u) - P_{ij}(t-u)]\,du.$$

State x membership can change only by departing from or entering x. The probability of such a change before u is at most $1 - e^{-k'_x u} \le k'_x u$, from any starting distribution. Consequently $|J_{\rho P_r(u)} - J_\rho| \le ak'_i u + bk'_j u$. For $h(v) = P_{jj}(v) - P_{ij}(v)$, one has $|h(v)| \le 1$ and

$$|h(v) - 1| \le (1 - e^{-\lambda_j v}) + (1 - e^{-\beta_j v}) \le (\lambda_j + \beta_j)v.$$

Subtract J_ρ inside the integral and integrate the two bounds. □

Without a sharper input, $J_+ = \max(a, b)$ is valid. This theorem targets the instantaneous current of the prepared ensemble. It does not identify an unspecified steady-state flux or extend the all-time sign theorem to arbitrary preparations. It is a finite-difference assay with an explicit bias bound.

8.7 Shared drift cancels to first order

The generic error certificate treats the arms separately. A genuinely shared additive generator error supports a stronger conclusion.

Theorem 8.5 (Common generator drift)**.** *Suppose the actual pulse generators are $Q+E$ and Q_r+E, all four generators being valid, and $\|E\|_{\mathrm{rowTV}} \le \kappa$. With the same arbitrary initial distribution and terminal effect $f \in [0,1]$, let D^E, D denote actual and reference matched contrasts. Then, with $M = \max(a, b)$,*

$$\begin{aligned} |D^E - D| &\le s(a+b+2M) \int_0^t (1 - e^{-\kappa u})\, du \\ &\le \tfrac{1}{2} s(a+b+2M)\kappa t^2. \end{aligned}$$

Proof. Mark actual semigroups by a superscript E. Subtract the two Duhamel response expressions to obtain

$$\begin{aligned} D^E - D = s \int_0^t \rho\big[&(P_r^E(u) - P_r(u)) L P^E(t-u) \\ &+ P_r(u) L (P^E(t-u) - P(t-u))\big] f\, du. \end{aligned}$$

The first row difference has TV at most $1 - e^{-\kappa u}$ and the column $LP^E f$ has oscillation at most $a + b$. For the second term, $\rho P_r(u) L$ is its instantaneous channel current, of absolute value at most M, times $e_j - e_i$. The remaining column has oscillation at most $2(1 - e^{-\kappa(t-u)})$. Integration proves the first inequality and $1 - e^{-\kappa u} \le \kappa u$ proves the second. □

For $\kappa > 0$ the exact integral is $t - (1 - e^{-\kappa t})/\kappa$; it is zero at $\kappa = 0$. After division by st, shared drift contributes at most $(a + b + 2M)\kappa t/2$ to current error, which vanishes as the pulse shrinks. The same norm bound for two different errors is insufficient: their matrices must be the same. A common terminal operation is allowed, but an actual versus reference change of the postpulse delay semigroup needs its own bound. This pulse-only improvement is provided as a separate helper so that it cannot be silently substituted into the complete independent-arm certificate.

8.8 Evidence and operational limits

The exact module `robust_physics.py` implements generator discrepancy, matched preparation, delayed kernels, current intervals, short-pulse bias, and shared-drift bounds. Fourteen tests cover the arithmetic and stated contracts. An additional seeded numerical sweep checks all five principal bounds on 180 constructed positive networks with two through eight states; matrix exponentials are numerical checks, not the proofs. The fixed seed is 71293. All generated records and models are synthetic. No empirical physical validation is reported.

The main experimental improvement is a choice between two explicit targets: native stationary current with a calibrated preparation envelope, or instantaneous prepared-ensemble current with a short-pulse bias bound. Both need matched preparation, selective control or a calibrated spillover

bound, and a justified singleton or protected detector. The delayed safe window and shared-error cancellation are useful only with their corresponding physical rate and matching certificates. The formulas identify the missing calibration when inference fails; they do not replace that calibration.

Chapter 9

A driven-qubit boundary for the classical pulse law

The classical sign restriction concerns a stationary preparation, a calibrated change to one physical edge channel, and observation of a single endpoint state. A driven qubit violates this restriction while retaining positive jump rates. The comparison is conditional on that physical contract. A coarse classical detector or a control that changes additional channels can also produce sign reversals; the result is not a device-independent test of quantumness.

The symbol γ in this chapter denotes positive damping; it is distinct from the signed forcing coefficient called γ in the classical proof.

Write $\mathcal{D}[L](\rho) = L\rho L^\dagger - \{L^\dagger L, \rho\}/2$, use $\hbar = 1$, and let $\sigma_+ = |1\rangle\langle 0|$, $\sigma_- = |0\rangle\langle 1|$. Consider the native generator

$$\mathcal{L}\rho = -i[gX, \rho] + \sum_e (a_e \mathcal{D}[\sigma_+] + b_e \mathcal{D}[\sigma_-])\rho, \qquad \sum_e a_e = \sum_e b_e = K > 0. \tag{9.1}$$

Take $g \geq 0$ without loss of generality: conjugation by Z reverses the sign of X while leaving the populations and jump dissipators unchanged. The stationary preparation is $\rho_* = I/2$. A selected channel has rates (a, b) and native directed current $J = (a - b)/2$. Retain a fraction $r \in [0, 1)$ of its two rates, set $\eta = 1 - r$, and leave all other terms unchanged. Assume $\Gamma = 2K - \eta(a + b) > 0$ and define $\gamma = \Gamma/4$. The generator is of GKSL form. These driven jump channels need not be thermal reservoirs in the Hamiltonian energy basis; no heat interpretation is needed.

With $z = \mathrm{Tr}(Z\rho)$ and $y = \mathrm{Tr}(Y\rho)$,

$$\frac{d}{dt}\begin{pmatrix} z \\ y \end{pmatrix} = A\begin{pmatrix} z \\ y \end{pmatrix} + \begin{pmatrix} 2\eta J \\ 0 \end{pmatrix}, \qquad A = \begin{pmatrix} -4\gamma & 2g \\ -2g & -2\gamma \end{pmatrix}, \qquad z(0) = y(0) = 0. \tag{9.2}$$

Thus the endpoint response is

$$\Delta_B(t) = \mathrm{Tr}(|1\rangle\langle 1|\rho(t)) - \tfrac{1}{2} = -\eta J I(t), \qquad I(t) = \int_0^t (e^{Au})_{11}\, du, \qquad \Delta_B'(0) = -\eta J. \tag{9.3}$$

Theorem 9.1 (Exact reversal threshold)**.** *For $J \neq 0$, the response in (9.3) has the opposite sign to its initial derivative at some positive time if and only if $q = g/\gamma > q_c$, where $q_c > 1$ is the unique solution of*

$$\log q_c = \frac{3\{\pi + \arctan\sqrt{4q_c^2 - 1}\}}{\sqrt{4q_c^2 - 1}}. \tag{9.4}$$

Exact rational bounds give $4.57066453253 < q_c < 4.57066453254$. *For* $g > \gamma/2$, *put* $\Omega = \sqrt{4g^2 - \gamma^2}$ *and*

$$t_* = \frac{\pi + \arctan(\Omega/\gamma)}{\Omega}. \tag{9.5}$$

When reversal occurs, t_* *gives the largest reversal amplitude for these fixed physical parameters.*

Proof. Writing $A = -3\gamma I + B$ gives $B^2 = (\gamma^2 - 4g^2)I$. In the underdamped case,

$$I'(t) = e^{-3\gamma t}\left[\cos(\Omega t) - \frac{\gamma}{\Omega}\sin(\Omega t)\right], \tag{9.6}$$

$$I(t) = \frac{2\gamma + e^{-3\gamma t}[-2\gamma\cos(\Omega t) + (\Omega + 3\gamma^2/\Omega)\sin(\Omega t)]}{4g^2 + 8\gamma^2}. \tag{9.7}$$

Critical points satisfy $\Omega t = \arctan(\Omega/\gamma) + n\pi$, $n \geq 0$. The odd n points are minima. Their values are

$$\frac{2\gamma - 2ge^{-3\gamma t}}{4g^2 + 8\gamma^2}, \tag{9.8}$$

so successive minima strictly increase. Since $I(0) = 0$ and $I(\infty) = 2\gamma/(4g^2 + 8\gamma^2) > 0$, a negative value exists exactly when $ge^{-3\gamma t_*} > \gamma$.

If $0 < g < \gamma/2$, set $h = \sqrt{\gamma^2 - 4g^2}$. Then $I'(t) = e^{-3\gamma t}[\cosh(ht) - (\gamma/h)\sinh(ht)]$ has exactly one zero, from positive to negative. The integral rises from zero and then decreases to a positive limit. At $g = \gamma/2$, replace the bracket by $1 - \gamma t$ and use the same argument. At $g = 0$, $I(t) = (1 - e^{-4\gamma t})/(4\gamma) > 0$.

For $q > 1/2$, the negative-minimum condition is $\log q > 3(\pi + \arctan w)/w$, where $w = \sqrt{4q^2 - 1}$. It cannot hold for $q \leq 1$. For $q > 1$, the left side increases strictly and the right side decreases strictly because

$$\frac{d}{dw}\frac{\pi + \arctan w}{w} = \frac{w/(1 + w^2) - \pi - \arctan w}{w^2} < 0. \tag{9.9}$$

Their difference is negative at $q = 1$ and tends to infinity, proving the unique threshold. Equality merely touches zero. The numerical bracket in the statement is certified using rational enclosures of logarithm, arctangent, square root, and Machin's identity for π. □

A finite protocol with positive rates. Choose channels $(3\kappa, \kappa)$ and $(\kappa, 3\kappa)$, with $\kappa = 1/10$, $g = 1$, and $r = 1/10$. Then $J = 1/10$, $\gamma = 11/100$, and $q = 100/11 > q_c$. All channel rates remain positive. Exact Taylor enclosures yield

$$\Delta_B(1/2) = -0.03416833483914909\ldots, \tag{9.10}$$

$$\Delta_B(9/4) = +0.01523153262660271\ldots. \tag{9.11}$$

For a common affine detector with gain $9/10$, the corresponding absolute short-pulse response is strictly greater than $3/100$, and the positive late-pulse response is strictly greater than $13/1000$. These conservative inequalities are checked with rational arithmetic. Offsets, drift, preparation error, and uncertainty in the intervention require their separate calibration bounds.

For a rational matrix A, the implementation evaluates $S_N = \sum_{k=0}^{N}(A^k)_{11}t^{k+1}/(k+1)!$. With $s = t\|A\|_\infty$ and $N + 3 > s$,

$$|I(t) - S_N| \leq \frac{ts^{N+1}}{(N+2)!}\,\frac{1}{1 - s/(N+3)}. \tag{9.12}$$

This is the geometric bound on the omitted matrix-norm series. At $t = 9/4$ and $N = 80$, its radius is below 4.1×10^{-63}. The incoherent control $g = 0$ has the same native stationary population and channel currents and no sign reversal. It does not have the same full temporal record law.

The time $9/4$ also maximizes the positive response and its square divided by pulse duration within the explicitly tested menu $t = j/4$, $1 \le j \le 16$. This finite-menu acquisition-time comparison omits apparatus overhead and is not a claim of a global energy optimum.

Calibrated pure dephasing. Adding $d\mathcal{D}[Z]$ changes the lower diagonal element of A to $-2\gamma - 2d$. Put $\alpha = 3\gamma + d$, $\beta = \gamma - d$. If $\Omega = \sqrt{4g^2 - \beta^2} > 0$, the same proof gives the exact reversal criterion

$$2g \exp\left[-\frac{\alpha\{\pi + \operatorname{atan2}(\Omega, \beta)\}}{\Omega}\right] > \alpha - \beta. \tag{9.13}$$

The angle is in $(0, \pi)$. If $2g \le |\beta|$, reversal is absent by the hyperbolic version of the preceding proof. This extension describes a specified Markovian dephasing channel, not arbitrary noise or memory.

Chapter 10

A calibrated finite-record response test

An exact response difference becomes an experiment only after preparation, control, detection and sampling have been specified. This chapter turns the preceding classical response constraint into a conservative finite-record test. The conclusion is rejection of a declared classical model class. It is not a device-independent certification of quantum behavior.

10.1 Three settings and one predeclared direction

Use independent fresh preparations in three settings: an unperturbed baseline, a selective pulse held through readout at time t_1, and the same pulse held through readout at time t_2. Each record is a binary detector outcome. Let p_0, p_1, p_2 denote nominal outcome probabilities. The protected classical contract implies

$$(p_1 - p_0)(p_2 - p_0) \geq 0. \tag{10.1}$$

This follows for a singleton endpoint or a validated entrance-gateway region, with native stationary preparation and a constant selective channel change. The two pulse times are fixed in advance. The condition also survives independent positive timing jitter when the same control remains applied. A useful robustness is that the two arms need not have exactly the same retention factor. Each may have its own constant $0 \leq r_s < 1$, provided it suppresses the same native channel and leaves the other dynamics unchanged. Each endpoint population displacement is a positive multiple of the same signed quantity $-J_e$. Exact retention calibration is needed for quantitative power and current estimation, rather than for this qualitative same-sign consequence. A delay after releasing the control requires a new calculation.

A common binary detector with false-positive rate d_0 and gain k maps population p to $d_0 + kp$. It preserves (10.1), even if $k < 0$; $k = 0$ removes the contrast. The known lower contrast used for power must include this gain. A fixed shared offset cancels from both contrasts. Setting-dependent offsets, preparation errors, control cross-talk and unmodeled readout delays do not automatically cancel.

Let actual fixed Bernoulli probabilities $\widetilde{p}_s$ obey $|\widetilde{p}_s - p_s| \leq \eta_s$, $s = 0, 1, 2$. These are independently justified complete-setting envelopes. Take n independent records per arm, also independent across arms, and form empirical frequencies $\widehat{p}_s$. Predeclare the alternative orientation $p_1 < p_0 < p_2$. Write

$$\widehat{d}_1 = \widehat{p}_0 - \widehat{p}_1, \qquad \widehat{d}_2 = \widehat{p}_2 - \widehat{p}_0.$$

Reject the protected classical contract only when both

$$\widehat{d}_1 > \eta_0 + \eta_1 + h_\alpha, \qquad \widehat{d}_2 > \eta_0 + \eta_2 + h_\alpha. \tag{10.2}$$

Otherwise report an inconclusive result. Do not switch the orientation after examining the data without a corresponding multiple-testing adjustment.

Theorem 10.1 (False rejection and power)**.** *Suppose the stated preparation, intervention, detector, independence and calibration contract holds. If*

$$nh_\alpha^2 \geq \log(1/\alpha),$$

the test (10.2) *has false-rejection probability at most* α *uniformly over the protected classical class. If a specified alternative has certified nominal gaps* $p_0 - p_1 \geq g_1 > 0$*,* $p_2 - p_0 \geq g_2 > 0$*, and*

$$h_\alpha + h_\beta \leq m := \min\{g_1 - 2(\eta_0 + \eta_1),\ g_2 - 2(\eta_0 + \eta_2)\}, \qquad nh_\beta^2 \geq \log(2/\beta),$$

then its probability of rejection is at least $1 - \beta$*.*

Proof. A difference of independent empirical Bernoulli frequencies, each based on n records, is a sum of $2n$ bounded terms with squared range sum $2/n$. Hoeffding's inequality therefore gives each one-sided deviation bound $\exp(-nh^2)$. Under (10.1), at least one of the two nominal directed contrasts is nonpositive. Its actual mean is at most its two calibration radii. Rejecting both inequalities implies an upper-tail deviation of this particular contrast, hence the bound α. This is an intersection–union test; sharing the baseline creates no extra union factor for false rejection.

Under the alternative, the actual mean of contrast i is at least its nominal gap minus its two calibration radii. Its rejection threshold contains the same radii once more. Thus failing that threshold requires a downward deviation of at least h_β. A union bound over the two possible failures gives $2\exp(-nh_\beta^2) \leq \beta$. Independence between the contrasts is not required. □

If calibration is itself random, these statements apply conditional on a fixed calibration result whose envelopes hold and for which the stated independence remains valid. A valid joint calibration failure probability δ is added once to an unconditional error bound. A stationary average envelope is insufficient for arbitrarily dependent or drifting records.

10.2 Exact design and the quantum benchmark

When $m > 0$, a sufficient real-valued choice has

$$n \geq \frac{(\sqrt{\log(1/\alpha)} + \sqrt{\log(2/\beta)})^2}{m^2}.$$

The implementation replaces both logarithms and square roots by rigorous rational upper bounds and rounds n upward. It exports h_α, h_β and checks the inequalities against a separately supplied contract. If $m \leq 0$, this particular design reports a calibration limitation; it does not claim that every possible experiment must fail.

For the driven-qubit benchmark, use $\kappa = 1/10$, $g = 1$, $r = 1/10$, $t_1 = 1/2$, $t_2 = 9/4$ and common detector gain $9/10$, with false-positive offset $1/20$ in the synthetic count example. Exact Taylor enclosures certify conservative detected gaps

$$g_1 = 3/100, \qquad g_2 = 13/1000.$$

With $\eta_0 = \eta_1 = \eta_2 = 1/2000$ and $\alpha = \beta = 1/20$, $m = 11/1000$. The exact record count and proof inequalities are exported in `completion/results/pulse_test.json`. The deliberately conservative design needs 110192 records per setting, 330576 in total. This quantifies a practical constraint: a clean algebraic effect can still require substantial data once detector gain and calibration are included. No optimal sample complexity is claimed.

```
from fractions import Fraction as F
from constructive_access.completion import pulse_test

contract = dict(gap_early=F(3,100), gap_late=F(13,1000),
                eta_baseline=F(1,2000), eta_early=F(1,2000),
                eta_late=F(1,2000))
design = pulse_test.plan(**contract)
assert pulse_test.verify_plan(design, **contract)
```

The decision routine accepts measured success counts. Its outputs are

```
reject_declared_classical_contract
inconclusive
```

The packaged count example uses rounded model means and is explicitly synthetic. It is a software demonstration, not a physical experiment or a sampled estimate of power.

10.3 Physical expenditure and model checks

In the benchmark, total pulse exposure is $n(t_1 + t_2) = 11n/4$ in the declared time unit. All $3n$ records also require preparation and readout. If validated apparatus bounds are $E_{\mathrm{reset+read}}$ per record and $P_{\max}$ during the pulse, a corresponding upper bound includes

$$3nE_{\mathrm{reset+read}} + P_{\max}n(t_1 + t_2)$$

and any remaining control, baseline-wait or calibration expenditure. This expression is an accounting rule, not a Landauer lower bound. Pulse duration alone is not an energy measurement.

Before interpreting a rejection, the relevant physical checks are native stationary preparation, a detector resolving a protected state or gateway, selectivity of the changed channel, and control persistence through readout. The classical coarse-region and delayed-readout counterexamples show why these checks matter. If they fail, a rejection can diagnose the access model rather than quantum dynamics. Conversely, no rejection does not establish classicality: coherent models below the response threshold provide explicit negative controls.

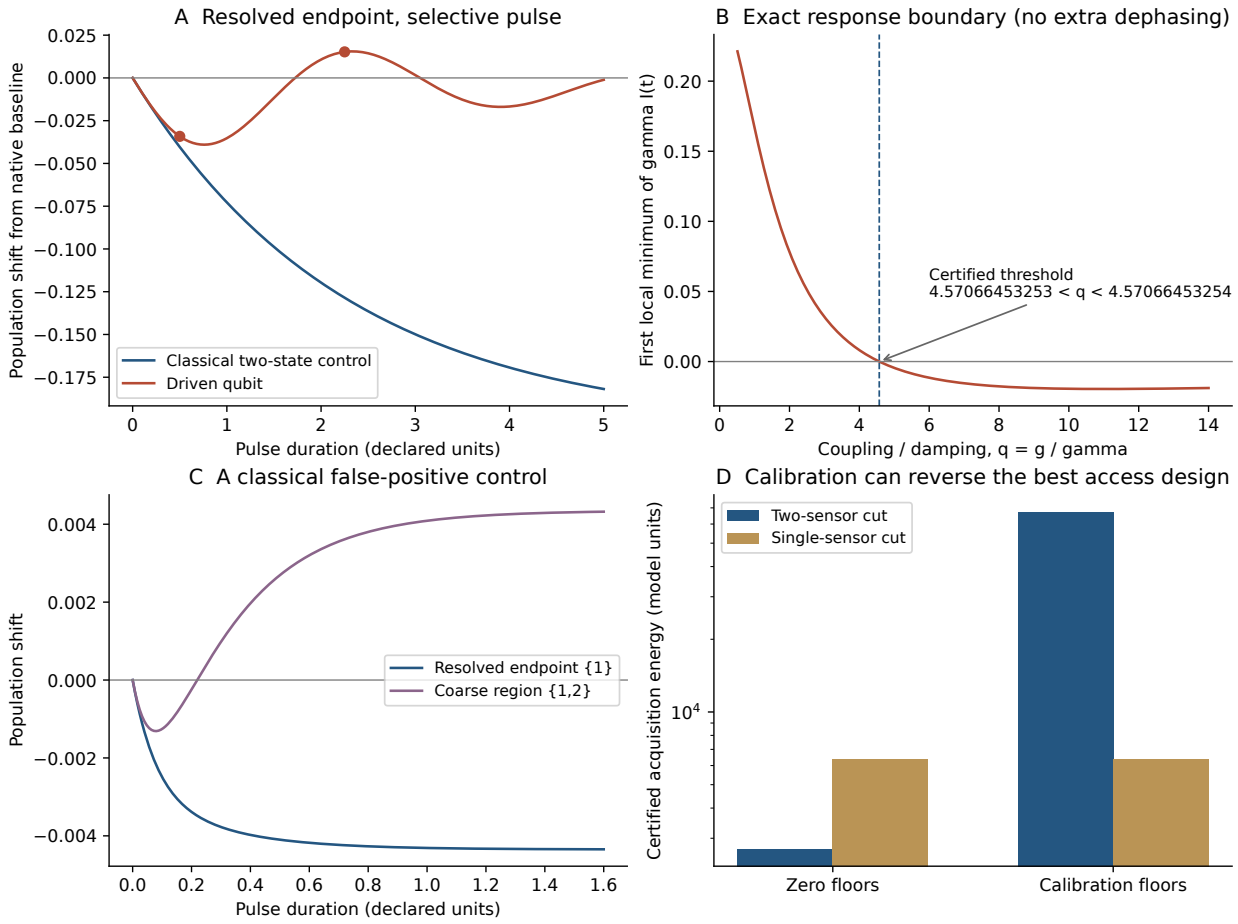


Figure 10.1: Four consequences of the declared access contracts. A: a driven qubit crosses its native endpoint baseline, whereas its incoherent control does not. B: the exact no-extra-dephasing response threshold separates the two regimes. C: a coarse detector also crosses in a fully classical network; its resolved endpoint remains protected. D: calibrated floors reverse the preferred current sensor design at fixed certified accuracy. Curves illustrate proved models; exact rational certificates, proofs and scope conditions accompany the release. No panel represents measured experimental data.

Chapter 11

A preparation and randomized-readout assay

The previous finite-pulse result used a stationary preparation and a selective change to one physical transition channel. Its positive first-hitting kernel also supports a different experiment: prepare a detected state, compare it with another preparation, and randomize the time of a single endpoint reading. This removes the stationary-preparation and selective-control requirements. It replaces them with a trusted reset and a common, time-homogeneous evolution. Neither access contract is automatically easier to establish in an apparatus.

The mathematical result below is a Markov occupation maximum principle. Converting it into an operational assay is useful, but it does not establish the first test of arbitrary-size classical dynamics. Established survival inequalities already address closely related questions [72, 73, 74]. Section 11.5 compares the present assay with those alternatives and records a baseline that outperforms its simplest version.

11.1 The classical preparation inequality

Let Q be a finite, time-homogeneous row generator and $P(t) = e^{tQ}$. A reference arm prepares state j exactly. A comparison arm prepares a fixed probability distribution ρ over the same states. Both arms evolve under Q. Independently of the trajectory and preparation arm, draw a readout time uniformly from $[0, T]$, measured from preparation. Record whether the system occupies the singleton state j.

Theorem 11.1 (Integrated return maximum)**.** *For $i \neq j$ and $T > 0$,*

$$\begin{aligned} K_{ij}(T) &= \int_0^T \left(P_{jj}(t) - P_{ij}(t)\right) dt \\ &= \int_0^T P_{jj}(s) \Pr_i(\tau_j > T - s)\, ds > 0, \end{aligned} \tag{11.1}$$

where τ_j is the first hitting time of j from i. Consequently, for every initial distribution ρ,

$$C_{\rho,j}(T) = \frac{1}{T} \int_0^T \left(P_{jj}(t) - [\rho P(t)]_j\right)\, dt \geq 0. \tag{11.2}$$

The inequality is strict if $\rho_j < 1$. Irreducibility and a stationary preparation are unnecessary. If the exit rates from i, j are at most Λ, then $K_{ij}(T)/T \geq e^{-\Lambda T}$.

Proof. The strong Markov property at τ_j gives

$$P_{ij}(t) = \int_{[0,t]} P_{jj}(t-u)\,\Pr_i(\tau_j \in du).$$

Integrating this identity and interchanging nonnegative integrals gives (11.1). No departure from j for s time units has probability at least $e^{-\Lambda s}$, and no departure from i for $T-s$ units has probability at least $e^{-\Lambda(T-s)}$. These events give the claimed lower bound and strict positivity. Finally, $TC_{\rho,j} = \sum_i \rho_i K_{ij}$, with $K_{jj} = 0$. □

Only the reference preparation must be a trusted singleton. The comparison preparation may be an unknown fixed distribution. This distinction matters for calibration: uncertainty about that comparison distribution need not enlarge the classical null, although it affects predicted power against a specified alternative. A computational-basis label in a quantum experiment does not by itself establish a singleton classical hidden state.

A sufficient coarse-target extension is a genuine reset interface. Let B be the detected set, with reference preparation ν_B supported in B and a comparison preparation outside B. Suppose the conditional state at first entry into B is ν_B, independently of the entry time and previous history. The strong Markov argument then replaces $P_{jj}(s)$ by $\Pr_{\nu_B}(X_s \in B)$ and proves the same inequality. Mixtures of an outside preparation with ν_B are also allowed. Arbitrary states initially inside B are not covered by this sufficient condition.

11.2 Protected clocks and failure controls

Theorem 11.2 (Decreasing readout densities)**.** *Let w be a nonnegative, normalized, nonincreasing readout-time density on $[0,\infty)$, shared by both preparations and independent of the dynamics. Under Theorem 11.1,*

$$\int_0^\infty w(t)\,(P_{jj}(t) - [\rho P(t)]_j)\,dt \geq 0.$$

Every such density is, up to null sets, a mixture of uniform densities:

$$w(t) = \int_{[t,\infty)} \frac{\mu(dT)}{T}, \qquad \mu(dT) = -T\,dw(T), \tag{11.3}$$

where μ is a probability measure.

Proof. For a decreasing integrable density, $tw(t) \to 0$ at both endpoints. Stieltjes integration by parts therefore gives $\int \mu = \int_0^\infty w(t)\,dt = 1$ and $\int_{[t,\infty)} \mu(dT)/T = w(t)$, with endpoint conventions irrelevant to probability densities. Tonelli's theorem expresses the asserted contrast as $\int C_{\rho,j}(T)\mu(dT)$, which is nonnegative by Theorem 11.1. Equivalently, integration by parts against the nonnegative cumulative kernel K proves the sign directly. □

The mixture representation is established mathematics [77]; its role here is to expose experimental timing choices. Randomizing a cutoff T and then drawing time uniformly in $[0,T]$ preserves the inequality exactly. For a specified alternative, no decreasing-density clock can beat the best uniform horizon unless an additional constraint changes the optimization problem. Mean-time constraints, for example, lead to a standard convex design problem.

The following controls show why the assumptions cannot be omitted.

1. *A fixed or delayed readout can reverse the classical contrast.* A directed three-cycle has $P_{jj}(t) < P_{ij}(t)$ at some times. Its uniform average over a delayed interval can also be negative. Uniform sampling must start at preparation. Increasing time densities have no general guarantee.

2. *A coarse detector can reverse the contrast.* Take four states j, i, k, a, with rates $j \to a$ and $i \to k$ both equal to 2, and a, k absorbing. Detect $B = \{j, k\}$, prepare j or i, and average over $[0, 1]$. The contrast is
$$\int_0^1 (2e^{-2t} - 1)\, dt = -e^{-2} < 0.$$
This wholly classical model violates the singleton conclusion because entry into B does not reset to the reference state.

3. *Scheduled dynamics can reverse the contrast.* During $[0, 1]$ allow only $j \to a$ at rate 4; during $[1, 2]$ allow only $i \to j$ at rate 4. Preparing j or i and averaging the singleton-j readout over $[0, 2]$ gives
$$\frac{1}{2}\left[\frac{2(1 - e^{-4})}{4} + e^{-4} - 1\right] < 0.$$
The generator is time dependent. Augmenting the system with an internal clock does not restore the required singleton or reset observation contract.

Nonincreasing density is a sufficient condition, not a necessary characterization of all safe clocks. For example, placing at least half of the readout probability at time zero protects the sign against any remaining stochastic transition kernel. The general clock cone is not characterized by this work. A rejection of the declared classical contract is not a device-independent identification of quantum dynamics.

11.3 A damped quantum model and an exact threshold

Consider the valid qubit Lindblad generator

$$\mathcal{L}(\rho) = -i\frac{\omega}{2}[X, \rho] + \gamma\left(\mathrm{Tr}(\rho)\frac{I}{2} - \rho\right), \qquad \omega > 0, \quad \gamma \geq 0. \tag{11.4}$$

The depolarizing term has Pauli jump operators $\sqrt{\gamma/4}\,X$, $\sqrt{\gamma/4}\,Y$, and $\sqrt{\gamma/4}\,Z$. Prepare $j = |0\rangle$ or $i = |1\rangle$ and detect $\Pi_j = |0\rangle\langle 0|$. The instantaneous contrast is $e^{-\gamma t}\cos(\omega t)$, so uniform timing yields

$$C(T) = \frac{\gamma + e^{-\gamma T}[-\gamma\cos(\omega T) + \omega\sin(\omega T)]}{(\gamma^2 + \omega^2)T}. \tag{11.5}$$

For $\gamma = 0$, this is $\sin(\omega T)/(\omega T)$.

Theorem 11.3 (Orthogonal-preparation damping threshold)**.** *For $\gamma > 0$, a horizon with $C(T) < 0$ exists if and only if*

$$\frac{\omega}{\gamma}\log\frac{\omega}{\gamma} > \frac{3\pi}{2}, \quad \textit{equivalently} \quad \frac{\omega}{\gamma} > e^{W(3\pi/2)} = 3.644173671645632\ldots. \tag{11.6}$$

Here W is the principal real Lambert function. This threshold concerns (11.4), the stated orthogonal preparations, and the decreasing-clock class. It is not a universal coherence threshold.

Proof. Write $I(T) = TC(T)$. Then $I'(T) = e^{-\gamma T}\cos(\omega T)$. Its first local minimum following the initial maximum occurs at $T_* = 3\pi/(2\omega)$; subsequent minima occur one full period apart and have values

$$I(T_n) = \frac{\gamma - \omega e^{-\gamma(3\pi/2+2\pi n)/\omega}}{\gamma^2+\omega^2}, \qquad n = 0, 1, \dots.$$

These values increase with n. Since $I(0) = 0$, a negative value exists exactly when $\omega e^{-3\pi\gamma/(2\omega)} > \gamma$. Setting $r = \omega/\gamma$ gives $r \log r > 3\pi/2$, whose unique root above one is $e^{W(3\pi/2)}$. The mixture theorem proves that decreasing clocks cannot produce a reversal when every uniform horizon is nonnegative. □

When reversal exists, the most negative uniform contrast occurs within $(\pi/\omega, 3\pi/(2\omega))$. Indeed $C'(T) = [TI'(T) - I(T)]/T^2$, and the numerator is strictly increasing throughout that interval because its derivative is $TI''(T) > 0$. It changes sign between the first zero of I and its first negative minimum. Later periods increase a negative numerator I toward its positive limiting value while also increasing T, so they cannot improve the negative average.

With $\gamma = 1/10$, $\omega = 2$, $T = 9/4$, and a common affine detector of gain $9/10$, the observed contrast is

$$C_{\mathrm{obs}} = -0.1440713137250376\ldots < -\frac{18}{125}.$$

The strict rational inequality is certified by an 80-degree Taylor enclosure, independent of floating-point agreement. Mean evolution time is $9/8$ per record. Preparation, readout, reset, and control energy remain separate costs. An exponentially distributed readout with rate λ, by contrast, yields

$$C_{\mathrm{exp}} = \frac{\lambda(\lambda+\gamma)}{(\lambda+\gamma)^2+\omega^2} > 0.$$

It respects the classical protection but has no power against this alternative.

11.4 The best comparison preparation

For a quantum channel semigroup Φ_t and a trusted reference state $|j\rangle$, define the averaged accessible effect

$$A_T = \frac{1}{T}\int_0^T \Phi_t^\dagger(\Pi_j)\, dt.$$

The reference score is $\langle j|A_T|j\rangle$; arbitrary quantum state preparation attains the maximal score $\lambda_{\max}(A_T)$. The Rayleigh variational principle therefore gives the optimal reversal gap

$$V_T = \lambda_{\max}(A_T) - \langle j|A_T|j\rangle. \tag{11.7}$$

This is established spectral optimization applied to experimental design, not a new eigenvalue theorem. A nonzero matrix element connecting j to its orthogonal subspace guarantees $V_T > 0$.

For (11.4), write

$$C = \frac{1}{T}\int_0^T e^{-\gamma t}\cos(\omega t)\, dt,$$
$$S = \frac{1}{T}\int_0^T e^{-\gamma t}\sin(\omega t)\, dt = \frac{\omega - e^{-\gamma T}[\omega\cos(\omega T) + \gamma\sin(\omega T)]}{(\gamma^2+\omega^2)T}.$$

Then $A_T = (I + CZ + SY)/2$ for the Hamiltonian convention in (11.4), and

$$V_T = \frac{\sqrt{C^2 + S^2} - C}{2}.$$

For $\gamma > 0$, the integrated sine S is positive at every $T > 0$: successive positive sine lobes precede their negative partners and carry larger exponential weight. Within a negative lobe the smallest accumulated value is its endpoint, where that paired contribution remains positive. Thus $V_T > 0$ for every $\gamma, \omega, T > 0$ when the comparison state can be coherently optimized. This does not promise useful precision under arbitrarily strong damping; for fixed T and $\gamma \gg \omega$, the gap scales approximately as $\omega^2/(4\gamma^3 T)$.

A fixed $+Y$ eigenstate is often nearly optimal and avoids a model-dependent preparation angle. Its gap is $(S - C)/2$. At $\gamma = 1/10$, $\omega = 2$, $T = 7/4$, and detector gain $9/10$, exact rational enclosures give

$$V_{+Y,\mathrm{obs}} = 0.257271133487835\ldots > \frac{257}{1000}.$$

Mean evolution time is $7/8$. Coherent comparison preparation is an additional physical resource, and already presumes the relevant coherent controls; this example is not evidence for device-independent quantum certification.

11.5 Calibration and established experimental baselines

A common fixed binary detector with response $m(p) = d_0 + kp$, $k > 0$, preserves the classical sign and cancels its offset. Reference-preparation total-variation error ϵ_j weakens the classical null by at most ϵ_j. Comparison-preparation error is unnecessary for that null when any initial distribution is allowed, but contributes to predicted alternative power. A common readout-law discrepancy of total variation δ changes a contrast by at most 2δ. Separate detector probability errors add their two radii. A joint calibrated record-law bound should replace, rather than duplicate, errors already accounted for individually. Statistical confidence cannot establish the reset or clock assumptions from the same test records.

A close prior comparator is the survival inequality [72]

$$|2P_{jj}(t) - P_{jj}(2t)| \le 1.$$

The classical semigroup actually gives the stronger relation

$$P_{jj}(2t) = \sum_k P_{jk}(t) P_{kj}(t) \ge P_{jj}(t)^2. \tag{11.8}$$

The usual upper inequality follows from $2p - p^2 \le 1$. To estimate (11.8) directly, multiply the binary outcomes of two independent length-t records and compare that product with a length-$2t$ record. The product is Bernoulli with mean $P_{jj}(t)^2$ and consumes two physical preparations.

For a linear contrast $\sum_k a_k p_k$ with positive alternative gap g, a common sufficient Hoeffding design has resource proxy

$$\mathcal{B} = \frac{\left(\sum_k |a_k| \sqrt{c_k}\right)^2}{g^2}, \tag{11.9}$$

where c_k is the resource cost of one independent observation of arm k. The same confidence factor multiplies this proxy for all methods. This is a comparison of sufficient concentration-bound designs, not their exact minimum sample complexities. For randomized clocks, evolution-time cost below

means *expected* evolution time; it is not a hard bound on elapsed laboratory time or a measurement of energy.

Table 11.1 fixes $\gamma = 0.1$, $\omega = 2$, ideal known readout, independent records, and zero preparation overhead. Each column optimizes its own time parameter numerically on $[0.001, 5]$. Physical record costs for the product arm include both preparations. The coherent-preparation variants use an additional control resource.

Method	Record proxy	Expected-time proxy
Uniform clock, orthogonal preparation	156.08	174.02
Established linear survival inequality	219.68	133.64
Classical survival-product inequality	75.84	58.02
Uniform clock, fixed quadrature preparation	48.81	40.15
Uniform clock, optimized preparation	48.39	40.02

Table 11.1: Matched sufficient-bound comparisons. Lower is better. These are numerical design results, not physical measurements or certified global statistical optima. The survival-product baseline beats the simplest new assay; extra coherent preparation changes the available operations.

The comparison rejects a claim of universal superiority for the randomized assay. Its contribution is a different exact access contract, protected timing families, and a route to calibrated design. Which method is experimentally preferable depends on preparation access, detector errors, timing, and actual apparatus costs. A framework becomes more useful when it selects among these methods and records their failure cases, not when it relabels every modeled quantum signal as a breakthrough.

11.6 Preparation design for a baseline-subtracted dimension witness

The dimension result developed later, Proposition 12.4, changes the null assumptions again. For a homogeneous CTMC with at most three hidden states, it permits arbitrary fixed preparations μ, ν and an arbitrary fixed stochastic detector $f \in [0, 1]$. Writing

$$C_0 = (\mu - \nu)f, \qquad C_T = (\mu - \nu)\frac{1}{T}\int_0^T e^{tQ} f \, dt,$$

its bound is $|C_0 - C_T| \leq 1$. Separately measuring the two preparations at baseline and at randomized time therefore removes the exact hidden-reset assumption of the earlier assay, at the price of a restriction on classical hidden-state dimension. Section 12.6 exhibits the importance of that restriction. The quantum design calculation is independent of the proof of the classical bound.

Lemma 11.4 (Quantum preparation design for the dimension witness)**.** *For a quantum averaged effect A_T and the same baseline effect Π_j, let $D_T = \Pi_j - A_T$. The largest value of*

$$W = \mathrm{Tr}[D_T(\rho_+ - \rho_-)]$$

over two arbitrary quantum preparations is the spectral diameter $\lambda_{\max}(D_T) - \lambda_{\min}(D_T)$. For the qubit model (11.4), it is

$$W_{\max}(T) = \sqrt{(1 - C)^2 + S^2}, \tag{11.10}$$

attained by antipodal pure Bloch vectors parallel to $(0, -S, 1 - C)$.

Proof. The maximum and minimum expectations of a Hermitian matrix are its largest and smallest eigenvalues, independently attainable by the two preparations. Here $D_T = [(1-C)Z - SY]/2$, whose eigenvalues are $\pm\sqrt{(1-C)^2+S^2}/2$. □

This is a standard spectral variational calculation used for experimental selection. Orthogonal Z preparations instead give $W = 1 - C$, whose optimal horizon is exactly the most negative uniform contrast studied above. A common affine detector of gain k multiplies these modeled quantum contrasts by k. For those orthogonal preparations, a predicted violation requires $k > 1/(1-C)$. The classical dimension bound itself does *not* require this affine detector model or a known positive gain: it already permits any fixed stochastic f. Gain enters prediction of experimental power, not validity of that null.

A simple rational preparation gives a larger certified margin than the orthogonal pair. Set

$$\gamma = \frac{1}{10}, \quad \omega = 2, \quad T = 2, \quad k = \frac{9}{10}, \qquad n = \left(0, -\frac{12}{37}, \frac{35}{37}\right).$$

The vectors n and $-n$ are pure because $12^2 + 35^2 = 37^2$. Exact rational Taylor enclosures certify

$$W = k\left[\frac{35}{37}(1-C) + \frac{12}{37}S\right] = 1.0806050936582667\ldots > \frac{27}{25}. \tag{11.11}$$

Each randomized-time arm has mean evolution time 1; each baseline arm has zero evolution time in the ideal timing convention. Four equally sampled arms therefore have mean evolution time 1/2 per physical record. All four arms still incur preparation, measurement, and reset costs. A common uncontrolled readout delay must be included consistently in the baseline and random-time effects; it is not automatically covered by the zero-delay alternative prediction above.

Numerical optimization gives an ideal-state value about 1.20123 at $T \simeq 2.04524$ for this damping and frequency, or 1.08111 after gain 0.9. The rational design is close to that numerical value, but the reported search is not a certificate of a global optimum. Established dimension witnesses and survival tests remain relevant comparators. The present result is an explicit physical alternative and a reproducible preparation choice for the stated finite-dimension null, not an unrestricted exclusion of classical hidden mechanisms.

Chapter 12

A sharp boundary for hidden classical explanations

A response reversal can establish that a declared experimental model is inadequate without identifying its replacement. Hidden detector states, imperfect preparation, and uncharacterized crossing rates can all matter. This chapter turns two of those alternatives into quantitative requirements. Its main result interpolates sharply between a protected regenerative interface and unrestricted coarse observation. It is a theorem about specified stochastic models, with no assumption of historical priority or empirical confirmation.

12.1 The measured quantity and the physical assumptions

Let R be a finite continuous-time Markov generator and B a detector region. An initial distribution ν_A is supported outside B and ν_B inside B. For any initial distribution ν, define

$$U_\nu(T) = \int_0^T \Pr_\nu(X_s \in B)\, ds, \qquad H(T) = U_{\nu_B}(T) - U_{\nu_A}(T).$$

The normalized contrast $H(T)/T$ is directly observable by preparing either initial distribution and reading the same binary detector at an independently chosen uniform time in $[0, T]$. This formulation requires neither stationary preparation nor a current measurement. Preparation errors and detector errors must still be propagated into the contrast interval.

In the stationary selective-channel experiment developed earlier, the same quantity enters through

$$D_B(T) = [\pi e^{TR}](B) - \pi(B) = \gamma H(T), \qquad \pi R = \gamma(\nu_B - \nu_A).$$

Thus the present result can also interpret a finite-pulse response, provided the forcing identity and any division by γ have been separately justified. The direct preparation experiment and this response-based experiment are separate access contracts.

We assume uniform *crossing-hazard* bounds

$$\sum_{y \in B} R_{xy} \le a \quad (x \notin B), \qquad \sum_{y \notin B} R_{xy} \le b \quad (x \in B). \tag{12.1}$$

These are bounds on every relevant hidden state, not averages inferred from steady occupancy. Rates of motion within either detector region need not be bounded. In particular, a proposed classical hidden-state model is not entitled to use only the dissipative jump rates of a Lindblad description as its hazard caps: that substitution would itself need physical justification.

Let τ be the first entrance into B starting from ν_A. Write μ for the possibly defective distribution of τ and ζ_u for the entrance state distribution conditional on $\tau = u$. A reset-defect bound η means

$$\mathrm{TV}(\zeta_u, \nu_B) \le \eta, \qquad \mu\text{-almost every } u \le T, \qquad 0 \le \eta \le 1. \tag{12.2}$$

The conditioning on entrance time is substantive. An unconditional average entrance distribution does not establish this condition. When ν_B is the single state j, the defect at time u is precisely the conditional probability of entering through a different hidden state. A nonzero defect can arise in a perfectly Markovian classical model; it is not synonymous with non-Markovianity.

12.2 A sharp hazard–reset tradeoff

For $c \ge 0$ set

$$A_c(T) = \begin{cases} (1 - e^{-cT})/c, & c > 0, \\ T, & c = 0, \end{cases}$$

and define

$$G_{a,b}(T) = \int_0^T e^{-bs - a(T-s)}\, ds, \tag{12.3}$$

$$K_{a,b}(T) = T - A_a(T) - A_b(T) + G_{a,b}(T). \tag{12.4}$$

Here $G_{a,a}(T) = Te^{-aT}$ and, for $a \ne b$, $G_{a,b}(T) = (e^{-aT} - e^{-bT})/(b - a)$.

Theorem 12.1 (Sharp coarse-detector response bound)**.** *Under* (12.1) *and* (12.2),

$$H(T) \;\ge\; G_{a,b}(T) - \eta K_{a,b}(T) = (1 - \eta) G_{a,b}(T) + \eta\{A_a(T) + A_b(T) - T\}. \tag{12.5}$$

The lower bound is attained by a finite classical Markov model for every $a, b > 0$, $T > 0$, and $\eta \in [0, 1]$.

Proof. Write $p_B(s) = \Pr_{\nu_B}(X_s \in B)$ and $S(u) = \Pr_{\nu_A}(\tau > u)$. The strong Markov property gives

$$U_{\nu_A}(T) = \int_{[0,T]} U_{\zeta_u}(T - u)\, \mu(du).$$

Adding and subtracting the same expression with ζ_u replaced by ν_B yields the exact decomposition

$$\begin{aligned} H(T) &= G_{\mathrm{actual}}(T) - \mathcal{R}(T), \\ G_{\mathrm{actual}}(T) &= \int_0^T p_B(s) S(T - s)\, ds, \\ \mathcal{R}(T) &= \int_{[0,T]} [U_{\zeta_u}(T - u) - U_{\nu_B}(T - u)]\, \mu(du). \end{aligned}$$

The hazard caps imply $p_B(s) \ge e^{-bs}$ and $S(u) \ge e^{-au}$ by considering trajectories that have not yet left their initial region. Consequently $G_{\mathrm{actual}}(T) \ge G_{a,b}(T)$.

For every initial state $x \in B$, $A_b(v) \le U_x(v) \le v$. Total variation therefore bounds the reset term by

$$\mathcal{R}(T) \le \eta\, \mathbb{E}[(T - \tau)_+ - A_b((T - \tau)_+)].$$

The function $v - A_b(v)$ is nonnegative and increasing. Since the entrance hazard is at most a, τ stochastically dominates an exponential variable of rate a; mass at $\tau = \infty$ contributes zero. For $a > 0$, the last expectation is at most

$$\begin{aligned}\int_0^T ae^{-au}[T - u - A_b(T-u)]\,du &= \int_0^T [1 - e^{-a(T-s)}][1 - e^{-bs}]\,ds \\ &= K_{a,b}(T).\end{aligned}$$

The formula extends continuously to $a = 0$ or $b = 0$, when $K = 0$. These estimates prove the inequality.

For attainment, use four states i, j, k, ℓ, detector $B = \{j, k\}$, $\nu_A = \delta_i$, $\nu_B = \delta_j$, and the only transitions

$$i \longrightarrow j : \ a(1-\eta), \qquad i \longrightarrow k : \ a\eta, \qquad j \longrightarrow \ell : \ b.$$

States k and ℓ are absorbing. The entrance time is exponential with rate a and its independent entrance law is $(1-\eta)\delta_j + \eta\delta_k$. Moreover, $U_j(v) = A_b(v)$ and $U_k(v) = v$. Every inequality above is an equality. □

The small-time defect penalty is $K_{a,b}(T) = abT^3/6 + O(T^4)$, whereas $G_{a,b}(T) = T - (a + b)T^2/2 + O(T^3)$. The bounds are therefore sensitive to the time available for both entrance and exit. The sharpness claim concerns the stated class, not the precision attainable at fixed energy on an unspecified apparatus.

For $0 < \eta < 1$, the same bound is the infimum within irreducible models compatible with the stationary selective-channel experiment. To see this, add $j \to i$, $k \to j$, and $\ell \to i$ at rate ε, and replace $j \to \ell$ by $b - \varepsilon$. For sufficiently small positive ε, the chain is irreducible, both hazard caps and the entrance-reset law remain unchanged, and its kernel tends to the attaining kernel. Double the two $i \leftrightarrow j$ rates to define the native generator Q. It has a positive stationary law π, and suppressing that channel by one half returns R. The response identity is then exact. The native current tends to zero in this limit; approximate sharpness does not confer a finite-sample advantage for measuring a shrinking current.

12.3 Comparison preparations that already occupy the detector

The outside-support condition cannot be inferred from a preparation's name or from an analogy with a qubit basis. Suppose instead that the comparison state is

$$\rho = p\nu_{\text{in}} + (1-p)\nu_{\text{out}}, \quad \operatorname{supp}\nu_{\text{in}} \subseteq B, \quad \operatorname{supp}\nu_{\text{out}} \subseteq B^c,$$

with $\mathrm{TV}(\nu_{\text{in}}, \nu_B) \le \eta_0$. Let the first-entrance reset defect from ν_{out} be bounded by η_1.

Proposition 12.2 (Sharp mixed-preparation extension)**.** *Under the same hazard caps,*

$$U_{\nu_B}(T) - U_\rho(T) \ge (1-p)\{G_{a,b}(T) - \eta_1 K_{a,b}(T)\} - p\eta_0\{T - A_b(T)\}.$$

Proof. Decompose the contrast into its inside and outside contributions. Apply the previous theorem to the outside contribution and $U_{\nu_B} - U_{\nu_{\text{in}}} \ge -\eta_0(T - A_b)$ to the inside contribution. The same four-state extremizer, with entrance mixture parameter η_1 and $\nu_{\text{in}} = (1-\eta_0)\delta_j + \eta_0\delta_k$, attains the result. □

With no reset assumptions, $\eta_0 = \eta_1 = 1$, the sharp bound simplifies to

$$U_{\nu_B}(T) - U_\rho(T) \geq A_b(T) + (1-p)A_a(T) - T.$$

This expression also follows directly from conditional crossing-hazard survival and therefore has the broader classical-process scope described next. If only an upper bound on p is known, use that upper bound. For $p = 1$ there need not be any positive protection interval. A preparation with one-half detector occupancy must not be analyzed using the $p = 0$ formula.

12.4 An explanation bound without a protected reset assumption

Setting $\eta = 1$ removes the reset assumption entirely:

$$H(T) \geq A_a(T) + A_b(T) - T. \tag{12.6}$$

This special case has a still broader interpretation. Consider any classical binary process with the stated initial detector value and with uniformly bounded conditional crossing intensities: at most a from outside to inside and at most b in the reverse direction, under every admissible history. Survival before the first crossing obeys the same exponential bounds. At each time s,

$$\Pr_{\nu_B}(B, s) \geq e^{-bs}, \qquad \Pr_{\nu_A}(B, s) \leq 1 - e^{-as}.$$

Integration proves (12.6). No finite hidden dimension, Markov property, stationarity, or specified first-entrance reset is needed for this case. The hypothesis of finite uniformly bounded crossing intensities remains essential; deterministic jumps and unbounded boundary chatter do not satisfy it.

With a common cap $a, b \leq \lambda$, define

$$F(z) = \frac{2(1 - e^{-z})}{z} - 1, \qquad F(0) = 1.$$

Then $H(T)/T \geq F(\lambda T)$. The function F decreases strictly from 1 to -1. Its zero is $z_0 = 1.593624260\ldots$, defined exactly by $z_0 = 2(1 - e^{-z_0})$. Hence an integrated contrast cannot become negative before $T = z_0/\lambda$, regardless of the number of hidden classical states. The supplied rational certificates place z_0 strictly between 1.593 and 1.594.

This result can be used without pretending that a classical hazard cap has already been experimentally certified. If $c_U \in (-1, 1)$ is a calibrated upper confidence bound on $H(T)/T$, every classical explanation with finite conditional crossing intensities must have

$$\lambda \geq \frac{F^{-1}(c_U)}{T}. \tag{12.7}$$

This is a lower bound on the peak crossing intensity required of that explanation. It becomes a rejection test only when an independently justified upper cap is smaller. Observing a negative contrast by itself neither excludes all classical models nor establishes a thermodynamic energy cost for their activity.

12.5 What finite data can establish

Under calibrated caps a, b with $K_{a,b}(T) > 0$, an upper confidence bound H_U on $H(T)$ implies

$$\eta \geq \max\left\{0, \frac{G_{a,b}(T) - H_U}{K_{a,b}(T)}\right\}. \tag{12.8}$$

A value exceeding one is inconsistent with the declared bounded-hazard classical contract. A positive value below one states a minimum required reset defect. It does not identify which microscopic state or physical mechanism caused that defect. If $K = 0$, the response bound is independent of η and one must test that bound directly instead of dividing by zero.

For a simple direct protocol, collect N independent binary observations from each initial preparation, choosing the observation time independently and uniformly in $[0, T]$ for every record. Let $\widehat{c}$ be the difference of the two empirical means in the order B minus A. Weighted Hoeffding gives

$$\Pr\{H(T)/T > \widehat{c} + u\} \le e^{-Nu^2}$$

under the ideal model. If separately calibrated per-arm record laws differ in total variation by at most ϵ_B and ϵ_A, use $c_U = \widehat{c} + u + \epsilon_B + \epsilon_A$. A joint calibration failure allowance is added once. The bound assumes independent records and predetermined sample counts; adaptive stopping requires a valid sequential procedure. The exact binomial procedure in the statistical chapter offers a complementary encoded single-bit implementation.

The software implements the sharp bound and its inversions with rational exponential enclosures. An output certificate establishes a conditional mathematical statement, not the correctness of the supplied hardware calibration. The hazard-inversion helper gives both endpoints of a certified root bracket. It does not report the numerical root as an exact physical rate.

12.6 A sharp four-state requirement without a hazard calibration

The geometry of a binary detector gives an additional explanation bound that does not require a kinetic cap or a protected reset interface.

Theorem 12.3 (Minimum classical state count for a negative integrated contrast)**.** *Consider a common finite, time-homogeneous classical Markov generator, a fixed binary detector region B, and initial distributions supported respectively inside and outside B. If the hidden state space has at most three states, then $H(T) > 0$ for every finite $T > 0$. Thus $H(T) < 0$ requires at least four classical Markov states. Four states suffice to realize any prescribed value $H(T)/T \in (-1, 0)$ at a prescribed positive T.*

Proof. Both detector regions are nonempty. With at most three states, one of them is a singleton. If $B = \{j\}$, the first-hitting return argument gives

$$U_j(T) - U_{\nu_A}(T) = \int_0^T P_j(X_s = j) \Pr_{\nu_A}(\tau_j > T - s)\, ds > 0.$$

The inequality is strict because a finite generator permits positive no-jump probabilities at every finite time. If instead $B^c = \{i\}$, complement the detector:

$$U^B_{\nu_B}(T) - U^B_i(T) = U^{B^c}_i(T) - U^{B^c}_{\nu_B}(T) > 0,$$

and apply the same argument. This covers arbitrary mixtures within the allowed initial supports.

For sufficiency, use the four-state attaining model with $\eta = 1$ and $a = b = \lambda$. Its contrast is $F(\lambda T)$, which decreases continuously from 1 to -1. Every specified contrast in $(-1, 0)$ therefore determines a unique positive λ and is attained exactly. □

The matching claim concerns one integrated contrast. Four states need not reproduce the full time-resolved record, multiple intervention responses, or a complete quantum process. State count

also does not determine stationary Shannon entropy, physical memory cost, or thermodynamic dissipation.

A qubit supplies a direct mathematical comparison. Under unitary evolution with Hamiltonian gX, initial states $|1\rangle$ and $|0\rangle$, and detector $|1\rangle\langle 1|$, the contrast is

$$\frac{H(T)}{T} = \frac{\sin(2gT)}{2gT}.$$

At $T = 3\pi/(4g)$ it equals $-2/(3\pi)$. A two-dimensional quantum system can therefore produce a contrast that requires at least four states in the stated classical Markov representation. This is a conditional state-count separation, not a device-independent proof of quantumness or a claim that classical four-state models are inadequate for that scalar observation.

Proposition 12.4 (A conservative preparation-robust state-count witness)**.** *Allow a fixed stochastic binary detector with response $f(x) \in [0,1]$ and arbitrary initial preparations ν_B, ν_A. Write*

$$m_B = \nu_B f, \quad m_A = \nu_A f, \quad C_f(T) = \frac{1}{T}\int_0^T (\nu_B - \nu_A)e^{sR} f \, ds.$$

Write $C_0 = m_B - m_A$. Every homogeneous classical model with at most three states satisfies the directly observable bound

$$\boxed{|C_0 - C_f(T)| \le 1.} \tag{12.9}$$

In particular,

$$C_f(T) \ge m_B - m_A - 1. \tag{12.10}$$

Proof. First consider a deterministic detector region D. Conditional preparations inside and outside D have nonnegative integrated contrast by the preceding theorem. Replacing the actual preparations by these conditional preparations changes a normalized occupation expectation by at most $\nu_B(D^c) + \nu_A(D)$ in total. If a conditional preparation is undefined, this sum is at least one and the bound follows from the range $[-1,1]$. Thus the normalized contrast for D is at least $-\nu_B(D^c) - \nu_A(D)$.

Use the layer-cake representation $f(x) = \int_0^1 \mathbf{1}_{\{f(x)>u\}}\, du$ and integrate that bound over the threshold detector regions. The two integrated impurity terms are $1-\nu_B f$ and $\nu_A f$, proving (12.10). Swapping the two preparations proves the opposite bound and hence (12.9). □

The bound in (12.9) is sharp as a supremum for the three-state class. Let a reference state be absorbing with detector response $c \in [0,1]$. Let the comparison state have detector response zero and jump at rate λ to a third absorbing state with detector response one. Then $C_0 = c$, $C_f(T) = c - 1 + A_\lambda(T)/T$, and $C_0 - C_f(T) = 1 - A_\lambda(T)/T \to 1$ as $\lambda \to \infty$. The more specific impurity form is not claimed optimal for every separately specified pair of initial means.

With simultaneous confidence bounds C_U, $m_{B,L}$, and $m_{A,U}$, the condition

$$C_U < m_{B,L} - m_{A,U} - 1$$

excludes every classical model with at most three homogeneous Markov states under the fixed-detector contract. It requires no statement that either preparation has exactly zero hidden impurity. The same observable bound applies to every independent nonincreasing observation-clock density starting at zero. The clock chapter represents such a density as a mixture of uniform clocks, so

$$C_w = \int C_f(T)\, \mu(dT), \qquad |C_0 - C_w| \le \int |C_0 - C_f(T)|\, \mu(dT) \le 1.$$

A clock chosen using the evolving hidden trajectory is not covered by this argument.

The directly observable score $C_0 - C_f(T)$ needs no assumption of exactly pure hidden preparations. It requires the same detector response function at baseline and at the random observation time, and the same preparations in the corresponding arms. Common fixed readout noise is already covered by $f \in [0,1]$. If a detector is permitted to reverse its labels between baseline and readout, a motionless two-state system can give $C_0 = 1$ and $C_f(T) = -1$; readout stability is therefore essential.

A common known delay d before both baseline and random-time observations does not invalidate (12.9). Replace the preparations by $\nu_B e^{dR}$ and $\nu_A e^{dR}$, which are allowed to be arbitrary. The same argument covers an independent common delay distribution used in every arm, by averaging these effective preparations; the delay must also be independent of the observation-clock draw. Different delay distributions, changed generators, or different detector calibrations require separate error bounds.

The four-state equal-hazard construction can match both averaged arm probabilities of a unital qubit with orthogonal preparations and a projective binary detector: they are $A_\lambda(T)/T$ and $1 - A_\lambda(T)/T$, equivalently $(1 + C)/2$ and $(1 - C)/2$ once $F(\lambda T) = C$. This statement applies when the random timestamps are discarded. Keeping the timestamp, intervening within records, or comparing several clocks yields additional data that the scalar matching argument does not reproduce.

Time homogeneity is essential. A three-state classical counterexample starts in j or i with detector $\{j\}$. First allow $j \to k$ at rate ten for time $1/10$; next allow $i \to j$ at rate ten for time $1/10$; then freeze all rates for time one. The integrated contrast over the whole interval is

$$H(6/5) = \frac{19}{10e} - \frac{9}{10} < 0.$$

Thus an external control clock, unmodeled memory, or preparation-dependent kinetics can invalidate the state-count conclusion. The theorem compares common autonomous Markov representations; these restrictions must be checked before interpreting the result as a hidden-state requirement.

12.7 Protected graph interfaces beyond one entrance gateway

For a known directed support graph, additional protections are available without numerical rate identification. Write $U_x(T)$ for the occupation integral from state x. Two elementary first-hitting comparisons hold for every rate choice:

1. If $x \in B$ and every directed path from x to B^c visits y, then $U_x(T) \geq U_y(T)$.

2. If $y \notin B$ and every directed path from y to B visits x, then $U_x(T) \geq U_y(T)$.

For the first, all time before hitting y is spent in B, and $U_y(T) - U_y(T - s) \leq s$. For the second, no time before hitting x is spent in B, and U_x is increasing. The strong Markov property gives each comparison, including trajectories that never reach the gateway.

These inequalities compose transitively. A reachability test after deleting a candidate gateway checks every primitive, and transitive closure supplies a verifiable chain of comparisons. For example, on the graph $i \to k$, $k \to \ell$, $j \to k$, $\ell \to k$ with $B = \{j, k\}$, neither j as the entrance gateway from i nor i as the exit gateway from j is valid. Nevertheless $U_j \geq U_k \geq U_i$. A closed detector region is also protected by the exit condition. The implemented preorder is a sufficient certificate. A complete graph characterization of universal occupation order is not claimed, and failure to find a certificate is not evidence of a counterexample.

12.8 Why exact hidden structure cannot be certified by finite records

A finite calibration experiment cannot uniformly rule out arbitrarily weak unmodeled transitions. Suppose extra transitions have total intensity at most δ at every state and under every allowed control. Coupling the original and perturbed processes until the first extra jump gives, for records with total pathwise horizon at most L,

$$\mathrm{TV}(P_{\mathrm{original}}, P_{\mathrm{perturbed}}) \leq 1 - e^{-\delta L} \leq \delta L.$$

This statement includes adapted controls that obey the same intensity bound; any detector or downstream processing preserves the inequality. Consequently, a finite-record rule that accepts the original structural explanation with probability p accepts a sufficiently small perturbation with probability at least $p - \delta L$.

The issue can affect all-time protection, not merely exact graph labels. Take $i \to k$ at rate one and $j \to \ell$ at rate δ, with $B = \{j, k\}$ and k, ℓ absorbing. At $\delta = 0$, $H(T) = 1 - e^{-T} > 0$ for every positive T. At every $\delta > 0$,

$$H(T) = A_\delta(T) + A_1(T) - T$$

eventually becomes negative, while finite-horizon records approach those of the protected model as $\delta \downarrow 0$. Structural physical knowledge, a quantitative tolerance, or a positive separation assumption is therefore needed. Taking additional finite records alone cannot establish exact absence of all arbitrarily weak hidden alternatives.

The practical output is an explicit choice: certify a physical interface, report a quantitative minimum reset defect, or report the hidden peak crossing intensity any competing classical explanation would require. Each result says what the evidence establishes and what remains open.

Chapter 13

Designing a protected random observation clock

Timing is part of experimental access. A fixed-time population comparison can have either sign in a classical network, while its uniform average starting at preparation has a protected sign. This chapter describes clocks that retain that guarantee and makes a limited resource-design problem exactly solvable. The mixture representation and linear-programming geometry are established mathematical tools; their use here does not assert new generic optimization theory.

13.1 Decreasing densities are mixtures of uniform clocks

For a protected classical reset state j and any other initial distribution ρ, write

$$f(t) = P_j(X_t = j) - P_\rho(X_t = j), \qquad H(T) = \int_0^T f(t)\,dt \geq 0.$$

The same statement applies to a region with the required regenerative reset property. A clock is drawn independently of the physical trajectory, with a common probability density w for both preparations. Its contrast is $C_w = \int_0^\infty w(t)f(t)\,dt$.

Proposition 13.1 (A constructive protected clock class)**.** *Every nonincreasing probability density on $[0,\infty)$ is, up to changes on a set of measure zero, a mixture of uniform densities starting at zero:*

$$w(t) = \int_{(t,\infty)} \frac{1}{T}\,\mu(dT), \qquad \mu \text{ a probability measure on } (0,\infty).$$

Consequently $C_w = \int H(T)/T\,\mu(dT) \geq 0$. When the expected duration exists, $\mathbb{E}_w t = \frac{1}{2}\int T\,\mu(dT)$.

Proof. Take a right-continuous nonincreasing representative and its nonnegative Stieltjes measure $-dw$. Set $\mu(dT) = -T\,dw(T)$. Integrability and monotonicity imply $Tw(T) \to 0$ at both endpoints. Integration by parts gives $\int \mu(dT) = \int_0^\infty w(t)\,dt = 1$, and Stieltjes integration gives the displayed representation. Since $|f| \leq 1$, Fubini applies to the contrast; each mixture component contributes $H(T)/T$. The duration formula follows from the mean of a uniform distribution. □

The construction gives a direct implementation: choose a horizon T using μ, then draw the actual readout time uniformly from $[0,T]$. The horizon mixture and readout time are chosen independently of the current system trajectory. A common exponential clock belongs to this class, but some

clocks erase a particular quantum signal. Classical validity and signal strength are separate design questions.

This is a sufficient clock class, not a characterization of every density valid for every classical model. A shifted uniform window or an increasing density need not preserve the guarantee. The independent tests supply fully classical counterexamples for both.

13.2 Exact design on a finite menu

For each available uniform horizon T_k, suppose a model calculation supplies an enclosing signed-contrast interval $[l_k, u_k]$. The desired reversal advantage is $v_k = -C_k$. Let $c_k \geq 0$ be a calibrated expected acquisition cost and B an expected-cost budget. A randomized horizon choice has weights $p_k \geq 0$, $\sum_k p_k = 1$. Its guaranteed advantage is at least $\sum_k p_k(-u_k)$. The finite design problem is

$$\max_p \sum_k p_k v_k \quad \text{subject to} \quad \sum_k p_k c_k \leq B, \quad \sum_k p_k = 1, \quad p_k \geq 0, \tag{13.1}$$

first with $v_k = -u_k$ and then with $v_k = -l_k$.

Proposition 13.2 (Two horizons and a checkable global certificate)**.** *If the menu is finite and feasible, problem* (13.1) *has an optimum supported on at most two horizons. An admissible p, a slope $a \geq 0$ and an intercept b certify global optimality if*

$$b + ac_k \geq v_k \quad \textit{for every } k, \qquad \sum_k p_k v_k = b + aB.$$

Proof. An extreme feasible point has at most two positive coordinates: with three, a nonzero supported perturbation can preserve both total probability and total cost, giving a feasible line segment through the point. If the cost constraint is slack, an extreme point has one positive coordinate. A linear objective attains an optimum at an extreme point of this compact polytope. For any feasible p', the supporting inequalities give $\sum_k p'_k v_k \leq b + a\sum_k p'_k c_k \leq b + aB$. Equality at p proves optimality without repeating the optimization. □

The implementation enumerates feasible singletons and pairs, then exports the supporting line. A separate checker verifies feasibility and the dual inequalities using exact rational arithmetic. Solving with $-u_k$ gives an achievable guaranteed advantage; solving with $-l_k$ gives an upper bound on the unknown true optimum. Their difference is an explicit numerical-enclosure gap. An interval covering negative advantage is not declared a successful reversal experiment.

The quantum example uses sixteen horizons $T = k/4$, $k = 1, \ldots, 16$, depolarization rate $1/10$, rotation frequency 2, and detector gain $9/10$. Exact enclosures are rounded outward, never to the nearest favorable value. The supplied budgets constrain mean active evolution time $T/2$. A physical energy model must additionally account for preparation, control, measurement and any duration-dependent supplied power. The code accepts such calibrated costs but does not infer them from the horizon labels.

13.3 When mixing does not improve an energy objective

A timing mixture can improve a constrained tradeoff while providing no gain for a different objective. Suppose $c_k > 0$ and the chosen sufficient precision bound depends on advantage squared per expected energy. For any mixture with positive average advantage,

$$\frac{(\sum_k p_k v_k)_+^2}{\sum_k p_k c_k} \leq \sum_k p_k \frac{(v_k)_+^2}{c_k} \leq \max_k \frac{(v_k)_+^2}{c_k}.$$

Here $x_+ = \max\{0, x\}$. Replace v_k by its positive part and apply Cauchy–Schwarz to $\sqrt{p_k c_k}$ and $\sqrt{p_k}(v_k)_+/\sqrt{c_k}$. Thus a mixture cannot beat the best single horizon for this particular proxy. Exact binomial power, preparation-dependent overheads, simultaneous objectives or additional constraints require their own calculation. Neither this proxy nor the mean-cost budget supplies a pathwise expenditure guarantee for a stopped sequential experiment.

13.4 Clock calibration and adaptive use

If the actual common clock law differs from its declared protected law by total variation at most ζ, then its contrast can change by at most 2ζ, because $f \in [-1, 1]$. If each arm has its own discrepancy radius ζ_j, ζ_ρ, a bound is $\zeta_j + \zeta_\rho$, since each population lies in $[0, 1]$. A fixed nonnegative readout delay d shifts a uniform $[0, T]$ clock; the total variation from the original law is $\min\{1, d/T\}$. Delay therefore belongs in the physical model or the calibration budget. It cannot be dismissed as an innocuous change of origin.

For sequential testing, select the horizon mixture and betting rule using completed earlier trials. The current arm and readout time must then be drawn without allowing those fresh values to select the bet. The classical null is valid after averaging the clock; it need not be valid conditional on its realized time. Later trials may use all earlier timing records. This distinction permits adaptive design while preventing selection of the very time intervals in which a classical instantaneous contrast is negative.

```
from constructive_access.robust import clock_design

certificate = clock_design.design(menu, budget)
assert clock_design.verify(menu, budget, certificate)
```

The menu is a supplied physical-model and calibration contract. The checker certifies the declared design problem, not the truth of the apparatus model.

Chapter 14

A calibrated experiment with optional stopping

The occupation inequality becomes experimentally useful when a measured reversal can be separated from calibration error at a feasible record count. This chapter supplies fixed-record and sequential procedures that use one preparation and one endpoint readout per record. Binomial testing, likelihood-ratio betting, and confidence inversion are established statistical methods. The contribution here is their explicit connection to the protected physical contrast, with an exact computational certificate.

14.1 Randomizing preparations removes a baseline nuisance parameter

Let p_i and p_j be endpoint detection probabilities averaged over the same protected observation clock. The ideal classical contract implies $p_i - p_j \le 0$. A record begins by drawing a fresh fair preparation arm: $A = 0$ prepares i, and $A = 1$ prepares j. After evolution and binary readout Y, retain the encoded bit

$$R = \begin{cases} Y, & A = 0, \\ 1 - Y, & A = 1. \end{cases} \qquad \Pr(R = 1) = \frac{1 + p_i - p_j}{2}.$$

Thus a population difference becomes a one-sided Bernoulli experiment without estimating either population separately. This does not supply a physical reset, a trusted detector, or the required clock; these remain independent experimental assumptions.

Let $\mathcal{F}_{n-1}$ contain completed records and controller decisions before the current arm and clock draws. A contrast error envelope b_n gives the statistical null

$$\mathbb{E}(R_n \mid \mathcal{F}_{n-1}) \le \theta_{0,n} := \frac{1 + b_n}{2}, \qquad 0 \le b_n < 1. \tag{14.1}$$

The next horizon, admissible generator, and calibration bound may depend on completed records if each chosen experiment preserves this conditional inequality. Marginal validity alone is insufficient under shared hidden drift.

For a declared ideal reversal $p_i - p_j \ge g$ and a constant absolute contrast error radius b, the alternative has

$$\mathbb{E}(R_n \mid \mathcal{F}_{n-1}) \ge \theta_1 := \frac{1 + g - b}{2}, \qquad \theta_0 := \frac{1 + b}{2}.$$

The sufficient separation condition is $g > 2b$. A common positive affine detector gain belongs in g and its common offset cancels. Setting-dependent detector changes and imperfect common control belong in the error envelope. A comparison preparation allowed to be arbitrary by the physical theorem need not contribute to the null error, although it can weaken predicted power. The implementation uses the same conservative envelope for null and alternative; this is sufficient rather than universally optimal.

14.2 An exact fixed-record certificate

Write

$$B(N, k, p) = \sum_{s=k}^{N} \binom{N}{s} p^s (1-p)^{N-s}.$$

Proposition 14.1 (Fixed-record validity and power)**.** *Choose N in advance and reject when $S_N = \sum_{n=1}^{N} R_n \geq k$. Under the conditional null with constant θ_0,*

$$\Pr(\text{reject}) \leq B(N, k, \theta_0).$$

If every conditional success probability is at least θ_1, then

$$\Pr(\text{no rejection}) \leq 1 - B(N, k, \theta_1).$$

Proof. For an adapted Bernoulli sequence with conditional probabilities q_n, use independent uniforms U_n to represent recursively $R_n = \mathbf{1}\{U_n \leq q_n(\text{past})\}$. If $q_n \leq \theta_0$, then $R_n \leq \mathbf{1}\{U_n \leq \theta_0\}$ coordinate by coordinate. The sum on the right is binomial. The reverse coupling with $q_n \geq \theta_1$ proves the power bound. Identically distributed and independent records are sufficient, but the stated conditional bounds are enough. □

For rational $p = a/d$, the certificate evaluates the binomial polynomial using integers

$$T_s = \binom{N}{s} a^s (d-a)^{N-s}, \qquad T_{s+1} = T_s \frac{(N-s)a}{(s+1)(d-a)}.$$

The division is exact. Summing and dividing by d^N, followed by outward rational rounding, encloses the tail without a floating-point probability library. The verifier recomputes both tails against an independently supplied contract; changing a threshold, record count, or calibration bound requires a new certificate.

The qubit example with $\gamma = 1/10$, $\omega = 2$, a uniform clock on $[0, 9/4]$, and detector gain $9/10$ has independently certified reversal $g \geq 18/125$. With contrast radius $b = 1/1000$, take

$$\theta_0 = \frac{1001}{2000}, \qquad \theta_1 = \frac{1143}{2000}, \qquad N = 600, \qquad k = 321.$$

The exact rational tails imply

$$\Pr_{H_0}(\text{reject}) < 0.049498, \qquad \Pr_{H_1}(\text{no rejection}) < 0.032603.$$

These are prospective guarantees, not an experimental result. No physical data are supplied in this release. The design uses 600 endpoint records; it does not claim that 600 is globally minimal.

A simpler Chernoff design chooses $\theta_0 < c < \theta_1$ and verifies

$$ND_{\mathrm{KL}}(c\|\theta_0) \geq \log(1/\alpha), \qquad ND_{\mathrm{KL}}(c\|\theta_1) \geq \log(1/\beta).$$

Rational logarithm bounds certify both inequalities. At the same physical gap, radius, and $\alpha = \beta = 1/20$, this sufficient design uses 1181 records. Exact binomial evaluation reduces this requirement to 600. That is a comparison at matched inputs. The much larger reduction from the preceding three-arm pulse experiment comes principally from a stronger physical contrast and different preparation access, not from a comparable improvement in statistical methodology.

14.3 Safe optional stopping

For a previsible betting fraction $0 < f < 1$, put

$$q_n = \theta_{0,n} + f(1 - \theta_{0,n}), \qquad L_n = \left(\frac{q_n}{\theta_{0,n}}\right)^{R_n} \left(\frac{1 - q_n}{1 - \theta_{0,n}}\right)^{1-R_n}, \qquad M_N = \prod_{n=1}^{N} L_n.$$

Proposition 14.2 (Anytime-valid rejection)**.** *Under* (14.1)*, M_N is a nonnegative supermartingale starting at one. So is any fixed convex mixture of such processes. Rejecting on the first crossing of $1/\alpha$ has false-rejection probability at most α, over all record counts.*

Proof. For a conditional success probability $p \leq \theta_{0,n}$,

$$\mathbb{E}(L_n \mid \mathcal{F}_{n-1}) = p\frac{q_n}{\theta_{0,n}} + (1 - p)\frac{1 - q_n}{1 - \theta_{0,n}} \leq 1.$$

The affine expression increases with p and equals one at the null boundary. Products are therefore nonnegative supermartingales, and fixed convex mixing preserves this property. Ville's maximal inequality yields the crossing bound [20, 21]. □

This construction permits optional stopping, continued acquisition after inconclusive results, and predictable adaptation of future valid protocols. It does not allow retrospective selection of the most favorable mixture component without its predeclared weight. The implementation maintains capital with exact rational arithmetic.

14.3.1 The current random clock cannot choose the current bet

For this experiment the null is protected after averaging the observation clock. A fixed-time contrast can reverse in a classical model. Betting parameters must therefore be chosen before the current arm *and* current clock are drawn. The realized clock may be stored and may inform future experimental choices; it cannot select the current bet under the proof above.

For example, two equally likely clock strata with encoded success probabilities one and zero have average success probability $1/2$. A likelihood bet with factors $3/2$ and $1/2$ is fair beforehand. Betting only in the current success-one stratum gives mean factor $(3/2 + 1)/2 = 5/4 > 1$. This negative control identifies a practical failure mode that a generic optional-stopping recipe would otherwise miss. Likewise, dynamics responding to the selected readout time within the record can violate the common physical evolution assumption.

14.4 A guaranteed stopping horizon

With constant θ_0 and a single fixed alternative $q > \theta_0$, terminal capital is

$$M_N(q) = \left(\frac{q}{\theta_0}\right)^{S_N} \left(\frac{1-q}{1-\theta_0}\right)^{N-S_N}.$$

It increases with S_N. If this component has mixture weight w, find the least k for which $wM_N(q) \geq 1/\alpha$. Under the conditional alternative, the probability of failing to cross by N is at most $1 - B(N, k, \theta_1)$. Earlier crossing can only increase power.

For the same calibrated qubit example, choose the single component $q = \theta_1$, $w = 1$. At $N = 1100$, $S_N \geq 601$ suffices, and the exact miss probability is below 0.043395. This is a sufficient horizon with anytime false-rejection control 0.05; it is not an expected stopping time, and some paths stop earlier.

For a less costly arithmetic certificate at large horizons, let

$$\ell_1 = \log(q/\theta_0), \quad \ell_0 = \log((1-q)/(1-\theta_0)), \quad K = \theta_1\ell_1 + (1-\theta_1)\ell_0, \quad D = \ell_1 - \ell_0.$$

If $K > 0$, conditional Hoeffding gives power at least $1 - \beta$ whenever

$$NK - D\sqrt{\tfrac{N}{2}\log(1/\beta)} \geq \log[1/(\alpha w)].$$

Rational lower bounds on K and upper bounds on D and the logarithms turn this into a checkable squared inequality, including its positive-sign condition. The example gives the more conservative horizon 1739.

14.5 From rejection to a bound on hidden classical explanations

At a fixed record count and observed successes s, choose a rational θ_L with $B(N, s, \theta_L) \leq \alpha$. Monotone binomial-tail inversion gives a conservative lower confidence bound on a stable common success probability. For nonidentical adapted trials, the exact statement is exclusion of models whose every conditional success probability lies below that bound. It is not automatically a confidence interval for an unspecified average of changing probabilities.

A time-uniform version fixes f before the experiment and inverts

$$M_N(\theta) = \left[1 + f\frac{1-\theta}{\theta}\right]^{S_N} (1-f)^{N-S_N} \geq \frac{1}{\alpha}.$$

This capital decreases with θ. For a true stable mean, or any uniform conditional upper bound θ_*, Ville's inequality controls the probability that an inverted lower bound ever exceeds θ_*. Selecting f after seeing the current counts is not covered. The module implements both inversions using rational bisection.

Set $d_{\mathrm{det},L} = 2\theta_L - 1$. For known common positive detector gain κ and contrast drift at most e, the physical reversal has lower bound

$$d_L = \frac{d_{\mathrm{det},L} - e}{\kappa}, \qquad d = -H/T.$$

Suppose the separately calibrated entrance and exit hazard caps are a, b, and the detector-boundary theorem gives

$$H \geq G - \eta K, \quad G = \int_0^T e^{-bs-a(T-s)}ds, \quad K = T - A_a - A_b + G, \quad A_c = \frac{1-e^{-cT}}{c}, \quad A_0 = T.$$

For $K > 0$, every compatible classical explanation must satisfy

$$\eta \geq \frac{G + Td_L}{K}. \tag{14.2}$$

A negative lower bound is replaced by zero. A lower bound greater than one excludes the declared bounded-hazard class. When $K = 0$, use the original inequality instead of dividing.

With common cap λ and unrestricted reset defect, write $F(z) = 2(1 - e^{-z})/z - 1$. The classical bound $H/T \geq F(\lambda T)$ implies, within the inverse's physical range,

$$\lambda \geq F^{-1}(-d_L)/T.$$

Thus a contrast confidence bound can quantify the minimum hidden classical peak crossing intensity or reset defect needed to explain the record. This is a bound on a declared model class, not a device-independent proof of quantumness. Adaptive protocols need a common declared apparatus constraint for a single-parameter interpretation of these bounds.

14.6 Resource accounting and reproducibility

For the uniform clock on $[0, 9/4]$, expected evolution time is $9/8$ per record and worst-case time is $9/4$. The 600-record design therefore has expected evolution time 675 and deterministic upper bound 1350 in model time units. Preparation, reset, readout, controller, and calibration costs must be added. Converting time into energy requires an apparatus model.

Statistical validity is conditional on the calibration and physical contract. If an independently justified joint calibration event fails with probability at most δ, add δ once. Conditioning on an event selected from the same outcomes must not destroy the conditional trial law; an unsupported high-probability calibration statement does not resolve this issue.

The module `statistics_design.py` exports exact design certificates, independent checkers, encoded-record handling, rational sequential capital, and fixed or time-uniform confidence inversion. Tests include independent small binomial sums, adapted conditional-null policies, exhaustive finite-path optional-stopping controls, calibration and certificate tampering, and the invalid clock-dependent-betting example. The evidence file records prospective mathematical guarantees; it contains no claimed empirical detection.

14.7 A four-arm test without trusted preparation support

The dimension theorem supplies a different physical null. For any common homogeneous CTMC with at most three states, arbitrary two fixed initial distributions μ, ν, and any fixed stochastic binary detector f, write

$$C_0 = \mu f - \nu f, \qquad C_T = \frac{1}{T}\int_0^T (\mu P_t f - \nu P_t f)\, dt, \qquad W = C_0 - C_T.$$

The proved inequality is $|W| \leq 1$. It imposes no exact initial support and no extremal detector calibration. Common homogeneous dynamics, preparation repeatability, and a fixed detector remain necessary. Rejection excludes this declared three-state class; it does not exclude arbitrary classical models with additional states or time-dependent dynamics.

Choose each record's arm uniformly from

$$(\mu, 0), \quad (\nu, 0), \quad (\mu, U), \quad (\nu, U), \qquad U \sim \mathrm{Unif}[0, T],$$

independently of the current apparatus state. Encode $R = Y$ in the first and fourth arms, and $R = 1 - Y$ in the second and third. Then

$$\mathbb{E}R = \frac{1}{4}\left[\mu f + (1 - \nu f) + (1 - \overline{\mu P f}) + \overline{\nu P f}\right] = \frac{2 + W}{4}.$$

Consequently an absolute systematic error radius e on the complete four-term witness gives

$$\theta_0 = \frac{3 + e}{4}, \qquad \theta_1 = \frac{2 + w - e}{4}$$

when the ideal alternative is certified to have $W \geq w > 1$. The same exact-binomial proposition applies, including its conditional version when the common arm-independent model changes across completed trials. The radius e is the total error in W; it must not be mistaken for one arm's probability uncertainty. Fixed imperfections already covered by the theorem's arbitrary detector do not themselves violate the null.

For the improved qubit design, take $\gamma = 1/10$, $\omega = 2$, $T = 2$, gain $9/10$, and antipodal pure preparations along the Bloch direction

$$\boldsymbol{n} = (0, -12/37, 35/37).$$

An independent rational quantum certificate gives $W > 27/25 = 1.08$. With full-witness radius $e = 1/500$, the exact design is

$$\theta_0 = \frac{1501}{2000}, \qquad \theta_1 = \frac{1539}{2000}, \qquad N = 6000, \qquad k = 4560.$$

Reject when at least 4560 of the 6000 encoded bits are successes. Integer binomial-tail evaluation certifies

$$\Pr_{H_0}(\text{reject}) < 0.045483, \qquad \Pr_{H_1}(\text{no rejection}) < 0.039465.$$

Four setting-specific probability errors bounded by $1/2000$ each suffice for this full-witness radius, matching the per-setting error scale of the earlier two-arm design. This four-arm experiment pays a larger record cost for a null that allows arbitrary preparations and a fixed imperfect detector.

Half of the records have zero evolution duration and half have mean duration $T/2$. Expected evolution time is therefore $T/4 = 1/2$ per record, or 3000 for this design. Preparation and readout costs still apply to all 6000 records. There is no claim that total physical energy is less than the 600-record experiment, whose protected preparation requirements are stronger.

The original untilted $T = 9/4$ model remains a comparison: $w = 261/250$, $e = 1/1000$, $N = 20000$, and $k = 15107$ give false-rejection probability below 0.048412 and miss probability below 0.036316. The improvement to 6000 records comes from choosing preparations that expose a larger dimension-witness contrast, not from changing the exact-binomial method.

The four-arm experiment uses the same statistical precautions as the two-arm experiment: current arm and clock randomization must precede the outcome, betting cannot depend on the current realized clock, and a fixed record certificate cannot be reused as an uncorrected stopping rule. The generic exact-binomial API verifies supplied thresholds and budgets; neither the 6000 nor 20000 record choice is asserted to be globally minimal.

Chapter 15

From recorded counts to a classical explanation bound

An observed failure of a model should say more than "the model is wrong." The sharp reset bound provides a quantitative alternative: a classical explanation must permit a certain reset mismatch or a sufficiently large peak rate of crossing the observed boundary. This chapter connects that statement to finite data. It does not infer entropy production, total jump activity or quantumness from a binary record.

15.1 The contract for inference

The reference preparation is supported inside an observed region B and the comparison preparation outside B. Both use the same independent uniform readout clock on $[0, T]$. A fresh fair coin selects the arm on each trial. Encode success as Y in the comparison arm and $1 - Y$ in the reference arm. With observed contrast $C_{\mathrm{obs}} = p_B^{\mathrm{reference}} - p_B^{\mathrm{comparison}}$, the encoded success probability is $\theta = (1 - C_{\mathrm{obs}})/2$.

The common detector gain $g \in (0, 1]$ is calibrated. Let $\delta \geq 0$ bound the remaining discrepancy of the observed contrast from gH/T, including any propagated preparation, clock and setting-dependent detector errors. Thus $C_{\mathrm{obs}} \geq gH/T - \delta$. A shared fixed detector offset cancels. For reset inference assume uniform entrance and exit caps a, b and the Markov or regenerative entrance assumptions of the sharp reset theorem. These caps concern every relevant hidden state or history; observed average rates do not establish them.

The outside-B condition matters. A coherent quadrature comparison preparation generally has nonzero initial occupancy of B and needs the partially occupied preparation version of the bound. It cannot be inserted into the outside-B formula solely because it gives a larger signal.

15.2 A fixed-horizon confidence statement

Fix n before data collection and let S be the encoded success count. Compute a rational lower confidence endpoint $q(S)$ by inverting the upper binomial tail. The implementation returns a conservative dyadic endpoint with

$$\Pr_{\mathrm{Bin}(n,q(S))}(S' \geq S) \leq \alpha$$

whenever the endpoint is positive. For a stable Bernoulli mean, standard one-sided test inversion gives $\Pr\{q(S) > \theta\} \leq \alpha$. More generally, a candidate uniform conditional upper bound $\Pr(R_k = 1 \mid$

$\mathcal{F}_{k-1}) \le \theta_0$ is stochastically dominated by $\mathrm{Bin}(n, \theta_0)$. The same procedure therefore excludes such upper bounds with type-I error at most α, without claiming that an arbitrary time-varying process has a single unknown mean.

On the corresponding confidence event,

$$C_{\mathrm{obs}} \le 1 - 2q(S), \qquad H \le H_+ := \frac{T[1 - 2q(S) + \delta]}{g}.$$

Let G and K have the meanings established in the sharp reset theorem, so that $H \ge G - \eta K$. If $K > 0$, every compatible explanation must obey

$$\eta \ge \max\left\{0, \frac{G - H_+}{K}\right\}. \tag{15.1}$$

The software uses a rational lower enclosure G_- and an upper enclosure K_+, giving the conservative numerator $(G_- - H_+)_+$ divided by K_+. If the resulting lower bound exceeds one, the declared bounded-hazard classical class is incompatible at the stated error level. A zero result is inconclusive. When $K = 0$, a strictly positive numerator already contradicts the contract.

This remains an exclusion of nested physical model classes when conditional probabilities vary between trials: a proposed common defect cap η_0 implies a uniform bound on each encoded success probability, to which the binomial domination argument applies. Calibration envelopes must hold in the conditional form needed for that argument. A separately justified joint calibration failure probability is added once.

15.3 A required peak intensity without assuming a reset law

Dropping the reset condition gives the sharp inequality

$$\frac{H}{T} \ge \frac{A_a(T) + A_b(T)}{T} - 1, \qquad A_c(T) = \int_0^T e^{-cs}\, ds.$$

If both macro-crossing intensities are at most λ, then

$$\frac{H}{T} \ge f(\lambda T), \qquad f(z) = \frac{2(1 - e^{-z})}{z} - 1, \qquad f(0) = 1.$$

The function decreases strictly from one to minus one. Indeed its derivative has the sign of $(z+1)e^{-z} - 1 < 0$ for $z > 0$. Hence a calibrated upper contrast $C_+ = H_+/T \in (-1, 1)$ forces

$$\lambda \ge f^{-1}(C_+)/T.$$

The implementation brackets this root using rational exponential bounds. This is a necessary peak macro-crossing intensity for a classical explanation. It is not a bound on average total dynamical activity, power or dissipation. This unrestricted-reset inequality also holds for time-dependent and history-dependent classical processes with the specified conditional crossing-hazard bounds. Memory alone does not evade a valid hazard cap.

15.4 A fully reproducible count example

Take the explicitly synthetic count $S = 343$ out of $n = 600$ records, with $T = 9/4$, $g = 9/10$, $\delta = 1/1000$ and $\alpha = 1/20$. The count was chosen as an illustrative rounded model mean; it is not a physical measurement. The exported exact certificates imply the following conservative consequences:

Supplied classical contract	Consequence of these synthetic counts
$a = b = 1/2$	No compatible reset defect in $[0, 1]$
$a = b = 1$	Reset defect at least 0.604
$a = b = 2$	Reset defect at least 0.162
No supplied cap; unrestricted reset	Required peak intensity above 0.812

Rates use the reciprocal of the declared time unit. Rounded numbers here are weaker than the rational certified bounds. They demonstrate how a single record can constrain competing explanations, rather than assigning an unconditional physical interpretation to a rejected ideal model.

```
from fractions import Fraction as F
from constructive_access.robust import explanation

result = explanation.from_counts(
    records=600, successes=343,
    entrance_cap=1, exit_cap=1, time=F(9,4),
    gain=F(9,10), radius=F(1,1000), alpha=F(1,20))
```

The record count must be fixed in advance for this routine. The sequential statistics module supplies separate time-uniform procedures; repeatedly inspecting the fixed-horizon endpoint and stopping opportunistically is not covered. Physical assumptions and the provenance of actual counts remain the user's responsibility at the apparatus boundary.

15.5 A laboratory route that does not assume hidden reset purity

The realization-size test asks a different question from reset-defect inversion. Its target is whether a single homogeneous classical Markov description with at most three states can explain the records. It needs two repeatable preparation commands, one fixed binary readout, a common evolution setting, and an independent clock. The preparations may be mixed and the readout may be noisy. Hidden environmental and controller states that affect the dynamics belong to the realization being bounded.

There are four equally randomized arms: each preparation is measured either at the common reference time or after an independent uniform hold in $[0, T]$. Use fresh trials and perform only the terminal measurement. The immediate and delayed observations come from different trials; a baseline measurement must not be inserted into the delayed evolution. Common latency is allowed when it is shared by all arms and independent of the hold-time draw. Calibration must bound changes between the arms, including preparation drift, detector drift, and arm-dependent dynamics. A fixed detector's intrinsic noise is already covered by the theorem.

The statistical chapter gives a complete four-arm encoding and a predeclared record-count certificate. Its exact qubit alternative is a feasibility prediction. The laboratory decision is a rejection of the calibrated three-state description, not confirmation of a unique quantum mechanism. A classical four-state control and a time-dependent three-state control test how the interpretation fails when the relevant model assumptions change. Time stamps should be retained for diagnostics, even when the stated decision uses only the integrated statistic; no time-bin postselection belongs in that decision rule.

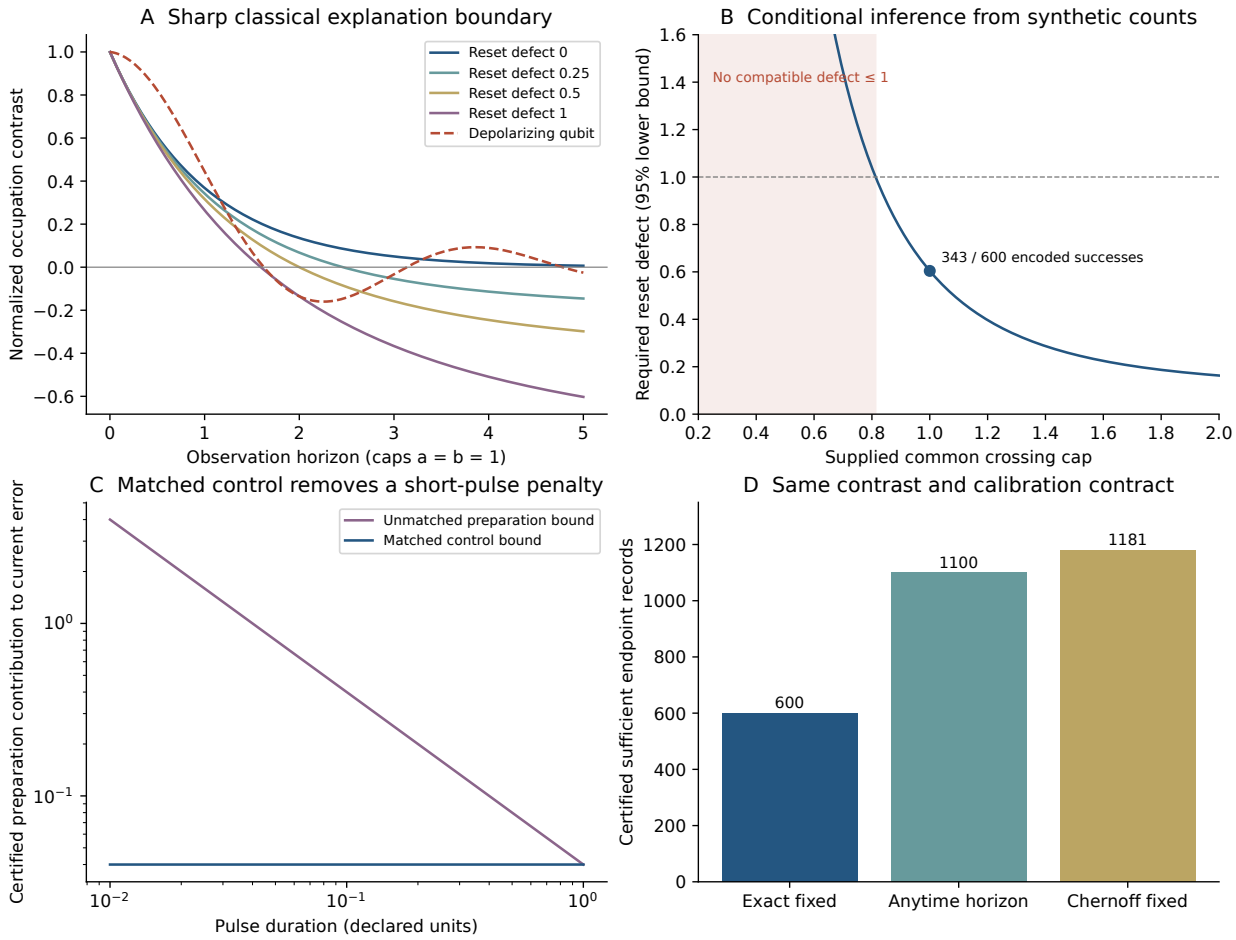


Figure 15.1: A: the sharp classical lower envelopes for several reset-defect caps, with a depolarizing-qubit contrast as a model illustration. B: the required reset defect from explicitly synthetic counts, conditional on the supplied common crossing cap; values above one exclude that cap. C: preparation-error contributions alone, for discrepancy 0.01, channel rates $3, 1$ and retention $1/2$; matching the native control removes the $1/t$ penalty in this bound. D: sufficient fixed and anytime record budgets for the same declared contrast, calibration and 5% error targets. The anytime bar is a guaranteed horizon, not an expected stopping time. No panel contains physical measurements.

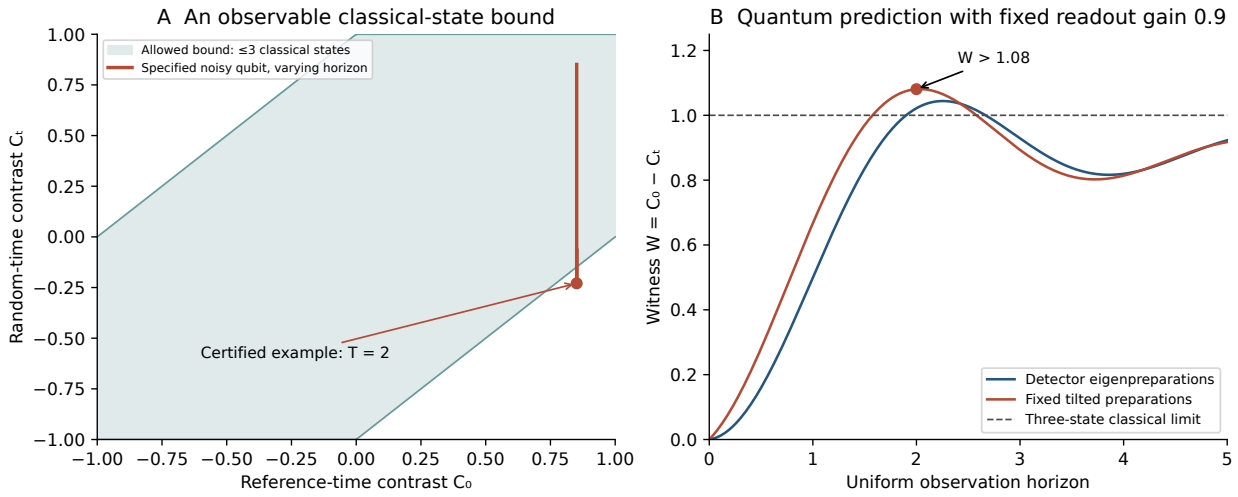


Figure 15.2: The integrated realization-size witness. The shaded strip contains all common homogeneous classical models with at most three states, arbitrary preparations and a fixed stochastic binary detector. The displayed qubit model leaves this strip. The two quantum curves use different fixed preparation pairs; neither is an experimental dataset or a claim of global optimality.

15.6 From required records to a physical energy budget

Record count, elapsed evolution, and energy answer different questions. Suppose an independently calibrated apparatus contract supplies a gross preparation and readout energy cap E_0 per trial and a gross holding-power cap $P_{\max}$. Include any clock or controller expenditure covered by these caps, and write E_{cal} for separate calibration and setup expenditure. A fixed N-record uniform-hold protocol then satisfies the deterministic bound

$$E_{\mathrm{total}} \le E_{\mathrm{cal}} + NE_0 + NP_{\max}T.$$

For the four-arm realization-size protocol, half the trials have no additional hold and half have mean hold $T/2$. Therefore

$$\mathbb{E}E_{\mathrm{total}} \le E_{\mathrm{cal}} + NE_0 + NP_{\max}T/4.$$

The expectation is not a pathwise guarantee. Common latency and other overhead must be added if they are absent from E_0.

The 600-record protected-preparation example has expected additional evolution 675 in its declared time unit. The 6000-record four-arm example at $T = 2$ has expected additional evolution 3000. The latter purchases freedom from exact hidden preparation assumptions at a measurable acquisition cost. Neither number is a joule value until the apparatus caps and time unit are calibrated. No thermodynamic minimum or energy advantage over other dimension witnesses is implied by these conditional accounting formulas.

Chapter 16

What a compatible explanation must contain

The access problem has a useful next form. Instead of asking only whether two known mechanisms can be distinguished, ask what *every* compatible explanation must contain. The answer may be a minimum number of classical states, a sufficiently large preparation resource, or sustained dissipation. These are different physical quantities. Their inference requires different experimental access, and they must not be collapsed into a universal scalar "information cap."

The following chapters establish complementary boundaries of this question. A robust nonnegative-factorization argument turns one preparation and one repeated channel into a finite-data hierarchy of classical state requirements. Erlang observation clocks give another route: finite measured probabilities certify a spectral obstruction without estimating exact hidden poles or bounding unseen transition rates. This certificate works at lower coherence than the explicit hierarchy example, but its sufficient record bound is much larger.

A constructive converse prevents overinterpretation. Finite fresh-reset prepare–wait–read records with continuous target probabilities can be approximated arbitrarily well by finite, irreducible, detailed-balance classical models when hidden dimension, rates and preparation resources are unrestricted. The construction retains timestamps and accommodates adaptive choices between reset trials. It does not reproduce arbitrary interventions within a persistent trajectory. Its explicit state and preparation requirements explain why an unrestricted classical null cannot be rejected uniformly from that restricted access.

Stationary pair observations impose additional structure. Reversible Markov autocorrelations are nonnegative mixtures of decaying exponentials. A measured departure from this class yields a quantitative preparation–housekeeping requirement; stationary preparation strengthens it to a sustained entropy-rate bound. Exact dual certificates turn uncertain finite observations into such requirements without selecting a hidden network architecture.

Access and premise	Certified conclusion	Essential qualification
Repeated common channel, fixed preparation and readout	Minimum classical state count	Hidden control and detector memory count as states
Finite reset records, unrestricted hidden resources	Constructive classical approximation	No nontrivial uniform test against the full null
Stationary noninvasive binary pairs	Entropy-production lower bound	Ordinary reversal and thermodynamic embedding required
Nonstationary pairs and a preparation budget	Preparation–housekeeping tradeoff	Reference mean cannot be inferred from transient mean

This is an extension of the common access contract, not a replacement for established system identification, positive realization or stochastic thermodynamics. The useful contribution is an explicit chain from observations to requirements on all members of a declared explanatory class, with exact countermodels where that chain cannot proceed. Candidate historical novelty, mathematical verification and laboratory validation remain separate questions.

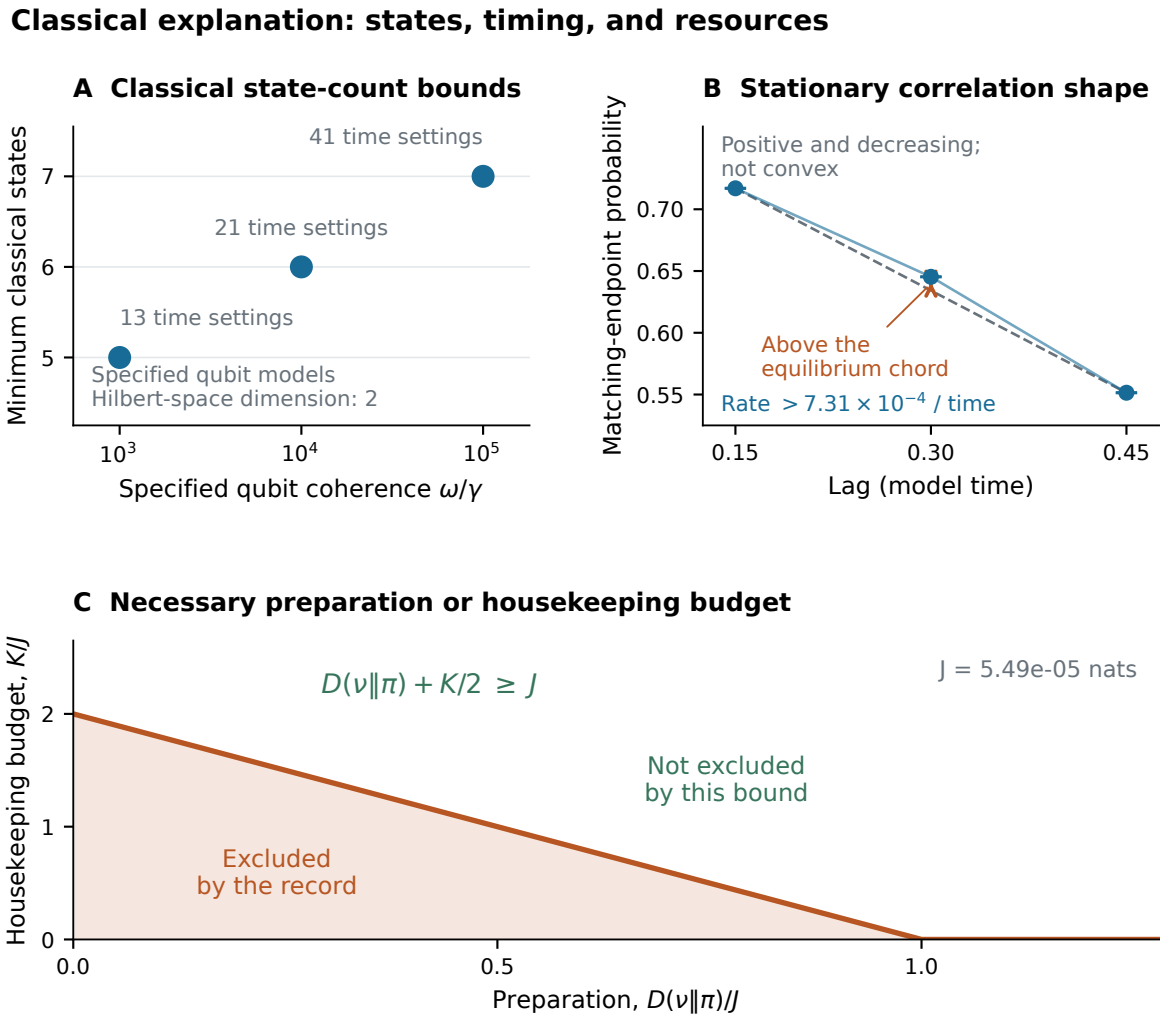


Figure 16.1: Three types of requirements on a classical explanation. (A) Specified finite-noise qubit examples force increasing classical state counts under one repeated-channel contract; these points are not an optimized coherence threshold. (B) The stationary ring example has positive decreasing agreement probabilities but violates reversible convex decay. Error bars show a synthetic uncertainty box. (C) A certified record separation J excludes preparation–housekeeping budgets below $D + K/2 = J$. The region above the line is merely not excluded by this bound; attainability is not asserted. All panels are model-based and require their respective access assumptions.

Chapter 17

A finite-record hierarchy of classical realization size

The three-state occupation bound can exclude a small classical explanation, but one scalar contrast cannot establish an unbounded hierarchy. Retaining several observation times changes the mathematical object. A single prepared state, a single binary detector, and a common repeated evolution produce a matrix with a positive factorization through the hidden state space. This chapter turns a robust obstruction to that factorization into a finite-record experiment. The dimension is the total number of states of the proposed classical Markov explanation, including any hidden controller or clock.

The foundations are established: positive realization and nonnegative-rank bounds, Sperner's antichain theorem, and quantum dimension witnesses already connect small quantum systems to larger classical realizations [91, 93]. In particular, the zero-noise sine matrix used below has appeared in prior work, including a stronger six-factor lower bound for its seven-by-seven instance [94]. The scoped addition developed here is a nonzero-noise certificate, its realization through one repeated channel and finitely many observation times, and an exactly checked finite-record statistical protocol. Historical priority of that combination is not established by this internal derivation.

17.1 The operational model

Let P be any $n \times n$ stochastic matrix, α a probability row, and $f \in [0,1]^n$ a fixed stochastic binary readout. Independent reset experiments observe a binary outcome after k applications of the same channel, with mean

$$p_k = \alpha P^k f. \tag{17.1}$$

A continuous-time homogeneous chain is included by $P = e^{\Delta Q}$. The result also covers repeated stochastic interventions, and independent, identically distributed nonnegative timing increments with $P = \mathbb{E}[e^{QZ}]$. It does not require a generator, stationarity, reversibility, known rates, pure preparations, or a noiseless detector.

Fix $N \geq 2$ and measure $p_1, \ldots, p_{2N-1}$. Form the matrix

$$M_{ij} = p_{N+i-j}, \qquad 0 \leq i, j < N. \tag{17.2}$$

It factors as $M = AB$, where $A_i = \alpha P^i$ and $B_j = P^{N-j} f$. Every row of A is a probability vector and every entry of B lies in $[0,1]$. The reversal of the column index is only a permutation of a finite Hankel matrix. There are $2N-1$ distinct measurement settings, not N^2 distinct experiments. The diagonal entries all equal p_N.

17.2 A robust antichain certificate

Theorem 17.1 (Finite-noise positive-factorization obstruction)**.** *Suppose an $N \times N$ matrix admits $M = AB$ with n columns in A, probability rows in A, and $0 \le B_{kj} \le 1$. If*

$$M_{ii} \le \epsilon, \qquad M_{ij} \ge \beta \quad (i \ne j), \qquad \beta > \tau + (1-\tau)\frac{\epsilon}{\tau} \tag{17.3}$$

for some $0 < \tau < 1$, then

$$N \le \binom{n}{\lfloor n/2 \rfloor}. \tag{17.4}$$

For $0 < \epsilon < 1$, minimizing the right side of (17.3) over τ gives the sufficient condition $\beta > 2\sqrt{\epsilon} - \epsilon$. For $\epsilon = 0$, any $\beta > 0$ suffices.

Proof. Write a_i for row i of A, and define $S_i = \{k : B_{ki} < \tau\}$. The diagonal bound gives $a_i(S_i^c) \le \epsilon/\tau$. If $S_i \subseteq S_j$, then

$$M_{ij} \le \tau a_i(S_j) + a_i(S_j^c) \le \tau + (1-\tau)\epsilon/\tau,$$

contradicting (17.3). Thus the N subsets S_i are pairwise incomparable. Sperner's theorem bounds their number by the largest binomial coefficient. The minimum of $\tau + \epsilon/\tau - \epsilon$ is obtained at $\tau = \sqrt{\epsilon}$. When $\epsilon = 0$, take $S_i = \{k : B_{ki} = 0\}$. The diagonal condition puts all of a_i on S_i, and inclusion $S_i \subseteq S_j$ would give $M_{ij} = 0$. □

This certificate supplies a sufficient lower bound, not the nonnegative rank itself and not the minimum dimension of a positive dynamical realization. Those quantities need not coincide. The antichain cardinality bound is sharp for unrestricted stochastic factors: use all subsets of size $\lfloor n/2 \rfloor$, let a_i be uniform on subset S_i, and let B_{kj} indicate $k \notin S_j$. The diagonal is zero and every off-diagonal entry is positive. This construction does not establish sharpness for a single-channel Hankel matrix.

Corollary 17.2 (One-channel finite-time dimension hierarchy)**.** *If the records (17.1) satisfy $p_N \le \epsilon$ and $p_k \ge \beta$ for every $1 \le k \le 2N-1$, $k \ne N$, with (17.3), their classical homogeneous Markov realization requires at least*

$$d_*(N) = \min\left\{ n : \binom{n}{\lfloor n/2 \rfloor} \ge N \right\}$$

hidden states. In particular $N = 7, 11, 21, 71$ require respectively at least $5, 6, 7, 9$ states.

17.3 An explicit qubit family

Prepare a qubit in $|0\rangle$, use a fixed $|1\rangle$ detector, and evolve with

$$\mathcal{L}(\rho) = -i\frac{\omega}{2}[X, \rho] + \gamma(\mathrm{tr}(\rho)I/2 - \rho).$$

Set $r = \omega/\gamma$ and $\Delta = 2\pi/(N\omega)$. The measured means are

$$p_k = \frac{1 - e^{-2\pi k/(rN)}\cos(2\pi k/N)}{2}. \tag{17.5}$$

For $N \ge 4$, their diagonal value is $\epsilon_r = (1 - e^{-2\pi/r})/2$, and all the off-diagonal values are at least $\sin^2(\pi/N)$. If the cosine is nonnegative, damping increases the probability above its unitary value;

if it is negative, the probability remains at least $1/2 \geq \sin^2(\pi/N)$. It follows that any fixed desired finite state lower bound can be reached at a sufficiently large, but finite, coherence ratio. This is a hierarchy of finite-resource experiments, not one finite experiment that excludes all finite classical models.

The sufficient coherence condition can be written

$$\epsilon_r < \left(1 - \cos(\pi/N)\right)^2. \tag{17.6}$$

For large N its right side is of order N^{-4}. Since N itself grows roughly exponentially in the desired state count, this certificate is not an efficient route to arbitrarily large dimension exclusions. Its transparent finite-noise guarantee is the benefit; efficiency and optimality remain separate questions. No relation between r and thermodynamic energy is assumed.

The exact-arithmetic implementation gives the following demonstrations:

N	observation settings	ω/γ	certified states
7	13	10^3	at least 5
11	21	10^4	at least 6
21	41	10^5	at least 7

They are calculated qubit probabilities. None is a laboratory observation.

17.4 A finite-record test without a multiplicity penalty on type I error

Choose a forbidden rectangular probability region $p_N \leq \epsilon_0$ and $p_k \geq \beta_0$ for $k \neq N$, satisfying (17.3). For each setting collect an independent fixed number n_k of reset records, and let S_k be the success count. Reject the proposed classical dimension only when

$$S_N \leq c_N \quad \text{and} \quad S_k \geq c_k \quad (k \neq N). \tag{17.7}$$

Calibrate each individual tail at its boundary to be at most α. Every model in the excluded classical class violates at least one of the probability constraints. The probability of passing every count criterion is therefore at most the rejection probability for that one violated constraint, which is at most α. A Bonferroni division of α among settings is unnecessary. This is an intersection–union test, an established statistical construction. For a specified alternative, the probability of failing any criterion is bounded by the sum of its individual miss probabilities.

Here is a fully specified example. Set $N = 7$, $r = 1000$, $\epsilon_0 = 3/500$, $\beta_0 = 3/20$, and $\tau = 2/25$. The classical comparison bound is $149/1000 < \beta_0$. Use the following counts and acceptance conditions:

Step k	Records n_k	Critical count c_k	Requirement
1	1900	313	$S_1 \geq 313$
2–5	100 each	22	$S_k \geq 22$
6	1800	297	$S_6 \geq 297$
7	10300	48	$S_7 \leq 48$
8	1800	297	$S_8 \geq 297$
9–12	100 each	22	$S_k \geq 22$
13	1700	282	$S_{13} \geq 282$

At $k = 7$ require $S_7 \le 48$; elsewhere require $S_k \ge c_k$. The total is 18 300 binary records and 126 900 channel applications. The exact certificates imply

$$\Pr_{n\le 4}(\text{reject}) < 0.049048, \qquad \Pr_{\text{specified qubit}}(\text{do not reject}) < 0.015708. \tag{17.8}$$

The elapsed evolution time is 126900Δ, excluding preparation, reset, readout, and controller overhead. An energy cost cannot be inferred from this operation count without a physical implementation and calibration. The record count is a checked sufficient design; no global optimality is claimed.

The probability intervals are certified with rational Machin-formula bounds on π and Taylor remainders. A binomial tail is bounded using its boundary mass and a geometric series whose ratio bounds all successive masses in the tail. All reported error comparisons use integer or rational arithmetic. Floating-point quantiles only propose critical counts; they do not certify them.

17.5 Calibration, timing and failure cases

If recorded means differ from a common-model mean by at most a known absolute amount δ, use null boundaries $\epsilon_0 - \delta$ and $\beta_0 + \delta$ in the two count tails. For a predicted alternative with the same uncertainty, its adverse bounds become $p_N + \delta$ and $p_k - \delta$. Calibration failure probability, if any, is added separately. The supplied routine implements this conservative extension, although the numerical design (17.8) uses $\delta = 0$.

No hidden-state preparation purity or rate bound is required. What must be stable is the preparation distribution, repeated channel, and readout across settings. A one-state device with a readout chosen according to k can fake any sequence p_k; a time-dependent or history-dependent classical process can also invalidate the factorization. Controller state must either be reset or included in the count. Arbitrary independent jitter at each total wait time need not preserve the additive grid when hidden rates are unbounded; a common random preparation latency, or independent identical increments composing each step, does preserve the repeated-channel model. General timing uncertainty requires a separate calibrated record-probability bound.

For context, an N-state deterministic discrete cycle with detector values $\sin^2(\pi j/N)$ reproduces the ideal unitary sequence exactly. Thus the certificate demonstrably excludes small classical models while allowing a larger classical explanation. It is neither a proof of nonclassicality against all models nor a new universal physical law.

Chapter 18

Finite random-clock certificates for classical realization size

An observed damped oscillation can require a large classical realization even when a qubit produces it. This fact and the relevant spectral obstruction are established [83]. The practical difficulty is different: finite, noisy observations do not determine an exact pole. We therefore measure a finite list of bounded probabilities whose distance from every small classical model can be certified directly. The resulting tolerance is conservative; at high dimension it is much too small for a practical experiment.

18.1 The operational contract

Let Q be a conservative, homogeneous generator with at most n hidden states, ρ an unknown but fixed initial probability row, and $f \in [0,1]^n$ an unknown but fixed binary response vector. The detector may be probabilistic. Neither stationarity, reversibility nor a bound on the exit rates is assumed. Each trial starts from ρ and measures f at a time chosen independently of the physical process. The clock does not change the generator, preparation or detector.

Fix a rate $s > 0$. For $k \geq 1$, choose an Erlang clock T_k with shape k and rate s; set $T_0 = 0$. Resolvent integration gives

$$m_k = \mathbb{E}[\rho e^{QT_k} f] = \rho R^k f, \qquad R = s(sI - Q)^{-1}. \tag{18.1}$$

The matrix R is stochastic. Thus independently randomized observation times convert a continuous-time record into finitely many powers of one stochastic matrix, without estimating derivatives or the hidden exit rates.

The comparison signal is the $+Z$ occupation probability of a qubit prepared in $+Z$, rotating under $\omega X/2$ with isotropic depolarization rate γ:

$$p_*(t) = \frac{1 + e^{-\gamma t}\cos(\omega t)}{2}, \qquad m_k^* = \frac{1 + \operatorname{Re}\mu^k}{2}, \qquad \mu = \frac{s}{s + \gamma - i\omega} = u + iv. \tag{18.2}$$

Here $\omega, s > 0$, $\gamma \geq 0$, and $v > 0$. Rational rates give rational probabilities m_k^*, even though the observation times are continuous.

18.2 A finite robustness theorem

The established generator spectral bound is

$$|\operatorname{Im}\lambda| \le -\operatorname{Re}\lambda \cot(\pi/n). \tag{18.3}$$

Writing $t_n = \tan(\pi/n)$ shows that every eigenvalue z of R lies in the lens

$$|z|^2 - \operatorname{Re} z + t_n |\operatorname{Im} z| \le 0. \tag{18.4}$$

In particular, for any certified $0 < t_0 \le t_n$, the lens is contained in both reflected disks with centers $1/2 \pm it_0/2$ and radius $\sqrt{1+t_0^2}/2$.

For the sharper bound it is useful to normalize the characteristic polynomial. Define

$$\nu = \frac{1-\mu}{1+\mu} = a - ib, \qquad b > 0, \qquad K = \sqrt{1 + (1+u)^2/v^2}, \qquad q = \lfloor (n-1)/2 \rfloor.$$

If $\gamma/\omega < t_0$, then ν is outside the upper-centered bounding disk. Choose

$$0 < d \le \min\left\{ b, \left|\nu - \left(\tfrac{1}{2} + \tfrac{it_0}{2}\right)\right| - \frac{\sqrt{1+t_0^2}}{2} \right\}, \qquad r_0 = \frac{v}{|1+\mu|}, \qquad h = \frac{|1+\mu|^2}{4} d(2b-d).$$

Notice that $h \le r_0^2$.

Theorem 18.1 (Finite Erlang exclusion)**.** *Under the preceding assumptions, every classical model with at most $n \ge 3$ states satisfies*

$$\max_{0 \le k \le n+1} |m_k - m_k^*| \ge \Delta_n, \qquad \Delta_n = \frac{|1-\mu|\, r_0^{n-1-2q} h^q}{4K} > 0. \tag{18.5}$$

Only $n+2$ observation settings enter this inequality. No fitted pole, exact rank determination, spectral visibility assumption, or exit-rate bound is used.

Proof. Pad a smaller model by unreachable absorbing states. Let P be the real monic characteristic polynomial of its $n \times n$ stochastic resolvent R, and let C be the sum of the absolute values of its coefficients. The resolvent roots have positive real parts. Each real-linear and conjugate-quadratic factor has alternating coefficients, and products preserve those signs. Hence $C = |P(-1)|$. Also $P(1) = 0$ and $P(R) = 0$.

If the maximum moment error is ε, Cayley–Hamilton applied at shifts zero and one gives

$$|\operatorname{Re} P(\mu)| \le 2\varepsilon C, \qquad |\operatorname{Re}(\mu P(\mu))| \le 2\varepsilon C.$$

The constant half of (18.2) cancels because $P(1) = 0$. Writing $P(\mu) = A + iB$ yields $|B| \le 2\varepsilon C(1+u)/v$ and therefore $|P(\mu)|/C \le 2\varepsilon K$.

For any root $z = s/(s-\lambda)$, put

$$w = \frac{1-z}{1+z} = 1 - \frac{2s}{2s-\lambda}.$$

The lens (18.4) is invariant under $z \mapsto 1-z$, so w lies in the same lens. Direct algebra gives

$$\frac{|\mu - z|}{|1+z|} = \frac{|1+\mu|}{2} |\nu - w|.$$

A real root therefore contributes at least r_0. For a conjugate pair write $y = |\operatorname{Im} w|$. If $y \geq b - d$, the near factor is at least d and the far factor at least $b + y \geq 2b - d$. If $y \leq b - d$, their product is at least $(b - y)(b + y) \geq d(2b - d)$. Thus each normalized conjugate pair contributes at least h. One root $z = 1$ contributes $|1 - \mu|/2$. Because $h \leq r_0^2$, the most conservative product uses the maximum number q of pairs:

$$\frac{|P(\mu)|}{|P(-1)|} \geq \frac{|1 - \mu|}{2} r_0^{n-1-2q} h^q.$$

Combining the two inequalities proves the result. Defective matrices cause no problem because the argument uses the characteristic polynomial, not an eigenvector decomposition. □

The distance calculation is entirely reproducible with rational interval arithmetic. If $F = |\nu - c|^2 - r^2 > 0$, then

$$|\nu - c| - r = \frac{F}{|\nu - c| + r}.$$

Rational upper bounds on the denominator give a lower bound on the distance. The implementation obtains t_0 from Machin's formula for π, alternating arctangent bounds and $\tan x \geq x + x^3/3$; for $n = 4$ it uses $t_0 = 1$ exactly. Integer square roots give directed dyadic bounds for the remaining radicals.

18.3 Finite duration, calibration and sampling

The ideal Erlang clock is unbounded. A laboratory implementation can return an artificial zero when $T_k > L$, without waiting beyond L. Its moment changes by at most

$$\tau_k(L) = \Pr(T_k > L) = e^{-sL} \sum_{j=0}^{k-1} \frac{(sL)^j}{j!}. \tag{18.6}$$

This tail is bounded above by replacing e^{-sL} with the reciprocal of a positive partial sum for e^{sL}. No hidden dynamical rate appears in the bound.

Let $\widehat{m}_k$ be sample means from N independent trials per setting, r_N a simultaneous confidence radius, b a calibrated probability error, and $\tau \geq \max_k \tau_k(L)$. A valid rejection rule is

$$\max_k |\widehat{m}_k - m_k^*| + r_N + b + \tau < \Delta_n. \tag{18.7}$$

Hoeffding and a union bound give confidence at least $1 - 2(n + 2)e^{-2Nr_N^2}$. The same data cannot first select an unrestricted target and then use a confidence guarantee proved only for a predeclared target.

The supplied design takes $r_N = \Delta_n/4$ and $\tau \leq \Delta_n/8$. With $b = 0$, both false rejection of the classical class and missed detection of the nominal qubit target are at most the declared α. Under the target, the confidence event gives total distance at most $2r_N + 2\tau < \Delta_n$. For an additional calibration radius, the corresponding power guarantee requires $2r_N + 2\tau + 2b < \Delta_n$.

18.4 What the certificate achieves, and its cost

For $\gamma = 1/10$, $\omega = 2$ and $s = 13/10$, exact arithmetic certifies

$$n = 4 : \qquad \Delta_4 > 0.0016816962795.$$

There are six clock settings. A fully bounded design with duration at most $320/13$ and $\alpha = 0.05$ uses a conservative total of 135,780,096 records. This large count is a limitation of the current sufficient bound, not a claim of practical superiority. The result closes a qualitative gap: a strictly positive finite-data exclusion exists without assuming an exact pole or bounding the rates of invisible modes.

Excluded state count n	Clock settings	Certified uniform gap
4	6	$1.6816962795 \times 10^{-3}$
5	7	$3.2905352735 \times 10^{-4}$
6	8	$8.2677897333 \times 10^{-5}$
8	10	$2.7407114770 \times 10^{-6}$
16	18	$1.5863939618 \times 10^{-13}$
62	64	$1.1657896420 \times 10^{-117}$

All entries concern the same damped qubit; the Erlang rate is chosen separately for each declared n and recorded in the reproducibility data. The last entries are constructive mathematical bounds, not recommended experimental plans.

A four-state directed cycle of rate 2, with fixed response probabilities $(1, 1/2, 0, 1/2)$, exactly produces $(1 + e^{-2t} \cos 2t)/2$. The certificate correctly stops separating at $\gamma = \omega$ for $n = 4$. Larger cyclic classical realizations exist for every strictly damped sinusoid when sufficiently many states are allowed. Thus the witness bounds classical realization size; it does not exclude every classical explanation or convert a state count into a thermodynamic energy cost.

The spectral obstruction, resolvent representation and linear recurrence machinery are established ingredients. The contribution investigated here is the explicit finite-setting, bounded-duration robustness certificate and its checkable error accounting. Historical priority remains open. Improving its sample complexity is a separate research problem; the finite probability-table witness elsewhere in this reference supplies a substantially stronger practical comparison in its effective regime.

Chapter 19

What finite reset experiments cannot exclude

An observable that excludes a three-state classical model does not exclude classical models with unrestricted hidden resources. This distinction survives precise timestamps, adaptive scheduling, and arbitrarily accurate finite records. This chapter constructs the alternative models explicitly. It also shows that a transient coherent-looking signal need not identify nonzero *stationary* dissipation: a classical model satisfying detailed balance can approximate the same records by starting far from equilibrium.

The approximation machinery is established. In particular, the Poisson operator used below is the Szász–Mirakjan operator, introduced in the classical approximation literature[97]. The role of the result here is to locate the operational boundary of inference and count the hidden resources needed by a sufficient countermodel. It is not a claim of a new universal approximation principle.

19.1 The access contract

Let $a \in \{1, \dots, m\}$ label finitely many preparations, and let $p_a(t) \in [0, 1]$ be the probability of a fixed binary detector outcome after waiting for time t. Each record begins with a fresh preparation. Its arm and waiting time may depend on earlier records, but are chosen independently of the subsequent hidden noise in that record. The apparatus retains the arm, waiting time, and outcome. The initial theorem concerns $t \in [0, T]$.

The classical comparison class has one common homogeneous finite Markov generator, one fixed detector, and a distribution for each preparation. All hidden clock and preparation-label states count toward its dimension. No intermediate measurement or control during the wait is included. The theorem does not reconstruct joint laws of repeated nondestructive measurements on a single persistent system, nor does it match independently measured energy or preparation-resource observables.

19.2 A finite homogeneous classical realization

Theorem 19.1 (Explicit finite-window approximation)**.** *Suppose all $p_a : [0, T] \to [0, 1]$ have Lipschitz constant at most $L > 0$. For every $r > 0$, there is a common homogeneous finite Markov generator with deterministic binary readout, at most*

$$n = 2m(\lceil rT \rceil + 1)$$

states, and peak exit rate at most r, whose output probabilities satisfy

$$\sup_{a,\,0\le t\le T} |q_a(t) - p_a(t)| \le L\sqrt{T/r}. \tag{19.1}$$

Thus $r = L^2T/\varepsilon^2$ is sufficient for error ε.

Proof. Set $K = \lceil rT \rceil$ and $f_a(k) = p_a(\min(k/r, T))$. A pure-birth clock $k = 0, \dots, K$ starts at zero, jumps forward at rate r, and stays at K thereafter. Introduce states (a, k, y) with $y \in \{0, 1\}$ and detector value y. On moving to phase $k+1$, choose the new bit according to Bernoulli$(f_a(k+1))$. Initially choose the bit according to Bernoulli$(f_a(0))$. This is a homogeneous Markov model; the label a selects an initial component of its common generator.

If N_{rt} is Poisson with mean rt, its detector probability is

$$q_a(t) = \mathbb{E}\, p_a(\min(N_{rt}/r, T)).$$

Projection onto $[0, T]$ cannot increase distance from t. The Lipschitz bound and Cauchy–Schwarz therefore give

$$|q_a(t) - p_a(t)| \le L\mathbb{E}|N_{rt}/r - t| \le L\sqrt{\mathrm{Var}(N_{rt}/r)} = L\sqrt{t/r}.$$

The claimed state and rate bounds follow directly from the construction. □

Zero-probability bit states may be deleted. Uniform error η in tabulating the phase probabilities adds at most η to the output bound. Preparations with $p_a(0) = 0$ or 1 remain supported entirely on the appropriate detector region: the construction does not rely on disguising an error at time zero.

Proposition 19.2 (Smooth approximation and finite truncation)**.** *Suppose $p_a : [0, \infty) \to [0, 1]$ is twice differentiable and $|p_a''(t)| \le M$ for every a, t. A clock frozen at phase K, tabulated using $f_a(k) = p_a(k/r)$, satisfies*

$$\sup_{a,\,0\le t\le T} |q_a(t) - p_a(t)| \le \frac{MT}{2r} + \Pr\{\mathrm{Poisson}(rT) \ge K\}. \tag{19.2}$$

Proof. For the untruncated clock, Taylor's linear term averages to zero because $\mathbb{E}(N_{rt}/r) = t$. The second-order remainder is at most $M\mathbb{E}(N_{rt}/r - t)^2/2 = Mt/(2r)$. Freezing changes a bounded detector mean by at most the probability of reaching the truncation region. Poisson tail monotonicity in t gives the stated bound. □

The global derivative assumption matters: the untruncated clock can query times beyond T. For rational $\mu = rT$, an exact finite tail certificate is

$$\Pr\{N \ge K\} \le \min_{0\le s\le K} \frac{\mu^s}{(K)_s}, \qquad N \sim \mathrm{Poisson}(\mu), \tag{19.3}$$

where $(K)_s = K(K-1)\cdots(K-s+1)$ and the $s = 0$ term is one. It follows from the factorial moment $\mathbb{E}(N)_s = \mu^s$. For very large state bounds the implementation also uses the simpler Cantelli certificate

$$\Pr\{N \ge K\} \le \frac{\mu}{\mu + (K-\mu)^2}, \qquad K > \mu.$$

These are sufficient bounds, without an optimality claim.

For a finite-dimensional Lindblad generator $\mathcal{L}$ and fixed binary effect E, all preparations satisfy usable global bounds

$$L = \|\mathcal{L}^*(E)\|_\infty, \qquad M = \|(\mathcal{L}^*)^2(E)\|_\infty,$$

as chosen Lipschitz and curvature constants. Indeed the derivatives are expectations of these Hermitian operators in the evolved density matrix. Thus every finite set of preparations of a finite-dimensional Lindblad semigroup fits the construction. The constructed classical clock rates need not equal the Lindblad damping rates and are not inferred physical parameters.

19.3 A detailed-balance countermodel with a preparation cost

Theorem 19.3 (Reversible realization)**.** *Replace the clock's forward rate r by forward and backward rates $r, d > 0$, with reflecting endpoints. Keep the destination-phase bit sampling used above. The resulting classical model satisfies detailed balance and has zero stationary entropy production. Its output probabilities differ from those of the one-way clock by at most*

$$1 - e^{-dT} \le dT \tag{19.4}$$

uniformly on $[0, T]$. Let $R = r/d$ and $Z = \sum_{k=0}^{K} R^k$. Relative to the stationary law with equally weighted preparation components, the prescribed initial state has relative entropy

$$D(\nu_a \| \pi) = \log(mZ). \tag{19.5}$$

Proof. The phase chain has stationary weights R^k/Z. Write $b_a(k, y) = f_a(k)^y [1 - f_a(k)]^{1-y}$, with its usual Bernoulli interpretation at endpoints. The lifted stationary law is

$$\pi(a, k, y) = \frac{R^k}{mZ} b_a(k, y).$$

For neighboring phases k, ℓ, the jump rate is $q_{k\ell} b_a(\ell, z)$. The two stationary edge flows agree because

$$R^k b_a(k, y) q_{k\ell} b_a(\ell, z) = R^\ell b_a(\ell, z) q_{\ell k} b_a(k, y).$$

Delete zero-weight bit states. Every retained state has positive stationary weight within its component. Couple the forward jumps and sampled bits with the one-way process until a backward jump occurs. The backward-jump counting process is dominated by a rate-d Poisson process, which proves (19.4). Finally, $\nu_a(a, 0, y) = b_a(0, y)$, so its likelihood ratio relative to π equals mZ on its support. This proves (19.5). □

If a single irreducible generator is required, add $c(\mathbf{1}\pi - I)$ for any $c > 0$ on the retained states. This introduces rare resets to π, connects all components, and preserves detailed balance. A reset-event coupling adds at most cT to the approximation error. Thus arbitrarily accurate reversible approximations do not depend on reducibility or absorbing states.

Zero *stationary* entropy production does not imply zero transient entropy production, zero reset cost, or free hardware. For an isothermal detailed-balance realization at temperature Θ, $k_{\mathrm{B}} \Theta \log(mZ)$ is the nonequilibrium free-energy difference from the specified equilibrium reference. It provides the corresponding preparation-work lower bound from equilibrium when suitable reversible controls are available. If the component label is conditioned on, the within-component divergence is $\log Z$. The result assigns no universal preparation charge to an arbitrary initial state supplied without a preparation model. It does expose where a reversible classical imitation stores its resource: highly nonequilibrium initialization of a large hidden clock.

19.4 Retained timestamps, adaptive experiments, and a testing limit

Theorem 19.4 (Transcript approximation)**.** *Suppose the one-record output probabilities of the target and classical model differ by at most ε for every allowed preparation and waiting time. For any fresh-reset protocol using at most N records, including adaptive choices and bounded stopping, their complete retained transcript laws obey*

$$\mathrm{TV}(P, Q) \le 1 - (1 - \varepsilon)^N \le N\varepsilon. \tag{19.6}$$

Proof. While the histories agree, share the random choices of next preparation and waiting time. Conditional on those choices, couple the two Bernoulli outputs maximally. The mismatch probability is at most ε at every step. Thus all records agree with probability at least $(1 - \varepsilon)^N$. The coupling inequality proves the result, including the retained timestamps. □

Corollary 19.5 (No uniform quantum test against an unrestricted reset class)**.** *Let $\mathcal{C}$ include every finite detailed-balance classical Markov model with the access contract above and no bound on hidden dimension, rates, or preparation resources. For a finite-record test $0 \le \phi \le 1$, suppose*

$$\sup_{Q \in \mathcal{C}} \mathbb{E}_Q \phi \le \alpha.$$

Then every smooth quantum prepare–wait–read law P in the same contract satisfies $\mathbb{E}_P \phi \le \alpha$.

Proof. For every $\varepsilon > 0$, the preceding construction gives a classical $Q_\varepsilon \in \mathcal{C}$ whose finite transcript law is within $N\varepsilon$ of P. Expectations of a test valued in $[0, 1]$ differ by at most total variation. Therefore $\mathbb{E}_P \phi \le \alpha + N\varepsilon$; let ε tend to zero. □

The same statement applies to eventual finite rejection by a sequential test. First restrict to rejection within N records with every wait at most T; then let N, T increase. Every finite rejection is included, assuming each chosen wait is finite and the target functions are regular on compact windows. This argument permits the approximating classical model to grow. It does not claim that a single fixed finite model matches infinitely many records.

The conclusion is a boundary on the null class, not a rejection of quantum experiments. State-size, rate, preparation, energy, or intervention restrictions can make a meaningful witness possible. Without such a restriction, the unrestricted classical class is dense in the observed experiment. Increasing the sample count alone cannot produce a test with controlled error and nontrivial power against that entire class.

19.5 A checked coherent-signal example

For the damped qubit

$$p_0(t) = \frac{1 + e^{-0.1t} \cos(2t)}{2}, \qquad p_1(t) = 1 - p_0(t), \qquad T = \frac{9}{4},$$

a global curvature bound is $M = 401/200$. The reproducible reversible construction for error $1/20$ uses

$$r = \frac{10827}{80}, \qquad d = \frac{1}{135}, \qquad K = 356.$$

Its deterministic lift has 1,426 retained states, below the certified bound 1,428. The exact uniform error certificate is below 0.048250; a numerical 101-point check finds maximum error approximately 0.009875. The latter is an illustration, while the former controls the full continuum of times. The code checks 2,848 exact detailed-balance flow identities. Its preparation relative entropy is approximately 3494.14 nats, and its stationary entropy production is zero.

For 6,000 records, a conservative Cantelli construction at per-record error 1/120000 gives a complete-transcript total-variation bound of 0.05. It uses $r = 812025$, $d = 1/810000$, and $K = 2638068$, with a sufficient upper bound of 10,552,276 deterministic hidden states. These are sufficient resources, without a minimality or practical-hardware claim. They explain precisely how a finite experiment can exclude a small classical model while remaining compatible with a much larger equilibrium model supplied with a nonequilibrium preparation. All of these checks concern synthetic models; no laboratory measurements enter this example.

19.6 A smaller exact realization and the discontinuity of exact inference

The reversible approximation is a sufficient countermodel with a particular thermodynamic property. It is not a minimum-state realization. For the damped cosine, an exact and much smaller classical construction is available. Consider a biased cycle with $n \geq 3$ phases and $\theta = 2\pi/n$. Choose forward and backward rates r, d by

$$r + d = \frac{\gamma}{1 - \cos\theta}, \qquad r - d = \frac{\omega}{\sin\theta}. \tag{19.7}$$

Its first Fourier mode has eigenvalue $-\gamma + i\omega$. From phase zero, the fixed stochastic binary detector

$$f(k) = \frac{1 + \cos(k\theta)}{2}$$

therefore gives exactly $[1 + e^{-\gamma t}\cos(\omega t)]/2$ for every $t \geq 0$. When n is even, preparation in phase $n/2$ gives the complementary trace. Both rates are nonnegative precisely when

$$n \arctan(\gamma/\omega) \geq \pi. \tag{19.8}$$

These facts follow by applying the cycle generator to $e^{ik\theta}$.

For $\gamma = 0.1$ and $\omega = 2$, rational arctangent and Machin-formula bounds certify that 62 fails (19.8) and 63 satisfies it. Together with the classical generator-sector lower bound, this gives an exact continuum minimum of 63 states for the single trace when a fixed stochastic detector is allowed. The 63-state rates are approximately 20.10533 and 0.018529. A 64-state cycle realizes both complementary preparations; no minimality of 64 for that pair is asserted. If the detector must be deterministic within the counted state space, destination-bit resampling supplies a lift with 125 states for the 63-cycle and 126 for the 64-cycle.

This model has stationary entropy production $(r - d)\log(r/d)$ in units of k_{B} per unit time, and preparation relative entropy $\log n$ from its uniform stationary phase distribution. This law does not satisfy detailed balance when $r \neq d$; no equilibrium thermodynamic interpretation of that reference is implied here. The small exact cycle and the large reversible approximation make different resource commitments.

Every finite detailed-balance generator has only real eigenvalues. It cannot exactly produce the nonzero complex poles of the damped cosine. Exact agreement on an open time interval would extend by analyticity. Nevertheless, the reversible construction approximates this same signal

arbitrarily closely on every finite interval. Exact continuum exclusion and robust finite-precision exclusion are therefore different inference problems. The latter requires additional resource or access restrictions.

Chapter 20

Dissipation and preparation from temporal records

The size of a classical explanation and its energy expenditure are different resources. A detailed-balanced network can produce a complicated transient after a costly preparation while producing no entropy in its stationary state. The next result identifies a stronger experimental contract under which a temporal record does constrain dissipation. Its measured input is a set of binary correlation probabilities, rather than a reconstructed generator. When initial stationarity is not guaranteed, the same construction yields a quantitative alternative: a compatible classical explanation must supply either a minimum departure from stationary preparation or a minimum housekeeping entropy budget. No hidden-state count needs to be specified.

20.1 Physical contract and normalization

Let Q be a finite, homogeneous, row-generator Markov chain with stationary law $\pi_i > 0$. States are even under physical time reversal. Observations are noninvasive: conditional on the physical states, the same fixed detector channel is applied at the beginning and end of each record, with independent readout noise. Write its binary output as $Y \in \{-1, 1\}$ and its conditional mean as $f_i = \mathbb{E}[Y \mid X = i] \in [-1, 1]$. Each record starts in π. Its independently selected lag U equals $t_k > 0$ with probability $w_k > 0$, where $\sum_k w_k = 1$. These choices are fixed before seeing the record.

Define the dimensionless stationary entropy-production rate

$$\sigma = \sum_{i<j} (a_{ij} - a_{ji}) \log \frac{a_{ij}}{a_{ji}}, \qquad a_{ij} = \pi_i Q_{ij}. \tag{20.1}$$

Zero fluxes are treated by limits; an unpaired positive flux gives infinite σ. With local detailed balance this is physical entropy production in units of k_B. For an isothermal stationary realization whose dissipated free energy is converted to heat at temperature T_{bath}, the corresponding dissipated power is $k_B T_{\mathrm{bath}} \sigma$. A conversion to watts therefore needs a calibrated time unit and an appropriate thermal model.

This is not a physical entropy formula for arbitrary hidden dynamics. In particular, equilibrium underdamped oscillators have odd momentum variables and can exhibit oscillating position correlations with zero physical entropy production. Quantum measurement backaction also lies outside this contract.

20.2 Comparison with a reversible process

The stationary adjoint and additive reversibilization are

$$Q^*_{ij} = \frac{\pi_j Q_{ji}}{\pi_i}, \qquad S = \frac{Q + Q^*}{2}. \tag{20.2}$$

S is a generator, has stationary law π, and obeys detailed balance. Both processes start in π. Their trajectory relative entropy on a fixed interval is

$$D(\mathbb{P}_Q^{[0,t]} \| \mathbb{P}_S^{[0,t]}) = t\, d(Q, S), \tag{20.3}$$

$$d(Q, S) = \sum_{i<j} \left[a_{ij} \log \frac{2a_{ij}}{a_{ij} + a_{ji}} + a_{ji} \log \frac{2a_{ji}}{a_{ij} + a_{ji}} \right] \le \frac{\sigma}{4}. \tag{20.4}$$

The escape-rate terms cancel pointwise because stationarity implies $Q_{ii} = Q^*_{ii} = S_{ii}$. The last inequality is the standard Jensen–Shannon versus Jeffreys divergence comparison [104]; the reversible information projection is established prior work [103].

For completeness, set $x = (a - b)/(a + b)$ for one unordered edge. The difference between one quarter of its entropy-production term and its contribution to d is $(a + b)/2$ times

$$F(x) = x \operatorname{atanh} x - (1 + x) \log(1 + x) - (1 - x) \log(1 - x).$$

This function is even, $F(0) = 0$, and $F'(x) = x/(1 - x^2) - \operatorname{atanh} x \ge 0$ for $0 \le x < 1$. The constant $1/4$ cannot be reduced uniformly: their ratio tends to $1/4$ as $a/b \to 1$. This does not imply that the final observed-data bound is tight.

20.3 A finite-lag entropy certificate

Let

$$C_Q(t) = \mathbb{E}_Q[Y_0 Y_t], \qquad p_k = \Pr_Q(Y_0 = Y_{t_k}) = \frac{1 + C_Q(t_k)}{2}. \tag{20.5}$$

For S, the spectral theorem in $L^2(\pi)$ gives

$$C_S(t) = m^2 + \sum_\ell b_\ell e^{-\lambda_\ell t}, \qquad m = \sum_i \pi_i f_i, \quad b_\ell \ge 0, \quad \lambda_\ell \ge 0, \quad \sum_\ell b_\ell \le 1 - m^2. \tag{20.6}$$

The inequality in the total mass allows a noisy detector. It follows that $q_k = \Pr_S(Y_0 = Y_{t_k})$ is nonincreasing with lag and is at least $(1 + m^2)/2$. At commensurate positive lags $t_k = n_k \Delta$, these probabilities lie in the closed moment set

$$\mathcal{M}_m = \left\{ q_k = \frac{1 + m^2}{2} + \frac{1 - m^2}{2} \int_0^1 x^{n_k}\, \mu(dx) : \ \mu \text{ a probability measure} \right\}. \tag{20.7}$$

Unused spectral mass can be placed at $x = 0$ because all n_k are positive. This is a closure: an atom at 0 can require an unbounded relaxation rate.

Recording only $(U, \mathbf{1}\{Y_0 = Y_U\})$ and applying data processing to (20.4) gives the inference theorem

$$\boxed{\sigma \ge \frac{4}{\bar{t}} \inf_{q \in \mathcal{M}_m} \sum_k w_k\, d_{\mathrm{B}}(p_k \| q_k), \qquad \bar{t} = \sum_k w_k t_k.} \tag{20.8}$$

Here d_{B} is Bernoulli relative entropy in natural logarithms. The same conclusion holds with any certified outer approximation of $\mathcal{M}_m$. The proof uses a joint record with a common lag distribution; it does not assume that all lag experiments last $\bar{t}$ individually.

If m is unknown, $q_k \geq 1/2$ remains valid. If the measured p_k are known only through a simultaneous confidence region, minimize over that region as well. Choosing a best fitting curve without a certified lower bound on this minimization does not certify dissipation.

20.4 Preparation or sustained irreversibility

Initial stationarity can be removed if preparation is included as a resource. Let the same homogeneous Q now start in an arbitrary distribution ν, while the reference process S starts in π. Define

$$h_i = \sum_{j \neq i} Q_{ij} \log \frac{Q_{ij}}{Q^*_{ij}}, \qquad \sigma_{\mathrm{hk}}(\nu_s) = \sum_i (\nu e^{sQ})_i h_i. \tag{20.9}$$

Because the exit rates of Q and Q^* agree for each state, h_i is a nonnegative rate multiplied by a relative entropy of jump-destination distributions. This is the standard adiabatic or housekeeping entropy-production rate [105]. It is not the total transient entropy-production rate.

The path relative-entropy chain rule and concavity of the logarithm give

$$D(\mathbb{P}^{[0,t]}_{Q,\nu} \| \mathbb{P}^{[0,t]}_{S,\pi}) = D(\nu\|\pi) + \int_0^t \sum_i (\nu e^{sQ})_i \sum_{j \neq i} Q_{ij} \log \frac{Q_{ij}}{S_{ij}}\, ds \tag{20.10}$$

$$\leq D(\nu\|\pi) + \frac{1}{2}\int_0^t \sigma_{\mathrm{hk}}(\nu_s)\, ds. \tag{20.11}$$

The pointwise bound follows from $\log[(Q_{ij} + Q^*_{ij})/2] \geq (\log Q_{ij} + \log Q^*_{ij})/2$. Therefore any certified record separation J from the stationary reversible reference set obeys

$$\boxed{D(\nu\|\pi) + \frac{1}{2}\mathbb{E}\left[\int_0^U \sigma_{\mathrm{hk}}(\nu_s)\, ds\right] \geq J.} \tag{20.12}$$

For the endpoint record, J can be the confidence-corrected moment-set separation in (20.8). If the stationary detector mean has not been independently calibrated, the reference set must include every possible m; the measured transient mean cannot replace it. The monotonicity relaxation with floor $1/2$ safely avoids this issue.

This is an explicit alternative between preparation and sustained irreversibility. A calibrated preparation budget $D(\nu\|\pi) \leq D_{\max}$ implies

$$\mathbb{E}\left[\int_0^U \sigma_{\mathrm{hk}}(\nu_s)\, ds\right] \geq 2\max\{0, J - D_{\max}\}. \tag{20.13}$$

For detailed-balanced Q, the housekeeping term vanishes and $D(\nu\|\pi) \geq J$. For stationary preparation $\nu = \pi$, the sharper factor $1/4$ in (20.4) applies instead. If π is a calibrated isothermal Gibbs state, $k_B T_{\mathrm{bath}} D(\nu\|\pi)$ is nonequilibrium free energy relative to that equilibrium. For a general nonequilibrium stationary law it remains an information resource; no unqualified conversion to preparation work is made.

The factor $1/2$ in the nonstationary path comparison is optimal uniformly. For jump-destination laws $p = (1-\varepsilon^2, \varepsilon^2)$ and $q = (1-\varepsilon, \varepsilon)$, $D(p\|(p+q)/2)/D(p\|q) \to 1/2$ as $\varepsilon \downarrow 0$. These can be the outgoing laws of Q and Q^* at state 0 in a positive three-state generator: set

$$Q_{01} = 1-\varepsilon^2, \quad Q_{02} = \varepsilon^2, \quad Q_{10} = 1-\varepsilon, \quad Q_{20} = \varepsilon, \quad Q_{12} = 1, \quad Q_{21} = 1-\varepsilon+\varepsilon^2.$$

Choose diagonal entries to give zero row sums. The stationary law is uniform. Preparation at state 0 and a sufficiently short observation interval approach the rowwise ratio. This sharpness concerns the path comparison, not every observed-data certificate.

20.5 Two closed-form cases

For one lag, an upper confidence bound $p \le p^+ < r$, where $r = (1 + m^2)/2$, implies

$$\sigma \ge \frac{4}{t} d_{\mathrm{B}}(p^+ \| r). \tag{20.14}$$

Taking $r = 1/2$ avoids the need to know m.

Two lags reveal something this test can miss. Let $t_1 < t_2$ and suppose simultaneous bounds give $p_1 \le u < v \le p_2$. Define $\bar{p} = wu + (1 - w)v$, with $0 < w < 1$. Then

$$\sigma \ge \frac{4\{w d_{\mathrm{B}}(u\|\bar{p}) + (1-w) d_{\mathrm{B}}(v\|\bar{p})\}}{w t_1 + (1-w) t_2}. \tag{20.15}$$

Indeed, minimize the two Bernoulli divergences subject to the relaxed reversible constraint $q_1 \ge q_2$. When $p_1 < p_2$, convexity forces the optimum to $q_1 = q_2 = wp_1 + (1 - w)p_2$. The resulting weighted Jensen–Shannon divergence decreases when the smaller probability is increased, and increases when the larger is increased. Thus the least favorable pair in the stated confidence region is (u, v).

For more lags, the order relaxation $q_1 \ge \cdots \ge q_K \ge 1/2$ is solved exactly by weighted binomial isotonic regression: pool neighboring blocks whose weighted means increase, then raise block means below $1/2$ to that floor. This is a convenient lower bound. The full moment set can provide more information than monotonicity; the independently checked dual certificates developed alongside this chapter retain those constraints.

There is a useful distinction from observable time-reversal tests. Every stationary binary process satisfies $\Pr(Y_0 = 1, Y_t = -1) = \Pr(Y_0 = -1, Y_t = 1)$, because its endpoint marginals agree. Consequently its binary two-time distribution equals its time reverse, even when (20.15) is strictly positive. The inference comes from the temporal restrictions on an even-state equilibrium Markov explanation, not from an asymmetry in this particular observed pair record.

20.6 A reproducible model and confidence calculation

Consider a 40-state ring with forward rate 20 and backward rate 1 in the stated time unit. Its stationary distribution is uniform. A deterministic detector reports $+1$ on 20 consecutive states and -1 on the others. At lags $t_1 = 8/5$ and $t_2 = 21/10$, rational uniformization certifies

$$p_1 \in [0.511989466390,\ 0.511989466391],$$
$$p_2 \in [0.736044845959,\ 0.736044845960].$$

Both single-lag tests with floor $1/2$ return zero. Equal lag weights in (20.15) certify

$$\sigma > 0.058501035648.$$

The model's true rate is $19 \log 20 \simeq 56.9189132$. The bound is therefore informative but loose; this example establishes a detection capability, not accurate reconstruction of the total dissipation.

For a finite-record illustration, take *synthetic* counts 2560/5000 and 3680/5000 of matching endpoints in two independent stationary-pair experiments. A one-sided Hoeffding radius 0.02 in

each group has combined failure probability at most $2e^{-2(5000)(0.02)^2} < 0.05$. Thus $p_1 \le 0.532$ and $p_2 \ge 0.716$ simultaneously with confidence at least 95%. Equation (20.15) gives

$$\sigma > 0.039302147509$$

at that confidence level. This is a worked inference from declared counts; no laboratory experiment generated them. Consecutive overlapping pairs cannot be inserted into this independent-sample calculation without a separate dependence analysis. A known bound on initial nonstationarity or detector drift must enlarge the probability confidence intervals.

20.7 What survives the counterexamples

The test implementation includes an irreducible four-state detailed-balanced chain whose two specially prepared transient populations reverse contrast. Every stationary current is zero. Hence the earlier preparation witness cannot be converted directly into a positive housekeeping-power bound. The reversible clock construction elsewhere in this reference extends this obstruction across arbitrary finite observation windows, while explicitly accounting for nonequilibrium preparation.

An equilibrium underdamped oscillator provides a different control: its position autocorrelation can be negative without dissipation. It fails the even-state reversal assumption, not the spectral theorem for S. These controls identify the physical access needed for the inference: stationary records, a common noninvasive detector, independent clock choice, and the appropriate class of microscopic time reversal.

Thermodynamic inference from time-symmetric fluctuations already has a substantial literature, including correlation-time bounds [101]. Cross-correlation asymmetry bounds [106] provide a different data reduction. Recent counterexamples to eigenvalue-only entropy bounds [108] reinforce the need to keep observable amplitudes and preparation contracts. An anonymous, unrefereed October 2026 technical release [110] also uses reversibilization and separation from reversible spectral mixtures for hidden-dissipation inference. The general bridge and its factor 4 are therefore not claimed here as first discoveries. The deliverable is an explicit finite-lag endpoint-correlation inference procedure, with exact certificates, confidence regions, counterexamples and runnable comparisons.

Chapter 21

Certified resource requirements from correlation data

The preceding comparison theorem turns a departure from stationary reversible behavior into a resource requirement. This chapter makes that statement usable with a finite set of uncertain measurements. Its output is a checkable lower bound, together with the assumptions needed to interpret it as sustained entropy production or as a preparation–housekeeping tradeoff.

The method combines established variational relative entropy and polynomial bounds with the reversible correlation representation. A numerical fit may propose a useful test, but the exported certificate is checked over the entire continuous comparison class. A finite spectral grid alone would not justify that conclusion.

21.1 The comparison class and observable records

At positive integer multiples $k\Delta$ of a known time step, let p_k be the probability that two binary readouts agree. The readouts must use the same fixed noninvasive detector. Conditional detector noise at the two times is independent; its state-dependent mean may be any function $g \in [-1, 1]$. Choose the lag with known positive weights w_k summing to one, independently of the trajectory. The complete reduced record retains the lag and the agreement bit.

For a reversible process started in its stationary distribution π, write $m = \pi g$. Spectral decomposition of its self-adjoint semigroup gives

$$q_k = r + (1 - r) \int_0^1 x^k \, \mu(dx), \qquad r = \frac{1 + m^2}{2}, \qquad \mu \text{ a probability measure.} \tag{21.1}$$

This comparison set may include limits of finite-state models; enlarging the set makes a lower bound conservative. For a stochastic detector the spectral mass can be smaller than $1 - m^2$. Missing mass is placed at $x = 0$, which leaves all positive-lag moments unchanged. This is why zero lag is excluded from the software's moment menu. Additional mass at $x = 1$ covers conserved components and arbitrarily slow modes.

An independently valid interval for m supplies an interval $[r_L, r_U]$. Without such information, use $[1/2, 1]$. In the nonstationary resource theorem, m is the *stationary reference* mean, not the mean of the transient preparation. Substituting the latter would invalidate the inference.

21.2 An exact dual certificate

Choose positive numbers $h_{k,0}, h_{k,1}$, interpreted as test scores for the two agreement outcomes. Define

$$B(r,x) = \sum_k w_k \big[h_{k,0} + (h_{k,1} - h_{k,0})\{r + (1-r)x^k\}\big], \tag{21.2}$$

$$U \geq \sup_{r\in[r_L,r_U],\ 0\leq x\leq 1} B(r,x). \tag{21.3}$$

Since B is affine in r, it suffices to check the two endpoints. For each endpoint it is a polynomial in x. The implementation converts its rational power coefficients into Bernstein coefficients on $[0,1]$. Their maximum is an upper bound; repeated exact subdivision tightens it. Thus U can be certified using rational arithmetic without discretizing away hidden rates.

Suppose simultaneous confidence bounds give $p_k \in [\ell_k, u_k]$. Set

$$J = \max\left\{0, \sum_k w_k \min_{p\in[\ell_k,u_k]} [(1-p)\log h_{k,0} + p\log h_{k,1}] - \log U\right\}. \tag{21.4}$$

The inner minimum occurs at an interval endpoint. Rational logarithm bounds make the numerical result a rigorous lower enclosure.

Proposition 21.1 (Observable separation certificate)**.** *For every record law P in the supplied confidence box and every stationary reversible comparison law Q in* (21.1)*,*

$$D(P\|Q) \geq J.$$

Proof. The variational relative-entropy inequality gives $D(P\|Q) \geq \mathbb{E}_P \log h - \log \mathbb{E}_Q h$. Equation (21.1) expresses $\mathbb{E}_Q h$ as a mixture of $B(r,x)$, so it is at most U. Minimize the first term over the entire confidence box. Nonnegativity of relative entropy permits truncation at zero. □

Scores, and weights on a predeclared lag menu, may be chosen after a valid simultaneous confidence box has been constructed. The bound holds for every probability vector in that box. An interval valid only for a test chosen before seeing the data does not justify unrestricted later selection. Undeclared searches over additional lags need their own coverage guarantee.

21.3 What the certificate requires of a physical explanation

For a stationary ordinary-reversal Markov explanation, the preceding chapter's path comparison implies

$$\sigma \geq \frac{4J}{\mathbb{E}U_{\mathrm{lag}}}, \qquad \mathbb{E}U_{\mathrm{lag}} = \Delta \sum_k w_k k. \tag{21.5}$$

Here σ is entropy production in natural units per model time. The lag variable U_{lag} is distinct from the score normalizer U.

If the initial law is instead ν, while the homogeneous generator has positive stationary law π, the more general conclusion is

$$D(\nu\|\pi) + \frac{1}{2}\mathbb{E}\int_0^{U_{\mathrm{lag}}} \sigma_{\mathrm{hk}}(\nu_s)\, ds \geq J. \tag{21.6}$$

Thus a separately calibrated upper preparation budget B_0 gives expected integrated housekeeping entropy at least $2\max\{0, J - B_0\}$. A detailed-balance explanation has zero housekeeping term and

must instead supply preparation divergence at least J. Neither interpretation assumes a known hidden-state count. The code refuses to use a narrowed stationary-reference mean interval for this general conclusion unless its separate calibration is declared.

These are model requirements. Physical heat interpretation additionally needs ordinary even-variable reversal and a valid thermodynamic embedding. In an isothermal stationary jump model with local detailed balance, $k_B T\sigma$ is the dissipated power. If π is a specified Gibbs law, $k_B T D(\nu\|\pi)$ is excess nonequilibrium free energy. Those identifications are not implied by a probability fit. Equilibrium systems with odd variables and invasive detector backaction provide explicit alternatives outside the contract.

21.4 Information beyond a reversal or monotonicity test

A stationary binary two-time distribution is symmetric under exchanging its times. Its forward–reverse relative entropy is therefore zero, even when the underlying process dissipates. The comparison class in this chapter imposes additional reversible Markov structure, so J can nevertheless be positive.

The two-lag ring example in the preceding chapter has increasing agreement probabilities, both above one half. A rational score certificate for a wider synthetic uncertainty box gives an entropy-production rate above 0.0468 in model units. Its true model rate is approximately 56.9: this is a valid but loose lower bound, not a near-complete heat reconstruction.

A further example distinguishes the full moment constraint from monotonicity. Take a 40-state ring with clockwise rate 20 and counterclockwise rate 1. Use a fixed stochastic binary detector whose conditional mean at state j is the exact rational number

$$g_j = \frac{\operatorname{round}(10^6 \cos(2\pi j/40))}{10^6}.$$

The rounded values define the model; no exact sinusoidal identity is assumed. The code exports all values and evaluates the three agreement probabilities at lags $3/20, 6/20, 9/20$ by exact uniformization. All three probabilities exceed one half and decrease with lag. They nevertheless violate convexity: $p_1 - 2p_2 + p_3 < 0$.

Choose $w_k = 1/3$, $v = (-1, 2, -1)$, $q = 2/3$, and $a = 1/200$, with scores

$$h_{k,0} = 1 - a v_k q / w_k, \qquad h_{k,1} = 1 + a v_k (1-q)/w_k.$$

Their reversible normalizer is at most one because $-x + 2x^2 - x^3 = -x(1-x)^2 \le 0$. Even after widening each exact model probability interval by $1/5000$, the exported certificate gives a rate above 0.000731. Negative-correlation and monotonic-revival bounds are both zero for this box. This is an exact model illustration of additional information in several lags; it is not a laboratory result or a claim of competitive sample efficiency.

21.5 Implementation and finite records

The module `energy_certificate.py` provides deterministic certificate construction, binding to an independently supplied contract, a fixed-count confidence interface, and a numerical score proposal followed by exact checking. The proposal fits a finite mixture to nominal probabilities. Its grid and optimizer status never certify the result: rational score rounding and the continuous polynomial upper bound are authoritative.

For M predeclared lag settings with N_k independent pair records each, a common radius a satisfying

$$2 \min_k N_k \, a^2 \geq \log(2M/\alpha)$$

gives simultaneous two-sided Hoeffding coverage at least $1 - \alpha$. Whole-record calibration radii enlarge the intervals. A calibration failure allowance must be added separately; overlapping pairs from one trajectory are not independent samples. Scores can then be selected using the entire box, and the exported certificate remains valid on its joint coverage event.

The implementation currently accepts distinct positive integer lags up to 256 and bounded polynomial subdivision depth. These are declared computational limits. A nonpositive result means that the supplied certificate finds no positive bound, not that the mechanism is reversible. The historical status of the specific calibrated construction remains open; its mathematical ingredients and the closest prior comparisons are recorded in the positioning chapter.

Chapter 22

What a virtual laboratory can establish

A proof identifies what follows from an experimental contract. A virtual laboratory asks whether a proposed apparatus and analysis actually satisfy that contract, whether the signal survives realistic forms of imperfection, and what a failed experiment would look like. It cannot establish that an unmeasured physical apparatus follows the simulated equations.

This extension supplies three connected laboratories. A driven-ring experiment generates individual physical jump counts and noisy endpoint readings. A qubit experiment propagates prepared states through independently implemented Bloch and Lindblad dynamics, with preparation, control and detector errors. A metastable classical experiment generates a chronological series of dependent pairs, exposing the difference between stationary marginal probabilities and independent records. The inference routines receive their declared counts and calibrations, not the hidden generator used to score performance.

The simulations have already changed the theory and protocol design. Timing noise sometimes preserves the exact comparison structure, while an apparently innocent exposure convention can create false equilibrium violations. A conditional-mean envelope corrects dependent sampling without an unjustified effective sample size. A fixed polynomial contrast uses the sampling budget more effectively for a known target, while the general simultaneous-box procedure retains greater freedom to choose targets after acquisition. A common clock correction repairs phase mismatch in the dimension experiment; its calibration expense must be compared with simply collecting more records.

22.1 The evidence contract

Each study declares its data-generating model, target, acquisition procedure, inference rule, cost accounting and simulation repetitions. Frozen JSON files and seeded streams make the reported samples reproducible. Pilot and held-out experiments use separate streams. Refinement loops are identified explicitly: a revised protocol is frozen before its new evaluation data are generated. The simulations retain unfavorable regimes and competing explanations.

The reported error guarantees and Monte Carlo intervals answer different questions. A theorem may guarantee at most 5% erroneous inference under its premises. Repeated simulations estimate the actual operating probability in a specified model, with finite Monte Carlo uncertainty. Zero errors among a finite number of trials does not establish zero error probability. A numerical solver agreement checks implementation, while a mathematical confidence bound checks an inference rule;

neither validates a real detector.

All time units and physical rates in this extension are illustrative. Entropy rates are reported in natural units per model time. Converting them into watts requires the thermodynamic embedding, temperature and time calibration specified earlier. Evolution time, pilot measurements, reset and detector overhead are reported separately. Unmodeled overhead is not assigned zero physical cost.

22.2 How to use the extension

Begin with the timing chapter before applying a correlation test to finite exposures. Use the sampling chapter when records share a physical trajectory. For a predeclared target, the contrast chapter provides a direct finite-record certificate; for later exploratory selection, retain the simultaneous-box workflow. The two apparatus chapters report complete simulation designs and controls. The final comparison chapter separates practical gains from failures and connects them to the next independent physical validation.

The software package includes exact certificates, numerical simulators, retained records, analysis code, independent checks and reproduction commands. It is intended to let another scientist replace a model or apparatus contract, rerun the same falsification tests and determine whether a claimed conclusion survives that change.

Chapter 23

When timing noise and finite exposure preserve an inference

A timing error need not be a small perturbation, and a finite exposure need not behave like an instantaneous detector. Some large timing fluctuations preserve the inference exactly. Other schedules produce apparent equilibrium violations even when their mean lags are perfectly calibrated. This chapter identifies the relevant contracts and gives exact controls for their failure.

23.1 A universal clock criterion

For positive integer settings $k = 1, \dots, K$, let $T_k \geq 0$ be a random observation time independent of the physical process. Define

$$\mathcal{H}_K = \left\{ \left(\int_0^1 x^k \, \mu(dx) \right)_{k=1}^K : \mu \text{ is a probability measure} \right\}.$$

The set is closed and convex. A reversible stationary process, observed through a fixed binary endpoint detector, has a correlation which is a positive mixture of decaying exponentials. This standard spectral fact underlies the following apparatus criterion [118, 119].

Theorem 23.1 (Clock compatibility)**.** *The clock family preserves $\mathcal{H}_K$ for every finite stationary reversible Markov chain and every fixed, noninvasive binary endpoint detector with independent readout noise if and only if*

$$\left(\mathbb{E} e^{-\lambda T_k} \right)_{k=1}^K \in \mathcal{H}_K \quad \textit{for every } \lambda \geq 0. \tag{23.1}$$

Proof. Write the reversible correlation as $C(t) = \sum_\ell b_\ell e^{-\lambda_\ell t}$, where $b_\ell \geq 0$ and $\sum_\ell b_\ell \leq 1$. If (23.1) holds, combine the representing measures with weights b_ℓ and place the unused mass at zero. All indices are positive, so the resulting measure represents the sampled correlations. Conversely, a symmetric two-state chain with rates $\lambda/2$ and detector values ± 1 has exactly $C(t) = e^{-\lambda t}$. It tests each scalar condition in (23.1). □

The theorem concerns a known clock family, not a finite numerical test over a few relaxation rates. For an endpoint detector with calibrated stationary mean m, its spectral atom m^2 at one is retained and the earlier mean-conditioned cone applies. Finite signed differences are useful necessary checks; they are not a complete finite moment-membership algorithm.

23.2 Two safe families with different dimension implications

If $T_k = X_1 + \cdots + X_k$ for independent, identically distributed nonnegative increments, then

$$\mathbb{E}e^{-\lambda T_k} = \left(\mathbb{E}e^{-\lambda X}\right)^k, \qquad \mathbb{E}e^{QT_k} = P^k, \qquad P = \mathbb{E}e^{QX}.$$

The stochastic matrix P acts on the same hidden state space. Thus arbitrary common increment noise preserves both the reversible moment cone and the classical dimension factorization. Neither exponential increments nor a hidden rate cap is required.

A common random scale $T_k = kV$ also satisfies the clock criterion, by the pushforward $x = e^{-\lambda V}$. It need not produce a common channel on the same state space. For example, a two-state chain with decay 2^{-t} and a fair scale choice $V \in \{1, 2\}$ gives prepared occupation probabilities

$$p_k = \frac{1}{2} + \frac{2^{-k} + 4^{-k}}{4}.$$

The 3×3 Hankel matrix $(p_{i+j})_{i,j=0}^{2}$ has determinant $9/32768 > 0$. A two-state common-channel representation is impossible. A three-state representation exists: two transient states survive each step with probabilities $1/2$ and $1/4$, then enter an absorbing state; use equal initial weights on the transient states and detector values $(1, 1, 1/2)$. The pooled record includes a persistent scale variable. Preserving a thermodynamic null therefore does not automatically preserve a dimension null.

23.3 The actual duration belongs in the entropy bound

Keep the full joint record consisting of the setting, the realized clock and the physical trajectory. Its clock law is common to the actual generator Q and its additive reversibilization S. The stationary path comparison gives

$$D(\mathbb{P}_Q \| \mathbb{P}_S) = \mathbb{E}[T]\, d(Q, S) \leq \frac{\mathbb{E}[T]}{4}\sigma, \qquad \mathbb{E}[T] = \sum_k w_k \mathbb{E}[T_k].$$

Discarding the realized time only decreases relative entropy. Consequently, for a certified record divergence J from the compatible reversible cone,

$$\sigma \geq \frac{4J}{\sum_k w_k \mathbb{E}[T_k]}. \tag{23.2}$$

The mean duration must be finite and positive; a calibrated upper bound on it can be used conservatively. For common increments it equals $\mathbb{E}[X] \sum_k w_k k$, and for common scale it equals $\mathbb{E}[V] \sum_k w_k k$. This remains a stationary even-state classical inference with the physical assumptions of the earlier entropy chapter.

Mean timing alone is insufficient. In an equilibrium two-state process with $C(t) = 2^{-t}$, take $T_1 = 1$ and let T_2 be zero with probability $4/5$ and ten otherwise. The mean lags are one and two, but the correlations increase from $1/2$ to $4097/5120$. Even stochastic ordering is insufficient for the full moment cone: the coupled schedules $(1, 1, 2)$ and $(1, 11, 12)$, chosen with probabilities $9/10$ and $1/10$, have means $(1, 2, 3)$ and pointwise ordered lags, yet their second correlation difference is $-7171/40960$. Both examples have zero physical entropy production.

23.4 Finite-duration symmetric detectors

Let a block last L and have a bounded conditional binary mean $F(X_{[0,L]}) \in [-1, 1]$. Assume F is invariant under reversing the path within the block, the detector is noninvasive, and different block outputs have independent noise conditional on the physical paths. Use identical blocks on $[0, L]$ and $[L + T, 2L + T]$, where $T \geq 0$ is the independent gap.

Theorem 23.2 (Disjoint symmetric exposure)**.** *For the reversible reference process, put* $g(x) = \mathbb{E}_S[F(X_{[0,L]}) \mid X_0 = x]$*. Then*

$$\mathbb{E}_S[Y_{\text{left}} Y_{\text{right}} \mid T] = \langle g, e^{ST} g \rangle_\pi. \tag{23.3}$$

Thus a compatible gap clock preserves the reversible moment cone. The corresponding stationary entropy bound is

$$\sigma \geq \frac{4J}{2L + \mathbb{E}[T]}. \tag{23.4}$$

Proof. Stationary path reversal and symmetry of F imply that its conditional mean given the final block state is also g. Conditioning on the state at the end of the first block and the beginning of the second, the Markov property yields (23.3). Since $|g| \leq 1$, its spectral measure has mass at most one. The observed record uses a path of length $2L + T$; applying the same path divergence comparison gives (23.4). □

For a nonlinear block functional, the mean under S can differ from the mean under Q. The mean-unrestricted outer cone is safe: its agreement baseline is $1/2$, and any conserved spectral mass is included. This does not assume that the physical mean is zero. Measured mean information cannot simply be transferred between the models. A symmetric linear exposure,

$$F = \int_0^L w(u) f(X_u)\, du, \qquad w \geq 0, \quad \int_0^L w(u)\, du = 1, \quad w(u) = w(L - u),$$

has mean πf under either generator, so the calibrated-mean refinement is valid. Rectangular exposure is included at the level of this bounded conditional mean. Invasiveness, additional detector memory and conditional noise dependence still require separate treatment.

23.5 Three exposure controls that prevent false discoveries

All three controls use an equilibrium two-state process with $C(t) = 2^{-t}$. A symmetric block channel reports a fair random choice of its two endpoint signs. For length L and start separation D, its correlation is exactly

$$C_L(D) = \frac{2\, 2^{-D} + 2^{-(D+L)} + 2^{-|D-L|}}{4}.$$

For disjoint blocks, $D = L + T$, this is $((1 + 2^{-L})/2)^2 2^{-T}$, confirming the gap formula.

First, overlap can create a false revival. For $L = 4$, start separations two and four give $49/256$ and $289/1024$, an increase of $93/1024$.

Second, even disjoint blocks require the correct time coordinate. For $L = 2$, start separations two and four give $C_1 = 25/64$ and $C_2 = 25/256$. Assigning them to the first and second normalized moments yields $C_2 - C_1^2 = -225/4096 < 0$. The actual gaps are zero and two. A gap-zero correlation has unknown spectral mass at most one and must not be assigned the normalization of an ideal noiseless instantaneous detector.

Third, reversing the detector can reverse the sign test. The antisymmetric channel $F = (X_{\mathrm{end}} - X_{\mathrm{start}})/2$, implemented through an independent binary encoding, gives correlation $-9/128$ at length two and gap one. Its agreement probability is $119/256 < 1/2$ despite equilibrium. The required time-reversal symmetry of the block channel is absent.

These examples are exact synthetic population calculations, not laboratory measurements. They show why the apparatus contract is part of the theorem. The clock criterion is a direct consequence of established positive spectral calculus; the present contribution is its integration with the inference contracts, complete duration accounting and explicit failure controls.

Chapter 24

Sampling dependent records without inventing independent trials

Stationary marginal probabilities do not justify treating consecutive records as independent. This matters physically: an otherwise valid entropy-production certificate can falsely reject an equilibrium process when its input probability box has been computed from strongly dependent counts. The virtual laboratory below exhibits that failure and tests a correction with an explicit time cost. The concentration tools are established conditional Hoeffding and martingale methods[17, 120]; the work here connects their assumptions to a usable sampling chronology and the continuous moment certificate.

24.1 A conditional-mean contract

There are m settings with deterministic record quotas $n_k > 0$. Before each outcome, choose a setting whose quota is unfinished. This choice may depend on earlier records or fresh randomization, but precedes the current outcome. After $N = \sum_k n_k$ records every quota is filled. Let $Z_i \in [0,1]$ denote the whole-record statistic; for same-outcome pairs it is binary. Write K_i for the selected setting and μ_i for its conditional mean given all previous information and the selected setting.

Theorem 24.1 (A simultaneous box for dependent records)**.** *Suppose fixed target probabilities p_k satisfy*

$$|\mu_i - p_{K_i}| \le \delta_{K_i} \quad \textit{almost surely.} \tag{24.1}$$

Let $\widehat{p}_k$ be the average of the n_k records assigned to setting k. For positive statistical radii r_k satisfying

$$2n_k r_k^2 \ge \log \frac{2m}{\alpha}, \tag{24.2}$$

the simultaneous coverage statement is

$$\Pr\{\forall k : |\widehat{p}_k - p_k| \le r_k + \delta_k\} \ge 1 - \alpha. \tag{24.3}$$

No independence between records or settings is required.

Proof. Conditional Hoeffding's lemma applies to $Z_i - \mu_i$, whose conditional range has length one. For each setting k and real λ, the process

$$\exp\left[\lambda \sum_{i \le j} \mathbf{1}_{\{K_i = k\}}(Z_i - \mu_i) - \frac{\lambda^2}{8} \sum_{i \le j} \mathbf{1}_{\{K_i = k\}}\right]$$

is a nonnegative supermartingale. The setting indicator is known before its outcome. At the final record its count equals the deterministic n_k. Exponential Markov inequality, optimized at $\lambda = 4r_k$, bounds either centered tail by $e^{-2n_k r_k^2}$. The accumulated drift in (24.1) is at most $n_k\delta_k$; dividing by n_k adds δ_k to the radius. A union bound over both tails and all settings proves (24.3). □

The bias correction is δ_k, not $n_k\delta_k$. This concerns an average of bounded conditional means. It differs from a total-variation comparison of entire transcripts, where per-record approximation errors can accumulate. Merely knowing that each record has unconditional mean p_k does not establish (24.1). Outcome-dependent acceptance of records and stopping when significance first appears are outside this fixed-quota theorem.

A uniform whole-record calibration radius η_k adds to the half-width. If an external pretest contract for the required envelope has failure probability at most β, a union bound gives total failure at most $\alpha + \beta$. A fitted favorable model or postselected subset does not establish that external guarantee.

The implementation encloses $\log(2m/\alpha)$ with rational arithmetic and checks (24.2) exactly. A radius designer uses an outward dyadic square root. The resulting probability box can be passed directly to the continuous reversible-moment certificate. Its scores may be selected after constructing the simultaneous box, because the certificate minimizes over that entire box. A direct fixed-score test can be statistically tighter, but its score must be fixed before the held-out observations.

24.2 How a blind gap supplies the contract

Consider the following sequential apparatus protocol. Finish the previous pair; select the next setting and a blind gap from past information; evolve without observing during that gap; then read the starting detector, wait the selected lag, and read the ending detector. Retain the same/not-same pair indicator. Current-pair readings are not part of the history on which its pre-record conditional mean is conditioned.

Suppose the hidden dynamics has stationary distribution π and satisfies the uniform mixing envelope

$$\sup_x \mathrm{TV}\Big(\delta_x e^{GQ}, \pi\Big) \le \delta. \tag{24.4}$$

Conditioning on any past history leaves an arbitrary initial hidden-state distribution. After the blind gap it lies within δ of π. The complete ensuing pair is a stochastic channel of that state, so data processing for total variation bounds its probability shift by δ. This proves the needed conditional-mean contract. Detector and controller memory must be included in the mixed state, reset, or separately bounded. Noninvasive readout is still required for the declared generator interpretation.

Two established mixing forms give usable arithmetic certificates. A Doeblin minorization $P_h(x,\cdot) \ge \varepsilon\nu(\cdot)$ for every hidden state implies a j-block bound $(1-\varepsilon)^j$. Alternatively, let $Q^\dagger$ denote the adjoint in $L^2(\pi)$ and set $S = (Q+Q^\dagger)/2$. If its spectral gap is at least $\lambda > 0$ and $\pi_x \ge \pi_{\min} > 0$, then

$$\sup_x \mathrm{TV}\Big(\delta_x e^{GQ}, \pi\Big) \le \min\left\{1, \frac{1}{2}\sqrt{\frac{1-\pi_{\min}}{\pi_{\min}}}\, e^{-\lambda G}\right\}. \tag{24.5}$$

This does not assume that Q itself is reversible. Differentiating the squared $L^2(\pi)$ norm of a mean-zero density perturbation yields the quadratic form of S, hence exponential contraction at rate λ; Cauchy–Schwarz converts it to total variation. A point mass has squared initial norm $1/\pi_x - 1$.

General concentration methods for Markov chains provide other possible designs[121]; the present blind-gap construction is sufficient, without a time-optimality claim.

The code rounds the square root outward and bounds $e^{-\lambda G}$ by the reciprocal of a positive Taylor lower bound for $e^{\lambda G}$. It checks the stated numbers. It does not infer a uniform mixing rate for all hidden models from a good fit to one favored model.

24.3 An equilibrium apparatus that produces false discoveries

The frozen virtual-laboratory model has four states (s, b), where s is a slow hidden sector and b is the observed bit. The sector flips at rate $1/1000$. The bit flips at rate $1/50$ in sector zero and rate 2 in sector one. Every transition has the same rate as its reverse, so the uniform distribution is stationary and detailed balance holds exactly. The detector reads b perfectly and noninvasively.

The true same-outcome probabilities are approximately 0.74465056 at lag one and 0.73051808 at lag two. Shared-endpoint pairs initialized once in stationarity have these correct marginal means. Their conditional agreement probabilities remain strongly influenced by the slowly changing sector.

Before generating data, the contract fixed seed 918273, 4,000 independent replicate trajectories, 600 records at each lag, the order of the settings, $\alpha = 1/20$, and statistical radius $61/1000$. Three sampling chronologies were tested. The first reused each pair endpoint as the next pair start. The second inserted a blind gap of 3,000 time units before every pair. The third used ideal independent stationary preparations before each pair. No parameter was retuned after production.

For the gap design, comparison of Dirichlet forms with independent bit flips at the minimum rate and sector flips proves a gap lower bound $1/500$. Equation (24.5), with $\pi_{\min} = 1/4$, gives a certified conditional bias below 0.002146663, so the declared correction $1/400$ is valid. The unseparated records admit the much broader generic correction $3/4$, since every hidden-state distribution is within $3/4$ of the uniform four-state law.

Sampling and analysis	Joint coverage	False equilibrium exclusion
Shared endpoints, unqualified iid box	2.30%	24.70%
Same counts, honest $\delta = 3/4$	100%	0/4000
Blind gap, corrected $\delta = 1/400$	99.90%	0/4000
Independent stationary pairs	99.875%	0/4000

The false-exclusion rule is that the lower confidence bound at lag two exceeds the upper bound at lag one. Such an increase is impossible for the complete stationary reversible autocorrelation class. The rule is valid only when its probability box has the advertised coverage. For the unqualified shared-endpoint analysis, the approximate 95% Wilson Monte Carlo interval for its false exclusion rate is $[23.388\%, 26.060\%]$. For zero exclusions in 4,000 replicates, the corresponding approximate upper limit is 0.095945%. These Monte Carlo intervals describe simulation uncertainty; the inference thresholds themselves are checked with exact rational arithmetic.

The correction has a substantial time cost. Pair evolution takes 1,800 model time units per replicate; adding the blind gaps raises this to 3,601,800. The independent-preparation benchmark does not model the external reset's duration or energy, so its pair-evolution time is not a complete hardware-cost comparison. The broad no-gap correction is retained even though it is uninformative: honest uncertainty need not yield a positive certificate.

Snapshot transitions were sampled from the CTMC transition semigroup. Pair semigroups were independently checked by uniformization, with maximum numerical discrepancy below 5.28×10^{-15}; the long-gap semigroup was cross-checked by symmetric eigendecomposition to 1.19×10^{-13}. The released results retain the frozen contract hash, count histograms with exact multiplicities, seeds,

solver checks, mixing certificate, and model times. These are reproducible synthetic experiments, not measurements from a physical apparatus.

Chapter 25

A direct contrast for finite-record energy inference

The virtual laboratory exposes a statistical cost in the earlier procedure. A simultaneous interval for every measured probability supports arbitrary later score selection, but it can be unnecessarily wide when the relevant physical contrast is known before acquisition. This chapter supplies a direct alternative. It uses the same observations and physical assumptions, and credits its ingredients to standard concentration, Pinsker's inequality and optimal allocation [17, 120, 127]. The practical question is whether their combination improves a complete inference procedure at matched acquisition cost.

25.1 A contrast certified against every reversible explanation

At positive integer gap indices k, let p_k be the probability of matching binary observations. The comparison set is the reversible moment class

$$q_k = r + (1-r)\int_0^1 x^k \, \mu(dx), \qquad r \in [1/2, 1].$$

The timing and detector chapter gives physical implementations of this class. Choose real coefficients a_k before observing the test records. An exact polynomial bound gives

$$b \geq \sup_{r\in[1/2,1],\, x\in[0,1]} \sum_k a_k\{r + (1-r)x^k\}.$$

Then $a \cdot q \leq b$ for every comparison law, including limits of hidden networks with unbounded rates. Rational Bernstein subdivision supplies a checkable b. With $a = (-1, 2, -1)$ at indices $(1, 2, 3)$, one can take $b = 0$ because $-x + 2x^2 - x^3 = -x(1-x)^2 \leq 0$.

A stronger predeclared family is

$$a_s = (-s^2, 2s, -1), \qquad b_s = -\frac{(1-s)^2}{2}, \qquad 0 \leq s \leq 1.$$

The support is exact because the polynomial is $-x(x-s)^2 \leq 0$; the largest baseline contribution occurs at $r = 1/2$. In correlation coordinates $C_k = 2p_k - 1$, the contrast is $(-s^2C_1 + 2sC_2 - C_3)/2$. When $C_1 > 0$, its unconstrained maximum is at $s = C_2/C_1$, exposing the Hankel log-convexity condition $C_2^2 \leq C_1C_3$. A pilot may propose a rational s and freeze it for later data. The refinement study fixes $s = 2/3$ before its independently seeded production sample. This is an established moment inequality compiled into the present apparatus and confidence contracts.

25.2 From a contrast to a physical resource requirement

Suppose the true probabilities satisfy $a \cdot p - b \geq C > 0$. For positive mixture weights w_k summing to one, Bernoulli Pinsker and Cauchy–Schwarz give

$$\sum_k w_k d_{\mathrm{B}}(p_k \| q_k) \geq 2 \sum_k w_k (p_k - q_k)^2 \tag{25.1}$$

$$\geq \frac{2C^2}{\sum_k a_k^2 / w_k}. \tag{25.2}$$

All logarithms are natural. Settings with $a_k = 0$ can be omitted. The stationary path comparison from Chapter 20 therefore implies

$$\sigma \geq \frac{8C^2}{(\sum_k w_k t_k)(\sum_k a_k^2 / w_k)}, \tag{25.3}$$

where $t_k > 0$ is the expected full path duration for setting k, or a calibrated upper bound on it. Finite exposure intervals must be included.

Proposition 25.1 (Optimized comparison mixture)**.** *Under the stationary ordinary-reversal Markov and noninvasive observation contract, a contrast lower bound C certifies*

$$\boxed{\sigma \geq \frac{8C^2}{\left(\sum_k |a_k| \sqrt{t_k}\right)^2}.} \tag{25.4}$$

Proof. Cauchy–Schwarz gives $(\sum_k w_k t_k)(\sum_k a_k^2 / w_k) \geq (\sum_k |a_k| \sqrt{t_k})^2$. Equality holds at w_k proportional to $|a_k| / \sqrt{t_k}$. Insert this valid choice into (25.3). The weights describe a mixture of record laws for the proof. They need not equal the proportions of samples collected at each setting. □

This optimizes the stated lower-bound formula, not all experimental inference procedures. For a preparation–housekeeping statement, the alternative rational weights $w_k = |a_k| / \sum_j |a_j|$ maximize the bound (25.2), yielding

$$D(\nu \| \pi) + \frac{1}{2} \mathbb{E}_w \Sigma_{\mathrm{hk}} \geq \frac{2C^2}{(\sum_k |a_k|)^2}.$$

This expectation uses those stated weights. A narrowed stationary-reference mean requires separate calibration and cannot be replaced by a transient mean.

25.3 A confidence statement from complete records

Collect fixed quotas N_k of agreement bits at each setting. A record may depend on earlier records, provided its conditional agreement probability is within a declared δ_k of one fixed target probability p_k, conditional on the full preceding history. Independent repetitions correspond to $\delta_k = 0$. Predictable routing between fixed quotas is allowed; stopping at a favorable count is not.

Let S_k be the number of agreements. Conditional Hoeffding's inequality applied to the weighted increments $a_k Z / N_k$ gives, with probability at least $1 - \alpha$,

$$a \cdot p - b \geq \sum_k a_k \frac{S_k}{N_k} - b - \sqrt{\frac{\log(1/\alpha)}{2} \sum_k \frac{a_k^2}{N_k}} - \sum_k |a_k| \delta_k. \tag{25.5}$$

Each increment has conditional range length $|a_k|/N_k$; summing its squared range over the fixed quotas proves the concentration term. The conditional bias contributes at most the final term. No multiplication of δ_k by the total number of records is required. Any failure probability of an externally supplied calibration event must be added separately.

Truncate the right-hand side at zero and substitute it into (25.4). The software bounds logarithms and square roots outward with rational arithmetic. The support, contrast and confidence rule are fixed before the test sample, or chosen from an independent pilot. Selecting an arbitrary contrast from these same counts would invalidate this single-contrast guarantee. The older simultaneous-box method remains useful when such post-acquisition exploration is needed.

25.4 Acquisition allocation and the matched experiment

For a fixed total record count, minimizing the concentration proxy $\sum_k a_k^2/N_k$ gives N_k proportional to $|a_k|$ in the continuous relaxation. If record k costs $c_k > 0$ units of calibrated acquisition time, then at fixed total time the relaxed optimum has N_k proportional to $|a_k|/\sqrt{c_k}$. This is standard cost-adjusted allocation. The code proposes rounded integer quotas and checks their actual cost; it does not claim a globally optimal integer design. Evolution time, reset overhead, exposure and any independently calibrated energy charge must be accounted for consistently.

The held-out virtual-laboratory comparison uses the fixed curvature contrast $(-1, 2, -1)$, the previous continuous-moment box certificate and the same trajectory-generated datasets. Each receives identical settings, counts and physical exposure. The inference method changes; the acquired data do not. No detector model or hidden generator is supplied to either inference routine. The generator is retained by the simulator solely to evaluate the known-truth performance and failure controls. Results and Monte Carlo intervals appear in the simulation chapter.

The API is `constructive_access.virtual.contrast.certify_counts`; its inputs include quotas, counts, coefficients, conditional probability radii and full expected path durations. A positive output is a model-bound confidence statement. It does not validate the detector, the physical reversal convention, or stationarity by calculation alone.

25.5 An exact check on the confidence-region limitation

The module also searches for a two-state equilibrium curve inside a supplied probability box. For rational $0 < x < 1$ and $0 \le A \le 1$, set

$$q_k = \frac{1 + Ax^k}{2}.$$

At deterministic times $k\Delta$, this is realized by a stationary symmetric two-state chain with jump rate $-\log x/(2\Delta)$ and fixed detector means $\pm\sqrt{A}$. For each trial x, the box inequalities are linear intervals for A; a nonempty intersection gives an exact countermodel. A separate checker verifies the probabilities directly. Failure of this limited search proves no infeasibility.

If such a curve lies inside a rectangular confidence region, no strictly positive dissipation bound can hold for every vector in that region. This does not prevent another confidence procedure from using the empirical counts: the direct contrast has a different coverage event. Nor is information necessarily lost by storing the box, whose midpoint can retain the counts. The limitation concerns treating the entire rectangle as the admissible set.

For a random-scale clock, the same curve is a point in the conservative moment comparison class, but it need not be generated by a two-state chain under that particular clock law. The benchmark

distinguishes these claims. Independent calibration of zero-lag detector variance would also narrow the comparison class and can remove countermodels; that extra access is not assumed here.

Chapter 26

A stochastic laboratory for temporal entropy inference

This laboratory generates hidden trajectories, noisy detector outputs and experimental timing before analyzing the observed counts. Its purpose is to test the full inference chain under declared apparatus models. It supplies simulation evidence; none of the following records came from a physical laboratory. The time unit and transition rates are illustrative.

The experiment specification, independent pilot and held-out seeds, record budgets, detector laws, confidence radii and original inference scores were saved in `contracts/thermodynamic_lab.json` before production. Twenty pilot datasets and 200 held-out datasets per case use disjoint random streams. Pilot outcomes remain in the release. The subsequent comparison with a direct, predeclared curvature contrast uses the same held-out records and is reported separately. This separation of design, data generation and evaluation follows established simulation-study practice [112].

26.1 Trajectory and apparatus construction

The main physical model is a ring with $N = 40$ states and translation-invariant forward and backward rates a, b. Each independent pair record begins with X_0 uniform on the ring, which is its exact stationary distribution. For the selected duration U, generate independent counts

$$N_+ \sim \mathrm{Poisson}(aU), \qquad N_- \sim \mathrm{Poisson}(bU), \qquad X_U = (X_0 + N_+ - N_-) \bmod 40. \tag{26.1}$$

This samples the continuous-time random walk exactly: independent forward and backward Poisson clocks generate its jumps. The simulator does not draw matching bits from an analytically supplied endpoint probability.

A fixed detector has conditional mean $f_j \in [-1, 1]$. Independent readouts are generated at both physical endpoints from

$$\Pr(Y = +1 \mid X = j) = \frac{1 + g f_j}{2}, \qquad 0 \le g \le 1.$$

Thus a gain g multiplies the stationary two-time correlation by g^2. The first menu uses a deterministic half-ring detector, $f_j = +1$ on 20 consecutive states and -1 on the others. The second uses the fixed rational approximation to a cosine detector specified in the frozen contract. These are declared detector models, not empirical calibrations.

The revival menu observes lags 1.6 and 2.1, with 5,000 independent records per lag. The curvature menu observes 0.15, 0.30 and 0.45, with 50,000 records per lag. They are separate

experiments with different detectors and observation windows; their results are not presented as a matched efficiency comparison. Within each menu, every inference procedure receives exactly the same records.

The driven ring has $(a, b) = (20, 1)$ and stationary entropy-production rate $19 \log 20 \simeq 56.9189$ in natural entropy units per model time. The equilibrium control has $(a, b) = (10.5, 10.5)$, preserving total jump activity while setting stationary entropy production to zero. A noisy detector control uses $g = 0.2$, equivalent to an independent bit flip with probability 0.4 after each ideal readout.

An independent timing control replaces each nominal lag t_k by $U_k = t_k V$, where V is uniform on $[0.7, 1.3]$ and independent of physical paths and detector noise. The same scale law is used at every lag. For a reversible process at $t_k = k\Delta$,

$$C(U_k) = m^2 + \sum_{\ell} c_\ell \mathbb{E}[e^{-\lambda_\ell k \Delta V}], \qquad c_\ell \geq 0.$$

After averaging over V, this is still a nonnegative moment mixture with $x = e^{-\lambda_\ell \Delta V} \in [0, 1]$. Consequently the reversible correlation cone is preserved, and the expected duration remains nominal because $\mathbb{E}[V] = 1$. Arbitrary setting-dependent timing errors do not inherit this conclusion.

26.2 Blind inference and matched accounting

Inference receives the number of matching endpoint pairs at each lag, not hidden states or jump counts. With K settings, n records at each, and empirical matching probabilities $\widehat{p}_k$, the simultaneous box

$$p_k \in [\max(0, \widehat{p}_k - r), \min(1, \widehat{p}_k + r)] \quad \text{for all } k \tag{26.2}$$

has failure probability at most $2Ke^{-2nr^2}$. The chosen radii are $r = 0.021$ for revival and $r = 0.007$ for curvature. Rational logarithm enclosures verify that both failure bounds are at most 0.05.

Three original procedures use this identical simultaneous box. A negative-correlation test rejects when an upper probability endpoint is below $1/2$. A revival test rejects when the upper endpoint at an earlier lag is below the lower endpoint at a later lag. A fixed dual witness evaluates the previously specified positive likelihood scores against the full reversible moment cone. Its exact polynomial normalizer is checked once, and each dataset is evaluated using the same rational logarithm enclosures as the original certificate. The latter is a fixed witness from the full cone; it is not an optimization over every valid witness.

These tests need no correction merely because they are compared: each rejection is impossible on the same joint confidence event under its reversible null. Accordingly even the union of their null rejections is bounded by the box failure probability. This does not extend to changes in the measurement contract.

The revival menu uses 10,000 pair records and nominal observation exposure $5000(1.6 + 2.1) = 18,500$ per dataset. The curvature menu uses 150,000 pair records and exposure $50,000(0.15 + 0.30 + 0.45) = 45,000$. Actual elapsed exposure is stored for every dataset, including jitter. Physical preparation, stationary resetting, detector operation and data processing costs are not calibrated or included in these exposure totals. The calculation does not equate wall-clock runtime with thermodynamic energy.

26.3 Held-out results and a useful failure

Table 26.1 gives rejection counts in 200 independently generated held-out datasets per case. All entries are simulated outcomes. The JSON report also gives exact-binomial Clopper–Pearson

confidence intervals for each Monte Carlo rejection probability [122], evaluated numerically. For example, 0/200 gives an upper 95% endpoint of approximately 0.0183, while 200/200 gives a lower endpoint of approximately 0.9817. Neither outcome establishes zero or unit rejection probability.

Menu	Simulated case	Negative	Revival	Fixed dual
Curvature	Driven	0	0	1
	Equilibrium	0	0	0
	High detector noise	0	0	0
	Independent scale jitter	0	0	0
	Detector memory*	0	0	1
Revival	Driven	0	200	200
	Equilibrium	0	0	0
	High detector noise	1	0	0
	Independent scale jitter	2	200	0
	Detector memory*	0	200	200
	Prepared equilibrium*	200	0	0

Table 26.1: Rejections among 200 held-out simulated datasets, with matched records and exposure within each menu. Asterisks identify controls outside the stationary, independently read-out detector contract. Their rejections must not be interpreted as dissipation of the simulated physical sample. The fixed dual uses the original predeclared scores, not all possible moment-cone witnesses.

The favorable revival experiment works with the planned record budget, and equilibrium controls produce no rejections in these runs. Independent readout noise substantially suppresses useful information although the sample's hidden stationary entropy production is unchanged.

The curvature experiment exposes a practical limitation. Its model curve has the required departure from convex equilibrium decay, but the combination of coordinate confidence intervals and the original score choice removes most of this witness's usable separation at this budget. The fixed witness rejects only once in 200 held-out datasets. This is not a successful practical detection design at that record count. It does not establish that optimized moment-cone inference has low power. The subsequent matched comparison includes both safely optimized box witnesses and direct-contrast inference, which concentrates uncertainty in a predeclared informative direction.

Timing jitter reveals a second distinction between validity and power. Common random scaling preserves the reversible cone, yet the original likelihood scores are poorly positioned for the shifted matching probabilities: the fixed dual never rejects, while the broader revival inequality rejects in all 200 datasets. A cone can remain the correct null class while one selected witness becomes ineffective. Simultaneous confidence boxes permit safe subsequent witness optimization, but these frozen-score results are retained without replacing the failed regime.

26.4 Controls that defeat an unqualified interpretation

The memory control evolves the sample with the equilibrium ring generator, but the detector stores its first bit and repeats it with a setting-dependent probability specified in the contract. This deliberately violates the assumption that each readout is a fixed channel of the current physical state with independent noise. The resulting revival record imitates the driven record and is rejected in all 200 datasets by both revival and the fixed dual. Sampling confidence about those observed probabilities cannot establish the missing detector contract. The sample's generator has zero stationary entropy production; the separate memory device's physical costs have not been modeled.

The preparation control is a detailed-balanced two-state process with rates $0 \to 1$ equal to 9 and $1 \to 0$ equal to 1. Direct Gillespie simulation starts every record in state 0, rather than in its

stationary law $(0.1, 0.9)$. The matching probability is $0.1 + 0.9e^{-10t}$, and the negative-correlation test rejects every held-out dataset. The stationary dissipation interpretation is inapplicable. The preparation-relative entropy is $D(\delta_0\|\pi) = \log 10$; the earlier preparation-or-housekeeping theorem explicitly allows this source of apparent temporal separation.

26.5 Reproducibility and remaining physical work

The release retains the full 150,000-record first held-out curvature dataset in a compressed archive: initial and final hidden states, forward and backward jump counts, elapsed times, both detector bits and setting labels. These hidden arrays support auditing only. Separate compact archives retain counts and exposure for every pilot and held-out dataset. The frozen contract and deterministic archives have recorded SHA-256 digests.

An independent matrix-exponential calculation checks the mean matching probabilities. For jitter, 32- and 64-node quadratures must agree within 10^{-11}. Reports include empirical joint confidence-box coverage against these numerical oracles: all 11 cases cover the model probability vector in between 198 and 200 of their 200 held-out datasets. This includes the outside-contract controls, emphasizing that correct statistical coverage does not establish the physical interpretation. Tests also compare the direct Gillespie control with its independent analytic transition law and compare compiled inference with the original exact certificate implementation.

This adds an executable apparatus stress test to existing thermodynamic inference research. Finite resolution and finite statistics have already been studied explicitly [114]; entropy inference from time-symmetric observations is also established [115]. The comparison here is deliberately confined to the shared endpoint record. Methods requiring transition labels, waiting times or resolved currents have different access and are not ranked as inferior from this experiment.

Finite-duration detector integration, imperfect physical resets, unmodeled sample drift and independently calibrated hardware energy remain experimental dependencies. The point-readout simulator and independent stationary preparation do not validate them.

Chapter 27

A virtual laboratory for the dimension witness

The finite-record dimension test has a useful separation between validity and power. Fixed preparation error, detector confusion, and independent noise on each repeated step can substantially weaken a quantum signal while preserving the classical null theorem. A small coherent frequency error is particularly costly: it moves the recurrence away from the prescribed observation grid. An independently calibrated *common* clock correction repairs this failure without changing the classical dimension bound. The following experiments establish that statement within an independently implemented simulator. They are not observations from a physical device.

27.1 Physical simulation and frozen experiments

The base system is a driven qubit with angular drive frequency $\omega = 1$ and isotropic depolarization rate $\gamma = 10^{-3}$, in declared model time units. The observation step is $\Delta = 2\pi/7$. A fixed erroneous preparation places probability e_{prep} on $|1\rangle$. A fixed binary detector has false-positive and false-negative probabilities e_0, e_1. The drive can have a fixed amplitude error, fixed detuning, and independent two-point amplitude and duration noise refreshed on every step.

The primary simulator exponentiates the three-dimensional Bloch generator, averages its four independently refreshed amplitude/duration channels, and then propagates the state. It does not call the dimension certificate's sinusoidal probability helper. A separate four-dimensional vectorized Lindblad calculation reproduces the probabilities to within 1.45×10^{-15} across the initial 24 declared regimes. One additional record stream simulates the preparation bit, every step's noise choice, final Born measurement, and detector confusion explicitly. Its 18 300 records and noise history are retained in `dimension_lab_shots.npz`.

The initial contract was saved before generating its pilot or production data. It fixes all settings, seeds, noise regimes, and the original 18 300-record critical-count test. Each power estimate uses 20 000 independently generated pseudoexperiments. A pilot of 20 000 records at each of 13 settings selects an alternative critical-count plan from a frozen candidate family. Its 260 000 additional records are separate from the production data. Later clock-repair and total-budget comparisons have separate frozen contracts and new seeds. All declared cases, including failed regimes, are retained.

The saved shot ordering is a random permutation of independent simulated records. It represents randomized acquisition under this simulator; it is not evidence that an actual acquisition was stable or robust against drift.

27.2 Which apparatus errors preserve the null

Let α, P, and f denote the actual prepared distribution, repeated classical channel, and stochastic binary detector. The null theorem concerns $p_k = \alpha P^k f$, regardless of whether these quantities match nominal values. Consequently a fixed imperfect preparation or readout is already covered. Independent identical step errors replace P by their average channel and are also covered. No assumed qubit frequency, pure preparation, or detector fidelity enters this classical argument.

An error drawn once and held for an entire run is different. Its record is generally $\mathbb{E}[\alpha P(V)^k f]$, not $\alpha(\mathbb{E}P(V))^k f$. The retained noise label carries memory and must be included in a classical state count. In the declared quasi-static control, these two probability sequences differ by as much as 0.20164. Likewise, allowing the detector response itself to depend on k lets one frozen classical state reproduce the full target sequence. The resulting rejection rate is high, as intended for this outside-contract control. A dimension claim therefore needs a common interface and channel, with any controller memory included.

27.3 Noise tolerance and a matched production comparison

The original test's simulated power is 98.565% in the nominal model. Representative single-error regimes show the following behavior:

Illustrative error	Original-test power
Detector false-positive probability 0.0005	95.245%
Detector false-positive probability 0.001	81.260%
Detector false-positive probability 0.002	28.050%
Preparation error probability 0.001	81.330%
Independent step amplitude jitter ±1%	98.145%
Independent step duration jitter ±1%	98.190%
Fixed amplitude offset +0.5%	63.920%
Fixed amplitude offset +1%	1.595%

These are illustrative dimensionless settings, not a claim about the calibration or feasibility of any particular hardware platform. Complete marginal Clopper–Pearson intervals appear in the machine-readable results. Monte Carlo uncertainty and unknown real-device systematics are separate matters.

The combined design regime has preparation error 0.0005, detector errors $(e_0, e_1) = (0.0005, 0.003)$, amplitude offset 0.003, detuning 0.02, independent amplitude jitter ±0.005, and duration jitter ±0.002. Its original-test power is 67.66%, with a 95% Monte Carlo interval $[67.01\%, 68.31\%]$. The independent pilot selects

$$\epsilon_0 = 61/10000, \qquad \beta_0 = 1503/10000, \qquad \tau = 781/10000.$$

The resulting critical counts are

$$(314, 22, 22, 22, 22, 298, 49, 298, 22, 22, 22, 22, 282),$$

with the low-count inequality at setting seven and high-count inequalities elsewhere. Exact rational tail checks give a type-I bound below 0.049144. At the unchanged production allocation it obtains 71.28% power, with interval $[70.65\%, 71.91\%]$. The paired improvement is 3.62 percentage points; a conservative Hoeffding interval is $[1.70, 5.54]$ points. This is a production-budget comparison conditional on an additional pilot, not a total-cost superiority claim.

A library of 400 homogeneous classical models, spanning one through four states and both continuous-time and discrete stochastic channels, supplies 12 adversarially selected null examples. None rejects in 20 000 repetitions per selected example. A zero count has marginal upper 95% limit 0.018443%. These finite checks are supplementary: the exact type-I certificate, rather than this library, establishes validity over the full null class.

27.4 Repairing coherent calibration error with one common clock

The fixed-amplitude failure is not merely a larger diagonal leakage. At a 1% amplitude error the last setting has mean probability approximately 0.148983, below the original required off-diagonal boundary 0.15. Averaging this particular criterion more accurately will not repair that mismatch. The observation grid itself needs attention.

In a second independently frozen loop, a 260 000-record pilot fits one drive amplitude offset with the nominal Bloch model. The production step is then multiplied by $1/(1+\widehat{a})$. All 13 observation times use this *same* step; none is individually corrected. The fit proposes a design and can be wrong. It does not supply a premise for the null theorem.

Proposition 27.1 (Independent common-clock calibration preserves validity)**.** *Condition on arbitrary pilot data D, and choose a positive common step $\Delta(D)$ and a production plan whose fixed-design null error is at most α. If the independent production records share one actual preparation, channel, and detector, and every production setting repeats that chosen step, then their unconditional null error is at most α.*

Proof. For each fixed pilot outcome, the production means factor through one actual channel P_D as $p_k = \alpha_D P_D^k f_D$. Apply the fixed-design dimension certificate and count-tail guarantee conditionally on D, and average the resulting bound over D. No accuracy condition on the frequency fit is used. □

The second holdout uses the original 18 300 record quotas and original critical counts. Its outcomes are:

Regime	Original clock	Common-clock repair	Repair's 95% interval
Amplitude +0.5%	63.740%	98.610%	[98.44, 98.77]%
Amplitude +1%	1.575%	98.720%	[98.55, 98.87]%
Amplitude +2%	0%	98.595%	[98.42, 98.75]%
Combined design noise	67.855%	80.380%	[79.82, 80.93]%

The four positive fitted offsets shorten the production evolution time by approximately 0.30% to 1.99%. The shot count and 126 900 repeated channel applications remain identical. Reset, readout, and controller overhead are not included in this evolution-time accounting.

27.5 Calibration is a resource, and sometimes more averaging wins

Each frequency pilot costs 260 000 records and 1 820 000 channel applications. Combined with one production experiment, its total is 278 300 records. If one calibration remains valid for 100 later production experiments, its record overhead amortizes to about 14.21% per experiment; for 1000 uses, about 1.42%. Such reuse assumes drive stability and is not established by the simulator.

An additional frozen comparison spends all 278 300 records on the original clock, proportionally enlarges the original quotas, and recalibrates the original forbidden-region test at the same 5%

type-I level. Its measured powers are 100%, 1.31%, 0%, and 100% for the four respective regimes in the preceding table. Thus extra averaging wins the mild-offset and combined-noise cases at this total record budget. Common retiming repairs the two severe phase failures. The equal-record baseline retains the original probability region and is not an oracle or an optimal competing protocol. The total evolution durations differ and are included in the result files. These comparisons support choosing an intervention according to the actual failure mechanism; they do not establish universal cost superiority.

27.6 Reproduction and interpretation

The module `dimension_lab.py` exposes the physical channel simulators, exact plan checks, pilot design, record generation, and three frozen runs. The contracts, selected thresholds, all noise-regime results, count intervals, raw shot stream, and seeds are retained. Twelve producer tests include an independent density-matrix solver, direct measurement-noise checks, timing-memory counterexamples, exact null checks, malformed-plan rejection, reproducible shot streams, and common-clock repair.

The concrete result is an experimentally relevant failure and repair: coherent grid error can dominate the power of a valid dimension witness, and independent calibration of one shared clock can restore power while preserving its target-independent null guarantee. Physical device validation, external review, and any claim of historical originality remain separate work.

Chapter 28

What the matched simulations changed

The virtual laboratory produced useful gains and useful failures. The important comparison holds observations and acquisition cost fixed while changing the inference or control design. The reported improvements apply to the specified models and regimes; the controls prevent extrapolation to arbitrary hardware or universally optimal experimental design.

28.1 The first contrast loses to a stronger baseline

The original curvature score and simultaneous probability box detect the 40-state driven-ring effect in only one of 200 held-out datasets, each with 50,000 records at each of three lags. A direct convexity contrast $(-1, 2, -1)$ detects it in 189 of the same datasets. This alone would be an incomplete comparison: the fourth-edition numerical proposal routine can optimize its scores and then check them over the entire continuous moment class. That stronger box-based procedure detects all 200 datasets.

The result is retained. A more targeted confidence calculation does not compensate for every weak choice of physical contrast. The simultaneous box also permits score selection after acquisition, a flexibility the direct single-contrast test does not share.

28.2 A revised contrast succeeds at a smaller matched budget

The second loop fixes the shifted-cubic contrast $(-4/9, 4/3, -1)$ before generating a newly seeded evaluation sample. It uses 5,000 independently simulated trajectory pairs per lag: 15,000 records and 4,500 units of expected evolution time per complete experiment. Both methods receive exactly the same counts and settings. No pilot records are added for this analytically chosen second-loop contrast.

Simulated case	Direct fixed contrast	Optimized moment-box method
Driven ring	367/400	0/400
Equilibrium ring	0/400	0/400
Strong independent readout noise	6/400	0/400
Common random-scale clock noise	370/400	0/400

The driven-case detection probability estimate is 91.75%, with a pointwise 95% Clopper–Pearson Monte Carlo interval of 88.61–94.25%. Zero detections in 400 equilibrium runs gives a corresponding upper interval endpoint of 0.918%, not a proof of zero error probability. The theoretical false-positive guarantee remains 5% under the declared contract. Strong readout noise remains a failure regime for useful power.

The revised certificate's mean entropy-rate lower bound in the driven case is about 0.00115, whereas the model's true rate is about 56.9. The improvement is in detecting a positive requirement at a finite budget, not accurately recovering total dissipation. Neither reset energy nor detector/controller energy is calibrated by this numerical model.

The comparison uses the implemented optimized-box procedure, whose numerical search proposes scores and whose rational polynomial calculation verifies them. A zero result alone does not prove that every possible score fails. We therefore also construct exact equilibrium points inside the boxes. In 329 of the 400 driven datasets, a symmetric two-state model with a suitable fixed stochastic detector lies inside the whole rectangular confidence region. In 296 of those same datasets the direct-contrast certificate is positive.

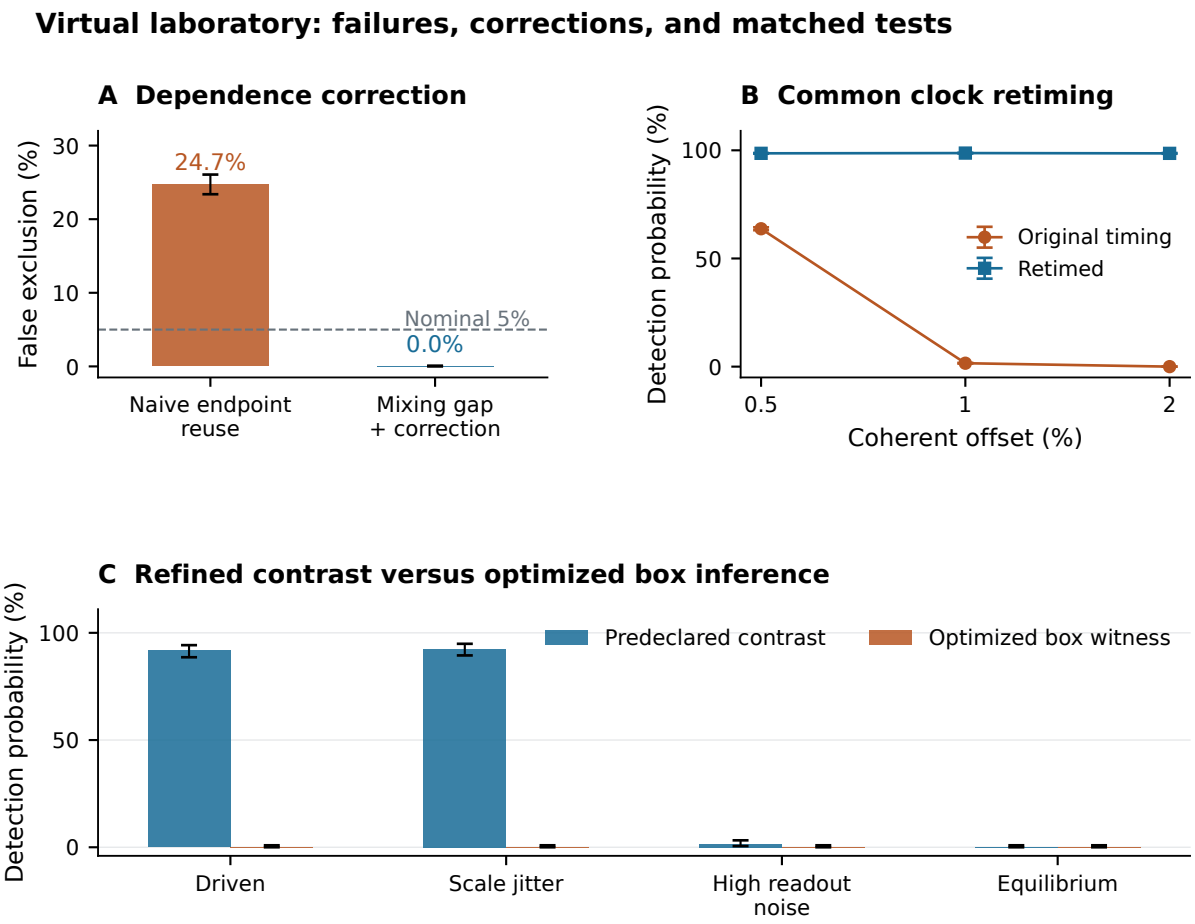
There is no contradiction. A lower bound valid uniformly over each such rectangle must allow its equilibrium point; the direct test uses a different, predeclared confidence region. The countermodels establish a limitation of that rectangular admissible set, rather than a universal limit on processing the original counts. They rely on the stated unknown-detector-variance contract. Independent zero-lag readout calibration could remove some of these alternatives.

28.3 Sampling and calibration costs determine the recommendation

For the original convexity contrast, reallocating the same 150,000 records from $(50{,}000, 50{,}000, 50{,}000)$ to $(37{,}500, 75{,}000, 37{,}500)$ reduces the concentration proxy. Expected evolution time remains 45,000 units. An independent 200-dataset cohort gives 195 detections, compared with 189 in the original cohort. Their Monte Carlo intervals overlap; this small difference is not presented as a precisely established performance gain. The exact reduction in the confidence proxy is the supported design statement.

The dependent-record experiment shows why favorable power is insufficient. A stationary equilibrium model yields 24.7% false equilibrium exclusions under an unjustified independent-record calculation. A known mixing envelope repairs coverage, but the supplied blind-gap design increases evolution time from 1,800 to 3,601,800 units for the same record count. That large cost is part of the result. An actual apparatus would need a justified faster reset, a sharper dependence analysis, or a different observation protocol.

The dimension laboratory finds a complementary limitation. A coherent one percent frequency error moves an intended oscillation minimum off the sampling grid, reducing the old production protocol's power to 1.575%. One common clock correction selected from an independent calibration sample restores 98.72% conditional production power. It preserves the classical common- channel guarantee, but uses 260,000 additional pilot records. At an equal total record budget, simple extra averaging beats the repair in some mild-noise regimes; it fails to repair the larger phase offsets. Repeated use can amortize calibration only while the apparatus remains stable.



A: mixing-gap correction uses 2,001 times the observation time.
B: production records are matched; retiming adds 260,000 pilot records.
C: 15,000 records and 4,500 expected model-time exposure per dataset.
Error bars: pointwise 95% Monte Carlo intervals. Simulations, not hardware measurements.

Figure 28.1: Virtual-laboratory hardening under declared model and apparatus contracts. Dependence corrections reduce false exclusions while increasing sampling time. A common clock calibration repairs selected coherent-control errors with an explicitly charged pilot. The lower panel compares the revised fixed contrast and optimized moment-box procedure on identical 15,000-record datasets. Error bars are pointwise Monte Carlo intervals. These are simulated experiments; the panels have different access and cost contracts.

28.4 What is ready for physical transfer

The transferable output is a procedure: specify the physical record, check clock and exposure compatibility, justify conditional sampling envelopes, freeze a targeted contrast or retain an exploratory joint region, and report a certificate with its cost and failure controls. Exact simulator checks, retained synthetic records and independent implementations make that procedure inspectable by another researcher.

The next physical test should measure detector memory and exposure behavior, validate stationarity or a preparation budget, and establish a time and energy calibration before interpreting a positive bound. Matched comparisons should include existing thermodynamic estimators when their required records are actually available. The present endpoint-only experiments cannot establish superiority over methods supplied with richer transition or waiting-time data. Historical priority of the particular operational synthesis and physical confirmation remain open.

Chapter 29

What additional calibration can buy

Detecting a departure from equilibrium and measuring total dissipation are different tasks. The previous virtual laboratories made detection more reliable. The next question is which further observations make a physical requirement stronger, and which missing information remains inaccessible even after the statistics improve.

This development follows that question through a complete chain. A calibrated detector variance shrinks the family of reversible explanations. A nonlinear information projection measures separation from that smaller family, with independently checked upper and lower bounds on the numerical relaxation. A separately calibrated transition-activity cap strengthens the conversion from record information to required entropy production. Sequential learning and dependent-record methods address the cost of collecting the evidence. Explicit countermodels identify the conclusions these improvements cannot buy.

The record-information argument also transfers to a stated class of stationary overdamped diffusions. Its path-probability identity follows from established stochastic calculus, while the reversible comparison retains the same positive spectral structure. This extends the applicable dynamics beyond finite jump networks. The additional finite-jump activity bound has a narrower domain and is not transferred to a diffusion by renaming its parameters.

29.1 Three boundaries that should not be conflated

The first boundary is observational. Two physical mechanisms may produce identical accessible records. More samples cannot distinguish those exact alternatives. The second is statistical: different record laws may be too close at the available precision or acquisition budget. The third is a physical inverse problem: a sharp information certificate need not equal the smallest entropy production compatible with every dynamical constraint. Improving an optimization algorithm addresses none of these boundaries by itself.

The new calibration results supply a concrete example of the first boundary. Finite positive-lag binary pairs can agree exactly while the detector's conditional second moment differs. Genuine additional calibration can therefore change which explanation is admissible. A noisy analog record can sometimes provide a useful upper variance bound without being a perfect measurement of the hidden state.

The upper-dissipation obstruction is stronger. A connected four-state cycle can expose every jump time while concealing the directional marks carrying its entropy. Even complete microscopic records of bounded duration cannot uniformly furnish finite upper entropy bounds over models with arbitrarily rare reverse transitions. This is not a universal cap on scientific data. It is a

target-specific limitation with explicit access and calibration repairs.

29.2 From proofs to acquisition choices

Calibrated comparison methods, longer observation windows, more readouts and additional control information have different physical costs. The experiments therefore report time and readout counts separately. A hypothetical conversion between these resources is labeled as such; unmeasured hardware expenditure is not assigned zero cost. Failed regimes and stronger comparators remain in the evaluation.

The last step moves beyond simulated data. Publicly released optical-tweezer trajectories are loaded with verified provenance and analyzed using the same access questions. Their descriptive correlations and detector comparisons are genuine reanalyses of existing measurements. The available metadata do not establish all premises of a thermodynamic confidence certificate, so the analysis records that limitation instead of manufacturing a dissipation claim. The result is a precise account of what the next laboratory calibration must supply.

The broad contribution remains operational: connect accessible evidence to what every admissible physical explanation requires, expose the assumptions that make the connection valid, and construct the next measurement when those assumptions are insufficient. Established probability, spectral and thermodynamic ingredients are credited throughout. Historical priority of the specific synthesis and independent physical confirmation remain open.

29.3 Choosing the next observation

The most useful next step depends on the obstruction already established. The following sequence makes that diagnosis explicit.

1. Specify the target. Detecting departure from detailed balance, lower bounding entropy production, and upper bounding total entropy are distinct inverse problems. A method adequate for the first can fail at the other two.

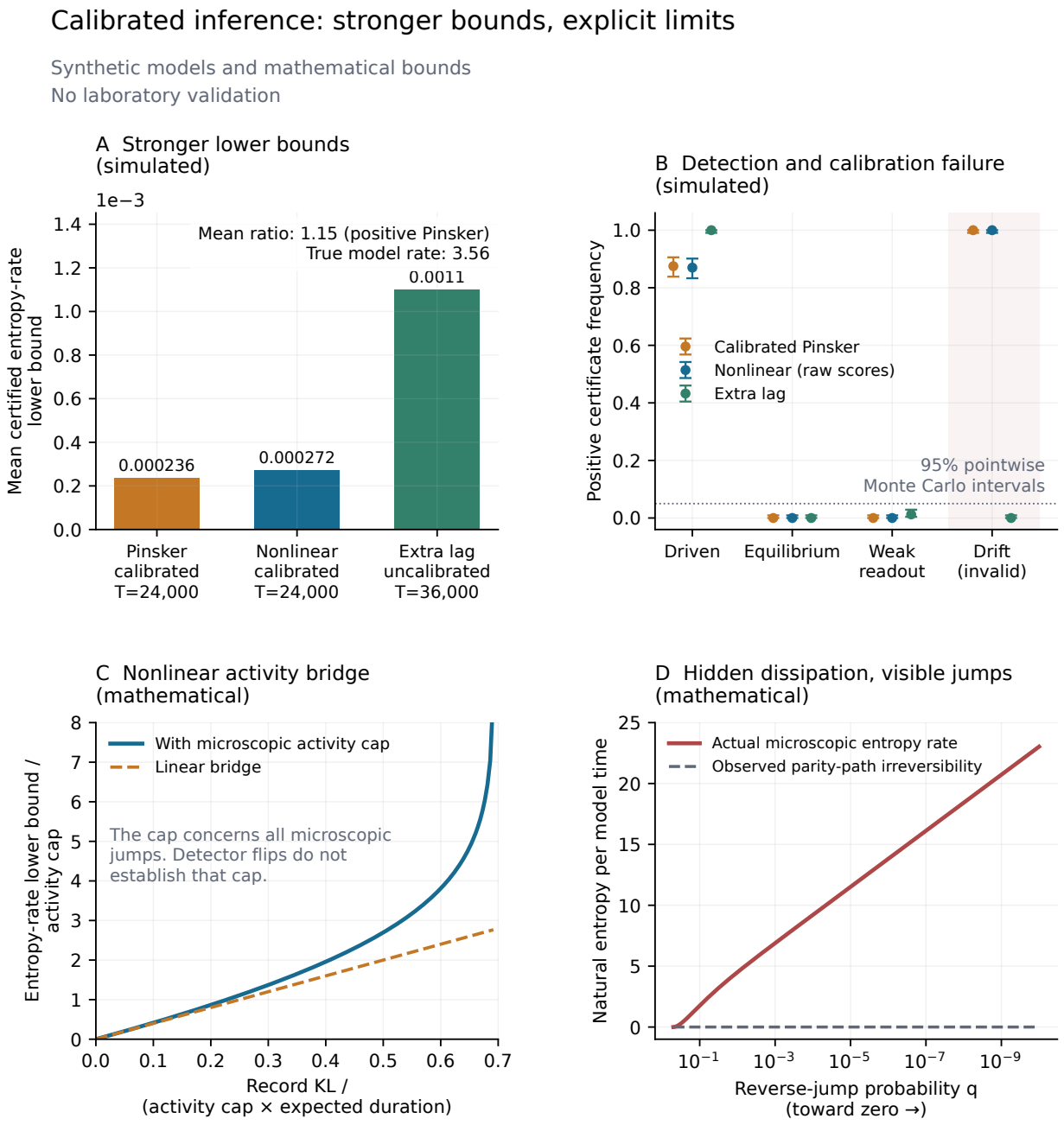
2. Validate the observation contract. Establish how the detector responds, whether readings disturb the state, how time is recorded, and which source of independence or dependence control justifies the statistical calculation. If those conditions are missing, collect calibration evidence before interpreting a positive contrast physically.

3. Test whether additional access removes an exact ambiguity. Same-state twins, bounded analog values, directional transition marks and controlled interventions reveal different information. More repetitions of an exactly uninformative record cannot substitute for the missing observation.

4. Use all the retained evidence under a valid confidence construction. A predeclared joint contrast can combine calibration and dynamics. A nonlinear projection can extract a stronger bound from a covered region. Predictable sequential tests can stop when sufficient evidence accumulates. Their conclusions remain limited by the physical comparison theorem.

5. Compare acquisition designs using the resources the apparatus actually consumes. Report evolution time and readout count before converting to energy. A microscopic activity cap can sharpen a bound on dissipation; it does not measure detector power, laser expenditure or initialization work.

This is also a way to decide when a branch is finished. An exact compatible countermodel settles an impossibility claim for its stated record class. A covered positive certificate settles a conditional lower-bound claim. Neither establishes the complete hidden dynamics. A physical application is complete only when the apparatus supports the premises and the result answers the stated target at the required precision.



A–B: two-stage calibration benchmark, 400 simulated datasets per case; 20,000 pairs and 40,000 reads per arm. Means include zero bounds; nonlinear scores are raw.
Counts include calibration. Evolution times differ; reset, detector latency, hardware and energy overhead remain unmeasured.
C–D: conditional mathematical statements. D fixes the four-state graph and unit activity; every jump is visible, yet the entire parity path law agrees.

Figure 29.1: Calibrated inference and its limitations. The panels distinguish an exact physical boundary, numerical information bounds and finite simulated comparisons. The underlying records, equations and uncertainty calculations are supplied in the following chapters and the release.

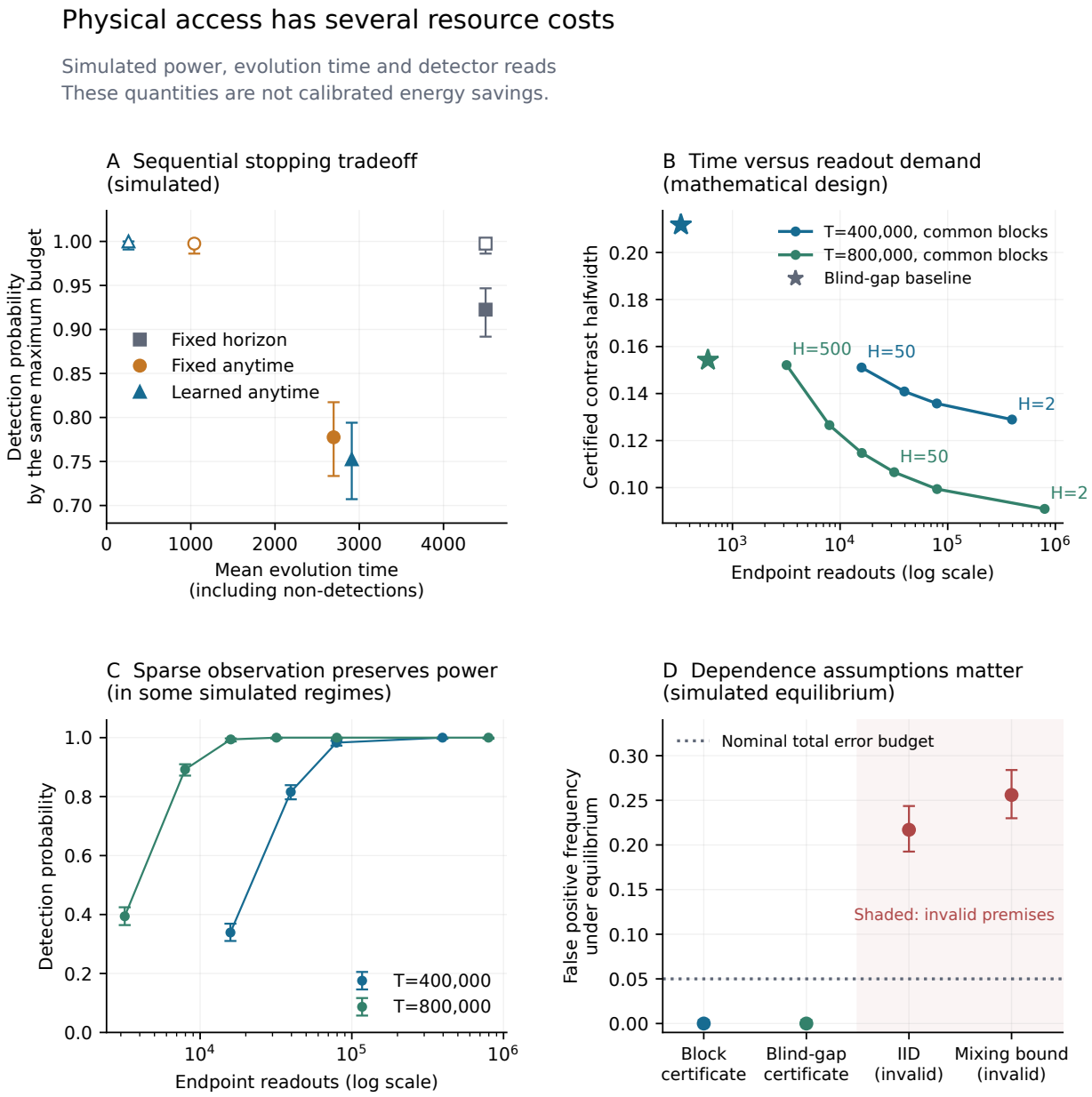


Figure 29.2: Acquisition choices at declared time and readout budgets. Each simulated comparison is restricted to its frozen design. Faster stopping or fewer readings does not imply a measured energy saving, and deliberately invalid calibration controls remain part of the evaluation.

Chapter 30

Detector calibration and the zero-lag access boundary

The size of an observed fluctuation depends on the detector as well as on the physical process. Calibrating that dependence can reveal irreversibility in a positive, monotonically decreasing correlation, before a negative correlation or revival appears. The same analysis also identifies a limit: arbitrarily precise measurements at a fixed positive-lag pair menu need not determine the calibration quantity itself.

30.1 The conditional second moment

Let a fixed, noninvasive binary detector have conditional mean $f_i = \mathbb{E}[Y \mid X = i] \in [-1, 1]$. Under the stationary law π, define

$$m = \sum_i \pi_i f_i, \qquad A = \sum_i \pi_i f_i^2, \qquad u = m^2, \qquad 0 \le u \le A \le 1. \tag{30.1}$$

Two readings with conditionally independent noise at the same unchanged hidden state obey

$$\mathbb{E}[Y^{(1)}Y^{(2)}] = A, \qquad \Pr(Y^{(1)} = Y^{(2)}) = \frac{1 + A}{2}. \tag{30.2}$$

Squaring one binary reading instead gives $Y^2 = 1$ and does not measure A. The two readings are generally dependent through their common hidden state.

For an instantaneous detector, m and A agree under Q and its additive reversibilization because π and f are unchanged. This transfer need not hold for nonlinear finite-exposure path functionals. A deterministic binary state detector already has $A = 1$. If $f(X)$ itself is reliably observed, its square directly measures A; this is additional access, outside the binary-only obstruction proved below.

30.2 Bounded analog observations supply an upper calibration

A trusted instantaneous analog record can improve the comparison class even without a same-state replicate. Let $Z \in [-1, 1]$ be the output of a fixed, noninvasive memoryless channel, and write $f_i = \mathbb{E}[Z \mid X = i]$. Then

$$m = \mathbb{E}Z, \qquad A = \mathbb{E}f(X)^2 \le B := \mathbb{E}Z^2, \qquad B - A = \mathbb{E}\,\mathrm{Var}(Z \mid X). \tag{30.3}$$

At distinct times, conditionally independent readout noise gives $\mathbb{E}[Z_0 Z_t] = \langle f, e^{Qt} f \rangle_\pi$. Thus a valid upper confidence bound on B is also an upper bound on A. A lower bound on B does not transfer. Both m and B are unchanged under the additive reversible reference because its stationary law and instantaneous detector channel are the same.

For example, let $Z = X/4 + \epsilon/2$, with independent fair signs X, ϵ. Then $A = 1/16$ while $B = 5/16$. The observable second moment gives a useful upper bound, but it is not the latent conditional second moment. Randomizing two binary encodings from the same noisy Z gives product mean B, not A, because they share the analog measurement noise. Independently encoding readings at distinct times is compatible with the correlation model.

The API accepts any valid simultaneous bounds on m and B and sets $A \in [0, B_U]$. Its optional iid summary helper uses a two-sided Hoeffding bound for the mean and a one-sided bound for the squared observation. A dependent trajectory requires a separate valid calibration procedure; the iid helper cannot be applied to every raw sample merely because the detector noise is memoryless. Neither this upper-bound transfer nor the earlier exact transfer is automatic for nonlinear finite-exposure functionals.

30.3 A sharp calibrated reversible class

For commensurate positive lags $k\Delta$, define

$$\mathcal{M}_{m,A} = \left\{ q_k = \frac{1 + u + (A - u) \int_0^1 x^k \, \mu(dx)}{2} : \mu \text{ a probability measure} \right\}. \tag{30.4}$$

Theorem 30.1 (Sharp calibrated comparison class)**.** *For a finite menu of positive integer lag indices, the closure of the agreement vectors of finite stationary reversible chains with detector mean m and conditional second moment A is exactly $\mathcal{M}_{m,A}$. The closure is unchanged if the approximating chains must be irreducible.*

Proof. Necessity follows from the established reversible spectral representation: $C(t) = m^2 + \sum_\ell b_\ell e^{-\lambda_\ell t}$, with $b_\ell \geq 0$ and $\sum_\ell b_\ell = A - m^2$ [118]. Additional conserved modes can be included at $\lambda = 0$.

For sufficiency, a finite moment vector has an atomic representing measure with at most $K + 1$ atoms, where K is the number of measured coordinates. If $A = m^2$, use a constant detector. Otherwise put

$$a = \frac{m - A}{1 - m}, \qquad r = \frac{m - a}{1 - a}.$$

Then $-1 \leq a < m < 1$ and $0 < r < 1$. A two-state block with stationary law $(r, 1 - r)$, detector values $(1, a)$, and generator

$$Q_x = \frac{-\log x}{\Delta} (\mathbf{1}\pi - I)$$

has mean m, second moment A, and correlation $m^2 + (A - m^2)x^k$. Mix such blocks with the desired atomic weights. Replace any atom $x = 0$ by positive atoms approaching zero. Finally add $\varepsilon(\mathbf{1}\pi - I)$ to the full generator. This preserves π, m, A, makes the model irreducible and converges to the required moments as $\varepsilon \downarrow 0$. □

The sharpness statement concerns deterministic commensurate lags. A compatible random-clock family preserves this calibrated set as an outer comparison class, but its particular clock law can impose further restrictions.

30.4 Calibration uncertainty and exact support

A simultaneous interval for m maps to an interval $[u_L, u_U]$ for m^2. Its lower endpoint is zero if the mean interval crosses zero; otherwise use the smaller squared endpoint. Together with $A \in [A_L, A_U]$, this produces the exact polygon

$$\mathcal{P} = ([u_L, u_U] \times [A_L, A_U]) \cap \{0 \le u \le A \le 1\}.$$

For a contrast $c = (c_k)$, write $S_c = \sum_k c_k$ and $p_c(x) = \sum_k c_k x^k$. Its reversible support is

$$\sup_{q \in \mathcal{M}_{\mathcal{P}}} \sum_k c_k q_k = \frac{1}{2} \max_{(u,A) \in \operatorname{vert} \mathcal{P}} \left[(1+u) S_c + (A-u) \max_{0 \le x \le 1} p_c(x) \right]. \tag{30.5}$$

The full class is the convex hull of the vertex curves. A mixture has effective parameters in $\mathcal{P}$, and reweighting its atomic measures by $A - u$ gives another admissible measure. The implementation uses exact rational Bernstein bounds for the continuous polynomial maximum; a numerical grid is only a lower comparison.

For fixed m, these closed positive-lag classes are nested in A: extra variance can be placed at $x = 0$. Thus an upper calibration of A sharpens the comparison set. A lower calibration alone does not force visible variance when arbitrarily fast hidden modes are allowed.

For N independent same-state pair records, the count table gives

$$\widehat{m} = \frac{n_{++} - n_{--}}{N}, \qquad \widehat{A} = \frac{n_{++} + n_{--} - n_{+-} - n_{-+}}{N}.$$

Both are averages of N variables in $[-1, 1]$. Hoeffding and a union bound give a simultaneous radius $\sqrt{2 \log(4/\alpha)/N}$ [17]. This does not treat the $2N$ readings as independent. Calibration error probability must be combined with the subsequent inference guarantee. An empty physical calibration region is reported explicitly rather than converted into an entropy estimate.

30.5 Detecting excessively steep positive decay

Since $x - x^2 \le 1/4$, every calibrated reversible model satisfies the sharp bound

$$q(t) - q(2t) \le \frac{A - m^2}{8}. \tag{30.6}$$

A two-state block with $x = 1/2$ attains it. This is a restriction on the rate of change relative to the calibrated fluctuation amplitude, not merely on whether the correlation is positive or decreasing.

Consider a synthetic four-state cycle with clockwise rate three, reverse rate one and conditional detector means $(1/2, 1/2, -1/2, -1/2)$. Its stationary law is uniform, $m = 0$, $A = 1/4$, and

$$C(t) = \frac{1}{4} e^{-4t} \cos(2t).$$

At $t = 7/40$, exact series enclosures certify

$$q(t) - q(2t) > 0.03473386533798 > \frac{1}{32} = 0.03125.$$

Both correlations are positive and decrease. The uncalibrated support ceiling $1/8$ does not detect this example. Its true generator entropy rate is $2 \log 3$; that known model value is distinct from any lower bound inferred from its restricted records. This is a population certificate, not sampled experimental evidence.

30.6 Exact positive-lag ambiguity

Theorem 30.2 (Finite pair menus do not identify the missing spectral mass)**.** *For every integer $K \geq 1$, there are finite irreducible reversible models with zero detector mean, identical stationary binary-pair laws at each lag $1, \ldots, K$, and different values of A.*

Proof. Choose $x_j = j/(K+2)$ and $v_j = (-1)^j \binom{K+1}{j}$ for $j = 1, \ldots, K+1$. Finite differences give

$$\sum_{j=1}^{K+1} v_j x_j^k = 0 \quad (1 \leq k \leq K), \qquad \sum_{j=1}^{K+1} v_j = -1.$$

Split positive and negative coefficients and scale their magnitudes by $s = (2^{K+1} - 1)^{-2}$. The two positive spectral measures have equal measured moments and unequal total mass.

For either measure, use independent symmetric telegraph coordinates Z_j with correlation x_j^t, and detector mean $f(Z) = \sum_j \sqrt{s|v_j|}\, Z_j$. Its absolute value is at most one because the nonzero $|v_j|$ are integers, so $\sqrt{s} \sum_j \sqrt{|v_j|} \leq \sqrt{s} \sum_j |v_j| \leq 1$. The product chain is finite and irreducible, and its correlation is exactly the desired spectral sum. Its mean is zero. The binary pair law is $(1 + abC(t))/4$ for outputs $a, b \in \{-1, 1\}$, so the measured laws agree exactly while $A = C(0)$ differs. □

For a small explicit example, compare spectral weights $3/16$ at $x = 1/4$ and $1/16$ at $x = 3/4$ with weight $3/16$ at $x = 1/2$. Their second moments are $1/4$ and $3/16$, but both give $C(1) = 3/32$ and $C(2) = 3/64$. They have irreducible four-state and two-state realizations, respectively, with exit rates below one. Even a finite rate cap therefore does not imply exact identification from an unchanged finite pair menu.

The result does not equate their full time-resolved trajectories or all higher-order temporal records. Those observations, a known detector law or true same-state calibration supply additional access. Repeating the unchanged pair measurements cannot remove their exact ambiguity.

30.7 Fast modes and the quantitative kinetic repair

A second obstruction concerns uniform precision over unknown rates. Compare independent fair binary outputs with a reversible two-state model having correlation $Ae^{-\lambda t}$. Their pair total variation distance is $Ae^{-\lambda t}/2$. For N independent records at lags at least $t_{\min} > 0$,

$$\mathrm{TV}(P_0^{(N)}, P_\lambda^{(N)}) \leq \frac{NA}{2} e^{-\lambda t_{\min}}. \tag{30.7}$$

An order-one difference in A can therefore be hidden at any finite precision by a sufficiently fast reversible mode. Every model in this comparison has zero stationary entropy production. For any finite independent clock menu with no atom at zero, dominated convergence gives the same conclusion using $\mathbb{E}e^{-\lambda T}$. Exact continuum data with unlimited precision are a different observation contract.

A calibrated kinetic bound permits a controlled approximation. For any stationary finite Markov chain, including a nonreversible one,

$$A - C(t) = \frac{1}{2}\mathbb{E}[(f(X_t) - f(X_0))^2]. \tag{30.8}$$

If every exit rate is at most Λ, then

$$0 \le A - C(t) \le 2(1 - e^{-\Lambda t}). \tag{30.9}$$

Indeed the integrand is at most four and vanishes when no jump occurs. A bound J on stationary jump activity similarly gives $A - C(t) \le 2Jt$. These bounds make delayed calibration uniformly accurate as the accessible delay approaches zero. They do not promise exact identification at fixed positive lags. An energy-dissipation bound alone cannot supply either kinetic bound: arbitrarily fast reversible chains dissipate no stationary heat.

Finally, the same-state noise premise is testable only with additional apparatus knowledge. Conditionally anti-correlated readouts can each have means $\pm 1/2$ while their product has mean zero, although $A = 1/4$. For example use conditional count proportions $(1/2, 1/4, 1/4, 0)$ and $(0, 1/4, 1/4, 1/2)$. An apparently feasible calibration can then be too small and falsely reject an equilibrium record. Marginal counts alone do not establish conditional independence or absence of measurement backaction.

The ingredients of this chapter—reversible spectral mixtures, truncated moments, replicate covariance and concentration—are established. The scoped advance is their integration into a sharp calibrated comparison class, an executable confidence contract and explicit physical countermodels marking the remaining access boundary. Historical originality and empirical validation remain separate questions.

Chapter 31

Quantitative inference with certified nonlinear bounds

A positive contrast answers whether the specified equilibrium comparison class can explain the observations. Its quadratic entropy bound can discard much of the available probability information. This chapter gives a nonlinear replacement and an independently checkable bracket on the resulting optimization problem. It also identifies a second source of looseness: even an exact observational KL projection need not determine physical dissipation. An additional calibration of microscopic jump activity can sharpen that separate physical step.

The variational representation of relative entropy, convex information projection and likelihood-score testing are established ingredients. The modern composite-null testing literature gives closely related log-optimal score constructions[129, 131, 130]. The contribution here is their explicitly certified use with a calibrated continuous moment class, a nonrectangular finite-data confidence region, and a separate physical-resource calculation. Simulations and numerical optimization do not establish the physical premises.

31.1 The comparison problem and its scope

Let $k_1, \dots, k_m$ be positive integer lag indices, $w_i > 0$ sum to one, and $t_i > 0$ bound the full expected path duration associated with each target record. Write p_i for the probability that the two binary readings agree. Suppose a valid joint confidence event places the target vector in

$$\mathcal{P} = \{p \in \textstyle\prod_i [\ell_i, h_i] : a \cdot p \geq b\}.$$

The contrast a must have been declared before its test records, or selected using independent pilot data. The halfspace can use the one-sided conditional Hoeffding bound proved in the preceding work; the default box is $[0,1]^m$. An empty region is rejected as an inconsistent input, not interpreted as a scientific certificate.

Let V be rational vertices of a calibrated polygon in $\{(u, A) : 0 \leq u \leq A \leq 1\}$. Here u bounds the square of the stationary detector mean and A bounds the stationary second moment of its conditional mean. Define the closed reversible comparison class

$$\mathcal{K}_V = \operatorname{conv}\left\{ q_i = \frac{1 + u + (A - u)x^{k_i}}{2} : (u, A) \in V,\ 0 \leq x \leq 1 \right\}.$$

The preceding calibration chapter explains the access needed to measure A. The convex-hull representation allows distinct spectral atoms and arbitrary calibration points inside the polygon.

Indeed, a convex combination has effective $u = \sum_j \theta_j u_j$, $A = \sum_j \theta_j A_j$, and spectral weights proportional to $\theta_j(A_j - u_j)$. Conversely, decomposing (u, A) into vertices gives such a convex mixture. The atom $x = 0$ is part of the closed comparison class and may represent an infinite-rate limit; it need not be an exact finite-rate physical realization.

The quantity to bracket is

$$B(\mathcal{P}, \mathcal{K}_V; w) = \inf_{p\in\mathcal{P}, q\in\mathcal{K}_V} \sum_i w_i \ \mathrm{kl}(p_i\|q_i), \tag{31.1}$$

where kl is Bernoulli relative entropy in nats. For a stationary ordinary-reversal CTMC, the earlier additive-reversibilization bridge gives

$$\sigma \geq \frac{4B}{\sum_i w_i t_i}.$$

This comparison class generally contains reference models that cannot be the additive reversibilization of the same hidden generator producing p. Consequently, B is an observational relaxation. A feasible upper bound on B is not an upper bound on physical entropy production, or proof that its lower bound is physically attainable. Odd-variable physical reversal, measurement disturbance and an uncalibrated initial transient remain outside this stationary interpretation.

31.2 An exact dual certificate and a primal bracket

Choose arbitrary positive rational scores h_{i0}, h_{i1}. Let

$$L(h) = \inf_{p\in\mathcal{P}} \sum_i w_i[(1-p_i)\log h_{i0} + p_i \log h_{i1}],$$
$$U(h) = \sup_{q\in\mathcal{K}_V} \sum_i w_i[(1-q_i)h_{i0} + q_i h_{i1}].$$

Then

$$B \geq \max\{0, L(h) - \log U(h)\}. \tag{31.2}$$

For each fixed pair (p, q), regard (i, y) as a joint record with probabilities $w_i \,\mathrm{Ber}(p_i)_y$ and $w_i \,\mathrm{Ber}(q_i)_y$. The relative-entropy variational inequality gives $D(P\|Q) \geq E_P \log h - \log E_Q h$. Taking the worst cases over the two convex sets proves (31.2).

Both optimization steps in a supplied certificate are checked on their whole domains. For a calibration vertex (u, A), the normalization is the maximum on $[0, 1]$ of the rational polynomial

$$\sum_i w_i \left[h_{i0} + (h_{i1} - h_{i0})\frac{1 + u + (A-u)x^{k_i}}{2}\right].$$

Rational Bernstein coefficients after interval subdivision give a rigorous upper bound. No sampling grid is used as a proof of this maximum. The logarithms are enclosed by the inherited rational atanh-series routine. Substituting their lower endpoints gives a linear lower objective on $\mathcal{P}$. Its minimum is an exact fractional-knapsack calculation: begin at a box minimizer, and if the contrast constraint fails, move coordinates in the direction increasing $a \cdot p$, ordered by objective cost per unit increase. These rational moves attain the minimum of the enclosed linear objective. The continuous normalization and outward logarithm enclosures then give a rational lower bound on B.

Conversely, supply rational $p \in \mathcal{P}$ and finitely many rational atoms (v_j, x_j, θ_j) with $v_j \in V$, $x_j \in [0, 1]$, $\theta_j \geq 0$ and $\sum_j \theta_j = 1$. They produce an exactly feasible $q \in \mathcal{K}_V$. Upper logarithm enclosures give

$$B \leq \sum_i w_i \,\mathrm{kl}(p_i\|q_i) \leq B_{\mathrm{upper}}.$$

Thus the reported gap $B_{\text{upper}} - B_{\text{lower}}$ measures the remaining uncertainty of the relaxation calculation. The numerical optimizer proposes scores and mixtures; the rational checks retain their validity even when it fails to converge or uses a coarse proposal grid.

For an interior minimizing pair (p^*, q^*), the dual has no gap. Set $h_{i1} = p_i^*/q_i^*$ and $h_{i0} = (1 - p_i^*)/(1 - q_i^*)$. First-order optimality in q implies $E_Q h \le 1$ for all comparison vectors, with equality at q^*. First-order optimality in p implies $E_P \log h \ge E_{P^*} \log h$ throughout $\mathcal{P}$. Substitution gives equality in (31.2). Boundary minimizers can require limiting scores; the implementation claims only the brackets it actually checks, not attainment at finite scores in every case.

For a fixed alternative, normalizing h by $U(h)$ gives a nonnegative score with expectation at most one throughout the composite equilibrium null. Its expected logarithm is the familiar information-projection testing objective. This connection does not make a score fitted to its own data an automatically valid significance test. Here data-dependent scores are safe because they are minimized over a previously valid confidence region that contains the entire true probability vector on the same event. Calibration failure probabilities must be included in that event separately.

31.3 A realizable calibration example

Consider a stationary twenty-state ring with conditional detector mean $f_j = \cos(2\pi j/20)$. Its uniform stationary law gives $m = 0$, $A = 1/2$. Put $r = \sqrt{497}/25$, $\omega = \arccos(19/\sqrt{497})$, $\gamma = -\log r$, and $\phi = 2\pi/20$. Choose clockwise and anticlockwise rates

$$a = \frac{1}{2}\left(\frac{\gamma}{1-\cos\phi} + \frac{\omega}{\sin\phi}\right), \qquad b = \frac{1}{2}\left(\frac{\gamma}{1-\cos\phi} - \frac{\omega}{\sin\phi}\right).$$

Both are positive: numerically $a \simeq 2.06125923$ and $b \simeq 0.27982488$. Independent binary readings with means f_j have agreement probability $p(t) = 1/2 + e^{-\gamma t}\cos(\omega t)/4$. The specified constants give $p(1) = 0.69$, $p(2) = 0.59$ exactly. The predeclared contrast $(38/25, -1)$ therefore takes the value $1147/2500 = 0.4588$. Its calibrated reversible support is $19/25 - 1/2 + (19/25)^2/4 = 0.4044$; allowing A to rise to one raises that support to 0.5488. Positive and decreasing correlations alone do not expose the discrepancy; their scale relative to calibrated detector variance does.

Using the idealized confidence halfspace with threshold 0.4588, equal mixture weights and durations $(1, 2)$, the saved deterministic calculation brackets the calibrated record-KL relaxation between 0.0010251000031 and 0.0010265638733. The uncalibrated bracket is $[0, 6.43 \times 10^{-14}]$. These are model calculations, not finite-record experimental confidence statements. The true model entropy rate is about 3.55736, whereas the certified lower rate is about 0.002734. The narrow optimization bracket therefore reveals a substantial evidential and physical-bridge limitation, not an unresolved numerical calculation.

31.4 The physical bridge has its own limit

For an unordered edge, put $a = \pi_i q_{ij}$, $b = \pi_j q_{ji}$, $c = a + b$, and $x = |a - b|/(a + b)$. Edges with zero traffic contribute zero. For finite entropy assume positive reverse flux whenever forward flux is positive. Define

$$j(x) = \frac{(1+x)\log(1+x) + (1-x)\log(1-x)}{2}, \qquad e(x) = x \log\frac{1+x}{1-x}.$$

The path-KL rate from Q to $S = (Q + Q^*)/2$ and the entropy-production rate are exactly

$$d = \sum_{\{i,j\}} c_{ij} j(x_{ij}), \qquad \sigma = \sum_{\{i,j\}} c_{ij} e(x_{ij}).$$

The escape rates of Q and S agree pointwise, so there is no omitted waiting-time term in d. Total stationary microscopic jump activity is $K_0 = \sum c_{ij}$.

As $x \to 1$, $j(x) \to \log 2$ while $e(x) \to \infty$. A fixed-activity driven ring therefore has bounded path-KL rate but unbounded entropy production. An arbitrarily accurate endpoint projection cannot eliminate this physical loss of information. The data do not determine a finite upper bound on total dissipation without further assumptions.

There is, however, a sharper lower bridge if a separate physical calibration gives $K_0 \le K$. Write $G = e \circ j^{-1}$ on $[0, \log 2)$. Then

$$\sigma \ge K\, G\left(\frac{B}{K\bar{t}}\right), \qquad \bar{t} = \sum_i w_i t_i, \quad 0 \le \frac{B}{K\bar{t}} < \log 2. \tag{31.3}$$

To prove it, $j'(x) = \operatorname{atanh} x$ and

$$\frac{e'(x)}{j'(x)} = 2 + \frac{2x}{(1-x^2)\operatorname{atanh} x}$$

increase strictly for $0 < x < 1$. The derivative of the second term has the sign of $(1+x^2)\operatorname{atanh} x - x$, which is positive because its value at zero is zero and its derivative is $2x \operatorname{atanh} x + 2x^2/(1-x^2) > 0$. Thus G is increasing and convex, with $G(0) = 0$ and $G'(0) = 4$. Jensen's inequality gives $\sigma \ge K_0 G(d/K_0)$. The perspective $K \mapsto KG(d/K)$ is nonincreasing: convexity gives $G(z) - zG'(z) \le 0$. Finally data processing gives $d \ge B/\bar{t}$. These three inequalities prove (31.3). It improves $\sigma \ge 4B/\bar{t}$, with strict improvement for nonzero evidence and finite activity cap except for numerical rounding.

The activity-calibrated envelope is sharp for full-path observations of a uniform ring with clockwise and anticlockwise rates $K(1+x)/2$ and $K(1-x)/2$: every edge has the same bias, the stationary law is uniform, total activity equals K, and Jensen is an equality. This is a sharpness statement about the full-path physical bridge. Restricted endpoint observations usually lose information, and the moment relaxation can lose more; no endpoint sharpness follows from the ring example.

After normalization by activity, d/K_0 is the Jensen–Shannon divergence between the stationary directed-flux distribution and its transpose, while σ/K_0 is half their Jeffreys divergence when Jeffreys is defined as the sum of the two directed KL divergences. Bounds between these divergences and general finite-support optimization methods are established [140, 141, 142]. The explicit envelope above is proved here for this inference contract; historical priority of the scalar divergence inequality is not claimed.

The implementation inverts j by rational bisection. If an upper logarithm enclosure proves $j(x_-) \le B/(K\bar{t})$, then $Ke(x_-)$ with a lower logarithm enclosure is a certified lower entropy bound. The previous linear bound is retained if it is numerically larger. If the supplied evidence exceeds $K\bar{t}\log 2$, the routine reports an incompatible joint contract rather than an entropy estimate. The microscopic activity cap cannot be inferred from a count of coarse detector changes: unobserved jumps still contribute activity. Its measurement cost, preparation cost and timing calibration belong in the experimental resource account.

31.5 A scoped transfer to overdamped diffusions

The observational certificate is not restricted to a finite state space. Consider the stationary diffusion

$$dX_t = b(X_t)\, dt + \sqrt{2D}\, dW_t$$

on a connected compact periodic domain, with smooth drift, constant symmetric positive-definite D, and smooth strictly positive stationary density π. All state coordinates have even physical time-reversal parity. Define

$$b_s = D\nabla \log \pi, \qquad v = b - b_s, \qquad \sigma = \int v^T D^{-1} v \, \pi \, dx.$$

This is the standard steady current expression for overdamped entropy production under the stated physical convention[146]. The reversible comparison diffusion has drift b_s and the same diffusion matrix and initial law π.

The drift likelihood formula[145] gives

$$D(\mathbb{P}_b^{[0,t]} \| \mathbb{P}_{b_s}^{[0,t]}) = \frac{1}{4} E_b \int_0^t v(X_s)^T D^{-1} v(X_s) \, ds = \frac{t\sigma}{4}. \tag{31.4}$$

The factor follows from the noise covariance $2D$: the control in the Brownian likelihood is $(\sqrt{2D})^{-1}v$, and relative entropy is one half its expected squared norm. On the compact periodic domain the drift difference is bounded, so Novikov's condition and finite energy hold. On a noncompact domain the conclusion instead requires nonexplosion, the appropriate boundary or decay conditions, a valid true likelihood martingale and finite expected energy. Novikov is a sufficient condition, not a claim automatically valid for every linear example and every horizon.

The reference generator is $L_s = \pi^{-1}\nabla\cdot(\pi D\nabla)$. Periodic integration by parts makes it self-adjoint and nonpositive in $L^2(\pi)$. For a fixed bounded detector conditional mean f, the spectral theorem therefore gives

$$E_s[f(X_0)f(X_t)] = m^2 + \int_{[0,\infty)} e^{-\lambda t} \, d\nu(\lambda), \quad \nu \geq 0, \quad \nu([0,\infty)) = A - m^2,$$

where $m = E_\pi f$ and $A = E_\pi f^2$. Independent noninvasive binary readout noise at the two endpoints has the same correlation. At times $k\Delta$, putting $x = e^{-\lambda\Delta}$ yields the calibrated moment comparison class already used above. Data processing of (31.4) proves the same record-KL lower certificate $\sigma \geq 4B/\bar{t}$.

This corollary composes established likelihood and spectral theory. It is not a new physical law. It supplies a legitimate mathematical route to suitable overdamped position measurements, including model classes used for optical traps; it does not certify that any particular experimental dataset satisfies the assumptions. Sharpness or finite realization within the CTMC comparison class does not establish sharpness in a diffusion-only class. The optional finite-jump activity envelope does not transfer to continuous diffusions.

For an independently computable check, take a two-dimensional rotational Ornstein–Uhlenbeck process

$$dX_t = (-I + \omega J)X_t \, dt + \sqrt{2} \, dW_t, \qquad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Its stationary covariance is I, the reversible drift is $-x$, and $v = \omega J x$. Hence $\sigma = 2\omega^2$ and the full-path relative entropy to the reversible reference is $t\omega^2/2$. This unbounded linear drift does not need a global Novikov assertion: smooth radial cutoffs of the rotational drift have bounded likelihood controls and preserve the Gaussian stationary density, since the rotation is orthogonal to radial gradients. Their expected control energies are uniformly bounded by the Gaussian second moment. The resulting likelihoods have uniformly bounded relative entropy and are uniformly integrable. Localization, nonexplosion and uniqueness of the linear diffusion identify the limiting true likelihood; its finite-energy Girsanov expression gives the identity above.

Choose the bounded detector mean $f(x) = \sin(x_1)$ and independent binary readout noise. The stationary mean is zero and $A = (1 - e^{-2})/2$. Since the two scalar Gaussian coordinates at times zero and t have correlation $\rho = e^{-t}\cos(\omega t)$,

$$C(t) = E[f(X_0)f(X_t)] = \tfrac{1}{2}\{e^{-(1-\rho)} - e^{-(1+\rho)}\} = e^{-1}\sinh\rho.$$

The first equality follows directly from the Gaussian characteristic function and $2\sin x \sin y = \cos(x - y) - \cos(x + y)$. For $\omega = 2$, $t = 1$, existing rational exponential and cosine enclosures certify $C(t) < 0$, hence a positive entropy lower certificate, while the same detector and score yield zero in the $\omega = 0$ equilibrium control. The model entropy rate is exactly eight and the full-path KL is exactly two. This is an exact Gaussian-model calculation with certified transcendental enclosures, not a new trajectory experiment or an empirical measurement.

31.6 Implementation and checks

The API `constructive_access.calibrated.nonlinear.design` returns the exact dual lower certificate, a feasible primal upper bracket, and their gap. The faster `certify` accepts supplied rational scores and may reuse them across independently constructed confidence regions. `confidence_region` constructs the fixed-contrast halfspace and `activity_bridge` supplies the optional activity-calibrated physical step. All four routines distinguish an assumed observation contract from evidence that a physical apparatus satisfies it.

The tests compare the rational linear-program solver with an independent linear optimizer, recover an analytic Bernoulli KL projection, exercise an equilibrium control, check off-grid normalization and reject malformed primal points. A positive, decreasing two-lag signal is accepted by the uncalibrated class but separated after second-moment calibration. Independent scalar root finding checks the activity inversion, including the equal-bias ring equality case. Deterministic example outputs are mathematical checks, not measured laboratory data. Matched held-out performance comparisons are reported in the integrated benchmark chapter.

Chapter 32

What restricted access cannot upper-bound

Sharper lower certificates do not automatically make total dissipation identifiable. This chapter separates three obstructions: an exactly hidden current, an identifiable parameter with a singular thermodynamic dependence, and incomplete physical channel resolution. Its main construction uses one connected four-state cycle. Every microscopic jump changes the detector output. There is no independent hidden subsystem, no parallel microscopic edge, and no unobserved transition time.

Strong lumpability and loss of entropy production under coarse-graining are established results [68, 134]. Optimizing over compatible hidden mechanisms is also an established approach to sharp lower bounds [132, 133]. The contribution developed here is the explicit obstruction-and-repair chain: complete observed path equivalence at fixed dimension and activity, a quantitative finite-exposure impossibility that survives full microscopic observation, and calibrated routes to usable upper intervals. These proofs do not establish historical priority or a new physical law.

32.1 Four states, visible jumps, unbounded hidden dissipation

Number the states $0, 1, 2, 3$ around a simple cycle. Let $r_i > 0$ be known exit rates and $0 < q \le 1/2$ an unknown reverse probability. Set

$$Q_{i,i+1} = r_i(1-q), \qquad Q_{i,i-1} = r_i q, \qquad Q_{ii} = -r_i, \tag{32.1}$$

with indices modulo four. The stationary law, mean jump activity, and total stationary entropy-production rate are

$$\pi_i = \frac{r_i^{-1}}{\sum_j r_j^{-1}}, \qquad K = \frac{4}{\sum_j r_j^{-1}}, \tag{32.2}$$

$$\sigma(q) = K(1-2q)\log\frac{1-q}{q}. \tag{32.3}$$

Indeed the stationary clockwise and counterclockwise fluxes across each undirected edge are respectively $(1-q)/\sum_j r_j^{-1}$ and $q/\sum_j r_j^{-1}$. Summing their current times flux affinity gives (32.3). The chain is irreducible for every allowed q, and all four states participate in the observed dynamics.

Let the detector report only the parity $Z = i \bmod 2$. Every microscopic jump flips Z. If all $r_i = s$, the full observed process is exactly the symmetric two-state chain with jump rate s, independently of q. Its waiting times are independent exponentials of rate s, including the entire

timed jump record. At stationarity its binary autocorrelation is e^{-2st} and its zero-lag amplitude is exactly one. Thus even perfect calibration of that amplitude, activity, state count, and graph does not distinguish the models.

Meanwhile $\sigma(1/2) = 0$, and $\sigma(q) \to \infty$ as $q \downarrow 0$. Every intermediate nonnegative entropy rate occurs. The obstruction persists under arbitrary feedback that multiplies both outgoing rates by a bounded factor chosen from the observed parity history: the observed jump intensity remains s times that factor and still contains no q.

This is an ordinary lumping obstruction, made explicit in a case where all jumps are visible. It does not say that the microscopic rates are all known: the exit rates are known, while their division between clockwise and counterclockwise motion is not. A known exit-rate floor is not a known reverse-transition floor.

32.2 Breaking exact blindness by kinetic tags

Distinct exit rates can encode microscopic state information in the waiting times without changing the binary detector. Write $f_i(t) = r_i e^{-r_i t}$. At a typical stationary jump into even parity, the microscopic entry state is uniform on $\{0, 2\}$ because stationary jump fluxes are uniform. The next two holding times have joint density

$$g_q(u, v) = \frac{(f_0 + f_2)(u)(f_1 + f_3)(v)}{4} + \frac{1 - 2q}{4}(f_0 - f_2)(u)(f_1 - f_3)(v). \tag{32.4}$$

Consequently

$$\mathrm{TV}(g_q, g_{q'}) = |q - q'| \,\mathrm{TV}(\mathrm{Exp}\, r_0, \mathrm{Exp}\, r_2)\,\mathrm{TV}(\mathrm{Exp}\, r_1, \mathrm{Exp}\, r_3). \tag{32.5}$$

This follows directly by subtracting the densities and factoring the absolute integral. The bias is therefore identifiable within this calibrated four-state family whenever both parity classes have distinct exit rates.

The distribution in (32.4) is a Palm distribution: a typical event sampled from the stationary jump process. It need not be the distribution of the first even entry after an arbitrary clock-time origin. Successive pairs in one trajectory are dependent. Neither event selection nor independence is silently granted by this identity.

The repair has a quantitative weakness. For tags

$$(r_0, r_1, r_2, r_3) = (1 + \eta, 1 + 2\eta, 1 - \eta, 1 - 2\eta), \qquad 0 < \eta < 1/2,$$

both exponential-law distances are of order η, so pair-law separation is of order η^2. For the event $u \le 1, v \le 1$, the absolute derivative of its probability with respect to q is exactly

$$2e^{-2} \sinh \eta \sinh(2\eta) = 4e^{-2}\eta^2 + O(\eta^4). \tag{32.6}$$

An independently sampled binary-window experiment based on that difference would have a conservative Hoeffding record requirement of order η^{-4} at fixed parameter separation. Obtaining independent Palm pairs is an additional experimental requirement. No free reset or independent sampling assumption is included in this scaling statement.

More generally, the conditional parity jump intensity always lies between the two exit rates of the current parity class. For any two biases with the same r_i, the entire coarse record over total exposure at most L obeys

$$\mathrm{TV}(P_q^Z, P_{q'}^Z) \le \min\{1, L \max(|r_0 - r_2|, |r_1 - r_3|)\}. \tag{32.7}$$

Couple the two predictable jump intensities while their observed histories agree. Their difference is bounded by the displayed rate spread; the probability of any unmatched jump is at most its integrated bound. Initial parity laws agree because π does not depend on q. This shows explicitly how nearly lumpable connected models can remain close over a fixed experimental budget.

32.3 Full microscopic access does not remove a finite-budget singularity

Even after kinetic tags make the parameter identifiable, the entropy function is singular near a vanishing reverse rate. The following result concerns the *complete microscopic record*, so every coarser observation inherits it.

Theorem 32.1 (Finite-exposure dissipation ambiguity)**.** *Fix any four positive exit rates r_i, a total pathwise exposure budget $L > 0$, a tolerance $0 < \varepsilon < 1$, and any desired entropy gap $M > 0$. There are two strictly positive reverse probabilities q_0, q_1 in (32.1), with the same state count, graph, stationary law, exit rates, and activity, such that*

$$|\sigma(q_1) - \sigma(q_0)| > M, \qquad \mathrm{TV}(P_{q_0}, P_{q_1}) \le \varepsilon.$$

The record may contain all microscopic states, directions and jump times, including multiple independent resets with the common initial law and total exposure at most L.

Proof. Use one common uniformization clock of rate $R = \max_i r_i$. While the two microscopic paths agree, their holding and acceptance decisions can be identical. An accepted jump has different direction with probability $|q_0 - q_1|$ under a common uniform direction draw. Hence

$$\mathrm{TV}(P_q, P_{q'}) \le \min\{1, RL|q - q'|\}. \tag{32.8}$$

Take $q_0 = \min\{1/4, \varepsilon/(2RL)\}$ and then let $q_1 > 0$ approach zero. The record bound remains at most $\varepsilon/2$, whereas $\sigma(q_1) \to \infty$. All chosen finite models retain strictly positive reverse rates. Bounded controls that preserve this common direction parameter and a common total exit-rate cap admit the same coupling argument. □

A standard two-point testing argument now gives a minimax consequence. For any estimator and either member of a pair separated by M, at least one error probability for absolute error at least $M/2$ is at least $(1 - \varepsilon)/2$. Finite dimension, a known graph, bounded activity and a positive exit-rate floor do not provide a uniform additive-error estimate of total dissipation.

The exact executable example fixes

$$(r_0, r_1, r_2, r_3) = (1.02, 1.04, 0.98, 0.96), \qquad L = 10.$$

Two finite models have a certified entropy-rate gap exceeding 209.8058 while their full-record total variation is at most 0.005. The units are natural entropy per declared model time. Large entropy differences arise from an uncalibrated large affinity, not from a larger state space, faster activity, or an appended hidden subsystem.

32.4 Why a valid upper confidence bound must sometimes be infinite

Theorem 32.2 (Unavoidable unbounded upper intervals)**.** *Suppose U is an extended-real upper confidence bound computed from an allowed record with exposure at most L, and $\Pr_q[U \geq \sigma(q)] \geq 1-\alpha$ for every $0 < q \leq 1/2$. Then, for each such finite model,*

$$\Pr_q[U = \infty] \geq \max\{0, 1 - \alpha - RLq\}. \tag{32.9}$$

Proof. Introduce only as a mathematical limit the clockwise process at $q = 0$. Equation (32.8) gives $\mathrm{TV}(P_q, P_0) \leq RLq$. Along $q_j \downarrow 0$, coverage implies $\Pr_0[U \geq \sigma(q_j)] \geq 1 - \alpha - RLq_j$. Since $\sigma(q_j) \uparrow \infty$, continuity from above gives $\Pr_0[U = \infty] \geq 1 - \alpha$. Transfer this event back to a finite q with the same total-variation bound. □

For the finite reference model in the example, every uniformly valid 95% upper-bound procedure must return infinity with probability at least 94.5%. This does *not* mean that every data-dependent upper bound must always be infinite. Enough resolved reverse events can exclude very small q and produce a finite upper interval. The obstruction is to uniformly finite guarantees without further restrictions on the admissible reverse rates.

32.5 Small access repairs and calibrated physical caps

One additional binary tag identifying a single physical edge already repairs structural identifiability within this ring. The existing parity before and after that tagged edge determines its direction. Its stationary fluxes F_+, F_- give

$$\sigma = 4(F_+ - F_-)\log(F_+/F_-). \tag{32.10}$$

Every edge has the same stationary cycle current and flux ratio. This is an instance of established unicyclic inference from a resolved edge, closely related to visible-transition methods [143]. Finite-time edge events remain dependent, and (32.10) does not itself supply an independent binomial sampling law.

A calibrated affinity cap provides a different repair. If every stationary undirected edge flux pair F_e, B_e obeys $|\log(F_e/B_e)| \leq A$, then

$$\sigma \leq KA\tanh(A/2). \tag{32.11}$$

Indeed its contribution is $(F_e + B_e)a_e \tanh(a_e/2)$, where $a_e = \log(F_e/B_e)$, and this even function increases with $|a_e|$. Sum over edges. The four-cycle attains equality when all edge flux affinities have magnitude A, so this bound is sharp under the stated information.

A stationary *flux* affinity includes the stationary population ratio; it is not automatically a calibrated bound on $\log(Q_{ij}/Q_{ji})$. If instead the latter bare rate ratios have a known bound B, stationarity cancels the population-potential term and gives the safe, generally looser bound $\sigma \leq KB$. In particular, known bounds on *every individual forward and reverse rate*, $m \leq Q_{ij} \leq M$, imply $\sigma \leq K\log(M/m)$. Merely knowing the exit rates r_i does not supply such a reverse-rate floor.

32.6 A finite confidence interval with declared observation time

For an executable finite-count repair, suppose the direction of every jump is resolved. Collect a predetermined quota of n jumps. Their reverse indicators are independent Bernoulli(q) draws,

even with unequal exit rates. This statement must not be transferred to the number of directions observed by a fixed time: conditioning on that random jump count can bias directions because rates depend on the visited states.

Exact binomial inversion produces simultaneous bounds $q \in [q_-, q_+]$. Intersect these with any independently calibrated floor $q \geq \delta$ and with $q \leq 1/2$, giving $q_L = \max\{q_-, \delta\}$ and $q_U = \min\{q_+, 1/2\}$. An empty intersection flags incompatibility of the record and the declared parameter class. Otherwise, since (32.3) decreases on $(0, 1/2]$, the entropy interval is

$$[\sigma(q_U), \sigma(q_L)].$$

If the lower endpoint for q is zero, the upper entropy endpoint is infinity. With a positive floor it is finite for every dataset compatible with the model; without a floor, positive reverse evidence can still make it finite.

For the tagged rates above, $n = 1000$, ten reverse observations, and a calibrated floor $\delta = 0.001$, exact rational binomial and logarithm bounds give a 95% interval contained in

$$[3.83196,\ 5.27668]$$

natural entropy units per model time. The prior floor alone only supplies the upper bound 6.88605. These counts are an explicitly synthetic worked input. With the same positive reverse evidence the interval is also finite without a prior floor; with zero reverse counts and no floor it is unbounded above.

Observation duration is a separate resource. If $r_{\min} = \min_i r_i$, the n holding times are dominated by independent exponentials of rate $r_{\min}$, so $\mathbb{E}T_n \leq n/r_{\min}$. A timeout at $2n/r_{\min}$ has failure probability at most

$$(2/e)^n \leq (3/4)^n.$$

For this example the expected-time bound is 1041.67, the timeout is 2083.33, and the timeout bound is below 1.16×10^{-125}. A timeout returns the full prior interval. Selecting only completed runs and calling the result an unbiased conditional binomial interval is not justified. Preparation, reset, detector and control overheads remain additional unmeasured costs.

32.7 Scope and reproducibility

The exact generator identities, entropy intervals, finite-budget pairs, count inversion, and timing bounds are reproduced by `dissipation_limits.py`. Frozen-seed uniformization simulations check complete and parity record couplings. Their mismatch frequencies are properties of the chosen coupling, not estimates of total variation. Thirteen producer tests check exact stationarity, full parity-path likelihoods, independent waiting-density integrals, weak-tag scaling, finite-budget ambiguity, and resolved-count inference.

The thermodynamic interpretation concerns a specified finite Markov jump network with physically resolved reverse transitions and the appropriate local detailed-balance accounting. A compatible embedding takes state-energy differences proportional to $\log r_i$ and a uniform clockwise driving work proportional to $\log[(1-q)/q]$. If additional channels or unresolved physical states exist, they introduce further ambiguities already discussed elsewhere in the reference. The present obstruction occurs before those complications are added.

Chapter 33

Learning an experiment without discarding its past

A fixed contrast is valid when selected before its evaluation records. An independent pilot permits selection, but its measurements have a physical cost. There is a third option: use past records to choose the next contrast and bet, and charge every choice to a single sequential evidence process. This chapter applies established predictable betting and conditional Hoeffding concentration to the thermodynamic contrast [17, 20, 120]. It does not claim a new concentration inequality.

33.1 The acquisition contract

At batch j, select coefficients a_{jk}, a valid reversible support b_j, nonnegative bet λ_j, and positive full-path duration bounds t_{jk}, using only completed past batches. Collect one binary agreement record Y_{jk} at each selected setting. Given all earlier readouts, including those already collected in the same batch, its conditional mean lies within δ_{jk} of the corresponding target law p_{jk}. All target laws concern the same stationary physical mechanism and entropy rate σ. Each menu satisfies the earlier clock and detector contract.

This conditional-mean premise permits the exponential bound below; stationary marginals alone do not. Independent stationary preparations are one way to satisfy it. The coefficients, support, bet and radii are held fixed throughout the current batch. Physical calibration is independently justified before the run; uncertainty in that justification adds its failure allowance. Selecting favorable calibration after observing the current data is unsupported.

Write

$$K_j = \sum_k |a_{jk}|\sqrt{t_{jk}}, \qquad B_j = \sum_k |a_{jk}|\delta_{jk}.$$

The physical contrast bound gives $a_j \cdot p_j - b_j \le K_j\sqrt{\sigma/8}$. Define

$$S_n = \sum_{j\le n} \lambda_j(a_j \cdot Y_j - b_j - B_j) - \frac{1}{8}\sum_{j\le n} \lambda_j^2 \|a_j\|_2^2, \qquad H_n = \sum_{j\le n} \lambda_j K_j.$$

Theorem 33.1 (Anytime physical lower bound)**.** *Under this contract, for $0 < \alpha < 1$, with probability at least $1-\alpha$, simultaneously at every completed batch with $H_n > 0$,*

$$\sigma \ge L_n := 8\left(\frac{[S_n - \log(1/\alpha)]_+}{H_n}\right)^2.$$

When $H_n = 0$, set $L_n = 0$. Stopping at a data-dependent completed batch preserves the statement. The theorem does not make an arbitrary learned contrast or stopping rule valid outside the specified evidence process.

Proof. For a candidate upper entropy value s, consider $M_n(s) = \exp\{S_n - H_n\sqrt{s/8}\}$. Conditional Hoeffding bounds can be applied successively to the bounded binary readouts of a batch. Their variance proxies add because each centered increment obeys its own conditional moment-generating bound; unconditional independence is not invoked. The conditional bias costs at most B_j. If the physical entropy is at most s, the contrast bridge bounds the remaining conditional drift by $K_j\sqrt{s/8}$. Thus $M_n(s)$ is a nonnegative supermartingale starting at one. Apply Ville's inequality at the true value $s = \sigma$ and rearrange its simultaneous upper bound $M_n(\sigma) < 1/\alpha$. No union bound over candidate s is needed: coverage is proved at the single true physical value. If all bets vanish, the convention returns no positive information. □

The denominator is a mathematical comparison-weight quantity, not the total experimental expenditure. Actual observation time sums every acquired path; preparation, calibration and detector costs remain additional. The same applies when a stopping rule reduces the number of records acquired.

33.2 A replayable policy

For three lag indices, the implementation uses $a_s = (-s^2, 2s, -1)$ and $b_s = -(1-s)^2/2$. It estimates the useful location $s \simeq C_2/C_1$ from completed past counts, rounds to a fixed rational grid and clips to $[0, 1]$. A nonnegative, bounded plug-in bet uses those same past counts. The first batch has zero bet. This is one transparent policy, not an optimal adaptive controller. A fixed-s policy supplies a matched sequential comparison, and the earlier fixed-horizon contrast remains another baseline.

The exact replay checker recomputes every decision before reading the current batch. Exported lower bounds use outward rational logarithm and square-root enclosures. Replaying the code checks arithmetic and decision dependence; it does not verify the provenance or physical timing of an externally supplied record. Those belong to the acquisition contract.

33.3 A held-out test of adaptation and stopping

The frozen experiment uses 400 independent datasets in each of four specified 40-state ring scenarios. A batch acquires three independent stationary pairs, at times $0.15, 0.30, 0.45$. Each method receives the same bank, but sequential decisions can access only completed prefixes. The maximum is 5,000 batches, or 15,000 pairs and 4,500 units of evolution time. No separate pilot is used. Preparation and readout energy remain uncalibrated.

Scenario	Learned sequential	Fixed sequential	Fixed horizon
Original drive	301/400	311/400	369/400
Changed drive	400/400	399/400	399/400
Equilibrium	4/400	0/400	0/400
High readout noise	10/400	17/400	5/400

In the changed-drive model, the learned policy uses a mean of 869.175 pairs, including the full cost of any non-detection. The fixed sequential policy uses 3,471.3525 pairs and the fixed-horizon test 15,000. Their corresponding mean evolution times are 260.7525, 1,041.40575 and 4,500. The

pointwise 95% Monte Carlo interval for the learned detection probability is $[99.082\%, 100\%]$; 400 successes do not prove certain detection.

The fixed contrast is the previously selected $s = 2/3$, and the fixed sequential policy learns only its bet. This comparison demonstrates adaptation to the changed mechanism without receiving its parameters. It does not compare against every possible fixed contrast or an oracle supplied with the true mechanism. In the original drive regime the fixed-horizon method has higher power. The adaptive equilibrium false-exclusion estimate is 1%, with a pointwise 95% interval of approximately $[0.273\%, 2.540\%]$, consistent with the 5% guarantee. Strong readout noise remains a low-power regime.

The retained bank and exported prefix certificates bind every reported stopping decision to its preceding observations. Independent checks compare vectorized integer scores with exact rational replay. These numerical results establish neither universal policy optimality nor an empirical hardware advantage.

Chapter 34

Continuous records without repeated long reset gaps

A long blind interval before every measurement makes conditional means nearly stationary, but can waste most of the physical evolution time. We replace that acquisition rule by equally spaced blocks and use a calibrated mixing envelope to control their dependence. The improvement concerns declared evolution time; it does not make readout energy or mixing calibration free. The concentration ingredient is established Markov-chain theory, combined here with a whole-record martingale bound and the continuous reversible comparison class.

34.1 An apparatus contract for a dependent record

Let X_t be a finite stationary continuous-time Markov chain with law π and generator Q. Write $Q^\dagger$ for its adjoint in $L^2(\pi)$. Suppose an *externally justified* lower bound $\lambda > 0$ on the spectral gap of $-(Q + Q^\dagger)/2$ is available. No complete generator need be supplied. The bound is required for every model to which the confidence claim applies; fitting it to the same coarse record does not establish this contract.

Divide acquisition into deterministic blocks of duration H. The hidden block starts $X_i = X_{(i-1)H}$ have transition $P = e^{HQ}$. In block i, a fixed setting produces a statistic $Z_i \in [0, 1]$ from a noninvasive record contained in that block. Unobserved padding is physical evolution and counts in H. The record kernel depends on the current hidden start and its predetermined setting, with fresh conditional detector noise. A detector with memory must instead be included in the state for which the mixing guarantee holds.

For an agreement pair at lag $t_i \le H$, two fresh detector readings may occur at a shared boundary between blocks. Reusing the same noisy detector bit as both the previous terminal and next initial reading generally violates the contract. A deterministic state readout has no such extra random memory. Settings, quotas and contrast coefficients are fixed before the inference data, or using an independent pilot. The result below does not cover adaptive routing of settings based on earlier outcomes.

Theorem 34.1 (Whole-block concentration under calibrated contraction)**.** *Define*

$$\rho = e^{-\lambda H}, \qquad A_H = \frac{1+\rho}{1-\rho}, \qquad K_H = (\sqrt{A_H} + 1)^2. \tag{34.1}$$

For deterministic real weights c_i, let $\mu_i = \mathbb{E}_\pi Z_i$. Then

$$\mathbb{E}_\pi \exp\left(\theta \sum_i c_i(Z_i - \mu_i)\right) \le \exp\left(\frac{\theta^2 K_H}{8} \sum_i c_i^2\right). \tag{34.2}$$

Consequently, for $\alpha \in (0,1)$,

$$\Pr_\pi\left\{\sum_i c_i(Z_i - \mu_i) > \sqrt{\frac{K_H \log(1/\alpha)}{2} \sum_i c_i^2}\right\} \le \alpha. \tag{34.3}$$

If the initial hidden law has total variation distance at most β from π, the same tail event has probability at most $\alpha + \beta$.

Proof. For centered f, dissipativity gives

$$\frac{d}{dt}\|e^{tQ} f\|_\pi^2 = 2\langle e^{tQ} f, \tfrac{1}{2}(Q + Q^\dagger) e^{tQ} f\rangle_\pi \le -2\lambda \|e^{tQ} f\|_\pi^2.$$

Thus $\|P - \Pi\|_{L^2(\pi)} \le \rho$, also for nonreversible Q. Set $g_i(x) = \mathbb{E}[Z_i \mid X_i = x]$ and $D_i = Z_i - g_i(X_i)$. Relative to the filtration containing the past blocks and the current hidden start, D_i is conditionally centered with range length at most one. Conditional Hoeffding therefore bounds the moment generating function of $\sum c_i D_i$ by $\exp(\theta^2 \sum c_i^2/8)$.

The stationary absolute-contraction result of Fan, Jiang and Sun, Theorem 1 [128], applies to the deterministic, possibly time-dependent state functions $c_i g_i$. It bounds the moment generating function of $\sum c_i(g_i(X_i) - \pi g_i)$ by $\exp(\theta^2 A_H \sum c_i^2/8)$. The two sums need not be independent. Apply Hölder with conjugate exponents $p = 1 + A_H^{-1/2}$ and $q = 1 + A_H^{1/2}$. The resulting coefficient is $pA_H + q = (\sqrt{A_H} + 1)^2$, proving the moment bound. Chernoff optimization gives the one-sided tail.

Finally, the entire future observed path is a Markov kernel of the initial state. Total variation contraction transfers initial distance β to the full path law. It is paid once, not once per record. □

This is a conservative bound for a statistic of an entire block. In particular, $K_H \to 4$ as $\rho \to 0$; it does not recover the sharp independent-record constant. Shared hidden endpoints can create dependence even with fresh detector noise. Exact independent preparations retain their separate, stronger concentration guarantee.

34.2 From counts to physical inference

For m fixed settings with quotas n_k, probabilities p_k and coefficients a_k, apply the theorem with $c_i = a_k/n_k$ in setting k. If $\widehat{p}_k = S_k/n_k$, a valid stationary contrast lower bound is

$$L_a = \sum_k a_k \widehat{p}_k - \sqrt{\frac{K_H \log(1/\alpha)}{2} \sum_k \frac{a_k^2}{n_k}} - \sum_k |a_k| \delta_k, \tag{34.4}$$

where δ_k are separately valid whole-record calibration radii. Subtracting the continuous reversible support $h_{\mathcal{M}}(a)$ gives $g = [L_a - h_{\mathcal{M}}(a)]_+$. Under the stationary ordinary-time-reversal and physical entropy assumptions of the preceding construction, this yields

$$\sigma \ge \frac{8g^2}{(\sum_k |a_k| \sqrt{t_k})^2}. \tag{34.5}$$

Here t_k is the observed path duration; the sampling cost uses the complete block duration H, including padding. The mixing assumption licenses a confidence calculation. It does not itself establish the thermodynamic interpretation or local detailed balance.

Alternatively, a simultaneous probability box uses radii $\sqrt{K_H \log(2m/\alpha)/(2n_k)} + \delta_k$. A score or contrast may be selected after seeing that jointly valid box. The smaller fixed-contrast radius in (34.4) does not permit such reuse. The code evaluates exponentials, logarithms and square roots with outward rational bounds inherited from the certified arithmetic layer.

A single initial blind interval G suffices if a stationary mass lower bound $\pi_{\min} \geq p_* > 0$ is available. The contraction estimate implies

$$\sup_x \|\delta_x e^{GQ} - \pi\|_{\mathrm{TV}} \leq \min\left\{1, \frac{1}{2}\sqrt{\frac{1-p_*}{p_*}} e^{-\lambda G}\right\}. \tag{34.6}$$

This is an initial-law distance, not a record-by-record conditional bias and not an observed event that a reset succeeded. A separate, externally valid calibration failure allowance γ increases total failure to $\alpha + \beta + \gamma$. A postselected claim that calibration looked adequate cannot replace this requirement.

34.3 A frozen comparison at equal evolution time

The initial evaluation fixes $\lambda = 1/500$, $p_* = 1/8$, lags $1, 2$, $a = (-1, 1)$, and $H = 2$. New acquisition pays $G = 4000$ once, with $\alpha = 0.049$ and $\beta = 0.001$. Its total failure allowance is 0.05. The old conditional-mean method pays a blind gap before every pair. To avoid a weak baseline, its gap is optimized before any data over $\{0, 50, \ldots, 10000\}$, minimizing its certified contrast halfwidth at the same total evolution time. Both methods count padding and every blind gap.

The valid equilibrium model has four states (s, b). The sector flips at 10^{-3}; the observed bit flips at rates 0.02 and 2 in the two sectors. Its uniform stationary law is reversible and its additive gap is 0.002. The driven alternative is an eight-state ring with clockwise rate 4, counterclockwise rate 0.1, and a detector separating the two half-rings. Its true agreement contrast is 0.1469539766. Both start at a fixed hidden state before acquisition. A third equilibrium model has a sector rate 10^{-5}, violating the advertised gap by a factor of 100; it starts stationary to isolate the incorrect mixing calibration.

Evolution	Halfwidth		Endpoint readouts		Driven exclusions /1000	
budget	New	Old	New	Old	New	Old
200000	.18327	.29032	196000	176	0	84
400000	.12894	.21176	396000	332	1000	212
800000	.09094	.15422	796000	592	1000	465

The new method gains substantial power at the two larger budgets but does not dominate every operating point: at the smallest budget its certified threshold exceeds the true contrast and all runs fail to certify. On the valid equilibrium model, new false exclusions are 0/1000 at each budget; the old counts are $1, 0, 1$. Treating exactly the same new dependent counts as independent gives $271, 217, 131$ false exclusions. Correct stationary marginals alone therefore do not justify the independent analysis.

The wrong-envelope control produces $173, 256, 258$ false exclusions under the new calculation, and $97, 131, 132$ under the old one. Neither method repairs an invalid external mixing guarantee. These controls are outside the theorem, not failures of its stated guarantee.

Reported Monte Carlo intervals are approximate 95% Wilson intervals, separate from the rigorous statistical confidence guarantee. For example, 1000/1000 corresponds to $[0.99617, 1]$ and 0/1000 to $[0, 0.00383]$. The old driven powers at the larger budgets are 0.212 and 0.465, with intervals $[0.18779, 0.23841]$ and $[0.43428, 0.49599]$. These simulations are not physical experimental validation.

34.4 Trading readouts against waiting

The dense-block improvement requires many more readouts. A second evaluation, frozen separately after the first and using independent seeds, varies H while leaving the observed lags and confidence contract unchanged. For total evolution budget T, equal quotas are

$$n = \left\lfloor \frac{T - G}{2H} \right\rfloor, \qquad r_H = \sqrt{K_H \log(1/\alpha)/n}.$$

Increasing H reduces the retained data and weakens this bound, but can leave enough margin for a useful test. It also reduces the number of detector operations. All settings in the frozen menu are retained, including underpowered choices; no score is selected using its own inference data.

Evolution budget	H	Endpoint readouts	Halfwidth	Driven exclusions /1000
400000	10	79200	.13576	983
400000	20	39600	.14088	816
400000	50	15840	.15107	339
800000	20	79600	.09937	1000
800000	50	31840	.10655	1000
800000	100	15920	.11472	994
800000	200	7960	.12656	892
800000	500	3184	.15210	394

At budget 400000, $H = 10$ retains 98.3% observed power with five times fewer endpoint readouts than the original $H = 2$ acquisition. At budget 800000, $H = 100$ retains 99.4% with fifty times fewer readouts. Their approximate Wilson intervals are $[0.97294, 0.98936]$ and $[0.98697, 0.99725]$. Every valid equilibrium refinement row has zero false exclusions in 1000 trials. The wrong-mixing control still fails; for example, $H = 100$ at budget 800000 produces 198/1000 false exclusions. Sparse acquisition reduces detector operations without eliminating the external mixing assumption. Nor does it guarantee high power: the last row retains only 39.4% observed power.

For apparatus design, suppose a constant waiting power P_{wait} and energy per endpoint readout E_{read} have been measured. Relative to one lag-unit t_0, define $\zeta = E_{\mathrm{read}}/(P_{\mathrm{wait}} t_0)$. In units of $P_{\mathrm{wait}} t_0$, the accounting model is

$$C = T + \zeta R, \qquad n = \left\lfloor \frac{C - G}{2(H + 2\zeta)} \right\rfloor, \qquad R = 4n. \tag{34.7}$$

The code minimizes the outward-certified radius over the predeclared finite grid $H = 2, 3, \ldots, 1000$. Without integer quotas the objective is proportional to $(H + 2\zeta)K_H$. This supplies a reproducible tradeoff once costs are calibrated, not a globally optimal protocol or a measured energy advantage. Detector duration, preparation overhead and external calibration must also be included when they are material. The current simulation records those quantities as unmeasured rather than zero.

For example, at normalized budget 800000, the finite-grid choices for $\zeta = 0, 1, 10, 100$ are respectively $H = 2, 16, 74, 307$, with certified halfwidths $0.09094, 0.10401, 0.12489, 0.17558$. The $\zeta = 10$ plan uses 16936 endpoint readouts and evolution time 630632, consuming normalized cost 799992. These are prospective design calculations, not additional held-out power estimates at those settings.

34.5 Why some mixing cost cannot be removed

A simple experiment-independent obstruction complements the usable bound. Consider two fully observed stationary two-state chains. Both have rate $0 \to 1$ equal to κ; their rates $1 \to 0$ are 3κ and $\kappa/3$. Their stationary probabilities of state 1 are $1/4$ and $3/4$. Even a detector retaining every transition for duration T sees the identical constant-zero path under both models with common mass $\frac{1}{4}e^{-\kappa T}$, and the identical constant-one path with common mass $\frac{1}{4}e^{-3\kappa T}$. Hence equal-prior classification error obeys

$$\varepsilon^*(T) \geq \frac{1}{8}\left(e^{-\kappa T} + e^{-3\kappa T}\right). \tag{34.8}$$

An estimator of stationary occupation with error strictly below $1/4$ under both hypotheses would also classify them by thresholding at $1/2$. Thus uniformly small failure requires time of order $1/\kappa$ in this class, even with full continuous observation. This overlap argument does not prove the present bound optimal, but rules out a universal removal of slow-mixing cost by simply sampling one trajectory more frequently.

The contribution of this chapter is the integrated acquisition and inference contract: whole-record dependence, one initial calibration payment, exact continuous-class support, explicit physical time, and a readout-cost design frontier. Markov concentration, conditional Hoeffding and the overlap argument are established tools rather than new physical laws.

Chapter 35

Direct inference from bounded analog observations

An analog detector need not discard its amplitude information before the binary-observation certificates can be used. A conceptual randomization proves the physical comparison; the acquired analog values can estimate its probability directly. This removes avoidable encoding noise without changing the physical observation model or claiming that every statistical test becomes more powerful.

35.1 A virtual binary channel and a directly measured mean

Let $Z \in [-1, 1]$ be the output of the instantaneous, noninvasive memoryless detector of Chapter 30. Conditional endpoint noise is independent, and its conditional mean is $f_i = \mathbb{E}[Z \mid X = i]$. For one stationary pair record define

$$W_t = \frac{1 + Z_0 Z_t}{2}, \qquad p_t = \mathbb{E}W_t = \frac{1 + \langle f, e^{Qt} f \rangle_\pi}{2}. \tag{35.1}$$

The score W_t belongs to $[0, 1]$ but need not be an integer. Given the complete analog pair, introduce a virtual bit $B_t \sim \text{Bernoulli}(W_t)$. Equivalently, encode each endpoint z independently as a sign with mean z and retain their agreement. The bit has marginal success probability p_t.

Proposition 35.1 (The physical binary comparison needs no generated bits)**.** *Under the same stationary Markov, reversal and detector assumptions as the earlier entropy-production comparison, the Bernoulli relative entropy between the analog-score means is bounded by the relative entropy of the full physical path. Every previous lower bound depending only on these means remains valid when the means are inferred directly from W_t.*

Proof. Use the same virtual randomization kernel under Q and its additive reversible reference S. Data processing gives

$$d_{\mathrm{B}}(p_t^Q \| p_t^S) \le D(P^Q_{\text{analog}} \| P^S_{\text{analog}}) \le D(P^Q_{\text{path}} \| P^S_{\text{path}}).$$

The last inequality includes the common detector channel. No virtual bit must actually be drawn to estimate its marginal mean. The reference retains the same stationary law and detector, so the calibrated moment comparison from Chapter 30 applies. □

This is a conditional statement for the stated physical model, not a way to validate a detector or a reversal convention from analog data alone. Bounded analog readouts also supply $A \le \mathbb{E}Z^2$; equating these quantities would discard the conditional measurement-noise variance.

35.2 A certificate for rational sums and means

Fix lag indices k, coefficients a_k, quotas N_k, and full physical path durations t_k. Let b be a certified upper bound on the calibrated reversible support $\sup_q a \cdot q$. Suppose score k lies in a trusted interval of width $r_k \le 1$. Its conditional mean, given preceding records, differs from the fixed target p_k by at most δ_k. Let $\widehat{p}_k$ be the supplied empirical mean, with numerical error at most ϵ_k. Conditional Hoeffding concentration [17] yields, with failure probability at most α,

$$C = \left[\sum_k a_k \widehat{p}_k - b - \sqrt{\frac{\log(1/\alpha)}{2} \sum_k \frac{a_k^2 r_k^2}{N_k}} - \sum_k |a_k|(\delta_k + \epsilon_k) \right]_+ \le [a \cdot p - b]_+. \tag{35.2}$$

Add the failure probability of the calibration region by a union bound; its data need not be independent of the science sample for that union bound. The contrast and valid concentration contract must still be fixed beforehand or selected on an independent pilot. The inherited physical consequence is

$$\sigma \ge \frac{8C^2}{(\sum_k |a_k| \sqrt{t_k})^2}. \tag{35.3}$$

The full durations include every exposure interval needed by the comparison. Acquisition resets and detector operations require separate resource accounting.

The implementations `certify_means` and `certify_sums` accept exact rational inputs; $N_k \widehat{p}_k$ is not required to be an integer. Outward rational bounds cover the polynomial support, logarithms and square roots. The descriptive helper `quantized_pair_summary` first interprets stored numeric readings exactly, forms each score, and rounds it to the nearest multiple of $1/M$. Its mean error is at most $1/(2M)$, regardless of record count. This is an arithmetic guarantee about the stored data. An unknown sensor error is a different quantity and is not covered by that rounding radius. If both true and quantized endpoint values are in $[-1, 1]$, endpoint errors η_0, η_t instead imply score error at most $(\eta_0 + \eta_t)/2$.

35.3 What avoiding randomization improves

For a fixed signed contrast of a complete analog bank, let D use its direct scores and H use independently encoded bits. Conditional on the analog bank,

$$D = \mathbb{E}[H \mid \text{analog bank}].$$

Conditional Jensen therefore gives $\mathbb{E}\phi(D) \le \mathbb{E}\phi(H)$ for every convex ϕ for which the expectations exist. This is the established Rao–Blackwell conditional-expectation principle [144]. In particular it reduces variance and the corresponding exponential moments. With independent records,

$$\operatorname{Var}(D) = \sum_k \frac{a_k^2}{N_k} \{\mathbb{E} W_k^2 - p_k^2\}, \tag{35.4}$$

$$\operatorname{Var}(H) - \operatorname{Var}(D) = \sum_k \frac{a_k^2}{N_k} \mathbb{E}[W_k(1 - W_k)] \ge 0. \tag{35.5}$$

The conditional-expectation statement itself does not require independent physical records. A confidence statement still needs its declared dependence contract. A rejection indicator is not generally convex, so this argument does not establish universal power dominance. Nor does the universal $[0, 1]$ Hoeffding radius shrink merely because the actual variance decreases.

35.4 A frozen comparison at the same physical access

The benchmark fixes a twenty-state stationary cycle, two lags 0.1 and 0.2, 3,000 independently prepared pair records per lag, and 4,000 independent single-reading calibration records. The detector is a cosine rounded to a grid of denominator 4096. The driven rates are 20 and 1; the equilibrium control has both rates 10.5. A second driven control reports half the cosine plus independent fair noise of amplitude one half. These choices, seed, sample sizes and contrast $(1, -1)$ were frozen before generating 400 replicates per case.

Each comparison uses exactly the same analog bank, calibration region, $[0, 1]$ Hoeffding radius, and confidence allocation $0.025 + 0.025$. Only the extra independent encoding coins differ. The calibration uses the upper bound on $\mathbb{E}Z^2$, not access to latent detector means.

Case	Direct certificates (out of 400)	Encoded certificates (out of 400)	MSE ratio encoded/direct
Driven	380	311	6.04
Equilibrium	0	0	5.87
Noisy driven	0	0	7.06

For the driven case, the direct certificate fraction is 95.0% with nominal Wilson 95% interval $[92.4, 96.7]$%; the encoded fraction is 77.75% with interval $[73.4, 81.6]$%. The theoretical contrast-variance ratio is 5.45; the finite-replicate squared-error ratio is 6.04. The two quantities differ because the latter is estimated from a finite simulation ensemble.

The noisy driven case is a useful limitation. It has conditional second moment $A \simeq 0.1250$ but observable second moment $B \simeq 0.3750$. The observed contrast is about 0.0273, below the relaxed reversible support $B/8 \simeq 0.0469$ even before adding statistical error. Removing encoding noise does not repair this calibration obstruction. Neither zero observed rejection fraction establishes a zero population rejection probability.

Each replicate comprises 10,000 acquisition records of two different types: 6,000 endpoint pairs plus 4,000 calibration singles. It uses 16,000 detector readouts and 900 units of declared science evolution time. Preparation, detector and calibration overhead are unmeasured, not zero. Results retain all replicates, exact sufficient statistics for every certificate and the first raw bank from each case. Adding those audit fields after review did not change any data-generation or decision choices. These are synthetic demonstrations of statistical efficiency; no laboratory performance claim follows.

Chapter 36

A matched virtual laboratory for calibrated inference

This study asks two separate questions. Does genuine detector calibration expose a distinction hidden from a specified positive-lag menu? Once the same finite-data confidence region is available, does nonlinear inference give a stronger quantitative lower bound than the optimized quadratic calculation? A third-lag alternative tests whether calibration is the best use of the same number of additional pair records. Its result is retained even if it outperforms the calibrated arm. All evidence here comes from simulated trajectories and readouts, not a physical experiment.

36.1 Frozen model, records and comparisons

The model is the twenty-state ring from Chapter 31, with $p(1) = 0.69$, $p(2) = 0.59$, $p(3) = 0.485712$, mean zero and conditional detector second moment $A = 1/2$. A record starts at an independently sampled uniform state. Independent Poisson forward and backward counts give its exact continuous-time random-walk displacement at the declared duration. The two binary readings are then drawn independently from their state-dependent probabilities. Thus the simulator generates hidden paths and detector outputs; it does not sample agreement bits from the analytic endpoint formula. Floating-point rates and random-number generation define the numerical simulation. An independent matrix-exponential calculation checks its endpoint laws and supplies known-truth coverage diagnostics; it is not supplied to inference as a measured calibration.

The contract was frozen before production under seed 73024119. It specifies 400 independent datasets in each of four cases: the driven ring; a reversible ring with the same total jump activity; the driven ring with detector gain 0.2; and a reversible ring whose calibration readouts have gain 0.2 while its science readouts have gain one. The last case deliberately violates the fixed-detector contract. It tests whether an incorrect calibration premise can produce an apparently convincing certificate, not the advertised statistical coverage under valid premises.

Each calibrated dataset contains 8000 pairs at lag one, 8000 at lag two, and 4000 same-state calibration pairs. A calibration pair uses two genuinely independent noise draws conditioned on the same unchanged hidden state. The mean and second-moment intervals jointly fail with probability at most 0.025; the predeclared science contrast $(38/25, -1)$ has one-sided failure probability 0.025. Their union therefore gives a 0.05 total bound. The nonlinear certificate and the optimized Pinsker certificate use the same covered halfspace, calibration polygon, counts and path durations. Nonlinear mixture weights are fixed rational approximations to the quadratic method's optimal time weights. Its continuous-domain dual check and feasible primal bracket remain authoritative if numerical

score optimization is imperfect.

The alternative arm keeps the same first two settings and spends its 4000 additional pair records at lag three. It receives the same analytical prior model knowledge used to choose the calibrated contrast. Before any production sample, the fixed grid $s = j/100$, $j = 1, \dots, 99$ selected the contrast

$$a_s = (-s^2, 2s, -1)$$

maximizing its model-centered finite-data Pinsker lower rate at the actual quotas and durations. The support follows from $-x(x-s)^2 \le 0$, giving $-(1-s)^2/2$ for the uncalibrated class. The selected value is $s = 17/50$, with coefficients $(-289/2500, 17/25, -1)$. The inherited $s = 2/3$ comparison is also reported. These are separate predeclared tests at level 0.05, not an uncorrected data-selected union test.

36.2 Additional access and its cost

Both arms use 20000 endpoint pairs and 40000 readings. The calibrated arm's declared evolution time is 24000 model units; the extra-lag arm's is 36000. This matches pair/readout counts, not total experimental time. Ideal same-state calibration has zero intervening evolution in this simulator; its detector exposure, reset time, two-channel hardware and energy are not calibrated and cannot be treated as zero physical cost. Conversely, the extra-lag arm requires a longer evolution window. A hardware conclusion would require these overheads and access constraints to be measured.

The missing distinction in the original two-lag menu is exact, independent of Monte Carlo power. A stationary symmetric two-state chain with jump rate $-\log(9/19)/2$ and detector means $\pm\sqrt{361/450}$ has zero entropy production and agreement probabilities

$$q_k = \frac{1 + (361/450)(9/19)^k}{2}.$$

It gives $q_1 = 0.69$, $q_2 = 0.59$, exactly matching both joint binary-pair laws because their means are zero. Its detector second moment is $361/450$, which differs from the driven model's $1/2$. Genuine same-state calibration can exclude it. A third positive lag can also distinguish these examples, so this construction proves a limitation of the stated two-lag access, not of every uncalibrated experiment.

36.3 Held-out findings, including the failed comparison

The following counts are positive certificates among 400 independently generated datasets per case. "Nonlinear" means the raw frozen numerical proposal and exact dual checker, before any fallback to the quadratic result.

Case	Cal. Pinsker	Cal. nonlinear	Extra lag	Legacy extra lag
Driven	350	348	400	392
Equilibrium	0	0	0	0
Readout gain 0.2	0	0	5	5
Changed calibration channel	400	400	0	0

The last row is outside the fixed-detector contract: those calibrated outputs are apparent dissipation in a system whose true entropy rate is zero. They are not successful physical detections. The calibration region contains the science detector's true moments in none of those 400 datasets. In

the three valid cases it contains them in all 400, and the joint science and calibration event covers in 399/400 per case. Such Monte Carlo coverage is a check of these specified simulation laws, not validation of a laboratory's detector stability or the Markov assumption.

For the driven case, Pinsker's detection rate is 87.5% with pointwise 95% Clopper–Pearson interval $[83.85\%, 90.58\%]$; the raw nonlinear method gives 87.0% with interval $[83.30\%, 90.14\%]$. The designed extra-lag method detects in all 400 datasets, with interval $[99.08\%, 100\%]$, and the legacy version in 392/400. A zero count in 400 has upper interval endpoint 0.918%; it is not proof of zero false-positive probability. The comparison demonstrates that this two-stage calibration protocol is not the best of the tested uses of a fixed pair budget. The stronger extra-lag arm also consumes 50% more declared evolution time. Neither comparison determines total energy or hardware cost.

Despite losing two marginal positive decisions through numerical approximation, the raw nonlinear method raises the all-dataset mean entropy lower bound from 0.0002355009853 to 0.0002716097823, a 15.33% increase in the ratio of means. This is a same-record quantitative improvement, not improved detection power in the frozen implementation. All 350 driven numerical proposals report solver success, but success is not a proof of zero optimization error. Their certified relaxation-gap mean is 1.15×10^{-7}, with maximum 2.40×10^{-6}. When the conservative support check yields no positive gap, the implementation returns no certificate without solving the projection; such skipped cases are not reported as exact zero KL projections.

The true driven entropy rate is approximately 3.55736. The mean nonlinear bound certifies only 0.007635% of that value. The designed extra-lag mean bound is larger, 0.0010998649243, but still only about 0.03092% of the true rate. Thus the study improves quantitative inference while exposing a substantial remaining loss of information. It does not reconstruct hidden dissipation.

Because the calibrated quadratic and nonlinear bounds hold on the same covered event, their maximum is also a valid lower bound without another failure-probability allocation. A separately labeled audit supplement reports this practical fallback while preserving every frozen raw result. It replays the retained counts, stores each computed primal–dual gap, and uses a finer proposal grid only to diagnose the marginal numerical sign losses. These post-production diagnostics are not substituted into the frozen comparison.

36.4 Reproducibility and limits

The contract and seed are recorded in `contracts/calibrated_benchmark.json`. The count archive retains every dataset's science and calibration counts and every method's rate estimate. The first driven dataset additionally retains hidden endpoints, forward and backward Poisson counts, elapsed durations and both raw detector readings. The JSON report gives pointwise 95% Clopper–Pearson Monte Carlo intervals for detection and coverage, representative exact certificates, all optimized relaxation gaps, and the simulator's true entropy rate.

A confidence region that covers the observations under an assumed detector does not validate that detector contract. The gain-mismatch control makes this distinction explicit. Even a positive, accurately optimized nonlinear lower bound does not reconstruct the total dissipation of the driven ring. The quantitative report compares its magnitude with known simulated entropy and retains the gap. No result here establishes a new universal physical law, empirical heat dissipation, or a ranking across all sensing instruments.

Chapter 37

Calibrating and testing in one contrast

The same-state measurement supplies information about the detector, but it does not have to be converted into a separate calibration interval before testing dynamics. A predetermined linear contrast can combine its raw agreement count with the positive-lag counts under one confidence allowance. This avoids paying separate worst-case statistical margins for an intermediate quantity that the final decision does not need on its own. Its proof uses the same established reversible moment inequality and Hoeffding bound as the earlier construction; the change is how the available records are used.

37.1 The lag-zero quadratic inequality

Let a stationary hidden CTMC have a fixed noninvasive binary detector with conditional mean $f(x) \in [-1, 1]$. At each setting, the two detector draws have independent noise conditional on their hidden states. At lag zero they are fresh readings of *one unchanged hidden state*. They use the same detector channel as the positive-lag settings. Write

$$C_k = \langle f, e^{k\Delta Q} f \rangle_\pi, \qquad q_k = \frac{1 + C_k}{2}, \qquad C_0 = A = \langle f, f \rangle_\pi.$$

These definitions do not replace the twin reading by the square of one noisy bit, whose value would always be one.

Proposition 37.1 (Joint reversible support)**.** *For $0 \le s \le 1$, define*

$$a_s = (-s^2, 2s, -1), \qquad b_s = -\frac{(1-s)^2}{2}. \tag{37.1}$$

Every stationary reversible explanation with this access contract obeys $a_s \cdot (q_0, q_1, q_2) \le b_s$. The support b_s is exact even when the hidden dimension, rates, detector mean and A are unknown.

Proof. For reversible Q, $P = e^{\Delta Q}$ is self-adjoint in $L^2(\pi)$. Thus

$$-s^2 C_0 + 2sC_1 - C_2 = -\|(P - sI)f\|_\pi^2 \le 0.$$

The constant part of $a_s \cdot q$ is b_s. A fair state-independent detector has $f = 0$ and attains equality, so the support is exact. For $0 < s < 1$, a mean-zero reversible one-mode detector with decay s and nonzero A also attains equality. Equivalently, the reversible spectral measure satisfies $2sC_1 - C_2 - s^2 C_0 = -\int (x - s)^2 \, \mu(dx)$. □

This is a quadratic moment inequality, equivalently a 2×2 Hankel positivity condition. It is not a new law of reversibility or a new concentration theorem [118, 17]. With $C_0 > 0$, the population gap is largest at $s = C_1/C_0$ when that ratio is in $[0, 1]$, and a positive optimized gap means $C_1^2 > C_0 C_2$. For finite data, s must be fixed in advance or selected using independent pilot records; optimizing it on the inference data requires a different, simultaneously valid confidence procedure.

37.2 One confidence budget, with lag-zero uncertainty retained

Collect independent stationary agreement records with fixed counts n_0, n_1, n_2 and success fractions $\widehat{p}_k$. At level $1 - \alpha$, a valid lower bound on the population contrast is

$$L_s = -s^2 \widehat{p}_0 + 2s\widehat{p}_1 - \widehat{p}_2 - \sqrt{\frac{\log(1/\alpha)}{2}\left(\frac{s^4}{n_0} + \frac{4s^2}{n_1} + \frac{1}{n_2}\right)}. \tag{37.2}$$

This follows directly from Hoeffding for the weighted sum of all independent bounded observations. A declared probability error δ_k between the observed and ideal target law subtracts a further $s^2\delta_0 + 2s\delta_1 + \delta_2$. An externally valid calibration-contract failure probability γ adds to statistical failure, giving $\alpha + \gamma$. Setting $\gamma = 0$ in a mathematical model does not verify the corresponding hardware assumptions.

The statistical saving can be isolated without any physical approximation. Put $v_0 = s^4/n_0$ and $v_+ = 4s^2/n_1 + 1/n_2$. If two linear parts of the same contrast are bounded separately at failures α_0, α_+ with $\alpha_0 + \alpha_+ = \alpha$, their combined Hoeffding penalty is

$$\sqrt{v_0 \log(1/\alpha_0)/2} + \sqrt{v_+ \log(1/\alpha_+)/2} \;\geq\; \sqrt{(v_0 + v_+)\log(1/\alpha)/2}.$$

The right side is the joint penalty. This comparison concerns that fixed linear target; it does not order a richer calibration region and every possible subsequent test.

The zero-lag term remains in the variance. An exact counterexample shows why. Take a reversible mean-zero one-mode detector with $A = 1/2$, decay $x = 1/2$, and $s = 1/2$. Then $(p_0, p_1, p_2) = (3/4, 5/8, 9/16)$ saturates the support. If only one zero-lag pair is collected, its agreement is zero with probability $1/4$. Treating that empirical value as exact, even with perfectly known positive-lag means, creates a false gap $3/16$. The correct radius in (37.2) prevents that error. Zero lag is a statement about the ideal physical evolution interval, not zero measurement uncertainty or zero apparatus cost.

37.3 Why the physical comparison omits lag zero

Let p be generated by a stationary, possibly irreversible Q, and let q be generated by $S = (Q + Q^\dagger)/2$ using the same stationary law and detector. Then q is reversible and *exactly* $p_0 = q_0$. Hence

$$a_s \cdot p - b_s \leq a_s \cdot (p - q) = 2s(p_1 - q_1) - (p_2 - q_2). \tag{37.3}$$

Under the previously established ordinary-time-reversal stationary Markov entropy bridge, $D(P_Q^t \| P_S^t) \leq \sigma t/4$. Data processing and Pinsker imply $|p_k - q_k| \leq \sqrt{\sigma k \Delta/8}$. Thus a confidence-certified $g = [L_s - b_s]_+$ yields

$$\boxed{\sigma \geq \frac{8g^2}{(2s\sqrt{\Delta} + \sqrt{2\Delta})^2}.} \tag{37.4}$$

The cancellation uses a common ideal twin law. If the detector changes between settings, or finite exposure changes the state between the alleged twins, it requires an explicit error bound before the cancellation can be used. The zero-lag count still contributes two endpoint readings, a stationary preparation and its uncertainty. Odd-variable physical reversal, invasive readout and uncalibrated transients remain outside this interpretation.

The cancellation has a general form. Suppose a comparison construction gives $a \cdot q \le b$, and a set of measured coordinates I satisfies $p_j = q_j$ exactly for every allowed original/reference pair. If the remaining coordinates obey $|p_j - q_j| \le \sqrt{\sigma t_j/8}$ and their weighted duration denominator below is positive, then any joint confidence gap g implies

$$\sigma \ge \frac{8g^2}{(\sum_{j\notin I} |a_j|\sqrt{t_j})^2}.$$

All coordinates still enter the statistical variance and acquisition cost. This is a bookkeeping rule for physically invariant measurements, not a new divergence inequality. For example, a fixed instantaneous detector's mean and second moment are unchanged by additive reversibilization because its stationary law and detector channel are unchanged. Whether a particular apparatus measures such an invariant must be established independently.

For the fixed example $s = 19/25$, the coefficients are $(-361/625, 38/25, -1)$ and $b_s = -18/625$. At population probabilities $(3/4, 69/100, 59/100)$, the exact reversible gap is $34/625 = 0.0544$. Illustrative mean-centered counts $(3000, 5520, 4720)$ out of $(4000, 8000, 8000)$ at $\alpha = 0.05$ give a sampling radius 0.02729006, gap 0.02710994, and entropy-rate lower bound 0.00068291. These illustrative counts are deterministic arithmetic examples, not an additional simulation or physical measurement.

The extra measurement addresses a real ambiguity in this example. The positive-lag pair $(p_1, p_2) = (0.69, 0.59)$ also has an exactly reversible two-state explanation with $A = 361/450$ and decay $x = 9/19$. That explanation predicts $p_0 = 811/900$, whereas the driven example predicts $p_0 = 3/4$. The joint contrast uses this additional observable to exclude the hidden reversible explanation. It does not reconstruct all hidden currents or determine total dissipation.

37.4 A paired, independently seeded evaluation

The evaluation contract was frozen before new data generation. It uses 400 datasets for each of the four previous cases: the driven twenty-state ring, its reversible same-activity counterpart, a driven low-gain detector, and a deliberately invalid calibration in which the detector gain changes between the twin and positive-lag measurements. The new seed is independent of all earlier evaluations. The parameter $s = 0.76$ is inherited from the prior model and fixed before these records.

The joint method and actual two-stage implementations use the identical 4000 twin pairs and 8000 pairs at each of lags 1 and 2. The latter retain their original separate 0.025 calibration and 0.025 science allowances, calibrated polygon, Pinsker bound and certified nonlinear projection. The optimized and legacy extra-lag implementations use the same lag-1/2 data, replacing the twins by 4000 independent lag-3 pairs. The optimized extra-lag parameter remains the earlier fixed 0.34; neither method is redesigned using these held-out observations.

Each arm has 20000 pair records and 40000 detector readings. The joint and two-stage arms use 24000 declared evolution units; the extra-lag arm uses 36000. Generating all menus for the paired simulation requires 24000 distinct pairs and 48000 readings, with the shared records reused only for comparison of methods. Preparation energy, detector duration, finite exposure and hardware

verification of the twin contract remain unmeasured. Equal record counts do not imply equal total physical energy.

Full sufficient counts, an illustrative full record dataset, exact certificate examples and pointwise 95% Clopper–Pearson Monte Carlo intervals are retained. The actual earlier inference functions are called unchanged, and their specification is checked against its saved frozen contract before sampling. The result records hashes of both contracts and the count archives. The computation uses outward rational bounds for certificate arithmetic; floating-point simulation is separately identified.

Method	Driven detections /400	Mean entropy lower	Evolution time
Joint calibration	399	0.000800513	24000
Two-stage Pinsker	353	0.000244061	24000
Two-stage nonlinear	352	0.000281353	24000
Optimized extra lag	400	0.001122925	36000
Legacy extra lag	393	0.000302670	36000

Joint detection is 99.75%, with pointwise 95% Clopper–Pearson interval $[0.98615, 0.99994]$. The two-stage Pinsker and nonlinear rates are 88.25% and 88%, with intervals $[0.84683, 0.91238]$ and $[0.84406, 0.91018]$. The joint method adds 46 and 47 paired detections, respectively, with no paired losses on these driven datasets. Its mean lower bound is 3.28 times the two-stage Pinsker mean and 2.85 times the two-stage nonlinear mean. These are ratios of dataset means, not guaranteed improvement factors for an arbitrary model.

The optimized extra-lag method remains stronger on mean certified rate and detects all 400 datasets; its additional evolution time is explicitly reported. The true simulated entropy rate is about 3.55736, so even the joint lower-bound mean recovers only about 0.023% of it. Improving statistical use of the records does not remove the observational and physical bridge limitations on quantitative dissipation inference.

Every method produces zero exclusions among the 400 valid equilibrium datasets. With high readout noise, the joint, two-stage Pinsker, two-stage nonlinear, optimized extra-lag and legacy methods detect respectively $6, 0, 0, 8, 6$ of 400 driven datasets. This remains a low-power regime. Joint-contrast coverage is 397/400, 399/400 and 396/400 in the driven, equilibrium and high-noise cases; the nominal guarantee is at least 95%. These empirical coverage frequencies are simulation diagnostics, separate from the theorem.

In the deliberately invalid detector-change control, all three calibration-dependent methods falsely exclude equilibrium in 400/400 datasets. Both extra-lag methods, which do not use the changed calibration channel, give zero exclusions. The control is outside the stated twin contract and demonstrates that increased statistical precision cannot repair a systematically wrong calibration channel. It is not omitted from the reported comparison.

The methods are not universally ordered. A two-stage confidence region can support many later scientific questions and exploit a tightly calibrated nonzero detector mean; the predetermined scalar contrast addresses one specific question. Additional lags can expose failures that this quadratic cannot see. The joint method's intended advantage is avoiding unnecessary statistical separation when the final contrast is already fixed.

Chapter 38

An exploratory analysis of published optical-trap measurements

38.1 The experimental question

Published measurements allow a practical test of the distinction between observing correlations and having enough calibration to certify dissipation. We analyze three records from Bryan and colleagues' optical-trap study [138, 139]. Two records measure the same particle simultaneously through detectors with different noise levels. The third is a separate harmonic-well record. The source describes harmonic confinement without feedback, while feedback creates other potentials in the study.

The useful result is a detector-calibration diagnostic. The simultaneous records support a conditional decomposition of position and measurement variance, and show why the lower-noise channel cannot simply be treated as exact position. They also provide an empirical example in which small correlation-moment departures do not justify a thermodynamic certificate. No entropy-production confidence bound is issued for these data.

The repository is distributed under CC-BY-4.0 and credits J. S. Bryan IV, P. Basak, J. Bechhoefer and S. Pressé. We preserve the original files and verify their published checksums. The source filename `harmwell_n1_groundtruth.h5` denotes the simultaneous lower-noise measurement. It does not provide exact latent positions. Serialized force functions in that file are also distinct from measured force trajectories; the loader never executes them.

38.2 Fixed processing and its limits

Each file contains 49 999 positions. The authors' code documents a nominal sampling interval of $10\,\mu$s. Individual timestamps are absent. We use the first 10 000 points for centering, scale selection and a simple covariance fit, and the remaining 39 999 for evaluation. The fixed lag menu, in sample intervals, is

$$0, 1, 2, 4, 8, 16, 32, 64, 128, 256.$$

These choices preceded the new correlation calculations, but an earlier audit had inspected the sources and whole-file descriptive statistics. This is an exploratory reanalysis, not a prospectively blinded experiment.

For the simultaneous detectors, each training mean supplies its center and twice the square root of the average training variance supplies a common scale. For the separate record, we use its own

training mean and twice its training standard deviation. The bounded observation is

$$Z_i = \tanh\left(\frac{X_i - c_{\text{train}}}{s_{\text{train}}}\right).$$

Neither center nor scale is refitted on the evaluation segment. The common scale for the simultaneous detectors is 29.0285 nm; the scale for the separate record is 30.9707 nm.

Every reported covariance uses the mean of its complete evaluation segment and averages the $N - k$ available ordered pairs at lag k. We repeat the calculation in eight contiguous blocks, using each block's own mean and excluding pairs that cross a block boundary. These block ranges measure sensitivity to the selected segment. They are not confidence intervals, independent replications, or a calibration of hidden-state mixing. Mean estimation itself can bias finite-trajectory covariances.

The evaluation span is nominally 0.39998 s per record, with an additional 0.09999 s training span. The two simultaneous detectors share the same physical exposure. Adding their sample counts would not double that exposure. Overlapping lag pairs share observations and remain dependent; the separate record's alignment relative to the first experiment is unknown.

38.3 What two detectors can and cannot calibrate

Consider the explicitly conditional model

$$X_t = S_t + \epsilon_t, \qquad Y_t = S_t + \eta_t,$$

with unit gains and zero-mean errors independent of each other and of the particle. Under these assumptions,

$$\operatorname{Var}(S) = \operatorname{Cov}(X, Y), \tag{38.1}$$

$$\operatorname{Var}(\epsilon) = \operatorname{Var}(X) - \operatorname{Cov}(X, Y), \tag{38.2}$$

$$\operatorname{Var}(\eta) = \operatorname{Var}(Y) - \operatorname{Cov}(X, Y). \tag{38.3}$$

The evaluation data give the following plug-in decomposition. These values are conditional model diagnostics, not independently calibrated variances.

Quantity	Estimate (nm^2)
Variance of detector X	234.411
Variance of lower-noise detector Y	226.424
Same-time covariance	218.387
Candidate noise variance of X	16.024
Candidate noise variance of Y	8.037
Variance of $X - Y$	24.061

Treating Y as exact position would assign the entire difference variance 24.061 to the first detector. Under the stated independent-error model, that is about 50.2% larger than its candidate variance 16.024. The repository's distinction between a lower-noise reference and latent truth therefore has a measurable consequence for this analysis. Correlated errors, unequal gains or detector memory could change the interpretation; the decomposition does not establish their absence.

Adding temporal whiteness to the model predicts zero positive-lag covariance of $X - Y$ and equality of the two auto-covariances and two ordered cross-covariances. The table reports the difference-record autocorrelation and the difference of ordered cross-covariances at every declared positive lag. The latter is $\widehat{\operatorname{Cov}}(X_t, Y_{t+k}) - \widehat{\operatorname{Cov}}(Y_t, X_{t+k})$.

Lag (μs)	Difference correlation	Block range	Cross difference (nm^2)
10	0.0092	[−0.0151, 0.0259]	0.195
20	0.0214	[−0.0064, 0.0541]	0.096
40	0.0101	[−0.0162, 0.0288]	0.160
80	0.0150	[−0.0063, 0.0319]	-0.052
160	0.0190	[0.0030, 0.0364]	-0.168
320	0.0076	[−0.0147, 0.0270]	0.174
640	0.0010	[−0.0295, 0.0310]	0.146
1280	0.0127	[−0.0115, 0.0401]	-0.155
2560	-0.0009	[−0.0105, 0.0078]	0.410

The residual correlation at 160 μs is positive in all eight blocks, so exact white measurement error should remain an assumption to check. These descriptive signs do not constitute a statistical rejection: block dependence and selection across the lag menu are uncalibrated. The auto-covariance difference at the shortest lag is $3.881\,\mathrm{nm}^2$, with a block range $[-0.692, 7.712]$. Ordered cross-covariance differences also have block ranges spanning zero at each tested positive lag.

There is a separate limitation for bounded observations. Even conditional independence of the two channels would make their same-time product estimate $\mathbb{E}[f_X(S)f_Y(S)]$, where f_X, f_Y are their conditional means. Different noise levels generally make these functions different after the nonlinear transform. Consequently, this product is not automatically the replicate calibration $A = \mathbb{E}[f_X(S)^2]$ used elsewhere in this work. Matching detector responses is an additional requirement.

38.4 A relaxation model and reversible moment checks

A naive Ornstein–Uhlenbeck covariance model, $C(t) = a\exp(-\lambda t)$, was fitted only to the symmetrized training cross-covariances at lags $1, 2, 4, 8, 16$, using least squares on their logarithms. It gives $a = 198.967\,\mathrm{nm}^2$ and $\lambda = 1536.63\,\mathrm{s}^{-1}$, a relaxation time of 0.6508 ms. On evaluation data its shortest-lag prediction is $195.933\,\mathrm{nm}^2$, compared with $215.798\,\mathrm{nm}^2$ observed. The discrepancy is 9.10% of the evaluation zero-lag cross-covariance. This out-of-segment discrepancy matters if a fitted relaxation model is used as calibration. It can reflect finite-trajectory fluctuation, drift or model mismatch; these records alone do not distinguish them. In particular, the fitted relaxation rate is not a bound on all hidden modes.

For each bounded detector, let $g(t)$ denote its centered positive-lag covariance and v its observed variance. A stationary reversible Markov model with a memoryless detector gives a positive spectral mixture for g. The detector's conditional variance contribution is at most v. The tested necessary population inequalities are

$$g(t) \ge 0, \qquad g(t) - g(2t) \ge 0, \tag{38.4}$$

$$g(t) - g(2t) \le v/4, \qquad g(2t)^3 \le g(t)^2 g(4t). \tag{38.5}$$

The last inequality uses the weights 2/3 and 1/3 in Hölder's inequality, because $2t = (2/3)t + (1/3)4t$. It is not the equal-weight midpoint inequality. The pure-exponential and spectral-mixture tests check this convention explicitly.

All three full evaluation segments satisfy the first three checks at every applicable declared lag. The smallest upper-drop margins are 0.00466, 0.00119 and 0.00293 for the first detector, reference and separate record, respectively. Some empirical weighted log-convexity margins are negative, but every such negative full-segment margin has a block range spanning zero. Individual blocks also give negative values for some of the simpler checks. These small-sample sensitivity results provide

no certified violation of the reversible class. Passing a finite menu would not establish equilibrium or identify total dissipation either.

38.5 What would enable a physical certificate

The executable analysis returns `abstain` for thermodynamic inference. Converting this workflow into a valid lower bound requires a physical and statistical contract beyond a good fitted covariance:

1. Calibrate detector gain, temporal response and error uniformly over the working position range. For same-state replicate calibration, match conditional mean responses and justify conditional independence.

2. Supply actual timestamps or a verified timing-error bound, including channel synchronization and exposure averaging.

3. Establish stationary preparation and an independently justified dependence bound, or collect genuinely independent reset trials with their preparation cost included. Fitting the observed correlation does not establish hidden-state contraction.

4. Specify the physical state, time-reversal parity and relevant dynamical approximation. For a driven comparison, separate particle-bath entropy from the unrecorded energy of the detector, laser and controller.

5. Freeze the calibrated lag menu and statistical procedure before the next evaluation record. Then apply the dependence-aware finite-record confidence procedure to the bounded outputs.

These requirements specify measurements that would make the theoretical certificate usable. The present reanalysis already prevents two concrete errors: treating a measured reference as exact latent position, and treating small empirical moment departures as thermodynamic discoveries.

Chapter 39

From an observable correlation to necessary dissipation

A stationary scalar autocorrelation looks the same when time is reversed. It can nevertheless rule out every reversible hidden Markov explanation within a declared physical class. The reason is that reversible semigroups impose relations between correlations at different lags. This chapter develops a direct quantitative use of that constraint.

Let f be a bounded instantaneous observable of a stationary Markov process, $C(t) = \mathbb{E}[f(X_0)f(X_t)]$, and let R bound the range of f. For a chosen real s, define

$$\Delta_s = -s^2 C(0) + 2sC(t) - C(2t).$$

The following chapter proves that, under ordinary reversal of the full hidden state and the stated path-entropy contract,

$$\sigma \geq \frac{1}{t}\,\phi\!\left(\frac{4[\Delta_s]_+}{R^2}\right), \qquad \phi(u) = 2\sqrt{u}\,\operatorname{artanh}\sqrt{u} \geq 2u.$$

The finite expression applies for $0 \leq u < 1$. Its linear consequence is $\sigma \geq 8[\Delta_s]_+/(R^2 t)$. No hidden-state reconstruction is required for this implication. The assumptions still concern the hidden process: an even recorded observable does not establish the required reversal rule.

This result improves the conversion of a covered correlation contrast into a physical lower bound. It does not increase the statistical evidence in the record. Replaying the sixth edition's frozen driven example, with no change to its records or confidence event, raises the mean certified rate from 0.0008005133 to 0.11224493, a factor of 140.216. The actual simulated rate is 3.55736, so the remaining gap is substantial. Detection remains 399/400: the improvement is quantitative, not a larger number of discoveries.

The next chapters supply a usable acquisition scheme. Bounded analog readings provide both lagged products and a conservative zero-lag calibration from squared readings. One joint confidence contrast accounts for their sampling uncertainty. In a separate frozen simulated experiment at identical data and resources, this joint method produces 399/400 positive certificates versus 377/400 for the earlier split-calibration analog method. The experiment uses 16,000 scalar detector readings and 900 units of evolution time per replicate. Preparation and detector energy need additional calibration.

The noisy failure then motivates a new experiment. Exact calculation shows that the earlier two-lag family has no positive population witness, while a predeclared later lag does. On fresh noisy records, retiming restores 386/400 positive certificates versus 0/400 for both earlier fixed

contrasts. It uses the same 900 units of evolution time and 3,600 readings instead of 16,000. This changes experimental access; it is not a reanalysis of the same acquired records or a universal timing prescription.

There are three different optimization problems here:

1. Optimize a lower bound on a discrepancy between observed laws.
2. Minimize physical entropy over all hidden models compatible with those laws.
3. Minimize it within a specified physical architecture.

The first does not automatically solve the second. An exact finite-size circulant calculation solves a version of the third and supplies realizable comparison systems. A sampling theorem also shows why, even with perfect spatial resolution, a finite menu of time gaps cannot generally provide a finite upper bound on dissipation without further restrictions.

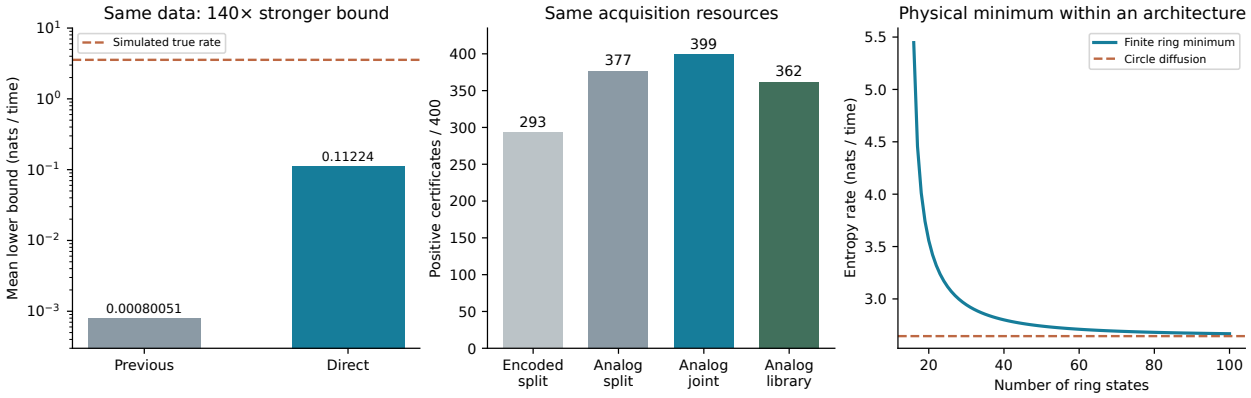


Figure 39.1: Distinct meanings of improvement. Left: the new physical bridge strengthens a lower bound using the same frozen observations. Middle: joint analog confidence increases detection at matched acquisition cost; selecting from a library pays an explicit statistical penalty. Right: physically realizable ring minima approach the known circle value inside a specified architecture. All numerical panels describe mathematical models and simulated records, not new laboratory measurements.

For practical use, start with the acquisition contract in the joint analog chapter and the executable `attainability_tutorial.py`. The proofs, negative controls, retained records and independent internal review then make each assumption and comparison inspectable. The direct scalar inequality is distinguished from the closest existing correlation-based inference results in the final positioning section. Historical priority and empirical validation remain open.

Chapter 40

Dissipation from a finite-lag correlation defect

A positive departure from a reversible correlation inequality contains more information than a generic distance-to-equilibrium argument retains. The operator identity below turns that departure directly into a lower bound on stationary entropy production. The improvement is linear in a small positive correlation defect; the previous comparison followed by Pinsker's inequality was quadratic in its confidence gap. Both statements remain valid under their respective contracts.

This is a theorem for a specified stochastic model class. It does not identify the hidden generator, establish the exact minimum dissipation over all compatible continuous-time models, or infer physical entropy from an arbitrary oscillatory record. The distinction between ordinary and generalized time reversal is essential.

40.1 Population theorem and its operational assumptions

Let X_t be a stationary, time-homogeneous Markov process with invariant law π and transition semigroup P_t. Initially consider a finite irreducible CTMC with positive stationary probabilities. Its physical time reversal must be the *ordinary reversal of the complete hidden state*, so that its reverse generator is the $L^2(\pi)$ adjoint $Q^\dagger$. Define

$$\sigma = \sum_{x \neq y} \pi_x Q_{xy} \log \frac{\pi_x Q_{xy}}{\pi_y Q_{yx}}.$$

The usual conventions give infinite entropy production for an unpaired positive transition. The finite-entropy case is the informative one below. Physical heat or power additionally requires a thermodynamic embedding; all rates here are dimensionless entropy per unit physical time.

Let f be a real bounded function with a valid range bound $\sup f - \inf f \leq R$, $R > 0$. This may be the conditional mean of a noninvasive memoryless detector. Detector noise must be conditionally independent between retained readings. Its range bound concerns the whole admissible channel, not the largest and smallest values in a finite sample. Set

$$C(t) = \langle f, P_t f \rangle_\pi, \qquad \Delta_s(t) = -s^2 C(0) + 2sC(t) - C(2t), \quad t > 0.$$

The population identity allows any real s. Selecting s from inference data requires a simultaneous confidence argument or an independent pilot.

Theorem 40.1 (Finite-lag dissipation bound)**.** *Put* $d = [\Delta_s(t)]_+$ *and* $u = 4d/R^2$*. Then* $0 \le u \le 1$*. If* σ *is finite,* $u < 1$ *whenever* $d > 0$*, and*

$$\boxed{\sigma \;\ge\; \frac{\Phi(u)}{t} \;\ge\; \frac{8[\Delta_s(t)]_+}{R^2 t}, \qquad \Phi(u) = 2\sqrt{u}\,\operatorname{atanh}\sqrt{u}.} \tag{40.1}$$

Here $\Phi(0) = 0$ *and* $\Phi(1) = +\infty$*. The theorem requires no bound on the number of hidden states or transition rates.*

Proof. Write $P = P_t$, $H = (P + P^\dagger)/2$ and $B = (P - P^\dagger)/2$. These are the self-adjoint and skew-adjoint parts of the transition operator, not of the generator. In a real Hilbert space,

$$C(2t) = \langle P^\dagger f, Pf\rangle_\pi = \|Hf\|_\pi^2 - \|Bf\|_\pi^2, \tag{40.2}$$

$$\Delta_s(t) = \|Bf\|_\pi^2 - \|(H - sI)f\|_\pi^2 \le \|Bf\|_\pi^2. \tag{40.3}$$

In particular a reversible process, for which $B = 0$, has no positive defect.

For each state x, let v_x be the total variation between the probability rows $P(x,\cdot)$ and $P^\dagger(x,\cdot)$. The bounded-expectation inequality gives

$$|Bf(x)| \le \frac{R}{2} v_x, \qquad u \le \sum_x \pi_x v_x^2 \le 1.$$

Let $\mu(x,y) = \pi_x P(x,y)$ and $\mu^{\leftarrow}(x,y) = \mu(y,x)$ be the forward and reverse stationary endpoint laws. Transposition exchanges the two laws, so their relative entropies in opposite directions are equal. Since both first marginals are π,

$$D(\mu\|\mu^{\leftarrow}) = \frac{1}{2}\sum_x \pi_x \{D(P_x\|P_x^\dagger) + D(P_x^\dagger\|P_x)\} \tag{40.4}$$

$$\ge \sum_x \pi_x \, 2v_x \operatorname{atanh} v_x. \tag{40.5}$$

The last inequality is the established sharp Jeffreys–total-variation inequality [148]. The power series

$$\Phi(u) = 2\sum_{k=1}^{\infty} \frac{u^k}{2k-1}$$

shows that Φ is increasing and convex on $[0,1)$, with $\Phi(u) \ge 2u$. Jensen's inequality therefore gives

$$D(\mu\|\mu^{\leftarrow}) \ge \Phi\left(\sum_x \pi_x v_x^2\right) \ge \Phi(u).$$

Finally, retaining only the endpoints cannot increase relative entropy. The forward and ordinarily reversed stationary CTMC path laws over $[0,t]$ have relative entropy $t\sigma$. Consequently $t\sigma \ge D(\mu\|\mu^{\leftarrow}) \ge \Phi(u)$. □

The elementary linear version follows already from conditional Pinsker and the same identity. The nonlinear version retains the sharp symmetric divergence ingredient. Neither version assumes that the observed scalar record is itself Markovian: the semigroup acts on the full hidden state.

40.2 What is sharp, and what remains a realization problem

The endpoint bound is sharp over stationary discrete-time Markov kernels. To see this, take a four-state cycle with uniform stationary law and

$$P_{i,i+1} = \frac{1+z}{2}, \qquad P_{i,i-1} = \frac{1-z}{2}, \qquad 0 < z < 1,$$

with indices modulo four. For $f = (1, 1, -1, -1)$ and $s = 0$,

$$C_0 = 1, \quad C_1 = 0, \quad C_2 = -z^2, \quad \Delta_0 = z^2, \quad R = 2.$$

Every row has total variation z from its reverse, and

$$D(\mu \| \mu^{\leftarrow}) = z \log \frac{1+z}{1-z} = \Phi(z^2).$$

Thus equality holds throughout the endpoint argument.

This example does *not* establish sharpness among finite CTMCs: $\det P = -z^2 < 0$, whereas $\det e^{tQ} = e^{t \operatorname{tr} Q} > 0$. The sharp kernel cannot be the finite-time transition matrix of a finite CTMC. The contraction from a full continuous path to its endpoints can lose further information. Determining the best CTMC-compatible rate from a given finite correlation record remains a separate physical realization problem.

For a nearest-neighbor ring with uniform stationary law and the cosine detector, write its Fourier eigenvalue as $-\gamma + i\omega$. Then

$$C(t) = \tfrac{1}{2} e^{-\gamma t} \cos(\omega t), \qquad s = e^{-\gamma t} \cos(\omega t), \qquad \Delta_s(t) = \tfrac{1}{2} e^{-2\gamma t} \sin^2(\omega t).$$

The nonnegative residual in (40.3) vanishes for this mode, but the subsequent information inequalities need not be saturated. This avoids interpreting a convenient exact spectral example as a universal minimum-dissipation model.

40.3 Use with finite-confidence calibration

Suppose an existing confidence construction returns $g \geq 0$ such that, on an event of probability at least $1 - \alpha$,

$$2g \leq [\Delta_s(t)]_+.$$

Both the same-state binary-twin construction and the bounded analog joint construction provide such a gap. For the analog detector, writing $f(x) = \mathbb{E}[Z \mid X = x]$ gives

$$B_{\mathrm{obs}} = \mathbb{E} Z^2 \geq \mathbb{E} f(X)^2 = C(0).$$

The coefficient of the diagonal moment is $-s^2$, so its excess measurement noise makes the contrast conservative. With $q_0 = (1 + B_{\mathrm{obs}})/2$, $q_1 = (1 + C(t))/2$, $q_2 = (1 + C(2t))/2$,

$$a_s = (-s^2, 2s, -1), \qquad b_s = -\frac{(1-s)^2}{2}, \qquad 2(a_s \cdot q - b_s) \leq \Delta_s(t).$$

Consequently the same confidence event yields

$$\sigma \geq \frac{1}{t} \Phi\left(\frac{8g}{R^2}\right) \geq \frac{16g}{R^2 t}. \tag{40.6}$$

For a detector mean in $[-1, 1]$, $R = 2$ gives the particularly simple lower bound $4g/t$. There is no new statistical failure allowance for replacing the old physical implication by this stronger one. Taking a maximum with another bound is justified without an extra penalty only when both follow on the same already-covered event, or have separate simultaneous coverage.

The implementation returns exact rational bounds. For $0 \le u < 1$ and an integer $K \ge 1$,

$$2\sum_{k=1}^{K} \frac{u^k}{2k-1} \le \Phi(u) \le 2\sum_{k=1}^{K} \frac{u^k}{2k-1} + \frac{2u^{K+1}}{(2K+1)(1-u)}.$$

The upper endpoint encloses the *lower-bound formula*, not the actual entropy production. If a confidence gap implies $u \ge 1$, the API flags incompatibility with finite entropy *on the coverage event*; it retains the valid linear inequality and does not silently clip u or treat a possible confidence failure as an empirical refutation of the model.

40.4 Reanalysis of the frozen joint-calibration experiment

The sixth-edition joint-calibration count archive contains 400 independent simulated datasets in each of four declared cases. Its contrast, confidence level, counts, and acquisition resources were fixed before those datasets were generated. Recomputing the old certificates reproduces all saved rates exactly. Applying (40.6) then changes only the mathematical physical bound, without collecting data or selecting a new contrast.

Case	Positive datasets	Previous mean bound	New nonlinear mean bound
Driven ring	399/400	0.0008005133	0.1122449304
Equilibrium	0/400	0	0
High readout noise	6/400	0.0000002836	0.0002243291
Invalid calibration	400/400	0.0105740727	0.4597513517

The driven mean is 140.216 times the former mean, while its actual model entropy production is about 3.55736. The new mean is about 3.16% of that value: the improved inference still leaves a substantial gap. Positive detection counts are unchanged because the same positive confidence gap is used. Each dataset still has 20,000 pair acquisitions, 40,000 endpoint readouts and 24,000 units of declared evolution time; preparation and detector overheads require their existing separate calibration.

The invalid-calibration row is a failure control, not a successful inference. Changing the detector between calibration and science measurements violates the shared-channel premise and produces false positive certificates even for equilibrium. A stronger valid inequality cannot repair an invalid measurement contract. The archived reanalysis is retrospective simulated-data evidence, not new physical experimental validation.

40.5 Vector observations and diffusion processes

The same proof holds for a bounded detector map F into a real Hilbert space, with image diameter at most R, if

$$C(t) = \mathbb{E}\langle F(X_0), F(X_t)\rangle.$$

For each row, the difference between the two vector expectations is total variation times the difference of two barycenters of the detector image. Its norm is at most Rv_x, since taking the convex hull does

not enlarge diameter. The operator identity holds componentwise, or directly in the corresponding Hilbert-valued L^2 space. All remaining inequalities are unchanged. This proves a joint-diameter bound, rather than adding scalar entropy lower bounds, an addition which would generally be invalid.

For example, replacing the cosine channel by $F(\theta) = (\cos\theta, \sin\theta)$ doubles the ring correlation and its defect while retaining diameter two. The linear population bound doubles; the nonlinear one improves further. This is a changed observation channel, whose full vector acquisition and readout costs must be recorded. It is not additional information obtained by processing the original cosine alone. For a noisy vector detector Z confined to the unit ball, $\mathbb{E}\|Z\|^2 \geq \mathbb{E}\|\mathbb{E}(Z \mid X)\|^2$, and the scores $(1 + \|Z\|^2)/2$ and $(1 + \langle Z_0, Z_t\rangle)/2$ have ranges $[1/2, 1]$ and $[0, 1]$. This supplies the analogous joint-score construction; the executable scalar benchmark does not claim a measured vector advantage.

The result also transfers to a stationary ordinary-even overdamped diffusion whenever the forward and reversed path laws exist with $D(\mathbb{P}_{[0,t]} \| \mathbb{P}^{\leftarrow}_{[0,t]}) = t\sigma$. For the smooth, compact, uniformly elliptic setting, or the separately justified absolute-continuity setting of the diffusion chapter, the proof uses the same transition-kernel identity and conditional divergences. Here the reverse comparison is $Q^\dagger$, and its path divergence is $t\sigma$, rather than the earlier reversible comparison's $t\sigma/4$. Those constants belong to different comparisons and cannot be interchanged.

40.6 Counterexamples that delimit the inference

Stationarity cannot be replaced by a reversible generator alone. Take $Q = \left(\begin{smallmatrix} -1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$, $f = (1/2, 1)$ and initialize the process in state zero. Its nonstationary products obey $C(t) = 3/8 - e^{-2t}/8$, so

$$-C(0) + 2C(t) - C(2t) = \tfrac{1}{8}(1 - e^{-2t})^2 > 0.$$

The generator's stationary entropy production is zero. The positive record defect therefore does not imply a positive stationary entropy rate without the stationary-preparation assumption.

Likewise, observing an even physical quantity is insufficient to establish ordinary time reversal for the hidden state. An equilibrium inertial oscillator has an even position and odd velocity; its bounded position readouts can have oscillatory correlations and a positive defect while its physical entropy production is zero. Another explicit model is circle drift–diffusion with the physical involution $\theta \mapsto -\theta$. Its generator $Q = \omega\partial_\theta + D\partial_\theta^2$ obeys $\Theta Q \Theta = Q^\dagger$ and is generalized reversible under that involution. The even observable $\cos\theta$ has $C(t) = e^{-Dt}\cos(\omega t)/2$; at $\omega t = \pi/2$ and $s = 0$, $\Delta_s = e^{-2Dt}/2 > 0$. Its ordinary-reversal mathematical entropy and its generalized-reversal physical entropy are different quantities. The theorem applies to the former unless ordinary full-state reversal has been physically justified.

Finally, related spectral, cross-correlation and variance-sum-rule dissipation bounds are established prior work. The specific contribution tested here is the finite-lag scalar-defect implication, its sharp endpoint envelope with explicit nonembedding caveat, and its integration with the already-covered calibration record. The adjacent literature assessment identifies the closest results and does not claim historical priority.

Chapter 41

Joint inference with one-reading analog calibration

A bounded analog detector can supply its calibration coordinate with a single reading. No unchanged-state twin is needed for the conservative construction below. Combining that coordinate with the two dynamical scores gives a single confidence bound. The direct semigroup dissipation inequality then converts the same covered contrast into a substantially stronger physical certificate. The statistical construction extends the joint-calibration and direct-analog chapters; its ingredients remain the reversible quadratic moment inequality, conditional Jensen, and Hoeffding concentration.

41.1 The observable second moment has the useful inequality sign

Let $Z \in [-1, 1]$ be the output of a fixed noninvasive memoryless detector, with conditional mean $f(x) = \mathbb{E}[Z \mid X = x]$. Use the stationary notation

$$A = \mathbb{E}_\pi f^2, \qquad B = \mathbb{E}_\pi Z^2, \qquad C_k = \langle f, P^k f\rangle_\pi, \qquad P = e^{\Delta Q}.$$

Conditional detector noise at the two endpoints of a positive-lag pair is independent. Consequently

$$B = A + \mathbb{E}_\pi \operatorname{Var}(Z \mid X) \geq A.$$

$$W_0 = \frac{1 + Z^2}{2} \in [1/2, 1], \qquad W_k = \frac{1 + Z_0 Z_{k\Delta}}{2} \in [0, 1] \quad (k = 1, 2),$$

and their means are $p_0 = (1 + B)/2$ and $p_k = (1 + C_k)/2$. Here $C_0 = A$; the observable squared reading estimates B, not A.

Proposition 41.1 (One-reading analog reversible support)**.** *For fixed $0 \leq s \leq 1$, let $a_s = (-s^2, 2s, -1)$ and $b_s = -(1 - s)^2/2$. Every reversible model satisfying this detector contract obeys*

$$a_s \cdot p - b_s = \frac{-s^2 B + 2sC_1 - C_2}{2} = -\frac{\|(P - sI)f\|_\pi^2 + s^2(B - A)}{2} \leq 0. \tag{41.1}$$

The support b_s is sharp across the allowed detector class.

Proof. For reversible Q, the operator P is self-adjoint, so $\|(P - sI)f\|_\pi^2 = C_2 - 2sC_1 + s^2 A$. The additional conditional-variance term is nonnegative. The deterministic zero-output detector attains equality for every s. □

The detector's noise costs population signal:

$$g_s^{\text{analog}} = g_s^{\text{ideal twins}} - \frac{s^2}{2}(B - A).$$

Avoiding twins therefore does not guarantee better power. When the detector noise dominates its state-dependent signal, the conservative square may erase the violation entirely. For a state-independent fair detector with outputs ± 1, $A = 0$, $B = 1$, and the analog gap is $-s^2/2$, even though its true conditional mean contains no state information. This example also shows why one cannot silently reinterpret a squared noisy reading as A.

41.2 One confidence event, with every acquisition counted

Take n_0 independent stationary single readings and n_1, n_2 independent stationary pairs at the two lags. Let $\widehat{p}_k$ denote their score means. For a fixed or independently piloted s, define

$$r_s = \sqrt{\frac{\log(1/\alpha)}{2}\left(\frac{s^4}{4n_0} + \frac{4s^2}{n_1} + \frac{1}{n_2}\right)}, \tag{41.2}$$

$$\underline{g}_s = \left[-s^2\widehat{p}_0 + 2s\widehat{p}_1 - \widehat{p}_2 - b_s - r_s - \sum_{k=0}^{2} |a_{s,k}|(e_k + \delta_k)\right]_+ . \tag{41.3}$$

Here e_k bounds numerical error in an empirical mean, and δ_k bounds the conditional mean discrepancy from its ideal target given the entire past. Independent stationary acquisitions have $\delta_k = 0$. The conditional version requires a valid external bound, not merely marginal stationarity. Hoeffding's bound, or its conditional exponential-moment proof, gives the stated event with failure at most α. Independently established contract failure can be added by a union bound. The API records these premises without verifying them from the score summaries.

The factor $1/4$ in the first variance term is the square of the score's range length $1/2$. It does not eliminate that term. For example, use a reversible reset generator with stationary distribution $(1/2, 1/4, 1/4)$, detector values $(0, 1, -1)$, decay $1/2$, and $s = 1/2$. Then $A = 1/2$ and the ideal score means are $(3/4, 5/8, 9/16)$, exactly on the support. One calibration reading is zero with probability $1/2$, giving $\widehat{p}_0 = 1/2$. Even if the two dynamical means were known exactly, treating that calibration reading as exact would create a false gap $1/16$. Equation (41.2) retains the uncertainty and prevents this example from giving a positive certificate.

This design uses

$$N_{\text{acquisitions}} = n_0 + n_1 + n_2, \qquad N_{\text{readouts}} = n_0 + 2n_1 + 2n_2, \qquad T_{\text{evolution}} = \Delta n_1 + 2\Delta n_2.$$

Preparation, reset, sensor and calibration overhead must also be counted when known. None is assigned zero physical energy here. The exact square-summary helper rounds the score with a declared numerical radius; it bounds the rounding error of stored digital readings, not unknown sensor calibration error.

41.3 A direct physical conversion on the same event

Under the stationary even-variable Markov assumptions of the direct spectral dissipation theorem, the detector mean has range at most two and

$$-s^2B + 2sC_1 - C_2 \le -s^2A + 2sC_1 - C_2.$$

Thus the covered gap in (41.3) supplies

$$\sigma \geq \frac{4\underline{g}_s}{\Delta}, \qquad \sigma \geq \frac{\phi(2\underline{g}_s)}{\Delta}, \qquad \phi(u) = 2\sqrt{u}\,\operatorname{atanh}\sqrt{u}, \quad 0 \leq u < 1. \tag{41.4}$$

The implementation encloses ϕ with a positive rational series. The interval is an enclosure of this lower-bound formula, not an interval for the true entropy production. An incompatible normalized defect is flagged on the coverage event; it is not silently clipped or presented as empirical falsification.

For comparison, the earlier additive-reversible record argument gives

$$\sigma \geq \frac{8\underline{g}_s^2}{(2s\sqrt{\Delta} + \sqrt{2\Delta})^2}. \tag{41.5}$$

The zero-lag squared-reading law is unchanged by additive reversibilization because the stationary law and detector channel are unchanged. It therefore contributes sampling uncertainty and apparatus expenditure but no comparison divergence in (41.5). The stronger direct argument uses the semigroup identity and forward/reverse comparison instead. Both physical conversions reuse the same confidence event; they spend no additional α. Different choices of s still require independent selection or a simultaneously valid procedure.

41.4 Frozen comparison on the same analog data

The benchmark fixes all models, parameters, counts, rules and random seed before drawing its 1,600 held-out synthetic datasets. Each of four cases has 400 independent datasets. A 20-state circulant CTMC is observed through a cosine detector rounded to denominator 4096. Each dataset contains 4,000 single-reading calibration acquisitions and 3,000 pairs at each of lags 1/10 and 1/5. Every compared method therefore receives the same 10,000 acquisitions, 16,000 detector readings and 900 declared units of evolution time. These comparisons match acquisition resources. Their analysis-computation time and energy are unmeasured and are not asserted equal.

The base contrast has $s = 1/2$, reproducing the earlier difference of positive-lag scores. A second contrast fixes $s = 3/4$ before production; "tuned" refers to this predetermined design, not a fit to the held-out records. The split methods call the sixth edition's actual analog calibration and score APIs, with $\alpha/2 = 1/40$ for each stage. The joint methods use $\alpha = 1/20$. Virtual coin methods retain a draw from Bernoulli(W_k) for each same analog pair. The calibration singles are common to every method.

Method	Driven	Equilibrium	Noisy driven
Joint analog, $s = 1/2$	399/400	0/400	0/400
Joint analog, $s = 3/4$	399/400	0/400	0/400
Joint coin, $s = 3/4$	361/400	1/400	0/400
Split analog, $s = 1/2$	377/400	0/400	0/400
Split analog, $s = 3/4$	379/400	0/400	0/400
Split coin, $s = 1/2$	293/400	0/400	0/400

The driven rates are 20 forward and 1 backward; equilibrium rates are both 21/2. The noisy detector outputs half the rounded cosine plus an independent fair noise of amplitude 1/2. Its squared-reading calibration is too conservative here, and every method returns zero in all 400

datasets. An exact population audit explains this failure. For $B > 0$, completing the square gives

$$\sup_{s\in\mathbb{R}} g_s^{\mathrm{analog}} = \frac{1}{2}\left(\frac{C_1^2}{B} - C_2\right).$$

Rational uniformization of this fixed noisy ring encloses both correlations and certifies that the right side is below -0.007. Its numerical value is approximately -0.00789686. The entire two-lag quadratic family therefore lacks a positive population gap here. More samples at these same settings cannot repair this obstruction; different lags, additional noise calibration or different observables remain possible. This retrospective exact audit does not change the frozen inference design or any reported decisions. The single equilibrium rejection by the joint coin method is reported rather than removed; the methods have individual 5% failure allowances.

Keeping the older quadratic physical conversion isolates the statistical improvement. At $s = 1/2$, the driven mean lower bound rises from 0.00145847 for split analog to 0.00351542 for joint analog. At $s = 3/4$, it rises from 0.00164098 to 0.00494257. Applying the new direct bridge to that same joint $s = 3/4$ event gives a mean linear bound 0.870376 and a mean nonlinear bound 0.884824: the latter is about 179 times the older quadratic mean. The simulated model's actual rate is 56.9189, so even the improved mean is only 1.55% of the microscopic value. This large remaining gap is explicit.

The virtual coin's greater variance can increase the average of a clipped quadratic lower bound even while reducing detection probability. Indeed its joint $s = 3/4$ quadratic mean is 0.00639217, above the analog value, despite 361 rather than 399 detections. This is why an average positive-part bound alone does not establish estimator quality or universal statistical dominance. The matched table reports detection counts, means, medians and Monte Carlo intervals in the accompanying results.

41.5 A detector change remains a physical failure

The fourth case uses an equilibrium ring with rates 70 in both directions, but replaces the calibration detector by zero gain while retaining the cosine science detector. This deliberately violates the common-channel contract. Both joint analog methods and both split analog methods falsely return positive values in all 400 datasets. The split coin returns 395 positives and the joint coin 400. Correcting the zero-lag conditional-mean discrepancy by the general bound $\delta_0 = 1/2$ makes the reported repair return zero in all datasets. This repair is deliberately conservative; obtaining a useful smaller bound requires actual detector calibration.

The archive retains every dataset's exact integer sufficient statistics and the first raw analog bank in each case. Rational score means can be replayed without drawing new coins or trajectories. The six methods are separate predeclared analyses; their unadjusted maximum is not advertised as a 95% certificate. These are reproducible simulations, not new measured physical data or a universal noise-removal procedure. Physical use requires stationary preparation, the appropriate reversal convention, detector independence and memory control, time calibration, and a justified thermodynamic embedding for conversion from entropy rate to dissipated power.

Chapter 42

Repairing a noise obstruction by changing the measurement lag

The noisy example in Chapter 41 cannot produce a positive population gap anywhere in its chosen two-lag quadratic family. That obstruction concerns a specific set of observation times. Retaining the same detector and changing those times exposes a negative correlation, which does admit a dissipation certificate. The new experiment below recovers 386 positive certificates in 400 fresh noisy datasets, compared with zero for either former fixed contrast, at the same declared evolution time and with substantially fewer detector readings.

This is a complete example of the research loop: certify why the present protocol fails, change an identifiable feature of physical access, and test the resulting prediction on new data. Negative-correlation constraints and the use of temporal correlations to study irreversibility have established antecedents [118, 135, 136]. The contribution here is the explicit repair, its direct finite-data dissipation bound and its frozen comparison. It is not a claim that useful timing is new or that every unknown system has a similarly favorable lag.

42.1 One observed lag is sufficient

Assume the stationary homogeneous Markov and ordinary even-variable reversal conditions of Theorem 40.1. Let $Z \in [-1, 1]$ be the same noninvasive memoryless analog detector, with conditional mean f and independent endpoint noise. Write

$$C(\tau) = \mathbb{E}[Z_0 Z_\tau] = \langle f, e^{\tau Q} f \rangle_\pi.$$

Under a reversible explanation,

$$C(\tau) = \|e^{\tau Q/2} f\|_\pi^2 \geq 0.$$

This condition includes a nonzero detector mean; no mean subtraction or zero-lag second-moment estimate is needed for this particular test.

Corollary 42.1 (Negative-correlation dissipation certificate)**.** *Set $t = \tau/2$ and $s = 0$ in the direct finite-lag theorem. If an already covered lower bound $g \geq 0$ obeys $2g \leq [-C(\tau)]_+$, then*

$$\sigma \geq \frac{2\Phi(2g)}{\tau} \geq \frac{8g}{\tau}, \qquad \Phi(u) = 2\sqrt{u}\,\operatorname{atanh}\sqrt{u}, \quad 0 \leq u < 1. \tag{42.1}$$

The conclusion concerns the same physical model and confidence event.

Proof. The semigroup defect becomes $-s^2C(0) + 2sC(t) - C(2t) = -C(\tau)$. The two unused coordinates have zero coefficients, so they need not be measured. The detector mean has range at most two. Substituting $t = \tau/2$ and normalized covered defect $2g$ into the direct theorem gives the result. □

The hidden semigroup identity uses $\tau/2$, but the instrument does not perform a half-lag measurement. It records only the two endpoints separated by τ. There are no dedicated zero-lag calibration acquisitions. Bounds on the detector, its channel stability and memory, the clock and stationary preparation remain physical calibration requirements.

For n independent stationary pairs, let $W_i = (1 + Z_{i,0}Z_{i,\tau})/2 \in [0, 1]$. At failure probability α,

$$g_n = \left[\frac{1}{2} - \overline{W} - \sqrt{\frac{\log(1/\alpha)}{2n}} - e - \delta\right]_+ \tag{42.2}$$

is valid. The quantity e bounds numerical mean error, and δ bounds a declared conditional mean discrepancy given the complete preceding record. The independent stationary-pair design has $\delta = 0$. This is Hoeffding's one-sided bound; it does not assume that an analog score is a Bernoulli count. The implementation accepts exact product sums, propagates the declared errors, and applies the same rational enclosure of Φ as the preceding chapter. A lag chosen on these same inference records requires a separate selection correction.

42.2 An exact reason to move the lag

The earlier ring has 20 states, clockwise rate 20 and counterclockwise rate 1. For an ideal cosine detector, its fundamental-mode correlation has the form

$$C(\tau) = Ae^{-\gamma\tau}\cos(\omega\tau), \qquad \gamma = 21\big(1 - \cos(2\pi/20)\big), \quad \omega = 19\sin(2\pi/20).$$

The implemented detector is rounded to denominator 4096; the noisy channel adds an independent fair noise of amplitude 1/2 to half that cosine. The exact audit therefore uses the actual rational detector and generator, not the ideal single-mode approximation.

At the former lags 0.1 and 0.2, completing the square and exact uniformization showed that the greatest analog quadratic gap over every real s is below -0.007. At the newly fixed lag $\tau = 0.5$, exact uniformization instead encloses

$$C(0.5) \simeq -0.0732037920 < -0.07.$$

Its population linear dissipation bound is approximately 0.585630; the nonlinear bound is 0.600583. Both remain far below the actual microscopic rate $19\log 20 \simeq 56.9189$.

The rational calculation expands $e^{\tau Q}$ by uniformization with rate 21. Each discrete-step correlation has absolute value at most A, by stationary Markov contraction in $L^2(\pi)$. A geometric bound on the positive exponential tail therefore encloses the omitted correlation contribution, including either sign. Eighty terms give the reported rigorous interval. An independent 100-digit Fourier calculation checks that interval. This calculation and the choice of the new lag use the known model after the earlier failed protocol; the subsequent evaluation uses a new random seed and new held-out records.

42.3 Fresh evaluation with equal evolution time

Before generating the new evaluation bank, the design fixes $\tau = 1/2$, $n = 1800$ and $\alpha = 1/20$. The new method uses 1,800 stationary pair preparations, 3,600 detector readings and 900 units of evolution time. The comparison retains both former fixed values $s = 1/2$ and $s = 3/4$, each using the actual joint-analog API and its stronger direct dissipation bridge. Those methods receive fresh banks of 4,000 stationary single readings and 3,000 pairs at each of lags 0.1 and 0.2: 10,000 acquisitions, 16,000 readings and the same 900 units of evolution time.

Thus the retimed design uses 22.5% as many readings, a factor 40/9 reduction. Its preparation count also changes. Preparation, reset, detector operation and processing overhead are unmeasured and are not assigned zero. The comparison matches declared evolution time; it does not claim equal total wall time, equal processing energy or a calibrated joule saving.

Four cases each receive 400 new datasets. The driven rates are 20 and 1; the equilibrium rates are 21/2 in each direction. Both clean and noisy versions use the same channels defined above.

Case	Old $s = 1/2$	Old $s = 3/4$	Retimed
Driven	400/400	400/400	400/400
Equilibrium	0/400	0/400	0/400
Noisy driven	0/400	0/400	386/400
Noisy equilibrium	0/400	0/400	0/400

The noisy detection fraction is 96.5%, with a nominal Wilson 95% Monte Carlo interval of 94.21–97.90%. Its mean nonlinear lower bound is 0.124947, only 0.2195% of the actual rate. Retiming restores a usable certificate in this example; it does not reconstruct the microscopic dissipation accurately. For the clean driven case, the mean bound is 2.05450, compared with 0.621640 and 0.900411 for the two earlier fixed designs. None of the 1,600 datasets produces a bound above its model's true rate. That simulation observation is additional checking, not a replacement for the confidence proof.

All three methods have individual 95% certificates. Their unadjusted maximum is not advertised as another 95% certificate. The archive preserves every dataset's exact sufficient statistics and the first raw bank of each case. Independent code reconstructs every reported rate and checks the raw-bank summaries, physical counts and population intervals.

42.4 What this repair does and does not identify

The timing change avoids the conservative $B - A$ calibration penalty because the zero-lag coefficient is zero. Independent detector noise still reduces the conditional mean signal and hence the accessible correlation. It does not vanish from the experiment, and many driven systems or available lag ranges may have no statistically useful negative correlation.

The ordinary-reversal and detector assumptions also remain necessary. An equilibrium one-state target coupled to an unobserved driven sensor ring can produce exactly the same recorded law as the noisy driven example. The full target-plus-sensor system dissipates, while the target alone need not. Assigning the inferred lower bound to the target requires excluding such sensor memory or including it in the physical system being bounded. Equilibrium systems with odd hidden variables, such as inertial dynamics, likewise require the correct reversal operation and a separate argument.

The result establishes a concrete change in experimental access: one protocol family provably has no positive population gap, while a redesigned timing choice provides a finite-data dissipation certificate with fewer readings at the same evolution budget. It is a model-assisted, reproducible

repair under stated assumptions, rather than a universal rule for selecting lags without prior information about a system.

Chapter 43

Choosing a contrast from the observations

The direct bound needs a correlation contrast parameter s. A model can suggest one, but requiring the correct hidden rates would undermine its practical use. A finite predeclared library provides a simple alternative with explicit coverage. This is an application of simultaneous confidence bounds, not a new statistical principle.

Let $\mathcal{S} \subset [0, 1]$ contain M distinct rational values chosen before collecting the inference records (an independent pilot is also allowed). For each s, compute the joint analog lower gap g_s with statistical failure budget α/M. The same calibration and pair records can be reused in every contrast. The union bound gives

$$\Pr\{\text{all } g_s \le \tfrac{1}{2}[\Delta_s]_+\} \ge 1 - \alpha.$$

Thus selecting the largest nonnegative gap after observing the data is valid:

$$\sigma \ge \frac{4}{t} \max_{s \in \mathcal{S}} g_s$$

for a detector conditional mean of range at most two and the physical contract of the direct semigroup theorem. The nonlinear bridge may likewise be applied to this selected gap. An external calibration-contract failure η is added once, giving total failure at most $\alpha + \eta$. Independence between candidate estimates is unnecessary. Acquisition costs count the shared records once.

The price is a larger confidence radius, with $\log(M/\alpha)$ in place of $\log(1/\alpha)$. A correctly chosen fixed contrast can outperform the library. A library picked or enlarged using these inference data does not satisfy the proof; it requires a new coverage argument. Likewise, reporting the maximum of several methods that each individually use all of α is invalid without simultaneous coverage.

The executable `certify_library` function removes duplicate parameters, allocates the error budget, retains the individual certificates, and returns the selected contrast and its physical bound. A frozen library of seventeen values $\{0, 1/16, \dots, 1\}$ is evaluated on the same records as the joint analog study. The full result is retained in `attainability/results/design.json`; it reports the selection penalty and comparisons even where a fixed contrast wins.

On the driven analog benchmark, the predeclared library detects 362/400, compared with 399/400 for each of the fixed $s = 1/2$ and $s = 3/4$ methods. Its mean linear rate lower bound is 0.346861, versus 0.600275 and 0.870376 for those fixed contrasts. The equilibrium and strong-noise controls both give 0/400; the invalid detector-calibration control gives 400/400 false positives and remains invalid. These results exhibit the cost of protecting selection from a larger library. They do not justify replacing a well-supported fixed design by a library in every experiment.

Chapter 44

Physical realizations and sampling limits

44.1 A physical minimum in a declared architecture

An upper bound on a relaxed record divergence is not a realizable upper bound on the least dissipation compatible with a measurement. This section gives an explicit physical construction and an exact optimization within one architecture. The architecture assumption is substantial: translation-invariant Markov dynamics on a ring, with the first Fourier mode observed. It is not inferred merely from two measured correlations.

Let $n \geq 3$ and let $k_j \geq 0$ be the rate of the displacement $x \mapsto x + j \pmod{n}$. Assume the displacement support is irreducible. The stationary distribution is uniform. Write the first Fourier eigenvalue as $-\gamma + i\omega$, where $\gamma > 0$, so that

$$\gamma = \sum_{j=1}^{n-1} k_j \{1 - \cos(2\pi j/n)\}, \tag{44.1}$$

$$\omega = \sum_{j=1}^{n-1} k_j \sin(2\pi j/n), \tag{44.2}$$

$$\sigma = \sum_{j=1}^{\lfloor (n-1)/2 \rfloor} (k_j - k_{n-j}) \log(k_j/k_{n-j}). \tag{44.3}$$

Terms with both rates zero contribute zero; a positive rate with zero reverse rate contributes infinite entropy production. A self-inverse displacement $j = n/2$ contributes no entropy. Entropy is dimensionless per unit time; converting it to physical power requires the thermodynamic embedding.

The Fourier formulas, the general delocalized-mode inequality $\sigma \geq \omega^2/\gamma$, and its saturation by circle drift–diffusion are established results, discussed explicitly by Gu [107]. The finite-n minimization below supplies a more restrictive, architecture-specific answer. No historical-priority claim is made.

Theorem: finite-size circulant minimum. Fix n, γ, ω. A finite-entropy circulant realization exists if and only if

$$\frac{|\omega|}{\gamma} < \cot(\pi/n). \tag{44.4}$$

Within that class the minimum is

$$\boxed{\sigma_n^{\min}(\gamma,\omega) = \frac{2\omega}{\sin(2\pi/n)}\operatorname{artanh}\!\left(\frac{\omega}{\gamma}\tan(\pi/n)\right).} \tag{44.5}$$

The expression is even in ω and is zero at $\omega = 0$. The minimizer uses only the two nearest-neighbor displacements, with rates

$$k_\pm = \frac{1}{2}\left\{\frac{\gamma}{1-\cos(2\pi/n)} \pm \frac{\omega}{\sin(2\pi/n)}\right\}. \tag{44.6}$$

At equality in (44.4), a one-way ring is possible but has infinite ordinary-reversal entropy production. Above the boundary there is no such circulant realization, even with infinite entropy.

Proof. Pair the displacements with angle $\theta_j = 2\pi j/n \in (0,\pi)$. Set $x_j = \tan(\theta_j/2)$, traffic $s_j = k_j + k_{n-j}$ and current $d_j = k_j - k_{n-j}$. A nonzero pair contributes

$$\gamma_j = s_j(1-\cos\theta_j), \qquad \omega_j = d_j\sin\theta_j, \qquad \sigma_j = 2d_j\operatorname{artanh}(d_j/s_j).$$

For $u_j = \omega_j/\gamma_j$ this is

$$\sigma_j = \gamma_j F_{x_j}(u_j), \qquad F_x(u) = \frac{u(1+x^2)}{x}\operatorname{artanh}(xu), \qquad |xu| < 1.$$

The function is even in u, convex because $F_x''(u) = 2(1+x^2)/(1-x^2u^2)^2$, and nondecreasing in x at fixed u. For the last assertion, $\operatorname{artanh}(xu)/x$ is nondecreasing for $u \geq 0$, as follows from its positive power series; evenness handles $u < 0$. Since $x_j \geq x_* = \tan(\pi/n)$, $F_{x_j}(u_j) \geq F_{x_*}(u_j)$. An antipodal displacement, when present, adds damping with $u_j = 0$ and zero entropy and can be included as another weight. Jensen's inequality with weights γ_j/γ gives

$$\sigma \geq \sum_j \gamma_j F_{x_*}(u_j) \geq \gamma F_{x_*}(\omega/\gamma),$$

which is (44.5). If a reverse rate vanishes the same inequality holds with infinite left side. Also, $|\omega_j|/\gamma_j \leq 1/x_j \leq 1/x_*$, strictly below the last boundary for a finite-entropy nonzero oscillation. This proves necessity of (44.4). The positive rates (44.6) prove sufficiency and attain equality. □

For fixed nonzero ω, the minimum decreases strictly as feasible n increases, and

$$\sigma_n^{\min} > \frac{\omega^2}{\gamma}, \qquad \lim_{n\to\infty}\sigma_n^{\min} = \frac{\omega^2}{\gamma}. \tag{44.7}$$

This follows from the same monotonicity of F_x and $x = \tan(\pi/n) \downarrow 0$. Equivalently,

$$\frac{\sigma_n^{\min}}{\omega^2/\gamma} = \sec^2(\pi/n)\,\frac{\operatorname{artanh}(r_n)}{r_n}, \qquad r_n = (\omega/\gamma)\tan(\pi/n).$$

The finite-size overhead is physical, not a sampling-confidence penalty.

A complete correlation law with explicit realizations. For $0 < A \le 1/2$, the bounded detector $f(x) = \sqrt{2A}\cos(2\pi x/n)$ has

$$C(t) = \mathbb{E}[f(X_0)f(X_t)] = Ae^{-\gamma t}\cos(\omega t). \tag{44.8}$$

Every ring constructed above realizes this same law for all $t \ge 0$. The circle diffusion

$$d\phi_t = \omega\, dt + \sqrt{2\gamma}\, dW_t \pmod{2\pi}, \qquad f(\phi) = \sqrt{2A}\cos\phi$$

has uniform stationary distribution, exactly the same correlation, and $\sigma = \omega^2/\gamma$. This is an attainable physical value, unlike an upper bound on a record-divergence relaxation. It provides an upper bound on the *infimum* of entropy among all compatible models in any larger class containing this diffusion. It is not an upper bound on the entropy of the unknown system. In the union of circulant finite-state architectures, it is an infimum approached by finite rings.

For the earlier calibrated example $C(0) = 1/2$, $C(1) = 0.38$ and $C(2) = 0.18$, impose the single-mode model and write $z = C(1)/A$, $w = C(2)/A$. Then

$$e^{-2\gamma} = 2z^2 - w = 497/625, \quad \gamma = 0.1145808118\ldots, \quad |\omega|_{\text{principal}} = 0.5504934894\ldots.$$

The smallest feasible ring has $n = 16$. Selected exact-form physical minima, evaluated numerically, are

n	16	20	40	100	1000	circle
$\sigma_n^{\min}$	5.44695	3.55736	2.80013	2.66781	2.64502	2.64480

These numbers sharpen the declared ring problem, not the unrestricted inverse problem. A hidden model outside this architecture can have the same measured moments. Eigenvalue-only entropy bounds require particular care outside the delocalized class; known counterexamples preclude replacing this declaration with a universal claim [108].

An explicit failure outside the architecture. Even the complete scalar correlation function does not by itself justify the circulant minimum. Take a four-state nearest-neighbor ring with damping $\gamma - \kappa$ and frequency ω, where $0 < \kappa < \gamma - |\omega|$. Add a fifth state, called the hub. From every ring state jump to the hub at rate κ; from the hub jump to each ring state at rate $\beta/4$, where $\beta = A\kappa/(1-A)$ and $0 < A < 1$. The stationary masses are $\pi_i = A/4$ on the ring and $\pi_{\text{hub}} = 1 - A$. The hub edges obey detailed balance. The detector is $f = (1, 1, -1, -1, 0)$, so $\mathbb{E}f = 0$ and $\mathbb{E}f^2 = A$. The ring Fourier subspace has zero hub component and its damping increases by κ, giving exactly

$$C(t) = Ae^{-\gamma t}\cos(\omega t), \qquad \sigma = 2A\omega\,\operatorname{artanh}\left(\frac{\omega}{\gamma - \kappa}\right).$$

The chain is irreducible and every nonzero edge has a positive reverse. At $A = 0.1$, $\gamma = 1$, $\omega = 0.5$, $\kappa = 0.01$, the entropy is $0.0556063004\ldots$, below the circle value 0.25, despite identical calibration and identical scalar covariance at every lag. Direct generator calculations reproduce both values. Thus a circle construction provides a compatible physical upper bound on the infimum; it does not prove the unrestricted infimum equals that construction. The detector law beyond second-order correlations may distinguish these architectures.

44.2 Sampling can hide arbitrarily large physical dissipation

The same realizable example exposes a second limitation. Even full spatial resolution at discrete times need not bound physical dissipation. This is a thermodynamic consequence of ordinary sampling aliasing, closely related to continuous-time system aliasing [149].

Exact grid aliasing. Fix $\gamma > 0$ and a sampling interval $\Delta > 0$. The circle transition kernel at time Δ depends on ω only through $\omega\Delta \pmod{2\pi}$. Therefore every finite or infinite stationary record sampled on that grid has exactly the same probability law under

$$\omega_k = \omega + 2\pi k/\Delta, \qquad k \in \mathbb{Z}. \tag{44.9}$$

This statement holds even if the full circle state is read at each instant. Any detector or subsequent processing retains the equivalence. Yet $\sigma_k = \omega_k^2/\gamma \to \infty$. Thus the known two-parameter architecture and arbitrarily many grid records do not supply a finite upper bound on dissipation without an additional frequency restriction or access to intervening motion.

For the scalar covariance, $C(0), C(\Delta), C(2\Delta)$ identify γ and the principal value

$$\omega_0 = \Delta^{-1} \arccos\left[\frac{C(\Delta)/A}{\sqrt{2(C(\Delta)/A)^2 - C(2\Delta)/A}}\right]$$

when the specified single-mode model is feasible. The frequencies $\pm\omega_0 + 2\pi k/\Delta$ remain possible. Their smallest circle entropy is ω_0^2/γ; the supremum is infinite. These conclusions concern the single-mode class and exact moments.

Exact identification need not give a stable entropy upper bound. If γ and A are known, three positive lags with pairwise irrational ratios determine $|\omega|$ from exact cosine correlations. Indeed, $\cos(\omega' t_j) = \cos(\omega t_j)$ implies $\omega' t_j = s_j \omega t_j + 2\pi k_j$, with $s_j \in \{-1, 1\}$. Among three signs, two agree. An irrational ratio between the associated lags forces both integers to vanish, and hence $\omega' = \omega$ or $\omega' = -\omega$. Two such lags do not always suffice: at $t_1 = 1, t_2 = \sqrt{2}$, the distinct positive frequencies $\pi(\sqrt{2} - 1)$ and $\pi(\sqrt{2} + 1)$ give identical cosines.

Exact uniqueness does not survive as a finite-precision upper bound. Fix a finite menu of positive observation gaps, a fixed finite number of intervals, and $\gamma > 0$. For any reference ω, there is a sequence $|\omega_k| \to \infty$ whose entire sampled-state record distributions converge in total variation to the reference distribution. This includes adaptive choices from the finite gap menu with a bounded number of steps.

To prove the statement, write the gaps as $t_1, \ldots, t_m$ and use simultaneous Diophantine approximation to choose arbitrarily large integers q such that the distances from qt_j/t_1 to integers approach zero. If an exact common period exists, its multiples give the conclusion immediately; otherwise the approximating denominators are unbounded. Set $\omega_k - \omega = 2\pi q/t_1$. All the wrapped phase shifts then approach zero. For a gap t, let δ_t be its phase shift reduced to $[-\pi, \pi]$. Comparing unwrapped normal increments and then wrapping gives

$$\mathrm{TV}(K_\omega^t, K_{\omega_k}^t) \le 2\Phi\left(\frac{|\delta_t|}{2\sqrt{2\gamma t}}\right) - 1 \le \frac{|\delta_t|}{2\sqrt{\pi\gamma t}}, \tag{44.10}$$

uniformly in the starting state. Sequential coupling bounds complete-record TV by the sum of these transition bounds, or by the number of steps times the largest menu bound. Independent replication multiplies the sum bound by the number of records. Stopping and further detector processing cannot increase it. Meanwhile ω_k^2/γ diverges.

Consequently every positive-TV neighborhood of the accessible law contains models with unbounded entropy. More statistically, any upper procedure U with uniform coverage $\Pr_\omega\{U \ge \omega^2/\gamma\} \ge 1 - \alpha$ for this experiment must satisfy

$$\Pr_\omega\{U = +\infty\} \ge 1 - \alpha \quad \text{at every reference } \omega. \tag{44.11}$$

For any finite M, choose near-aliases with entropy exceeding M and record TV tending to zero. Uniform coverage then implies $\Pr_\omega\{U \geq M\} \geq 1 - \alpha$; take $M \to \infty$. This statement assumes a fixed finite gap menu and bounded experiment. It does not address a continuously distributed random timing design whose exact realized times are retained, an unbounded observation horizon, or a separately calibrated frequency cap. Those changes alter access.

The implementation illustrates the near-alias mechanism constructively. At gaps 1 and $\sqrt{2}$, Pell solutions $p^2 - 2q^2 = 1$ give shifts $2\pi q$, with second-gap phase error $2\pi/(p + \sqrt{2}q)$. For $\gamma = 1$, reference $\omega = 1/2$, and $(p, q) = (665857, 470832)$, the alternative entropy is 8.751688×10^{12} while the two-interval full-state record TV is at most 1.119197×10^{-6}. This is a mathematical model calculation, not a physically achievable drive-rate claim. A calibrated bandwidth excludes such alternatives; absent that calibration the model allows them.

Reproduction and limits. `physical_minimum.py` reconstructs generators, compares matrix exponentials with (44.8), checks the inequality on 1,000 seeded nonlocal circulant generators, and saves the physical and aliasing examples. Fourteen producer tests cover realization, nonlocal jumps, equilibrium, one-way limits, invalid inputs, continuum convergence, and exact and near aliasing. Floating-point evaluation is not advertised as an interval certificate; the proofs establish the mathematical claims. No new measured-data entropy estimate, universal physical minimum, or historical priority follows from these calculations.

Chapter 45

What a reversible signal can require of its hidden dynamics

A recorded signal can have exactly the same law when played backwards and still require positive dissipation from every admissible hidden model. There is no contradiction: the recorded law and the physical class of its possible generators supply different constraints. This chapter develops that distinction for stationary, smooth, ordinary-reversal overdamped diffusions. It does not extend the conclusion to arbitrary physical realizations.

The earlier correlation certificates give lower bounds without reconstructing hidden dynamics. Their remaining gap raises a physical question: can any admissible system actually attain the inferred minimum? The present layer answers that question for a family of observed processes while allowing any finite hidden dimension. It also identifies the extra access needed for this answer. A covariance function alone is insufficient. Local fluctuation strength and a conditional past–future statistic retain information that global correlations discard.

45.1 The information that closes a minimum

Let $Y_t = f(X_t)$ be a scalar detector applied to a hidden diffusion. For ideal continuous observation, the local quadratic variation determines

$$V_t = \frac{1}{2}\frac{d[Y]_t}{dt}.$$

This is the local strength of the observed fluctuations, not a finite-window sample variance. Condition the symmetric short-time drift and the product of past and future increments on $U = (Y_0, V_0)$:

$$g(U) = \lim_{h\downarrow 0} \mathbb{E}\left[\frac{Y_h + Y_{-h} - 2Y_0}{2h}\,\middle|\, U\right],$$
$$K(U) = \lim_{h\downarrow 0} \mathbb{E}\left[\frac{(Y_h - Y_0)(Y_{-h} - Y_0)}{h^2}\,\middle|\, U\right].$$

The theorem proved next gives

$$\mathcal{I}(Y) = \mathbb{E}\left[\mathbf{1}_{\{V_0>0\}}\frac{[g(U)^2 - K(U)]_+}{V_0}\right] \le \sigma.$$

The difference is formed before division and integration. Its two terms need not be separately integrable. All rates use a calibrated physical clock; σ is entropy production in nats per unit

time under the stated thermodynamic embedding, not a heat or power measurement without that embedding.

For a stationary phase diffusing around a circle, $d\phi = \omega\, dt + \sqrt{2\gamma}\, dW$ and $Y = \cos\phi$, this bound equals ω^2/γ, the full hidden entropy production. Every compatible admissible hidden diffusion must have at least this value, and the circle attains it. The same reasoning closes a larger family with nonuniform even stationary densities and diffusivities. Phase reflection makes their scalar observations reversible. Thus the sign of circulation can remain unidentifiable while its minimum dissipation is determined exactly.

45.2 From a population theorem to an experiment

Three levels of access must be kept distinct.

Level	What is justified
Ideal observed law	Local quadratic variation and conditional derivative functions determine a population lower bound; certain laws attain it.
Finite time resolution	Frozen weighted triple scores become valid after subtracting an explicit semigroup remainder, using calibrated generator derivative bounds.
Finite records	Bounded-score concentration and exact quantized summaries give a confidence statement, retaining calibration and acquisition costs.

The implementation uses a predeclared central-state rule q and a pointwise upper envelope a for local diffusion strength. It scores independent stationary triples, including the zero scores of excluded central states. This avoids estimating an entire conditional distribution or dividing by an estimated bin occupancy. Its finite-time correction and confidence penalty are part of the reported result. A simulation compares it with the earlier endpoint certificate using the same records. Such a comparison establishes performance for the declared design, not optimality over all experimental protocols or laboratory validation.

A covered fallback recomputes the geometry and endpoint bounds with a shared failure budget before taking their maximum. It recovers much of the weaker-drive sensitivity in the same simulated records, with the loss from the stricter component budgets stated explicitly. A separate consistency result shows that a calibrated sequence of finite-record experiments approaches the sharp circle minimum. Its rate is a conservative sufficient construction, not an optimal sample-complexity claim.

45.3 Where the physical premise matters

A conditional average of local diffusion strength cannot replace a pointwise envelope: the connected-torus counterexample overestimates the true entropy under that substitution. An uncorrected finite-time replacement of g and K can also overestimate entropy, even after a smooth invertible detector recalibration. Measurement noise, finite exposure and physical odd-variable reversal change the relevant contract. The countermodels identify these failures explicitly.

The sharp minimum is therefore a solved compatible-model problem within a declared diffusion class. It is not a restriction on every conceivable realization of the signal. The underlying curvature and force-free inference identities have established predecessors; the literature chapter separates those results from the present local-geometry construction and its attainment claim. Historical priority and external scientific review remain open.

Chapter 46

Observed diffusion geometry and a compatible-model minimum

An observed continuous path contains local information that a list of correlations need not retain. In a diffusion, quadratic variation records the noise strength along the observed coordinate. A conditional comparison of past and future increments then separates a directed contribution from variation of the reversible drift inside an unresolved observation fiber. The result below bounds entropy production without fixing the dimension or architecture of the hidden diffusion. For one complete scalar process law, the bound determines and attains the minimum over the entire declared hidden model class.

These are population statements for ideal noiseless continuous observations. They do not certify that a finite dataset has a specified process law. Their physical interpretation requires ordinary reversal of the complete hidden state; in particular, an equilibrium inertial model is not covered merely because its measured position is even under reversal. The proof uses standard diffusion calculus, conditional expectation, and the stationary reverse semigroup. Historical priority for the combination and its consequences is not asserted.

46.1 Model class, entropy convention, and observed geometry

Let M be a compact connected smooth Riemannian manifold without boundary, of arbitrary finite dimension. Let π be a strictly positive smooth probability density relative to its volume form, D a smooth symmetric uniformly positive definite diffusion tensor, and v a smooth vector field such that $\operatorname{div}(\pi v) = 0$. On smooth real functions define

$$L = S + A, \qquad Sf = \pi^{-1} \operatorname{div}(\pi D \nabla f), \qquad Af = v \cdot \nabla f. \tag{46.1}$$

Consider the stationary two-sided diffusion $X = (X_t)_{t\in\mathbb{R}}$ with generator L and invariant density π. These assumptions imply $L^\dagger = S - A$ in $L^2(\pi)$ and smooth bounded generator derivatives of every smooth detector. Its stationary reverse process has generator $L^\dagger$.

The convention for the diffusion coefficient is local covariance $2D\,dt$. Write

$$e(x) = v(x)^{\mathsf{T}} D(x)^{-1} v(x), \qquad \sigma = \int_M e \, d\pi. \tag{46.2}$$

For ordinary reversal, Girsanov's formula gives the relative entropy of the forward and reverse stationary hidden path laws on an interval of length T as $T\sigma$: their drift difference is $2v$ and their

covariance is $2D$. Equation (46.2) is dimensionless entropy per unit physical time. A conversion into dissipated heat or power needs a valid thermodynamic embedding and its temperature convention.

Let $f \in C^\infty(M;\mathbb{R})$ be a deterministic noninvasive detector and $Y_t = f(X_t)$. Define the observed local diffusivity

$$V_t = \Gamma_f(X_t), \qquad \Gamma_f = \nabla f^{\mathsf{T}} D \nabla f, \qquad U_t = (Y_t, V_t). \tag{46.3}$$

The scalar semimartingale quadratic variation satisfies

$$[Y]_t - [Y]_s = 2\int_s^t V_r\, dr. \tag{46.4}$$

Since V has continuous paths, its value is half the time derivative of this quadratic variation, including at a fixed time such as zero. Thus V_t is determined by an ideal continuous record of Y and its calibrated clock; it need not be a function of Y_t alone.

This observation is a pathwise limiting statement. The square of one increment, a regression of squared increments given Y_t, and a measurement blurred by exposure averaging are not interchangeable with V_t. The theorem below cannot be applied to them without an additional access and error argument.

46.2 The conditional local-defect theorem

All conditional quantities in this chapter are defined up to the stationary law's null sets. For $h > 0$ put

$$g_h(U_0) = \frac{\mathbb{E}[Y_h + Y_{-h} - 2Y_0 \mid U_0]}{2h}, \tag{46.5}$$

$$K_h(U_0) = \frac{\mathbb{E}[(Y_h - Y_0)(Y_{-h} - Y_0) \mid U_0]}{h^2}. \tag{46.6}$$

The symmetric definition of g_h is essential. In general, $\mathbb{E}[Af \mid Y_0, V_0]$ need not vanish, so a forward conditional drift cannot silently replace it.

Theorem 46.1 (Observed-geometry dissipation bound)**.** *Under the preceding assumptions, g_h and K_h converge in L^∞ as $h \downarrow 0$, with limits*

$$g(U_0) = \mathbb{E}[Sf(X_0) \mid U_0], \tag{46.7}$$

$$K(U_0) = \mathbb{E}[(Sf(X_0))^2 - (Af(X_0))^2 \mid U_0]. \tag{46.8}$$

Set $d(U) = g(U)^2 - K(U)$, and define its ratio by zero on $V = 0$. Then

$$\boxed{\sigma \;\geq\; \mathcal{I}(Y) := \mathbb{E}\left[\mathbf{1}_{\{V_0>0\}} \frac{[d(U_0)]_+}{V_0}\right].} \tag{46.9}$$

Equivalently, the truncated bounds

$$\mathcal{I}_\varepsilon(Y) = \mathbb{E}\left[\mathbf{1}_{\{V_0\geq\varepsilon\}} \frac{[d(U_0)]_+}{V_0}\right] \leq \sigma, \qquad \varepsilon > 0,$$

increase to $\mathcal{I}(Y)$ as $\varepsilon \downarrow 0$. The number $\mathcal{I}(Y)$ depends only on the full stationary continuous path law of Y, with its time scale, and not on the particular compatible hidden diffusion used to calculate it.

Proof. Let P_h and $P_h^\dagger$ be the forward and reverse semigroups. The past and future are conditionally independent given X_0. Consequently,

$$\begin{aligned}\mathbb{E}[Y_h - Y_0 \mid X_0] &= (P_h f - f)(X_0),\\ \mathbb{E}[Y_{-h} - Y_0 \mid X_0] &= (P_h^\dagger f - f)(X_0),\\ \mathbb{E}[(Y_h - Y_0)(Y_{-h} - Y_0) \mid X_0] &= (P_h f - f)(X_0)(P_h^\dagger f - f)(X_0).\end{aligned}$$

For a smooth function on the compact manifold, the semigroup expansion is uniform in the hidden state:

$$\frac{P_h f - f}{h} = Lf + O(h), \qquad \frac{P_h^\dagger f - f}{h} = L^\dagger f + O(h).$$

For example, the first remainder has supremum norm at most $h\|L^2 f\|_\infty/2$, by the integrated semigroup identity and its supremum-norm contraction. Conditional expectation is an L^∞ contraction. Averaging the two expansions and multiplying them therefore gives (46.8) in L^∞. No Markov assumption on Y has been used.

Abbreviate $a = Sf(X_0)$ and $c = Af(X_0)$. The conditional defect has the exact form

$$d(U_0) = \mathbb{E}[c^2 \mid U_0] - \mathrm{Var}(a \mid U_0). \tag{46.10}$$

In particular,

$$[d(U_0)]_+ \le \mathbb{E}[c^2 \mid U_0].$$

Pointwise Cauchy–Schwarz in the diffusion metric gives

$$c^2 = (v \cdot \nabla f)^2 \le (\nabla f^{\mathsf{T}} D \nabla f)(v^{\mathsf{T}} D^{-1} v) = V_0 e(X_0).$$

Since V_0 is included in U_0,

$$[d(U_0)]_+ \le V_0 \mathbb{E}[e(X_0) \mid U_0].$$

Divide on $V_0 \ge \varepsilon$ and integrate to obtain the truncated bound. Monotone convergence gives (46.9). On $V_0 = 0$, uniform positive definiteness of D implies $\nabla f = 0$ and hence $c = 0$. Equation (46.10) then gives $[d]_+ = 0$ on this event; the declared zero value of the ratio is consistent.

Finally, continuous observation determines quadratic variation as a limiting path functional, and (46.4) determines its continuous derivative. One precise construction uses the sums of squared increments along dyadic partitions on each bounded interval. For the present bounded smooth Itô coefficients, these converge uniformly almost surely to quadratic variation: the usual martingale estimate gives a squared supremum error of order the mesh, and the dyadic meshes are summable. Defining the functional to be zero on the null set of nonconvergent paths makes it Borel measurable. Taking its right derivative at zero recovers V_0. Thus the continuous path law fixes the joint law of U_0, Y_{-h}, Y_0, Y_h and the integrated conditional quantities above for every h. Their unique L^∞ limits, followed by the nonnegative truncated limits, have the same value in every compatible model. Versions on null sets cannot change that value. □

Although the quadratic-variation construction uses a path neighborhood, under every admissible model its limit equals the instantaneous function $\Gamma_f(X_0)$. Conditioning on this ideal limit therefore does not spoil the conditional independence step given X_0. Replacing that limit with a finite-window estimate built from the same increments is a different statistical procedure and can introduce dependence and selection bias.

46.3 The exact sources of slack

The theorem provides more than an inequality: it separates two losses of information. On $V_0 > 0$ write

$$m(U_0) = \mathbb{E}[c^2 \mid U_0], \qquad w(U_0) = \mathrm{Var}(a \mid U_0).$$

Since $m - [m - w]_+ = \min(m, w)$, the following decomposition is exact:

$$\sigma - \mathcal{I}(Y) = \mathbb{E}[e\,\mathbf{1}_{\{V_0=0\}}] + \mathbb{E}\left[\left(e - \frac{c^2}{V_0}\right)\mathbf{1}_{\{V_0>0\}}\right] + \mathbb{E}\left[\frac{\min(m, w)}{V_0}\mathbf{1}_{\{V_0>0\}}\right]. \tag{46.11}$$

Every term is nonnegative and finite. The second term is dissipation in directions not detected by f in the local diffusion metric. The third records the loss caused by variation of the reversible drift inside a retained observation fiber. In particular, equality holds if and only if, up to stationary null sets:

1. $e = 0$ wherever $V_0 = 0$;
2. on $V_0 > 0$, v is a scalar multiple of $D\nabla f$;
3. on $V_0 > 0$, either $\mathbb{E}[c^2 \mid U_0] = 0$ or $\mathrm{Var}(a \mid U_0) = 0$.

The conditions concern a particular compatible model. Their satisfaction by one model, together with a path-law invariant lower bound, establishes attainment of the minimum over the compatible class.

46.4 Coordinate invariance and increasing physical access

Proposition 46.2 (Invertible scalar recalibration)**.** *Let ψ be a smooth one-to-one scalar transformation on an open interval containing the range of f, with a smooth inverse and nonzero derivative. Then the observed process $Z_t = \psi(Y_t)$ has $\mathcal{I}(Z) = \mathcal{I}(Y)$.*

Proof. Write $p = \psi'(Y_0)$ and $r = \psi''(Y_0)$. Diffusion calculus gives

$$\Gamma_{\psi\circ f} = p^2 V_0, \qquad S(\psi \circ f) = pa + rV_0, \qquad A(\psi \circ f) = pc.$$

The conditioning coordinates $(\psi(Y_0), p^2 V_0)$ determine, and are determined by, (Y_0, V_0). Thus

$$g_Z = pg + rV_0, \qquad K_Z = p^2 K + 2prV_0 g + r^2 V_0^2, \qquad g_Z^2 - K_Z = p^2(g^2 - K).$$

The ratio in (46.9) is unchanged pointwise, including its zero convention. The full bounds are invariant; truncations using the same numerical threshold on the two different diffusivities need not be. □

For a complementary access statement, let $\mathcal{G} \subseteq \mathcal{H} \subseteq \sigma(X_0)$ be instantaneous conditioning sigma-fields that both contain V_0. Define the same conditional defect using each field; assume the required conditioning variables are justified by the retained access. The tower property and total conditional variance give

$$d_{\mathcal{G}} = \mathbb{E}[d_{\mathcal{H}} \mid \mathcal{G}] - \mathrm{Var}(\mathbb{E}[a \mid \mathcal{H}] \mid \mathcal{G})\,.$$

Convexity of the positive part and the measurability of V_0 imply

$$\mathcal{I}_{\mathcal{G}} \le \mathcal{I}_{\mathcal{H}} \le \sigma. \tag{46.12}$$

Retaining more valid instantaneous information therefore cannot weaken this population certificate. This statement does not assume that the refined conditioning variables can be learned accurately from a finite dataset.

An exact variational form makes the connection with experimental witnesses explicit:

$$\mathcal{I}(Y) = \sup_{q,w} \mathbb{E}\Big[w(U_0)\{2q(U_0)g(U_0) - q(U_0)^2 - K(U_0)\}\Big], \tag{46.13}$$

where q, w are bounded measurable functions, $w \ge 0$, and $w(U_0)V_0 \le 1$. For fixed w, the identity $2qg - q^2 - K = d - (q-g)^2$ shows that $q = g$ is optimal. Taking $w = V_0^{-1}\mathbf{1}_{\{V_0 \ge \varepsilon,\, d>0\}}$ and letting $\varepsilon \downarrow 0$ proves equality. Each fixed admissible pair is already a lower bound; the supremum is a population characterization, not permission to optimize functions on the inference records without accounting for selection.

46.5 An attained minimum across hidden architectures

Let $\mathbb{P}^{\cos}_{\gamma,\omega}$ denote the complete stationary continuous path law of

$$d\phi_t = \omega\, dt + \sqrt{2\gamma}\, dW_t \pmod{2\pi}, \qquad Y_t = \cos\phi_t, \quad \phi_0 \text{ uniform}, \quad \gamma > 0. \tag{46.14}$$

Define $\mathfrak{C}(\mathbb{P}^{\cos}_{\gamma,\omega})$ to contain every hidden model satisfying the smooth compact ordinary-diffusion assumptions of this chapter, in any finite dimension, whose detector has precisely this continuous path law with the same clock.

Theorem 46.3 (Global compatible-model minimum for the cosine-circle law)**.** *Within the declared class,*

$$\boxed{\min_{(X,f)\in\mathfrak{C}(\mathbb{P}^{\cos}_{\gamma,\omega})} \sigma(X) = \frac{\omega^2}{\gamma}.} \tag{46.15}$$

The circle model itself attains the minimum. Hidden dimension, topology, and additional undetected coordinates are not fixed in this minimization.

Proof. For the circle, $Sf = -\gamma\cos\phi$, $Af = -\omega\sin\phi$, and $\Gamma_f = \gamma\sin^2\phi$. Hence, with $y = Y_0$,

$$V_0 = \gamma(1-y^2), \qquad g = -\gamma y, \qquad K = \gamma^2 y^2 - \omega^2(1-y^2).$$

The stationary law of Y_0 is the arcsine law; in particular its endpoints have probability zero. The ratio in (46.9) is therefore ω^2/γ almost surely. Path-law invariance implies this same lower bound for every member of the compatible class. For the circle's constant current velocity $v = \omega$ and diffusivity $D = \gamma$, equation (46.2) gives $\sigma = \omega^2/\gamma$. This feasible model attains the lower bound, proving the minimum. □

Full path-law agreement is sufficient but stronger than necessary for this particular minimum. Let the compatibility requirement instead fix the stationary joint law of (Y_0, V_0) and the two local conditional functions g and K appearing in (46.8). Every compatible model still has the same $\mathcal{I}(Y)$, and the reference circle is still feasible, so the minimum over this larger class remains ω^2/γ. These

local population functions are substantially more information than a finite list of lagged moments; the statement does not remove the cost of resolving or estimating them.

This minimum is determined by a reversible observed law. Reversing the stationary circle process changes ω to $-\omega$; reflecting the phase $\phi \mapsto -\phi$ makes the same change and leaves $\cos\phi$ unchanged. Thus the observed forward and backward path laws agree, and their relative-entropy rate is zero even when (46.15) is positive. The result does not contradict data processing: it uses additional restrictions on admissible hidden physical models and the diffusion geometry encoded in the observed law. It does not assert that Y is Markovian.

The order of operations matters even for this exact example. Near $|Y| = 1$, the separate weighted terms g^2/V and K/V need not be integrable; for the cosine-circle law their expectations are both positive infinity. Their pointwise difference is nevertheless finite and equals ω^2/γ. Equation (46.9) first forms the conditional local difference, then its positive part, and then the truncated integral. It is not a subtraction of divergent global integrals.

46.5.1 A family with nonuniform stationary density and diffusivity

The exact minimum is not confined to a constant-speed circle. Let $\rho(\phi)$ and $D(\phi)$ be positive smooth even 2π-periodic functions, with $\int_0^{2\pi} \rho(\phi)\, d\phi = 1$, and let J be a constant stationary probability flux. Define the circle diffusion by

$$S = \rho^{-1}\partial_\phi(\rho D \partial_\phi), \qquad A = \frac{J}{\rho}\partial_\phi, \qquad Y = \cos\phi. \tag{46.16}$$

For the complete observed law generated by this model, the minimum over all compatible hidden diffusions in the chapter's class is

$$\min \sigma = J^2 \int_0^{2\pi} \frac{d\phi}{\rho(\phi) D(\phi)}. \tag{46.17}$$

Indeed, Sf is even, as are $\Gamma_f = D\sin^2\phi$ and $e = J^2/(\rho^2 D)$. Every even function on the circle is a measurable function of $\cos\phi$. Thus $\mathrm{Var}(Sf \mid Y, V) = 0$. Away from the two detector critical points, which have zero stationary probability, the one-dimensional Cauchy–Schwarz step is equality. Equation (46.11) proves $\mathcal{I}(Y) = \sigma$ for this model; path-law invariance and the model's feasibility prove (46.17) across the compatible class. Reflection of the phase preserves the invariant density, commutes with S, and changes A into $-A$. It therefore maps the forward hidden generator to its stationary reverse while leaving Y fixed. Every scalar observed law in this family is reversible, including the positive-minimum cases $J \neq 0$.

More generally, a one-dimensional admissible hidden diffusion attains the bound whenever Sf is measurable with respect to (f, Γ_f) and $\nabla f \neq 0$ almost surely. These are sufficient conditions to verify in a model, rather than a claim that an arbitrary detector resolves its hidden drift. Nonuniform diffusion or stationary density alone does not prevent exact inference at the level of a full path law.

Equality of the full scalar covariance is a weaker specification than the path law used here. For example, the earlier hub construction matching an oscillatory covariance is a jump process and does not reproduce the cosine-circle law's continuous paths and local diffusivity. Conversely, this chapter gives no theorem that a fitted oscillatory covariance forces the minimum (46.15).

For reference, the circle's finite-h conditional quantities are exactly

$$g_h(y) = \frac{e^{-\gamma h}\cos(\omega h) - 1}{h}\, y, \tag{46.18}$$

$$K_h(y) = \frac{[e^{-\gamma h}\cos(\omega h) - 1]^2 y^2 - e^{-2\gamma h}\sin^2(\omega h)(1 - y^2)}{h^2}. \tag{46.19}$$

The untruncated finite-h population ratio integrates to

$$\frac{e^{-2\gamma h}\sin^2(\omega h)}{\gamma h^2} \le \frac{\omega^2}{\gamma},$$

with equality in the limit $h \downarrow 0$. This one-model identity does not license an uncorrected finite-h bound for arbitrary compatible diffusions. A finite observation experiment needs explicit approximation, detector, selection, and sampling guarantees.

46.6 A vector extension

The same argument has a matrix form. Let $F = (f_1, \dots, f_k)$ be a smooth vector detector and let

$$G_{ij}(X_0) = \nabla f_i^{\mathsf{T}} D \nabla f_j, \qquad U = (F(X_0), G(X_0)).$$

The matrix G is half the time derivative of the observed quadratic covariation matrix. Put $a = SF$, $c = AF$, $g = \mathbb{E}[a \mid U]$, and

$$K_{\mathrm{sym}} = \lim_{h\downarrow 0} \frac{1}{2h^2}\mathbb{E}\Big[(F_h - F_0)(F_{-h} - F_0)^{\mathsf{T}} + (F_{-h} - F_0)(F_h - F_0)^{\mathsf{T}} \mid U\Big].$$

Uniform componentwise semigroup expansion gives $K_{\mathrm{sym}} = \mathbb{E}[aa^{\mathsf{T}} - cc^{\mathsf{T}} \mid U]$. If G^+ is the Moore–Penrose inverse and H_+ denotes the positive spectral part of a symmetric matrix, then

$$\sigma \ge \mathbb{E}\,\mathrm{tr}\left[\Big((G^+)^{1/2}(gg^{\mathsf{T}} - K_{\mathrm{sym}})(G^+)^{1/2}\Big)_+\right]. \tag{46.20}$$

This includes singular G; it is an inequality, not a claim of general coordinate invariance under transformations of a singular vector detector.

To prove it, let J be the detector differential, so that $G = JDJ^{\mathsf{T}}$ and $c = Jv$. Linear algebra gives $c^{\mathsf{T}}G^+c \le e$ and $c \in \mathrm{range}(G)$. Conditionally,

$$gg^{\mathsf{T}} - K_{\mathrm{sym}} = \mathbb{E}[cc^{\mathsf{T}} \mid U] - \mathrm{Cov}(a \mid U) \preceq \mathbb{E}[cc^{\mathsf{T}} \mid U].$$

After whitening by $(G^+)^{1/2}$, the right side is positive semidefinite. For symmetric $H \preceq N$ with $N \succeq 0$, the variational identity $\mathrm{tr}\, H_+ = \max_{0\preceq P\preceq I} \mathrm{tr}(PH)$ gives $\mathrm{tr}\, H_+ \le \mathrm{tr}\, N$. Conditional expectation and the preceding pointwise metric inequality establish (46.20). Truncating to events where $\|G^+\|$ is bounded justifies every division; monotone convergence extends to their union, which includes every finite matrix. As with the scalar theorem, a physically available extra detector has an acquisition cost that must be recorded.

46.7 Boundary of the result

The minimum in (46.15) is global over a broad, explicit diffusion class, rather than over all conceivable physical descriptions. The proof requires stationarity, smooth continuous deterministic detection, the ordinary reversal convention for the complete hidden state, calibrated time, and enough path access to determine local quadratic variation. It does not cover arbitrary jump processes, singular latent dynamics, generalized odd-variable reversal, or hidden sensor dynamics. Conditioning on an averaged local diffusivity does not establish the required pointwise geometry. No finite record proves exact path-law equality, and no laboratory validation is supplied by this mathematical argument. These restrictions are part of the theorem and are needed when deciding which application or extension is scientifically justified.

Chapter 47

A finite-resolution certificate for local diffusion geometry

The ideal conditional theorem involves time derivatives. A laboratory record has a nonzero time step, a detector and a finite sample. This chapter gives a conservative certificate that pays for all three, under explicit calibration assumptions. It uses a single bounded score per stationary three-time record; no conditional regression, empirical squaring or division by estimated bin occupancies is required.

47.1 Physical and calibration contract

Let X be a stationary smooth ordinary-reversal diffusion in the model class of the preceding population theorem. Write $L = S + A$, let $L^\dagger = S - A$ be its stationary adjoint, and set $f(X_t) = Y_t$. Assume that f belongs to the domains needed below and that the Markov semigroups are contractions in the supremum norm. The following are calibrated bounds, not quantities silently inferred from the test records:

$$\|Lf\|_\infty \le M_{1,+}, \qquad \|L^\dagger f\|_\infty \le M_{1,-}, \tag{47.1}$$

$$\|L^2 f\|_\infty \le M_{2,+}, \qquad \|(L^\dagger)^2 f\|_\infty \le M_{2,-}. \tag{47.2}$$

Fix a time step $h > 0$. A stationary triple is acquired at times $0, h, 2h$, with the middle time relabelled zero. The complete central state determines an accessible coordinate U, a retention indicator $r(U) \in \{0, 1\}$, a real number $q(U)$ and an envelope $a(U)$. On retained states require

$$|q(U)| \le Q, \qquad a(U) \ge a_{\min} > 0, \qquad \Gamma_f(X_0) \le a(U) \quad \text{pointwise.} \tag{47.3}$$

The positive lower bound is on the *chosen upper envelope*; it is not an assumption that the true local diffusivity is bounded away from zero. The retention and weighting rules must be fixed before the test records or fitted on a genuinely independent pilot. A finite partition with constant q and a on each retained cell is one case. A continuous rule, such as $q(Y) = -\gamma Y$, is equally admissible.

47.2 Uniform control of the time-step bias

Define the conditional population quantities

$$g_h(U) = \mathbb{E}[(Y_h + Y_{-h} - 2Y_0)/(2h) \mid U], \tag{47.4}$$

$$K_h(U) = \mathbb{E}[(Y_h - Y_0)(Y_{-h} - Y_0)/h^2 \mid U]. \tag{47.5}$$

Write $g(U) = \mathbb{E}[Sf \mid U]$ and $K(U) = \mathbb{E}[(Sf)^2 - (Af)^2 \mid U]$. Then, almost surely,

$$|g_h - g| \le \beta_g := \frac{h}{4}(M_{2,+} + M_{2,-}), \tag{47.6}$$

$$|K_h - K| \le \beta_K := \frac{h}{2}(M_{1,-}M_{2,+} + M_{1,+}M_{2,-}). \tag{47.7}$$

For completeness, let $a_+ = (P_h f - f)/h$ and $a_- = (P_h^\dagger f - f)/h$. The integral semigroup identity gives

$$\|a_\pm\|_\infty \le M_{1,\pm}, \qquad \|a_+ - Lf\|_\infty \le hM_{2,+}/2, \qquad \|a_- - L^\dagger f\|_\infty \le hM_{2,-}/2. \tag{47.8}$$

Conditional independence of the stationary past and future given X_0 identifies the two-sided product with a_+a_-. Apply the exact identity

$$ab - cd = \tfrac{1}{2}(a - c)(b + d) + \tfrac{1}{2}(b - d)(a + c) \tag{47.9}$$

and the preceding norm bounds. This proves the second bound without an unnecessary term of order h^2. Conditioning on U preserves both bounds.

47.3 A weighted score with a physical lower-bound guarantee

For each ideal noiseless triple let

$$W_h = \frac{r(U)}{a(U)}\left[-\left(q(U) - \frac{Y_h - Y_0}{h}\right)\left(q(U) - \frac{Y_{-h} - Y_0}{h}\right) - 2|q(U)|\beta_g - \beta_K\right], \tag{47.10}$$

where excluded centers contribute zero. Then

$$\mathbb{E}[W_h] \le \sigma. \tag{47.11}$$

Indeed, the conditional expectation of the uncorrected numerator is $2qg_h - q^2 - K_h$. After the bias penalty it is at most $2qg - q^2 - K \le g^2 - K \le \mathbb{E}[(Af)^2 \mid U]$. The pointwise inequality $(Af)^2 \le \Gamma_f e \le a(U)e$, with $e = v^\top D^{-1}v$, completes the proof after integration. The integrand need not be positive; its positive part must not be taken record by record. Taking the positive part of the final covered lower bound is legitimate because $\sigma \ge 0$.

If Y lies in an interval of width R, a common deterministic score range is

$$B_- = -\frac{(R/h + Q)^2 + 2Q\beta_g + \beta_K}{a_{\min}}, \qquad B_+ = \frac{R^2}{4h^2 a_{\min}}. \tag{47.12}$$

For the upper bound put $u = q - (Y_h - Y_0)/h$ and $v = q - (Y_{-h} - Y_0)/h$. Then $-uv \le (u - v)^2/4 \le R^2/(4h^2)$. The lower bound uses $|u|, |v| \le Q + R/h$. The bias penalty is nonnegative and bounded above by the stated amount. Both bounds include the zero scores from excluded centers.

47.4 Finite-sample confidence with exact arithmetic

Let $W_1, \dots, W_n$ be independent stationary records under one frozen rule. Set $B = B_+ - B_-$. Hoeffding's inequality gives, with probability at least $1 - \alpha$,

$$\sigma \ge \left[\overline{W} - B\sqrt{\frac{\log(1/\alpha)}{2n}}\right]_+. \tag{47.13}$$

A dependence-aware version holds if the conditional mean of each bounded score, given the preceding complete record, differs from one fixed target mean by at most a calibrated radius; that radius is subtracted. This is not a claim that overlapping triples from one trajectory are independent.

For independent identically distributed triples, the empirical-Bernstein inequality of Maurer and Pontil, Theorem 4 of arXiv:0907.3740v1 [151], applied to the reflected affine-rescaled score, gives for $n \geq 2$

$$\sigma \geq \left[\overline{W} - \sqrt{\frac{2s_n^2 \log(2/\alpha)}{n}} - \frac{7B \log(2/\alpha)}{3(n-1)} \right]_+, \quad s_n^2 = \frac{1}{n-1} \sum_i (W_i - \overline{W})^2. \tag{47.14}$$

This is a proved empirical-variance inequality, not the substitution of a sample variance into an inequality requiring the unknown population variance. The choice of concentration rule is predeclared. Selecting the largest among several data-dependent lower bounds requires simultaneous coverage or a new argument; the implementation does not silently grant it.

The reference implementation accepts exact rational score sums and sums of squares. Integer quantizer codes therefore suffice even for a large record bank. Suppose each stored score $\widetilde{W}_i$ differs from the ideal score by at most δ_W. Apply the selected concentration inequality to the stored scores in $[B_- - \delta_W, B_+ + \delta_W]$, then subtract δ_W from its lower bound. Numerical errors in the empirical mean are separate penalties; the empirical-Bernstein path requires exact stored means and second moments. Logarithms and square roots are enclosed outward using rational arithmetic. An external calibration failure probability may be added to α only when the physical and concentration arguments hold conditional on frozen pre-test calibration information, for example from an independent calibration experiment, or when a separately proved joint guarantee supplies the needed coverage. A calibration event selected from the same test data does not automatically preserve independence or coverage. Calibration failure is not assigned zero merely because a numerical routine runs.

47.5 Optional variance calibration

For noiseless observations, a known global bound $\Gamma_f \leq G$ gives a second concentration route. Conditional Itô isometry and the bounded drift imply

$$\|Y_{\pm h} - Y_0\|_{L^2(\cdot|X_0)} \leq h M_{1,\pm} + \sqrt{2Gh}. \tag{47.15}$$

The two factors in the raw score are conditionally independent given X_0. Consequently the second moment of the corrected score is bounded by

$$V_* = \left[\frac{(Q + M_{1,+} + \sqrt{2G/h})(Q + M_{1,-} + \sqrt{2G/h}) + 2Q\beta_g + \beta_K}{a_{\min}} \right]^2. \tag{47.16}$$

It can be intersected with the squared maximum absolute range bound. The usual known-variance Bernstein radius is then

$$\frac{B\ell}{3n} + \sqrt{\frac{2V_*\ell}{n} + \left(\frac{B\ell}{3n}\right)^2}, \qquad \ell = \log(1/\alpha). \tag{47.17}$$

For fixed calibration bounds, $V_* = O(h^{-2})$ whereas the squared range is $O(h^{-4})$. This separates a genuine small-step sampling cost from an overly pessimistic bounded-range estimate. It does not promise useful constants for an arbitrary apparatus. Bounded score perturbations replace V_* by $(\sqrt{V_*} + \delta_W)^2$. This option is not used for the noisy-detector contract below.

47.6 Two distinct detector contracts

First, bounded conditionally unbiased readouts may replace Y_t if the noises at the three endpoints are conditionally independent given the complete hidden triple (or satisfy the corresponding zero cross-covariances), and if r, q, a are measured independently of those noises as functions of the true central state. The mean drift is unchanged. The product acquires the additional conditional center variance:

$$\mathbb{E}[(Z_h - Z_0)(Z_{-h} - Z_0) \mid X_0] = \mathbb{E}[(Y_h - Y_0)(Y_{-h} - Y_0) \mid X_0] + \mathrm{Var}(Z_0 \mid X_0). \tag{47.18}$$

Its minus sign in the score makes this noise conservative. Use the detector's full bounded range in the concentration inequality. This argument does not justify selecting the rule from the same noisy central reading or estimating local quadratic variation from a noisy record.

Second, deterministic bounded readout error can be paid as a score perturbation. Suppose ideal and stored readings are in the same interval and differ by at most δ_Y. The ideal and approximate scores must use *identical* r, q, a. The factorization of the score gives

$$|\widetilde{W}_h - W_h| \le \frac{4(R/h + Q)\delta_Y}{h a_{\min}}. \tag{47.19}$$

A useful circle example defines z_0 by deterministic rounding of the true center, fixes $q = -\gamma z_0$, and uses

$$a(z_0) = \gamma \left[1 - \max(0, |z_0| - \delta_Y)^2\right]. \tag{47.20}$$

This bounds the true $\Gamma_f = \gamma(1 - Y_0^2)$ on the calibrated error event. Retention based on z_0 is the actual fixed state rule, so there is no change of selector between ideal and stored scores. The reading error, output score rounding and arithmetic enclosure errors must all be included in δ_W. A fitted detector model or an uncalibrated numerical simulator is not a substitute for these premises.

47.7 Resources and limits

A quota of n independent stationary triples uses $3n$ endpoint readings, n independent stationary preparations and $2hn$ units of declared evolution time. Excluded centers still consume all these resources. Additional coordinate readings are counted separately. Geometry calibration, stationary preparation, resetting, sensor exposure, independent pilot experiments and energy expenditure require additional accounting. Equal declared evolution time is not equal total laboratory time or equal energy.

A nonzero step must retain its bias correction. The nonlinear-observable counterexamples in the adversarial chapter show that an uncorrected finite-h local ratio can diverge near a zero of Γ_f, or remain finite and substantially exceed the true entropy production even after an invertible scalar recalibration. The lower envelope floor and its sampling penalty are substantive access requirements. The certificate also retains the ordinary hidden-state reversal restriction; it does not turn reversible inertial oscillations into evidence of dissipated heat.

Chapter 48

A finite-observation route to the attained minimum

The population minimum is an exact statement about ideal process information. The finite-resolution certificate makes a different promise: a lower bound from a fixed finite experiment, with its approximation and sampling errors paid explicitly. This chapter connects the two for the cosine-circle law. It constructs a sequence of finite experiments whose covered lower bounds approach the exact minimum. The sufficient rate is deliberately conservative; neither optimality nor inexpensive laboratory implementation is asserted.

48.1 A fixed calibrated family and a sequence of experiments

Fix a physical reference time $h_0 > 0$, a calibrated diffusion coefficient $\gamma > 0$, and a calibrated drift bound $\Omega \geq 0$. The target process has the stationary law

$$Y_t = \cos \phi_t, \qquad d\phi_t = \omega \, dt + \sqrt{2\gamma} \, dW_t \pmod{2\pi}, \qquad |\omega| \leq \Omega.$$

The value of ω is not supplied to the score. Assume exact noiseless readings for the first theorem. Independent stationary triples, a calibrated clock, and ordinary hidden-state reversal remain part of the physical contract. In particular, the triples are not overlapping intervals silently treated as independent.

Define constants, with their physical time units retained,

$$M = \gamma + \Omega, \qquad R = \gamma^2 + \Omega^2, \qquad C = R(2\gamma + \Omega). \tag{48.1}$$

They satisfy $M_{1,+} = M_{1,-} = M$ and $M_{2,+} = M_{2,-} = R$ in the finite-resolution contract. Indeed, $\|Lf\|_\infty = \|L^\dagger f\|_\infty = \sqrt{\gamma^2 + \omega^2} \leq M$ and $\|L^2 f\|_\infty = \|(L^\dagger)^2 f\|_\infty = \gamma^2 + \omega^2 \leq R$. The known global diffusivity bound is $\Gamma_f \leq \gamma$.

For an integer $m \geq 1$, choose the experiment before its records:

$$\begin{aligned} n_m &= m^{12}, \qquad h_m = h_0 m^{-3}, \qquad \varepsilon_m = \gamma m^{-2}, \\ q(y) &= -\gamma y, \quad a(y) = \gamma(1 - y^2), r_m(y) = \mathbf{1}_{\{a(y) \geq \varepsilon_m\}}. \end{aligned} \tag{48.2}$$

This is the existing weighted three-time score with envelope floor ε_m, $Q = \gamma$, and noiseless range $[-1, 1]$. Every excluded center still consumes a complete record. The envelope is exact under the stated circle calibration; it is not estimated from the test increments.

Let $W_{m,i}$ be the corrected score in Equation (47.10), and let $\mu_m = \mathbb{E}W_{m,i}$. The correction uses

$$\beta_g = h_m R/2, \qquad \beta_K = h_m M R, \qquad 2|q|\beta_g + \beta_K \le C h_m.$$

The known-variance Bernstein option in the finite-resolution implementation applies directly. No new concentration algorithm is required.

48.2 An explicit population and confidence bound

The ideal truncated local target is

$$J_m = \sigma\, \mathbb{P}\{\Gamma_f \ge \varepsilon_m\}, \qquad \sigma = \omega^2/\gamma.$$

Uniform stationary phase gives

$$\sigma - J_m = \sigma \frac{2}{\pi} \arcsin(1/m) \le \frac{\sigma}{m}. \tag{48.3}$$

The last inequality uses $\arcsin u \le \pi u/2$ for $0 \le u \le 1$. Since $q = g$ and $a = \Gamma_f$, the corrected finite-step numerator lies between its ideal conditional numerator minus twice its bias envelope and that ideal numerator. Therefore

$$J_m - \frac{2Ch_m}{\varepsilon_m} \le \mu_m \le J_m \le \sigma, \qquad 0 \le \sigma - \mu_m \le \frac{\sigma + 2Ch_0/\gamma}{m}. \tag{48.4}$$

This deliberately uses the general calibrated bias bound rather than substituting the unknown ω into an exact finite-step correction.

For explicit concentration constants set

$$H = \frac{(2\gamma + \Omega + \sqrt{2\gamma/h_0})^2 + Ch_0}{\gamma}, \tag{48.5}$$

$$B_0 = \frac{h_0^{-2} + (2/h_0 + \gamma)^2 + Ch_0}{\gamma}. \tag{48.6}$$

The finite-resolution second-moment bound and range width satisfy

$$\mathbb{E}W_{m,i}^2 \le H^2 m^{10}, \qquad B_{m,+} - B_{m,-} \le B_0 m^8. \tag{48.7}$$

For example, the square root of the second-moment envelope is at most

$$\frac{(\gamma + M + \sqrt{2\gamma/h_m})^2 + Ch_m}{\varepsilon_m} \le H m^5,$$

because $m \ge 1$. The range bound follows from Equation (47.12) with $R = 2$ for the detector width; this detector-width symbol is separate from the calibrated generator constant R in (48.1).

Choose a desired two-sided failure probability $\alpha \in (0,1)$ and put $\ell = \log(2/\alpha)$. Choose the calibrated range-width and second-moment bounds $b_m = B_0 m^8$ and $v_m = H^2 m^{10}$, or any valid smaller bounds, and set, in exact real arithmetic,

$$\rho_m = \frac{b_m \ell}{3n_m} + \sqrt{\frac{2v_m \ell}{n_m} + \left(\frac{b_m \ell}{3n_m}\right)^2}, \qquad \widehat{L}_m = [\overline{W}_m - \rho_m]_+. \tag{48.8}$$

This uses the same known-variance Bernstein construction as the implementation, with one-sided allocation $\alpha/2$. Applying the same inequality to both signs and taking a union bound gives $|\overline{W}_m - \mu_m| \le \rho_m$ with probability at least $1 - \alpha$. The deterministic envelopes imply

$$\rho_m \le \frac{H\sqrt{2\ell}}{m} + \frac{2B_0 \ell}{3m^4}. \tag{48.9}$$

Theorem 48.1 (Constructive consistency at the attained circle minimum)**.** *For the calibrated experiment above, with probability at least* $1-\alpha$*,*

$$\boxed{0 \le \sigma - \widehat{L}_m \le \frac{\sigma + 2Ch_0/\gamma + 2H\sqrt{2\log(2/\alpha)}}{m} + \frac{4B_0\log(2/\alpha)}{3m^4}.} \tag{48.10}$$

Replacing σ *on the right by* Ω^2/γ *gives a completely calibrated uniform constant for* $|\omega| \le \Omega$*. For every fixed* α*,* $\widehat{L}_m \to \sigma$ *in probability as* $m \to \infty$*.*

Proof. On the two-sided Bernstein event, $\overline{W}_m - \rho_m \le \mu_m \le \sigma$ and $\overline{W}_m - \rho_m \ge \mu_m - 2\rho_m$. Taking the positive part preserves the upper inequality because $\sigma \ge 0$, and can only improve the lower inequality. Equations (48.4) and (48.9) establish the displayed bound.

Convergence in probability is not deduced merely by fixing a nonzero failure probability in that bound. Rather,

$$\operatorname{Var}(\overline{W}_m) \le \frac{H^2 m^{10}}{m^{12}} = \frac{H^2}{m^2} \longrightarrow 0.$$

Chebyshev's inequality gives $\overline{W}_m - \mu_m \to 0$ in probability, while $\mu_m \to \sigma$ and $\rho_m \to 0$ deterministically. Continuity of the positive-part map proves the stated convergence, including $\omega = 0$. □

The same result holds with a conservative rational radius that actually vanishes in an executable implementation. Choose once, by outward enclosure, rational constants $\overline{H} \ge H$, $\overline{B} \ge B_0$, $\overline{\ell} \ge \log(2/\alpha)$ and $\overline{k} \ge \sqrt{2\overline{\ell}}$. Use

$$\overline{\rho}_m = \frac{\overline{H}\,\overline{k}}{m} + \frac{2\overline{B}\,\overline{\ell}}{3m^4} \tag{48.11}$$

and exact rational division by the integer powers of m. This radius dominates the exact Bernstein radius, gives the same confidence argument, and tends to zero; the rate bound uses the corresponding enclosed constants. The reference wrapper implements this construction without changing the existing finite-resolution module. A fixed absolute rounding grid applied directly to the final square root would leave a nonzero numerical floor, so the literal fixed-precision final-radius calculation is not used to claim asymptotic consistency.

If almost-sure convergence is desired for a sequence of experiments, use a summable sequence of two-sided failure budgets, for example $\alpha_m = m^{-2}$ for $m \ge 2$. The bound then has order $\sqrt{\log m}/m$, and Borel–Cantelli gives eventual success and almost-sure convergence. Dependence between different experiments does not invalidate that summability argument; independence of records within each experiment remains required.

48.3 Controlled score perturbations and resource costs

The same conclusion survives a calibrated uniform perturbation of each already defined score. Suppose the stored iid scores satisfy $|\widetilde{W}_{m,i} - W_{m,i}| \le \eta_m$, with the same fixed selection, weight and drift rule. Expand the score range by η_m at both ends, replace the square root of the second-moment bound by $\sqrt{v_m} + \eta_m$, and subtract η_m from the final lower bound, as in the existing quantization interface. Applying the two-sided argument to the stored scores gives

$$0 \le \sigma - \widehat{L}_m^{\mathrm{stored}} \le \sigma - \mu_m + 2\widetilde{\rho}_m + 2\eta_m \tag{48.12}$$

with probability at least $1-\alpha$. In particular, $\eta_m = O(m^{-1})$ preserves the sufficient rate and consistency. This covers a proved score-rounding envelope. It does not assume that rounding the

central detector value leaves the selector, local diffusivity or weight unchanged. A finite-resolution detector must justify those changes, including the boundary of the retention region, before this result applies.

Alternatively, retain the radius computed for the ideal iid scores and assume a certified deterministic error $|\widetilde{\overline{W}}_m - \overline{W}_m| \le \eta_m$ in their stored mean. Subtracting η_m then gives the same performance inequality with the ideal-score radius and the term $2\eta_m$. This follows directly on the ideal-score concentration event and does not require a fresh independence claim about numerical errors. The rational consistency wrapper uses this version. The error envelope must still cover the actual fixed score rule.

For one experiment at level m, the declared costs are

$$\text{stationary preparations} = m^{12}, \qquad \text{endpoint readings} = 3m^{12}, \qquad T_m = 2n_m h_m = 2h_0 m^9. \tag{48.13}$$

Thus the conservative sufficient accuracy rate is $n_m^{-1/12}$ in the number of preparations, or $(T_m/(2h_0))^{-1/9}$ in declared evolution time. These are not minimax exponents. They account for increasingly fine timing, a shrinking geometry floor and growing sample size using broad uniform bounds. Preparation, reset, sensor bandwidth, calibration, data handling, and energy costs are additional; T_m is not total laboratory duration. If every level is run successively, the cumulative cost is the sum over those levels, rather than the single-level cost above.

The theorem closes a precise existence question: under the stated calibration and access assumptions, finite covered experiments can approach the population minimum. It does not establish a practical advantage at every resource budget, reconstruct the hidden architecture, or transfer the same rate to arbitrary nonlinear detectors and diffusions.

Chapter 49

Why local geometry and finite resolution cannot be interchanged

The local-volatility theorem depends on a physical access contract. The following exact counter-models delimit that contract within smooth, connected, compact, uniformly elliptic stationary diffusions. They also distinguish ordinary hidden-state reversal from a reversal involving odd variables. Numerical values below are reproducible checks; the rejection arguments are analytic.

49.1 A conditional average of volatility is insufficient

On the two-dimensional periodic torus let

$$d(z) = 1 + \tfrac{1}{2}\cos z, \qquad L = d(z)\partial_x^2 + \partial_z^2 + c\,d(z)\partial_x.$$

The uniform density is stationary, the diffusion matrix is $D = \operatorname{diag}(d, 1) \succeq I/2$, and the current velocity is $v = (cd, 0)$. Thus the ordinary entropy rate is $\sigma = c^2$. For $Y = \cos x$,

$$Sf = -dY, \quad Af = -cd\sin x, \quad \Gamma_f = d(1 - Y^2), \quad \mathbb{E}[d] = 1, \quad \operatorname{Var}(d) = \tfrac{1}{8}.$$

If only Y is retained, the limiting conditional quantities are

$$g(Y) = -Y, \qquad K(Y) = \tfrac{9}{8}Y^2 - \tfrac{9}{8}c^2(1 - Y^2).$$

Replacing actual local volatility by its conditional mean $\mathbb{E}[\Gamma_f \mid Y] = 1 - Y^2$ would give

$$\begin{aligned} B_{\text{mean}} &= \mathbb{E}\left[\left(\tfrac{9}{8}c^2 - \tfrac{1}{8}\cot^2 x\right)_+\right] \\ &= \frac{2}{\pi}\left[(A + B)\arctan\sqrt{A/B} - \sqrt{AB}\right], \qquad A = \tfrac{9}{8}c^2, \quad B = \tfrac{1}{8}. \end{aligned}$$

At $c = 8$ this is 68.30307165, exceeding the actual entropy 64. The failure does not rely on numerical precision: using $\arctan(24) = \pi/2 - \arctan(1/24) > \pi/2 - 1/24$ and $\pi > 3$ gives

$$B_{\text{mean}} > \frac{19619}{288} > 64.$$

All regularity assumptions are intact. The substituted denominator is the invalid step.

Retaining $U = (Y, V)$ with $V = \Gamma_f$ repairs the same model. Almost surely, U determines $d = V/(1 - Y^2)$; consequently Sf is U-measurable and the conditional Jensen loss vanishes. The correct functional is exactly

$$\mathbb{E}\frac{(g(U)^2 - K(U))_+}{V} = \mathbb{E}[c^2 d] = 64.$$

Local path geometry can therefore contain essential information that a conditional second-moment regression discards.

49.2 Finite-lag substitution can diverge at finite entropy

Consider the stationary circle $dX = dt + \sqrt{2}\, dW$ and the smooth bounded detector $f(x) = \cos x + \frac{1}{2}\cos 2x$. Its ordinary entropy rate is 1. Writing $c = \cos x$ gives

$$Y = c^2 + c - \tfrac{1}{2}, \qquad V = (1 - c^2)(1 + 2c)^2.$$

The pair $U = (Y, V)$ determines c: the only other possible solution to the same Y is $c' = -1 - c$, but $V(c) - V(c') = (1 + 2c)^3$, which vanishes only where the two solutions coincide.

Let g_h and K_h denote the two-sided conditional drift and product at a finite lag, using denominators $2h$ and h^2, respectively. Set $H_h = (P_h + P_h^\dagger)/2$ and $B_h = (P_h - P_h^\dagger)/2$. Conditional past–future independence gives

$$g_h = \mathbb{E}[(H_h f - f)/h \mid U], \qquad K_h = \mathbb{E}[((H_h f - f)^2 - (B_h f)^2)/h^2 \mid U].$$

Here $H_h f - f$ is U-measurable and

$$B_h f = -\sin x\, [e^{-h} \sin h + e^{-4h} \sin(2h)c].$$

The uncorrected finite-lag ratio is therefore

$$F_h(c) = \frac{g_h^2 - K_h}{V} = \frac{[e^{-h} \sin h + e^{-4h} \sin(2h)c]^2}{h^2(1 + 2c)^2}.$$

At $c = -1/2$ its numerator before squaring is $\sin h(e^{-h} - e^{-4h} \cos h)$, nonzero for every $0 < h < \pi$. The stationary density $1/(\pi\sqrt{1 - c^2})$ is positive there. Hence $\mathbb{E}[F_h] = +\infty$ at every such finite lag, although $\sigma = 1$. Scaling f to lie in $[-1, 1]$ leaves this ratio unchanged.

The failure persists with a fixed positive cutoff. Take $h = 1$ and $\epsilon = 1/15625$. The two intervals $1/200 \le |c + 1/2| \le 1/10$ lie inside $\{V \ge \epsilon\}$. Elementary bounds give $e^{-1} \sin 1 > 3/10$ and $|e^{-4} \sin 2| < 1/50$. Integrating over those intervals and using $\pi < 22/7$ yields the certified inequality

$$\mathbb{E}[\mathbf{1}_{\{V \ge \epsilon\}} F_1] > \left(\frac{7}{25}\right)^2 \frac{190}{2} \frac{7}{22} = \frac{6517}{2750} > 1.$$

Numerical quadrature gives approximately 3.610087355. For each noncritical state, $F_h \to 1$ as $h \downarrow 0$. Thus this example invalidates an interchange of the small-lag and small-volatility limits; it does not invalidate the truncated limiting theorem. A finite-resolution certificate must control the bias before removing the cutoff.

Under exact equilibrium, this particular failure mechanism is absent at all lags. If $L = L^\dagger$, then for every current-state conditioning variable U,

$$g_h^2 - K_h = -\operatorname{Var}((P_h f - f)/h \mid U) \le 0.$$

This is an exact negative control rather than an asymptotic approximation.

49.2.1 Failure under an invertible scalar recalibration

An additional example removes the internal critical point and has a pointwise volatility closure. On the same circle use $f(x) = \cos x + \kappa \cos^2 x$ with $\kappa = 49/100$. The map $\psi(c) = c + \kappa c^2$ has derivative at least $1/50$ on $[-1, 1]$ and is a smooth diffeomorphism on a slightly larger interval. Thus Y alone determines $c = \cos x$ and its local volatility. The naive finite-lag ratio is

$$F_h(c) = \frac{[e^{-h} \sin h + \kappa e^{-4h} \sin(2h) c]^2}{h^2 (1 + 2\kappa c)^2}.$$

Set $r = 2\kappa$, $\alpha = e^{-4h} \sin(2h)/2$ and $\beta = e^{-h} \sin h - \alpha$. Direct circle integration yields

$$\mathbb{E}[F_h] = \frac{1}{h^2} \left[\alpha^2 + \frac{2\alpha\beta}{\sqrt{1 - r^2}} + \frac{\beta^2}{(1 - r^2)^{3/2}} \right].$$

At $h = 1$ this is 11.54021337, while $\sigma = 1$. The violation has the rational certificate

$$\mathbb{E}[F_1] > \left(\frac{29}{100} \right)^2 \frac{12500}{99} = \frac{4205}{396} > 1,$$

because $\alpha > 0$, $\beta > 29/100$, $1 - r^2 = 99/2500$ and $\sqrt{99} < 10$. There is no divergent integral or cutoff limit in this example. The limiting functional is invariant under this scalar recalibration; its uncorrected finite-lag substitute is neither invariant nor generally an entropy bound.

49.3 An even observation does not determine physical reversal

On a torus (q, p), set

$$\pi \propto e^{\cos p}, \qquad S = \partial_q^2 + \partial_p^2 - \sin p \, \partial_p, \qquad A = c \sin p \, \partial_q, \qquad \Theta(q, p) = (q, -p).$$

The diffusion matrix is I. The density is invariant under Θ, and $\Theta S \Theta = S$, $\Theta A \Theta = -A$. Therefore $L^\dagger = \Theta L \Theta$: the stationary path entropy under time reversal combined with this parity is zero.

The detector $f = \sin q$ is even under Θ, yet

$$\Gamma_f = 1 - Y^2, \qquad Sf = -Y, \qquad Af = c \sin p \cos q.$$

The local-defect functional equals the positive ordinary-reversal entropy

$$\sigma_{\text{ordinary}} = c^2 \mathbb{E}_\pi[\sin^2 p] = c^2 \frac{I_1(1)}{I_0(1)},$$

where I_j are modified Bessel functions. At $c = 1$ its value is 0.4463899659, while the generalized-reversal entropy remains zero. A measured scalar being even is therefore insufficient to identify the reversal convention of inaccessible variables.

49.4 Noise and exposure change the path geometry

For a stationary cosine-circle trace recorded with independent additive noise of variance ν^2 at $n + 1$ equally spaced times on $[0, T]$,

$$\mathbb{E} \sum_{k=1}^{n} (Z_k - Z_{k-1})^2 = n[1 - e^{-\gamma T/n} \cos(\omega T/n)] + 2n\nu^2.$$

The clean term tends to γT; the noise term diverges with sampling frequency. Raw noisy quadratic variation cannot be substituted for the latent local volatility.

Finite exposure causes a different obstruction. If $Z_t = \delta^{-1} \int_{t-\delta}^{t} Y_s \, ds$, continuous latent paths make Z continuously differentiable, with $\dot{Z}_t = (Y_t - Y_{t-\delta})/\delta$. Thus $[Z]_t = 0$, although the two-sided product limit is $-\mathbb{E}[\dot{Z}_t^2 \mid Z_t]$ and need not vanish. This history-dependent detector violates the current-state observation contract. A separate calibrated apparatus model is required; finer sampling alone does not restore the theorem's assumptions.

Chapter 50

A frozen simulation at matched acquisition resources

The population result closes a physical realization problem with ideal path access. The finite-resolution theorem must instead tolerate nonzero sampling intervals, finite records and conservative calibration. This experiment tests that second claim against the preceding edition's endpoint certificate using exactly the same triples. It also exhibits a regime where the new certificate loses. All records here are simulated; no apparatus has been validated.

50.1 A predeclared design

The model is the stationary circle

$$d\phi_t = \omega\, dt + \sqrt{2}\, dB_t \pmod{2\pi}, \qquad Y_t = \cos\phi_t, \tag{50.1}$$

with uniform initial phase. Its physical entropy-production rate under ordinary reversal is $\sigma = \omega^2$. Three cases are generated: $\omega = 3$, 1 and 0. Inference receives the same bounds in every case:

$$|\omega| \le 3, \qquad M_{1,+} = M_{1,-} = 19/6, \qquad M_{2,+} = M_{2,-} = 10, \qquad \Gamma_f \le 1. \tag{50.2}$$

These are valid since $\|L^{\pm} f\|_\infty = \sqrt{1+\omega^2}$ and $\|(L^{\pm})^2 f\|_\infty = 1 + \omega^2$. The inference routine does not receive the actual frequency. The circle supplies exact transition sampling in principle; the implemented normal, trigonometric and random-number routines are numerical approximations to it.

Each of 400 independent datasets per case contains $n = 200000$ independent stationary triples at $-h, 0, h$, with $h = 1/25$. Conditional on a uniform central phase, the past and future phases have independent normal increments with means $-\omega h$ and ωh and variance $2h$. This has the same stationary law as a chronological record at $0, h, 2h$.

The design followed preliminary simulations at $h = .035$ and .05, $n = 200000$ and $\omega = 3$. Those pilot records and their seed are disclosed in the frozen contract and excluded from this bank. The new master seed is `202610130731`; scenario and replicate indices define separate `SeedSequence` inputs. The contract and source hashes were written before generation of any bank records. No bank result changes the timing, quota, generator bounds, rules, primary method or confidence allocation.

50.2 Numerical rounding preserves the geometry contract

Set $d = 2^{20}$ and store $z_j = j_j/d$ for each rounded reading. Assume the original smooth detector value differs from its stored approximation by at most $\delta = d^{-1}$. Nearest-grid rounding uses only half that allowance; the rest covers numerical evaluation under the stated simulator contract. The frozen central rule is

$$r = \mathbf{1}_{5|j_0|\le 4d}, \qquad q = -z_0, \qquad a = 1 - \max\{|z_0| - \delta, 0\}^2. \tag{50.3}$$

This is a deterministic function of the instantaneous underlying state. It does not treat a noisy central observation as that state. The error premise implies $1 - Y_0^2 \le a$, so the envelope holds pointwise. On retained centers, $|q| \le 4/5$ and $a \ge 9/25$. This small inflation is needed: simply substituting $1 - z_0^2$ could underestimate the true local diffusivity.

With $u_\pm = 25j_\pm - 24j_0$, the corrected score evaluated on the rounded readings is the exact rational number

$$\widetilde{W}_h^{\mathrm{pre}} = \frac{-15u_+u_- - 6|j_0|d - 19d^2}{15\{d^2 - \max(|j_0| - 1, 0)^2\}} \tag{50.4}$$

on retained centers, and zero otherwise. It is floored to a multiple of $1/256$. Integer division with quotient and remainder avoids multiplication overflow. No floating score evaluation enters the certified aggregate. The uniform error allowance relative to the ideal smooth-reading score is

$$\delta_W = \frac{4(2/h + 4/5)\delta}{h(9/25)} + \frac{1}{256} \simeq 0.017364. \tag{50.5}$$

The first term follows by telescoping the two score factors while keeping r, q, a fixed. The concentration calculation expands the score range by this allowance and subtracts it from the final lower bound. It does not assume that rounding is unbiased. Excluded centers still count toward the quota and cost.

The primary method uses the finite-resolution empirical-Bernstein certificate with $\alpha = 1/20$. We retain the corresponding Hoeffding result as a predeclared comparison, including its failures. Exact integer totals and sums of squares determine all confidence calculations. Sums of squares are formed in safe integer chunks when their total exceeds the 64-bit range.

50.3 The same-record endpoint comparison

The baseline fixes $s = 1 - h = 24/25$ before observing bank records and uses

$$T = -(Y_h - sY_0)(Y_{-h} - sY_0), \qquad \mathbb{E}[T] = \Delta_s = -s^2C(0) + 2sC(h) - C(2h). \tag{50.6}$$

Its deterministic range is $[-(1+s)^2, 1]$. The same empirical-Bernstein theorem, same α and same triples give a covered lower bound on Δ_s. The preceding edition's nonlinear bridge then returns

$$\sigma \ge \frac{\Phi([\Delta_{s,\mathrm{lower}}]_+)}{h}, \qquad \Phi(u) = 2\sqrt{u}\,\operatorname{atanh}\sqrt{u}. \tag{50.7}$$

The bridge is evaluated by the existing exact rational implementation. The endpoint score is floored at denominator 2^{24}, with a separate allowance $2(1+s)^2\delta + 2^{-24}$. This finer grid prevents numerical rounding from unnecessarily obscuring the smaller endpoint defect. It is exact integer arithmetic because $\lfloor 2^{24}T(z)\rfloor = \lfloor -u_+u_-/(625\,2^{16})\rfloor$.

The two methods use identical acquired data. The new method additionally uses calibrated local geometry and generator derivative bounds. Their acquisition cost is not included in the raw reading count; this is a test of the value of that additional physical information, not a claim of universal statistical dominance at an identical prior-information budget.

50.4 Results and the informative failure

Table 50.1 reports all cases. Values are mean lower certificates, not estimates of the actual dissipation. Positive counts are out of 400 independent datasets.

Case	True σ	Weighted mean	Positive	Endpoint mean	Positive
Driven	9	2.982329	400	0.303964	400
Weakly driven	1	0	0	0.011963	396
Equilibrium	0	0	0	0	0

Table 50.1: The frozen same-triple benchmark. Each column uses its own predeclared 95% lower certificate; taking their maximum is not automatically covered.

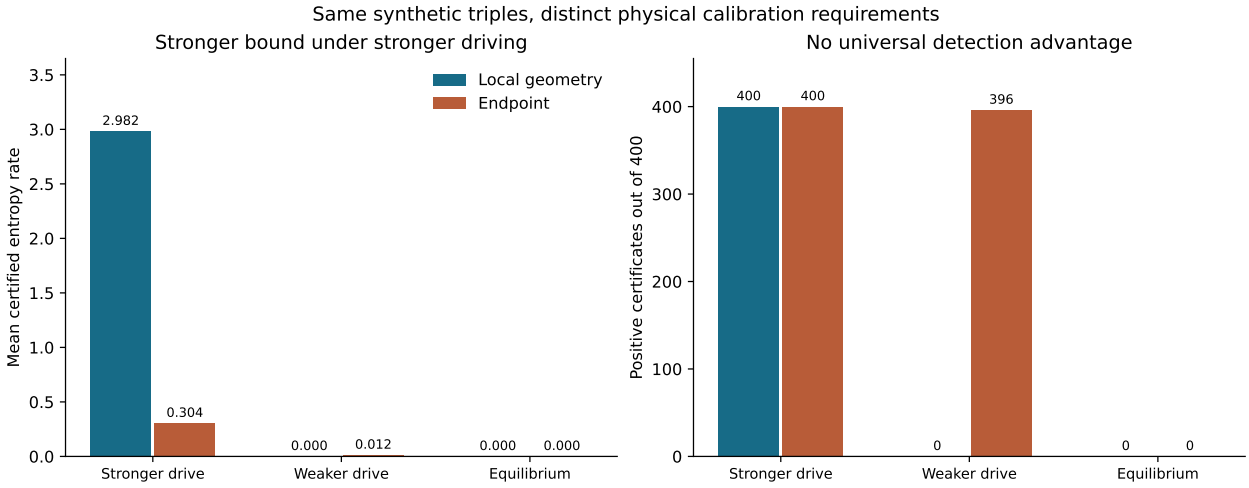


Figure 50.1: The new calibration improves the strong-drive lower bound, while its finite-time correction prevents a positive certificate for weaker driving. The two methods use identical synthetic triples.

In the driven case, the geometry certificate has a mean lower bound 9.811 times the endpoint mean, certifying on average 33.14% of the actual rate. The endpoint method certifies 3.38%. Neither method overestimates the known rate in this bank. The Hoeffding geometry certificate is zero in all 1200 datasets: its broad worst-case score range incurs a sampling penalty of about 24.38, far larger than the driven score mean. The predeclared empirical-variance inequality makes the finite-resolution result useful here.

The weakly driven case is an essential limitation. Keeping the same frequency cap and generator bounds incurs a bias penalty too large for the geometry certificate, while the endpoint method remains informative in 396 datasets. A method that resolves the ideal path-law minimum therefore need not dominate at a finite sampling interval with coarse derivative calibration. We do not replace the common bounds with values computed from the hidden true frequency. No equilibrium false positive occurs in this bank. These empirical counts check implementation behavior; mathematical coverage still rests on the physical and probabilistic assumptions, not on seeing zero violations.

50.5 Resources, reproducibility and application limits

Each dataset costs 600000 scalar readings, 200000 independent stationary preparations and $2hn = 16000$ units of modeled evolution time for either method. The complete bank comprises 240 million triples. Preparation and reset time, clock accuracy, exposure, calibration effort, detector energy and heat are unmeasured. Equal modeled evolution time is not a claim of equal total laboratory time or energy.

The release retains all 1200 exact aggregate records, their seeds, the frozen contract and a full raw triple exemplar for each case. The NPZ archive uses fixed entry timestamps. Producer checks verify score flooring against rational arithmetic, pointwise-envelope inflation, uniform rounding penalties, integer overflow handling, source bindings, aggregate consistency and seed replay. The independent review rederives the formulas and audits the bank separately.

The practical output is a covered finite-data route to exploit an independently justified diffusion-geometry envelope. It has not yet been validated for a noisy laboratory sensor, dependent overlapping records, an uncertain physical clock, or generator bounds learned from the same data. Those are additional access and calibration problems, rather than implementation details that can be assigned zero cost.

Chapter 51

A covered fallback on the same physical records

The local-geometry certificate improves the strong-drive benchmark, but its finite-step penalty removes all positive certificates in the weak-drive case. The endpoint certificate retains sensitivity there. These complementary responses suggest a practical composite inference rule. Its justification is the established union bound; this composition introduces no new statistical principle and requires no additional acquisition.

51.1 Two certificates, one failure budget

Let L_G and L_E be geometry and endpoint lower bounds computed on the same fixed independent stationary triples, with component statistical failure probabilities α_G and α_E. Assume the two physical/calibration contracts hold jointly. Then

$$\mathbb{P}\{\max(L_G, L_E) > \sigma\} \le \mathbb{P}\{L_G > \sigma\} + \mathbb{P}\{L_E > \sigma\} \le \alpha_G + \alpha_E. \tag{51.1}$$

The scores may be arbitrarily dependent. The failure allocation and the two component rules must be fixed for prospective use, or supplied by a separately justified simultaneous procedure. Here $\alpha_G = \alpha_E = 1/40$, so the covered maximum has statistical failure probability at most $1/20$. A shared pre-test calibration failure probability is counted once when one joint event validates both contracts; independent calibration failures can instead be conservatively summed before adding them. A same-test-data calibration event does not inherit this guarantee automatically.

The geometry component uses the corrected score of Chapter 47, including the calibrated reading error and score rounding. The endpoint component uses

$$Z_s = -(sY_0 - Y_{-h})(sY_0 - Y_h), \qquad 0 \le s \le 1, \qquad Y_t \in [-1, 1]. \tag{51.2}$$

Stationarity gives $\mathbb{E}Z_s = -s^2C(0) + 2sC(h) - C(2h)$, with $Z_s \in [-(1+s)^2, 1]$. The same empirical-Bernstein construction, with its uniform score-error correction, supplies a covered lower bound d on the positive endpoint defect. The seventh-edition physical bridge then yields

$$L_E = \frac{\Phi(d)}{h}, \quad 0 \le d < 1, \qquad \Phi(u) = 2\sqrt{u}\,\operatorname{atanh}\sqrt{u}, \tag{51.3}$$

with the implementation retaining the existing rational outward enclosure and its finite-entropy incompatibility handling. The constant follows from the observable range $R = 2$. The implementation requires the common normalized readout contract; it does not silently use the same constants for a wider sensor range.

51.2 Retrospective reanalysis of the frozen bank

The equal allocation was introduced after inspecting the primary benchmark. The following is therefore a retrospective evaluation of a fixed composite rule, not a new frozen confirmatory experiment. No timing, physical parameter, record, generator cap or original result was changed. The newly computed bounds use the exact stored sums and sums of squares from all 1,200 datasets. The original individually valid $\alpha = 0.05$ bounds remain in the primary report; maximizing those original bounds would not by itself retain a 0.05 failure guarantee.

Scenario	Geometry at 0.025	Endpoint at 0.025	Covered maximum
Strong drive, $\sigma = 9$	400/400	400/400	400/400
Weak drive, $\sigma = 1$	0/400	383/400	383/400
Equilibrium, $\sigma = 0$	0/400	0/400	0/400

The geometry component wins every strong-drive dataset; its mean certified lower bound is 2.887645, compared with 2.982329 at its original individual 0.05 allocation. The endpoint component supplies every positive weak-drive result. Its mean bound is 0.00880994, compared with 0.01196295 at its original allocation; detections fall from 396 to 383 out of 400. This is the observed cost of simultaneous coverage. No reported lower bound exceeds the known generating entropy in these simulations, which is a numerical check rather than a proof of coverage beyond the theorem's assumptions.

Every dataset still uses 200,000 independent stationary triples, 600,000 endpoint readings and 16,000 units of modeled evolution time. Both scores reuse these same records, so those resources are counted once. The physical cost of preparation, geometry and clock calibration, exposure, resetting and energy remains additional. The combined result preserves the strong-drive benefit while recovering much of the weak-drive endpoint sensitivity. It does not dominate an individually allocated endpoint test at every operating point, and it does not justify optimizing the allocation on the same data without further simultaneous coverage.

Chapter 52

From ideal access to a physical detector

The preceding geometry theorem answers a realization question: what must every admissible hidden diffusion dissipate to produce a given ideal observation law? A detector changes the law available to the scientist. Exposure averages a signal over time, sampling retains selected moments, and read noise adds uncertainty. Those operations must be included before an ideal-law theorem becomes an experimental claim.

This extension separates three questions. Does the complete detector law still identify the ideal law? Can a finite experiment estimate the desired property with a stated error? How do preparation, read precision, duration and measurement energy change that error? Different answers to these questions can hold for the same physical model.

52.1 One target, several kinds of access

Use the stationary cosine-circle family as a reference:

$$d\phi_t = \omega\, dt + \sqrt{2\gamma}\, dW_t, \qquad Y_t = \cos\phi_t, \qquad \gamma > 0.$$

Its ideal scalar law is reversible. Within the ordinary hidden-diffusion class of the preceding chapters, its exact compatible minimum is $\mathcal{M} = \omega^2/\gamma$. This target is a property of the ideal law, rather than an attempt to count arbitrary unobserved dissipative subsystems. The detector considered here retains an exposure average plus calibrated additive noise.

Available access	Supported conclusion
Complete densely sampled law; known exposure and independent noise	The ideal law can be identified under the stated mixing assumptions. The inverse is an infinite limiting construction.
Fixed sampling grid after known exposure	Distinct mixing diffusion mechanisms can have identical complete sampled laws; an explicit Gaussian example proves this ambiguity.
Finite stationary trials at fixed positive read noise	No uniformly finite upper confidence bound exists for the circle minimum over an unbounded frequency class.
Known binary hypotheses with fixed read precision	Passive stationary trials require and suffice at order ω^4; oriented phase preparation changes this order to ω^2.
Read precision purchased through the declared law $\nu^2 = \kappa/E$	Optimized passive and prepared measurement-energy requirements both have order ω^2. Other apparatus costs remain additional.

The count comparison concerns the known pair ω and 2ω along a specified resonant sequence, at fixed positive diffusion, exposure, read-noise level and target error. It does not solve estimation over an arbitrary frequency interval. Preparation means setting the actual phase to $\pi/2$; selecting a scalar value $Y = 0$ does not resolve the two possible orientations and does not supply the same operation.

52.2 Why the resource distinction matters

Stationary preparation makes the first moment of the exposure vanish. The leading contrast through a noisy detector then appears in a second moment. Oriented preparation supplies a first-moment contrast. This familiar statistical mechanism explains the different trial-count exponents in the proved example. It is not presented as a new general information law.

The same example shows why a smaller number of trials is insufficient to establish an energy advantage. Under the explicitly declared sensor model, passive experiments can spend more energy on read precision and use fewer trials. The two access classes then share the same measurement-energy exponent. The result compares orders, not equal optimal constants, and does not include preparation, resetting, maintaining the drive, exposure overhead or computation in a total-energy optimum.

The population proofs, finite-access theorems, exact countermodel and deterministic design table form one account. The new table contains evaluated analytic bounds, not additional simulated or measured experiments. The earlier frozen simulation bank remains a separate numerical validation of the finite-resolution certificate. Independent internal review checks both layers; external peer review and prospective detector calibration remain open.

The next practical dependency is a detector-specific joint calibration guarantee for local volatility and the short-time statistics used in the entropy certificate. Existing motion-blur correction and noisy-diffusion inference provide essential baselines. Their assumptions must be carried through the physical entropy bound, including selection and dependence between calibration and test records.

Chapter 53

What calibrated exposure preserves at the population level

Finite exposure changes a diffusion observation before measurement noise is added. In particular, a boxcar average of a continuous signal is continuously differentiable and has zero quadratic variation. The local geometry theorem therefore cannot be applied directly to an exposed trace as if it were the original diffusion coordinate. The question here is different: does the complete exposed observation law still determine the ideal law from which that trace came?

Under the assumptions below, the answer is yes, even after a known independent additive noise channel. The recovery uses dense temporal population access and an infinite limiting reconstruction. It does not provide a stable finite-data inverse or recover local quadratic variation by differentiating noisy samples. The constituent deconvolution and averaging arguments are established mathematical tools; historical priority is not asserted.

53.1 Detector and access model

Let $Y = (Y_t)_{t\in\mathbb{R}}$ be a real stationary continuous process, bounded by a finite deterministic constant, and write $\mu = \mathbb{E}Y_0$. Assume

$$\operatorname{Cov}(Y_0, Y_t) \longrightarrow 0 \quad \text{as } |t| \longrightarrow \infty. \tag{53.1}$$

This condition is satisfied by a smooth bounded coordinate of a stationary compact connected uniformly elliptic diffusion. A known exposure width $\tau > 0$ produces the causal boxcar signal

$$Z_t = (A_\tau Y)_t := \frac{1}{\tau}\int_{t-\tau}^{t} Y_s\, ds. \tag{53.2}$$

Let $\mathcal{T}$ be a known countable dense subset of the real time axis. At each distinct $t \in \mathcal{T}$, the retained measurement is

$$B_t = Z_t + \xi_t, \tag{53.3}$$

where the ξ_t are independent of Y, independent across distinct timestamps, and have a known probability law ν. The known noise law need not have finite variance or a nowhere-zero characteristic function for the population uniqueness statement. Its independence and calibration are part of the detector contract.

The accessible research object in this section is the complete joint law of $(B_t)_{t\in\mathcal{T}}$, rather than one finite record. Repeated samples of a finite collection of timestamps do not automatically supply this infinite family of joint distributions. The clock, exposure rule, and noise law are common to every model being compared.

53.2 Noise removal as an identifiability statement

Lemma 53.1 (Known additive convolution on a bounded latent law)**.** *Let P and Q be compactly supported probability laws on $\mathbb{R}^k$, and let N be any probability law on $\mathbb{R}^k$. If $P * N = Q * N$, then $P = Q$.*

Proof. The characteristic function $\widehat{N}$ is continuous and equals one at the origin, so it is nonzero on some neighborhood of zero. The convolution identity implies $\widehat{P} = \widehat{Q}$ on that neighborhood. Compact support makes both characteristic functions real analytic on $\mathbb{R}^k$ (indeed entire after complexification). Analytic uniqueness therefore extends the equality to all real arguments. Uniqueness of characteristic functions gives $P = Q$. □

Apply this lemma at any finite set of distinct timestamps, with the known noise law $N = \nu^{\otimes k}$. Equality of the noisy record laws implies equality of the corresponding finite-dimensional laws of Z. Since Z is continuous, its values on the countable dense set determine its law on $C(\mathbb{R})$, equipped with uniform convergence on compact intervals. Thus the complete noisy law identifies the complete exposure law.

This proof does not divide by the noise characteristic function at its zeros. It uses bounded latent support and uniqueness of analytic continuation; it supplies no stability estimate for that continuation. If bounded support is dropped, a standard sufficient replacement is a noise characteristic function that is nowhere zero. These are different identifiability contracts. Known correlated additive noise can also be treated if its complete joint laws are supplied and remain independent of the signal; knowing only its one-time marginal would not suffice.

53.2.1 What infinitely dense independent readouts can recover

For the iid noise contract, dense access permits a stronger statement that does not require bounded signal amplitude. Let Z have continuous finite real-valued paths and let $B_t = Z_t + \xi_t$ be observed on the countable dense set, with the same known iid noise independent of Z. Then the complete continuous exposure path is a measurable function of the infinitely dense noisy record, almost surely.

To prove this, fix a target t in the dense set and a deterministic sequence of distinct available timestamps $t_j \to t$. For any fixed real u, the bounded complex observations satisfy

$$\frac{1}{n}\sum_{j=1}^{n} e^{iuB_{t_j}} \longrightarrow e^{iuZ_t}\widehat{\nu}(u) \quad \text{almost surely.} \tag{53.4}$$

Conditioned on the entire continuous Z path, the centered summands are independent bounded variables. The independent-variable strong law applies because the sum of their variances divided by j^2 is finite. Their average conditional means converge to the right-hand side by continuity of Z. The argument uses no moment of ξ_t.

Choose a countable sequence $u_k \to 0$ with $u_k \neq 0$. The known noise characteristic function is nonzero for all sufficiently large k. Dividing the limits in (53.4) by $\widehat{\nu}(u_k)$ determines $e^{iu_kZ_t}$, and

$$Z_t = \lim_{k\to\infty} \frac{\operatorname{Im}(e^{iu_kZ_t})}{u_k}$$

removes the phase ambiguity. A countable intersection of probability-one events recovers every dense-set Z_t; continuity supplies the full path. All operations are limits of measurable coordinate functions, with an arbitrary default on the null set where the limits fail.

Consequently the main injectivity theorem below also holds with boundedness replaced by finite second moment, using this dense iid reconstruction and the same covariance-decay assumption. The compact diffusion application remains bounded. The proposition spends infinitely many independent readouts in arbitrarily small time neighborhoods. It is not a claim that faster sampling makes actual sensor errors independent: overlapping exposures, photon reuse, memory, or correlated readout noise can invalidate the stipulated iid contract. The earlier bounded-law convolution argument is still useful for known correlated noise at the population-law level, where pathwise separation from one continuous record need not hold.

53.3 An explicit finite-window reconstruction and its limit

Continuity of Y gives the pathwise identity

$$\tau Z'_t = Y_t - Y_{t-\tau}. \tag{53.5}$$

The derivative is a measurable functional of the ideal C^1 exposure path, for example the limit of rational difference quotients. For $N \geq 1$ set

$$\mathcal{R}_N Z(t) := \mathbb{E}Z_0 + \tau \sum_{k=0}^{N-1} \left(1 - \frac{k}{N}\right) Z'_{t-k\tau}. \tag{53.6}$$

Telescoping gives the exact relation

$$\mathcal{R}_N Z(t) = Y_t - \left\{ \frac{1}{N} \sum_{j=1}^{N} Y_{t-j\tau} - \mu \right\}. \tag{53.7}$$

Only a window whose length grows as $N\tau$ is used at each finite N, but exact derivatives of the exposed path are required.

Theorem 53.2 (Dense-law injectivity through boxcar exposure and known noise)**.** *Within the bounded continuous stationary class satisfying Equation (53.1), equality of the noisy dense-time laws in (53.3), for a common known boxcar width and noise law, implies equality of the ideal continuous path laws of Y.*

Proof. The preceding lemma identifies the common continuous law of Z. Stationarity gives $\mathbb{E}Z_0 = \mu$. By Equation (53.7),

$$\begin{aligned} \mathbb{E}|\mathcal{R}_N Z(t) - Y_t|^2 &= \frac{1}{N^2} \sum_{i,j=1}^{N} \operatorname{Cov}(Y_0, Y_{(i-j)\tau}) \\ &= \frac{\operatorname{Var}(Y_0)}{N} + \frac{2}{N^2} \sum_{j=1}^{N-1} (N-j) \operatorname{Cov}(Y_0, Y_{j\tau}) \longrightarrow 0. \end{aligned} \tag{53.8}$$

For the last limit, bound the finite initial part of the covariance sum by a constant times N^{-1}, and use arbitrarily small covariance on its remaining tail. The same conclusion holds jointly at every finite set of times, by summing their mean-square errors. The law of each $\mathcal{R}_N Z$ is fixed by the common exposure law, so the limiting finite-dimensional laws of Y agree. Continuity again identifies the full path laws. □

An explicitly calibrated covariance-decay bound gives a finite-window population estimate. If $|\operatorname{Cov}(Y_0, Y_t)| \le Ce^{-\kappa|t|}$ for $C < \infty$ and $\kappa > 0$, then

$$\mathbb{E}|\mathcal{R}_N Z(t) - Y_t|^2 \le \frac{C}{N} \coth(\kappa\tau/2). \tag{53.9}$$

This controls truncation of an ideal derivative reconstruction. It does not control estimation of Z' from finitely many noisy or exposed readings. For example, if a derivative approximation has root mean-square error at most δ at each of the retained times, and the stationary mean has error at most δ_μ, the triangle inequality gives only

$$\|\widehat{\mathcal{R}_N Z}(t) - Y_t\|_2 \le \sqrt{\frac{C}{N} \coth(\kappa\tau/2)} + \frac{\tau(N+1)}{2}\,\delta + \delta_\mu.$$

Increasing the reconstruction window can amplify derivative error. Any finite-data use must pay for both terms under an explicit detector model.

53.4 The spectral obstruction and a derivative-free population inverse

The exposure multiplier at frequency λ is

$$H_\tau(\lambda) = e^{-i\lambda\tau/2} \frac{\sin(\lambda\tau/2)}{\lambda\tau/2}, \qquad H_\tau(0) = 1. \tag{53.10}$$

It vanishes exactly at $\lambda = 2\pi k/\tau$, $k \in \mathbb{Z} \setminus \{0\}$. The decay assumption excludes spectral atoms at those frequencies. The minimal spectral premise for recovering a particular Y is that its spectral measure assign zero mass to those zeros; full mixing is a convenient sufficient assumption.

For completeness, the usual Landweber reconstruction gives a second proof that does not explicitly differentiate Z. On the stationary L^2 shift space let $A = A_\tau$ and $T = A^*A$, so $0 \preceq T \preceq I$. Define

$$\mathcal{L}_N Z = \sum_{j=0}^{N-1} (I - T)^j A^* Z. \tag{53.11}$$

Since $Z = AY$,

$$\mathcal{L}_N Z = [I - (I - T)^N] Y.$$

The spectral theorem gives L^2 convergence to Y exactly when Y has no component in $\ker A$. The kernel consists of the nonzero boxcar-resonant spectral modes above; the stationary mean is preserved because $H_\tau(0) = 1$. Each iterate is a finite linear combination of finite-window averages of Z and is therefore a fixed measurable function of its continuous path.

This inverse is also not uniformly stable as $N \to \infty$. Its operator norm is at most $\sqrt{N}$, because for $0 \le x \le 1$,

$$\sqrt{x} \sum_{j=0}^{N-1} (1 - x)^j = \frac{1 - (1 - x)^N}{\sqrt{x}} \le \min\{N\sqrt{x}, x^{-1/2}\} \le \sqrt{N}.$$

At $x = 0$ the expression is defined by its zero limit for fixed N. For a stationary L^2 perturbation of the entire input process, this bounds its amplification by $\sqrt{N}$. It is not an assertion that a finite collection of independent point-noise values forms a directly integrable continuous error process.

The spectral bias is explicitly

$$\|\mathcal{L}_N AY - Y\|_2^2 = \int (1 - |H_\tau(\lambda)|^2)^{2N} \, d\mu_Y(\lambda).$$

Vanishing mass at exact zeros ensures convergence; a useful finite-error rate additionally needs quantitative control of spectral mass near the zeros and at large frequencies. Injectivity alone supplies no such rate.

53.5 Consequences for the compatible diffusion minimum

Every admissible compact connected uniformly elliptic hidden diffusion in the observed-geometry chapter has a bounded smooth detector and exponentially decaying centered correlations. Consequently, if its ideal coordinate is observed through the detector contract here, the complete noisy dense-time law determines its ideal continuous observation law.

In particular, fix the noisy exposed law generated by the stationary cosine-circle process with parameters $\gamma > 0$ and ω, with the same calibrated detector channel used in all compatible models. Every compatible admissible hidden diffusion then has the same ideal law as that circle. The earlier attained minimum therefore remains

$$\min \sigma = \omega^2/\gamma. \tag{53.12}$$

The circle with the stated detector is feasible, so this is an attained minimum, not only a lower bound. The result concerns entropy production in the target hidden diffusion, not the unmodeled energy consumed by the detector or the readout process.

The identification uses infinitely rich population information. It does not show that a finite noisy experiment can attain the same accuracy as a noiseless high-bandwidth experiment. In particular, Z itself has zero quadratic variation. Local geometry is recovered only after an appropriate limiting reconstruction of the ideal law, rather than by applying the previous local formula directly to Z or B.

53.6 When the width itself is identifiable

Here and in the preceding compatible-minimum statement, the diffusion assumptions are those of the earlier observed-geometry class: a smooth compact connected manifold without boundary, smooth coefficients and strictly positive stationary density, uniform ellipticity, and ordinary reversal of the complete hidden state. No unspecified boundary condition or generalized reversal is included by the shorter phrase "compact elliptic" below.

Within the compact elliptic model class, a known boxcar shape has a further population property: its width can be identified from the exposed law even if that width was not initially supplied. Assume the detector is nonconstant. Let f now denote the centered detector, and use the complex $L^2(\pi)$ inner product. The symmetric part of the generator has a Poincaré gap, so its mean-zero semigroup decays exponentially and the resolvent exists at every $i\lambda$, $\lambda \in \mathbb{R}$. For $u = (i\lambda - L)^{-1} f$, the ideal power spectral density is

$$S_Y(\lambda) = 2\,\mathrm{Re}\langle f, (i\lambda - L)^{-1} f\rangle_\pi = 2\int_M \nabla \overline{u}^{\mathsf{T}} D \nabla u \, d\pi > 0. \tag{53.13}$$

The identity follows from $f = (i\lambda - L)u$ and skew-adjointness of the stationary current term. The Dirichlet form vanishes only for constant u; in the mean-zero subspace this would force $u = 0$ and then $f = 0$, contrary to the nonconstant detector assumption.

The centered exposed spectrum is $S_Z(\lambda) = |H_\tau(\lambda)|^2 S_Y(\lambda)$. Its zero set is therefore exactly the nonzero boxcar frequencies. The smallest positive zero equals $2\pi/\tau$ and identifies τ. Known-noise deconvolution may first be used to identify the exposure law; no finite-variance noise assumption is needed if the bounded-law uniqueness argument is used. This proposition assumes the boxcar shape, nonconstant diffusion detector, continuous population spectrum and calibrated time. A constant signal has no such width information. Estimating an exact spectral zero from a finite noisy record is a separate problem, and an unknown general response kernel is not identified by this argument.

53.7 Necessary distinctions

The mathematical ingredients have substantial prior art. Landweber's 1951 iteration is an established inverse method. Known-error deconvolution with characteristic-function zeros is treated, under quantitative assumptions, by Hall and Meister (2007). Nakakita and Uchida (2020) already estimate an unknown exposure-to-frame ratio for convolutionally observed diffusions in a shrinking-exposure, increasing-duration regime. The fixed-width full-law argument here is a different observation regime, not the first inference of an unknown shutter parameter. Markov-generator resolvent formulas for power spectra are also established; Gupta and Khammash (2022) provide a close explicit use for stochastic reaction networks. The focused primary-source audit is recorded separately in this research phase. These comparisons leave historical priority for any narrower combination open.

A stationary periodic signal with uniform random phase at frequency $2\pi/\tau$ is erased exactly by the boxcar. Such a process is stationary and can be ergodic under the full continuous time-shift group while failing mixing and ergodicity under the sampled τ-shift. Continuous-flow ergodicity alone is therefore not a sufficient replacement for the assumptions used here.

A fixed discrete sampling grid also supplies less information than the dense-time law in the theorem. Continuity does not reconstruct an exposed path between grid points. Exact aliasing or different hidden dynamics can survive fixed-grid exposure data, and no claim of injectivity for that access model follows from this chapter.

Finally, the inverse has no uniform stability bound over all admissible mixing signals. Boxcar multipliers become arbitrarily small near their zeros and at high frequency. Additional bandwidth, dynamical, detector or resource restrictions are needed for a useful finite-error conclusion. Population identifiability, physical attainability, finite-data confidence, and experimental energy are separate claims throughout this argument.

Chapter 54

Finite detector access to a sharp physical minimum

An ideal observation law can determine a physical minimum while a finite apparatus record remains unable to bound that minimum from above. The distinction can be quantified in the same circle family. At fixed exposure and positive read-noise variance, passive stationary trials and trials with an oriented phase preparation have different sample requirements for a specified binary comparison. Allowing read precision to consume a declared measurement energy changes that comparison again. This chapter proves all three statements in one explicit detector model.

The target is the already attained minimum compatible with the ideal cosine-circle law,

$$\mathcal{M}(\omega) = \frac{\omega^2}{\gamma}, \qquad \gamma > 0. \tag{54.1}$$

It is not an upper bound on the actual entropy of every hidden realization: an invisible independently dissipating subsystem would already rule out that stronger interpretation under perfect access. Here the detector obstructs inference of a functional that is sharply fixed by the ideal observed law itself.

54.1 One fixed exposure and its moments

Let the target diffusion satisfy

$$d\phi_t = \omega\, dt + \sqrt{2\gamma}\, dW_t \pmod{2\pi}.$$

An exposure of fixed length $\tau > 0$ yields

$$A_\omega = \frac{1}{\tau}\int_0^\tau \cos\phi_t\, dt, \qquad X_\omega = A_\omega + \xi, \qquad \xi \sim N(0, \nu^2), \quad \nu > 0, \tag{54.2}$$

where the read noise is independent of the entire target trajectory. The parameters γ, τ, and ν are known and fixed unless a different precision law is explicitly declared below. In every case $|A_\omega| \le 1$.

For passive stationary trials, the initial phase is uniform and $\mathbb{E}A_\omega = 0$. Define $V(\omega) = \mathbb{E}A_\omega^2$. The covariance $\mathbb{E}[\cos\phi_0 \cos\phi_t] = \frac{1}{2}e^{-\gamma|t|}\cos(\omega t)$ gives

$$V(\omega) = \frac{1}{\tau^2}\,\mathrm{Re}\left\{\frac{\tau}{z} - \frac{1 - e^{-z\tau}}{z^2}\right\}, \qquad z = \gamma - i\omega. \tag{54.3}$$

This follows by integrating the covariance over the exposure square, or equivalently by integrating $\tau^{-2}(\tau - t)e^{-\gamma t}\cos(\omega t)$ over $0 \le t \le \tau$.

For an explicit asymptotic comparison, use the resonant sequence

$$\omega_m = 2\pi m/\tau, \quad m = 1, 2, \dots, \qquad a_0 = 1 - e^{-\gamma\tau} > 0. \tag{54.4}$$

At these frequencies,

$$V(\omega) = \frac{(\gamma\tau + a_0)\omega^2 + (\gamma\tau - a_0)\gamma^2}{\tau^2(\gamma^2 + \omega^2)^2} \sim \frac{\gamma\tau + a_0}{\tau^2\omega^2}. \tag{54.5}$$

In particular,

$$\Delta_V(\omega) := |V(\omega) - V(2\omega)| \sim \frac{3(\gamma\tau + a_0)}{4\tau^2\omega^2}. \tag{54.6}$$

All asymptotic statements in this chapter fix the detector and confidence constants and take $m \to \infty$ along this sequence.

Two moment bounds hold uniformly over the initial phase distribution, and hence apply to both stationary and prepared trials. Set

$$H_2 = \frac{[2 + \gamma\tau + \sqrt{2\gamma\tau}]^2}{\tau^2}, \tag{54.7}$$

$$H_4 = \frac{8[(2 + \gamma\tau)^4 + 12\gamma^2\tau^2]}{\tau^4}. \tag{54.8}$$

For $\omega \neq 0$,

$$\mathbb{E}A_\omega^2 \le \frac{H_2}{\omega^2}, \qquad \mathbb{E}A_\omega^4 \le \frac{H_4}{\omega^4}. \tag{54.9}$$

To see this, Itô's formula gives

$$\omega\tau A_\omega = \sin\phi_\tau - \sin\phi_0 + \gamma\int_0^\tau \sin\phi_s\, ds - \sqrt{2\gamma}\int_0^\tau \cos\phi_s\, dW_s. \tag{54.10}$$

The bounded part has absolute value at most $2 + \gamma\tau$. For $M_t = \int_0^t \cos\phi_s\, dW_s$, Itô isometry gives $\mathbb{E}M_\tau^2 \le \tau$ and the fourth-moment identity gives

$$\mathbb{E}M_\tau^4 = 6\mathbb{E}\int_0^\tau M_s^2\cos^2\phi_s\, ds \le 6\int_0^\tau s\, ds = 3\tau^2.$$

Minkowski's inequality proves the second-moment bound. The elementary inequality $(x + y)^4 \le 8(x^4 + y^4)$ proves the fourth-moment bound. No independence between the bounded and martingale terms is assumed.

54.2 A finite record cannot provide a uniformly finite upper bound

Fix n predetermined exposure windows, each of width τ, from a stationary trajectory. The windows may overlap and their latent exposure values may be correlated. Let $P_\omega^{(n)}$ be the law of their noisy records under independent Gaussian read errors, and let $Q^{(n)} = N(0, \nu^2 I_n)$. Convexity of relative entropy under Gaussian mixing yields

$$D(P_\omega^{(n)}\|Q^{(n)}) \le \frac{\mathbb{E}\|A_\omega^{(n)}\|^2}{2\nu^2} = \frac{nV(\omega)}{2\nu^2} \le \frac{nH_2}{2\nu^2\omega^2}. \tag{54.11}$$

Indeed, conditional on the exposure vector, the readout is Gaussian with that vector as its mean and covariance $\nu^2 I_n$. Its relative entropy to $Q^{(n)}$ is the squared mean divided by $2\nu^2$; mixing can only decrease the relative entropy. Pinsker's inequality therefore implies $\mathrm{TV}(P_\omega^{(n)}, Q^{(n)}) \to 0$ as $\omega \to \infty$, whereas $\mathcal{M}(\omega) \to \infty$.

Theorem 54.1 (Upper-confidence obstruction for the sharp minimum)**.** *Suppose a measurable extended-valued statistic U of this finite record satisfies*

$$P_\omega^{(n)}\{U \geq \mathcal{M}(\omega)\} \geq 1-\alpha \quad \textit{for every } \omega > 0, \qquad 0 < \alpha < 1.$$

Then $Q^{(n)}\{U = +\infty\} \geq 1-\alpha$. In particular, no such uniformly covered upper bound is finite almost surely under even one of the finite-ω record laws. The conclusion concerns the fixed finite detector design and unbounded frequency class just stated.

Proof. For every finite K, sufficiently large ω has $\mathcal{M}(\omega) > K$, so

$$Q^{(n)}\{U > K\} \geq 1-\alpha-\mathrm{TV}(P_\omega^{(n)}, Q^{(n)}).$$

Let $\omega \to \infty$ and then $K \to \infty$ to obtain the first assertion. Each Gaussian mixture $P_\omega^{(n)}$ has a strictly positive density everywhere on $\mathbb{R}^n$, and is therefore mutually absolutely continuous with $Q^{(n)}$. If U were finite almost surely under any one $P_\omega^{(n)}$, it would also be finite almost surely under $Q^{(n)}$, contradicting the first assertion. □

The pure-noise law is a limiting reference, not necessarily a finite-parameter member of the model class. Accordingly, the theorem does not claim $P_\omega^{(n)}\{U = +\infty\} \geq 1-\alpha$ at every fixed ω. The general quantitative statement is

$$P_\omega^{(n)}\{U = +\infty\} \geq 1-\alpha-\mathrm{TV}(P_\omega^{(n)}, Q^{(n)}).$$

A calibrated frequency or generator envelope removes the unbounded class used in this argument. Adaptive timing, adaptive precision, or a growing number of measurements needs its own sequential analysis; it is not included in this fixed-design theorem.

54.3 Matching passive sample requirements at fixed precision

Now compare the known alternatives ω and 2ω using independent stationary trials, each with a fresh stationary preparation and one exposure. Define $n^*_{\mathrm{passive}}(\omega, \epsilon)$ as the minimum number of such trials needed for equal-prior binary error at most $\epsilon \in (0, 1/2)$. An upper construction below bounds each of the two conditional errors by ϵ, which is a stronger guarantee.

The zero stationary mean improves the mixture bound for independent trials. Let P be the one-record law of $A+\xi$, $Q = N(0, \nu^2)$, and let A' be an independent copy of A. Direct multiplication of Gaussian likelihood ratios gives

$$\chi^2(P\|Q) = \mathbb{E}e^{AA'/\nu^2} - 1 = \sum_{k=1}^{\infty} \frac{(\mathbb{E}A^k)^2}{k!\nu^{2k}}. \tag{54.12}$$

Bounded A justifies termwise expectation. Since $\mathbb{E}A = 0$ and $|\mathbb{E}A^k| \leq \mathbb{E}A^2 = V$ for every $k \geq 2$,

$$\chi^2(P\|Q) \leq C_{\nu,0}V^2, \qquad C_{\nu,0} = e^{1/\nu^2} - 1 - 1/\nu^2 > 0. \tag{54.13}$$

For the stationary circle the bound is stronger. Shifting every phase by π preserves its stationary path law and changes A to $-A$. All odd moments therefore vanish, and retaining only the even terms in Equation (54.12) gives

$$\chi^2(P\|Q) \leq C_\nu V^2, \qquad C_\nu = \cosh(1/\nu^2) - 1. \tag{54.14}$$

The sharper circle constant is used in the count bounds below. For independent trials, relative entropy tensorizes. Using $D \le \log(1+\chi^2) \le \chi^2$ and the triangle inequality for total variation,

$$\mathrm{TV}(P_\omega^{\otimes n}, P_{2\omega}^{\otimes n}) \le \sqrt{\frac{nC_\nu}{2}}\{V(\omega)+V(2\omega)\}.$$

The optimal equal-prior error is $(1-\mathrm{TV})/2$. Hence a necessary sample count is

$$n \ge \frac{2(1-2\epsilon)^2}{C_\nu\{V(\omega)+V(2\omega)\}^2}. \tag{54.15}$$

The constant can be numerically weak, especially for small ν; a bound below one is vacuous as an integer-count constraint. The asymptotic order remains informative at fixed positive ν.

For a matching upper bound, use the sample mean of X^2 and put its threshold midway between $\nu^2+V(\omega)$ and $\nu^2+V(2\omega)$. Gaussian moments and $|A| \le 1$ give

$$\begin{aligned}\mathrm{Var}(X^2) &= \mathbb{E}A^4 - V^2 + 4\nu^2 V + 2\nu^4 \\ &\le S_\nu(V) := (1+4\nu^2)V - V^2 + 2\nu^4.\end{aligned} \tag{54.16}$$

Chebyshev's inequality then bounds each conditional error by ϵ whenever

$$n \ge \left\lceil \frac{4\max\{S_\nu(V(\omega)), S_\nu(V(2\omega))\}}{\epsilon\,\Delta_V(\omega)^2} \right\rceil. \tag{54.17}$$

A simpler valid numerator is $4(1+6\nu^2+3\nu^4)$, obtained by bounding $\mathbb{E}X^4$ directly. The more precise expression above improves constants without changing the access or confidence criterion.

Equations (54.5)–(54.6) and the two sample bounds prove

$$n^*_{\mathrm{passive}}(\omega_m, \epsilon) = \Theta(\omega_m^4) = \Theta(\mathcal{M}(\omega_m)^2). \tag{54.18}$$

The constants may depend on fixed γ, τ, ν, and ϵ. This is a binary comparison along the declared sequence, not a minimax theorem for estimating an arbitrary unknown frequency from an arbitrary trajectory. Nonoverlapping windows from one stationary trajectory are not automatically independent stationary preparations.

54.4 Matching requirements with an oriented phase preparation

Suppose each independent trial can instead start from the actual known phase $\phi_0 = \pi/2$, under both hypotheses. Its expectation during the exposure is $\mathbb{E}\cos\phi_t = -e^{-\gamma t}\sin(\omega t)$. At resonance, the one-record mean is

$$\mu(\omega) = -\frac{a_0\omega}{\tau(\gamma^2+\omega^2)}, \qquad \Delta_\mu(\omega) = \frac{a_0|\omega|\,|2\omega^2-\gamma^2|}{\tau(\gamma^2+\omega^2)(\gamma^2+4\omega^2)} \sim \frac{a_0}{2\tau|\omega|}. \tag{54.19}$$

The gap is positive for sufficiently large resonant frequencies; at $\omega^2 = \gamma^2/2$ this particular mean test has zero gap and supplies no discrimination guarantee.

Since $A \in [-1,1]$, its centered version is sub-Gaussian with variance proxy one. Adding the independent Gaussian read noise gives proxy $1+\nu^2$. The sample-mean midpoint test therefore bounds each conditional error by ϵ with the sufficient count

$$n \ge \left\lceil \frac{8(1+\nu^2)\log(1/\epsilon)}{\Delta_\mu(\omega)^2} \right\rceil. \tag{54.20}$$

A lower bound of the same order follows without assuming stationarity. The uniform second-moment bound (54.9) and the Gaussian-mixture relative-entropy inequality give, for any common fixed phase-preparation rule,

$$D(P_\omega^{\otimes n}\|Q^{\otimes n}) \le \frac{nH_2}{2\nu^2\omega^2}.$$

For the pair $\omega, 2\omega$, Pinsker and the triangle inequality imply

$$\mathrm{TV}(P_\omega^{\otimes n}, P_{2\omega}^{\otimes n}) \le \frac{3\sqrt{nH_2}}{4\nu|\omega|}.$$

An equal-prior error at most ϵ consequently requires

$$n \ge \frac{16\nu^2(1-2\epsilon)^2\omega^2}{9H_2}. \tag{54.21}$$

The construction and obstruction together prove

$$n^*_{\text{prepared}}(\omega_m, \epsilon) = \Theta(\omega_m^2) = \Theta(\mathcal{M}(\omega_m)). \tag{54.22}$$

The preparation need not be a perfect point mass. A known initial phase law with $\mathbb{E}\cos\phi_0 = 0$ and a fixed nonzero $r = \mathbb{E}\sin\phi_0$ multiplies the mean in Equation (54.19) by r, preserving the exponent with a sufficient-count constant proportional to r^{-2}. Merely preparing $Y_0 = \cos\phi_0 = 0$ does not establish this condition: an equal mixture of $\pi/2$ and $3\pi/2$ has $r = 0$ and cancels the mean signal.

These prepared records are not stationary entropy-production records and are not inserted into the stationary local-defect certificate. They distinguish two generators whose stationary ideal laws have the target minima $\mathcal{M}(\omega)$ and $4\mathcal{M}(\omega)$. A phase preparation is a change in permitted access, with its own apparatus and energy requirements.

54.5 What changes when read precision consumes energy

The previous exponents count trials at fixed positive read-noise variance. They are not total-energy exponents. To make that distinction explicit, consider the following additional, declared sensor model:

$$\nu_i^2 = \frac{\kappa}{E_i}, \qquad \kappa > 0, \quad E_i > 0, \tag{54.23}$$

where E_i is the measurement energy allocated to record i. This is an assumed calibration law, not a universal thermodynamic law. The finite energy schedule is fixed before the data, and read errors remain independent of the target and each other. Preparation, reset, exposure duration, and actuation costs are not included in E_i.

Let $E_{\text{meas}} = \sum_i E_i$, and use the Gaussian reference with diagonal variances κ/E_i. Conditional Gaussian relative entropy and Equation (54.9) imply

$$D(P_\omega\|Q) \le \sum_i \frac{E_i\mathbb{E}A_{\omega,i}^2}{2\kappa} \le \frac{H_2E_{\text{meas}}}{2\kappa\omega^2}. \tag{54.24}$$

This bound permits correlated latent exposures at predetermined windows, provided each has the stated dynamics and duration. The initial phases can follow the same fixed preparation rule under

both hypotheses. The corresponding pairwise Pinsker bound gives the necessary measurement energy

$$\boxed{E_{\text{meas}} \geq \frac{16\kappa(1-2\epsilon)^2\omega^2}{9H_2}.} \tag{54.25}$$

The proof does not cover an energy schedule adaptively selected from noisy records or an intervention that changes the dynamics during exposure.

Passive stationary trials can attain the same exponent by buying more precision per record. Fix $c > 0$ and use $E_i = c\omega^2$ under both alternatives, so $\nu_i^2 = \kappa/(c\omega^2)$. This design is permitted because the binary candidate pair is known. Let

$$B_c = H_4 + \frac{6\kappa H_2}{c} + \frac{3\kappa^2}{c^2}. \tag{54.26}$$

Under either hypothesis, the Gaussian fourth-moment expansion and Equation (54.9) give

$$\mathbb{E}X^4 \leq \frac{B_c}{\omega^4}.$$

The midpoint test of the sample mean of X^2 consequently has both errors at most ϵ if

$$n \geq \left\lceil \frac{4B_c}{\epsilon\,\omega^4\Delta_V(\omega)^2} \right\rceil. \tag{54.27}$$

By Equation (54.6), the right-hand side remains bounded along the high-frequency resonant sequence. Thus a bounded number of passive trials, each with energy proportional to ω^2, suffices. Together with (54.25), this establishes

$$E^*_{\text{meas,passive}}(\omega_m, \epsilon) = \Theta(\omega_m^2). \tag{54.28}$$

Prepared trials at fixed ν already attain this measurement-energy order, because each such readout has fixed energy κ/ν^2 and their required count is $\Theta(\omega^2)$. Therefore, under the declared sensor law and allowing precision allocation,

$$\begin{aligned} E^*_{\text{meas,prepared}}(\omega_m, \epsilon) &= \Theta(\omega_m^2), \\ E^*_{\text{meas,passive}}(\omega_m, \epsilon) &= \Theta(\omega_m^2). \end{aligned} \tag{54.29}$$

Both are of order $\mathcal{M}(\omega_m)$ at fixed γ; their optimal constants and procedures have not been shown equal. Preparation does improve the fixed-precision sample exponent; this model supplies no improvement in the optimized measurement-energy exponent. A total-energy comparison would additionally need calibrated preparation, reset, control, exposure, and timing costs.

54.6 Operational scope and reproducible design

At fixed precision, n independent trials consume n preparations, n exposed scalar readouts, and $n\tau$ units of declared evolution time. The passive and prepared trial classes have different preparation access. Under the precision-energy model, passive inference can instead use a bounded trial count with increasing per-read energy. None of these accountings identifies declared evolution time with total laboratory time or with total energy.

The comparison closes a restricted operational problem: discrimination of known candidate generators along a specified high-frequency family, with fixed exposure and a declared observation channel. It is supported by exact moment identities, Gaussian-mixture information bounds, and

explicit tests. It does not establish a universal data cap, an optimal estimator for every unknown diffusion, or a new physical energy law. The elementary statistical ingredients are established; any narrower novelty lies in the particular physical target, access comparison, and assumptions, subject to prior-art review. Numerical design tables evaluate these proved formulas and their resource counts; they are not laboratory measurements or simulated detection successes unless separately accompanied by such records.

54.7 A reproducible design table

Table 54.1 evaluates the preceding exact bounds at $\gamma = \tau = 1$, $\nu = 1/2$, $\epsilon = 1/10$ and $\omega = 2\pi m$. The alternatives are the known pair ω and 2ω. Passive trials start independently from the uniform stationary phase. Prepared trials start independently from the actual oriented phase $\pi/2$. Each trial yields one exposed noisy scalar reading.

The passive necessary bound uses $C_\nu = \cosh(1/\nu^2) - 1$, justified by symmetry of the stationary exposure law. Its sufficient bound uses the tighter variance expression $S_\nu(V) = (1+4\nu^2)V - V^2 + 2\nu^4$. The prepared bounds are the information lower bound and sub-Gaussian sample-mean construction stated above. "Necessary" means every decision rule in that trial class requires at least that many records. "Sufficient" refers to the specified constructive test; the gap between the columns is not a determination of the exact optimum.

	Passive stationary		Prepared phase	
m	Necessary	Sufficient	Necessary	Sufficient
1	20	9 419	1	9 939
2	298	99 499	3	37 215
4	4 687	1 395 963	10	146 407
8	74 713	21 559 862	37	583 200
16	1 194 326	341 864 087	148	2 330 376
32	19 104 937	5 457 459 068	591	9 319 083

Table 54.1: Numerical ceilings of the proved count thresholds. Values use 100-decimal-digit arithmetic and their displayed ceilings agree at 180 digits. This is a deterministic evaluation of analytical bounds, not simulated success counts. The numerical implementation does not provide interval-certified rounding; the exact inequalities in the theorem are authoritative.

A separate calculation in the independent review uses exact rational inequalities to certify a finite separation at $m = 32$: 10 000 000 prepared exposures suffice, whereas 17 000 000 passive stationary exposures are insufficient for every decision rule in that trial class. This conclusion uses coarse proved enclosures for π, exponentials and logarithms; it does not rely on treating the table's numerical ceilings as certified intervals.

For completeness, put $w_- = 64(157/50)$ and $w_+ = 64(22/7)$. The elementary inequalities $157/50 < \pi < 22/7$, $8/3 < e < 11/4$, and $\log 10 < 7/3$ give

$$\Delta\mu \geq d_- := \frac{(5/8)w_-(2w_-^2 - 1)}{(1 + w_+^2)(1 + 4w_+^2)}, \qquad \frac{10 \log 10}{(\Delta\mu)^2} < \frac{70}{3d_-^2} < 10\,000\,000.$$

Also $C_\nu = \cosh 4 - 1 < 28$ and $V(\omega) + V(2\omega) < (5/4)(18/11)/w_-^2$, so the necessary passive threshold exceeds

$$\frac{2(4/5)^2 w_-^4}{28\big((5/4)(18/11)\big)^2} > 17\,000\,000.$$

The final comparisons involve rational numbers only. The independent test evaluates these inequalities with exact integer fractions.

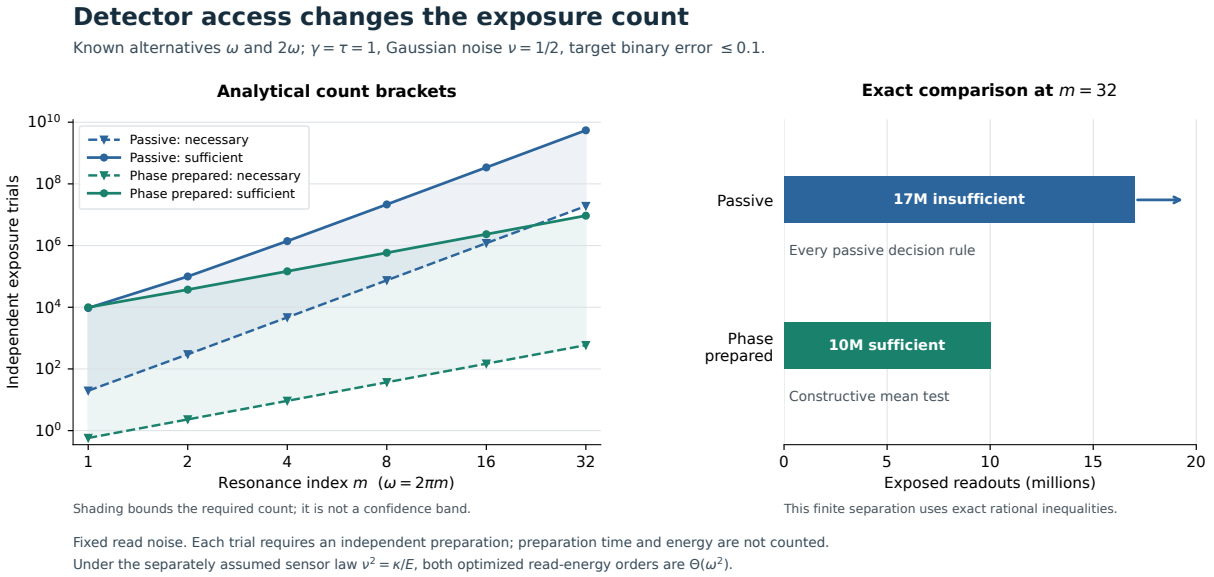


Figure 54.1: Analytical exposure-count bounds at fixed read noise. The shaded regions bracket the required number of trials; they are not statistical confidence bands. The right panel uses the separate exact finite-count certificate. Preparation and total apparatus energy are not matched.

The table makes the different fixed-precision count exponents visible. At the smallest displayed frequency, the prepared constructive guarantee is slightly more expensive than the passive constructive guarantee; the asymptotic statement does not promise improvement at every finite parameter. For a count n, both designs acquire n exposed readouts, require n independent preparations and spend $n\tau = n$ units of declared evolution time. The preparations require different physical access. Their duration, accuracy and energy have not been assigned a cost here.

These fixed-noise counts do not establish a total-energy advantage. Under the separately assumed sensor law $\nu_i^2 = \kappa/E_i$, optimizing read precision permits both trial classes to reach the measurement-energy order $\Theta(\omega^2)$, as proved above. Preparation, reset, driving, exposure, timing and calibration costs remain separate. No universal energy–precision law or apparatus performance is inferred from this table.

The implementation retains its exact formulas, 42-digit displays of the real thresholds, numerical ceilings and source hash in `detector/design/results.json`. Seven deterministic tests check the moments against complex integral formulas, information-bound constants, constructive error inequalities, asymptotic orders, precision stability, reproducibility and input guards.

Chapter 55

An exact ambiguity after exposure and fixed-rate sampling

A known exposure window does not remove the ambiguity caused by a fixed sampling interval. This chapter constructs mixing diffusion models whose complete exposed records agree at every integer time, although their continuous signal laws and ordinary hidden entropy rates differ. Equality holds for the entire sampled Gaussian law, so collecting more samples on the same grid cannot resolve it.

The example uses smooth, uniformly elliptic Ornstein–Uhlenbeck diffusions on Euclidean space and an unbounded linear detector. Its state space is *noncompact*. These assumptions differ from the compact, bounded-detector class used for the earlier sharp minimum theorem. The conclusion concerns an unbounded range of *actual compatible hidden entropy rates*; no claim about a change in the minimum compatible entropy follows from this construction.

55.1 An exposed Gaussian oscillator

Choose units in which the decay rate, exposure width and sampling interval are all one. Let

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \omega_m = 2\pi m, \qquad m \in \{0, 1, 2, \dots\}.$$

For $a > 0$, take the stationary Ornstein–Uhlenbeck diffusion in two dimensions

$$dX_t = (-I + \omega_m J) X_t \, dt + \sqrt{2a}\, dW_t, \qquad X_0 \sim N(0, aI). \tag{55.1}$$

Its diffusion matrix is $aI \succ 0$. Its invariant density is Gaussian, its coefficients are smooth, and its dynamics mix exponentially. If Y_t is the first coordinate, then

$$\operatorname{Cov}(Y_0, Y_t) = a e^{-|t|} \cos(\omega_m t). \tag{55.2}$$

The measured sequence consists only of the boxcar averages

$$Z_k = \int_{k-1}^{k} Y_t \, dt, \qquad k \in \mathbb{Z}. \tag{55.3}$$

Exposure is known exactly; no detector noise is needed for the ambiguity.

For a unit-variance oscillator ($a = 1$), set

$$x_m = (2\pi m)^2, \qquad r = e^{-1}, \qquad c_* = 2(\cosh 1 - 1).$$

Direct integration of (55.2) gives

$$V_m := \mathrm{Var}(Z_0) = \frac{2[r + (2-r)x_m]}{(1+x_m)^2}, \tag{55.4}$$

$$F_m = \frac{c_*(1-x_m)}{(1+x_m)^2}, \qquad \mathrm{Cov}(Z_0, Z_k) = F_m e^{-|k|}, \quad k \neq 0. \tag{55.5}$$

To verify the integration explicitly, put $\lambda_m = -1 + i\omega_m$. For $k \geq 1$,

$$V_m = 2\,\mathrm{Re}\,\frac{e^{\lambda_m} - 1 - \lambda_m}{\lambda_m^2},$$

$$\mathrm{Cov}(Z_0, Z_k) = \mathrm{Re}\left[e^{\lambda_m k}\frac{e^{\lambda_m} + e^{-\lambda_m} - 2}{\lambda_m^2}\right].$$

Since $e^{\lambda_m} = e^{-1}$ and $e^{\lambda_m k} = e^{-k}$ at integer k, every member has the same sampled exponential decay. The variance and its nonzero-lag coefficient retain the frequency dependence displayed in (55.5). For general a, both quantities are multiplied by a.

55.2 Matching complete sampled laws

Theorem 55.1 (Known exposure does not identify a fixed-grid diffusion law)**.** *For every integer $n \geq 2$, there are two stationary, exponentially mixing, smooth uniformly elliptic diffusion models with the following properties:*

1. *their known unit-width boxcar averages have identical complete probability laws on the integer grid;*
2. *their continuous latent scalar laws differ;*
3. *their ordinary hidden entropy rates are $8\pi^2$ and $8\pi^2 n^2$, respectively.*

The first model has two hidden coordinates; the second has four. Their state spaces are Euclidean and their scalar detectors are linear.

Proof. Let Model A be the $m = 1$, $a = 1$ oscillator. To construct the second model, define $R_m = F_m/V_m$. As a function of $x \geq 0$,

$$R(x) = \frac{c_*}{2}\frac{1-x}{r + (2-r)x}, \qquad R'(x) = -\frac{c_*}{[r + (2-r)x]^2} < 0.$$

Thus $R_0 > R_1 > R_n$ whenever $n \geq 2$. The coefficients

$$A_0 = \frac{V_1(R_1 - R_n)}{(R_0 - R_n)V_0}, \qquad A_n = \frac{V_1(R_0 - R_1)}{(R_0 - R_n)V_n} \tag{55.6}$$

are strictly positive. Their definitions imply the exact identities

$$A_0 V_0 + A_n V_n = V_1, \qquad A_0 F_0 + A_n F_n = F_1. \tag{55.7}$$

Indeed, the first is the sum of the two complementary interpolation weights multiplied by V_1; the second follows by multiplying them by R_0 and R_n.

Let Model B_n consist of two independent oscillators: an $m = 0$ oscillator with stationary coordinate variance A_0, and an $m = n$ oscillator with stationary coordinate variance A_n. Observe the sum of their first coordinates. The hidden diffusion matrix is $\operatorname{diag}(A_0 I_2, A_n I_2) \succ 0$, so this is an admissible smooth uniformly elliptic Euclidean diffusion. Independence and (55.7) imply equality of the exposed variance and of every nonzero-lag covariance with Model A. Both sampled sequences are centered Gaussian. Every finite-dimensional Gaussian distribution is determined by its mean and covariance matrix, so *all* finite-dimensional sampled laws agree, which establishes equality of the entire record laws.

The latent continuous covariance of Model B_n is

$$e^{-|t|}\big[A_0 + A_n \cos(2\pi n t)\big],$$

whereas Model A has covariance $e^{-|t|}\cos(2\pi t)$. These functions cannot agree: the trigonometric functions at frequencies 0, 1 and $n \geq 2$ are linearly independent, and both mixture coefficients are positive. Hence the continuous scalar laws differ.

Finally, specify ordinary reversal of both hidden coordinates in each oscillator. The Gaussian stationary density separates the drift in (55.1) into the reversible part $-X$ and current velocity $v = \omega_m J X$. Therefore

$$\sigma_m = \mathbb{E}\big[(\omega_m J X)^{\mathsf{T}} (aI)^{-1} (\omega_m J X)\big] = 2\omega_m^2. \tag{55.8}$$

Model A has entropy $8\pi^2$. Entropy is additive for the independent components of Model B_n: the zero-frequency component contributes zero, and the $m = n$ component contributes $8\pi^2 n^2$. This completes the proof. □

For $n = 2$, the positive weights are approximately

$$A_0 = 0.0004577350937, \qquad A_2 = 3.8531460671.$$

The latent variance is 1 in Model A and approximately 3.8536038022 in Model B_2. Nevertheless, both exposed integer-grid records have variance approximately 0.07909835314 and covariance $-0.02550729842\, e^{-|k|}$ at nonzero integer lag k. These decimal values illustrate the exact identities; they are not the basis of the proof.

55.3 What the ambiguity establishes

Letting n increase leaves the complete sampled observation law fixed while the constructed compatible hidden entropy rates grow without bound. Every individual model remains finite dimensional and uniformly elliptic, but there is no common frequency or diffusion-coefficient bound across the sequence. Such additional prior restrictions would define a different identification problem. The entropy convention is also substantive: (55.8) uses ordinary reversal, not a reversal declaring one oscillator coordinate odd.

The equality survives every common processing rule for the sampled record. It also survives adding an independent detector-noise process having the same specified law under both models. Further samples on the same grid cannot distinguish the alternatives. Changing the observation times, retaining the continuous exposed trace, or applying a physical intervention changes the access contract and is not covered by the equivalence.

There is no conflict with continuous-law recovery under mixing. The full continuously indexed exposed laws are more informative than their integer-grid restrictions. The recovery theorem concerns the former; Theorem 55.1 concerns the latter. Nor does this construction establish that the

compact-class minimum-entropy theorem extends to these noncompact OU models. Its conclusion is the stated fixed-grid ambiguity and the absence of a finite upper bound on actual compatible ordinary entropy within the declared OU family.

The companion implementation evaluates (55.5)–(55.6) and records numerical covariance residuals and entropy values. Five tests check the exposure formulas against independent integration, positivity of the mixture coefficients, covariance matching, and entropy scaling, and reject invalid indices. Those computations diagnose implementation errors. Exact Gaussian-law equivalence for arbitrarily long records follows from the analytic argument above; neither a finite covariance checklist nor simulated data alone would establish it. No laboratory validation or historical-priority claim is made for this construction.

Chapter 56

Learning the calibration that an entropy certificate needs

An inference method is usable only when its calibration assumptions can be obtained or justified. The preceding detector chapter established when a complete observation law retains the ideal diffusion information. A finite experiment must also account for uncertainty in the detector and in the diffusion scale. This chapter introduces a construction that does so, together with a precise obstruction to removing all structural assumptions.

56.1 The question answered here

Suppose a bounded signal is recorded through a shutter and a noisy detector. Can the same finite records both calibrate its local diffusion scale and certify positive entropy production, without treating either a fitted scale or a detector-noise estimate as exact?

The answer developed below is conditional on a specified local geometry:

$$|f| \le 1, \qquad \Gamma_f = \gamma(1 - f^2), \qquad \gamma > 0.$$

The unknown constant γ is estimated. The pointwise functional relation is a premise. Hidden dimension, hidden drift and stationary density are not fixed to the circle used in the numerical example. Uniform bounds on the relevant first and second generator derivatives remain additional premises. This distinction is substantive: a compact countermodel proves that finite passive records cannot uniformly certify those global bounds over an unrestricted smooth diffusion class.

56.2 What each part of the experiment supplies

Repeated detector readings of one exposure estimate powers of its underlying signal through products of distinct readings. An ordinary power of one noisy reading does not do this. Conditional independence of the detector randomizations, a common conditional mean and a bounded output range are the required noise assumptions; the method does not insert a known read-noise variance.

A constant reference signal and a timed step reference cover a common multiplicative gain and the width of a uniform shutter. These are actual additional reference operations, with their own observations and costs. They do not establish the local-geometry identity or the generator bounds. A detector with an unknown offset, changing gain or dependent replica noise requires a different contract.

Time-separated target records provide a finite-step estimate of the average local diffusivity. The structural relation then converts this into a covered bound on γ. A simultaneous region for fixed measured moments carries its uncertainty through the entropy calculation. Calibration and testing can use the same target records because their joint coverage is proved; neither step is silently treated as certain.

56.3 A hierarchy with a finite cost

The polynomial weights

$$w_m(y) = 1 + y^2 + \cdots + y^{2m}$$

obey $w_m(f)\Gamma_f \le \gamma$. They replace an inverse local-diffusivity weight by a finite polynomial that independent replicas can measure. Order m uses $2m + 2$ center readings and one reading at each endpoint. All readings, including additional replicas, consume physical access.

For the stationary cosine-circle example, the ideal limit at order m recovers the fraction

$$1 - \frac{\binom{2m+2}{m+1}}{4^{m+1}}$$

of the exact entropy minimum. This fraction tends to one. The constructive consistency theorem accounts for an unknown diffusion scale, gain calibration, finite time steps, shrinking exposure and increasing numerical precision. It does not claim consistency at a fixed nonzero shutter width or practical optimality of its conservative acquisition schedule.

The sharp result extends beyond a constant hidden current. Let a reversible hidden process Z modulate angular drift through $c(Z)$, while the measured signal remains $\cos\phi$. Within the stated ordinary time-reversal class, the compatible entropy minimum is exactly $\mathbb{E}[c(Z)^2]/\gamma$. It can be positive even when $\mathbb{E}[c(Z)] = 0$, and the complete measured signal law is time-reversible. Thus mean circulation and observed temporal asymmetry can both vanish while the observation and its local geometry still force dissipation. This is an analytic extension; the simulation below uses constant angular drift.

At a finite budget, more replicas and a higher polynomial order can reduce performance. The implementation therefore retains the lower orders and compares them under one simultaneous confidence event. Fixed linear contrasts preserve correlations between measured moments before concentration; remaining gain uncertainty is paid through the same covered region. Selecting the largest covered certificate is justified by that event, rather than by an uncorrected search through individually valid tests.

56.4 How to read the extension

The calibration-limit chapter explains why additional structure is necessary. The constructive chapter gives the detector contract, the exposure error bounds, the joint confidence theorem and the hierarchy. The virtual experiment then evaluates the actual finite procedure, including its simpler comparator and unsuccessful regimes. An independent audit checks the proofs and the implementation; the literature chapter identifies the established inference methods on which the construction depends.

The numerical experiments use a circle to generate records, but the reported geometry certificates use the broader stated structural class. A fit that assumes the complete circle dynamics would answer a different inverse problem. The mathematical results, simulated observations and requirements for a future physical experiment retain separate evidence status throughout.

Chapter 57

Joint detector calibration and an entropy certificate

Finite detector records need not supply a pointwise volatility estimate in order to certify dissipation. Under an explicit structural relation between the detector coordinate and its local diffusivity, simultaneous moment bounds can instead calibrate the unknown geometric scale and a polynomial entropy score. The construction below accounts for exposure, independent replica noise and a calibrated detector gain. It does not infer the structural relation itself from the records.

57.1 Physical and acquisition assumptions

The hidden process is a stationary smooth uniformly elliptic diffusion on a smooth compact connected manifold without boundary, with positive smooth stationary density π. Reversal is ordinary hidden-state reversal. Write $L = S + A$, $L^\dagger = S - A$, $Af = v \cdot \nabla f$, and

$$\Gamma(f, g) = \nabla f^{\mathsf{T}} D \nabla g, \qquad e = v^{\mathsf{T}} D^{-1} v, \qquad \sigma = \mathbb{E}_\pi e.$$

In this convention $d[f(X)]_t = 2\Gamma_f(X_t)\, dt$. For a smooth real coordinate f with $|f| \le 1$, assume the pointwise identity

$$\Gamma_f(x) = \gamma(1 - f(x)^2), \qquad \gamma > 0 \text{ unknown and constant.} \tag{57.1}$$

Hidden dimension and drift are otherwise unrestricted in the stated class. An average relation, a stationary marginal or an estimated conditional mean of Γ_f is not a substitute for (57.1). The generator bounds $M_{1,\pm}, M_{2,\pm}$ from the finite-resolution chapter are retained:

$$\|Lf\|_\infty \le M_{1,+}, \qquad \|L^\dagger f\|_\infty \le M_{1,-},$$
$$\|L^2 f\|_\infty \le M_{2,+}, \qquad \|(L^\dagger)^2 f\|_\infty \le M_{2,-}.$$

These are physical calibration assumptions. Set $M_* := \min(M_{1,+}, M_{1,-})$ and $M_{2,*} := \min(M_{2,+}, M_{2,-})$.

Each independent stationary trial has nominal centers $-h, 0, h$ and a common causal uniform exposure

$$A_t = \frac{1}{\tau} \int_{t-\tau}^{t} f(X_s)\, ds, \qquad 0 \le \tau \le \tau_U \le h. \tag{57.2}$$

At $\tau = 0$, interpret $A_t = f(X_t)$. Acquire $K = 2m + 2$ center replicas and one endpoint readout at each of $-h, h$. Conditional on the entire hidden path, all detector randomizations in a trial are

independent and each bounded readout Z has mean A_t. A binary shot with mean A_t is an example. Conditional noise independence does not assert independent hidden states. Unknown gain is treated separately below. Repeated powers of a single noisy readout do not satisfy this replica contract.

57.2 An exactly unbiased replica polynomial

Let e_k be the normalized elementary symmetric polynomial of the K center readouts: the average of all products indexed by distinct k-subsets. Set $e_0 = 1$. The replica assumptions imply $\mathbb{E}[e_k \mid X_{\mathbb{R}}] = A_0^k$. For fixed $\lambda \in \mathbb{R}$, put

$$b = 1 - \lambda h, \qquad \beta_g = \frac{h}{4}(M_{2,+} + M_{2,-}), \tag{57.3}$$

$$\beta_K = \frac{h}{2}(M_{1,-}M_{2,+} + M_{1,+}M_{2,-}), \qquad \beta = 2|\lambda|\beta_g + \beta_K. \tag{57.4}$$

Define the record score

$$T_m = \frac{1}{h^2}\sum_{j=0}^{m}\Big[-Z_-Z_+e_{2j} + b(Z_- + Z_+)e_{2j+1} - b^2 e_{2j+2}\Big] - \beta\sum_{j=0}^{m} e_{2j}. \tag{57.5}$$

Its expectation is exactly the displayed polynomial in A_-, A_0, A_+. Write T_m^{ideal} for that polynomial evaluated at $Y_t = f(X_t)$ instead.

Proposition 57.1 (Ideal polynomial inequality)**.** *Under* (57.1)*,*

$$\mathbb{E}T_m^{\mathrm{ideal}} \le \gamma\sigma. \tag{57.6}$$

Proof. Set $w_m(y) = \sum_{j=0}^{m} y^{2j}$ and $q(y) = -\lambda y$. The uncorrected ideal polynomial is

$$-w_m(Y_0)\left(q(Y_0) - \frac{Y_h - Y_0}{h}\right)\left(q(Y_0) - \frac{Y_{-h} - Y_0}{h}\right).$$

The two-sided semigroup bounds in the finite-resolution chapter control its conditional time-step bias by $2|q|\beta_g + \beta_K \le \beta$. Conditional on the complete center state, the corrected expectation is therefore at most

$$w_m(f)\{2qSf - q^2 - (Sf)^2 + (Af)^2\} = w_m(f)\{(Af)^2 - (q - Sf)^2\}.$$

Since $(Af)^2 \le \Gamma_f e$ and $w_m(f)\Gamma_f = \gamma(1 - f^{2m+2}) \le \gamma$, integration proves the claim. The argument does not clip individual records or intermediate expectations. □

57.3 A multitime moment bound for exposure

Lemma 57.2 (Bidirectional gap derivative)**.** *For the stationary ordered moment $F(t_1, \dots, t_d) = \mathbb{E}\prod_i f(X_{t_i})$, its derivative in gap j satisfies*

$$\left|\partial_{t_{j+1}-t_j} F\right| \le a_{d,j} := \min\{j^2 M_{1,-}, (d-j)^2 M_{1,+}\}, \qquad 1 \le j < d. \tag{57.7}$$

The assertion includes one-sided limits at coincident times.

Proof. A backward semigroup word g containing k factors f obeys $\|g\|_\infty \le 1$ and $\|Lg\|_{L^1(\pi)} \le k^2 M_{1,+}$. For the induction, semigroup application preserves this bound by stationary L^1 contraction and commutation with L. Multiplication uses

$$L(fg) = fLg + gLf + 2\Gamma(f,g),$$
$$\mathbb{E}\Gamma_f = -\mathbb{E}fLf \le M_{1,+}, \qquad \mathbb{E}\Gamma_g = -\mathbb{E}gLg \le \|Lg\|_1.$$

The Cauchy inequality for Γ gives the new bound $B + M_{1,+} + 2\sqrt{BM_{1,+}}$ when the old bound is B; if $B \le (k-1)^2 M_{1,+}$, the result is at most $k^2 M_{1,+}$. At a gap, the conditional product of preceding factors has absolute value at most one. Differentiating the suffix and integrating its generator in $L^1(\pi)$ yields the $(d-j)^2 M_{1,+}$ bound. Applying the same argument under stationary time reversal gives $j^2 M_{1,-}$. Smooth compactness justifies the generator identities and differentiation. □

For arbitrary causal exposure supported in $[t_i - \tau, t_i]$, sorted displacements from the nominal ordered times lie in $[-\tau, 0]$, so each gap changes by at most τ. Consequently a degree-d product changes by at most

$$\tau \sum_{j=1}^{d-1} a_{d,j}. \tag{57.8}$$

Products of exposure averages are integrals over auxiliary exposure times; these integration variables do not imply independence of signal values.

Common uniform boxes give explicit sharper constants. Suppose the nominal windows do not overlap and have chronological multiplicities $(p_1, \dots, p_r)$, deleting zero entries, with $d = \sum_i p_i$. Define $B(p_1, \dots, p_r)$ by summing the following contributions at each global prefix k. An internal gap of a cluster with p factors contributes

$$\frac{a_{d,k}}{p+1}. \tag{57.9}$$

A gap between consecutive clusters with p and q factors contributes

$$a_{d,k}\left[1 - \frac{1}{p+1} - \frac{1}{q+1} + \frac{2p!q!}{(p+q+1)!}\right]. \tag{57.10}$$

The resulting blur radius is $\tau_U B(p_1, \dots, p_r)$; degrees zero and one have radius zero. To verify the formula, internal uniform order-statistic spacings have mean $1/(p+1)$. A cross-gap displacement divided by τ is $U_{\min,q} - U_{\max,p}$. Its positive part has expectation

$$\int_0^1 t^p (1-t)^q \, dt = \frac{p!q!}{(p+q+1)!},$$

while the mean of its negative is $1 - 1/(p+1) - 1/(q+1)$. This yields its expected absolute value in (57.10). Integrate (57.7) along the segment between the two nonnegative gap vectors and then average.

57.4 Calibrating the geometric scale

Let

$$C(t) = \mathbb{E}[Y_0 Y_t], \quad D_0 = \mathbb{E}A_0^2, \quad D_h = \mathbb{E}[A_0 A_h], \quad q_0 = \mathbb{E}\Gamma_f.$$

Stationarity implies $C(-t) = C(t)$, $C'(0+) = -q_0$, $q_0 \le M_*$ and $|C''(t)| \le M_{2,*}$ for $t \ge 0$, with a right derivative at zero. Hence

$$0 \le C(0) - D_0 \le \delta_0 := \frac{M_* \tau_U}{3}, \tag{57.11}$$

$$|D_h - C(h)| \le \delta_h := \frac{M_{2,*} \tau_U^2}{12}. \tag{57.12}$$

The first bound uses $C(t) \le C(0)$, the Lipschitz constant M_* and $\mathbb{E}|U - V| = \tau/3$ for two uniform exposure offsets. The second uses their centered difference, whose variance is $\tau^2/6$, in the Taylor remainder; $h \ge \tau$ prevents passage through the covariance cusp. The same bound holds at $2h$. A more precise identity is $D_0 = C(0) - q_0\tau/3 + R_0$, with $|R_0| \le M_{2,*}\tau^2/12$; the displayed bound avoids an unknown q_0 in the calibration envelope.

Write $\mathcal{D}(p_1, \dots, p_r)$ for either valid general blur radius above, optionally replacing a degree-two radius by δ_0 for one window and by δ_h for distinct windows separated by at least h. Define

$$\delta_N = \frac{1}{h^2} \sum_{j=0}^{m} \{\mathcal{D}(1, 2j, 1) + |b|[\mathcal{D}(1, 2j+1) + \mathcal{D}(2j+1, 1)] + b^2 \mathcal{D}(2j+2)\} \tag{57.13}$$

$$+ \beta \sum_{j=0}^{m} \mathcal{D}(2j). \tag{57.14}$$

Delete zero clusters before evaluation. The last term accounts for blur of the bias-correction moments. The two cross terms need not have identical bounds when the forward and adjoint caps differ.

The following three population quantities now have compatible directions:

$$P = \mathbb{E}T_m - \delta_N \le \gamma\sigma, \tag{57.15}$$

$$\overline{A} = 1 - D_0 - \delta_0 \le A_{\text{true}} := 1 - C(0), \tag{57.16}$$

$$\overline{H} = \frac{D_0 - D_h + \delta_0 + \delta_h}{h} + \frac{M_{2,*} h}{2} \ge \gamma A_{\text{true}}. \tag{57.17}$$

The last inequality follows from $|C(0) - C(h) - hq_0| \le M_{2,*}h^2/2$ and the exact relation $q_0 = \gamma(1 - C(0))$. This step uses neither an arcsine marginal nor a local-volatility estimate.

57.5 A simultaneous finite-record certificate

Theorem 57.3 (Joint geometry and score calibration)**.** *Assume simultaneous bounds $p \le P$, $a \le \overline{A}$ and $u \ge \overline{H}$ hold with probability at least $1 - \alpha$ for the declared acquisition. Set*

$$\widehat{L} = \begin{cases} [p]_+[a]_+/u, & u > 0, \\ 0, & u \le 0, \end{cases} \tag{57.18}$$

marking the latter case inconclusive. Then $\Pr(\sigma \ge \widehat{L}) \ge 1 - \alpha$. On the same confidence event, if $a > 0$, then $\gamma \le u/a$.

Proof. If $p, a > 0$, then $A_{\text{true}} > 0$ and $u \ge \gamma A_{\text{true}} > 0$. Therefore $pa/u \le \sigma a/A_{\text{true}} \le \sigma$. Otherwise the displayed lower bound is zero. The geometric upper bound follows from $u \ge \gamma A_{\text{true}} \ge \gamma a$. □

One construction covers the means of T_m, e_2 and $(Z_- + Z_+)e_1/2$ simultaneously using the same independent trials. Their population means are respectively $\mathbb{E}T_m, D_0, D_h$. Interval arithmetic

supplies the required lower and upper endpoints. Unconditional bounded-mean concentration for each statistic, followed by a declared union bound, remains valid when the statistics share records. There is no need to split data merely because the geometric scale and the entropy score are calibrated together.

For $|Z| \le B$, $|e_k| \le B^k$, so valid finite score ranges follow directly. Hoeffding or a valid empirical-Bernstein inequality supplies the mean intervals. Numerical endpoints require outward enclosure. A variance estimate cannot be substituted into a theorem requiring a known variance. Selecting m or λ from the same records is valid only if the underlying moment region or a finite-family confidence event covers that selection. Applying an ordinary iid score inequality after an uncovered coefficient selection would not establish the theorem's premise.

57.6 Unknown detector gain and a physical reference protocol

Suppose a raw bounded readout R has mean κA_t conditional on the hidden path, with unknown common constant $\kappa > 0$. A raw degree-d product has mean κ^d times its exposed moment. Cover its raw mean and project that interval over a simultaneously covered positive gain region; this is different from inserting an estimated gain as if exact. In the formulas of this gain-calibration section and the raw-contrast projection below, e_k denotes the elementary polynomial of the raw center readouts R, before gain correction.

For example, the complete raw contrast

$$S_j = \frac{1}{h^2}\left[-R_-R_+e_{2j} + b(R_- + R_+)e_{2j+1} - b^2 e_{2j+2}\right] \tag{57.19}$$

has homogeneous degree $2j+2$. Its mean interval can be divided by κ^{2j+2} while retaining cancellations within the contrast. The degree-$2j$ bias-correction means are handled separately. Signed endpoints must be minimized or maximized over the gain interval: dividing every endpoint by its lower gain is invalid for negative means. A full joint region may preserve the common gain across degrees; rectangular projection is conservative. If the gain region reaches zero, finite inverse bounds may be unavailable and the procedure can abstain.

A concrete bounded reference protocol can calibrate both gain and exposure. Feed the same detector a known constant signal one, whose mean readout is κ. Also feed a known step from zero to one at time zero. If a coarse metrological bracket permits a fixed reference time $0 < s < \tau_{\min} \le \tau$, the step's mean readout is

$$p_{\text{step}} = \kappa s/\tau. \tag{57.20}$$

Simultaneous bounded-mean confidence intervals yield the joint constraints

$$\kappa \in [\kappa_L, \kappa_U], \qquad p_{\text{step}} \in [p_L, p_U], \qquad p_{\text{step}}\tau = \kappa s.$$

When $p_L > 0$, a conservative exposure interval is

$$\left[\frac{\kappa_L s}{p_U}, \frac{\kappa_U s}{p_L}\right] \tag{57.21}$$

intersected with the coarse bracket. Without the ramp condition the correct response is $\kappa \min(1, s/\tau)$, and that relation must be retained. An empty region is an inconclusive/model-check flag. A nonpositive lower step mean can leave the exposure upper bound inconclusive.

These references require a known input shape, a known time origin and common detector response between reference and target. They do not validate (57.1) or the generator caps. Cover all

raw target and reference means with one declared error allocation. Dependence among statistics from the same trials is harmless to a union bound; independence across the trials used by each concentration inequality is still required. Adaptive reference design requires its own valid confidence argument.

57.7 Retaining covariance before nuisance projection

Intervals for separate polynomial components can discard useful cancellations. A predetermined raw contrast can retain those cancellations before concentration, while still allowing an uncertain gain afterward. Choose a nominal $\kappa_0 > 0$ as a predeclared fixed design constant, or using a pilot independent of the inference records, and include in the same covered mean vector, all S_j, the required e_{2j}, the calibration contrast, and

$$R_\ell = \sum_{j=0}^{\ell} \frac{S_j}{\kappa_0^{2j+2}}, \qquad 0 \le \ell \le m. \tag{57.22}$$

Here the S_j include the h^{-2} factor in (57.19). A confidence interval for $\mathbb{E}R_\ell$ is computed from its actual record variance or range, not by summing the individual S_j margins. This is the established principle of concentrating a fixed linear contrast before projecting nuisance uncertainty.

Suppose $\mathbb{E}S_j \in [L_j, U_j]$ and $\mathbb{E}R_\ell \ge L_{R_\ell}$ on the common confidence event. For $d_j = 2j+2$, define

$$I_j = [\kappa_U^{-d_j} - \kappa_0^{-d_j}, \kappa_L^{-d_j} - \kappa_0^{-d_j}], \qquad c_j = \min\{xy : x \in \partial I_j,\ y \in \{L_j, U_j\}\}.$$

The four-corner minimum is valid because a bilinear function on a rectangle attains its extrema at corners. For every gain in the covered interval,

$$\sum_{j=0}^{\ell} \frac{\mathbb{E}S_j}{\kappa^{2j+2}} = \mathbb{E}R_\ell + \sum_{j=0}^{\ell} (\kappa^{-d_j} - \kappa_0^{-d_j})\mathbb{E}S_j \ge L_{R_\ell} + \sum_{j=0}^{\ell} c_j. \tag{57.23}$$

Subtract the covered upper bound on $\beta \sum_{j=0}^{\ell} \mathbb{E}e_{2j}/\kappa^{2j}$ and the appropriate δ_N to obtain a lower bound for P_ℓ. The maximum of this bound and the componentwise lower bound remains valid on the same event. All prefixes $\ell \le m$ use the same $K = 2m+2$ genuine replicas: their elementary averages remain unbiased. Once the common confidence vector covers every contrast actually used, choosing the largest certificate across these prefixes spends no further error probability. Adding an uncovered contrast after inspecting the data would require additional justification.

57.8 An endpoint entropy comparison

The order-zero raw contrast also provides a comparison that does not require the structural identity (57.1). Keep the stationary ordinary diffusion and detector assumptions and let $0 \le b \le 1$. Suppose the common confidence event gives s below the ideal, gain-corrected and blur-corrected mean

$$N_0 = \frac{2bC(h) - C(2h) - b^2 C(0)}{h^2}. \tag{57.24}$$

This endpoint comparison subtracts the raw order-zero blur radius, but not the β penalty used by the derivative-based entropy score. Then

$$\sigma \ge \frac{2h^3[s]_+^2}{(2b+\sqrt{2})^2}. \tag{57.25}$$

This is the earlier reversible-support and entropy-bridge construction, expressed in the current covariance normalization.

To check constants, compare L with $S = (L + L^\dagger)/2$ using the same stationary density, diffusion tensor and coordinate. The diffusion Girsanov identity gives $D(P_L^{[0,t]} \| P_S^{[0,t]}) = \sigma t/4$. Pinsker and the bounded endpoint product yield $|C_L(t) - C_S(t)| \le \sqrt{\sigma t/2}$. The zero-lag moments agree exactly, and reversibility gives

$$2bC_S(h) - C_S(2h) - b^2 C_S(0) = -\|(e^{hS} - bI)f\|^2_{L^2(\pi)} \le 0.$$

It follows that $h^2[N_0]_+ \le (2b + \sqrt{2})\sqrt{\sigma h/2}$, which proves (57.25). Exposure errors must be removed before using the ideal zero-lag cancellation. Both entropy bounds can be reported on the same simultaneous confidence event. Their difference then reflects different physical inequalities, with detector and calibration access held fixed, rather than a comparison against an uncorrected estimator.

57.9 A sharp limiting hierarchy with unknown scale

For the stationary circle $d\phi = \omega\, dt + \sqrt{2\gamma}\, dW$ and $f = \cos\phi$, one has $Sf = -\gamma f$ and $(Af)^2 = \omega^2(1 - f^2)$. The zero-step weighted target divided by γ is exactly

$$J_m(\lambda) = \frac{\omega^2}{\gamma}(1 - \mathbb{E}Y^{2m+2}) - \frac{(\lambda - \gamma)^2}{\gamma}\mathbb{E}[w_m(Y)Y^2]. \tag{57.26}$$

For $\lambda = \gamma$, this increases to the attained compatible-law minimum $\sigma = \omega^2/\gamma$, since

$$\mathbb{E}Y^{2m+2} = \frac{\binom{2m+2}{m+1}}{4^{m+1}} = O(m^{-1/2}).$$

A grid point with $|\lambda - \gamma| \le \eta$ loses at most $(m + 1)\eta^2/\gamma$. A deterministic finite grid with simultaneous coverage therefore permits choosing the largest certified lower bound without knowing γ. Independently, u/a converges to γ as the moment error, exposure error divided by h, and h vanish with $C(0) < 1$.

For clarity, a conservative sufficient sequence makes the statement constructive and removes the need for an oracle choice of λ. Use raw target and reference readouts bounded by one, as in the binary detector example. Fix positive parameter envelopes for the circle and valid resulting generator caps. Suppose a known integer $B \ge 0$ obeys $\kappa \ge \kappa_{\min} \ge 2^{-B}$. In a fixed time unit, at stage k take

$$\begin{aligned} m = k, \qquad h_k = 2^{-k}, \qquad \tau_k \le 2^{-4k}, \\ K_k = 2k + 2, \qquad n_k = n_{\mathrm{ref},k} = 2^{(8B+16)k}. \end{aligned} \tag{57.27}$$

Use the finite grid $[0, k] \cap 2^{-k}\mathbb{Z}$ for λ. References use a known coarse shutter bracket shrinking at the stated exposure scale, so their upper exposure bound also has that order. Allocate a fixed total error probability over all covered means and grid scores, or a summable stagewise allocation. Eventually the grid contains a point within 2^{-k} of the fixed true γ.

The raw score ranges are at most a polynomial in k times 2^{2k}. Correcting degree at most $2k + 2$ for gain multiplies interval widths by at most $2^{2B(k+1)}$. A bounded-mean radius with the declared sample size is therefore at most a polynomial in k times $2^{-(2B+6)k}$; the logarithm of the finite-family size is $O(k)$. Gain uncertainty has the same vanishing order up to fixed and polynomial factors, by differentiating κ^{-d} on the known positive gain interval. The calibration reference count is included in (57.27).

The generic exposure bound gives $O(k^4\tau_k/h_k^2) = O(k^4 2^{-2k})$ for raw-score blur. The bias-correction blur is smaller. Uniformly over the finite grid, the finite-step penalty is $O(k^2 2^{-k})$. The nearest grid point loses $O(k4^{-k})$ in (57.26), while the gamma upper bound converges to the true γ. Direct uniform concentration around the covered means now gives

$$\begin{aligned} \max_{\lambda \text{ on the grid}} \widehat{L}_k &\longrightarrow \sigma \quad \text{in probability,} \\ \sigma - \max \widehat{L}_k &= O_{\mathbb{P}}(k^{-1/2}) = O_{\mathbb{P}}((\log n_k)^{-1/2}). \end{aligned} \tag{57.28}$$

In the last expression the difference is signed; the stochastic error is smaller than the displayed order. This rate is a conservative sufficient rate, not a minimax assertion. Direct concentration proves convergence even at a fixed advertised confidence level; coverage alone would not.

Numerical enclosure error must also vanish after degree and time-step amplification. For example, carrying certified elementary-operation enclosures to $k^2 + (4B + 12)k$ binary digits is ample for the finite algebraic construction just described. This includes refining any logarithm-series truncation and square-root enclosure, rather than only rounding a fixed approximation to more digits. An implementation that retains a fixed absolute rounding floor remains a valid finite-stage certificate, but its unchanged output is not asserted to satisfy (57.28).

This is an existence construction with increasing physical access, not a practical acquisition recommendation. It requires shrinking exposure, increasing replica access and reference measurements. Fixed positive exposure is not removed by this finite-difference argument. Reset and reference energies remain separate acquisition costs.

57.10 Sharpness with an unobserved stochastic drive

The sharp hierarchy is not confined to constant angular drift. Let Z be any smooth reversible uniformly elliptic diffusion on a compact connected manifold without boundary, with generator L_Z and positive smooth stationary density π_Z. For a smooth real function c on that manifold, consider the joint generator

$$L = \gamma\partial_\phi^2 + c(z)\partial_\phi + L_Z, \qquad \phi \in \mathbb{S}^1, \qquad \gamma > 0. \tag{57.29}$$

All coordinates are subject to ordinary, even-variable reversal. The stationary measure is $(d\phi/2\pi)\pi_Z(dz)$: the angular drift is divergence-free with respect to the uniform angular measure, and L_Z preserves π_Z. The diffusion tensor is block diagonal with angular entry γ. Thus

$$S = \gamma\partial_\phi^2 + L_Z, \qquad A = c(z)\partial_\phi, \qquad \sigma = \frac{\mathbb{E}_{\pi_Z} c^2}{\gamma}. \tag{57.30}$$

Proposition 57.4 (Sharp minimum under reversible hidden modulation)**.** *For $f(\phi, z) = \cos\phi$, the full stationary scalar observation law is time-reversible. Nevertheless its minimum compatible entropy over the ordinary compact hidden-diffusion class of the observed-geometry theorem is exactly $\mathbb{E}_{\pi_Z} c^2/\gamma$. The actual joint diffusion attains that value. The finite polynomial hierarchy has the same sharp limit and the same ideal formula (57.26), with ω^2 replaced by $\mathbb{E}_{\pi_Z} c^2$.*

Proof. Direct computation gives

$$\Gamma_f = \gamma(1 - f^2), \qquad Sf = -\gamma f, \qquad Af = -c(z)\sin\phi.$$

At stationarity Z and ϕ are independent, and $U = (f, \Gamma_f)$ contains only $Y = f$. The conditional quantities in the population observed-geometry theorem therefore satisfy

$$\begin{aligned} g(Y) &= -\gamma Y, \\ K(Y) &= \gamma^2 Y^2 - \mathbb{E}_{\pi_Z} c^2\,(1 - Y^2). \end{aligned}$$

The integrated conditional bound is

$$\mathbb{E}\left[\frac{[g(Y)^2-K(Y)]_+}{\gamma(1-Y^2)}\mathbf{1}_{\{|Y|<1\}}\right]=\frac{\mathbb{E}_{\pi_Z}c^2}{\gamma}.$$

The excluded endpoints have zero stationary probability. This functional is fixed by the ideal continuous observation law, so it bounds every compatible ordinary hidden diffusion from below. Equation (57.30) provides a realization attaining it.

For reversibility of the observation, the reflection $\mathcal{R}(\phi,z)=(-\phi,z)$ preserves the stationary measure, fixes f and conjugates L to $L^\dagger$. Applying this reflection to the reversed stationary hidden path thus restores its forward law without changing any detector values. All finite-dimensional distributions of Y agree with their time reverses. Finally, stationarity and independence yield

$$\begin{aligned}\frac{1}{\gamma}\mathbb{E}\left[w_m(Y)\{(Af)^2-(-\lambda Y-Sf)^2\}\right]&=\frac{\mathbb{E}c^2}{\gamma}(1-\mathbb{E}Y^{2m+2})\\&\quad-\frac{(\lambda-\gamma)^2}{\gamma}\mathbb{E}[w_m(Y)Y^2],\end{aligned}$$

which proves the hierarchy assertion. □

The same sufficient acquisition and precision schedule applies under valid uniform generator and positive-gain envelopes: the stationary arcsine marginal and ideal hierarchy formula are unchanged, while the finite-lag and exposure bounds already hold for the general hidden class.

The generator calibration can also be written explicitly. Let $C=\|c\|_\infty$ and $G=\|L_Zc\|_\infty$. Then

$$Lf=-\gamma\cos\phi-c\sin\phi, \tag{57.31}$$

$$L^2f=(\gamma^2-c^2)\cos\phi+(2\gamma c-L_Zc)\sin\phi. \tag{57.32}$$

The adjoint formulas reverse the sine terms. Both first-order norms equal $\sqrt{\gamma^2+C^2}$, and both second-order norms equal

$$\sup_z\sqrt{(\gamma^2-c(z)^2)^2+(2\gamma c(z)-L_Zc(z))^2}.$$

The simple choices $M_{1,\pm}=\gamma+C$ and $M_{2,\pm}=(\gamma+C)^2+G$ consequently suffice. For unknown parameters, calibrated upper envelopes for these expressions are required, just as in the main finite certificate; the records do not automatically certify them.

For an explicit zero-mean example take a second angle z, $L_Z=\eta\partial_z^2$ with $\eta>0$, and $c(z)=\varepsilon\cos z$. Then

$$\begin{gathered}\mathbb{E}c=0,\qquad\sigma=\frac{\varepsilon^2}{2\gamma},\qquad M_{1,\pm}=\sqrt{\gamma^2+\varepsilon^2},\\M_{2,\pm}=\sqrt{(\gamma^2-\varepsilon^2)^2+(2\gamma+\eta)^2\varepsilon^2}.\end{gathered} \tag{57.33}$$

The last identity follows by maximizing the squared norm over $c^2\in[0,\varepsilon^2]$; its derivative in c^2 is positive. Mean angular current vanishes, but nonzero conditional angular currents remain and the ordinary-reversal entropy minimum is positive when $\varepsilon\neq0$. This is an analytically verified family, not an additional simulation or physical experiment. Interpreting a hidden coordinate as odd under physical reversal changes the admissible class and can change the entropy target; that interpretation is not included here.

57.11 Scope and failure modes

The theorem calibrates a scale in a declared pointwise geometry. It does not learn arbitrary hidden diffusivity or global generator suprema from a finite record. Rare-state countermodels obstruct that stronger inference. A false structural identity, correlated detector randomization, an altered reference response, nonstationary target records or uncharged trial dependence invalidates the certificate. A single noisy center value cannot replace the distinct replicas.

Replica inference, confidence-region inversion and the semigroup and Dirichlet identities used here are established tools. The result assembled here is their explicit finite-sample propagation into this entropy functional and its polynomial limiting hierarchy. The record count, replica channels, references, exposure and evolution time are acquisition resources. Total energy requires a separate calibrated sensor and preparation model; it is not inferred from the sample count.

Chapter 58

Why empirical averages cannot certify global calibration

A finite record can estimate the local fluctuations it samples without controlling fluctuations in every hidden state. The distinction matters when an entropy certificate requires a pointwise diffusivity envelope or a supremum bound on generator derivatives. Smoothness of each possible model, by itself, does not provide the common numerical constants needed for that extrapolation.

This chapter proves a precise obstruction. Over a class of smooth compact diffusions, any uniformly covered upper confidence bound for global local diffusivity must be infinite with probability at least its nominal coverage, even under a benign reference process. The result holds with a bounded detector, known uniform stationary density, fixed hidden dimension, a common ellipticity lower bound, and a common mixing gap. It even holds when the complete hidden trajectory is observed continuously and without noise. Two further constructions separate the regularity assumptions: an upper bound on diffusivity does not bound Lf, and upper bounds on both diffusivity and Lf do not bound $L^2 f$.

These are statements about global or stationary essential suprema. They do not deny that an ideal continuous path determines local volatility along that path. Every model constructed below is reversible and has zero ordinary entropy production; no change in an ideal-law entropy minimum is asserted.

58.1 A compact diffusion with an arbitrarily high rare bump

Let (x, z) lie on the torus $\mathbb{T}^4 = (\mathbb{R}/2\pi\mathbb{Z})^4$, where z has three coordinates. Near the origin in its fundamental cube $[-\pi, \pi]^3$, use the Euclidean norm $|z|$. Choose a smooth radial profile b satisfying

$$0 \le b \le 1, \qquad b(s) = 1 \ (0 \le s \le \tfrac{1}{2}), \qquad b(s) = 0 \ (s \ge 1).$$

One explicit choice on $1/2 < s < 1$ is

$$b(s) = \frac{h(1-s)}{h(1-s) + h(s-1/2)}, \qquad h(t) = \begin{cases} e^{-1/t}, & t > 0, \\ 0, & t \le 0. \end{cases} \tag{58.1}$$

Together with its constant pieces this profile is C^∞. Being constant near zero also makes $b(|z|/r)$ smooth at the origin.

For $M \geq 1$ and $0 < r \leq 1/2$, set

$$\begin{aligned} d_{M,r}(z) &= 1 + (M-1)b(|z|/r), \\ L_{M,r} &= d_{M,r}(z)\partial_x^2 + \Delta_z, \qquad D_{M,r} = \operatorname{diag}(d_{M,r}, 1, 1, 1), \\ f(x,z) &= \cos x. \end{aligned} \tag{58.2}$$

The bump is supported strictly inside the fundamental cube, so it extends smoothly and periodically. In Itô form the process satisfies

$$dx_t = \sqrt{2d_{M,r}(z_t)}\, dW_t^x, \qquad dz_t = \sqrt{2}\, dW_t^z,$$

with independent Brownian components and coordinates taken modulo 2π. The detector lies in $[-1, 1]$.

The normalized uniform density $\pi = (2\pi)^{-4}$ is stationary. Although d depends on z, its only nonconstant tensor entry is the xx entry, so $L = \operatorname{div}(D\nabla)$ and L is self-adjoint in $L^2(\pi)$. Each model is smooth, connected, compact and uniformly elliptic, with the common lower bound $D \succeq I$. Its stationary current vanishes, giving ordinary entropy rate $\sigma = 0$.

The mixing gap does not deteriorate with the bump. For every smooth mean-zero function g, the torus Poincaré inequality gives

$$-\langle g, Lg\rangle_\pi = \mathbb{E}_\pi[d|\partial_x g|^2 + |\nabla_z g|^2] \geq \mathbb{E}_\pi[|\nabla g|^2] \geq \mathbb{E}_\pi[g^2]. \tag{58.3}$$

Thus the L^2 spectral gap is at least one, and the mean-zero semigroup contracts at least as fast as e^{-t}. The construction does not rely on metastability or a disappearing lower ellipticity constant.

For the detector in (58.2), direct calculation yields

$$\Gamma_f = \nabla f^{\mathsf{T}} D \nabla f = d_{M,r} \sin^2 x, \tag{58.4}$$

$$Lf = -d_{M,r} \cos x, \tag{58.5}$$

$$L^2 f = (d_{M,r}^2 - \Delta_z d_{M,r}) \cos x. \tag{58.6}$$

On the open plateau $|z| < r/2$, the coefficient is M and all its derivatives vanish. Consequently,

$$\sup \Gamma_f = M, \qquad \|Lf\|_\infty = M, \qquad \|L^k f\|_\infty \geq M^k \quad (k \geq 1). \tag{58.7}$$

These global suprema equal their stationary essential versions where appropriate: the coefficients are continuous and π is positive on every open set.

The mean local diffusivity can nevertheless remain close to its reference value. The volume of the enclosing three-dimensional ball gives

$$\frac{1}{2} \leq \mathbb{E}_\pi[\Gamma_f] \leq \frac{1}{2} + \frac{(M-1)r^3}{12\pi^2}. \tag{58.8}$$

Indeed, $\mathbb{E}[\sin^2 x] = 1/2$ and the stationary mass of the bump support is at most $(4\pi r^3/3)/(2\pi)^3$. With $r = M^{-2}$, the global targets diverge while the mean tends to $1/2$. This is an explicit reason that a sample average cannot substitute for a global envelope.

58.2 An explicit bound on encountering the rare region

Lemma 58.1 (Stationary hitting bound)**.** *Let z_t be stationary Brownian motion with generator Δ on $\mathbb{T}^3$, and let $\tau_r = \inf\{t \geq 0 : |z_t| \leq r\}$. For $0 < r \leq 1/2$ and $T \geq 0$,*

$$\mathbb{P}(\tau_r \leq T) \leq \min\left\{1,\ r\left(\frac{9}{64} + 6T\right)\right\}. \tag{58.9}$$

Proof. Choose a smooth radial probability density ρ_r supported on $|y| < r/2$. Its scaled Newtonian potential is

$$p_r(z) = r \int_{\mathbb{R}^3} \frac{\rho_r(y)}{|z-y|}\, dy.$$

It is smooth and nonnegative, with $\Delta p_r = -4\pi r \rho_r \le 0$. The radial harmonic equation and unit total mass imply $p_r(z) = r/|z|$ for $|z| \ge r/2$. Inside that ball the radial potential is nonincreasing with radius, so $p_r \ge p_r(r/2) = 2$. In particular, $p_r \ge 1$ on $|z| \le r$.

Take $\chi(s) = 1$ for $s \le 1$, $\chi(s) = 0$ for $s \ge 2$, and

$$\chi(s) = q(s-1) \quad (1 < s < 2), \qquad q(t) = 1 - 10t^3 + 15t^4 - 6t^5.$$

The first two derivatives vanish at the joins, making χ a nonnegative C^2 cutoff. Differentiation gives $\max_{[0,1]} |q''| = 10\sqrt{3}/3 < 6$. Define $u_r(z) = \chi(|z|)p_r(z)$ and extend it periodically. Its support is contained in the radius-two ball, strictly inside the fundamental cube. Thus u_r is a nonnegative C^2 function on the torus and is at least one on the target ball.

For $|z| \le 1$, its Laplacian is nonpositive. On $1 < |z| < 2$, the radial identity in dimension three gives

$$\Delta\left(\frac{\chi(s)}{s}\right) = \frac{\chi''(s)}{s}, \qquad s = |z|.$$

Hence $\Delta u_r \le 6r$ everywhere; outside radius two it is zero. The initial stationary expectation satisfies

$$\begin{aligned}\mathbb{E}[u_r(z_0)] &\le \frac{r}{(2\pi)^3} \int \rho_r(y) \int_{|z|\le 2} \frac{dz}{|z-y|}\, dy \\ &\le \frac{r}{(2\pi)^3} \int_{|w|\le 2+r/2} \frac{dw}{|w|} = \frac{r(2+r/2)^2}{4\pi^2} \le \frac{81r}{64\pi^2} < \frac{9r}{64}.\end{aligned}$$

Here $r \le 1/2$ and $\pi > 3$ supply the final constants. Dynkin's formula at the bounded stopping time $\tau_r \wedge T$ now gives

$$\mathbb{P}(\tau_r \le T) \le \mathbb{E}[u_r(z_{\tau_r \wedge T})] \le \mathbb{E}[u_r(z_0)] + 6rT \le r\left(\frac{9}{64} + 6T\right).$$

Capping by one proves (58.9). The cutoff needs only C^2 regularity for this argument; the diffusion coefficients in the model remain C^∞. □

Let $\mathbb{P}_0$ denote the reference model $d = 1$. Couple it with (58.2) using the same uniform initial state and Brownian drivers. Their z paths agree at all times. Their x coefficients agree outside the bump, so pathwise uniqueness gives identical full hidden trajectories until τ_r. Therefore

$$\mathrm{TV}\left(\mathbb{P}_{M,r}^{\mathrm{full},[0,T]}, \mathbb{P}_0^{\mathrm{full},[0,T]}\right) \le \min\left\{1, r\left(\frac{9}{64} + 6T\right)\right\}. \tag{58.10}$$

This bound is independent of the bump height M and applies to complete continuous trajectories, not merely endpoint observations.

Every common observation channel contracts total variation. Thus the same bound applies after retaining only f, integrating an exposure window, adding common independent detector noise, or applying a common randomized processing rule. For N independent stationary records with durations summing to T_{total}, a union coupling gives

$$\mathrm{TV}(\mathbb{P}_{M,r}^{\mathrm{record}}, \mathbb{P}_0^{\mathrm{record}}) \le \min\left\{1, r\left(\frac{9N}{64} + 6T_{\mathrm{total}}\right)\right\}. \tag{58.11}$$

All latent intervals used by an exposure must be included in these durations. The access contract is passive stationary observation, not an intervention that prepares the hidden system inside a known bump.

58.3 The consequence for honest upper confidence bounds

Theorem 58.2 (A global diffusivity bound cannot be learned uniformly)**.** *Fix a finite passive protocol covered by* (58.11) *and* $0 < \alpha < 1$*. Let* U *be a measurable statistic of its record, including any common external randomization, taking values in* $[0, +\infty]$*. If*

$$\mathbb{P}_{M,r}\{U \geq \sup \Gamma_f\} = \mathbb{P}_{M,r}\{U \geq M\} \geq 1 - \alpha \tag{58.12}$$

for every $M \geq 1$ *and* $0 < r \leq 1/2$*, then*

$$\mathbb{P}_0\{U = +\infty\} \geq 1 - \alpha. \tag{58.13}$$

The same conclusion holds for a uniformly covered upper bound on $\|Lf\|_\infty$ *or* $\|L^2 f\|_\infty$ *over this class.*

Proof. For each fixed M, total variation and (58.12) give

$$\mathbb{P}_0\{U \geq M\} \geq 1 - \alpha - \min\left\{1, r\left(\frac{9N}{64} + 6T_{\text{total}}\right)\right\}.$$

Let $r \downarrow 0$. The resulting lower bound $1 - \alpha$ holds for every M. Let integer M increase without bound and apply continuity from above of probability to obtain (58.13). For $\|Lf\|_\infty$ the target equals M; for $\|L^2 f\|_\infty$ it is at least M^2. The same argument applies. □

In particular, a bound finite almost surely under the reference cannot have the stated uniform coverage. The obstruction concerns the breadth of the model class. It does not invalidate a confidence bound derived inside a separately justified parametric family.

The theorem also applies to a claimed pointwise envelope $\widehat{a}(y)$ satisfying, with uniform confidence,

$$\Gamma_f(x, z) \leq \widehat{a}(f(x, z)) \quad \text{for every hidden state } (x, z).$$

At $x = \pi/2$ and z in the plateau, this condition requires $\widehat{a}(0) \geq M$. Thus a uniformly covered pointwise envelope must satisfy $\mathbb{P}_0\{\widehat{a}(0) = +\infty\} \geq 1 - \alpha$. If validity is asserted only stationary-almost-everywhere, the same conclusion holds for continuous envelopes, by continuity and positivity of the stationary density. For arbitrary measurable almost-everywhere envelopes, their value at the null level $y = 0$ cannot be inferred this way; an integrated formulation would be needed.

A finite stopping rule does not remove the obstruction. For a common nonanticipating passive rule with reference stopping time $S < \infty$ almost surely, coupled stopped records agree unless $\tau_r \leq K$ or $S > K$. Their total variation is therefore bounded by

$$r\left(\frac{9}{64} + 6K\right) + \mathbb{P}_0(S > K), \qquad K < \infty.$$

First let $r \downarrow 0$, then $K \to \infty$. The same reference convergence and confidence argument follow. Infinite-time observation is not included as a finite resource.

58.4 A hierarchy of missing regularity assumptions

The preceding family allows an unbounded diffusion coefficient. The next two variants show that supplying an upper diffusion bound still does not supply the generator bounds used to control finite-lag approximation.

58.4.1 Bounded diffusion does not bound the first generator

Let N be a positive integer multiple of four, and define

$$\begin{aligned} d_{N,r}(x,z) &= 1 + b(|z|/r)\frac{1+\sin(Nx)}{2}, \\ L_{N,r} &= \partial_x(d_{N,r}\partial_x) + \Delta_z. \end{aligned} \tag{58.14}$$

Now $I \preceq D \preceq 2I$ uniformly. The divergence-form drift $\partial_x d_{N,r}$ is part of the generator and is essential to its uniform stationary density. The process remains reversible, and (58.3) still proves a spectral gap at least one. The z dynamics are unchanged; outside the same rare ball, both the x drift and diffusion match the reference. Thus the full-path coupling bounds remain valid.

For $f = \cos x$,

$$Lf = -d_{N,r}\cos x - \frac{N}{2}b(|z|/r)\cos(Nx)\sin x.$$

At $x = \pi/2$ and z in the plateau, $\cos(Nx) = 1$, giving $Lf = -N/2$. Consequently,

$$\Gamma_f \le 2, \qquad \|Lf\|_\infty \ge N/2. \tag{58.15}$$

Taking $N \to \infty$ and $r = N^{-2}$ makes the target diverge while the finite-record law converges to the reference. The confidence argument therefore remains valid for $\|Lf\|_\infty$ even when both lower and upper diffusion bounds are known.

58.4.2 Bounded diffusion and first generator do not bound the second

Return to (58.2) and fix $M = 2$. Then $I \preceq D \preceq 2I$ and $\sup \Gamma_f = \|Lf\|_\infty = 2$ for every r. For the explicit profile (58.1), differentiating at $s = 3/4$ gives

$$b(3/4) = \tfrac{1}{2}, \qquad b'(3/4) = -8, \qquad b''(3/4) = 0.$$

The radial Laplacian in three dimensions therefore satisfies

$$\Delta_3 b(3/4) = b''(3/4) + \frac{2b'(3/4)}{3/4} = -\frac{64}{3}.$$

At $x = 0$ and $|z| = 3r/4$, equation (58.6) gives

$$L^2 f = \frac{9}{4} + \frac{64}{3r^2}, \qquad \|L^2 f\|_\infty \ge \frac{9}{4} + \frac{64}{3r^2}. \tag{58.16}$$

This diverges as $r \downarrow 0$, while the observation law still approaches the reference. Hence no useful uniformly finite upper confidence bound for $\|L^2 f\|_\infty$ follows merely from known bounds on D and Lf. Additional regularity constants are substantive assumptions, not consequences of individual smoothness.

58.5 Interpretation and reproducible checks

The obstruction is extrapolation to hidden regions that a finite passive record need not visit. If the bump is encountered, ideal continuous access can reveal its increased volatility. If it is not encountered, even the full hidden trajectory can agree exactly with the reference under the coupling. There is no conflict with quadratic-variation reconstruction of $\Gamma_f(X_t)$ at observed times.

The examples also clarify what can change the conclusion: a bounded parametric family, known spatial derivative bounds, controlled preparation, or an independently justified envelope can exclude the alternatives. A common lower ellipticity bound, stationary density and mixing gap do not do so. The strengthened variants retain an upper diffusion bound as well and still separate the first and second generator requirements. All models have $\sigma = 0$; the result concerns calibration assumptions, not hidden dissipation or a new entropy minimum.

For scale, the main family at $T = 1$ and $r = M^{-2}$ has the exact analytic record-distance upper bound $(393/64)M^{-2}$. At $M = 10, 100, 1000$ this is approximately 0.06140625, 0.0006140625 and 0.000006140625, respectively, while the global diffusivity supremum is M. These are evaluations of the proved formulas, not simulated experiments.

The companion program implements the smooth profile and evaluates the coupling and mean-volatility formulas. Eleven checks cover the cutoff curvature, radial Laplacian identity, generator composition, independent radial integration, the strengthened variants and input contracts. Numerical outputs are non-outward-rounded diagnostics: a very small positive mean-volatility margin can disappear under floating-point rounding. The analytic formulas and proofs, rather than those rounded values or a finite collection of simulations, establish the claims. Internal independent review is distinct from external peer review or laboratory validation; no historical priority is asserted here.

Chapter 59

A frozen detector-calibration experiment

The finite certificate was evaluated on 1,600 independently seeded synthetic datasets, with unknown detector gain, exposure width and diffusion scale carried through the inference. The driven case produced a positive geometry certificate in all 400 repetitions; equilibrium produced none. The acquisition is expensive: each dataset represents more than 80 million detector readings. Higher polynomial order did not improve the selected bound in this design. These results demonstrate a working finite inference chain under its declared assumptions, with its sampling and calibration costs retained. They are not laboratory observations or an efficiency claim.

59.1 The frozen acquisition and inference

The data-producing process was the stationary circle diffusion $d\phi_t = \omega\,dt + \sqrt{2\gamma}\,dW_t$, observed through $f = \cos\phi$. Three uniform exposure windows start at $0, h, 2h$ and have width τ. Each binary detector output $R \in \{-1, 1\}$ has conditional mean κA, where A is the average of f over its window. The two endpoint windows have one readout each; the center has four conditionally independent readouts of the same exposure. Translating all window times makes this the causal-boxcar convention of the preceding theorem, without changing its stationary moment law. The detector is noninvasive in this model. No noise variance is supplied to inference; bounded outputs and the conditional replica contract justify the moment calculation.

The design was fixed at

$$h = 7/200 = 0.035, \quad \lambda = 1, \quad m_{\max} = 1, \quad \kappa_0 = 0.9, \quad n = 13{,}333{,}333.$$

Every dataset also has 100,000 constant-signal references and 100,000 step references. The constant reference has mean κ; the step is read $s = 1/500{,}000$ after its known transition and has mean $\kappa s/\tau$. Inference starts from $\kappa \in [0, 1]$ and $\tau \in [1/200{,}000, 1/50{,}000]$. It receives the common physical bounds $M_{1,+} = M_{1,-} = 4$ and $M_{2,+} = M_{2,-} = 13$.

The gain and width supplied to the simulator are not supplied as exact inference inputs. The diffusion scale is likewise unknown to the certificate: its upper bound comes from the covered twin and lag moments. The pointwise identity $\Gamma_f = \gamma(1 - f^2)$ and the generator caps remain assumptions. The inference does not fit the complete circle model or substitute its arcsine marginal as a measured second moment.

Two-sided empirical-Bernstein intervals cover the fixed target-statistic vector at total failure allocation 0.04. Bounded reference-mean intervals receive allocation 0.01. The resulting 0.05 per-

Case	γ	ω	κ	$10^5\tau$	σ
Driven	1	3	0.9	1	9
Equilibrium	1	0	0.9	1	0
Weak drive	1	1	0.9	1	1
Changed calibration	0.5	3	0.8	1.5	18

Table 59.1: Simulator parameters and exact entropy $\sigma = \omega^2/\gamma$. Each case has 400 datasets. The changed-calibration case uses fresh references with altered gain, exposure and diffusion; it is not an uncalibrated change of detector between target and reference.

dataset allowance covers gain and shutter projection, the geometry scale, and selection among the predeclared order-zero and order-one bounds. The fixed nominal-gain contrasts retain covariance before calibration projection; componentwise bounds are also retained, and their covered maximum is reported.

The endpoint bound (57.25) uses the same records, reference intervals and simultaneous event. Here $b = 1 - \lambda h = 0.965$. It is an established reversible-support and entropy-bridge comparison. It does not require the pointwise geometry identity, so it has broader structural scope. Comparing the two values measures what the additional geometry assumption provides in these examples; it does not establish dominance under identical assumptions or against all inference methods.

59.2 A finite-law simulator with an explicit error charge

The retained six-bit law is a finite exposure-moment polynomial:

$$\Pr(R = r) = 2^{-6}\mathbb{E}\prod_{i=1}^{6}(1 + \kappa r_i A_{w(i)}). \tag{59.1}$$

The simulator computes the needed integrated Fourier moments, rather than approximating a Brownian path on a time grid. A closed system with 140 real states supplies the moments through total degree six. Rational Taylor actions enclose the exposure and free-gap semigroups; if $x = t\|G\|_\infty$, the degree-N action remainder is bounded by

$$\|v\|_\infty \, 3^{\lceil x \rceil} \frac{x^{N+1}}{(N+1)!}.$$

Input uncertainty, exposure normalization and event-polynomial coefficients are propagated explicitly. In the driven case the exposure stages use Taylor degree 15 and the gaps degree 58.

The resulting law is rounded to a nonnegative 63-bit dyadic probability vector summing exactly to one. Conditional center exchangeability collapses the 64 labeled outcomes to 20 categories: left sign, center plus-count, and right sign. These counts retain all predeclared symmetric statistics; they would not support an arbitrary later score that distinguishes labeled center channels. Integer-CDF sampling avoids floating probability thresholds. Independent uniform random bits are the ideal sampling assumption; the implementation uses NumPy 2.3.5 and PCG64 pseudorandom streams.

The exact rational total-variation charges include Taylor enclosure, dyadic rounding, all target records and both reference batches. Rounded upward decimal bounds per dataset are

$$1.308\,10^{-11}, \quad 1.717\,10^{-11}, \quad 1.067\,10^{-11}, \quad 1.141\,10^{-11}$$

in the order of Table 59.1. The full bank's product-law charge is below $2.093\,10^{-8}$. Thus transferring a per-dataset probability statement between the ideal model and the sampled dyadic law adds its

specified charge. These bounds concern the probability-law approximation; they do not certify hardware or the literal randomness of a deterministic pseudorandom generator.

59.3 Results, inclusion checks and negative controls

Case	Geometry positive	Mean lower	Endpoint positive	Mean lower
Driven	400/400	0.57735	400/400	$8.1640\,10^{-5}$
Equilibrium	0/400	0	0/400	0
Weak drive	0/400	0	27/400	$6.8211\,10^{-9}$
Changed calibration	395/400	0.59905	400/400	$6.9611\,10^{-5}$

Table 59.2: Observed synthetic detection counts and mean lower bounds. The geometry and endpoint bounds use the same detector records but different structural assumptions. No method exceeded the true entropy in these runs.

The geometry lower bounds in the driven case have median 0.57649 and empirical 2.5th–97.5th percentiles $[0.29974, 0.84860]$. Under changed calibration the median is 0.59490; the corresponding percentiles are 0.05523 and 1.14799. These are descriptive quantiles across repeated datasets, not confidence intervals for the true entropy. The mean certificates represent only about 6.4% and 3.3% of the respective entropies 9 and 18, despite the large acquisition quota.

Case	Target event	Gain/width inclusion	Finite γ bounds	Mean upper
Driven	400/400	400/400	400/400	1.69770
Equilibrium	399/400	400/400	400/400	1.53917
Weak drive	399/400	400/400	400/400	1.55591
Changed calibration	397/400	400/400	400/400	1.20801

Table 59.3: Independent replay of the simultaneous target-mean event, reference-region inclusion and geometric-scale output. Every finite γ upper bound covered its simulator value; there were no unbounded geometry abstentions. The target event missed five times in total without causing an entropy overshoot.

All 1,600 count records were independently reconstructed and replayed. The target mean-event inclusion check used rational law enclosures and had no unresolved enclosure-boundary cases. Calibration inclusion and entropy overshoot checks are separate: failure of a sufficient simultaneous event need not cause the final lower bound to overshoot.

The equilibrium control gives no positive result, as required for a useful check against obvious false positives. The weak-drive case is an equally important negative result for sensitivity: the geometry certificate is zero throughout, while the endpoint certificate is positive in 27 runs. Under changed calibration the endpoint method also returns five more positive results. The geometry method therefore does not dominate even this one broader-class comparator in detection frequency.

A transparent derived fallback is to report the maximum of the geometry and endpoint lower bounds. Under the narrower geometry contract, both hold on the already covered event, so this maximum requires no additional error allocation. Its positive counts are $400, 0, 27, 400$ in the four cases. This is a deterministic combination made after the frozen benchmark, not a change to its per-method outputs or a separately preregistered experiment.

Zero overshoots in 400 independent repetitions do not prove zero failure probability. Under the ideal repeated-sampling model, their one-sided 95% binomial upper bound is $1 - 0.05^{1/400} \simeq 0.00746$, separately for each case and method. This diagnostic is not a simultaneous guarantee

across the comparisons. The finite-sample coverage claim comes from the preceding theorem and implementation, not from the observed frequencies.

59.4 What the higher-order and design comparisons did not show

The selected prefix was order zero in every dataset. Order one improved neither the final driven result nor the changed-calibration result. Retaining covariance did help its own unselected estimate: in the driven case the order-one mean rose from 0.02109 for the componentwise calculation to 0.06497 for the fixed-contrast projection. Both remained below the selected order-zero bound. The sharp infinite hierarchy is a theorem; useful finite-resource gains from its higher orders were not demonstrated here.

Before freezing the experiment, deterministic population calculations examined lags and replica counts. At $h = 0.035$, under a ceiling of 80 million target readouts, two, four and six center replicas permit respectively 20,000,000, 13,333,333 and 10,000,000 stationary records. The corresponding population-moment values inserted into the confidence formula were 0.54161, 0.55834 and 0.50535, with order zero selected in all three. All designs add 200,000 reference readouts. The prospective calculation used target/reference allocations 0.025/0.025, distinct from the final 0.04/0.01 allocation.

Those numbers are design diagnostics, not confidence-certified outcomes, power estimates or additional sampled experiments. They show why comparing replicas at fixed record count would conceal extra readout costs. Earlier $h = 0.02$ diagnostics returned zero for the strong-drive geometry score at two and ten million target records, and higher orders remained zero even at fifty million. A separately retained design with narrower exposure and tighter supplied generator caps looked better, but changed the apparatus contract and was not used for the primary experiment. Its gains cannot be attributed to the estimator alone.

59.5 Acquisition costs and reproducibility

Each dataset uses 79,999,998 target readings, 200,000 reference readings and 80,199,998 readings in total. It requires 13,333,333 independent stationary preparations and 200,000 controlled reference preparations. Four center shots share one physical exposure; they still count as four detector readings. There are 39,999,999 target windows and 200,000 reference windows.

The summed modeled target evolution duration is $n(2h + \tau)$: 933,466.64333 model-time units for the first three cases and 933,533.309995 for changed calibration. The summed target exposure durations are respectively 399.99999 and 599.999985; reference exposure durations add 2 and 3. These are sums over independent acquisitions, not wall-clock durations or measured energy. Stationary preparation, reset, replica construction, clock calibration and shutter-control costs remain unmeasured. Fast categorical simulation does not make those physical preparations inexpensive.

The authoritative files in `experiment/` are `contract.json`, `benchmark.json`, the compressed sufficient counts `benchmark_counts.json.gz`, and the first-generation clock record `benchmark_run.json`. The contract binds simulator, certifier and benchmark source hashes. It was frozen on 2026-10-13 at 01:45:27 UTC; generation started at 01:45:40 and completed at 01:48:01 with four workers. The choices preceding this freeze are retained as prospective population diagnostics, not concealed as a preselected universal design. No stochastic pilot target records were used.

The seed rule is $1032026400000 + 100000\,i + 10\,r + s$ for case index i, repetition r and stream $s \in \{0, 1, 2\}$ for target, gain and step records. Exact counts, law enclosures, source bindings and the independent replay provide the reproduction chain. The stochastic-drive family proved in the

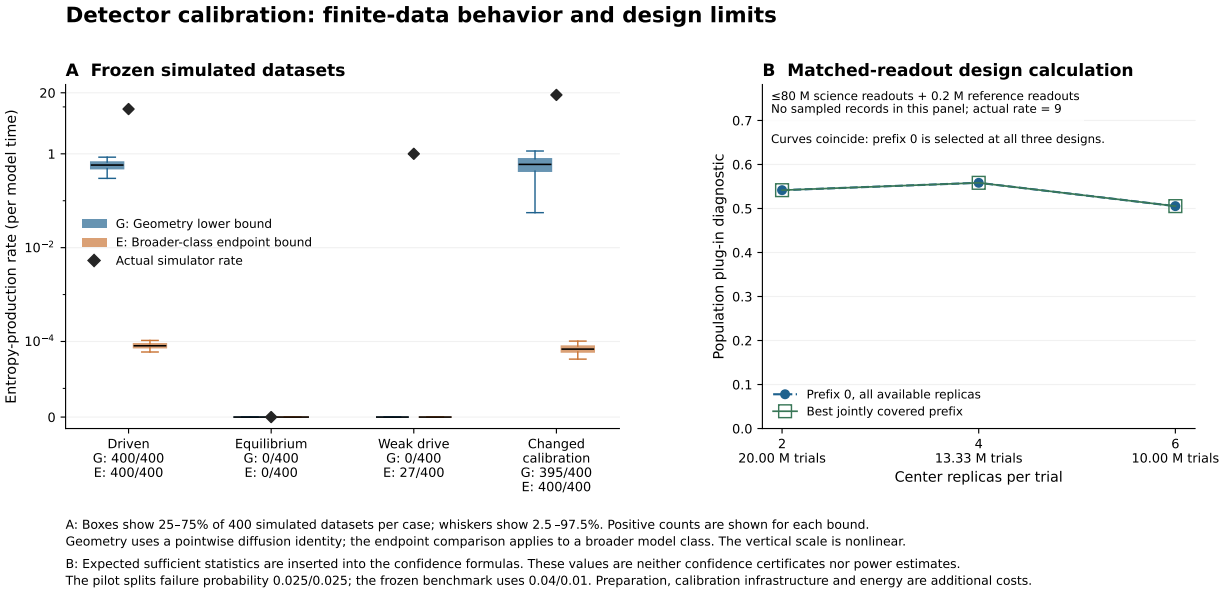


Figure 59.1: Left: distributions over the 400 simulated datasets in each case, with boxes at the interquartile range and whiskers at empirical 2.5th and 97.5th percentiles; the vertical axis uses a symmetric logarithmic scale. Black diamonds mark the simulator entropy. The endpoint bound has broader structural validity than the geometry bound. Right: the separate prospective population diagnostic at matched target readout ceilings, with the additional reference readings counted. This panel contains no random experimental outcomes or confidence certificates.

preceding chapter was not simulated in this bank. Its analytic result and this constant-drift detector benchmark remain distinct evidence.

Chapter 60

Completing access to stationary channel currents

Conservation turns some inaccessible currents into measurable combinations of other currents. It also has a definite limit: conservation of the aggregate flow cannot determine how distinct physical channels divide that flow. This chapter separates those two situations and gives exact sensor designs and independently checkable uncertainty certificates.

The graph, its physical channels, and their assignment to pairs of states are assumed known. The results do not reconstruct unknown hidden topology. They specialize established linear observability, max-flow/min-cut, and spanning-tree methods to a declared current-measurement contract. Conservation-based sensor placement and measured cotrees have substantial prior literature, including Bagajewicz and Sánchez (2000) and He (2013); no priority claim is made for these graph primitives. The purpose is to connect them explicitly to channel access, sharp error guarantees, and physical realizability.

60.1 The observation contract

Let $G = (V, E)$ be a finite undirected multigraph without self-loops. Each physical channel e has an arbitrary reference orientation $u_e \to v_e$. Parallel edges represent different transition mechanisms between the same pair of states. The incidence matrix $B \in \mathbb{R}^{V \times E}$ has -1 at a channel's tail and $+1$ at its head. A stationary current vector belongs to

$$\mathcal{J} = \ker B. \tag{60.1}$$

Reference orientation is bookkeeping: every component of $j \in \mathcal{J}$ may be positive, negative, or zero. Initially, the model class imposes no activity, rate, affinity, or current ceiling. Such additional restrictions can change the sharp uncertainty bounds below. The minimum-cut formulas require an incidence matrix. A general chemical stoichiometric balance matrix need not have that form; it requires the more general linear-identification and robust-recovery formulation rather than an automatic application of these graph formulas.

A sensor for channel s supplies either its exact native current j_s or a number z_s with a justified bound $|z_s - j_s| \le \epsilon_s$. The apparatus need not observe the full state trajectory. This distinction matters: complete state-resolved jump trajectories already identify aggregate transition rates and their aggregate currents, subject to statistical uncertainty. The problem here concerns sparse current access, including local intervention sensors that report selected physical-channel currents.

Every impossibility or minimax statement is relative to the retained sensor records. Richer raw observations may contain additional information. A sensor's construction, calibration, and resource accounting must be supplied separately; assigning an edge a cost does not by itself give that cost an energy meaning.

For a vertex set U, write

$$\sigma_U(e) = \mathbf{1}\{u_e \in U\} - \mathbf{1}\{v_e \in U\}. \tag{60.2}$$

Summing $Bj = 0$ over the vertices of U gives

$$\sum_{e \in E} \sigma_U(e) j_e = 0. \tag{60.3}$$

Fix a target $e_0 : u \to v$. For $u \in U$ and $v \notin U$, let $C(U) = \{e \neq e_0 : \sigma_U(e) \neq 0\}$. This is the corresponding u–v cut in $G \setminus \{e_0\}$.

60.2 Exact identification and the minimum additional access

Theorem 60.1 (A target current requires a measured separating cut)**.** *Let $S \subseteq E \setminus \{e_0\}$ be the channels observed exactly. The following conditions are equivalent:*

1. *$j_S = j'_S$ implies $j_{e_0} = j'_{e_0}$ for every $j, j' \in \mathcal{J}$.*
2. *e_0 is a bridge in $G \setminus S$.*
3. *S separates u from v in $G \setminus \{e_0\}$.*
4. *Some U containing u and excluding v satisfies $C(U) \subseteq S$.*

Whenever these conditions hold,

$$j_{e_0} = -\sum_{s \in C(U)} \sigma_U(s) j_s. \tag{60.4}$$

Proof. Suppose a path from v to u remains after deleting S and e_0. Send one unit of signed flow along that path, and one unit along e_0 from u to v. The resulting vector h obeys $Bh = 0$, $h_S = 0$, and $h_{e_0} = 1$. Thus $j = 0$ and $j' = h$ have identical sensor records and different target currents. Identification is impossible.

If no such path remains, let U be the component of u after those deletions. Then $v \notin U$, and every original crossing edge except e_0 belongs to S. Equation (60.3), with $\sigma_U(e_0) = 1$, gives (60.4). It determines the target from the record. The existence of the alternative path is exactly the failure of e_0 to be a bridge in $G \setminus S$, proving all equivalences. □

An original bridge has stationary current zero. The theorem then requires no sensor: conservation already fixes its current. Parallel channels must not be collapsed before applying the theorem. A second unsensed channel between u and v supplies an alternative path of length one and hence an ambiguity.

Corollary 60.2 (Minimum additional sensing cost)**.** *Give each eligible nontarget channel a nonnegative acquisition cost c_s. Other channels cannot be selected. The minimum total cost of an exact indirect target measurement is*

$$\min_{U:\, u \in U,\, v \notin U} \sum_{s \in C(U)} c_s, \tag{60.5}$$

where a cut containing an ineligible channel has infinite cost. If direct measurement of e_0 is also allowed, take the smaller of its cost and the indirect optimum.

Proof. Every successful sensor set contains $C(U)$ for some admissible U, by Theorem 60.1. Nonnegative costs make its total cost at least that of this cut. Conversely, measuring exactly the edges of a minimum cut supplies the estimator (60.4). □

This is an ordinary minimum-cut computation with a scientific interpretation. Its optimality is for the supplied sensor menu and additive cost model. It does not assert that the menu contains every physically possible experiment.

60.3 A sharp error bound, including nonlinear estimators

Let S now supply noisy current estimates

$$z_s = j_s + \eta_s, \qquad |\eta_s| \le \epsilon_s, \qquad s \in S. \tag{60.6}$$

The full Cartesian error box is allowed. This is an adversarial uncertainty set, not an assumption of probabilistic independence. Define the deterministic minimax radius

$$R_S = \inf_f \sup_{\substack{j \in \ker B \\ |\eta_s| \le \epsilon_s}} |f(j_S + \eta) - j_{e_0}|, \tag{60.7}$$

where f ranges over all estimators, including nonlinear ones.

Theorem 60.3 (The exact radius is an error-weighted cut)**.** *Under* (60.1) *and* (60.6)*,*

$$R_S = \min_{\substack{U:\, u \in U,\, v \notin U \\ C(U) \subseteq S}} \sum_{s \in C(U)} \epsilon_s. \tag{60.8}$$

The value is $+\infty$ when there is no admissible cut. For a finite minimum, the signed cut-sum estimator attains the radius globally.

Proof. For any admissible cut, use $f(z) = -\sum_{s \in C(U)} \sigma_U(s) z_s$. Subtracting the exact conservation identity shows that its error is a signed sum of the η_s. Its absolute value is at most $\sum_{s \in C(U)} \epsilon_s$. Minimizing over cuts gives the upper bound.

For the matching lower bound, remove e_0. Give each sensor edge capacity ϵ_s in either direction, and every unobserved edge infinite capacity. If a finite separating cut exists, undirected max-flow/min-cut gives a signed flow of value R from v to u, where R is the right side of (60.8). Close the flow with current R on e_0. The resulting circulation h satisfies

$$Bh = 0, \qquad h_{e_0} = R, \qquad |h_s| \le \epsilon_s \quad (s \in S). \tag{60.9}$$

Both $j = h$ with errors $\eta = -h_S$ and $j = -h$ with errors $\eta = h_S$ produce the identical record $z = 0$. Their target values are R and $-R$. Any single estimate $f(0)$ has error at least R in one of the two cases. This lower bound applies to every estimator and matches the upper bound.

For completeness, the infinite-capacity construction can be made finite. If there is no entirely unobserved v–u path, assign every unobserved edge a capacity strictly larger than $\sum_{s \in S} \epsilon_s$. At least one cut uses only sensor edges, so a minimum cut never crosses an unobserved edge. The finite-capacity max-flow theorem therefore gives exactly the required flow. If an entirely unobserved path does exist, the unit circulation in the proof of Theorem 60.1 can be scaled without bound while leaving every sensor record unchanged. Then $R_S = +\infty$. □

The theorem explains why averaging redundant sensors need not improve a worst-case box guarantee. It does not prohibit statistical improvement under a specified noise law or an uncertainty set that excludes the least-favorable error combinations. Known rate or activity bounds can also reduce the sharp radius by restricting the allowed currents.

A certificate requiring no optimization. A finite certificate consists of a vertex cut U and a circulation h. A checker verifies $C(U) \subseteq S$, the identities and bounds in (60.9), and $h_{e_0} = \sum_{s \in C(U)} \epsilon_s$. These checks establish matching lower and upper bounds without trusting or rerunning the search algorithm. An infinite certificate instead supplies $Bh = 0$, $h_{e_0} = 1$, and $h_S = 0$.

60.4 Aggregate access and the internal channel obstruction

Partition the physical channels into bundles $\mathcal{B}$, one for each unordered pair of endpoint states. Choose one canonical orientation per bundle. Let A take signed bundle sums, so $k = Aj$ is the vector of aggregate currents and

$$B = \bar{B}A, \tag{60.10}$$

where $\bar{B}$ is the incidence matrix of the underlying simple graph. Suppose bundle totals are known exactly for $T \subseteq \mathcal{B}$, and individual channels are sensed exactly for $S \subseteq E$. These aggregate readings are treated as already available. If their calibration errors are nonzero, a robust linear-observation formulation is needed instead of silently using the exact identities below.

Write $r_b = |b \setminus S|$ for the number of unsensed channels in bundle b. Form a graph H on all vertices V, including isolated vertices, with one edge for each bundle $b \notin T$ having $r_b > 0$. Let $c(H)$ denote its number of connected components.

Theorem 60.4 (Two distinct sources of current ambiguity)**.** *The homogeneous ambiguity space*

$$\mathcal{N} = \{h : Bh = 0,\ A_T h = 0,\ h_S = 0\} \tag{60.11}$$

has dimension

$$\dim \mathcal{N} = \sum_{b \in \mathcal{B}} \max(r_b - 1, 0) + |E(H)| - |V| + c(H). \tag{60.12}$$

An unsensed target e_0 in bundle b_0 is identifiable if and only if

$$r_{b_0} = 1 \quad \textit{and} \quad \big(b_0 \in T \ \textit{or}\ b_0 \ \textit{is a bridge of}\ H\big). \tag{60.13}$$

Proof. After imposing $h_S = 0$, each nonempty unsensed bundle has r_b free channel coordinates. Taking its signed sum maps these coordinates onto one aggregate coordinate. The kernel of that map has dimension $r_b - 1$. Because bundles have disjoint channel coordinates, their internal kernels form a direct sum of dimension $\sum_b \max(r_b - 1, 0)$.

Known aggregate variations vanish. All remaining aggregate variations must satisfy conservation on H, by (60.10); hence they belong to $\ker B_H$. Conversely, choose one unsensed representative channel in each nonempty bundle. Any circulation on H can be lifted by assigning its aggregate current, with the appropriate orientation sign, to that representative and assigning zero to every other channel. Known bundle variations remain zero, and $Bh = \bar{B}Ah = 0$. The map from $\mathcal{N}$ onto $\ker B_H$ is therefore surjective, with exactly the internal kernel described above. The dimension formula follows from $\dim \ker B_H = |E(H)| - |V| + c(H)$.

If $r_{b_0} \geq 2$, one can change the target and compensate on another unsensed channel in the same bundle. Choose the signs so that their aggregate variation is zero. This changes h_{e_0} while preserving all supplied measurements and conservation, so the target is unidentifiable. If $r_{b_0} = 1$, the target variation equals the aggregate variation of b_0 up to orientation. It vanishes when $b_0 \in T$. Otherwise, it vanishes for every circulation on H precisely when b_0 is a bridge, by Theorem 60.1. This proves both directions of (60.13). □

The first term in (60.12) is ambiguity between physical channels sharing the same endpoints. The second is circulation ambiguity among unknown aggregate edges. The decomposition need not select a preferred representative channel, but its dimension and target criterion are independent of that choice.

In particular, if every aggregate current is known, then $\ker A \subseteq \ker B$ by (60.10). Conservation supplies no further constraint on the internal split. It cannot distinguish two physical channels whose currents change oppositely while leaving the aggregate unchanged. This is a necessary-access obstruction: some measurement or intervention must distinguish those channels, unless additional physical assumptions already fix their relation.

60.5 Constructive completion with mixed access

Proposition 60.5 (Minimum indirect access for one target)**.** *Assume all nontarget channels can be sensed at nonnegative costs c_e, and the exact aggregate readings in T are already available. A minimum-cost indirect sensor set for target $e_0 \in b_0$ is obtained as follows:*

1. *Sense every other channel in b_0.*
2. *If $b_0 \in T$, stop.*
3. *Otherwise, build the simple graph of bundles not in T, remove b_0, and give each other bundle the cost $C_b = \sum_{e \in b} c_e$.*
4. *Find a minimum cut between the endpoints of b_0 and sense every physical channel in every bundle crossing that cut.*

Proof. Every other channel in b_0 must be sensed: otherwise the internal two-channel ambiguity in Theorem 60.4 survives. These costs are unavoidable. If the target aggregate is known, the forced sensors already determine the target by subtraction.

If the target aggregate is unknown, it must become a bridge of the residual unknown-bundle graph. A nontarget bundle disappears from that graph exactly when all its physical channels are sensed, at cost C_b. Partially sensing a bundle leaves at least one channel able to carry its arbitrary aggregate variation. By Theorem 60.1, every successful design must therefore remove the bundles of a separating cut. Its additional cost is at least the minimum cut cost. Sensing all channels on that minimum cut makes the target a bridge and attains the bound. These bundles are disjoint from b_0, so their costs add to the forced parallel-channel costs. □

Proposition 60.6 (Minimum access for every physical-channel current)**.** *Suppose every channel has a nonnegative sensing cost. Define $w_b = \max_{e \in b} c_e$. The minimum total cost for recovering every current, given exact aggregate access on T, is*

$$\sum_{e \in E} c_e - \sum_{b \in T} w_b - \max_{F \text{ forest in } \mathcal{B} \setminus T} \sum_{b \in F} w_b. \tag{60.14}$$

A maximum-weight spanning forest of the unknown-bundle graph supplies an optimal design.

Proof. Full recovery requires zero ambiguity dimension in Theorem 60.4. Thus at most one channel may remain unsensed in each bundle, and the unknown bundles retaining such a channel must form a forest. Conversely, these two conditions make both terms of the dimension formula zero and hence suffice.

To maximize saved sensing cost, retain a maximum-cost unsensed channel in each known-total bundle; these choices impose no cycle constraint. For an unknown-total bundle that is retained, the best saved cost is also w_b. The set of retained unknown bundles must be a forest, so a maximum-weight forest maximizes their total savings. Subtracting both savings from the cost of sensing all channels gives (60.14). Nonnegative weights allow a maximizing forest to be extended to a spanning forest in each component without decreasing its weight. □

With no aggregate access, this reduces to the familiar measured-cotree rule on the physical multigraph. With every aggregate known, one maximum-cost channel per bundle can be left unsensed, but all other channel splits require individual information. The algorithms are exact for their declared access and price assumptions; uncertain aggregate measurements or unavailable channels require the corresponding constraints to be retained explicitly.

60.6 The ambiguity witnesses are realizable by positive rates

Allowing signed currents does not require negative transition rates.

Proposition 60.7 (Realization at a prescribed stationary distribution)**.** *Let $j \in \ker B$ be finite, and let π be any strictly positive probability distribution on V. Choose $t_e > |j_e|/2$ for every channel and define*

$$q_e(u_e, v_e) = \frac{t_e + j_e/2}{\pi_{u_e}}, \qquad q_e(v_e, u_e) = \frac{t_e - j_e/2}{\pi_{v_e}}. \tag{60.15}$$

The resulting bidirectional channel rates are strictly positive. Their sum defines a continuous-time Markov chain with stationary distribution π and channel currents exactly j. If the support graph is connected, the chain is irreducible.

Proof. The strict traffic inequality makes both numerators positive. The stationary forward and reverse channel fluxes are respectively $t_e + j_e/2$ and $t_e - j_e/2$, whose difference is j_e. Summing their net contributions at each vertex gives $Bj = 0$, so incoming and outgoing stationary fluxes balance. Setting generator diagonals to minus outgoing rates therefore gives $\pi Q = 0$. Connected bidirectional support makes every state reachable from every other state and hence makes the chain irreducible. □

A finite pair of least-favorable currents h and $-h$ can use the same π and the same traffic $t_e > |h_e|/2$. Thus the sharp-error witnesses can have identical stationary occupations and positive finite rates. An unbounded impossibility result uses arbitrarily large finite rates across the model class, not an infinite rate in an individual model. A fixed physical device may impose traffic, fuel, or affinity limits that restrict this class. Fixed chemical stoichiometry and fixed reservoirs also introduce constraints not encoded by conservation alone.

60.7 Worked example and reproducible interface

Consider states A, B, C, D and six physical channels:

$$e_0 : A \to B, \quad p : A \to B, \quad a : A \to C, \quad b : B \to C, \quad c : C \to D, \quad d : D \to A. \tag{60.16}$$

Suppose only p, a, d are measured, with respective error radii $1/100$, $1/50$, and $3/100$. The cut $U = \{A\}$ gives

$$\widehat{j}_{e_0} = -z_p - z_a + z_d, \qquad R = \frac{3}{50}. \tag{60.17}$$

The following matching circulation certifies sharpness:

$$(h_{e_0}, h_p, h_a, h_b, h_c, h_d) = \left(\frac{3}{50}, -\frac{1}{100}, -\frac{1}{50}, \frac{1}{20}, \frac{3}{100}, \frac{3}{100}\right). \tag{60.18}$$

It obeys conservation, reaches the target error radius, and respects all sensor error bounds. Its two signs can be realized by (60.15) with common stationary occupations.

If all five bundle totals are instead known exactly, one ambiguity remains: the split between e_0 and p. Reading p resolves it. Conservation alone cannot. For a separate exact sensing-cost example, assign costs $c_p = 2, c_a = 3, c_b = 4, c_c = 5, c_d = 6$. The minimum indirect design for e_0 measures p and b, at total cost 6. Conservation gives $j_{e_0} = -j_p + j_b$. This acquisition-price example and the preceding error-radius example use different declared inputs; their numbers must not be interpreted as the same resource quantity.

The implementation is in `code/graph_access.py`. It accepts exact integers, rational strings, or `Fraction` values and rejects floating point capacities. Its main interfaces are:

`target_cut` Finds a minimum cut for supplied acquisition costs or sensor-error capacities, returning a cut and matching circulation.

`verify_target_cut` Checks the certificate directly, without invoking an optimization algorithm.

`mixed_observability` Returns the internal-channel and network-cycle ambiguity dimensions and all identifiable target channels.

`target_completion` Returns a minimum indirect sensor set with exact aggregate access; direct measurement of the target is excluded.

`all_target_completion` Returns an optimal full-current design using the maximum-weight forest construction.

`realize_currents` Constructs positive rational channel rates at a prescribed strictly positive stationary distribution.

The rational example certificates and rate realization are saved in `results/graph_access.json`. Reproduction commands are

```
python code/graph_access.py
python -m unittest discover -s tests -p test_graph_access.py -v
```

Author tests compare cut values with exhaustive vertex partitions, mixed observability with an independent rational rank calculation, and both sensor designs with exhaustive subsets. Separate internal review additionally checked 2,197 mixed-observation configurations, 65 completion instances, and 160 cut cases. These checks support the implementation; the proofs establish the general results. External peer review and experimental validation remain separate requirements.

Chapter 61

Precision, calibration, and acquisition energy

An intervention can make a current identifiable while leaving its numerical value too uncertain to be useful. Conversely, a very precise instrument can remain blind to the desired distinction. Experimental design must address both questions: which observations determine the target, and which attainable error bounds make the answer useful?

This chapter connects those questions for conserved currents. It first treats shared calibration errors, then optimizes a declared relation between sensor precision and acquisition energy. A sharp change occurs when irreducible calibration errors are admitted: a design problem solvable by one minimum cut becomes NP-complete in general. We nevertheless obtain inexpensive certificates that bound the optimum and sometimes determine it exactly.

The ingredients—linear-programming duality, max-flow/min-cut, Hölder's inequality, and supporting hyperplanes—are established mathematics. Their role here is to provide a connected, checkable experimental-design method. The energy results are conditional on an explicit sensor contract. They are not universal thermodynamic lower bounds.

61.1 Recovering a target in the presence of shared nuisance

Let $j \in \mathbb{R}^m$ denote a vector of currents and let $Bj = 0$ encode known conservation laws. For a graph incidence matrix, a column has -1 at the tail and $+1$ at the head of its oriented edge. Negative currents are permitted; the orientation only fixes the sign convention. Suppose the available observations satisfy

$$y = Hj + A\eta + e, \qquad |e_i| \le \varepsilon_i, \tag{61.1}$$

where $\eta \in \mathbb{R}^k$ is an unknown fixed calibration nuisance. Its effect across observations is the known matrix A. The scientific target is $b^T j$.

Throughout this section, the admissible parameter space is the entire linear space $\ker B$, the nuisance is unrestricted, and the error set is the full coordinatewise box. These assumptions define the minimax claim. Bounded traffic, known affinities, correlated errors, or prior parameter bounds can change the sharp answer.

Theorem 61.1 (Constrained linear minimax recovery)**.** *Under* (61.1)*, the minimum, over all*

estimators, of the worst-case absolute error is

$$R = \min_{w,\phi} \sum_i \varepsilon_i |w_i| \qquad \text{subject to } H^T w + B^T \phi = b, \quad A^T w = 0, \tag{61.2}$$

$$= \max_{h,z} \; b^T h \qquad \text{subject to } Bh = 0, \quad |Hh + Az| \le \varepsilon. \tag{61.3}$$

If the constraints in (61.2) have no solution, $R = +\infty$. Otherwise the finite optima are attained, and an optimal estimator is $w^T y$.

Proof. The identities in (61.2) give $w^T y - b^T j = w^T e$, whose absolute value is at most $\sum_i \varepsilon_i |w_i|$. For any feasible (h, z) in (61.3), the two parameter pairs $(j, \eta) = (h, z)$ and $(-h, -z)$ produce the same observation $y = 0$ when their errors are respectively $-(Hh + Az)$ and $Hh + Az$. Their targets are $b^T h$ and $-b^T h$, so every estimator incurs maximum error at least $|b^T h|$. Linear-programming duality equates the two bounds. Equivalently, append the exact observations $Bj = 0$ to (61.1) and apply linear optimal recovery to (j, η). If the primal identities are infeasible, the annihilator alternative provides $Bh = 0$, $Hh + Az = 0$, and $b^T h \neq 0$. Scaling this direction proves unbounded ambiguity. Finite-dimensional linear programs attain their finite optima, including the cases of redundant constraints and zero error radii. □

The certificate contains the estimator weights (w, ϕ) and the least-favorable perturbation (h, z). Checking linear identities, error bounds, and equality of the two objective values establishes optimality without rerunning the optimizer. The small exact solver supplied with this release enumerates supports and vertices; that implementation is intended for examples, not large-scale general linear programming.

A shared bias can be cancelled; an arbitrary bias cannot. On an oriented triangle, conservation gives $j_1 = j_2 = j_3 = J$. Consider

$$y_1 = J + \eta + e_1, \qquad y_2 = -J + \eta + e_2.$$

The estimator $(y_1 - y_2)/2$ has exact worst-case error $(\varepsilon_1 + \varepsilon_2)/2$. For radii $1/10$ and $1/5$, the result is $3/20$. If both responses are $J + \eta$, the current remains unidentifiable. It also remains unidentifiable if the two offsets are independently unknown. The useful operation is a physical response change that reverses the current contribution while preserving the shared offset. Negating an existing recorded number reverses both contributions and creates no additional access.

The lower-bound currents can have positive physical rates. For a connected bidirectional network, fix any positive stationary distribution π. Given a finite circulation j, choose symmetric edge traffics $t_e > |j_e|/2$. On an edge $e : u \to v$, set

$$q^e_{uv} = \frac{t_e + j_e/2}{\pi_u}, \qquad q^e_{vu} = \frac{t_e - j_e/2}{\pi_v}.$$

All rates are positive, π is stationary by conservation, and the channel current is j_e. A least-favorable pair $\pm h$ can use the same π and the same traffics $t_e > |h_e|/2$. Thus the circulation lower bound does not rely on negative transition rates. This construction does not imply that every fixed device with bounded rates can realize the entire current space.

61.2 The exact uncertainty of one hidden current

Let $e_0 : u \to v$ be an edge whose current cannot be measured directly. Remove that edge from the graph. Some remaining edges have sensor records $y_e = j_e + e_e$, with $|e_e| \le \varepsilon_e$; other edges are unobserved. The errors jointly range over the full box. No probabilistic independence assumption is involved.

Proposition 61.2 (Current uncertainty as a minimum cut)**.** *Over the full circulation space, the exact minimax absolute error for j_{e_0} is the minimum u–v cut capacity in the graph without e_0, assigning capacity ε_e to a measured edge and infinite capacity to an unobserved edge. An unobserved u–v path makes the error unbounded. If e_0 is a bridge, its stationary current is zero and needs no measurement.*

Proof. Conservation expresses the target current as a signed sum of the other currents crossing any separating cut. A cut containing only measured edges therefore gives an estimator with error at most the sum of its radii. A maximum flow from v to u through the remaining graph, closed by a flow through e_0, gives a circulation h with $|h_e| \le \varepsilon_e$ on every measured edge. The pair $\pm h$ has a common possible record and target separation $2h_{e_0}$. Max-flow/min-cut equates these bounds. An entirely unobserved return path supports an arbitrarily scaled invisible circulation. A bridge cannot carry a nonzero conserved current. □

The same minimum-cut algorithm can optimize installation cost, error, or a transformed parameter for acquisition energy. These are different inputs and different scientific claims. An error capacity must not be interpreted as an energy cost merely because the same graph algorithm processes it.

61.3 A declared relation between precision and acquisition energy

Assume that an available sensor has the calibrated error contract

$$\varepsilon_e(E_e) = \delta_e + \sqrt{\frac{a_e}{E_e}} = \delta_e + \sqrt{\frac{q_e^3}{E_e}}, \qquad q_e = a_e^{1/3} > 0, \quad \delta_e \ge 0, \tag{61.4}$$

where $E_e > 0$ is acquisition energy in consistently declared units. The floor δ_e represents an error that additional acquisition energy cannot remove under this contract. Zero allocation means no reading and hence unbounded uncertainty.

Equation (61.4) is an input to the design problem. It must be justified for the estimated *current*, including the response model that converts a detector output into that current. A square-root uncertainty law for raw photon or transition counts does not by itself establish the same law after an uncertain dynamical inversion. Finite pulses, poorly known gains, and systematic response errors can produce different exponents or additional floors.

The optimization below concerns fixed sensor selections and nonadaptive, continuous energy allocations. Integer record counts, power limits, minimum pulse durations, and history-dependent acquisition require additional constraints. Preparation, control, reset, calibration, installation, and thermal-support energy are additional unless explicitly included in E_e.

If (61.4) describes a simultaneous confidence envelope valid with probability at least $1 - \alpha$, the resulting target-error bound has that same confidence. Marginal guarantees for individual sensors do not automatically give a joint guarantee. If the envelope is conservative, the optimum below is the optimum certified by the declared adversarial box; it need not be the minimum energy of every physically conceivable experiment.

Theorem 61.3 (Optimal allocation on a cut)**.** *For a nonempty sensor cut C, define*

$$Q_C = \sum_{e \in C} q_e, \qquad \Delta_C = \sum_{e \in C} \delta_e.$$

The minimum acquisition energy giving target-current error at most $\tau > 0$ through that cut is

$$E_C^* = \frac{Q_C^3}{(\tau - \Delta_C)^2}, \qquad \tau > \Delta_C. \tag{61.5}$$

It is attained by

$$E_e = \frac{Q_C^2 q_e}{(\tau - \Delta_C)^2}, \qquad \varepsilon_e = \delta_e + \frac{(\tau - \Delta_C) q_e}{Q_C}. \tag{61.6}$$

No finite energy suffices if $\tau \le \Delta_C$. An empty cut has zero cost.

Proof. Put $x_e = \sqrt{q_e^3 / E_e} > 0$. Feasibility requires $\sum_{e \in C} x_e \le \tau - \Delta_C$. Hölder's inequality gives

$$Q_C = \sum_{e \in C} \left(\frac{q_e^3}{x_e^2} \right)^{1/3} x_e^{2/3} \le \left(\sum_{e \in C} E_e \right)^{1/3} \left(\sum_{e \in C} x_e \right)^{2/3}.$$

This proves the lower bound in (61.5), and (61.6) attains equality. □

Corollary 61.4 (Global sensor and energy design)**.** *Under the full circulation and box-error model, the optimum over fixed allocations, arbitrary sensor subsets, and all estimators is*

$$E^* = \min_C \frac{Q_C^3}{(\tau - \Delta_C)^2}, \tag{61.7}$$

where the minimum ranges over sensor-only separating cuts with $\Delta_C < \tau$, with the bridge exception assigned zero energy.

Proof. Every successful allocation has some separating cut whose summed error is at most τ, by Proposition 61.2. Its total acquisition energy is at least the optimal allocation on that cut. Conversely, a cut estimator with (61.6) attains the displayed value. □

The formula also supplies a finite-energy feasibility test. Let ρ be the minimum cut sum of δ_e, with unavailable sensors assigned infinite capacity. For a nonbridge target with strictly positive a_e, finite energy is possible exactly when $\tau > \rho$. At $\tau = \rho$, more acquisition can approach the floor but cannot attain it at finite energy.

Remark 61.5 (Other calibrated precision exponents). The allocation argument is not specific to a square-root law. For a declared contract $\varepsilon_e = \delta_e + (a_e / E_e)^{1/p}$ with $p > 0$, put $q_e = a_e^{1/(p+1)}$. The same Hölder argument gives $E_C^* = Q_C^{p+1} / (\tau - \Delta_C)^p$ and $E_e = Q_C^p q_e / (\tau - \Delta_C)^p$. This algebraic extension does not establish which exponent a physical sensor obeys. The exact allocation software in this release implements $p = 2$.

61.4 An efficient exact case and a complexity boundary

61.4.1 Zero calibration floors

If every $\delta_e = 0$, minimizing (61.7) amounts to minimizing Q_C. Thus one minimum cut with edge capacities q_e gives

$$E^* = \frac{\left(\min_C \sum_{e \in C} q_e \right)^3}{\tau^2}. \tag{61.8}$$

The cut chooses the sensors and (61.6) chooses their precision. A matching flow certifies that no alternative sensor set can have less energy under the contract.

For rational q_e and τ, all quantities in this certificate are rational. Equivalently, the input sensor family has $a_e = q_e^3$. General cube roots require algebraic or certified interval arithmetic; the exact implementation does not silently round them. The checker verifies the graph, capacity bounds, conservation, cut–flow equality, and the allocation identities, against the externally supplied sensor parameters.

61.4.2 Irreducible calibration floors

Consider the following decision problem. Given a graph, target edge, available sensors with rational $q_e > 0$ and $\delta_e \geq 0$, rational accuracy $\tau > 0$, and rational acquisition budget $L \geq 0$, does a fixed allocation achieve worst-case target error at most τ with total energy at most L?

Theorem 61.6 (Floor-aware design complexity)**.** *For binary-encoded rational inputs, this decision problem is NP-complete. NP-hardness holds even when all nontarget sensors are available, all calibration floors are positive, every cut is feasible at finite energy, and the graph is a target edge plus parallel two-edge paths with distinct internal vertices.*

Proof. A separating cut is a polynomial-size witness. Compute its rational Q_C and Δ_C and check

$$\tau > \Delta_C, \qquad Q_C^3 \leq L(\tau - \Delta_C)^2.$$

The corresponding allocation has polynomial encoding length. An empty cut is handled separately. Thus the decision problem is in NP.

For hardness, reduce PARTITION on positive integers $v_1, \ldots, v_n$ whose sum is V. Set $\beta = V$. For each i, introduce the path s–i–t, with sensor parameters

$$(q, \delta) = (\beta, \beta + v_i) \quad \text{on } s\text{–}i, \qquad (q, \delta) = (\beta + v_i, \beta) \quad \text{on } i\text{–}t.$$

The target is the direct edge s–t. Every separating cut in the graph without the target chooses one edge from each path. If S indexes its second-edge choices, then

$$Q = n\beta + \sum_{i \in S} v_i, \qquad \Delta = n\beta + V - \sum_{i \in S} v_i, \qquad Q + \Delta = K = 2n\beta + V.$$

Set

$$Q_* = n\beta + \frac{V}{2}, \qquad \tau = K - \frac{Q_*}{3}, \qquad L = \frac{9Q_*}{4}.$$

All numbers have polynomial bit length. Every cut has $\tau - \Delta = Q - Q_*/3 > 0$, because $Q \geq nV > Q_*/3$. Its energy is

$$f(Q) = \frac{Q^3}{(Q - Q_*/3)^2}.$$

The exact factorization

$$4x^3 - 9Q_*(x - Q_*/3)^2 = (x - Q_*)^2(4x - Q_*)$$

shows that $f(x) \geq 9Q_*/4$ for $x > Q_*/3$, with equality exactly at $x = Q_*$. There is therefore a feasible cut with energy at most L precisely when $\sum_{i \in S} v_i = V/2$. This completes the reduction. □

The reduction establishes NP-hardness through PARTITION; it does not establish strong NP-hardness or rule out useful approximation, pseudopolynomial methods, or tractable subclasses. It separates an efficiently checkable design from an efficient general search for the optimum. The result also does not mean that scientific access is uncomputable.

Budgeted terminal-cut problems already have a substantial complexity literature. For example, Aissi and Mahjoub's treatment of minimum s–t cuts with budget constraints discusses related hardness results and earlier work.[1] The direct reduction above addresses the particular nonlinear precision–energy objective used here; it is not a claim to have discovered the hardness of multiobjective cuts.

61.5 Global certificates when exact design is difficult

Computational hardness does not prevent a useful guarantee on a proposed design. For any chosen rational support parameter $\lambda > 0$, assign capacities

$$c_e(\lambda) = 3\lambda^2 q_e + 2\lambda^3 \delta_e, \tag{61.9}$$

and let $W(\lambda)$ be their minimum cut value.

Proposition 61.7 (Supporting-cut energy bracket)**.** *The optimum floor-aware acquisition energy obeys*

$$E^* \geq \max\{0, W(\lambda) - 2\tau\lambda^3\}. \tag{61.10}$$

If a minimum cut returned for (61.9) *satisfies $\Delta_C < \tau$, its allocation in* (61.6) *provides a feasible upper bound. Matching bounds certify global optimality.*

Proof. For a feasible cut, write $s = \tau - \Delta_C > 0$ and $x = Q_C/s$. Then

$$\frac{Q_C^3}{s^2} - \left(3\lambda^2 Q_C - 2\lambda^3 s\right) = s(x-\lambda)^2(x+2\lambda) \geq 0.$$

Minimizing the supporting expression $3\lambda^2 Q_C - 2\lambda^3 s$ over all cuts gives $W(\lambda) - 2\tau\lambda^3$. Allowing infeasible cuts in that minimum can only lower it, so it remains a valid lower bound. Energy is also nonnegative. The upper bound follows from the explicit feasible allocation. □

One max-flow computation supplies the lower-bound certificate. Its flow and cut can be checked without rerunning the optimization. Repeating this procedure for a declared finite set of support parameters gives the maximum certified lower bound and the minimum feasible upper bound encountered. This is a supporting-hyperplane bound, not a general exact solution of the NP-complete problem.

The hardness construction also supplies a useful failure control. At $\lambda = 3/2$, the weights of q_e and δ_e coincide. The lower bound is exactly the PARTITION threshold L for every instance. If the integers have no equal partition, the true optimum is strictly larger. Thus a valid global lower bound need not be attainable by any physical sensor cut.

[1] Primary manuscript: https://www.lamsade.dauphine.fr/~mahjoub/Papers/Paper-Aissi-Mahjoub-2023.pdf.

61.6 A complete design example

Consider three vertices s, a, t. Besides the inaccessible target s–t, there is one edge from s to a and two parallel physical channels from a to t. Let

$$q_{\mathrm{single}} = 4, \qquad q_{\mathrm{pair},1} = 1, \qquad q_{\mathrm{pair},2} = 2, \qquad \tau = \frac{1}{10}.$$

The two possible sensor cuts are the single first edge or the pair of parallel channels. With zero floors, the single sensor costs 6400 acquisition units. The pair costs 2700 when its energies are 900 and 1800, giving error radii 1/30 and 1/15. Requiring equal error radii 1/20 instead costs 3600.

Now give the paired sensors floors 3/50 and 1/50, retaining a zero floor for the single sensor. The pair requires 67500 units to attain the same target accuracy. The optimum switches to the single sensor at 6400.

Design at target error 1/10	Acquisition energy
Zero floors: optimal sensor and precision choice	2700
Zero floors: equal precision on the optimal pair	3600
Zero floors: fewest-sensor choice	6400
Nonzero paired floors: retaining the former pair	67500
Nonzero paired floors: revised optimal choice	6400

At support parameter $\lambda = 40$, one supporting cut certifies matching lower and upper values of 6400 for the floor-aware problem. This example illustrates why measurement count, precision allocation, and calibration robustness are different design objectives. Its values are exact results for a constructed model, not measurements of an experimental device.

61.6.1 Executable example and independent checking

From the release's `code` directory, the following example constructs the ideal design and checks its certificate:

```
from constructive_access.completion.graph_access import Edge, Graph
from constructive_access.completion.robust_completion import (
    minimum_ideal_energy, verify_ideal_energy,
    floor_energy_bound, verify_floor_energy_bound,
)

g = Graph(("s", "a", "t"), (
    Edge("target", "s", "t"),
    Edge("single", "s", "a"),
    Edge("pair_1", "a", "t"),
    Edge("pair_2", "a", "t"),
))
q = {"single": 4, "pair_1": 1, "pair_2": 2}
design = minimum_ideal_energy(g, "target", q, "1/10")
checked = verify_ideal_energy(g, "target", q, "1/10", design)
assert checked["total_energy"] == "2700"
```

```
floor = {"single": 0, "pair_1": "3/50", "pair_2": "1/50"}
bracket = floor_energy_bound(g, "target", q, floor, "1/10", 40)
checked = verify_floor_energy_bound(
    g, "target", q, floor, "1/10", bracket
)
assert checked["lower_energy"] == "6400"
assert checked["upper_energy"] == "6400"
```

The generic nuisance interface provides two separate calls:

```
certificate = robust_target_certificate(model)
checked = verify_robust_certificate(model, certificate)
```

The model fields specify conservation, observations, nuisance coefficients, the target, and error radii. The checker tests the primal and dual identities against that supplied model. Exact inputs are integers, rational strings, or Python `Fraction` values; floating-point inputs are rejected.

The test suite includes shared and independent bias, inaccessible targets, bridge targets, zero error, corrupted certificates, calibration-induced sensor choice reversal, PARTITION instances, comparisons with exhaustive cut search, and strict lower-bound gaps. Its purpose is to test both successful design and the conditions under which a conclusion must be withheld. Physical use additionally requires validation of the observation and energy contracts.

Chapter 62

Interfaces that preserve experimental meaning

A local equivalence certificate is useful in a larger apparatus only when the allowed coupling respects what that certificate preserves. This chapter states a finite classical interface contract, proves its error propagation, and gives an exact refinement algorithm. The ingredients are established stochastic lumpability, intertwining and partition refinement. Their role here is to make modular use of access certificates checkable.

62.1 Joint outcome and interface state

A finite classical instrument has nonnegative matrices $M_{a,y}$ with $\sum_y M_{a,y}$ stochastic. Let C be an $n \times k$ row-stochastic interface, and $B_{a,y}$ a k-state instrument with the same actions and outcomes. Define the local defect

$$\epsilon_a(x) = \frac{1}{2} \sum_{y,z} |(M_{a,y}C - CB_{a,y})_{xz}|.$$

The defect includes both the recorded outcome and the state passed to later operations. Agreement of immediate readout probabilities alone is not enough. Preparations, couplings, controls and readout updates belong to the contract.

Theorem 62.1 (Adaptive composition bound)**.** *Let P, Q be the complete terminal-record laws of the concrete and quotient instruments under the same finite adaptive policy tree, with respective initial rows μ, ν. At each internal history h, let $\mu_h = \mu M_h$ be the unnormalized concrete branch row and $a(h)$ its chosen action. Then*

$$\mathrm{TV}(P, Q) \le \min \left\{ 1, \mathrm{TV}(\mu C, \nu) + \sum_h \mu_h \epsilon_{a(h)} \right\}.$$

In particular, exact intertwining $M_{a,y}C = CB_{a,y}$ and matching interface preparations imply identical records for every permitted finite adaptive policy, including outcome-dependent stopping.

Proof. For a quotient branch row ν_h, put $d_h = \mu_h C - \nu_h$. At an action a,

$$d_{hy} = \mu_h(M_{a,y}C - CB_{a,y}) + d_h B_{a,y}.$$

Nonnegativity gives $\sum_y \|\mu_h(M_{a,y}C - CB_{a,y})\|_1 \le 2\mu_h\epsilon_a$. Outcome normalization gives $\sum_y \|d_h B_{a,y}\|_1 \le \|d_h\|_1$. Sum through the tree, retaining every outcome and stopped branch. The

leaf probability difference is $d_h\mathbf{1}$ because $C\mathbf{1} = \mathbf{1}$. Half the sum of absolute leaf probability differences proves the claim. □

The defect sum is an expected accumulated error under the concrete model; it can improve on horizon times a uniform worst-case defect. Exact intertwining is a sufficient contextual certificate. It is not asserted to characterize every trace-equivalent pair.

62.2 Independent components and new couplings

For product instruments with product interface $C_1 \otimes C_2$, the local joint defect at a state pair obeys

$$\epsilon_{12} \le \epsilon_1 + \epsilon_2 - \epsilon_1\epsilon_2.$$

Maximally couple each pair of normalized outcome/interface distributions independently. Both agree with probability at least $(1-\epsilon_1)(1-\epsilon_2)$, proving the bound. For several factors use $1 - \prod_i(1-\epsilon_i)$. A subsequent adaptive context is covered by the preceding theorem if every added interaction also has a checked joint defect.

Matching local preparation marginals does not establish matching joint preparations. The distributions $(00+11)/2$ and $(01+10)/2$ have identical one-bit marginals; a controlled-NOT followed by target-bit readout identifies their parity perfectly. Joint interface equality must be established, for example through an independently justified product preparation.

62.3 Constructing a sufficient deterministic interface

Given an initial observation partition $\mathcal{P}_0$, repeatedly split each block according to the signature

$$\left(\sum_{j\in B} M_{a,y}(i,j)\right)_{a,y,B\in\mathcal{P}}.$$

Proposition 62.2. *The iteration terminates after at most $n - |\mathcal{P}_0|$ strict rounds and returns the coarsest deterministic Markov interface refining $\mathcal{P}_0$ for all declared instruments and all initial preparations.*

Proof. Each strict round increases the number of blocks. At termination, states in one block send the same outcome/transition mass to every target block, exactly the required intertwining identity for the deterministic partition matrix. Any stable refinement of the current partition must separate unequal signatures, since each current block is a union of refined blocks. Induction shows that every stable refinement refines every iteration, and hence the terminal partition. This proves coarseness as well as closure. □

The software exports the forced refinements and independently checks every split and final closure. Minimality refers to deterministic Markov interfaces; it does not imply a smallest linear realization or a cheapest physical sensor. An algorithm naming a finer hidden block does not supply physical access to it.

In the four-state model (h, o), passive readout sees only o. Preparations $(0,0)$ and $(1,0)$ are passively identical. Adding the allowed reversible logical coupling $(h, o) \mapsto (h, o \,\mathrm{XOR}\, h)$ makes a subsequent visible readout distinguish them. The coarsest interface expands from two states to four. This illustrates an actual change in permitted access, while making no claim that physical

implementation of the logical gate costs zero energy. A second example has identical immediate output laws but different hidden updates; its later records separate. The joint defect detects that failure of a readout-only calibration.

The exact implementations are in `constructive_access/completion/composition.py`; reproducible examples and independent checks accompany the source. No theorem in this chapter is claimed for arbitrary quantum couplings or unmodeled shared environments.

Chapter 63

Complete application patterns

The examples in this chapter use the same sequence of questions. What is the target? Which distinct mechanisms fit all presently accessible evidence? Which permitted intervention changes that equivalence? How large is the resulting contrast after calibration error? What experiment and resource boundary support the conclusion? This order is useful across disciplines because each answer constrains the next. The physical ingredients remain different. All numerical results here are exact or certified model calculations, not new laboratory measurements.

63.1 Separating chemical channels hidden by an occupancy trace

Consider two states A, B connected by two distinct chemical channels. Channel one has forward/backward rates (a, b) and channel two (c, d), all strictly positive. An ideal occupancy detector sees

$$k = a + c, \quad \ell = b + d, \qquad Q = \begin{pmatrix} -k & k \\ \ell & -\ell \end{pmatrix}, \qquad \varpi = \frac{1}{k+\ell}(\ell, k). \tag{63.1}$$

The target may be the channel-one net current, its forward throughput, or total entropy production. These are different quantities:

$$J_1 = \varpi_A a - \varpi_B b = \frac{ad - bc}{k + \ell}, \qquad \Phi_1^+ = \varpi_A a, \qquad J_2 = -J_1. \tag{63.2}$$

With channel-resolved local detailed balance,

$$\frac{\dot{S}_{\mathrm{tot}}}{k_B} = J_1 \log \frac{\varpi_A a}{\varpi_B b} + J_2 \log \frac{\varpi_A c}{\varpi_B d} = J_1 \log \frac{ad}{bc}. \tag{63.3}$$

The rates encode a chemostatted effective model. Reservoir identities and chemical free energies are additional physical information [28].

63.1.1 An exact obstruction and the intervention it requires

The rates $(2, 2, 2, 2)$ and $(3, 1, 1, 3)$ give the same complete occupancy generator, with $k = \ell = 4$. Every occupancy-path law from a common preparation therefore agrees. Their native channel-one currents are respectively zero and one, and their entropy rates are zero and $2 \log 3$ in k_B per model time. This ambiguity survives ideal continuous observation; collecting more of the same record cannot resolve it.

Adding the same state-level generator L to both hypotheses leaves their aggregate generators equal. Coupling the same controller to that additive channel also leaves the complete controlled occupancy model equal. The gate in Chapter 6 exposes differences between generators hidden by a detector. It cannot expose a decomposition difference when the full permitted generators already coincide. The appropriate extra access must act differently on the two physical channels.

Suppose a calibrated intervention multiplies both directions of channel one by a known $r > 0$, $r \neq 1$, while leaving channel two unchanged. The observed rates at baseline and under intervention obey

$$k_0 = a + c, \quad k_1 = ra + c, \qquad \ell_0 = b + d, \quad \ell_1 = rb + d. \tag{63.4}$$

Consequently

$$a = \frac{k_0 - k_1}{1 - r}, \qquad c = \frac{k_1 - rk_0}{1 - r},$$
$$b = \frac{\ell_0 - \ell_1}{1 - r}, \qquad d = \frac{\ell_1 - r\ell_0}{1 - r}, \tag{63.5}$$
$$J_1 = \frac{k_0\ell_1 - \ell_0 k_1}{(1 - r)(k_0 + \ell_0)}. \tag{63.6}$$

This is the linear response fingerprint $F_r = \left(\begin{smallmatrix} 1 & 1 \\ r & 1 \end{smallmatrix}\right)$ applied separately to forward and backward rates. Its rank answers exact identifiability. Its inverse conditioning answers practical precision.

At $(a, b, c, d) = (3, 1, 1, 3)$ and $r = 1/2$, the perturbed totals are $(k_1, \ell_1) = (5/2, 7/2)$, and Eq. (63.5) recovers every native rate. The intervention changes the stationary distribution to $(7/12, 5/12)$ and the actual perturbed current to $2/3$. It estimates the native current through a calibrated intervention model; it does not preserve physiology automatically.

63.1.2 A finite precision claim

If each total-rate response has deterministic absolute error at most ε and r is known, the worst-case absolute error of the reconstructed a is $2\varepsilon/|1 - r|$. The upper bound follows directly from Eq. (63.5); opposite endpoint errors attain it. At $\varepsilon = 0.01$, choosing $r = 0.5, 0.9, 0.99, 0.999$ gives error $0.04, 0.2, 2, 20$ respectively. Exact rank alone therefore does not establish a useful assay. An independently calibrated second intervention can improve conditioning: the response rows $(1, 1), (1/2, 1), (2, 1)$ reconstruct a with weights $(0, -2/3, 2/3)$, giving worst-case error $1/75$ at the same response precision.

An explicit uncertain-selectivity example uses the simultaneous calibration box

$$k_0, \ell_0 \in [3.99, 4.01], \qquad k_1 \in [2.49, 2.51],$$
$$\ell_1 \in [3.49, 3.51], \qquad r \in [0.49, 0.51]. \tag{63.7}$$

Equation (63.6) has the sharp range

$$J_1 \in [193/204,\ 207/196] \simeq [0.9461, 1.0561]. \tag{63.8}$$

With the other coordinates held fixed, the expression is linear-fractional in each coordinate and has positive denominator. Its extrema over the box therefore occur among the 32 vertices. Exact vertex evaluation also checks strict positivity of the reconstructed rates. Bounding the positive current and the cycle-affinity logarithm separately gives the conservative entropy interval $[1.946079453, 2.483272369]$ in k_B per model time. This entropy interval need not be the sharp joint range. The rate box is a calibration premise; a confidence statement additionally needs its simultaneous coverage.

For a real biochemical cycle, the decisive work is to validate channel selectivity, off-target effects, bidirectional scaling, the effective state space, and detector time resolution. Molecular counts or concentration and volume are needed to convert single-system turnover into molar flux. A fluorescent binding reporter is not automatically the required selective reaction. Known chemical affinities can also reduce an ambiguity that exists when only occupancy rates are known; the analysis must include those priors. High-order fluorescence correlations and dynamic isotope-flux methods are established tools [27, 29]. This calculation specifies a complementary native-rate identification experiment, with a failure control at $r = 1$ and a quantitative selectivity requirement.

63.2 Quantum access: a symmetry obstruction and two ways to change it

The original quantum benchmark concerns a boundary-driven three-spin chain. It is useful because a target can remain hidden after one enlarges the instrument class substantially. It does not model coherent gate error in a quantum processor; that separate target is treated in the next section.

Let $s^- = |0\rangle\langle 1|$, $s^+ = (s^-)^\dagger$, and $n = s^+ s^- = (I - Z)/2$. In units $\hbar = 1$, define

$$\begin{aligned} H &= g \sum_{j=1}^{2} (s_j^+ s_{j+1}^- + s_{j+1}^+ s_j^-), \\ \mathcal{L}_\delta(\rho) &= -i[H, \rho] + \gamma \sum_{j \in \{1,3\}} \left[(1 - f_j)\mathcal{D}[s_j^-](\rho) + f_j \mathcal{D}[s_j^+](\rho) \right], \\ \mathcal{D}[A](\rho) &= A\rho A^\dagger - \tfrac{1}{2}\{A^\dagger A, \rho\}, \quad f_1 = \tfrac{1}{2} + \delta, \quad f_3 = \tfrac{1}{2} - \delta, \quad |\delta| < \tfrac{1}{2}, \quad g, \gamma > 0. \end{aligned} \tag{63.9}$$

Each record begins in its unique stationary state ρ_δ. Both raising and lowering operators act at the ends. Their commutant has the form $I \otimes A \otimes I$, and commutation with the connected Hamiltonian forces A to be scalar; the finite dynamics is irreducible with a faithful unique stationary state. An accessible operation acts only on site two unless stated otherwise. All ancillary states and experiment rules are independent of the unknown bias.

63.2.1 An exact accessible-effect closure

Central population measurements have projectors $P_s = (I + sZ_2)/2$. More generally consider instruments with real diagonal central Kraus operators $K = \alpha I + \beta Z_2$, with their outcome sums trace preserving. Define the Pauli strings

$$\begin{aligned} B_0 &= III, & B_1 &= IZI, \\ B_2 &= -IXY + IYX + XYI - YXI, & & \\ B_3 &= IIZ + ZII - XZX - YZY, & B_4 &= IZZ + ZZI - XIX - YIY. \end{aligned} \tag{63.10}$$

The span of these five observables is preserved by $\mathcal{L}_\delta^*$. With columns giving the images of basis observables, its matrix is

$$C = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -4g & 0 & 0 \\ 0 & g & -\gamma/2 & -2g & 0 \\ 0 & 0 & 2g & -\gamma & 0 \\ 0 & 0 & 0 & 0 & -\gamma \end{pmatrix}. \tag{63.11}$$

For $a_K = \alpha^2 + \beta^2$, $b_K = 2\alpha\beta$ and $c_K = \alpha^2 - \beta^2$, the adjoint outcome map $O \mapsto KOK$ has matrix

$$M_K = \begin{pmatrix} a_K & b_K & 0 & 0 & 0 \\ b_K & a_K & 0 & 0 & 0 \\ 0 & 0 & c_K & 0 & 0 \\ 0 & 0 & 0 & a_K & b_K \\ 0 & 0 & 0 & b_K & a_K \end{pmatrix}. \tag{63.12}$$

These identities follow by Pauli multiplication and contain no δ. The stationary expectation vector obeys $C^{\mathsf{T}} v = 0$, $v_0 = 1$. The lower block has determinant $4g^2\gamma^2$, so $v = (1, 0, 0, 0, 0)^{\mathsf{T}}$. Every record probability is therefore a product of e^{tC} and outcome matrices evaluated between this vector and the identity observable. It is independent of δ.

This proves complete blindness to both magnitude and orientation for that instrument class, including adaptive times and almost surely finite stopping: each terminal branch has the same probability. A central bit flip is outside the real diagonal Kraus class. Arbitrary complex diagonal Kraus operators have not been included in the closure. Measurements are invasive: inserting an unrecorded population measurement can change later statistics. The records at different schedules are thus different experiments, rather than samples of one undisturbed classical trajectory.

63.2.2 Magnitude can be activated; orientation has a stronger obstruction

At $g = \gamma = 1$, an X_2 readout, evolution for t, and another X_2 readout give

$$p^{XX}_{++}(\delta, t) - p^{XX}_{++}(0, t) = \frac{4}{25}\delta^2 t^2 + O_\delta(t^3). \tag{63.13}$$

A population readout, immediate X_2 bit flip independent of its result, evolution for t, and another population readout instead give

$$p^{ZXZ}_{++}(\delta, t) - p^{ZXZ}_{++}(0, t) = -\frac{16}{25}\delta^2 t^2 + O_\delta(t^3). \tag{63.14}$$

These coefficients can be checked without numerical fitting. The exact stationary solution is

$$\begin{aligned} 8\rho_\delta = {} & III + \tfrac{2\delta}{5}(IIZ - ZII) + \tfrac{4\delta}{5}(-IXY + IYX - XYI + YXI) \\ & - \tfrac{16\delta^2}{25}(IZZ + XIX + YIY + ZZI) - \tfrac{4\delta^2}{25} ZIZ \\ & + \tfrac{8\delta^2}{25}(-XYZ + YXZ + ZXY - ZYX). \end{aligned} \tag{63.15}$$

Substitution verifies unit trace and $\mathcal{L}_\delta(\rho_\delta) = 0$. Expanding the two completely positive instrument expressions through t^2 gives coefficients $(1/2, 0, -1/4 + 4\delta^2/25)$ for XX and $(0, 0, 1/2 - 16\delta^2/25)$ for ZXZ. At a sufficiently small fixed positive t, all four outcome probabilities have positive zero-bias limits. Their divergence from the zero-bias law is consequently $\Theta(\delta^4)$, with independent-record fixed-error cost $\Theta(\delta^{-4})$. A magnitude signal does not imply a sign signal.

Let S exchange sites one and three, and let $\mathcal{S}(\rho) = S\rho S$. Equal exchange bonds imply

$$\mathcal{L}_{-\delta}\mathcal{S} = \mathcal{S}\mathcal{L}_\delta, \qquad \rho_{-\delta} = S\rho_\delta S. \tag{63.16}$$

Every central operation commutes with $\mathcal{S}$. By induction, the unnormalized states on each outcome branch under opposite biases remain related by S and have equal trace. Replacing S by $S \otimes I_A$ includes arbitrary center–ancilla instruments, coherent control during evolution, feedback and stopping. Thus every central-only protocol remains blind to the orientation $+\delta$ versus $-\delta$. This is

the operational symmetry principle familiar from quantum identification [30, 31], instantiated with explicit stationary preparation. Ancillas already correlated with the hidden reservoirs would change the access contract.

One way to break the obstruction is to change the physical bonds to known positive values l, r. Solving the stationary covariance equations then gives

$$\langle n_2 \rangle = \frac{1}{2} + \delta \frac{l^2 - r^2}{l^2 + r^2}. \tag{63.17}$$

For a short derivation, the single-particle covariance satisfies $\dot{A} = -i[h, A] - \{\Gamma, A\}/2 + P$, where h has adjacent entries l, r, $\Gamma = \mathrm{diag}(\gamma, 0, \gamma)$, and $P = \mathrm{diag}(\gamma(1/2 + \delta), 0, \gamma(1/2 - \delta))$. The stationary current and central occupation solve

$$J = \frac{8\delta\gamma l^2 r^2}{(l^2 + r^2)[\gamma^2 + 2(l^2 + r^2)]}, \qquad A_{22} = \tfrac{1}{2} + \delta - \frac{J[\gamma^2 + 2(l^2 + r^2)]}{4\gamma l^2}.$$

Substitution proves Eq. (63.17). With $l = g(1 + \eta)$ and $r = g(1 - \eta)$, the Bernoulli shifts are $\pm 2\delta\eta/(1 + \eta^2)$ and their KL divergence is $32\delta^2\eta^2/(1 + \eta^2)^2 + O(\delta^4\eta^4)$. Unknown bond asymmetry would be a nuisance parameter confounded with bias. Changing bonds and re-preparing stationarity costs a physical resource; the sign activation does not come from center-only computation.

63.3 Targeted coherent-error diagnosis without full tomography

For a processor diagnostic, choose two explicit single-qubit alternatives:

$$U = aI - ibX, \quad a^2 + b^2 = 1, \qquad \mathcal{U}(\rho) = U\rho U^\dagger, \quad \mathcal{P}(\rho) = a^2\rho + b^2 X\rho X. \tag{63.18}$$

Follow each call by known depolarization with Bloch retention λ. The channels have equal average fidelity and equal ideal gate-independent randomized-benchmarking decay parameter. Preparing $|0\rangle$ and making one Z readout after one call also agrees. A phase-sensitive readout separates the coherent rotation from the stochastic alternative. This does not imply equality of all randomized-benchmarking records; unitarity and related methods already diagnose noise coherence [32].

Take $a = 399/401$, $b = 40/401$, $\lambda = 19/20$ and $\cos\vartheta = a^2 - b^2$, $\sin\vartheta = 2ab$. Starting on the Z axis, the stochastic alternative has zero Y expectation after any number of calls. The coherent alternative rotates the Bloch vector by $m\vartheta$ and contracts it by λ^m. A suitable calibrated phase-sensitive readout therefore has probability gap

$$\Delta_m = \frac{1}{2}\lambda^m |\sin(m\vartheta)|. \tag{63.19}$$

Assume per-hypothesis preparation, known rotation and readout discrepancy at most 10^{-3}, and per-call half-diamond discrepancy at most 10^{-4}. The record discrepancy is at most $10^{-3} + m10^{-4}$. A certified gap is

$$g_m = \Delta_m - 2(10^{-3} + m10^{-4}). \tag{63.20}$$

For $g_m > 0$, thresholding the average binary outcome at the midpoint gives per-hypothesis error at most $\exp(-Mg_m^2/2)$ with fresh independent preparations. Taking $M = \lceil 5/g_m^2 \rceil$ makes this smaller than 0.1. The exact rational menu search over $m = 1, \ldots, 80$ yields

Calls per record	Sufficient records	Unknown-channel calls	Role
1	590	590	single-call baseline
4	60	240	least call count in this menu
7	44	308	largest nominal gap in this menu

The count optimum and contrast maximum differ. These are sufficient counts under a declared bound, not an exact optimal test or a total energy comparison. Preparations, calibrated pulses, readouts and computation remain separate resources. Dephasing after every unknown-channel call, combined with Z-only preparation and detection and no intervening coherent control, is an exact negative control: the two accessible models then coincide.

This approach can support targeted monitoring of a promised finite local dictionary. If there are B independent calibrated blocks, a fixed number of candidate types per block, and a uniform positive witness gap g, a union allocation of error δ/B gives call cost of order $B\log(B/\delta)/g^2$ for fixed-depth witnesses. This assumes independent fresh records and excludes unmodeled cross-talk and memory. It does not characterize arbitrary n-qubit channels at constant cost: their Hermitian representation dimension is 4^n, and their interior parameter count is $16^n - 4^n$.

Even a fixed small dimension cannot bound the required precision cost. Consider replacer channels producing diagonal qubit states with probabilities $1/2 \pm \epsilon$, $0 < \epsilon \leq 1/4$. A two-action prepare/readout word distinguishes them, but the single-output divergence is

$$D = 2\epsilon \log \frac{1+2\epsilon}{1-2\epsilon} \leq 16\epsilon^2. \tag{63.21}$$

Any strategy using at most N calls, including collective output measurements and feedback, is a hypothesis-independent processing of N fresh commuting output states. Product additivity and data processing imply total variation at most $\sqrt{8N}\epsilon$. Error at most $\delta < 1/2$ consequently requires

$$N \geq \frac{(1-2\delta)^2}{8\epsilon^2}. \tag{63.22}$$

At $\epsilon = 1/1024$ and $\delta = 0.1$, this is at least 83887 calls. The bound concerns a maximum call count, not an expected stopping time. A short algebraic witness and reliable diagnosis are distinct achievements.

63.4 Ramsey sensing with an unknown fixed readout offset

An effective two-level Ramsey sensor has the model

$$p(+ \mid \omega, b, t, \phi) = \frac{1}{2} + b + \frac{C}{2} e^{-t/T_2^*} \cos(\omega t + \phi). \tag{63.23}$$

Here b is an unknown offset that remains fixed throughout the experiment. The target is the sign of ω, not its local differential sensitivity. For an NV implementation, the effective channel, fluorescence contrast, preparation, microwave phases, relaxation and detector response require independent calibration [33].

The finite benchmark uses four equally weighted hypotheses: $\omega = \pm\vartheta/\tau$, $b = \pm 1/20$, with $\cos\vartheta = 4/5$, $\sin\vartheta = 3/5$, $C = 4/5$ and $e^{-\tau/T_2^*} = 4/5$. The 16 actions choose $t = k\tau$, $k = 1, \ldots, 4$, and phase $\phi \in \{0, \pi/2, \pi, 3\pi/2\}$. Each record costs $k + 2$ time ticks, including overhead. All probabilities are rational. The signal map is a valid rotation and dephasing channel; the offset and contrast define valid binary effect operators. Conditional on the same fixed (ω, b), fresh preparations are independent. Redrawing b independently every shot would define a different experiment.

For unnormalized hypothesis weights z_h, the terminal sign-decision loss is

$$L(z) = \min\left\{ \sum_{\omega_h < 0} z_h, \sum_{\omega_h > 0} z_h \right\}. \tag{63.24}$$

With remaining budget B, the exact recursion is

$$V(z,B)=\min\left\{L(z),\quad \min_{a:c_a\le B}\sum_{y\in\{+,-\}}V\big((z_hp_h(y\mid a))_h,B-c_a\big)\right\}. \tag{63.25}$$

Positive action costs make the tree finite. This is a Bayes classification objective over four fixed candidates; it is not a continuous-nuisance minimax or a Fisher-information optimum. A cosine-only probe can have positive local Fisher information and remain exactly blind to the global alternatives $+\omega$ and $-\omega$.

At maximum budget 12 ticks, the optimal risk is $9754517453/61035156250 = 0.159818013949952$. The best fixed sequence attains the same value. Early stopping reduces expected duration in that example, but there is no benefit from feedback-dependent probe choice. At budget 15, the adaptive risk is 0.157898013949952, whereas exhaustive comparison with all 1500 admissible nonadaptive multisets gives 0.159818013949952. The absolute advantage is exactly $6/3125 = 0.00192$. This comparison matches maximum duration, not expected duration or physical heating.

The small exact improvement supplies a reproducible decision-design benchmark. Adaptive Ramsey sensing has experimental precedents [34, 35]. To use this construction on a device, replace the hypothetical four-point model with calibrated candidates or envelopes, include drift and readout memory if present, and compare protocols at the same inference target and complete resource boundary. Refocusing a DC signal and same-axis static noise removes both. Ideal sign modulation also does not remove white Markov dephasing. Noncommuting operations and asymmetric timing are tools for a specified signal model, not universal filters for biological noise.

63.5 Why complete computation does not imply complete energy inference

A final example supplies a limit that every proposed energy-diagnostic application must respect. Let the accessible state X be a two-state chain, and let Z be an independent unobserved three-state cycle. Its clockwise rate is $1+r$, its counterclockwise rate is one, and $r\ge 0$. The hidden stationary law is uniform. With a visible control a the joint generator is

$$G_r^a = Q_a\otimes I + I\otimes R_r. \tag{63.26}$$

Preparations, controls and detectors act on X alone; the hidden state is initially uniform. For $B = I\otimes\mathbf{1}$, the intertwining identity

$$G_r^aB = BQ_a \tag{63.27}$$

implies $e^{tG_r^a}B = Be^{tQ_a}$. Visible instruments and reset maps obey the corresponding identities. Multiplication proves equality of every finite accessible record law for all r. Conditioning on each history extends the equality to causal policies, including noncommuting visible controls. Thus even complete optimization over that access cannot learn r.

Yet the hidden entropy-production rate is

$$\frac{\dot S_{\mathrm{hidden}}}{k_B} = r\log(1+r). \tag{63.28}$$

Under an isothermal local-detailed-balance realization, restoring a rate scale κ gives net fuel power $k_BT_b\kappa r\log(1+r)$, unbounded across this family in the absence of a fuel or rate bound. Gross supply is at least the net power. All visible records are identical while physical expenditure can differ

without bound. This proves a target-specific nonidentifiability result, rather than a failure of an inference algorithm.

Forward/reverse relative entropy of a coarse record can provide a lower bound on entropy production under the relevant assumptions [36]. Exact reconstruction is possible in some more restricted settings, including particular observed-edge unicyclic networks [37]. Neither statement makes every coarse detector a complete heat meter. Landauer's principle constrains specified erasure operations under its thermodynamic premises [38]; it does not turn a current witness into a route that automatically attains the bound.

The practical route for AI hardware is a targeted diagnostic coupled to independently calibrated fuel, power, or heat measurements. A useful claim would identify a particular waste mechanism, intervene upon it, and verify lower total expenditure at matched computational performance. The access framework determines which diagnostic distinctions are possible and what additional measurement is necessary. Optimization of the physical hardware then needs a causal model of performance and expenditure.

63.6 What transfers across the examples

Chemical channel ambiguity, a quantum reflection symmetry, readout-offset uncertainty, and a hidden dissipative cycle are physically different. Their common structure is an equivalence of accessible record laws. A successful intervention changes that equivalence for the target; a reliable experiment also makes the contrast exceed calibration and sampling error.

The examples supply different remedies. Selective chemistry changes a rate fingerprint. Bond asymmetry breaks a quantum symmetry. A phase-sensitive readout preserves coherent information. Adaptive Ramsey design allocates a finite duration among globally useful probes. Independent physical energy measurement adds a distinction absent from the computational record. None of these remedies follows merely from increasing the complexity of an algorithm applied to unchanged evidence.

These are usable model-level application patterns. Deploying any one of them requires a declared target and candidate class, calibrated instruments and preparations, a checked finite-error decision rule, a matched comparison, and an explicit resource boundary. That common contract is the contribution of the integrated subject to scientific practice; successful physical transfer must be demonstrated separately for each apparatus.

Chapter 64

A complete target-specific scientific workflow

The pairwise algebraic engine answers whether two mechanisms can be separated. Many experiments involve several plausible mechanisms, some of which agree on the property of interest. Reconstructing every mechanism would then solve a harder problem than the scientific question requires. This chapter closes that interface with a finite-family design and inference procedure.

64.1 Compatible models are the common output

Let Θ be a model family and $g : \Theta \to \mathcal{G}$ the target. A data-dependent set $C_t \subseteq \Theta$ contains the mechanisms still compatible with the accessible evidence at time t. Define the retained target set

$$T_t = g(C_t).$$

A scalar target gives an interval or a more general subset of the real line; a discrete scientific question gives a set of labels. The output need not identify a single mechanism.

Theorem 64.1 (Target and resource composition)**.** *Suppose a fixed true mechanism θ_* belongs to the declared family. Let G be a joint calibration event with $P(G^c) \le \delta$. Suppose*

$$P\{G \text{ and } \theta_* \notin C_t \text{ for some } t\} \le \alpha.$$

An output made only when it is valid for every target in the nonempty set $g(C_t)$ is invalid at some time with probability at most $\alpha+\delta$. If a resource certificate additionally gives $P\{G \text{ and } W > B\} \le \beta$, then the probability of any invalid output or expenditure above B is at most $\alpha + \beta + \delta$.

Proof. On G, if $\theta_* \in C_t$, then $g(\theta_*) \in g(C_t)$ and every output valid for that whole set is valid for the actual target. Thus an invalid output on G implies the coverage failure in the premise. The first assertion follows by adding $P(G^c)$. For the second, take a union with the resource failure on G. Calibration failure is added once. No independence between the data, resource expenditure and calibration event is needed. □

The premise is an unconditional intersection bound. Conditioning an otherwise valid concentration inequality on a favorable calibration event does not automatically supply it. The adequacy chapter gives one construction using predictable tests; externally fixed valid record-law envelopes give another. An empty set triggers abstention, not a vacuous claim that every target has been proved. The theorem composes guarantees but does not establish that the family contains reality.

64.2 A finite experiment menu

Let H be a finite candidate set, with target labels $g(h)$. Each available experiment e has a finite complete record alphabet $\mathcal{Y}_e$, a nominal law $p_{h,e}$ for every candidate, and a nonnegative selection charge c_e. Suppose the actual law under candidate h belongs to the whole-record total-variation ball

$$\mathrm{TV}(\widetilde{p}_{h,e}, p_{h,e}) \le \eta_{h,e}.$$

Preparation, measurement backaction, control errors and detector effects must be included in this record-law envelope or separately transferred into it. A bound on an isolated readout probability is not a substitute.

For a pair i, j with different target labels, the event

$$A_{ij,e} = \{y : p_{i,e}(y) > p_{j,e}(y)\}$$

attains the nominal total-variation difference. Its robust separation is

$$d_{ij,e} = \mathrm{TV}(p_{i,e}, p_{j,e}) - \eta_{i,e} - \eta_{j,e}.$$

If $d_{ij,e} > 0$, the declared event probability intervals are disjoint under the two candidates. If the bound is nonpositive, this event does not provide a positive robust separation. Additional structure in the calibration set can support a stronger conclusion than independent TV balls; the present algorithm uses only the stated balls.

A suite S covers the target if every cross-target pair has a positive gap for at least one $e \in S$. Selecting a minimum-charge suite is the finite weighted set-cover problem

$$\min_{S \subseteq \mathcal{E}} \sum_{e \in S} c_e \quad \text{subject to} \quad \forall i, j : g(i) \neq g(j),\ \exists e \in S : d_{ij,e} > 0.$$

This is established combinatorial optimization. The implementation uses exact dynamic programming on covered pair masks, with explicit pair and state caps. A cap violation is an unsupported-size error, not an impossibility certificate. The objective counts each selected experiment once. Optimizing its repeated execution, elapsed time or full energy is a different problem.

If the menu does not cover the target, the software names the unresolved pairs. This is an obstruction for the declared robust menu. It does not establish that all physically possible experiments fail, nor that the target is unidentifiable under a more precise calibration model.

64.3 From the suite to a finite-error target decision

If every candidate has the same target, no acquisition is needed within that family. Otherwise assign every cross-target pair to one covering experiment. Retain the associated events, merging duplicate events within an experiment, and discard any experiment assigned no pair. Let $L > 0$ be the number of retained event tests. For each chosen experiment, choose

$$r_e = \frac{1}{4} \min\{d_{ij,e} : \text{the pair is assigned to } e\}.$$

The implementation chooses an integer k and record counts n_e satisfying

$$\frac{2L}{2^k} \le \alpha, \qquad n_e = \left\lceil \frac{k}{2r_e^2} \right\rceil.$$

These are sufficient, conservative bounds. They use exact rational arithmetic rather than a floating-point logarithm. Since $\exp(1) > 2$, Hoeffding's inequality gives

$$2\exp(-2n_e r_e^2) \le \frac{2}{2^k}.$$

The nominal laws, events, radii and suite are fixed before the classification records are collected. Within each experiment, the records must be independent and identically distributed under one fixed mechanism. Different experiments need not be mutually independent for the following union bound. If calibration data determine the inputs, the same iid premise must hold conditional on that pre-acquisition information, and the joint calibration event must bound these conditional record laws. Integrating the conditional concentration guarantee then gives the intersection bound on G required by Theorem 64.1; conditioning on an arbitrary favorable event is not sufficient.

For each retained event A, let $\widehat{p}_e(A)$ be its frequency. Retain candidate h precisely when, for every retained event,

$$|\widehat{p}_e(A) - p_{h,e}(A)| \le \eta_{h,e} + r_e.$$

Theorem 64.2 (Finite-family target decision)**.** *Suppose the selected suite covers every cross-target pair and the finite-family and record-law assumptions above hold. With probability at least $1-\alpha$, the retained set contains the fixed true candidate and all retained candidates share its target. Returning that target when the retained set is nonempty, and abstaining when it is empty, has probability at most α of either an incorrect target or an empty retained set. A jointly bounded calibration failure adds once as in Theorem 64.1.*

Proof. For each event, Hoeffding bounds the difference between empirical frequency and its actual probability by r_e except with probability at most $2/2^k$. The union bound over L events has probability at most α. On the complementary event, the actual record-law calibration bound and the triangle inequality retain the true candidate.

For any two retained candidates i, j with different targets, examine their assigned event and experiment. Their nominal event probabilities differ by more than $\eta_{i,e} + \eta_{j,e} + 2r_e$, since $d_{ij,e} \ge 4r_e > 2r_e$. But both candidates' retention would force the difference to be at most that quantity, by applying the triangle inequality through the same empirical frequency. This is a contradiction. Consequently, any nonempty retained set has a single target; on the simultaneous concentration event it is the true target. The failure assertion follows. □

The algorithm cannot recover an erased distinction between mechanisms that share the same target and the same laws. It does not need to do so. Nor does all-rejected abstention prove that a particular alternative model is correct. It identifies a failed inference attempt that warrants checking the contract.

64.4 A reproducible five-model example

Five candidate mechanisms have two target labels, "even" and "odd". Their binary-record probabilities are:

Model	Target	Left probe	Right probe	Direct probe
A	even	1/4	1/4	1/4
A copy	even	1/4	1/4	1/4
B	even	3/4	3/4	1/4
C	odd	1/4	3/4	3/4
D	odd	3/4	1/4	3/4

A copy is operationally identical to A. The left probe alone leaves A and C unresolved; the right probe alone leaves A and D unresolved. Together they separate every cross-target pair. If the left and right selection charges are one each and the direct probe charge is three, the exact minimum suite is left plus right, with charge two.

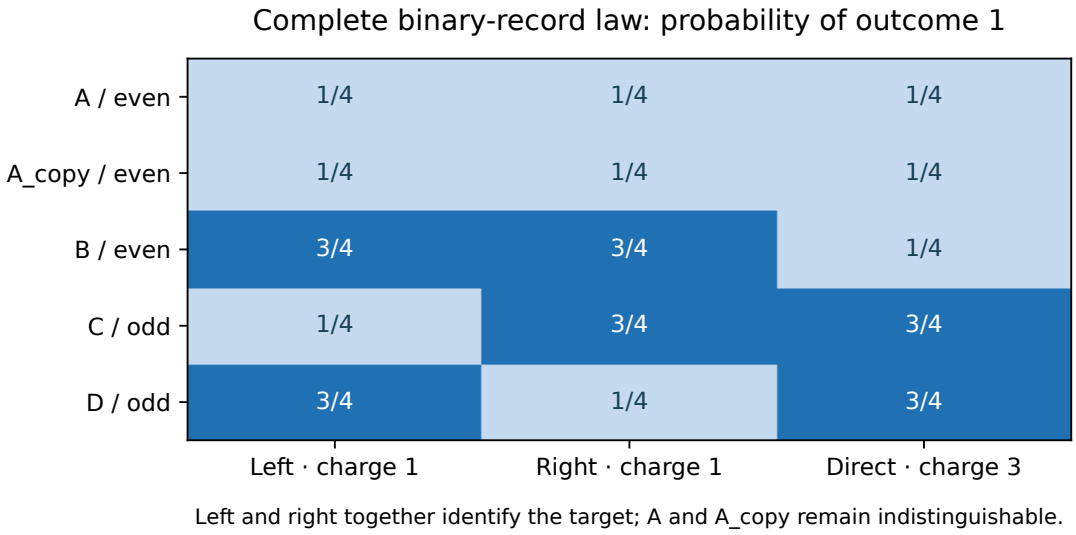


Figure 64.1: The complete laws in the finite-menu tutorial. The selected pair of probes resolves the target label while leaving two mechanisms indistinguishable. Every displayed probability is specified analytically.

Set every whole-record TV radius to 1/100 and the sampling failure budget to 1/20. The robust gaps are 12/25, the selected sampling radii are 3/25, and the implementation allocates 278 records to each of the two experiments: 556 records in total. Deliberately synthetic frequencies near 1/4 for both probes retain A and A copy and report "even". The complete mechanism remains unidentified. If both TV radii of a pair increase to 1/4, its nominal half-unit contrast no longer gives positive separation; the robust menu reports the unresolved pairs.

Run the example after installation:

```
python examples/suite_tutorial.py --output results/suite_tutorial.json
```

The JSON includes every nominal law, radius, selected event, record count, retained model and limitation. The supplied counts are synthetic and test the inference interface; they are not simulated discoveries or laboratory measurements.

64.5 Using a result in the laboratory

A useful deployment proceeds through one chain of evidence. First write the physical contract and the target. Then test structural access: either produce an equivalence certificate or generate a candidate experiment. Next verify that calibration leaves a nonzero target-relevant gap. Choose the statistical objective and repetition or feedback policy, and translate the proposed operations into a complete resource ledger. Execute the protocol only after this mapping is justified. Finally retain the models or target values compatible with the actual records, and report abstention or model failure when required.

A result becomes transferable when another group can substitute its calibrated model, run the same checks, execute the returned protocol, and reproduce the promised target coverage or error rate. Different laboratories need not share microscopic mechanisms. They must share the mathematical interface and justify its assumptions for their apparatus.

The next extensions should preserve that standard. Continuous parameter families require validated optimization or analytic envelopes. Large quantum devices require locality, sparsity or other justified structure. Nonstationary experiments require conditional predictions or explicit drift models. Resource comparisons require matched objectives and complete physical accounting. These are clear mathematical and experimental tasks, rather than claims already established by the finite examples.

Chapter 65

The reference implementation

The accompanying package turns a precisely specified comparison into a portable algebraic certificate and a complete experimental plan. Its public entry point begins with the scientific question, the two target labels, the available interventions, and the preparation. The program does not infer these ingredients from a dataset. A scientist supplies them and can inspect every consequence reported by the calculation.

Version 0.6.0 retains a strict JSON workflow for two finite classical instruments with rational coefficients. A second, Python-level interface exposes finite linear representations, continuous-time approximation, bounded adaptive design, and sequential model checks. These interfaces have different hypotheses; their availability in one package does not silently extend the scope of the JSON workflow.

65.1 Install and run the first comparison

From the root of the released reference package, use Python 3.10 or later:

```
python -m pip install .
constructive-access analyze examples/passive.json -o passive-report.json
constructive-access analyze examples/active.json -o active-report.json
constructive-access verify examples/active.json active-report.json
```

The installation requires SymPy and mpmath. The portable algebra checker itself uses only Python's standard library. For an existing environment with the dependencies installed, the same commands are available through `python -m constructive_access`. No code imports from an earlier research directory or requires an unreleased local file.

The passive example has two hidden states, initial distribution $(1/2, 1/2)$, and the same state-revealing read operation under each candidate. All passive transcript laws coincide. The program returns an exact equivalence certificate; collecting more passive records does not change that model conclusion.

The active example also permits an unobserved drive. Its transition matrix has both rows equal to $(3/4, 1/4)$ under candidate 0 and $(1/4, 3/4)$ under candidate 1. Reading costs one operation, driving costs two, and fresh preparation costs one additional operation per complete record. The least-action-charge distinguishing experiment is drive followed by read. It has action charge 3 and total declared charge 4 per record. Its complete two-outcome laws are

$$P_0 = (3/4, 1/4), \qquad P_1 = (1/4, 3/4).$$

The reported total variation is 1/2 and the one-record equal-prior Bayes error is 1/4. The outward-rounded affinity calculation certifies 12 independent complete records as sufficient for average error at most 1/10, with declared batch charge 48. This is a sufficient count, not a minimum sample count. Neither the operation charge nor its minimum is a measured energy consumption.

The witness contains an action–outcome word, but the implemented experiment retains *all* outcomes of the associated action sequence. A drive is not repeated for free until a desired outcome occurs. The report lists the complete normalized law, making that distinction inspectable.

65.2 What a contract contains

`load_contract` accepts a file path, a JSON-compatible mapping, or a validated contract. It rejects floating-point model coefficients, duplicate JSON keys, unsupported fields, invalid probability kernels, negative charges, and mismatched action/outcome alphabets. Integers and exact rational strings such as `"1/3"` are accepted. Decimals written as strings are interpreted as the exact rational number they specify, without a claim that the physical coefficient is exact.

Each contract contains the following information:

- A question, target name, and two distinct target labels, ordered with the candidate models.
- Two normalized classical instruments: a row initial distribution and one nonnegative substochastic matrix for each recorded outcome of each action. Outcome matrices sum to a row-stochastic matrix for each action. Missing probability mass must be represented by an explicit recorded outcome, such as abort.
- A declared additive action charge, units, resource interpretation, per-record overhead, calibration statement, and included and excluded items. Energy is an allowed interpretation of a calibrated bound, never an inference from the word length.
- A preparation description and an explicit declaration of whether complete records are independently and identically prepared.
- A desired equal-prior average error, rational precision settings, and a cap on explicit transcript enumeration.
- Optionally, a deterministic finite automaton restricting complete action sequences. Its transitions cannot depend on postselected outcomes.
- Optionally, a whole-record total-variation calibration radius for each candidate, a joint calibration failure probability, and the stated justification for those inputs.

The canonical contract hash binds these assumptions and target meanings to the report. It identifies the supplied specification. It does not establish the provenance of a physical calibration, authenticate a laboratory, or prove that the candidate family contains the true mechanism.

65.3 Separate synthesis from checking

The Python entry points are:

```
from constructive_access import load_contract, analyze, verify_report
```

```
contract = load_contract("examples/active.json")
report = analyze(contract)
verification = verify_report(contract, report)
assert verification["verified"]
```

Analysis builds a paired linear representation, runs an exact charge-priority span search, exports its filtered closure identities, and checks them. A witness is minimal in the lexicographic order (additive action charge, word length). The report checker independently reconstructs the paired model from the caller-supplied contract and checks every required expansion identity. It does not rerun the witness search, compute ranks, or trust the synthesizer's history. It then reevaluates the complete selected experiment and its reported statistical quantities. Modified conclusions, unsupported fields in the report, altered probabilities, missing closure identities, and certificates for a substituted model fail this check.

There are three distinct mathematical outputs. An equivalence certificate proves equality for every permitted finite protocol in the declared model. A witness proves a nonzero probability contrast. A finite-error bound provides a sufficient number of repeated complete records under its stated sampling assumptions. A nonzero contrast is not perfect discrimination.

The transcript cap is a computational guard, not an observability theorem. If explicit enumeration exceeds it, the program retains the exact access certificate and returns `transcript_limit_exceeded`; it does not issue a statistical plan. If iid preparation is not declared, the program reports one-record quantities without a repeated-record guarantee.

65.4 Calibration can change the operational answer

Run the uncertainty example with:

```
constructive-access analyze examples/calibrated.json -o calibrated.json
constructive-access analyze examples/calibration_gap_closed.json
```

The first example assumes whole-record TV radii $1/100$ under both candidates and a joint calibration failure probability $1/100$. These are illustrative assumptions, not measurements. For the event $E = \{y : P_0(y) > P_1(y)\}$, the program evaluates

$$a = \max\{0, P_0(E) - r_0\}, \qquad b = \min\{1, P_1(E) + r_1\}, \qquad g = a - b.$$

When $g > 0$, classify using the empirical event frequency with threshold $(a + b)/2$. Conditional on pre-acquisition calibration information, each fixed actual hypothesis must produce iid classification records. A simultaneous calibration event certifies the corresponding conditional whole-record laws. Classification records cannot be reused to choose or calibrate the event or threshold.

For a conditional error allocation α, the exact implementation chooses an integer M satisfying

$$\frac{Mg^2}{2} \geq L \geq \log(1/\alpha).$$

Here L is a rational upper enclosure obtained by range reduction and the positive series

$$\log x = 2\sum_{k=0}^{n-1} \frac{z^{2k+1}}{2k+1} + R_n, \quad z = \frac{x-1}{x+1}, \quad 0 \leq R_n \leq \frac{2z^{2n+1}}{(2n+1)(1-z^2)}$$

after reducing x to $[1, 2]$. Hoeffding's inequality gives a conditional error at most α for each hypothesis. Adding the stated joint calibration failure probability gives the unconditional bound. There is no claim that this conservative event test is optimal.

The supplied example has $g = 12/25$, threshold $1/2$, and $M = 21$. Its conditional allocation is $9/100$, and the unconditional bound is $1/10$. Declared batch charge is 84 operations, including the stated preparation overhead. The negative control increases both radii to $1/4$; the robust gap closes and the program returns `calibration_gap_not_certified`. This failure concerns the specified uncertainty certificate. It does not prove that every possible physical experiment fails.

65.5 Advanced interfaces and their boundaries

The top-level Python package also exposes the following preserved implementations:

Finite instruments. Use `RationalInstrument` for a normalized classical instrument. Use `LinearRepresentation` for a general weighted representation, and `ActionDFA` for an action grammar. The functions `compare_models`, `minimum_cost_witness`, and `experiment_law` support direct finite-model work. A signed linear representation is not automatically a physical quantum instrument; a separate complete-positivity and normalization justification is required.

Continuous time. `certified_wait` encloses a finite CTMC transition by an exact rational stochastic approximation and a row-TV error. `protocol_tv_radius` and `robust_word_gap` propagate the stated approximation error. Equality of approximate transition laws cannot prove equality of the true continuous-time laws.

Adaptive design. `optimal_design` optimizes the declared two-hypothesis classical Bayes risk under a finite integer budget and a mandatory finite horizon. `best_fixed_design` supplies a matched open-loop comparison. These routines do not optimize arbitrary quantum policies or unknown physical models.

Sequential adequacy. `AdequacyMonitor`, `Interval`, and `interval_bet` maintain e-process evidence against declared conditional predictive envelopes. Their validity requires the appropriate conditional predictions at every step; stationary marginal intervals cannot be substituted.

Linear targets. `identify_linear_target` checks the row-space criterion for a known fingerprint matrix. `minimax_certificate` and `verify_minimax_certificate` compute and check exact primal–dual recovery bounds for small rational instances over the full positive parameter orthant with a coordinatewise deterministic error box. The generator uses vertex enumeration and is not presented as a scalable general linear-programming solver.

Counting-channel energy. `count_cap` and `verify_count_cap` produce and check a rational Chernoff envelope under a separately supplied predictable compensator bound. `declared_energy_cap` combines a checked count cap with a gross-work-per-event bound and an apparatus bound in the same physical unit. The compensator, work, and apparatus assumptions must be justified separately from observation-law calibration.

The finite-family experiment-suite workflow is described separately in this reference. It addresses target-specific design across multiple candidates; the two-model JSON command remains deliberately explicit about its smaller input class.

For a concrete linear recovery example:

```
from constructive_access import minimax_certificate, verify_minimax_certificate
```

```
F = [[1, 1], ["1/2", 1]]
b = [1, 0]
epsilon = ["1/100", "1/100"]
certificate = minimax_certificate(F, b, epsilon)
assert certificate["absolute_error"] == "1/25"
assert verify_minimax_certificate(F, b, epsilon, certificate)["verified"]
```

This deterministic minimax guarantee assumes a known intervention response matrix and the specified full error box. It does not certify that a biochemical reporter implements the declared selective intervention. For a counting-channel illustration, `count_cap(10, "1/100")` returns 22. With gross event work at most 3/2 and apparatus work at most 4 in the same unit, the conservative declared energy cap is 37. The example does not establish those physical input bounds from sensor data.

65.6 Reproduction and provenance

The package contains source-level provenance for every preserved module in `constructive_access/_vendor/PROVENANCE.json`, including the original SHA-256, the vendored SHA-256, and any import-path substitution. Source copies are independent of earlier release directories. No software license is invented or implied by this packaging step.

The workflow tests exercise passive equivalence, active witnesses, charge minimality checking, zero-charge actions, action grammars, complete transcript normalization, calibration gap closure, non-iid declarations, enumeration limits, malformed exact inputs, report forgery, model substitution, and command-line failure handling. The independent review tests supplement these with separately derived examples. Run the complete original workflow test collection using:

```
python -m unittest discover -s tests -v
```

Passing these checks establishes reproducibility for the supported software contracts. It does not establish a sensor calibration, a successful laboratory intervention, an unmodeled dissipation estimate, or community acceptance of the proposed framework.

65.7 Response and access-completion modules

The `constructive_access.completion` subpackage adds six modules: `physical_access`, `quantum_completion`, `pulse_test`, `graph_access`, `robust_completion`, and `composition`. Their contracts and examples appear in the corresponding chapters.

```
python examples/completion_tutorial.py \\
    --output results/completion_tutorial.json
python verify.py
python research/clean_reproduction.py
```

The first command uses the installed package to verify the qubit response threshold, construct the calibrated three-arm test, and certify both an energy lower and upper bound of 6400 in the declared model units. The full reproduction additionally runs producer tests and separate adversarial checks, regenerates exact certificates and numerical illustrations, and builds this reference. The clean reproduction builds an offline wheel, installs it in a separate environment, and checks scientific outputs against the source run. The software can check a supplied physical contract; it cannot establish an apparatus calibration or an experimental result by executing a model.

65.8 Calibrated explanations and sequential experiments

The `constructive_access.robust` subpackage adds exact physical response bounds, protected clock design, fixed and anytime statistical tests, and counts-to-explanation inference. Run the complete installed example with

```
python examples/robust_tutorial.py --output results/robust_tutorial.json
```

The explanation routine receives an already specified preparation, detector, clock and calibration contract. Its synthetic demonstration is labeled in both source and output. For actual data the record count must be fixed before collection, or the separate time-uniform inference method must be used. The sequential betting fraction and null envelope must be selected before the current random preparation arm and clock are drawn. Marginal clock agreement without independence from the physical trajectory is insufficient.

The same example certifies the arbitrary-preparation realization-size witness and its 6000-record four-arm design. The functions `encode_dimension`, `dimension_plan`, and `verify_dimension` preserve the distinction between a complete witness-error radius and a single arm's error. The quantum certificate uses exact rational preparations and an independently checked open-system model.

65.9 Classical realization and thermodynamic certificates

The `constructive_access.frontier` subpackage exposes five separate contracts: `hierarchy`, `spectral`, `limits`, `thermodynamic`, and `energy_certificate`. Run their installed examples together with

```
python examples/frontier_tutorial.py --output results/frontier_tutorial.json
```

The tutorial checks three finite-noise state requirements, an Erlang-clock certificate, a matching cyclic classical model, a reversible approximation, a synthetic-count entropy bound, and the continuous-moment curvature witness. These are different access contracts, not interchangeable black-box tests.

For example, the following inference accepts two predeclared independent-pair counts. The numbers below are synthetic:

```
from fractions import Fraction as F
from constructive_access.frontier import thermodynamic

result = thermodynamic.from_counts(
    5000, 2560, 5000, 3680,
    F(1, 50), F(1, 20), F(8, 5), F(21, 10))
assert result["rate_lower"] > 0
```

The two lags are 8/5 and 21/10 in declared time units, the one-sided probability radius is 1/50, and the joint sampling failure allowance is 1/20. An entropy interpretation requires stationary sampling, common homogeneous dynamics, noninvasive fixed readout and ordinary time reversal. Sampling counts do not validate those physical assumptions.

For a positive-integer lag menu, `energy_certificate.certify` accepts simultaneous probability intervals and positive rational scores. Its exported certificate records all inputs, a continuous polynomial normalizer bound and a directed lower bound on relative-entropy separation. The

function `verify` recalculates that certificate against separately supplied inputs. It is a deterministic recheck, not a second proof implementation. Independent tests separately check generator models and polynomial bounds. The optional `design` score proposer requires NumPy and SciPy; it is followed by exact continuous-domain checking. A failed or weak proposal cannot be interpreted as evidence of equilibrium.

The complete cumulative test and figure collection is run by `python verify.py`; `-skip-pdf` omits typesetting only. Source provenance binds all five packaged modules to their preserved phase sources. No previous release directory is required. None of these certificate functions discovers an apparatus's hidden-state count, reversal parity or thermodynamic embedding from an undeclared dataset.

65.10 Virtual laboratories and apparatus-aware inference

The `constructive_access.virtual` subpackage adds exact contrast and dependent-record confidence interfaces, timing and exposure controls, and numerical laboratories for thermodynamic and qubit experiments. Simulations require the optional numerical dependencies:

```
python -m pip install '.[simulation]'
python examples/virtual_tutorial.py --output results/virtual_tutorial.json
```

The tutorial exercises the installed mathematical interfaces without replaying the production simulations. The complete frozen simulation contracts are in `virtual/contracts`. Standalone producer scripts and raw retained records are in `virtual/code` and `virtual/results`. A full `python verify.py` replay regenerates all model experiments, matched comparisons, exact countermodels, figures and the reference. The separate clean reproduction installs a wheel and compares scientific outputs byte for byte.

All parameters in these virtual laboratories are illustrative. The simulator knows a hidden mechanism to evaluate coverage and power; the inference functions do not receive that mechanism. Exact concentration certificates, numerical physical evolution, Monte Carlo intervals and measured physical evidence are separately labeled. Chapter 38 separately reanalyzes published measurements; those data are included with attribution and an explicit inference limit.

65.11 Calibrated inference and public-data reanalysis

The `constructive_access.calibrated` subpackage supplies calibrated moment supports, nonlinear certificates, activity bounds, dissipation countermodels, sequential replay, common-block confidence and bounded analog inference. Run the installed tutorial with

```
python examples/calibrated_tutorial.py --output results/calibrated_tutorial.json
```

Each function retains its own physical and sampling contract. In particular, stationary marginal records, independent pairs, dependent common blocks and predictable batches do not share interchangeable confidence corrections. The nonlinear primal upper bound concerns an observational relaxation, not an upper bound on physical entropy production. A null upper endpoint in the resolved-jump entropy helper means positive infinity.

The empirical extra installs HDF5 support. The three licensed, unmodified measurement files live in `calibrated/data/source_audit`; the wheel does not embed them. Standalone loader and analysis commands reproduce the data audit and exploratory analysis without network access. Hash and schema checks precede numerical reads. Neither embedded Python objects nor pickled

force models are evaluated. The empirical chapter records which unavailable calibrations prevent a thermodynamic certificate.

Chapter 66

Research lineage, scope, and routes to application

66.1 The subject and its place in existing science

Constructive access concerns a concrete experimental question: given a target property, a class of possible mechanisms, and an apparatus, which conclusions can be justified, and what additional experiment would justify more? The apparatus includes preparation, interventions, timing, detection, retained records, and resource limits. Its mathematical object is the family of transcript laws produced by the permitted protocols. This formulation supplies a common language for systems whose underlying physics can be quite different.

This subject lies at the intersection of system identification, statistical decision theory, controlled stochastic models, quantum characterization, and thermodynamic inference. Each has a substantial history. Aggregated Markov models and their indistinguishable realizations were studied in ion-channel kinetics [39, 40]. Finite linear representations and equivalence algorithms originate in automata and realization theory [2, 3, 5]. Observable operator models and predictive state representations already replace inaccessible coordinates by predictions of future experiments [6, 7]. A predictive coordinate need not name a unique molecule, hidden conformation, or physical degree of freedom.

The classical–quantum linear correspondence is also established. Quantum finite-state equivalence uses finite-dimensional operator spaces [8, 41]; a recent unified treatment constructs canonical generalized hidden Markov representations for classical and quantum models [44]. The present use of an accessible effect space inherits that lineage. Similarly, risk-based comparison of experiments and optimal recovery provide established foundations for deciding which distinctions matter and how errors limit recovery [1, 45].

The contribution developed in this reference is an explicit chain from those foundations to a checkable experimental conclusion. A user specifies a physical access model and a target; obtains an obstruction or a distinguishing protocol; verifies its algebraic certificate; assigns calibration and sampling errors; and states the physical resource boundary under which the conclusion holds. The same chain includes abstention when the supplied model class fails or the remaining models disagree about the target. The particular model results and software artifacts are useful parts of that chain. Their status is recorded separately from the established general principles.

66.2 How the earlier results fit together

The initial boundary question asked when a local observation must display an arrow of time inherited from driving elsewhere. A finite ring made the boundary mechanism explicit: one connection carries the driving, while a binary detector reports a region. This separated three targets that an informal arrow-of-time argument can conflate: a driven hidden dynamics, irreversibility of the recorded process, and incompatibility of that recording with every equilibrium explanation.

In the four-state adjacent-region example, the recording can be reversible even when the ring is driven; on the specified reciprocal branch an explicit equilibrium realization produces the same complete recording. Larger-ring classifications, detector placement, and weak-drive response orders then showed that access depends on geometry and acquisition. Retaining timestamps or changing readout spacing can change the leading information order without changing the hidden physical process. These are exact classifications of the stated models, rather than statements about the existence of time.

The next step changed access through interventions. In the driven triangle, each constant-control singleton record is blind to current orientation, whereas a sequence of two controls produces a current-dependent return contrast. An autonomous reversible kinetic gate provides a second route: a binary three-time correlation acquires a leading term proportional to the controlled edge current and the gate profile's Dirichlet form. The gate's own steady channel entropy contribution can vanish while the source remains driven. Preparation, sensing, and control implementation still require their own accounting.

The quantum branch showed why the target must remain explicit. For the specified three-site chain, one family of central instruments conceals the bias magnitude, while a larger family exposes it. Reflection symmetry still conceals the sign under every permitted central-only protocol. A calibrated spatial asymmetry changes that conclusion. Magnitude, orientation, and a complete microscopic model are different identification tasks.

The finite-instrument formulation subsequently provided a common algebraic interface for these examples. Its cost certificates concern the cheapest nonzero witness under a prescribed additive charge. Robust testing and finite-budget decision design concern reliable inference. These objectives require separate calculations: a very cheap nonzero witness may have much worse reliable-decision cost than a more informative experiment.

The channel-resolved application studies expose a further distinction between an inaccessible state and an inaccessible transition channel. Two reaction mechanisms can have the same full state generator yet different physical currents and dissipation. Adding the same state-level control preserves that equivalence. A calibrated channel-selective intervention can remove it in the worked two-channel model. This is a more precise rule for choosing access than the suggestion that any temporal modulation reveals hidden dynamics.

Table 66.1 summarizes the retained role of each stage. The machine-readable provenance record binds individual inherited claims to their source and result hashes; the table is a reading map, not a count of discoveries.

The subsequent stages extend this chain in two directions. Compatible-model results turn evidence into requirements on hidden state count, preparation and sustained dissipation, with explicit equilibrium countermodels. The virtual laboratories establish how timing, detector exposure and serial dependence alter those certificates. The calibrated-inference chapters then improve statistical use of detector moments, analog records and sequential acquisition, while proving limits on total-dissipation reconstruction. Their joint calibration test treats the extra measurement directly as part of the final scientific contrast. The public-data chapter tests this chain against an actual observation format and records precisely where calibration remains insufficient.

Stage	Retained contribution to the reference
Boundary and detector models	Explicit source, sensor, equilibrium mimics, sampling effects, and response-order examples.
Controlled hidden currents	Interventions that alter observable distinction; control placement and preparation failure cases.
Autonomous equilibrium gate	Current–correlation identity, triangle activation criterion, and the cost of hiding the controller record.
Finite instruments	A common exact representation, finite equivalence test, witness synthesis, and bounded decision design.
Certification and adequacy	Independently checked cost certificates, conditional prediction tests, calibrated abstention, and energy tails.
Physical diagnostic models	Selective channel identification, targeted quantum diagnosis, fixed-nuisance sensing, and hidden-energy obstructions.

Table 66.1: The dependency structure of the research. Each row supplies a different part of the experimental argument.

66.3 What is established, and what is specific to this work

Three-time correlation tests of nonequilibrium and current inference predate this work [27]. Equilibrium kinetic coupling can also carry trajectory information [42]. The specific gate result here is the factorized coefficient, its controlled assumptions, the triangle converse, and the stated information-order comparison. The general-network coefficient supplies a sufficient activation condition; a vanishing coefficient need not imply complete invisibility. Historical priority for this precise combination remains open.

The independent cost checker verifies explicit filtered closure identities, including those needed to exclude every cheaper word. Its useful distinction is the exported proof obligation: a user can check the claimed objective without trusting the search history. Exact equivalence, invariant-space certificates, and weighted witness methods already exist [3, 5]. The implementation does not establish that its supplied matrices describe an apparatus, nor does it convert arbitrary charges into joules.

The adequacy layer uses established nonnegative betting processes [21, 43]. Its application obligation is to supply valid conditional prediction envelopes under the actual interventions and histories. That is stronger than supplying stationary averages. The selective-response criterion and its deterministic error bounds use linear identifiability and optimal-recovery geometry. Their exact certificates make the assumptions and the amplification of calibration error visible.

Recent thermodynamic inference already extracts kinetic information from partially observed transitions, including periodically driven systems [46]. Care is needed when applying thermodynamic time reversal after coarse observation: a physically justified reversal operation and a defined observation map are part of the model [47]. This reference's instantaneous finite-state detectors have an explicit reversal convention; extensions to milestoned, filtered, or history-dependent detector outputs must re-establish the relevant correspondence.

Adaptive quantum design also has experimental precedents [48, 34]. The worked quantum and Ramsey calculations here provide transparent small models, explicit alternative hypotheses, and matched decision objectives. They are reference calculations against which an implementation can be checked. Demonstrating a hardware advantage requires comparison with appropriate existing methods on the same device.

66.4 A practical standard for extending the subject

A new application should identify the target before selecting its mathematics. For a current, the target must name the physical channel and its units. For a quantum fault, it must name a justified error dictionary and the operations permitted during diagnosis. For a sensor, it must specify whether the task is sign discrimination, local parameter precision, tracking, or a decision with unequal consequences. For energy inference, it must state whether the target is native-channel free-energy dissipation, gross supplied work, or total apparatus consumption.

The minimum useful transfer includes a negative control. Examples are a channel decomposition that produces the same state record, a control family that preserves a quantum symmetry, an insufficiently selective intervention, or a missing candidate that forces abstention. A successful case should then change exactly the access assumption responsible for that failure and predict the resulting effect before evaluating new data.

Four external milestones would establish practical maturity in a new setting:

1. An independent specialist checks the theorem, the software contract, and the closest prior methods for the stated model class.

2. Independent calibration and physical measurements test the preparation, intervention, detector, memory, and resource assumptions.

3. A comparison holds the scientific objective, error criterion, and full resource boundary fixed; an increase in contrast alone is insufficient.

4. A second implementation or apparatus reproduces the prediction and tests at least one intended transfer beyond the development example.

These are measurable acceptance conditions. The present release supplies mathematical derivations, executable models, and internal adversarial checks. It also includes the exploratory published-data reanalysis in Chapter 38, which identifies concrete apparatus-calibration gaps. It contains no new laboratory measurements, empirical entropy validation or external peer-review decisions.

66.5 Extensions with a clear scientific payoff

Several extensions now have a precise starting point. A validated nonlinear response family can replace the linear selective-response model, provided identifiability and uncertainty propagation are proved for that family. Chapter 6 supplies a finite CTMC adapter. Extensions beyond that finite-model, finite-menu contract to continuous controls or infinite-dimensional dynamics must also carry approximation error into the final decision bound, rather than report only solver tolerance. A structured many-body model can reduce computational burden, provided the structure is physically justified and the bound includes deviations from it. Detector memory, drift, and correlated preparation can be handled by enlarging the state or by conditional envelopes that actually cover those effects.

Chapter 62 now supplies a finite classical composition contract and a constructive deterministic interface refinement. Beyond that scoped result, composition remains a concrete research direction. Individually uninformative components can become informative when the allowed coupling changes their transcript laws; combining records that remain jointly equivalent cannot do so. A useful composition theorem must specify the joint preparation, shared hidden variables, allowed couplings, and target. This distinguishes a real increase in experimental capability from merely adding more components or more processing.

The broad claim is therefore an operational one: scientific conclusions can be organized around what specified experiments preserve, erase, and expose. The constructive finite theory makes a substantial part of that question computable and reviewable. Further generality is earned by extending those interfaces and proving their contracts, with the physical interpretation checked in each new setting.

66.6 Relation to earlier work and the next physical test

The contribution of access completion is a chain of usable certificates: identify an inaccessible target, specify the additional physical operation, bound what its data can establish, and expose the assumptions under which the conclusion fails. Several links in this chain are established mathematics. Separating those foundations from the results derived here makes the proposed physical test easier to assess and reproduce.

Established foundations. Conservation-law sensor placement, spanning-tree constructions, cost, and precision constraints have a substantial earlier literature [49, 50, 51]. The cotree and target-cut results are explicit network-flow specializations. Perturbation design based on graph-theoretic identifiability also predates this work [53]. Target-specific allocation belongs to optimal experimental design [54, 55]; adding resource constraints to terminal cuts has established complexity consequences [56]. The particular calibration-floor objective studied here therefore requires its stated error model and direct proof. It is not a universal thermodynamic energy law. Likewise, lumpability and approximate trace guarantees underpin the classical interface construction [68, 71]. Quantum equivalence under composition has its own established theory [69, 70].

The precise response result. Stationary barrier-response identities are established [57, 58]. Finite-amplitude stationary formulas already include parallel physical channels [59]; transient response relations are also available [60]. Monotonicity of a stationary response as a barrier strength varies [61] concerns a different variable from the pulse duration used here. First-passage response theory supplies further close connections [62, 63].

The candidate extension is the following all-time statement. Starting from native stationarity, suppress one bidirectional physical channel by a fixed factor $r < 1$ while holding the other rates fixed. For a singleton endpoint detector, the response factors as

$$\Delta p_j(t) = -(1-r)J_e H^r_{ij}(t), \qquad H^r_{ij}(t) > 0 \quad (t > 0). \tag{66.1}$$

The first-hit proof uses the strong Markov property and ordinary occupation potentials, within established Markov potential theory [65]. Its useful consequence is a current-sign certificate at every positive pulse duration, with explicit bounds using only endpoint exit-rate ceilings. It does not require reconstructing the hidden network. A protected coarse detector also qualifies when a separately justified entrance-gateway condition holds; arbitrary aggregation does not preserve the conclusion.

The driven-qubit calculation gives an exact parameter boundary at which this classical response restriction fails. Connections between coherent dynamics and failure of classical stochastic descriptions are already well developed [66, 67]. Here the distinguishing feature is the specified intervention and endpoint-response inequality. The result rejects the calibrated classical model class; its assumptions cannot be inferred from two response averages alone.

Priority and evidence limits. The proofs and computational certificates establish the stated mathematical claims within their models. They establish neither historical priority nor laboratory performance. The particularly close sensor-design article by Prakash [52] was available only through primary metadata and indexed excerpts in this audit. The higher-cumulant response article of Bao and Liang [64] was inspected at abstract level only. Their full texts remain necessary overlap checks. The search of older finite-time potential-kernel results is also incomplete. A detailed source map records which full texts, sections, or abstracts were inspected. These physical-response examples have no new empirical validation or external peer-review decision. Chapter 38 separately reports an exploratory reanalysis of published optical-tweezer measurements.

The next validation gate. The highest-leverage physical test is a controlled two-level platform with independently adjustable incoherent excitation and decay channels. It should first establish stationary preparation, population-readout calibration and leakage bounds; then independently verify that the pulse rescales both rates of the selected channel while leaving the Hamiltonian and other channels within declared tolerances. This calibration is part of the experiment, not an assumption justified by fitting the desired sign reversal.

The measurement then compares native and pulsed endpoint populations at two prespecified durations, with randomized acquisition order and independent repreparation. The incoherent control should obey the classical sign restriction. Increasing the calibrated coherent-drive ratio across the derived threshold should produce the predicted reversed response. Sampling, preparation, drift, detector, and intervention error budgets must be small enough that the two response intervals exclude zero with opposite signs. Repeating the experiment with deliberately nonselective control or coarse readout tests the documented failure modes rather than merely reproducing the favorable case.

Passing this gate would validate a practical way to test a structural model assumption using a small set of interventions. It would also determine which parts of the calibration dominate sensitivity. A failure would locate a specific mismatch between the physical platform and the model. Either outcome is more informative than projecting the framework directly onto unresolved problems of large-scale computing or biological flux.

66.7 Positioning and physical interpretation

The principal extension in this phase is a quantitative treatment of a coarse detector whose natural entrance distribution need not coincide with the reference preparation. A departure from ideal occupation ordering then has a measurable consequence: any admissible classical explanation must provide enough entrance mismatch, sufficiently rapid crossings of the observed boundary, or both. The comparison is against a specified set of physical capabilities, rather than against a fitted list of hidden networks.

Let B be the detected region, let the reference preparation lie in B, and let the comparison preparation lie outside it. Write $H(T)$ for the difference of their expected cumulative occupations in B. In the reset-conditioned theorem, η bounds the total-variation mismatch between the first-entrance law and the reference reset law, conditional on the entrance time. With the stated uniform entry and exit intensity bounds a, b, define

$$A_c(T) = \int_0^T e^{-ct}\,dt, \qquad G_{a,b}(T) = \int_0^T e^{-bs-a(T-s)}\,ds.$$

The sharp envelope is

$$H(T) \geq G_{a,b}(T) - \eta\{T - A_a(T) - A_b(T) + G_{a,b}(T)\}.$$

Sharpness means attainability within the declared mathematical class; it does not imply that an arbitrary apparatus realizes an extremizer. If the braces are positive, a calibrated upper confidence bound on $H(T)$ gives a necessary lower bound on η for any classical explanation satisfying the chosen rate caps. It gives neither an upper bound on the actual mismatch nor a unique hidden mechanism.

At $\eta = 1$, the envelope reduces to $H(T) \geq A_a(T) + A_b(T) - T$. This endpoint requires only the relevant survival envelopes and remains valid for history-dependent and time-dependent classical processes satisfying the conditional hazard bounds. The intermediate $0 \leq \eta < 1$ theorem additionally uses its stated reset and continuation conditions; the broader endpoint scope must not be transferred to it without those conditions. Here a and b are caps on crossing intensities. They are not the average total jump activity, heat production, or energy expenditure.

Classical speed and no-jump bounds are established. Shiraishi, Funo and Saito relate distributional motion to dissipation and dynamical activity [79]; Hasegawa and Nishiyama derive concentration and response bounds using no-jump probabilities [80]. The present inverse problem concerns a difference between two occupation experiments, with an explicit reset-defect parameter and a saturating construction. Coarse-grained memory and its effect on inference are also established concerns [81]. No claim of historical priority follows from these comparisons. The contribution offered for review is this particular sharp envelope, its operational interpretation, and the calibrated implementation.

Physical use requires evidence for the caps over the histories and preparations covered by the test. An observed mean switching rate is insufficient. A hidden entrance distribution cannot generally be certified from a binary detector alone; when independent reset information is unavailable, the $\eta = 1$ envelope is the applicable conservative alternative. Preparation errors, detector drift, timing discretization and finite-sample uncertainty must be included before inverting the bound. A violation excludes the calibrated class. It does not by itself establish quantum dynamics or identify the failed assumption. No apparatus-specific validation of these requirements has yet been performed.

66.7.1 Dimension rather than an assumed reset interface

A complementary interpretation replaces reset and rate information with a bound on hidden state count. Let C_0 denote the contrast between two arbitrary preparations measured immediately, and C_T their contrast with the same binary detector at an independent uniform time in $[0, T]$. Every common homogeneous classical Markov realization with at most three states obeys

$$|C_0 - C_T| \leq 1.$$

The detector may have any fixed state-dependent response probability in $[0, 1]$. Measuring the immediate contrast removes the need to assume perfect extremal preparations. The protected-sign rule follows when $C_0 = 1$. A violating qubit example and a four-state classical construction distinguish quantum attainability from the minimum classical realization size at this scalar statistic. They do not characterize the full quantum record or exclude classical explanations of unrestricted size.

Dynamical dimension inference already has a substantial literature. In particular, delayed-data and spectral methods establish classical/quantum dimension separations [83], while repeated-operation null tests provide experimentally implemented alternatives [87]. Temporal memory

witnesses also predate this work [85, 86]. The proposed distinction is a finite-window integrated inequality with an exact three-state threshold and a four-state construction, under its stated preparation and evolution assumptions. Historical priority and an experimental advantage over these methods remain unresolved.

Those assumptions supply real information. With two unrestricted preparations, general prepare-and-measure data can always use two classical messages [84]; the stronger state count here depends on requiring a single homogeneous continuous-time evolution and a common detector. Hidden controller or environmental states that affect the evolution belong in the classical realization being counted.

Chapter 67

Relation to established results and the remaining contribution

The useful question is not whether partial observations can hide a mechanism. That obstruction is established. The question pursued here is what a finite, calibrated temporal record forces *every compatible model* to contain or expend. The answer depends on the experimental contract: a common dynamics, repeatable preparation, detector, accessible times and an explicitly stated classical or quantum comparison class. These conditions belong to the theorem, not merely to its interpretation.

The work combines three tasks that are often separated: exclusion of models with too few states, exclusion of stationary equilibrium explanations, and constructive examples showing when neither conclusion is justified. The specific mathematical and computational results should be evaluated on their own statements. A shared presentation does not establish a new physical law, a complete classification of hidden dynamics, or priority over the literatures below.

67.1 State number, positive realization and finite records

Wolf and Pérez-García already derived dimension information from observable dynamics, including noisy Hankel-rank witnesses and a damped-qubit example with arbitrarily large classical dimension requirements [83]. Their spectral argument uses stochastic eigenvalue restrictions originating in work including Dmitriev–Dynkin and Karpelevich [88, 89]. An unbounded classical–qubit gap or a frequency-to-decay state bound is therefore established prior work. Statistical detection matrices also infer observable stochastic dimension from noisy data [96].

The extra issue is *positive realization*: a scalar signal can have small ordinary linear rank while requiring a much larger probabilistic realization. The established positive-realization literature makes this distinction explicit [90]. A nonnegative matrix factorization is a necessary condition for a finite temporal model; it need not supply a common stochastic semigroup realizing every time. Conversely, a low ordinary rank cannot certify a small positive realization.

Two finite-record constructions address this gap here. The random-clock construction supplies a quantitative distance from all generators below a specified state number, allowing unknown preparation, unknown fixed detector and arbitrarily large hidden rates. It does not infer an exact pole. The antichain construction uses a finite temporal probability matrix and an explicit positive-noise margin. The conservative record count of the first construction is part of its result; existence of a finite test does not imply experimental efficiency.

The antichain ingredient has especially close precedents. The exact support-based inverse-central-binomial bound follows from established Boolean rank and nonnegative-rank arguments

[92, 91] and has already been applied to quantum preparation–measurement tables [93]. Heinosaari and collaborators use the same sinusoidal matrix, up to normalization, to demonstrate unbounded quantum advantage; they also establish a stronger exact rank for the seven-point instance [94]. The candidate addition here is the analytic tolerance for nonzero diagonal entries, its single repeated-channel temporal realization, and a specified finite-count rejection rule. It is not the first sinusoidal matrix, quantum advantage or application of Sperner's theorem.

Temporal memory and environment-dimension witnesses provide another relevant body of work [86, 95]. Their measurement sequences can be invasive. A fresh preparation with one endpoint measurement imposes a different contract, so numerical comparisons require matching the operations, reset assumptions and counted memory. Quantum counting-process precision is also connected to finite memory [111]; an external randomized endpoint clock is not itself a first-tick counting process.

67.2 Classical approximation identifies the limit of the contract

The Poisson smoothing operator used by the classical compiler is the established Szász–Mirakjan construction [97]. Its approximation properties are not new. The role of the compiler is to expose the resources that can remain unobserved: enough hidden states, fast transitions and a specially prepared initial state can approximate a finite collection of prepare–wait–read curves over a bounded interval. An explicit finite Markov implementation, error budget and adaptive-record comparison make that obstruction usable in the present inference problem.

This yields a necessary distinction. A finite experiment can exclude every model below a state bound while failing to exclude the union over all state bounds. It can also reveal an expensive preparation without forcing persistent stationary dissipation. A claim that the same finite trace disproves all classical explanations would erase precisely the degrees of freedom that the compiler constructs.

67.3 Dissipation, reversible comparison and preparation

Thermodynamic restrictions on coherence and temporal correlations are established. They include affinity–coherence constraints, cross-correlation asymmetry bounds, oscillation bounds and correlation-time inequalities [98, 99, 100, 101, 102]. Their resource definitions matter: cycle affinity, stationary entropy production, activity and an eigenmode's weight are different quantities. Recent work further emphasizes these distinctions [107, 108]. Finite-lag cross-correlation bounds are also under active development [106].

The reversible comparison used here is not a new operation. Additive reversibilization has an established information-geometric interpretation [103], and the Jensen–Shannon/Jeffreys inequality underlying the quarter-factor estimate is established [104]. The inspected, unrefereed October 2026 manuscript [110] already states the exact trajectory bound

$$D(\mathbb{P}_Q^{[0,t]} \| \mathbb{P}_{(Q+Q^*)/2}^{[0,t]}) \leq t\sigma/4$$

and uses reversible persistence restrictions to derive an architecture-independent dissipation certificate. Its uninterrupted residence observations differ from endpoint equality observations here. Its existence precludes presenting the comparison inequality as an original discovery; its unrefereed status means it does not independently validate this work.

The operational contribution pursued here uses the common positive spectral measure of several endpoint autocorrelations. It certifies distance from that moment set while propagating finite counts

and detector calibration into an entropy-production lower bound. A dual polynomial supplies a checkable lower certificate, which a numerical best fit alone cannot provide. Binary endpoint-pair records can be time-reversal symmetric even when their dependence on lag violates the reversible spectral restrictions. The inference concerns compatibility with a specified equilibrium model class, rather than an observed pairwise current.

A further preparation–maintenance extension uses the standard adiabatic and nonadiabatic decomposition of entropy production [105]. Initial relative entropy and subsequent housekeeping dissipation are already known physical resources. The candidate result is their combined lower constraint from an accessible departure from equilibrium temporal laws. No claim is made that the decomposition, initial nonequilibrium free energy, or data processing is newly discovered.

Randomized observation times themselves also have close prior art. A recent unrefereed manuscript uses two randomized clocks to identify fully observed generators and studies the sampling cost of rejecting hidden equilibrium explanations [109]. This strengthens the need to state the specific finite-record certificate, comparison class and error guarantee, rather than claim first use of random clocks for hidden dynamics.

67.4 What has and has not been established

This part provides theorem statements, proofs and reproducible synthetic examples. Independent mathematical checks and exact arithmetic support their internal validity. The candidate additions are the particular robust temporal certificates, their resource consequences and the construction of matching obstructions under weaker experimental contracts. The literature audit identifies their closest known ingredients and overlapping results. It does not prove historical originality.

Applications to a physical apparatus require evidence for homogeneous dynamics, preparation, readout, stationarity where invoked, and the relevant time-reversal convention. In particular, even-state Markov entropy bounds do not apply without modification to equilibrium systems with odd momentum variables or invasive quantum measurements. No new empirical data or independent peer review is reported. A practical next application should test an explicit laboratory contract and compare against an existing method on the same records, rather than infer engineering solutions from the breadth of the formalism.

Chapter 68

What the virtual laboratory establishes

The virtual laboratory tests whether the inference procedure works when records are generated through a physical model and an imperfect measurement process. It separates the hidden process, apparatus and statistical analysis. This makes it possible to locate a failure: a detector can erase a signal, a clock can change the model class, a reset can leave dependence between trials, or a confidence calculation can overstate the evidence. Each calls for a different correction.

The results have two levels. The proofs specify conditions under which a certificate is valid. The simulations measure the operating characteristics of particular implementations under specified data laws. Agreement between an independent simulator and an analytical prediction verifies the numerical model. Establishing that an apparatus obeys that model requires physical measurements. This distinction follows established verification and validation practice [113].

68.1 The experimental contract includes the apparatus

A temporal trace is generated jointly by dynamics, preparation, detector and clock. A fixed imperfect detector can often be included in an unknown response function. A detector that changes with the selected waiting time generally cannot. Fixed preparation errors can similarly be absorbed into an unknown initial state, whereas an unrecorded preparation variable correlated with the controls can enlarge the hidden state space.

This distinction makes a useful robustness question precise: does an imperfection leave the null class unchanged, merely weaken detection power, or change the class the theorem describes? The laboratory includes examples of all three. A counterfeit signal produced by setting-dependent readout is a demonstration of a missing apparatus assumption. It is not evidence against a theorem that explicitly requires a fixed detector.

Practical dimension witnesses and processor characterization already have a substantial experimental literature. Hendrych and collaborators demonstrated preparation–measurement dimension witnesses [123]. Gate set tomography jointly characterizes states, operations and measurement [125]. Proctor and collaborators detected and tracked drift in processor experiments [124]. These results support checking apparatus stability and retaining acquisition timestamps. They do not establish those conditions for the illustrative apparatus used here.

The rates, decoherence ratios and readout errors in this laboratory are *illustrative parameters*. They are not calibrated performance figures for a superconducting, trapped-ion, photonic or spin platform. A physical implementation would need joint evidence for its coherence, drive stability,

reset, detector and clock under the intended protocol. A favorable ratio of frequency to damping alone is insufficient.

68.2 Finite-data dissipation inference has close precedents

Hidden dissipation can already be inferred from apparently time-symmetric observations. Skinner and Dunkel established such bounds through optimization over compatible hidden models and applied their framework to biological data [115]. Waiting-time methods supply related information [116]. Finite-sample uncertainty in kinetic inference has also been treated directly [117].

The effects of imperfect resolution and finite statistics are particularly close to the study by Fritz, Ertel and Seifert [114]. Their comparisons show that finite-data estimates can exceed the underlying entropy production and that finer resolution need not improve performance at a fixed observation duration. The practical issue addressed here is therefore a certified lower confidence bound under an explicit observation contract.

For fresh binary endpoint records, the single-lag floor, monotonicity constraint, full reversible moment set and predeclared linear contrast use the same accessible data. Comparisons among them can match record count, confidence and exposure. A waiting-time estimator requires observed events that this contract does not supply. Full tomography likewise asks a different inference question with a richer menu of operations. Neither is a fair performance baseline if its required observations are withheld.

Combining observables to tighten thermodynamic bounds is established [126]. The specific improvement studied here compiles a polynomially certified equilibrium constraint into a single contrast with a finite-count guarantee. Its validity depends on selecting the contrast before the held-out records are examined. Optimizing it on those same records requires an additional selection correction.

68.3 Timing errors have different mathematical effects

Positive exponential mixtures are a standard consequence of reversible spectral theory [118]. Random-time semigroups and their functional calculus belong to classical subordination theory [119]. The timing results here apply those structures to the measurement contract.

For independent, identically distributed nonnegative time increments X_j,

$$\mathbb{E}\Big[e^{Q(X_1+\cdots+X_k)}\Big] = \Big(\mathbb{E}[e^{QX_1}]\Big)^k .$$

The same finite hidden state space therefore supports a common stochastic channel. Both the repeated-channel dimension certificate and the reversible spectral restrictions survive this type of timing uncertainty.

A common random scale, $T_k = kV$, has a different consequence. For each reversible spectral mode, $\mathbb{E}[e^{-\lambda kV}]$ is the kth moment of $x = e^{-\lambda V}$ on $[0,1]$. It preserves the equilibrium moment set. It need not preserve the same finite-state channel representation: mixing two clock speeds can turn one decaying mode into two. The clock contributes unobserved memory if its realization is not retained.

Arbitrary setting-dependent timing laws do not inherit either guarantee merely because their mean is correct or their errors are bounded. The relevant condition is the entire family of Laplace profiles used in the theorem. This provides an apparatus design rule: specify the joint timing mechanism before assigning it a statistical error bar.

The denominator in a dissipation-rate certificate must also describe physical exposure. Expected actual waiting time, nominal time and a maximum duration are different resource statements. An unbounded clock with finite mean has no pathwise duration cap. Ordinary integration over a detector exposure window also requires its own observation model; it cannot automatically be replaced by a randomly selected endpoint.

68.4 Dependence is handled through a declared reset guarantee

Independent trials make binomial intervals appropriate [122]. Chronological records can instead remain dependent after an imperfect reset. The dependent-record extension uses a uniform bound on the conditional bias of each bounded observation relative to its target probability. Conditional moment-generating-function bounds then control the fluctuations around those conditional means, and the reset bias enters as a separate deterministic allowance.

This is an application of established concentration machinery [17, 120]; mixing-based Markov concentration also has an extensive theory [121]. The physical content lies in justifying the conditional-bias bound. A known worst-case mixing or minorization bound can supply it after a specified blind waiting interval. A fitted correlation time from a short record does not establish that uniform condition. Chapter 34 supplies a separate common-block argument under a calibrated hidden-process contraction bound.

The supplied argument specifies predictable settings and fixed per-setting quotas. It does not automatically cover unrestricted data-dependent stopping. Chapter 33 implements a separate anytime construction using established confidence-sequence ideas [120], conditional mean bounds and predictable acquisition. It does not license optional stopping for this fixed-quota result.

68.5 Matched design and reproducible evaluation

The simulation plan separates design data from evaluation data, fixes the analysis before held-out trials, and retains unsuccessful regimes. This follows standard simulation methodology [112]. The report should identify the generated model, accessible records, target quantity, competing methods, performance measures and random seeds. Independent implementations should cross-check physical propagation and statistical evaluation where possible.

A scalar contrast can reduce the loss from demanding simultaneous confidence for every coordinate. Allocating samples in proportion to coefficient size, and adjusting by the square root of per-record cost, follows classical optimal stratified allocation [127]. Its use here provides a direct connection between the chosen physical witness and the measurement budget; the allocation principle itself is established.

Finite simulation repetitions have uncertainty of their own. If no failures occur in R independent repetitions, the one-sided 95% exact binomial upper limit on the failure probability is $1 - 0.05^{1/R}$. A reported zero count must therefore retain that uncertainty. Power comparisons on common records should use paired outcomes, while comparisons with different acquisition menus must state the additional access and exposure.

The contribution of this laboratory is an auditable bridge from model theorem to executable protocol: it identifies which imperfections preserve a certificate, quantifies the resulting confidence and power, and gives explicit counterexamples when an assumption is removed. Physical validation, external review and historical priority remain separate stages of evidence.

Chapter 69

Calibrated inference in context

69.1 What stronger access adds

The distinction between detecting irreversibility and determining total entropy production is central to thermodynamic inference. Optimizing over hidden models compatible with observations is established [132, 133]. Coarse descriptions can omit microscopic currents [134]. An experimentally useful result must therefore specify which lower bound follows from the recorded evidence, which extra calibration tightens it, and which unobserved changes leave that evidence unchanged. A numerical minimum over a chosen hidden architecture is not a proof of the minimum over every physical realization.

Stationary means and variances are physical information. Fixing them can shrink a permissible correlation class. This use of calibration does not introduce a new spectral theorem: reversible autocorrelations already have positive spectral measures [118]. The contribution to assess is the resulting finite-observation certificate, including uncertainty in the calibration itself. The zero-lag response of two separate noisy readouts must not silently be replaced by the square of a single recorded value.

69.2 Closest thermodynamic comparators

Correlation-based entropy bounds are an active subject [100, 101, 106]. The Raux–Maier–Seifert preprint studies cross-correlation asymmetry and optimizes observable shifts and lag. Its numerator vanishes when both observables are the same; the single-observable moment constraints used here answer a different measurement question. Comparisons require the same observables and statistical budget, not merely a common underlying simulated process.

The variance sum rule gives a particularly relevant physical comparison [135, 136]. For overdamped steady dynamics, its full version uses displacement and force statistics together with mobility and thermal calibration. Its solvable switching-trap example already diagnoses strong nonequilibrium through position-correlation curvature. Consequently, a finite-lag curvature certificate would strengthen the observation and uncertainty treatment of an established effect. It would not constitute its first discovery. Force-assisted reconstruction is a valuable benchmark when those additional measurements are retained.

Finite resolution and finite statistics can alter the performance and bias of entropy estimators [114]. The present comparisons should retain these errors and report a lower confidence bound, rather than identify a positive fitted value with a certified physical quantity.

Activity calibration connects the path-divergence bridge to classical divergence comparisons.

Normalizing stationary directed edge fluxes by the total jump activity gives two probability distributions related by edge reversal. Their Jensen–Shannon divergence is the relative-entropy rate to the additive reversibilization divided by activity; their half-Jeffreys divergence $[D(P\|R) + D(R\|P)]/2$ is entropy production divided by activity. Linear and nonlinear comparisons between these divergences already exist [140, 141], and sharp f-divergence comparison is a developed subject [142]. The activity-calibrated envelope proved here is positioned as a precise specialization and operational composition with the record certificate; its proof does not establish historical priority over that literature. Full-path equality does not imply equality after a noisy endpoint observation.

The converse inference boundary also has established foundations. Lumpability can preserve an observed Markov law while microscopic mechanisms vary [68]. Resolving a single bidirectional edge can recover the complete steady entropy production in a unicyclic network [143, Sec. V.2]. This asymptotic identifiability does not by itself provide a finite-sample upper confidence bound when reverse events can be arbitrarily rare. The fixed-graph constructions in this part make that distinction quantitative.

69.3 Statistical tools and their conditions

Reverse information projections, log-optimal e-variables, and composite-null safe testing are established constructions [131, 129]. General constraint-generated null hypotheses have an extensive dual theory [130]. Using a calibrated reversible moment set as the null is an application of that theory. The mathematical and software questions here are whether its infinite family of physical constraints is represented soundly and whether the numerical certificate can be independently checked.

An e-variable valid for one stationary marginal distribution cannot simply be multiplied over an arbitrarily dependent trajectory. Conditional validity or a valid trajectory-level construction is required. Likewise, data-dependent selection of a contrast needs a held-out record, predictable construction, or an explicit selection correction.

For stationary Markov chains, Fan, Jiang and Sun give a Hoeffding inequality with dependence penalty $(1 + \rho)/(1 - \rho)$ when the mean-zero transition operator has norm at most $\rho < 1$ [128]. This applies to bounded functions and does not require reversibility. Such a calibrated contraction bound can replace long blind gaps. A lagged observation pair, however, is a function of a path segment, and its appropriate block-chain contraction must be justified. The observed autocorrelation of one detector does not bound mixing of every hidden degree of freedom.

69.4 Public experimental records and the transfer boundary

Two primary datasets were audited. The switching-trap data accompanying the variance sum rule list timestamps, position and force measurements [137]. Their documented apparatus and model make them a promising future comparison. The raw archive download was unavailable in this audit; metadata, the source manuscript with supplementary material, and the companion analysis notebook were accessible. No result from those unretrieved trajectories is claimed.

Selected records from the optical feedback-trap study of Bryan and colleagues were downloaded [138, 139]. The repository distinguishes experimental trajectories from simulations. Its files called `groundtruth` contain simultaneous lower-noise measurements, not exact latent positions. They support a reproducible detector and data-format audit. The selected harmonic-well experiments are described as using an optical tweezer without feedback; feedback generates other potentials in the same study. The authors' code documents a nominal $10\,\mu$s sampling interval, but the selected files contain no per-sample timestamps. Stationary preparation, hidden-state mixing, clock error

and detector calibration still require an explicit contract before an entropy-production confidence statement can be transferred to these records. Descriptive statistics of the downloaded measurements are kept separate from such a certificate.

A successful reanalysis would be evidence about published measurements under a documented model and calibration. It would not be a new measurement of the apparatus, nor a measurement of the unrecorded energy expenditure of its laser, controller, and detector.

Chapter 70

Positioning the direct correlation result

70.1 Relation to existing thermodynamic inference

The central practical question is how much dissipation is forced by a specified observation law. Optimizing over hidden models compatible with observations is an established approach. Ehrich constructs compatible masked Markov models and minimizes their entropy production [132]; the inversion assumes the correct hidden-state number. Nitzan, Ghosal and Bisker formulate an optimization over a specified CTMC topology using observed transitions and waiting-time moments [133]. Their moment hierarchy progressively tightens the compatible-model bound. Their numerical experiments with a smaller hidden topology do not establish a universal reduction to that topology; the paper explicitly leaves that question open.

This distinction matters for interpreting computations. A feasible physical model provides an upper bound on the *minimum compatible dissipation*. It need not bound the dissipation of the actual system from above. Conversely, a local numerical minimum in a selected architecture need not provide a valid lower bound over a larger physical model class. The direct correlation certificate developed here and the solvable circulant optimization serve different purposes: the former is valid without a specified hidden graph, whereas the latter resolves an explicitly restricted realization problem.

70.1.1 The particular role of a scalar finite-lag defect

Reversible autocorrelation functions have positive spectral measures, and their log-convexity follows from Cauchy–Schwarz. Those facts predate this work. Thermodynamic information in correlation functions is also well established. Ohga, Ito and Kolchinsky relate cross-correlation asymmetry to cycle affinity [99]. Shiraishi bounds fluctuation oscillations by entropy production [100]. Van Vu, Vo and Saito extend entropy and affinity bounds to finite-time cross-correlations [147]. In particular, their equation (13) bounds a squared cross-asymmetry divided by autocorrelation decay. Their Appendix D, equations (112)–(117), uses the contraction of path relative entropy to the joint endpoint law. Raux, Maier and Seifert give another arbitrary-lag bound for two-observable cross-asymmetry [106]; the consulted version is a preprint.

With one common observable, the numerator of each of these two-observable asymmetry tests vanishes. The finite-lag construction here instead uses

$$\Delta_s = -s^2 C(0) + 2sC(t) - C(2t).$$

Writing the stationary transition operator as $P_t = H_t + B_t$, with H_t self-adjoint and B_t skew-adjoint, gives

$$\Delta_s = \|B_t f\|_\pi^2 - \|(H_t - sI)f\|_\pi^2.$$

Thus a quantity available from one scalar autocorrelation bounds an unobserved antisymmetric response from below. The subsequent conversion to dissipation uses established relative-entropy contraction and bounded-response inequalities. The contribution to be assessed is this particular finite-lag combination and its calibrated confidence procedure. It is neither a new data-processing inequality nor the first use of correlations for thermodynamic inference.

The sharper nonlinear conversion also has an established statistical ingredient. With the convention $J(P,Q) = [D(P\|Q) + D(Q\|P)]/2$, Sason's Proposition 4 states

$$\min_{\mathrm{TV}(P,Q)=z} J(P,Q) = 2z \operatorname{atanh} z \qquad (0 \le z < 1)$$

[148]. Applying this inequality to conditional transition laws, then averaging with the appropriate convexity argument, is distinct from claiming the divergence inequality itself. General sharp-divergence and thermodynamic-response results provide further context [142, 141].

Time-symmetric scalar fluctuations already support other entropy estimators. The variance sum rule and its proofs derive a lower bound using the initial curvature of displacement variance in overdamped Langevin dynamics, with calibrated diffusion coefficients [135, 136]. Full force-assisted measurements can yield an equality. The present bounded observable uses specified nonzero lags and a stationary Markov-semigroup identity; it does not require estimating a derivative at zero. These are different access assumptions, and neither method should be described as universally stronger without a comparison under matched access and errors.

70.1.2 Circulant optima and the limits of a spectral interpretation

The thermodynamic cost of coherent oscillations has an extensive literature. The affinity-and-topology constraint of Barato and Seifert [98] and its proof by Ohga and coworkers [99] must be distinguished from an entropy-production bound using a relaxation eigenvalue alone. Gu proves

$$\sigma \ge \eta \frac{\omega^2}{\gamma}, \qquad \eta = \frac{\|v\|_{2,\pi}^2}{\|v\|_\infty^2},$$

for an oscillatory eigenmode v with eigenvalue $-\gamma + i\omega$ [107]. In that paper, Section V explicitly treats arbitrary translation-invariant jump processes on a ring, establishes $\eta = 1$ for Fourier modes, and gives the nearest-neighbor ring formulas. The drift-diffusion limit on a circle attains $\sigma = \omega^2/\gamma$ there. That continuum envelope and its saturation are established results. The finite-size minimization over all allowed circulant displacements is a more specific optimization problem, whose proof is supplied in this work.

Kolchinsky derives a dissipation-coherence inequality from a condition on phase-current fluctuations and proves it in the weak-noise Gaussian regime, with some extensions beyond Gaussian fluctuations [150]. This is not an unconditional eigenvalue-only theorem for arbitrary finite Markov networks. Shiraishi's counterexample retains a finite oscillatory eigenvalue while making its driven part increasingly rare, with stationary entropy production tending to zero [108]. The consulted counterexample is an unrefereed preprint. It illustrates why observable amplitude and stationary population cannot be removed from a universal inference claim. The scalar Δ_s preserves those weights.

70.1.3 Temporal access, uncertainty and implementation

Sampling a continuous-time system at a fixed interval can leave its generator ambiguous. System aliasing and the associated matrix-logarithm branches are established identification problems [149]. The circle diffusion construction makes their thermodynamic consequence explicit: different drifts can produce the same sampled transition kernel while dissipating very different amounts. A finite set of irregular lags requires a separate finite-precision analysis. Approximate aliases from simultaneous Diophantine approximation concern a fixed finite timing menu and a bounded record length; they do not automatically establish an impossibility result for continuously randomized timing, unlimited records or every adaptive experiment. Randomized clocks under detector deadtime are studied in the recent spectral-access preprint [109].

The analog acquisition scheme combines standard conditional-variance identities, bounded-observation concentration and direct use of the measured record. Its value is an explicit interface between the detector contract, calibration uncertainty and a physical lower bound. A squared noisy reading has second moment $B = \mathbb{E}Z^2$, while the latent conditional mean has $A = \mathbb{E}(\mathbb{E}[Z \mid X])^2 \leq B$. They are different quantities. The sign of the calibration coefficient makes that distinction usable in a conservative joint inequality, while retaining its potential loss of power. The implementation and acquisition-matched simulations evaluate this complete procedure; they do not replace laboratory validation.

The primary sources checked in this audit do not state the precise scalar finite-lag entropy certificate and joint analog procedure presented here. That finding supports a focused contribution claim, while leaving historical priority to broader specialist review. A formal result within the stated stationary classical models does not establish a new law for arbitrary quantum, inertial or nonstationary systems.

Chapter 71

Positioning the observed-geometry result

The new access contract belongs to kinetic inference and the variance-sum-rule line of work. Symmetric fluctuations can already reveal dissipation under appropriate overdamped assumptions. Minimum dissipation among compatible hidden models is also an established inference objective. The contribution to assess here is a conditional diffusion-geometry bound and its sharp compatible-realization consequence for a specified complete observed law.

71.1 The closest curvature identity

Di Terlizzi's force-free kinetic formula expresses stationary entropy in Cartesian overdamped coordinates with constant diffusion through covariance curvature and the score of the stationary density [152]. For diagonal diffusion, its component identity is

$$\sigma_i = -\frac{C_i''(0)}{D_i} + D_i\,\mathbb{E}[(\partial_i \log \pi(X))^2]. \tag{71.1}$$

The paper's Eqs. (15)–(18) derive partial-observation bounds from positive traffic components. Its full-state identity also gives a useful marginal corollary. When differentiation under the hidden-coordinate integral is valid and boundary terms vanish,

$$\mathbb{E}[\partial_i \log \pi(X) \mid X_i = y] = \partial_y \log p_i(y).$$

Conditional Jensen therefore yields

$$\sigma_i \geq -\frac{C_i''(0)}{D_i} + D_i\,\mathbb{E}[(\partial_y \log p_i(X_i))^2]. \tag{71.2}$$

This is a direct corollary of an established identity. It is the signed, constant-diffusion specialization of the present conditional argument, before taking a positive part locally. We do not claim a new underlying curvature–score identity.

The present result retains more local information: a nonlinear detector f, its instantaneous quadratic variation $V = \Gamma_f(X)$, and the conditional combination $g^2 - K$ before averaging. On the cosine circle, g^2/V and K/V have divergent separate averages near the endpoints, whereas $(g^2 - K)/V$ is finite. Truncation and cancellation before integration are essential. Treating this as the subtraction of two finite global averages would be incorrect. The transformation $\arccos Y$

also produces folding endpoints; a globally smooth constant-diffusion coordinate formula does not automatically apply to it.

The earlier variance sum rule relates displacement curvature, force variance and dissipation [135, 136]; the latter reference gives the entropy identity explicitly in Sec. 3.3, Eq. (30). Harada–Sasa instead relates dissipation to correlation and response spectra [153]. Its perturbation-response access differs from the passive conditional statistics used here. These established methods remain appropriate when their respective measurements and physical assumptions are available.

71.2 Standard ingredients and the extra information retained

The drift decomposition, generator adjoint and stationary diffusion entropy are standard. Da Costa and Pavliotis give these explicitly in Propositions 3.7 and 3.12 and Theorem 4.1; their Sec. 6.2 distinguishes ordinary reversal from reversal with a phase-space involution [154]. That distinction is required here. Observing an even scalar does not prove that all hidden degrees of freedom have even parity.

Short-time current optimization already attains diffusion entropy when the appropriate current is available [155]. Stochastic force inference already uses finite-basis regression, estimates spatially varying diffusion, and handles hidden variables through conditional projection [156]. Its Appendix E expressly limits the projected entropy to observable currents. The present conditional past–future product retains information that is absent from the mean projected current of a reversible scalar law.

There is a related access distinction. Regression of squared increments on Y estimates a conditional average of the hidden local diffusivity. It need not recover the pathwise coordinate $V = \Gamma_f(X)$ used in the weighted theorem. Replacing one by the other without a pointwise bound is not justified by a successful regression fit.

The completed-square inequality $2qg - q^2 \le g^2$ is elementary least-squares duality. Density-score matching has a long history, including Hyvärinen's integration-by-parts formulation [157]. The regression test function for a conditional drift is not automatically a density score. Neither duality nor the use of an independent training sample is claimed as a new mathematical principle.

71.3 The compatible-model minimum

Ehrich minimizes entropy over hidden Markov transition matrices reproducing observable jump statistics, with an explicit minimum in Eq. (51) and a comparison to reversed-record information in Eq. (52) [132]. His discussion identifies the dependence on the assumed number of hidden states. Nitzan, Ghosal and Bisker formulate full waiting-time-law compatibility and a moment hierarchy in Eqs. (16)–(18), assuming a specified underlying topology [133]. Their topology-reduction extension is tested numerically and retained as an open universal question.

The cosine-law theorem addresses a different instance: every compatible finite-dimensional smooth ordinary-reversal hidden diffusion in the stated class must obey the same local observed-path identities, and the circle attains the resulting lower bound. Its force is the absence of a prescribed hidden dimension or architecture within that class. It does not enlarge the claim to jumps, generalized reversal, noisy detectors or equality of a few covariance values.

The scalar Gaussian reversal obstruction discussed by Lucente and colleagues [26] concerns what a scalar record and broader equilibrium realizations can establish. The cosine law is non-Gaussian, and the current minimum explicitly restricts the realization class. A reversible observed law need

not have a reversible generator within every chosen class of hidden models. Reversed-record KL and minimum compatible physical dissipation are therefore different quantities.

71.4 What finite-resolution validation adds

Di Terlizzi's numerical procedure fits sums of exponentials to correlations and studies exclusion of noise-contaminated small lags; stochastic force inference also treats sampling and localization errors [152, 156]. Those are useful practical precedents. A uniform finite-lag remainder and a lower confidence guarantee for a frozen triple statistic are separate obligations. Generator-derivative envelopes, sensor calibration and independent acquisitions must be stated explicitly; they are not automatically learned from a coarse trace. A successful simulation using known calibration constants demonstrates a conditional procedure, not universal data-only recovery.

The audit consulted primary equations and supplements. Inspection notes and source locations are retained in `geometry/research/sources.json`. The audit does not certify historical priority. External review and physical experiments remain independent steps. The claim advanced here is a specific mathematical and computational extension with an explicit access contract, not a new universal law of matter.

Chapter 72

Positioning calibrated detector access

The detector results join an established inverse-problem literature to a specific thermodynamic inference target. Exposure correction, deconvolution, noisy volatility estimation and inference of dissipation from imperfect records all have substantial prior work. The contribution to assess is the connection proved here: the sharp compatible-model minimum can survive a calibrated detector at the complete-law level while its finite-data measurement requirements diverge. The matching count and detector-energy bounds concern stated binary experiments, not an unrestricted inference problem or a new law of measurement energy.

72.1 Exposure inversion and what its zeros mean

Landweber iteration is a classical inverse method [158]. Its application to the boxcar filter in Chapter 53 is standard spectral regularization. A zero of the transfer function loses an exact component only when the signal's spectral measure gives that zero positive mass. Covariance decay excludes the resonant atoms relevant here. Recovery from the complete continuous law can therefore hold despite isolated filter zeros, without providing a bounded inverse or stable reconstruction from finitely many measurements. The explicit averaging inverse and its error bound make this distinction checkable; fixed-grid aliasing remains a separate obstruction.

Known-error deconvolution also has a substantial theory when the error characteristic function has zeros. Hall and Meister's ridge construction and rate bounds treat such cases under quantitative assumptions [159], especially their Theorems 4.2–4.3. Those zeros are in the Fourier variable dual to measurement amplitude, whereas shutter zeros are in temporal frequency. They are different operators. The bounded-law convolution argument here uses analytic uniqueness; it neither introduces a new general deconvolution principle nor supplies stable analytic continuation. Dense independent readouts are an additional access assumption, not a consequence of knowing a sensor's one-time error distribution.

The generator-resolvent identity used for the spectrum is also established; Gupta and Khammash give an explicit Markov-process version in Eqs. (6)–(9) [160]. The narrower observation here combines that identity with strict ellipticity to identify a boxcar width from its population spectral zeros. It assumes a known boxcar shape, calibrated time and the specified diffusion class. No unknown-kernel or finite-record spectral-zero estimator is established.

72.2 Diffusion inference through cameras and noisy observations

Berglund derives the shutter and localization-error statistics of free Brownian tracking, including Fisher information and the effect of the illumination profile [161]. Vestergaard, Blainey and Flyvbjerg develop covariance estimators, calibration corrections and finite-sample confidence calculations for that model [162]; see Sec. III C and Appendix C. These give a strong practical baseline. Comparing a proposed detector correction only with naive realized variance would understate existing capabilities.

Integrated-diffusion estimation predates those camera applications. Gloter's Theorem 3.1 and Remark 3.2 show the factor 2/3 in quadratic variation computed from successive shrinking boxcar averages [163]. Nakakita and Uchida estimate an unknown exposure-to-frame ratio from quadratic-variation ratios and consistently estimate diffusion and drift parameters under explicit assumptions [164], Theorems 1–4. Their exposure width shrinks with the frame interval, while total observation duration grows. Unknown shutter calibration is therefore an established research topic. The fixed-width, complete-law result here concerns a different regime.

For noisy point observations, pre-averaging estimates integrated volatility with a stable central limit theorem [165], Theorem 3.1. Hoffmann, Munk and Schmidt-Hieber estimate the entire possibly random volatility path using pre-averaging and adaptive wavelets [166], Theorems 2.9 and 2.12. These results require their noise, regularity and asymptotic sampling assumptions. They do not directly supply the pathwise geometry $V = \Gamma_f(X)$, jointly with the conditional short-time statistics in this volume, under fixed nonzero shutter exposure. Averaging a continuous path over that fixed width gives zero ordinary quadratic variation. Its replacement by a point-sample model, or replacement of V by $\mathbb{E}[V \mid Y]$, would require a separate justification.

72.3 Noisy thermodynamic inference

Stochastic force inference already treats diffusion estimation, projected currents and measurement errors [156]. The force-free kinetic approach also examines the treatment of noise-contaminated short lags [152]. Their relation to the population geometry bound is discussed in Chapter 71.

Gingrich and colleagues directly study thermodynamic inference from noisy single-molecule records [167]. In Sec. II of the inspected preprint, Eq. (13) expresses the noisy two-point law using a memoryless measurement kernel; Eq. (15) inverts that transformation when its matrix is invertible. Eqs. (19)–(22) relate finite-block time-reversal information to lower bounds through data processing. The measured record generally has memory, and increasing temporal resolution alone can worsen particular estimates in the presence of noise. These are close precedents for the present access distinction.

That memoryless detector model does not include shutter convolution or calibration of local quadratic variation. Moreover, deconvolving a sampled transition law does not by itself identify its continuous-time generator. A matrix-log reconstruction requires its own uniqueness conditions; it cannot remove the fixed-grid aliases exhibited here. The present target is also distinct from the noisy record's reversed-path KL: it is the minimum ordinary-reversal diffusion entropy among models compatible with the ideal law. The stated realization and detector classes are indispensable to that conclusion.

72.4 The Gaussian testing mechanism is established

The second-moment method for Gaussian location mixtures is standard in statistical testing [14]. Cai and Wu's Lemma 3 in the inspected preprint states the precise identity used here, attributing it

to Ingster and Suslina [168]. For independent copies A, A' and Gaussian read error of variance ν^2, it gives

$$\chi^2\Big(\mathcal{L}(A+\xi)\,\|\,N(0,\nu^2)\Big) = \mathbb{E}e^{AA'/\nu^2} - 1 = \sum_{k\geq 1} \frac{(\mathbb{E}A^k)^2}{k!\nu^{2k}}$$

when boundedness justifies the expansion. A zero signal mean removes the $k = 1$ term. An oriented preparation can restore mean information; stationarity and sign symmetry can leave only even moments. This familiar distinction between mean and variance information is the mechanism behind the different fixed-precision count exponents. It is not a new Gaussian-channel principle.

The restricted result proved here connects that mechanism to exact exposure moments and the sharp ideal-law dissipation minimum, obtaining matching orders for a specified high-frequency binary family. Allowing precision to consume energy under the declared law $\nu^2 = \kappa/E$ changes the comparison: the passive and prepared detector-energy orders agree even though their fixed-precision count orders differ. This conclusion depends on that sensor law and omits preparation, reset and control energy. Neither equal optimal constants nor a universal total-energy limit is asserted.

72.5 The remaining experimental bridge

The next practical obligation is a finite-data joint confidence statement for the detector-corrected local geometry and the entropy statistic, with shutter, noise and regularity calibration errors propagated through both. An inverse at the level of complete laws is insufficient. A useful benchmark must match duration, exposure, readout precision and preparation access, and must include the established corrected estimators above.

The source ledger records the primary versions and equation locations inspected for this comparison. This targeted audit does not establish historical priority or exhaust the literature. The results support a specific, reproducible extension of an existing inference programme; external mathematical review and physical validation remain separate steps.

Chapter 73

Calibration, replicated measurements and the limits of extrapolation

The calibration extension addresses a specific inference problem. A bounded observable has the pointwise geometry

$$\Gamma_f = \gamma(1 - f^2), \qquad |f| \le 1, \tag{73.1}$$

with an unknown positive scale γ. Independent detector randomizations provide repeated readings of the same latent exposure. Constant and timed-step reference signals constrain a common multiplicative gain and a uniform shutter width. A finite collection of jointly covered moments then gives an entropy lower bound, including uncertainty in those calibration quantities. The structural identity and the supplied generator bounds remain premises of the physical model. The reference measurements do not establish either premise.

Several ingredients have substantial precedents. The result should be assessed as their explicit connection to this structural entropy problem, including its bias bounds, finite-data coverage and failure cases. Neither the use of replicas, confidence-set projection nor a search of the literature establishes priority for the complete construction.

73.1 What repeated readings identify

Deconvolution with repeated measurements is established statistical theory. Delaigle, Hall and Meister [169], Section 2.1, equations (2.1)–(2.4), use repeated additive measurements of one latent variable to estimate the error characteristic function and then the signal density. Their particular construction assumes a real, nowhere-zero error characteristic function; Theorems 3.1–3.3 compare risks with known-error procedures under smoothness restrictions. More recently, Capitao-Miniconi, Gassiat and Lehéricy [170], Theorem 1, identify the signal and two independent error laws up to translation under a signal Laplace-transform condition that includes bounded signals. Their identification theorem does not require error moments, symmetry or nonvanishing characteristic functions. Their quantitative estimation results, including Proposition 2, impose additional class restrictions.

The present acquisition uses a simpler finite-moment consequence of conditional independence. If each bounded readout Z_r has conditional mean A, the common latent exposure, then

$$\mathbb{E}\left[\prod_{r=1}^{k} Z_r \,\middle|\, \text{hidden path}\right] = A^k \tag{73.2}$$

for distinct independent detector randomizations. This identity also covers conditionally unbiased shot readouts whose distribution depends on A; it does not require an independent additive-noise representation. Boundedness supplies finite-sample concentration. With unrestricted unknown tails, unbiased products alone would supply no such uniform numerical guarantee.

These are genuine repeated observations of the same exposure, not independent systems with the same marginal distribution, and not copies of one stored number. Shared sensor errors can invalidate the factorization. Repeating an exposure does not undo its temporal averaging. For the stationary cosine circle, the one-time marginal is independent of both angular drift and diffusion scale, so repetitions at one instant cannot identify those dynamics. The temporal moments and reference signals are additional physical access.

73.2 From covered moments to a covered entropy bound

Inference through moment restrictions remains useful when point identification fails. Kato, Sasaki and Ura [171], Theorem 1 and equation (3), derive linear restrictions from repeated additive measurements without requiring the inversion used in conventional deconvolution. Their Theorem 2 gives uniform coverage of at least $1 - \alpha - Cn^{-c}$ under specified class assumptions. This is a quantitative guarantee, although its class-dependent constants and approximation conditions are not an immediately usable exact α-level guarantee for the present detector.

Here the target statistics form a declared finite family. Bounded-mean inequalities [17, 151] and a joint error allocation cover the target and reference moments simultaneously. Optimizing over the resulting nuisance region is legitimate because that region contains the true quantities on the same event. Sharing records across moments does not invalidate a union bound; treating a score fitted to those records as though it had been fixed in advance would require a separate argument. Confidence-set mapping itself is standard. For example, Söhl and Trabs [175], equation (3.6) and Proposition 3.4, transfer an invariant-density confidence set through the map from density to drift.

The geometric step specific to this construction is the finite polynomial hierarchy

$$w_m(f) = \sum_{j=0}^{m} f^{2j}, \qquad w_m(f)\Gamma_f = \gamma(1 - f^{2m+2}) \le \gamma. \tag{73.3}$$

This replaces division by the vanishing factor $1 - f^2$ with estimable polynomial moments. At degree m, the specified score uses $2m + 2$ independent center readouts. Temporal second moments cover the unknown scale γ; replicated higher moments provide the weighted entropy score. The argument does not substitute a conditional mean diffusivity for a pointwise geometry. On the cosine-circle model the ideal hierarchy approaches its full entropy rate. The constructive approach with estimated calibration also increases replica access, shrinks the time step and shutter width, and increases the number of trials. It is not a claim of useful sample complexity or consistency at fixed positive exposure.

73.3 Appropriate detector and diffusion baselines

Berglund [161] and Vestergaard, Blainey and Flyvbjerg [162] derive the effects of localization error and motion blur on free-Brownian trajectories. Vestergaard et al., Section III C and Appendix C, provide covariance estimators and finite-sample distribution and confidence calculations. These are substantive baselines, not merely uncorrected mean-square-displacement fits. In their centered

increment model, fixed frame spacing Δ gives

$$\mathbb{E}[(\Delta x_n)^2] = 2D\Delta + 2q, \qquad \mathbb{E}[\Delta x_n \Delta x_{n+1}] = -q, \\ q = \sigma_{\mathrm{loc}}^2 - 2DR\Delta. \tag{73.4}$$

Thus diffusion can be estimated without separately identifying the blur coefficient R and localization variance from these increments. Gaussianity makes the covariance the complete centered increment law. This confounding statement is specific to that observation model.

For confinement, Calderon [172], Section II A, equations (1)–(10), derives a likelihood for a blurred Ornstein–Uhlenbeck trajectory with localization error, retaining the correlation between transition noise and exposure noise. The exposure time is supplied; drift, confinement, diffusion and localization parameters are estimated. Nakakita and Uchida [164], Theorem 1, instead estimate an unknown exposure-to-frame ratio in a shrinking-exposure, increasing-duration regime; their model does not include the general independent localization-noise channel considered here. Lindén et al. [173], Methods equation (6), use an image model and a Laplace approximation to obtain pointwise localization uncertainty, with camera-noise calibration supplied. That method requires pixel information beyond a retained scalar readout.

A free-Brownian covariance estimator is a suitable control for the free-Brownian acquisition model. It is not an estimator of the same entropy target for a bounded cosine observation of a circulating circle. A comparison on that target requires an adapted circle forward model, matched observations and resources, and uncertainty calculations for its target functional. Likewise, an Ornstein–Uhlenbeck likelihood is a confined calibration control, not a hidden-circle entropy estimator without adaptation.

The constant and step references in the present protocol supply independent metrological information. A common gain κ gives mean κ on the unit reference and mean $\kappa \min(1, s/\tau)$ on a unit step read at known time $s > 0$ through a causal uniform shutter of width τ. Inferring (κ, τ) from these covered means is ordinary calibration. Its role here is to propagate a joint nuisance region through the entropy certificate. Common response between reference and target, known reference shape and time origin, and the uniform shutter model are explicit assumptions; they are not learned from the target trajectory alone.

73.4 Entropy inference and the scope of empirical calibration

Noise-aware inference of diffusion, currents and entropy already exists. Frishman and Ronceray [156], equations (12)–(14) and Appendices E–G, treat diffusion and force inference, localization corrections, projection and uncertainty. Gingrich et al. [167], Section II, equations (13), (15) and (19)–(22), treat noisy thermodynamic inference using a known memoryless error kernel. Neither comparison warrants claiming that noise correction or entropy inference starts here. A projected scalar current can vanish for the reversible cosine record while a bound conditional on the admissible hidden diffusion geometry is positive: these are different targets. Known-kernel pair-law recovery also does not establish unrestricted continuous-time generator identification from a fixed sampling grid.

Conversely, fixed-grid observations are informative in important restricted classes. Gobet, Hoffmann and Reiß [174], Section 3 and Theorems 2.4–2.5, estimate drift and diffusion of a reflected scalar diffusion on a compact interval using its self-adjoint transition operator and Sobolev regularity assumptions. Their result prevents a blanket claim that all fixed-grid diffusion inference is impossible. Söhl and Trabs [175], Assumption 3.1, impose drift regularity and recurrence conditions for their scalar constant-volatility model. Their adaptive bands add the self-similarity condition in Assumption 4.1; Theorem 4.6 gives asymptotic coverage and shrinking sup-norm diameter. These are

structured positive results, not permission to replace a required numerical envelope by an empirical average.

Chapter 58 supplies the complementary failure case: rare smooth regions can make global diffusivity or generator suprema arbitrarily large while leaving finite-duration trajectory laws arbitrarily close. Its confidence consequence uses the standard transfer of probabilities through total variation [14]. The construction retains fixed dimension, a common ellipticity lower bound and a common mixing gap; every individual model is smooth, but the class has no common regularity envelope. The obstruction concerns unseen global suprema, even with ideal continuous observations. It neither contradicts pathwise local-volatility recovery nor identifies the same problem as adaptation of confidence bands over smoothness classes. All those countermodels are reversible.

The resulting scope is concrete. Supplied geometry and regularity, genuine replicas, temporal moments and calibrated references can support a computable finite-data entropy certificate with simultaneous nuisance coverage. Unsupported global extrapolation cannot. Numerical experiments can test the implementation and show when a certificate is informative; they do not validate the physical premises or constitute an empirical discovery. The primary-source audit records the inspected versions and exact comparison points without asserting an exhaustive novelty search.

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